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Labokha et al.(10) **Patent No.: US 12,404,330 B2**(45) **Date of Patent: Sep. 2, 2025**(54) **ANTIBODIES TO ICOS**(71) Applicant: **KYMAB LIMITED**, Cambridge (GB)(72) Inventors: **Aksana Labokha**, Cambridge (GB);
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Arkininstall, Cambridge, MA (US)(73) Assignee: **KYMAB LIMITED**, Cambridge (GB)(*) Notice: Subject to any disclaimer, the term of this
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None

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Primary Examiner — Zachary S Skelding*Assistant Examiner* — James Ryland Melchior(74) *Attorney, Agent, or Firm* — LATHROP GPM LLP;
James H. Velema, Esq.(57) **ABSTRACT**Antibodies that bind ICOS (Inducible T cell Co-Stimulator).
Therapeutic use of anti-ICOS antibodies for modulating the
ratio between regulatory T cells and effector T cells, to
stimulate the immune system of patients, including use in
treating cancers. Combinations of anti-ICOS antibodies and
other drugs for immunooncology.**16 Claims, 71 Drawing Sheets****Specification includes a Sequence Listing.**

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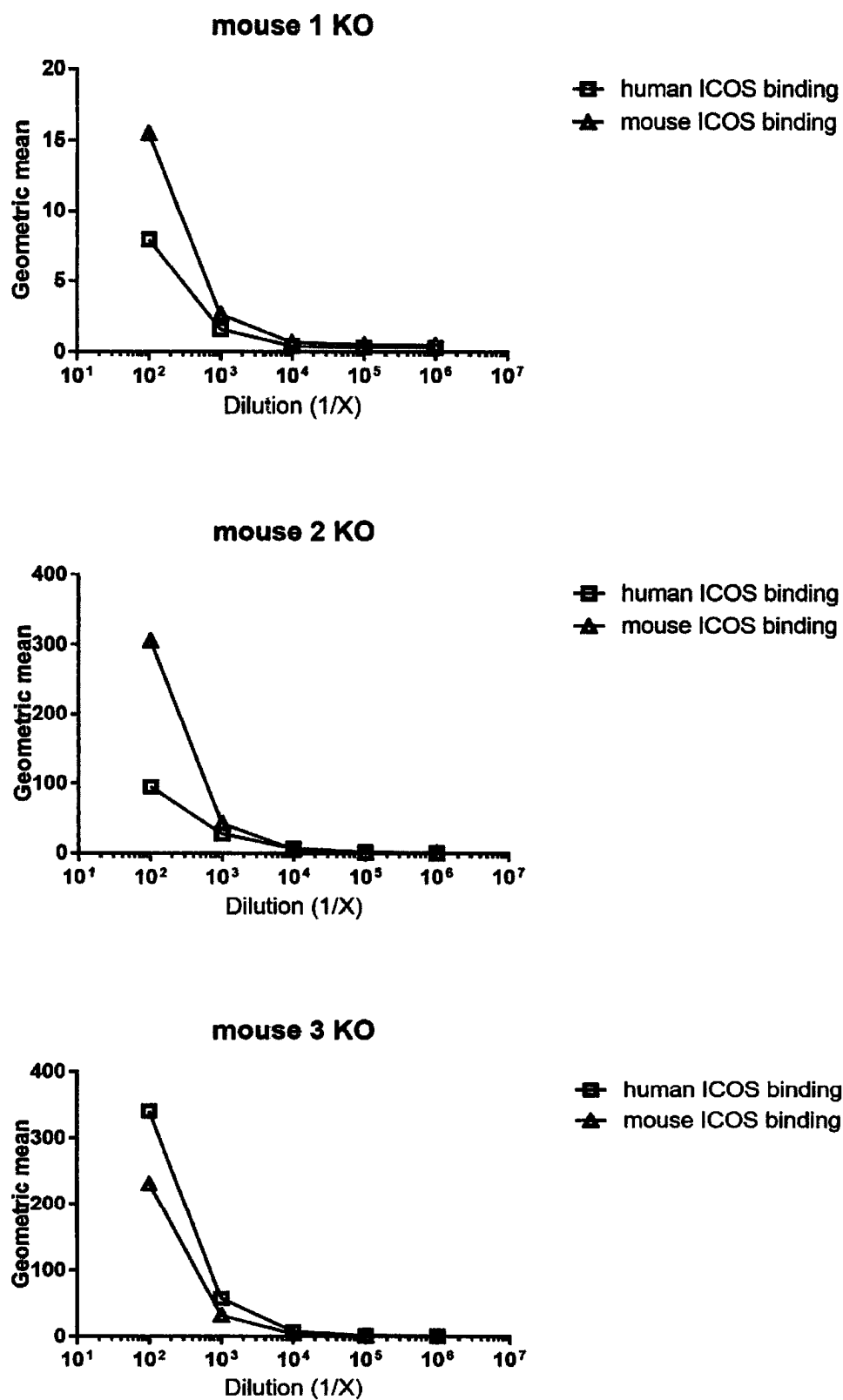


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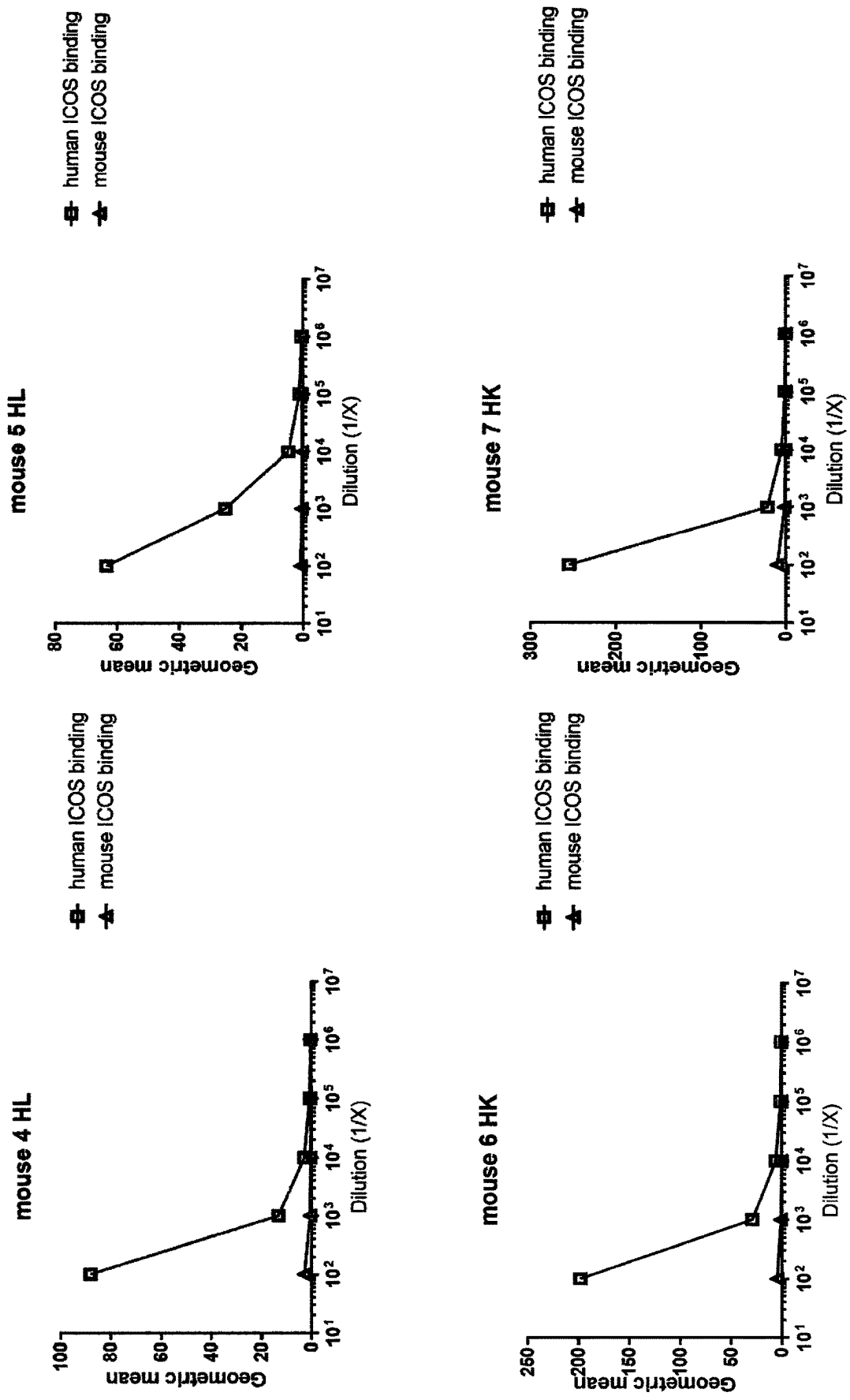


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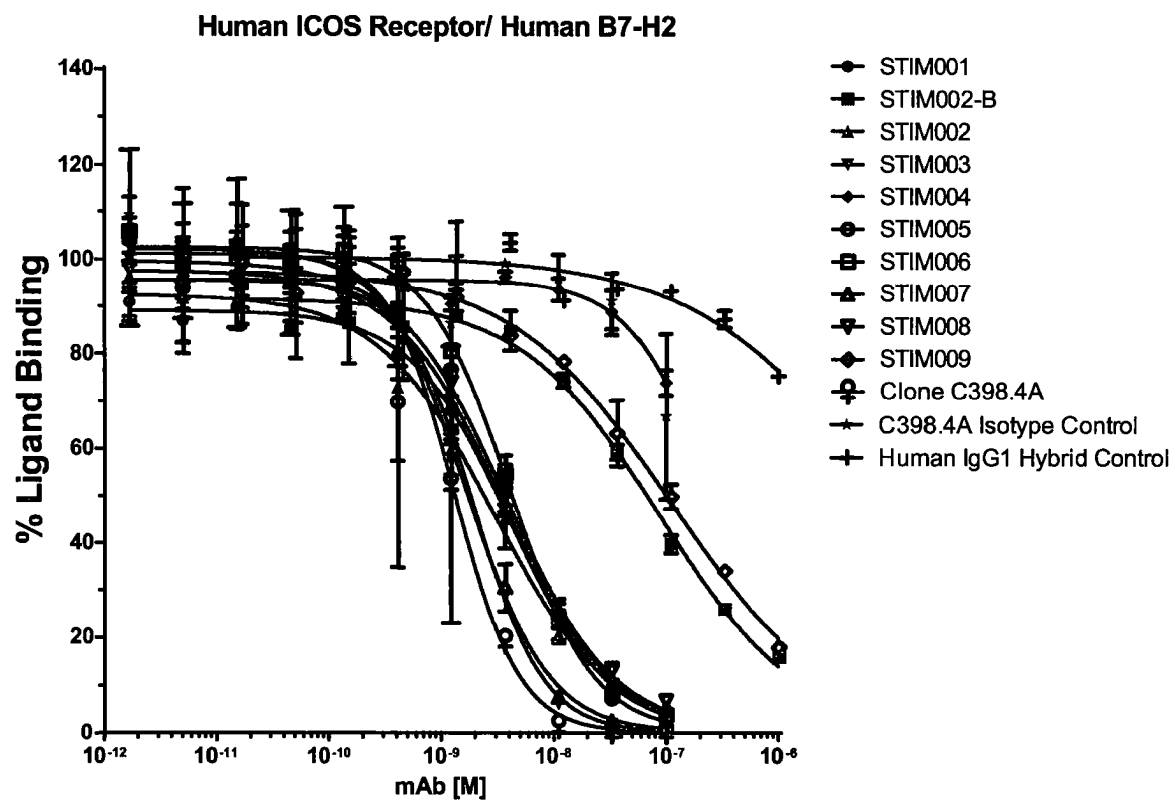


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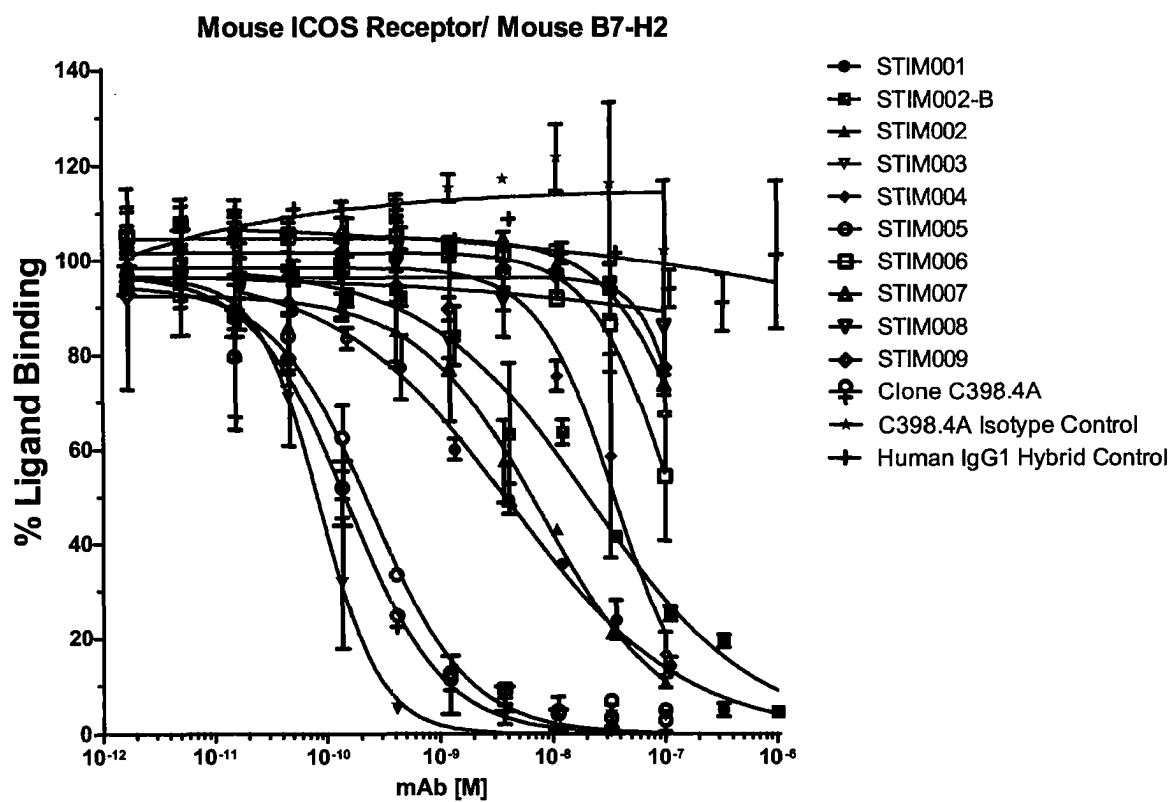


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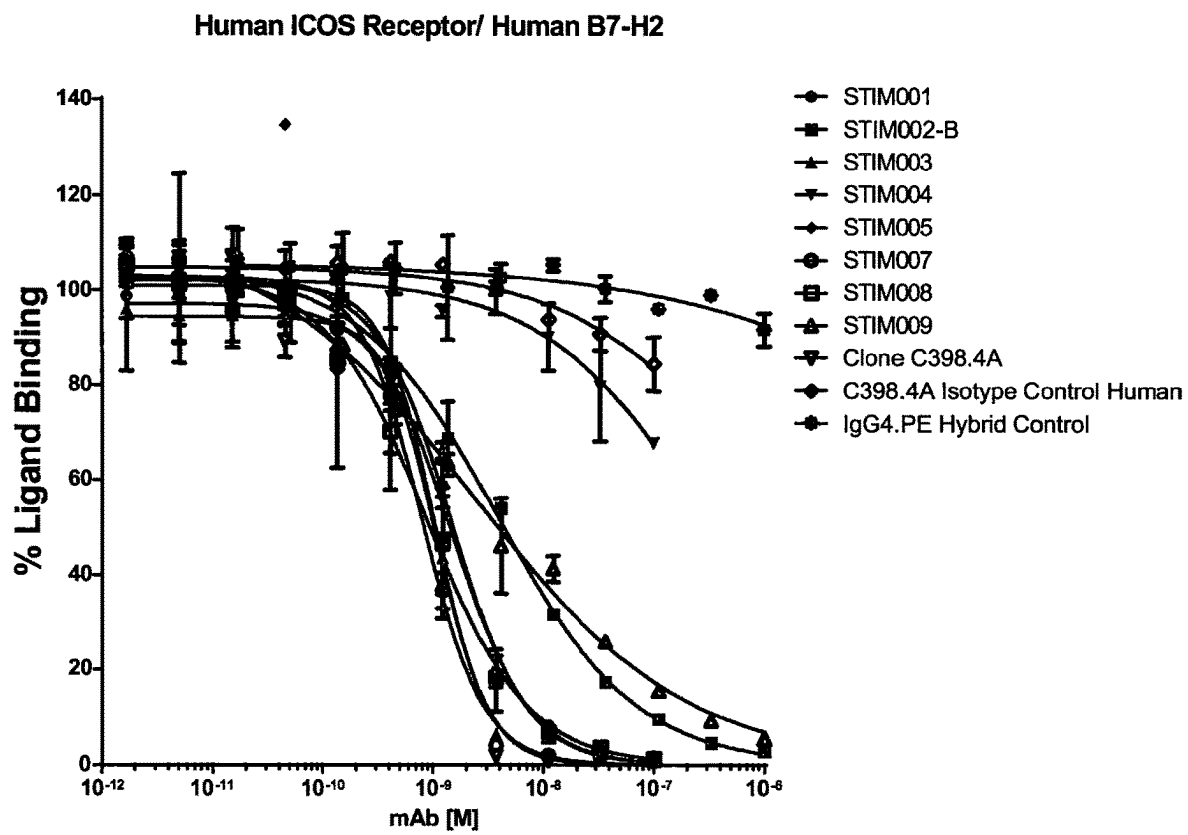


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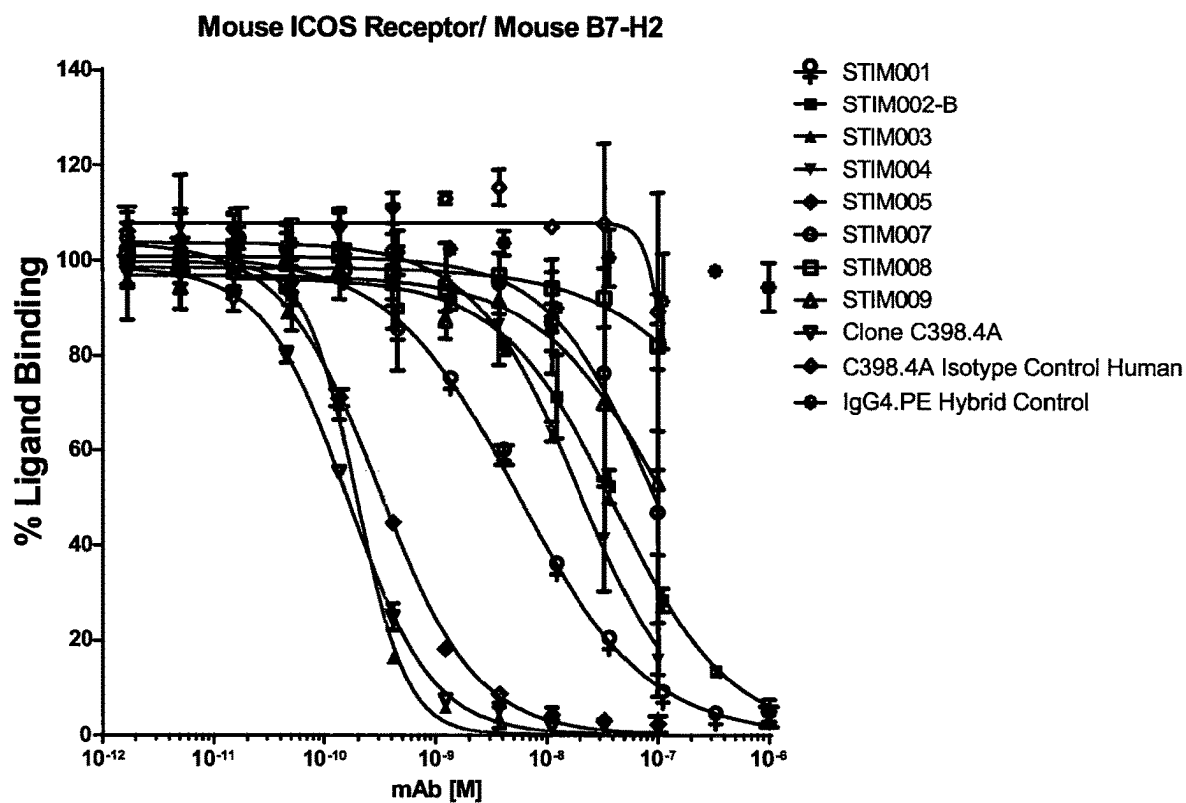
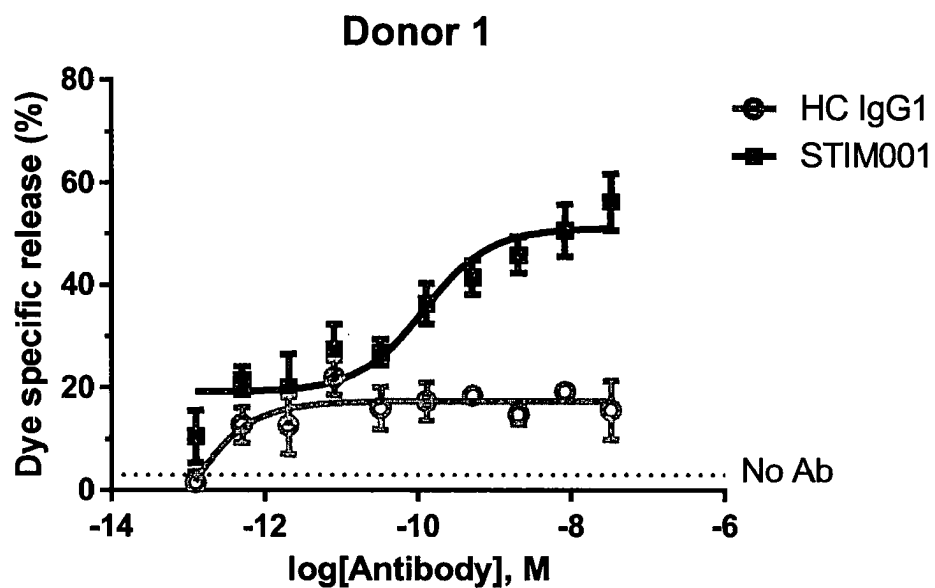


Figure 5

a



b

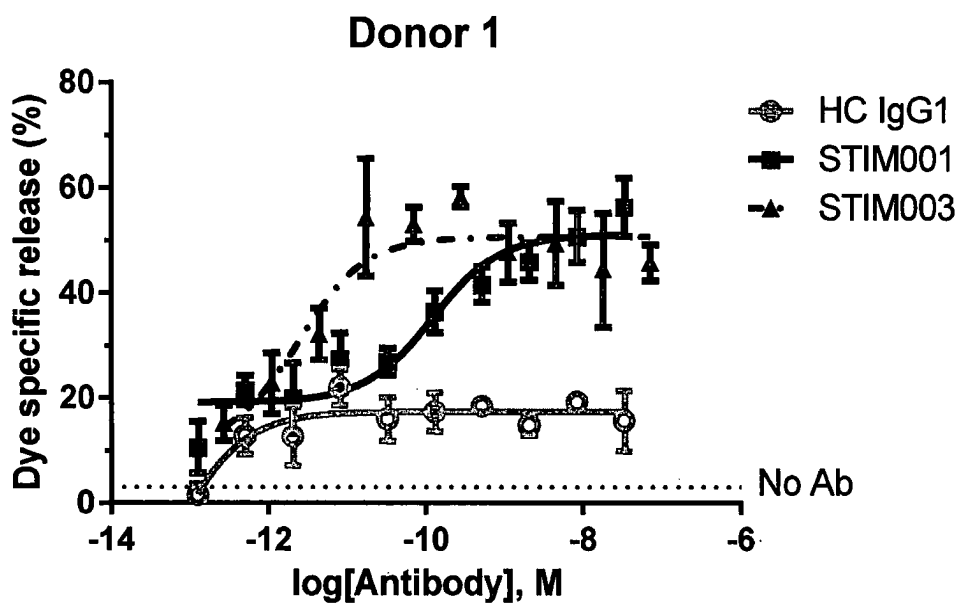
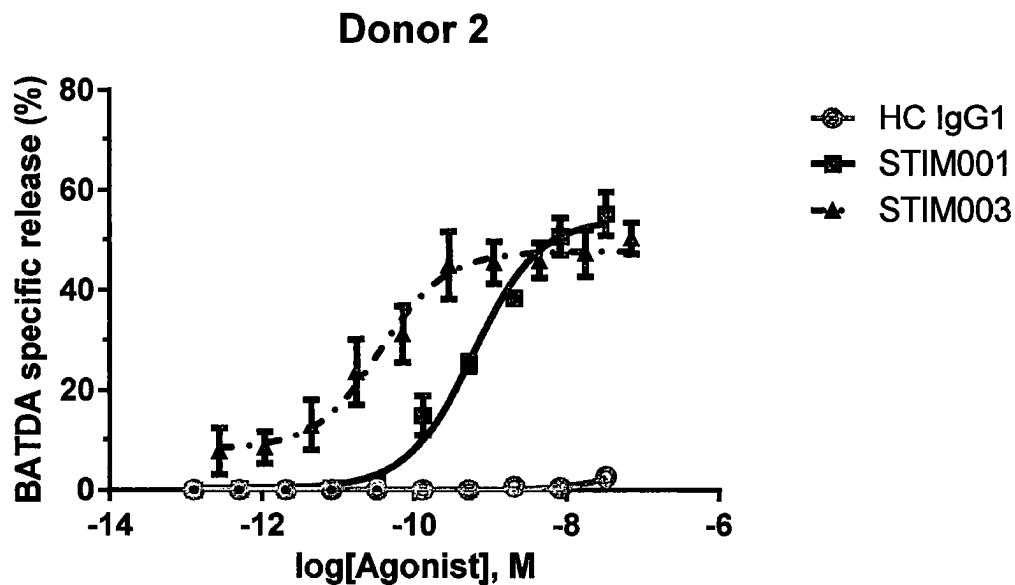


Figure 6 a and b

c



d

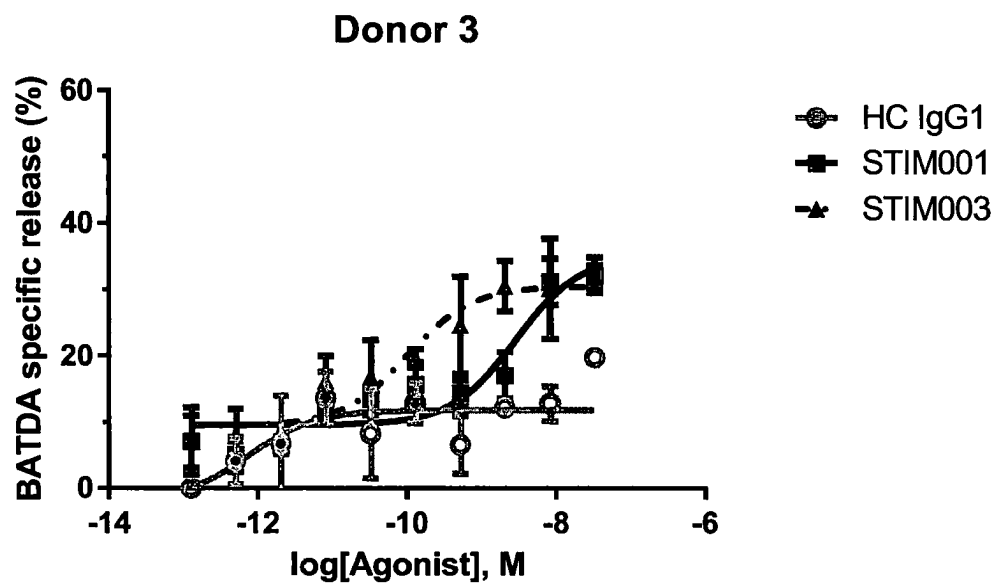
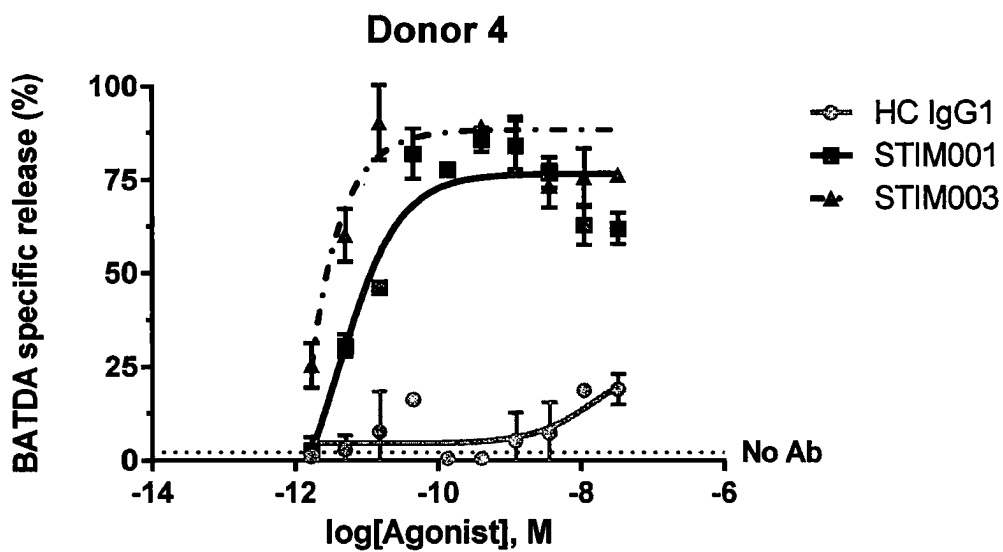


Figure 6 c and d

e



f

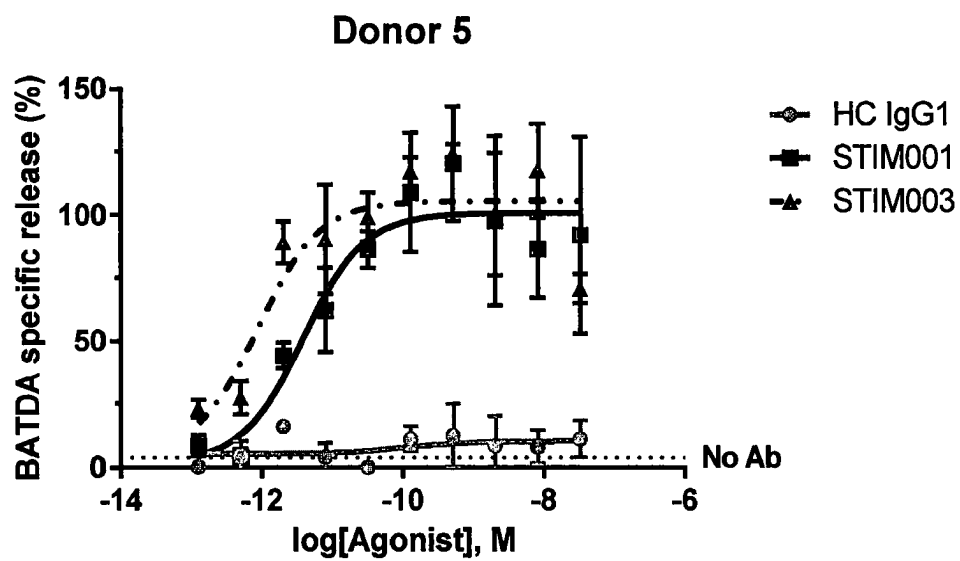


Figure 6 e and f

g

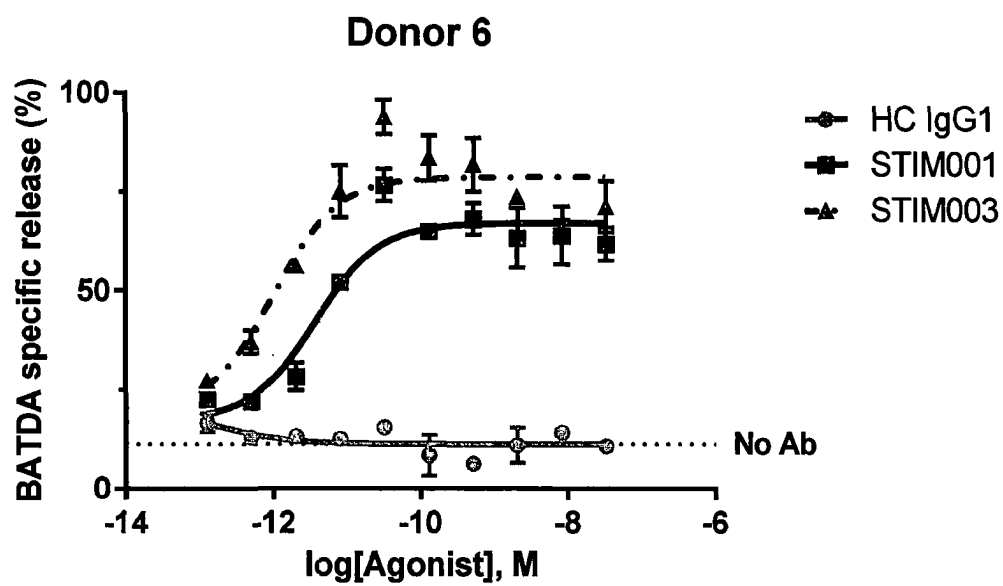


Figure 6 g

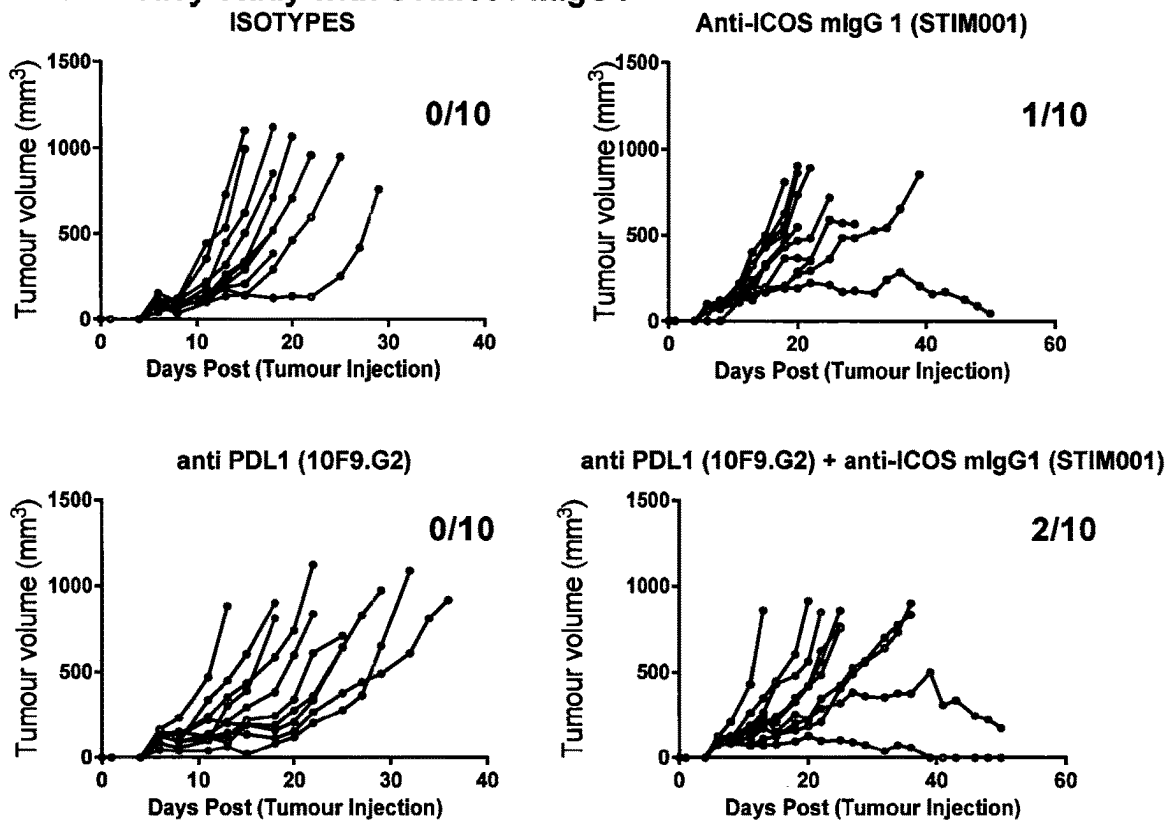
CT26 efficacy study with STIM001 mIgG1**Dosing from days 6-18**

Figure 7

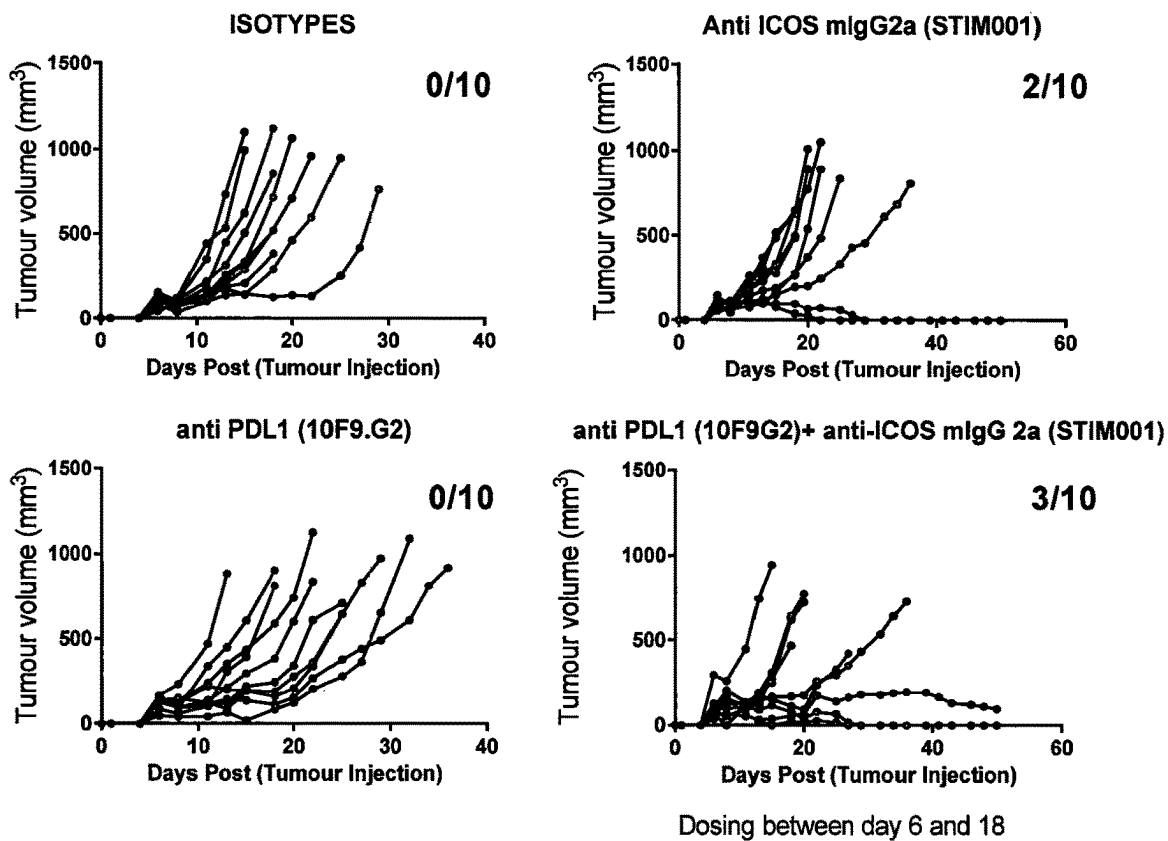
CT26 efficacy study with STIM001 mlgG2a

Figure 8

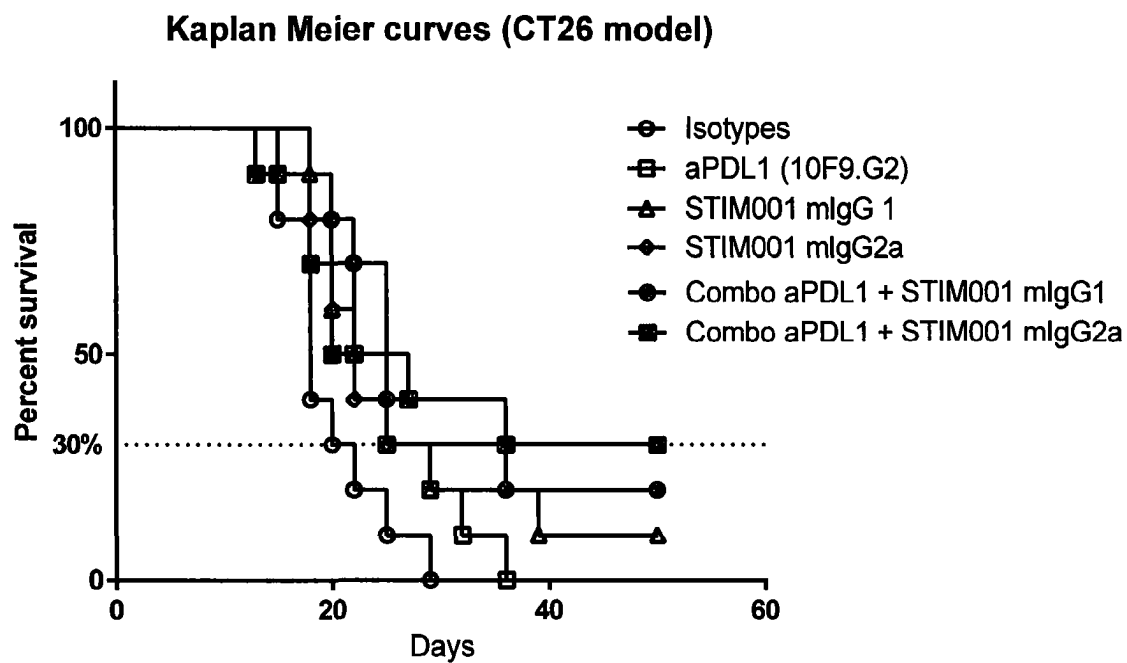
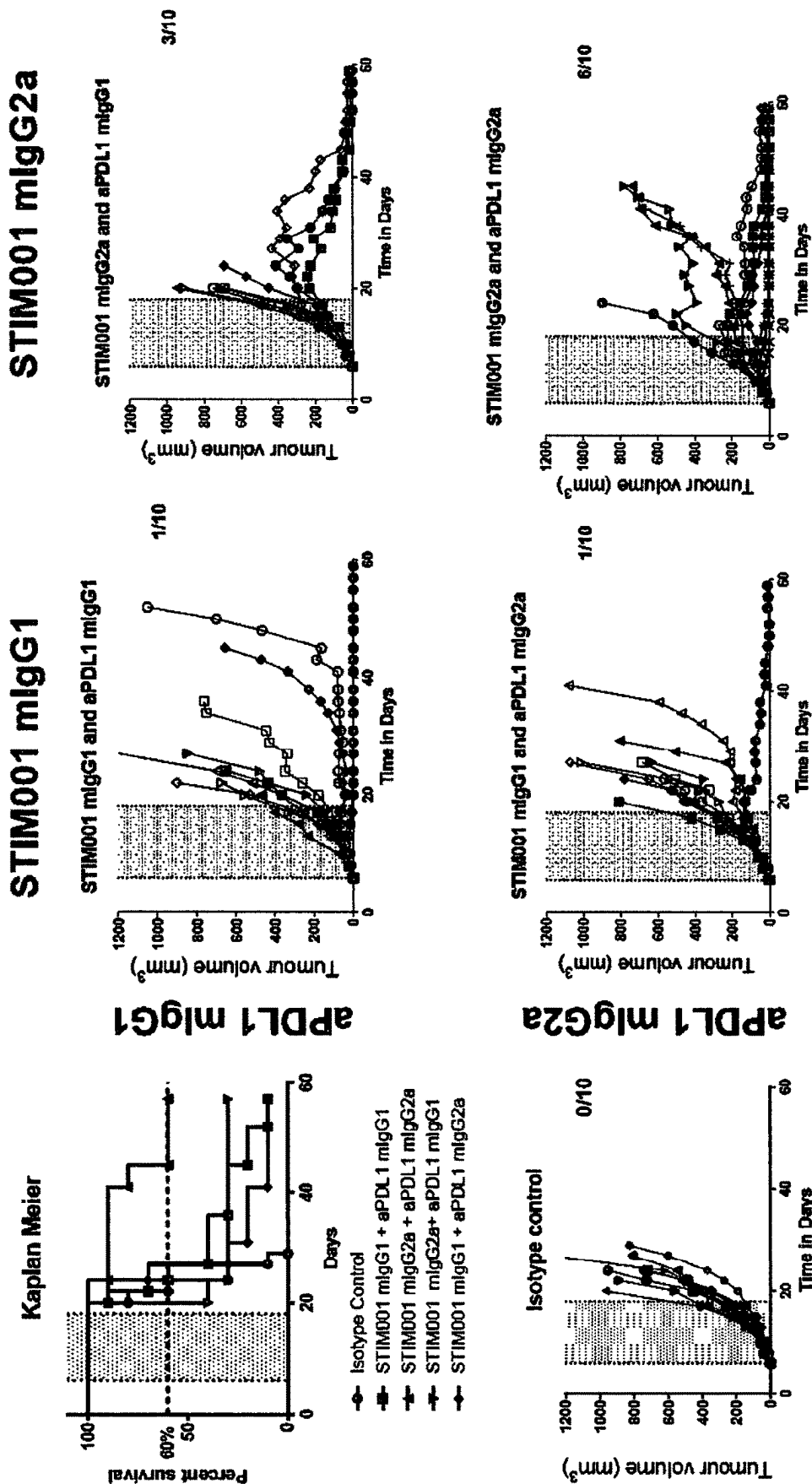


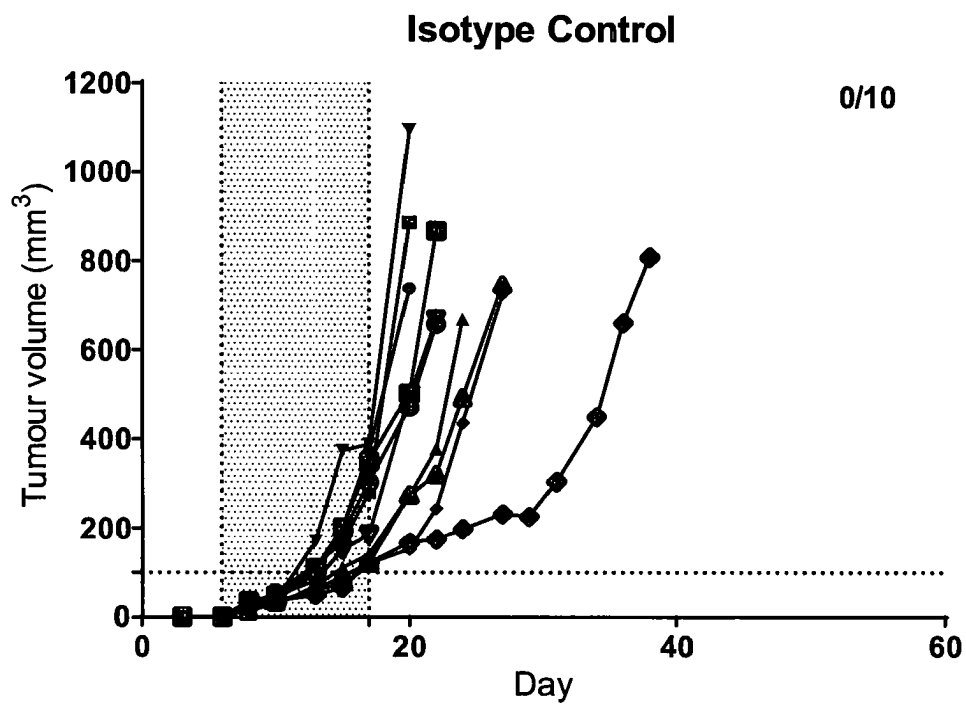
Figure 9



(*) Dosing time is highlighted by the shaded area
(**) Numbers in top right indicates number of animal(s) with no signs of disease at the end of the study.

Figure 10

a



b

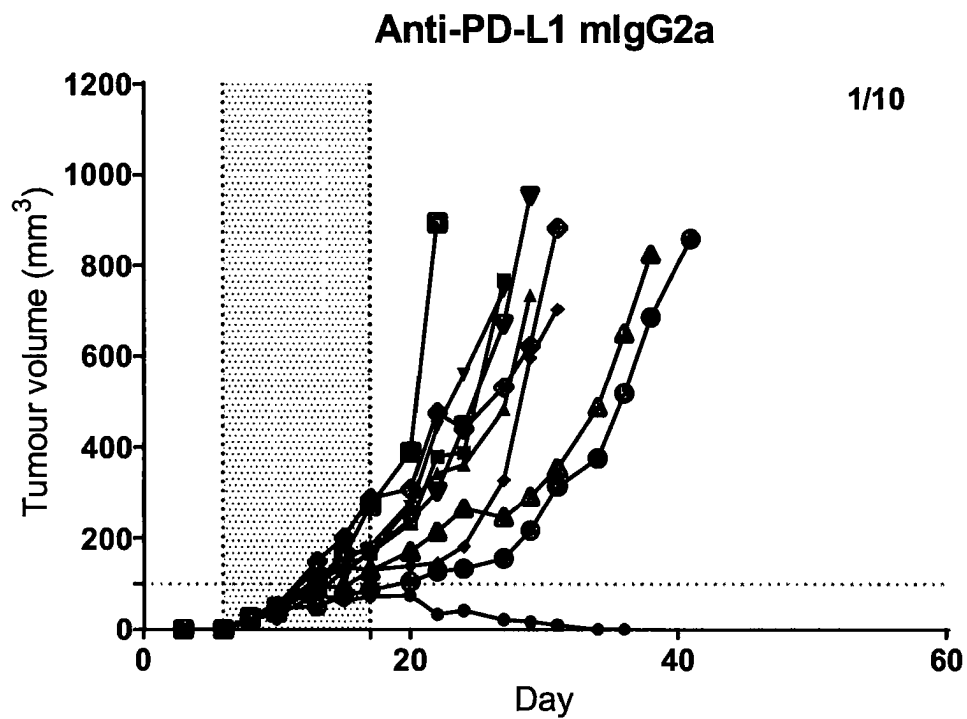
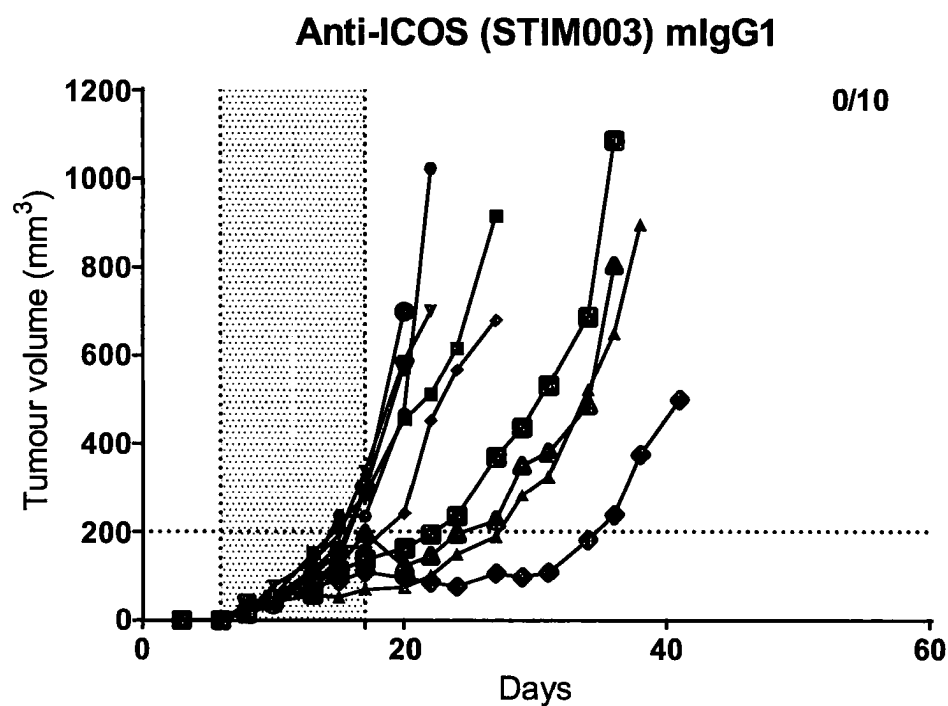


Figure 11 a and b

c



d

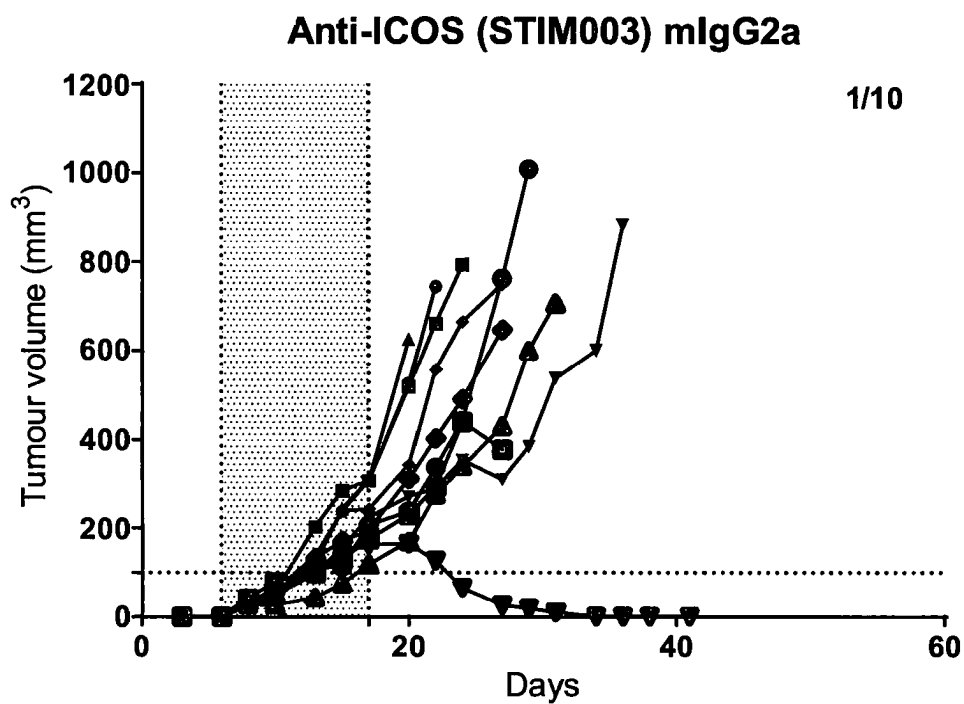
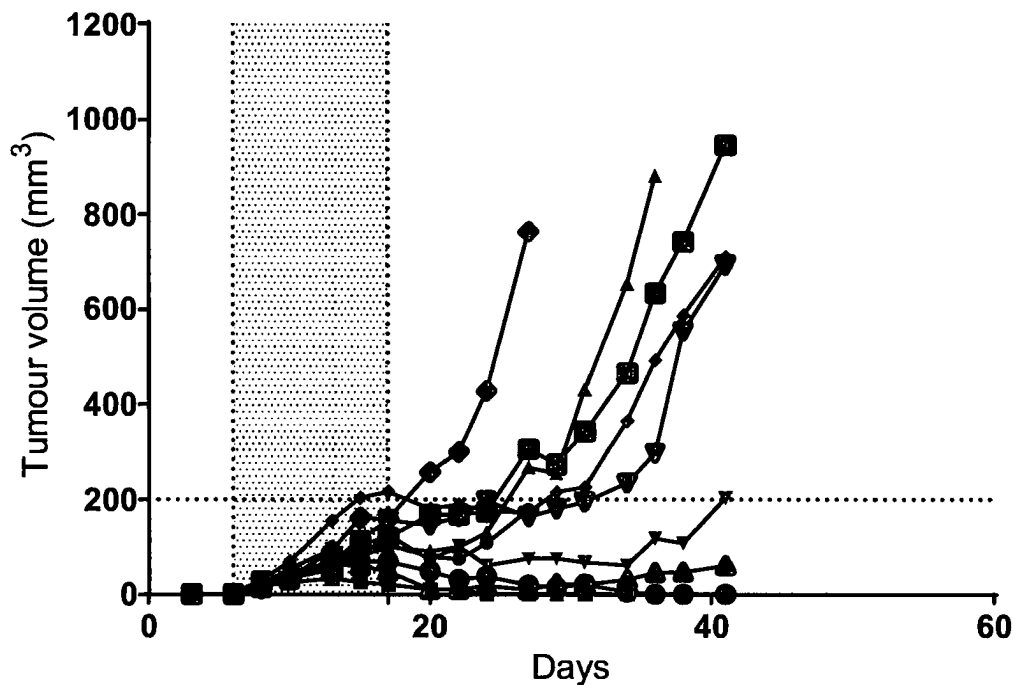


Figure 11 c and d

e

Anti-PD-L1 mlgG2a + STIM003 mlgG1

f

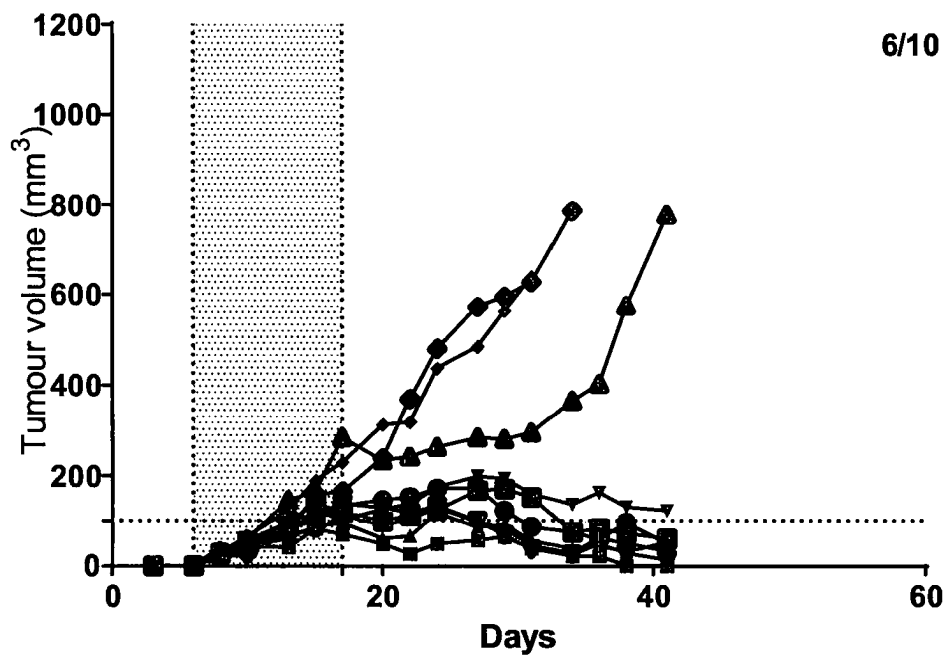
Anti-PD-L1 mlgG2a + STIM003 mlgG2a

Figure 11 e and f

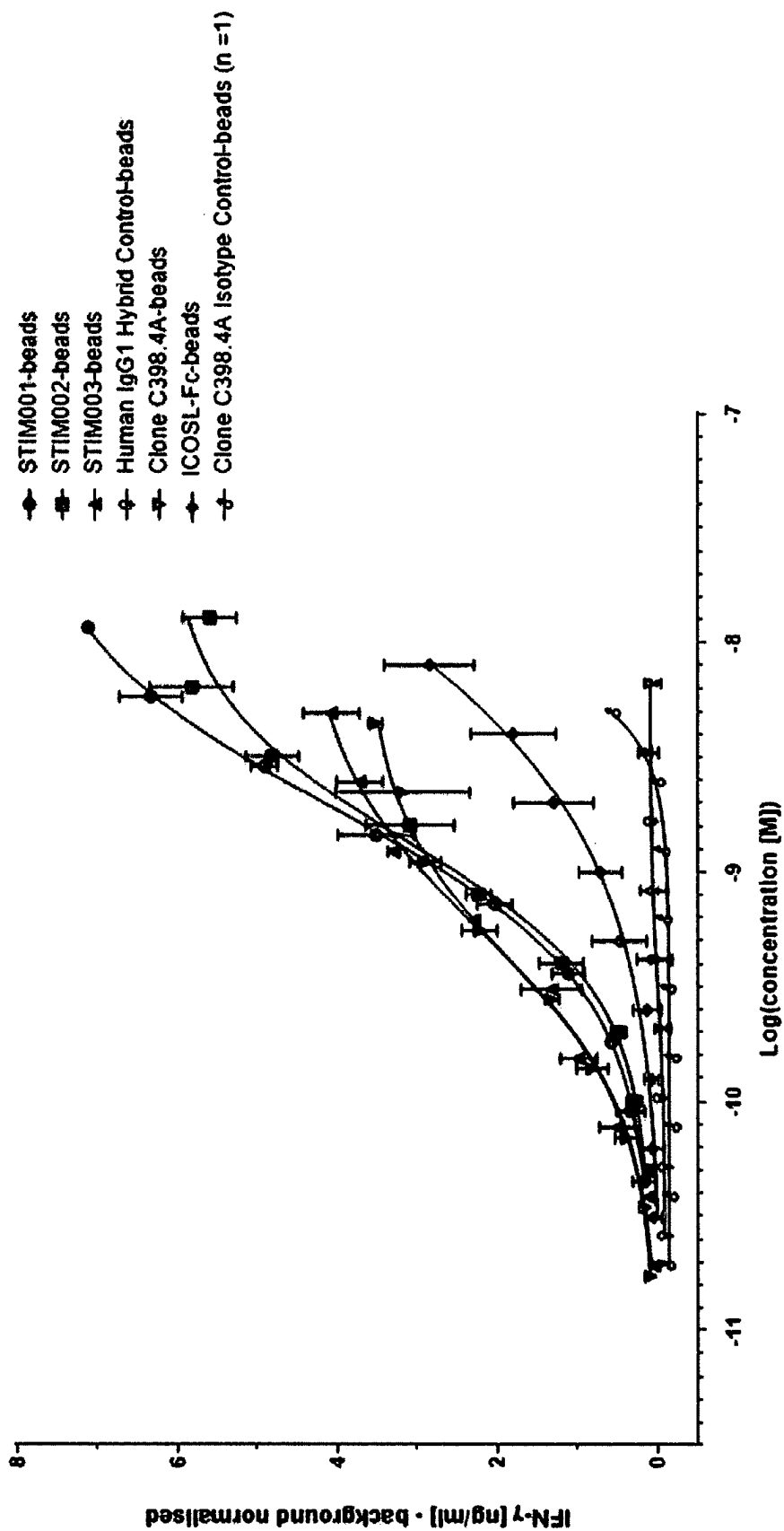


Figure 12

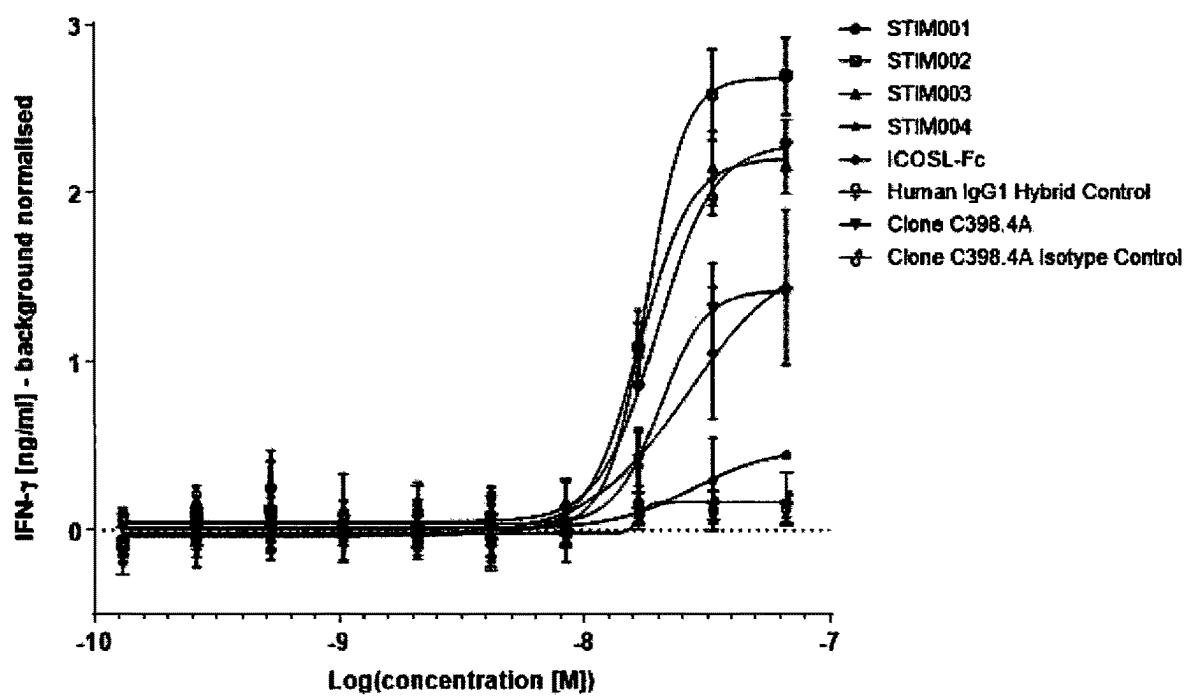


Figure 13

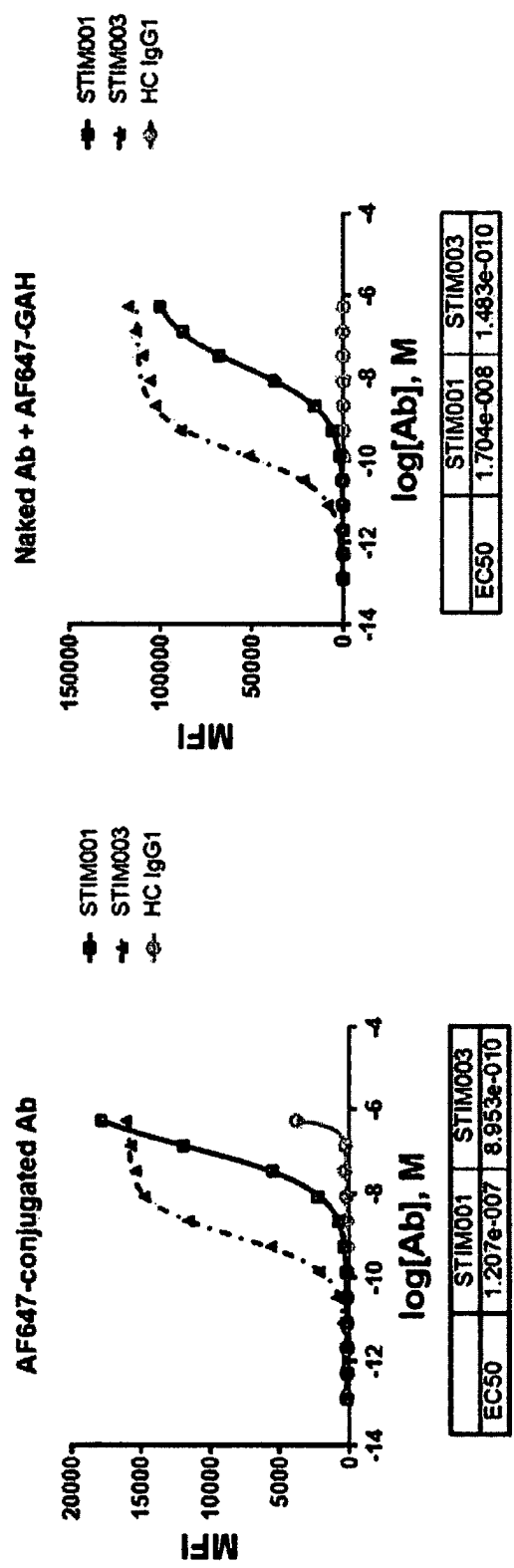


Figure 14

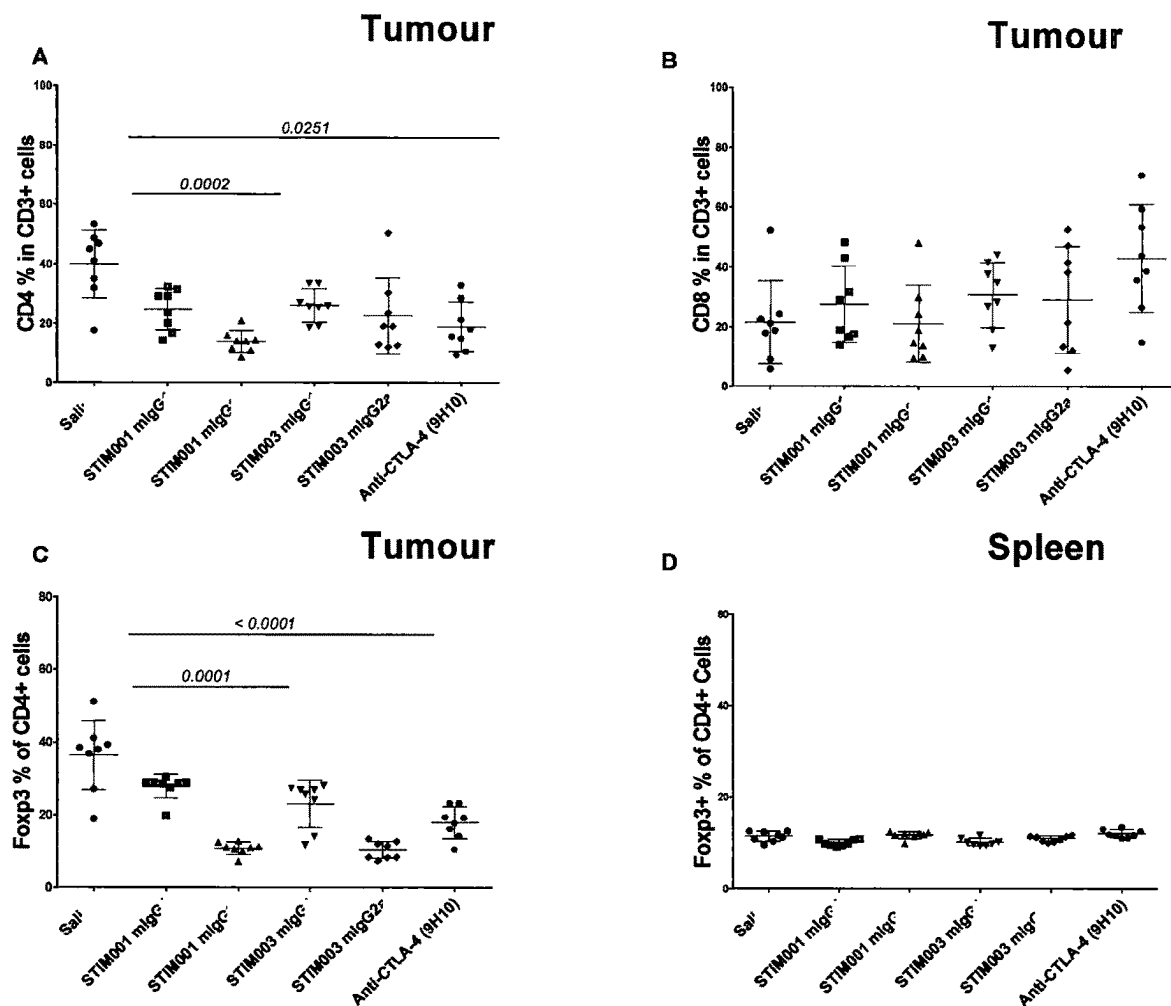


Figure 15 A-D

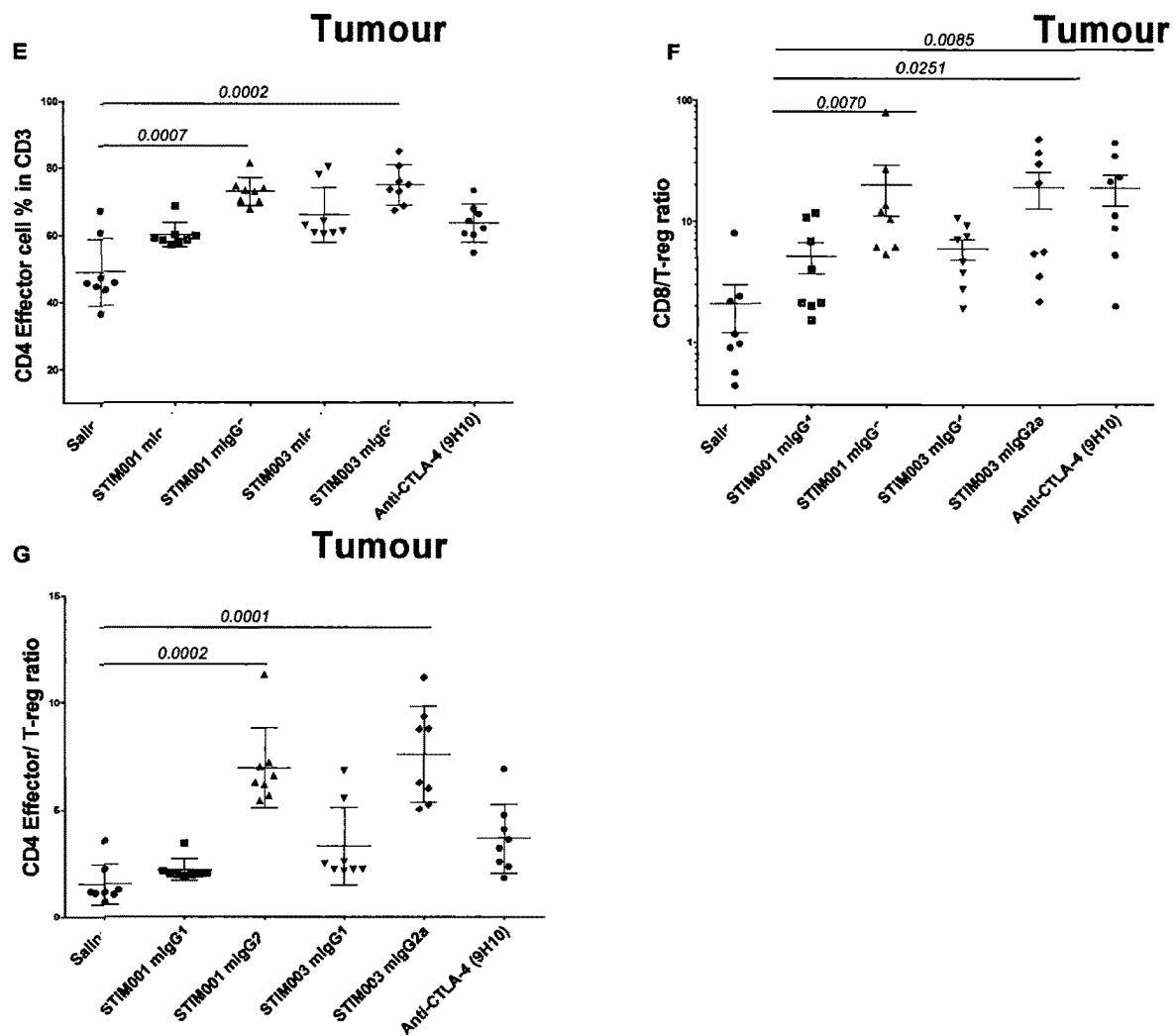


Figure 15 E-G

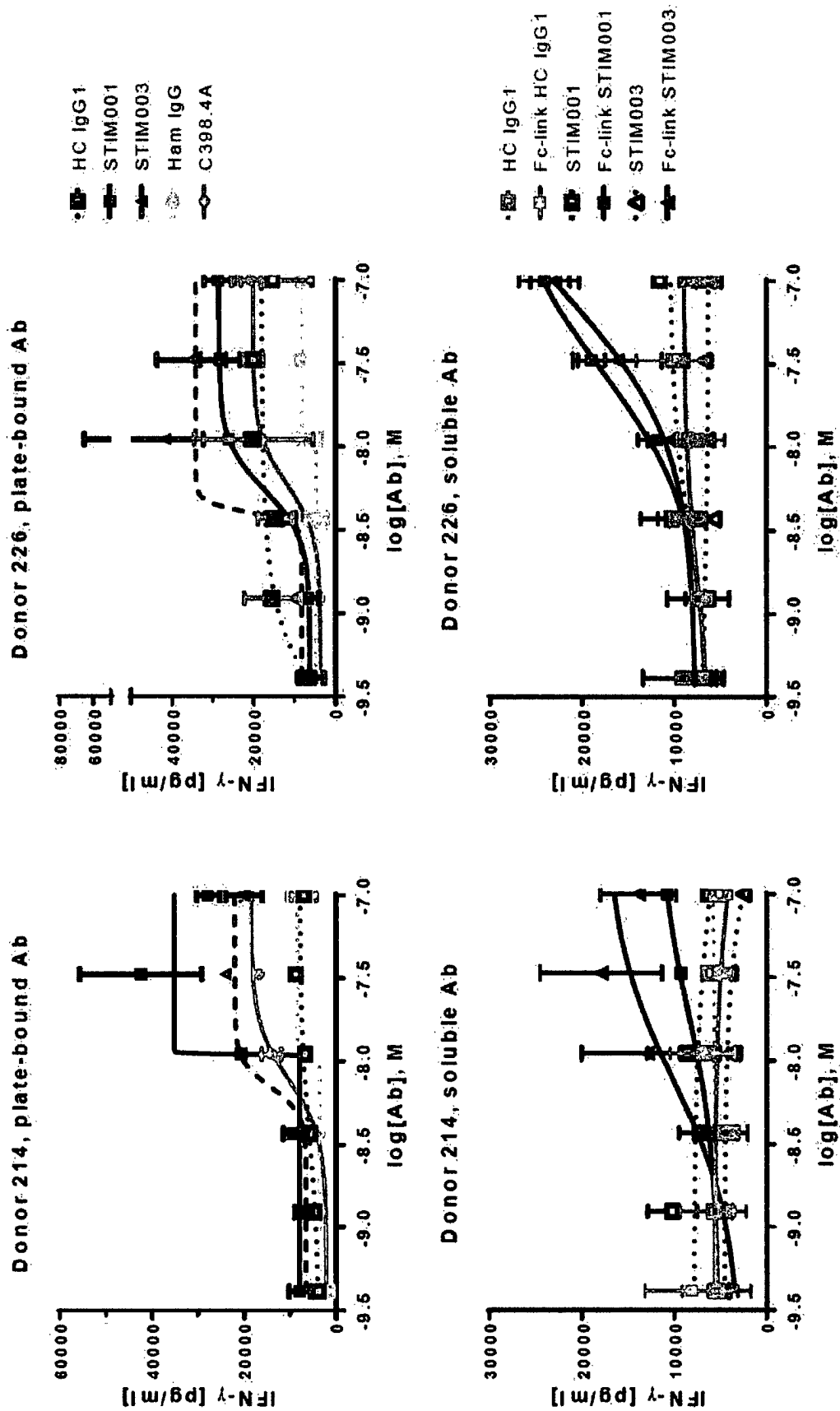


Figure 16

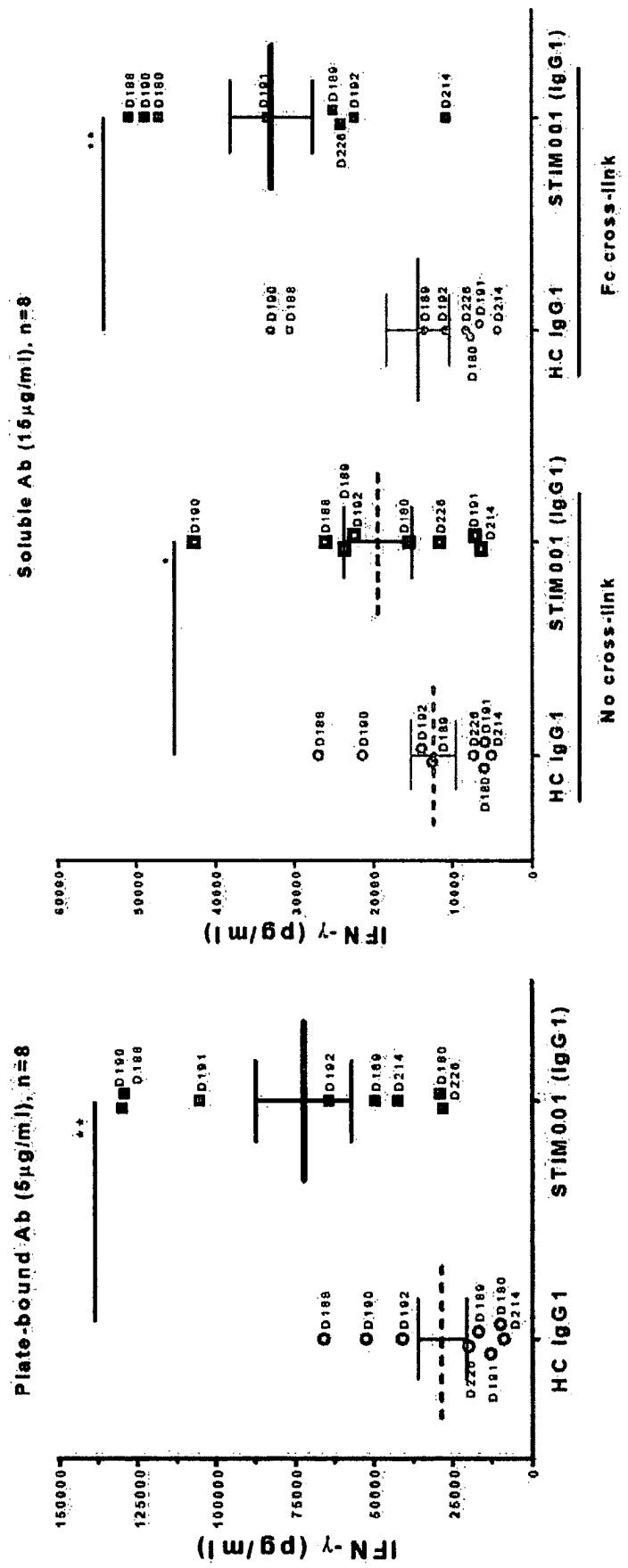


Figure 17

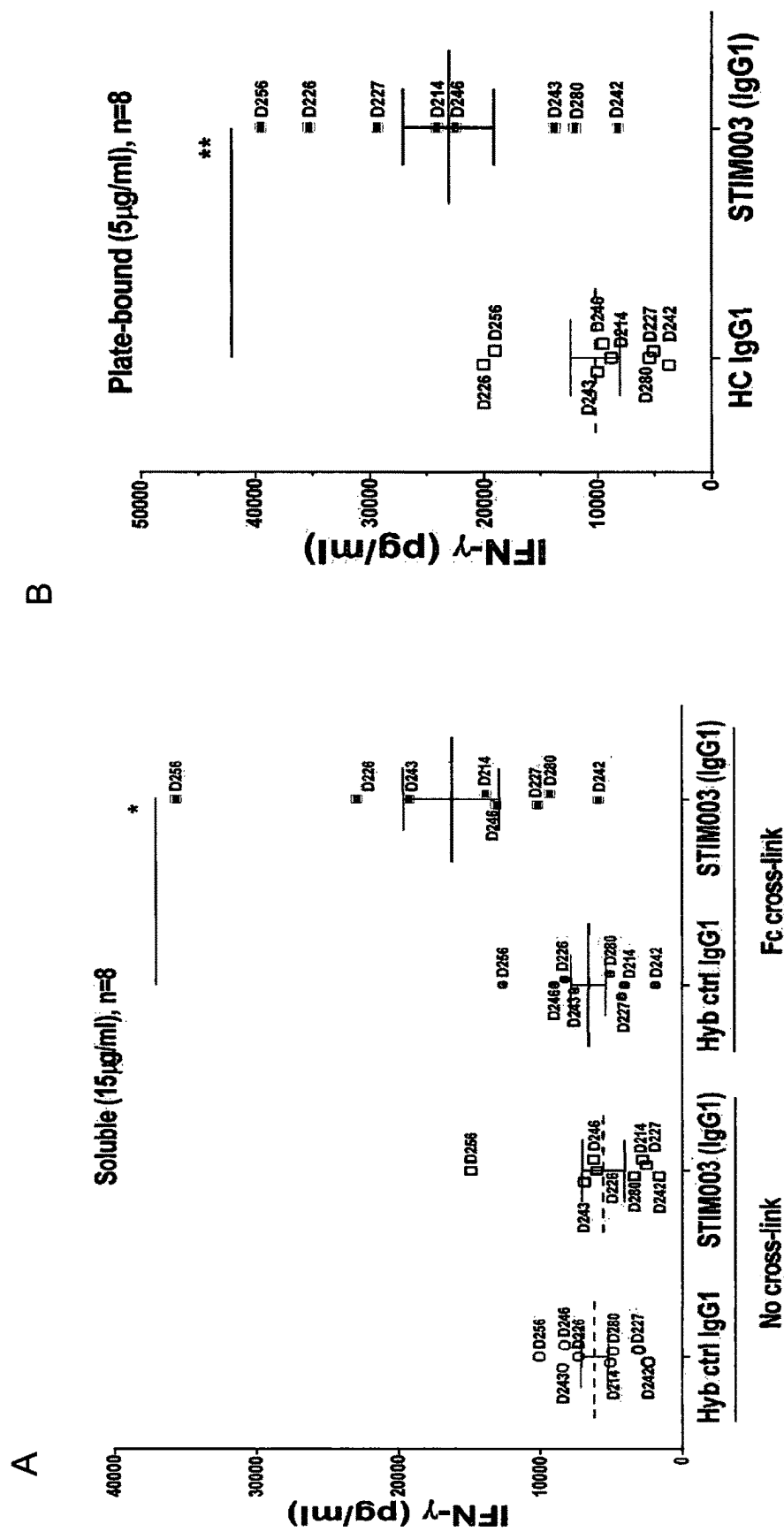


Figure 18

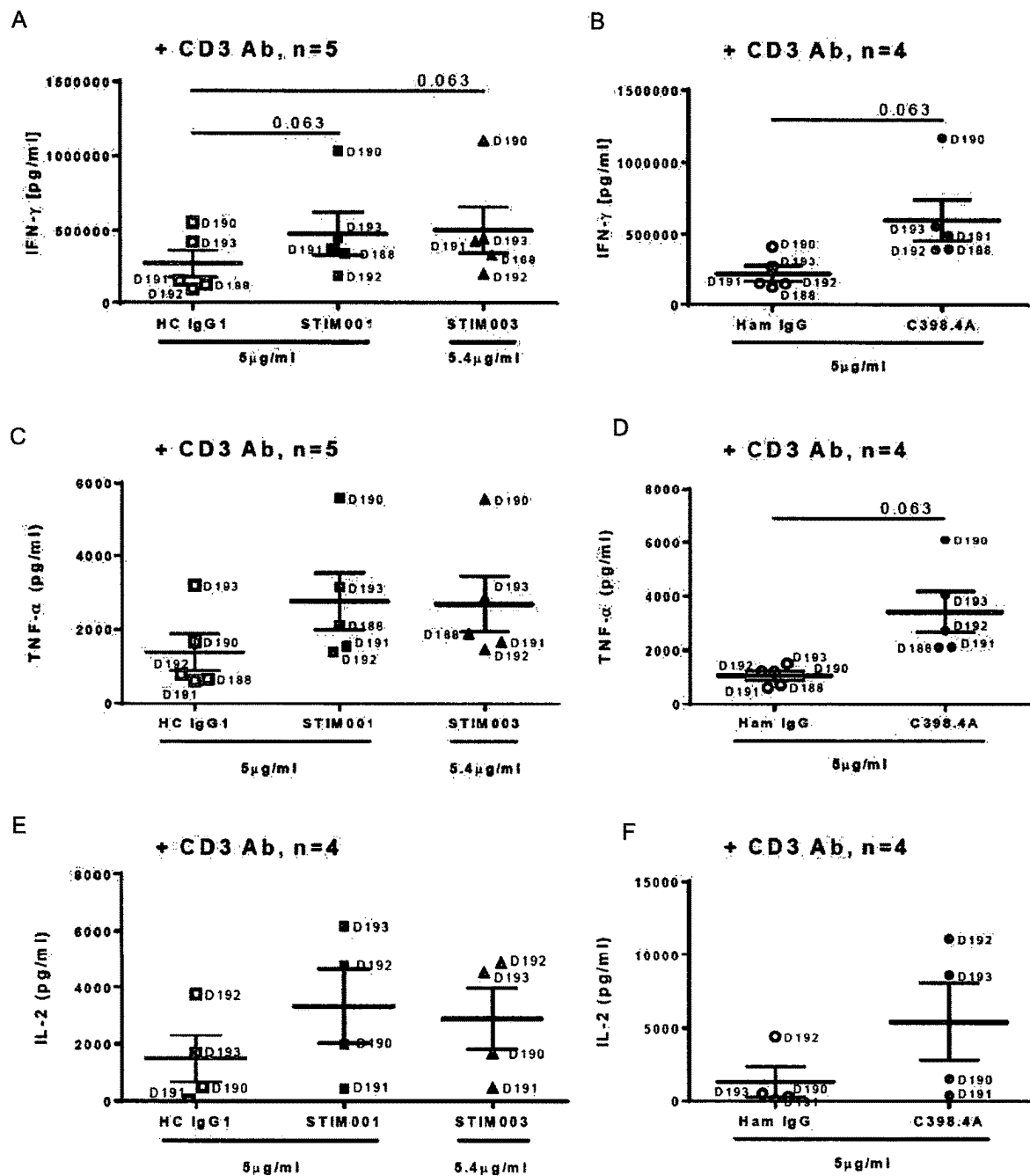


Figure 19

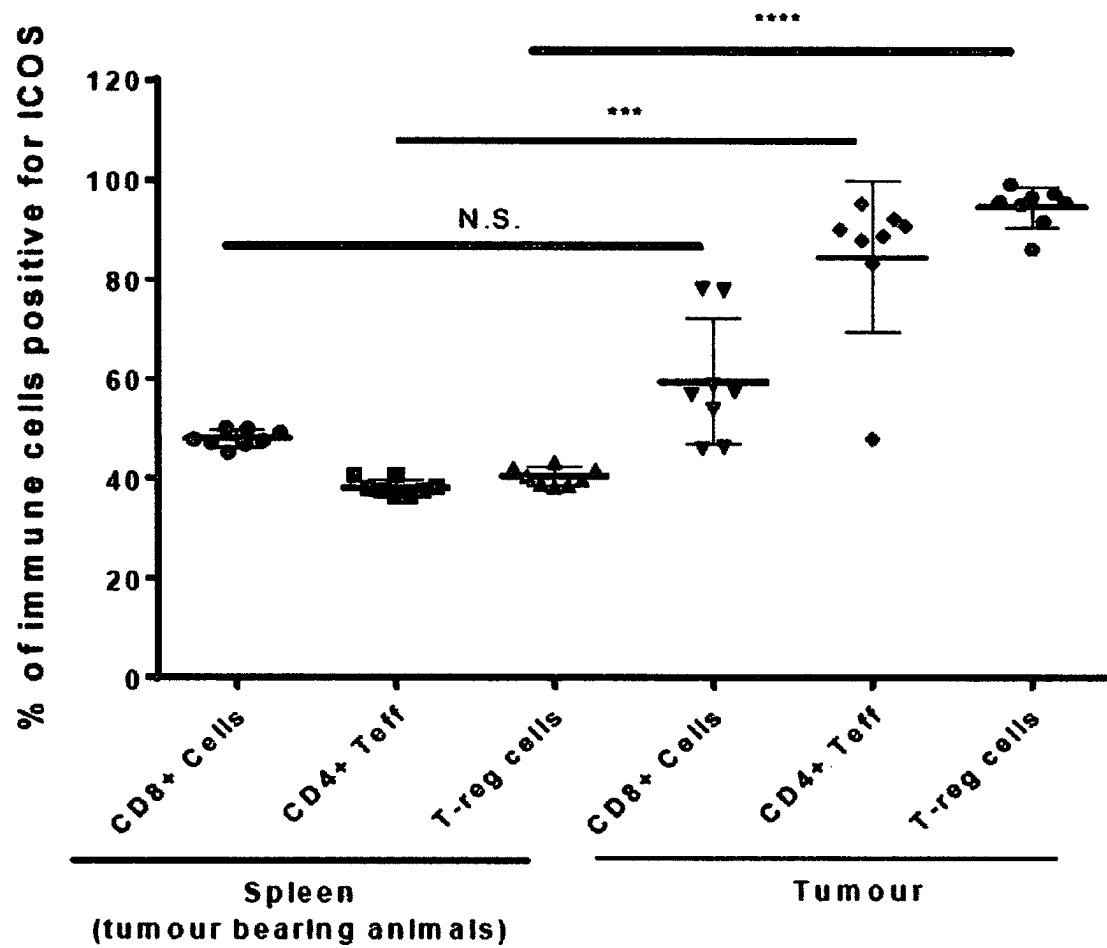


Figure 20

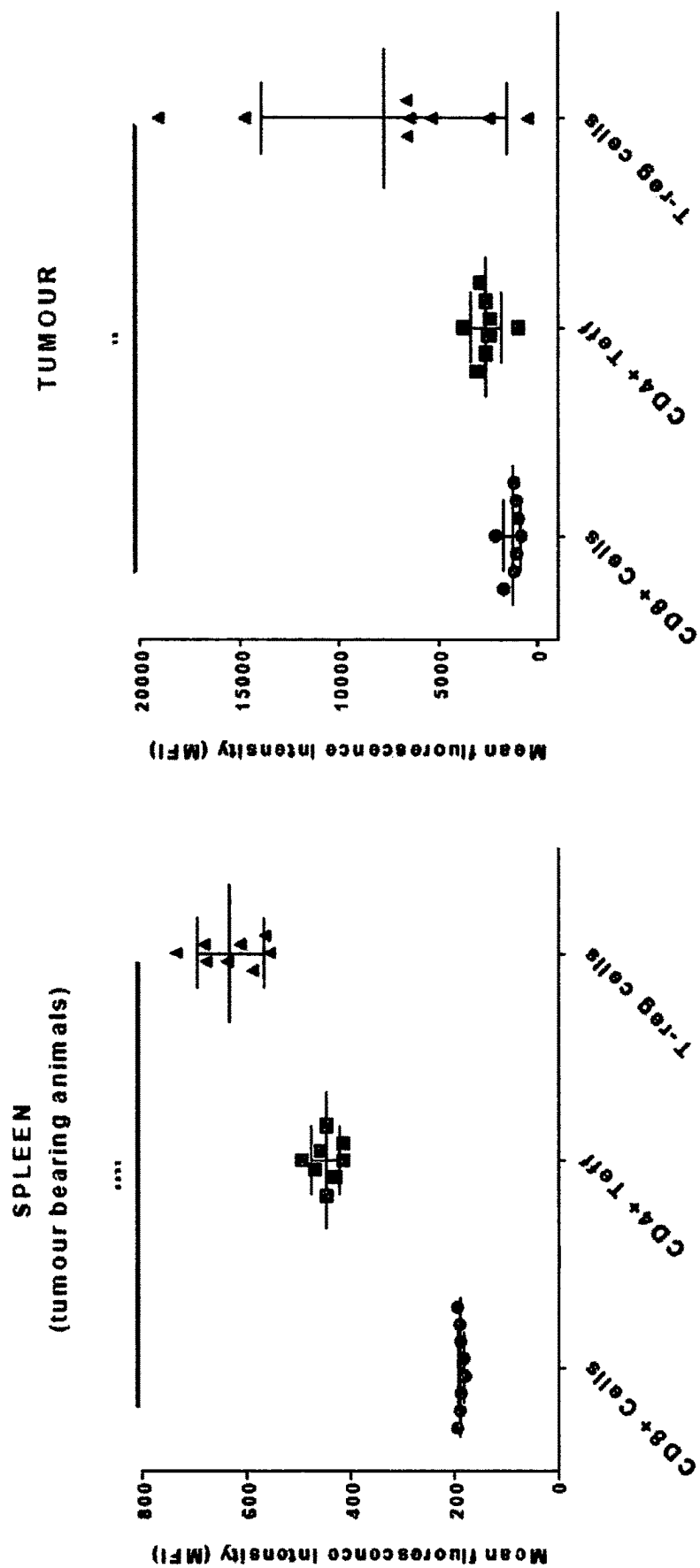


Figure 21

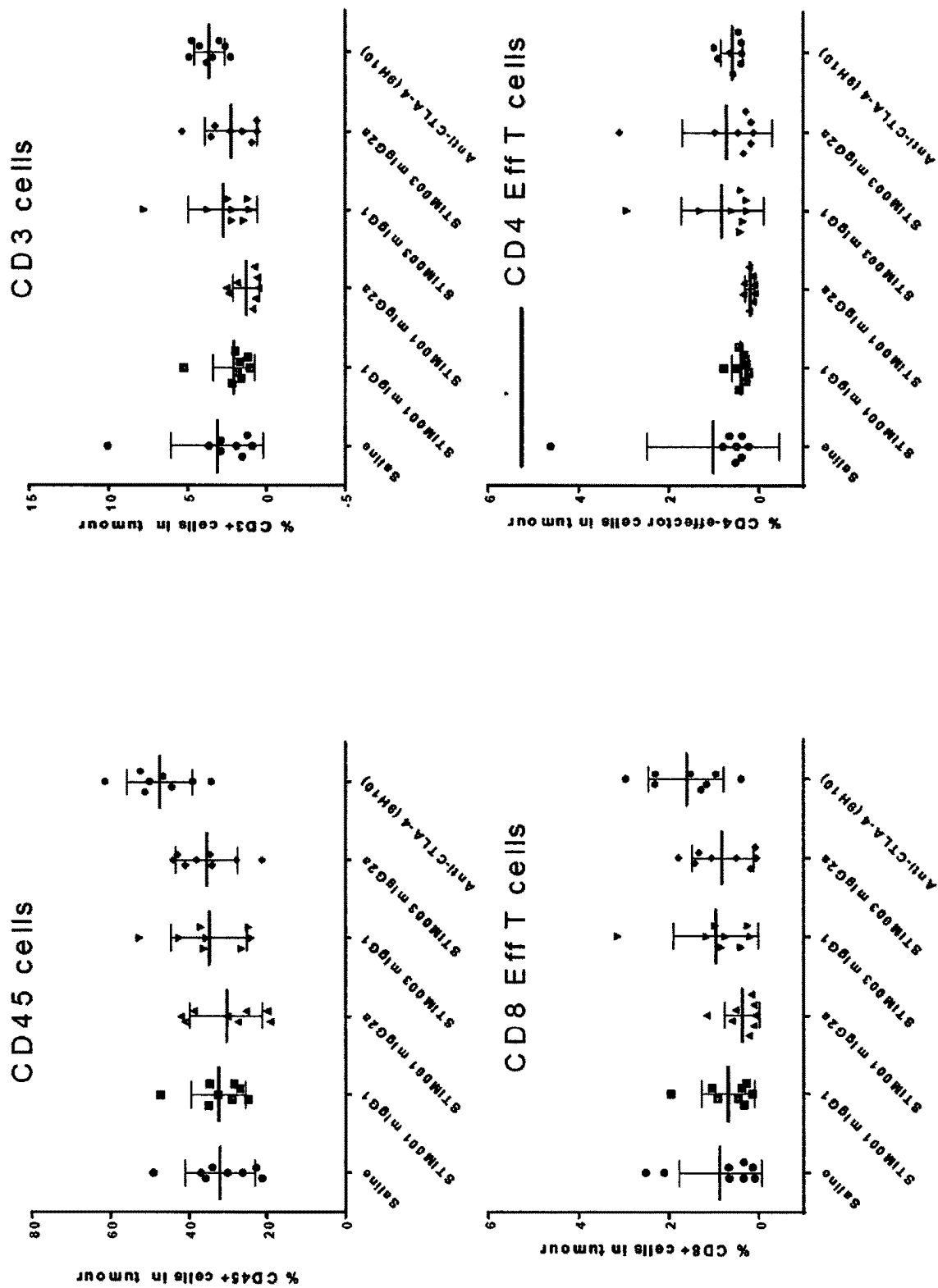


Figure 22

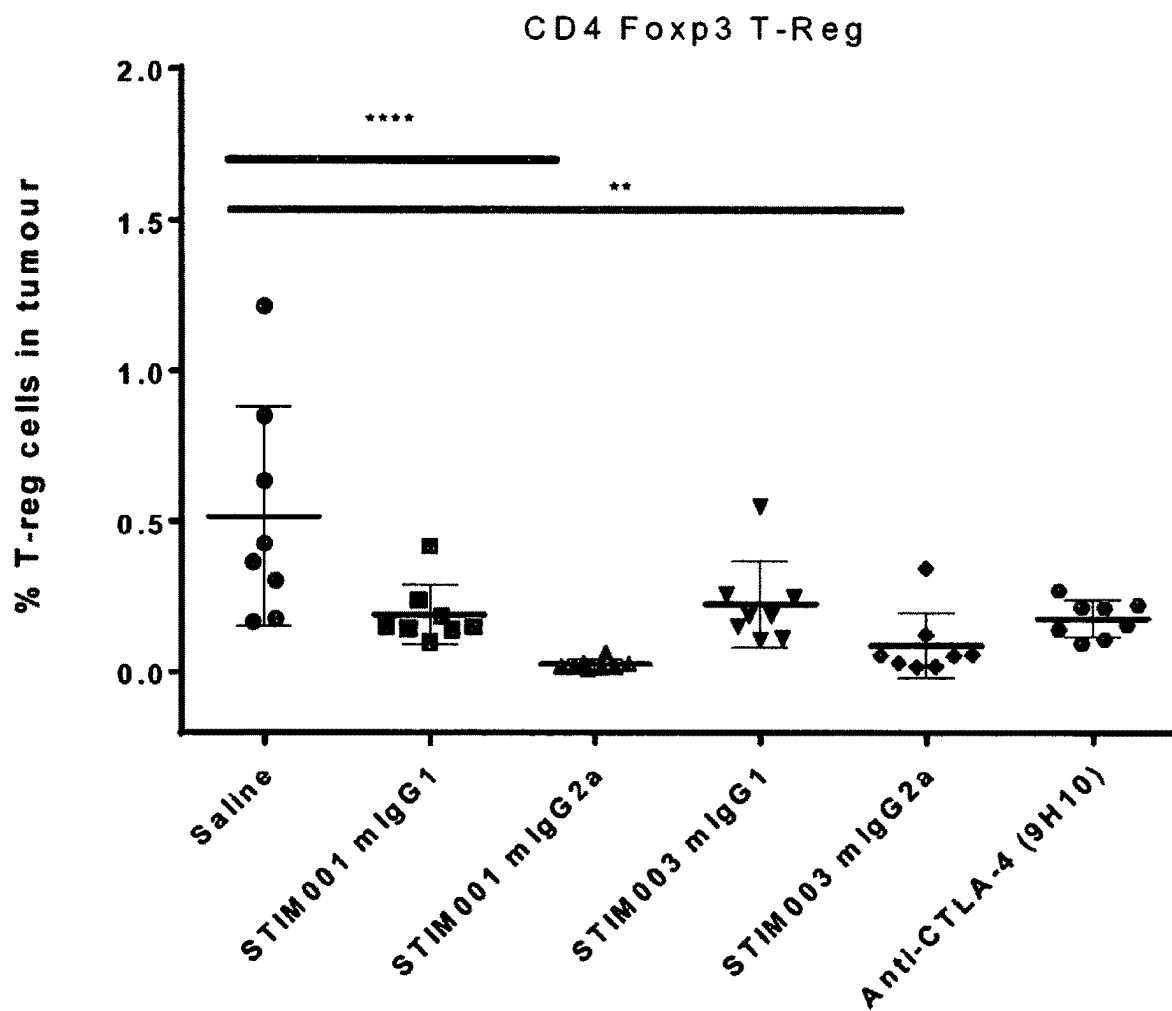


Figure 23

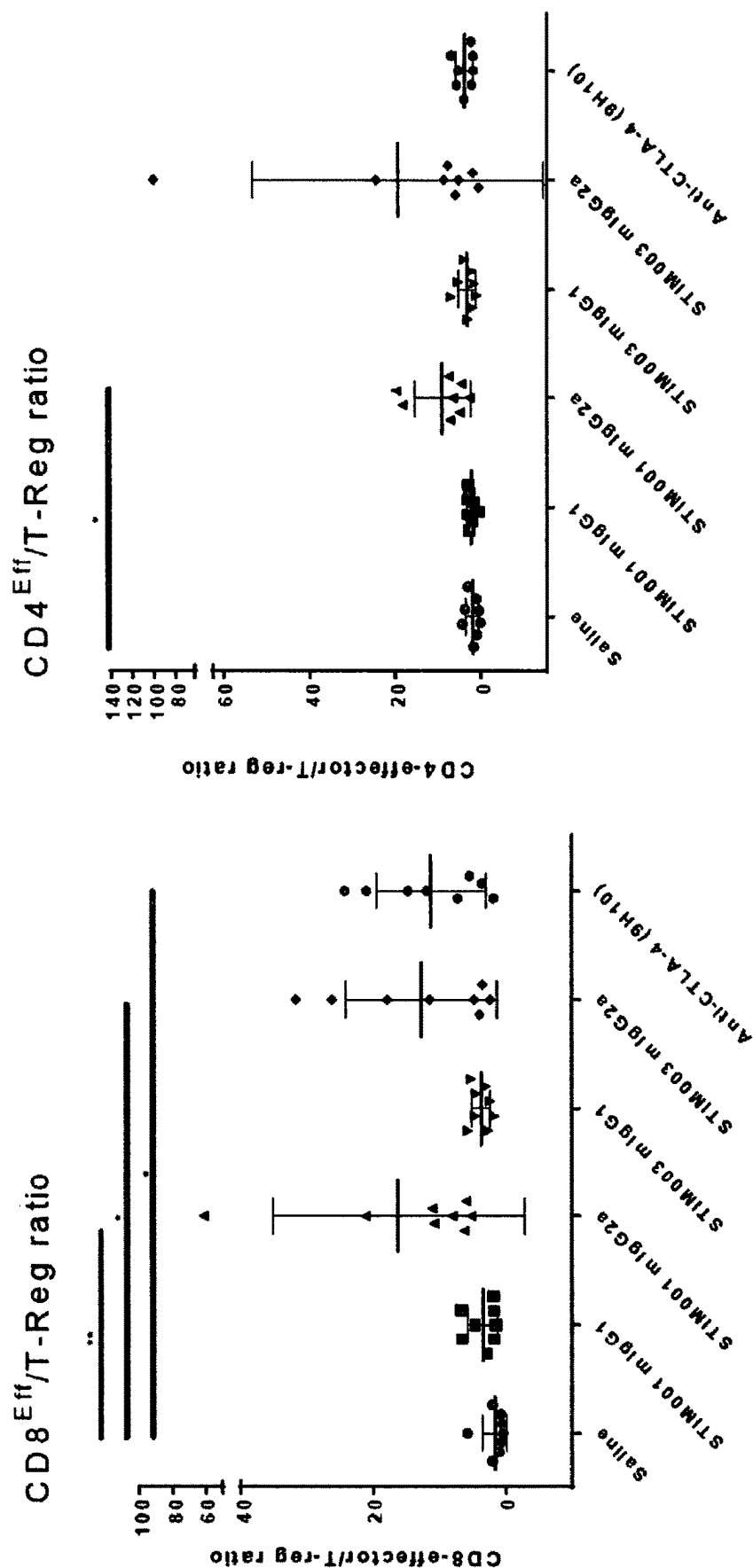


Figure 24

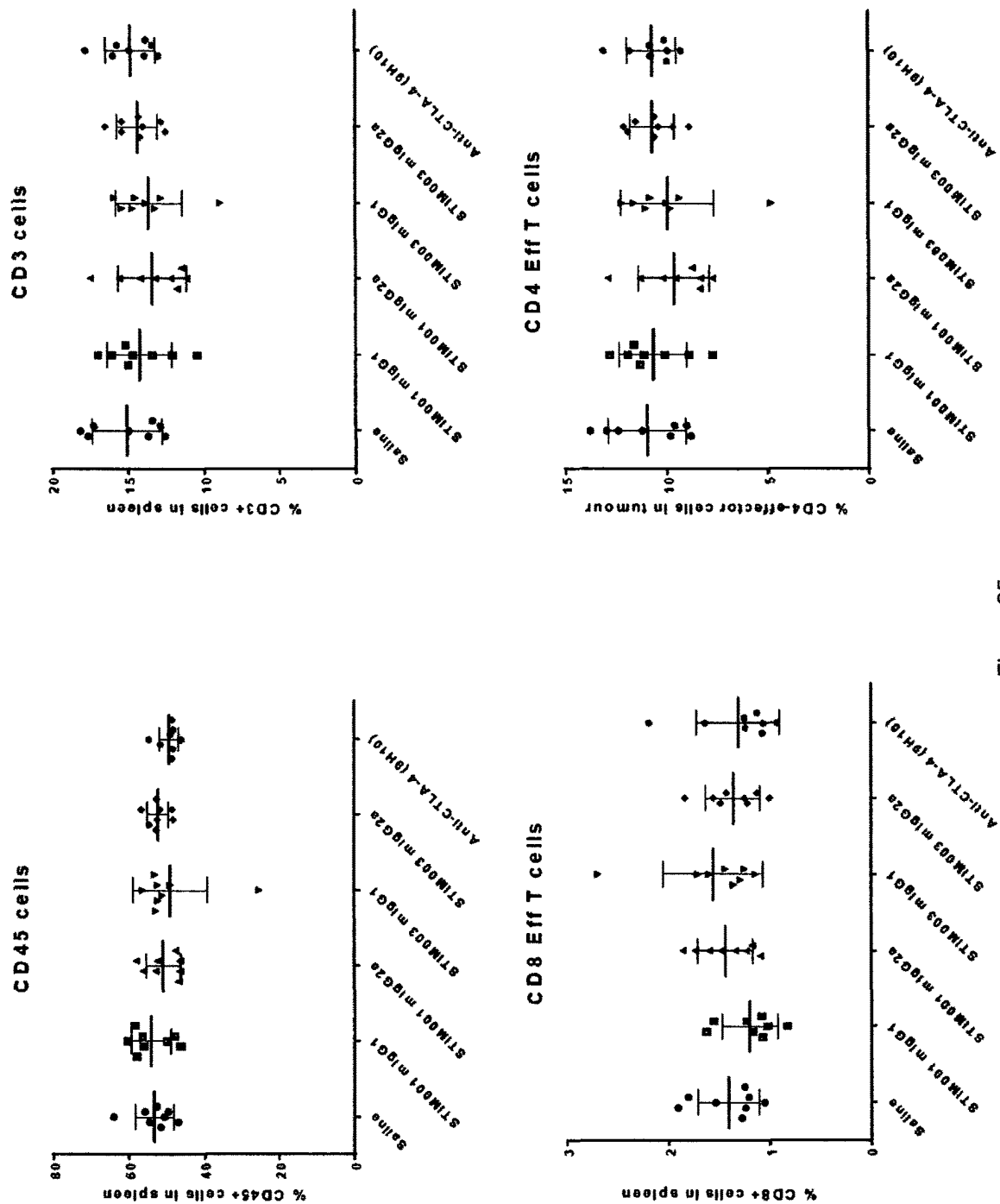


Figure 25

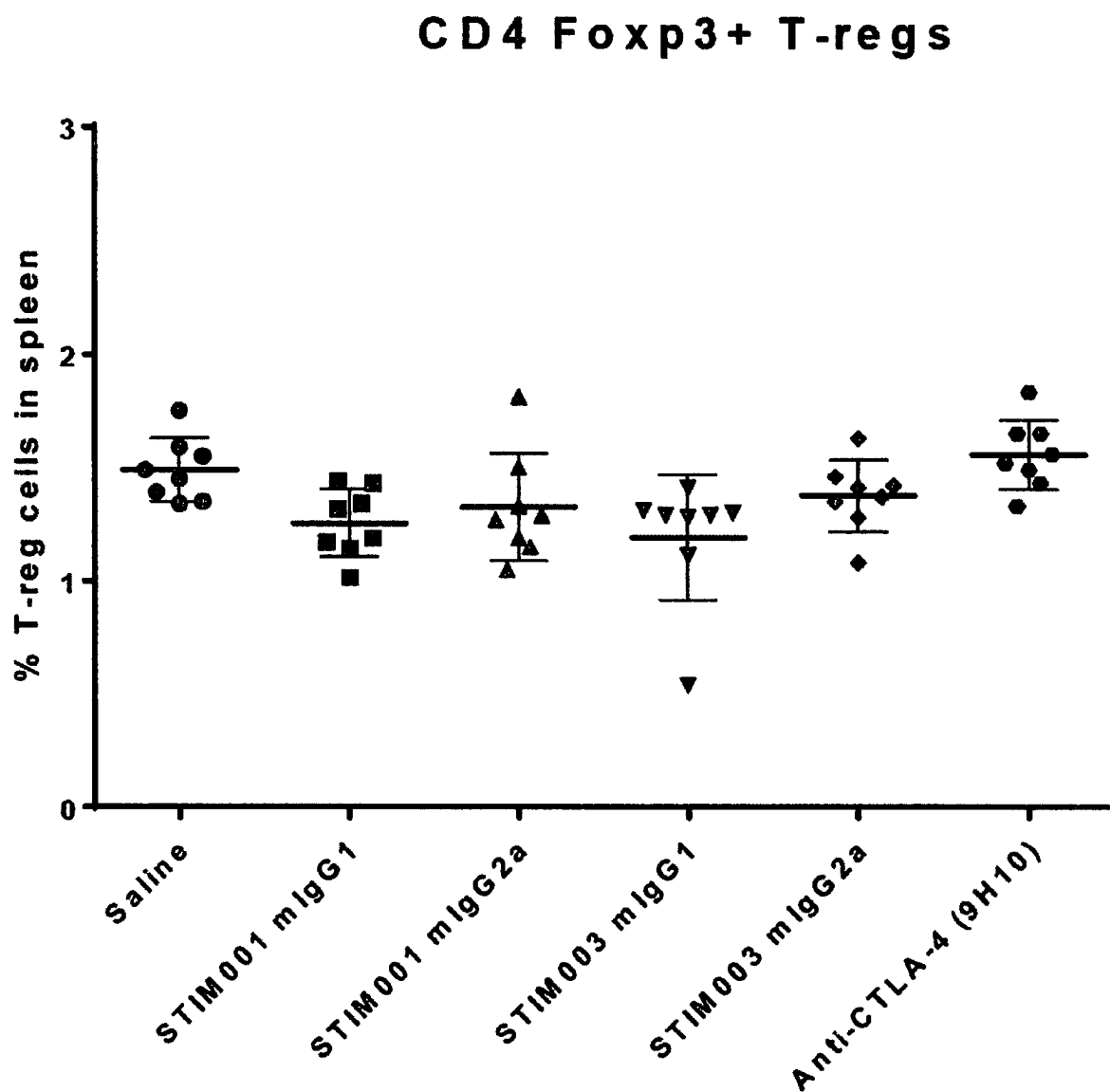
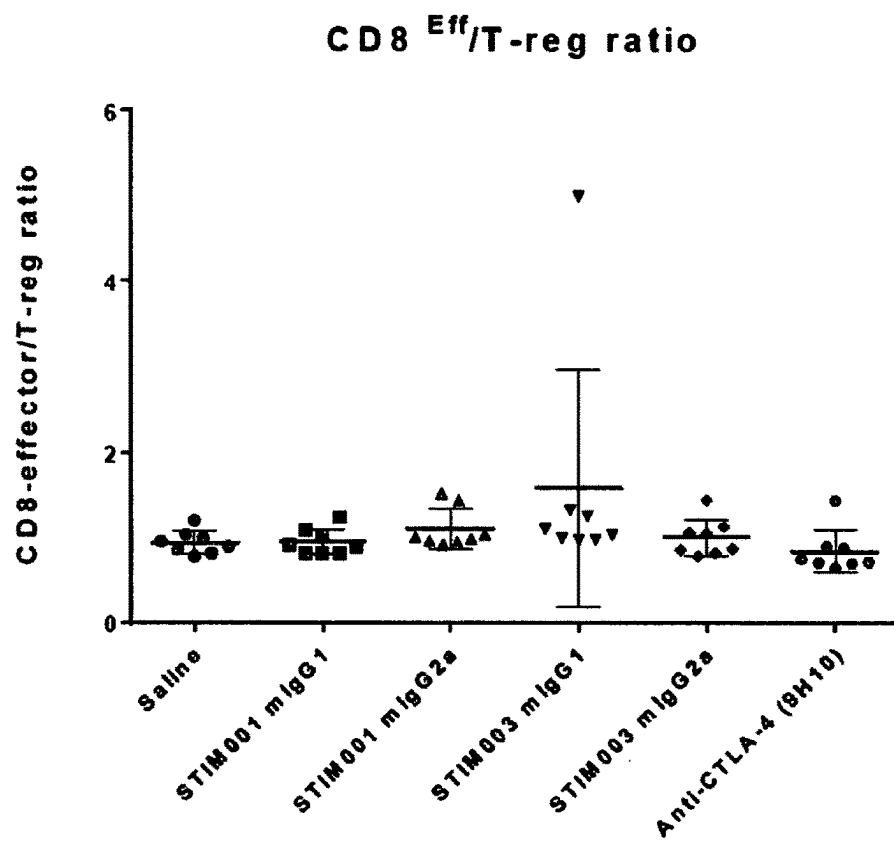


Figure 26

A



B

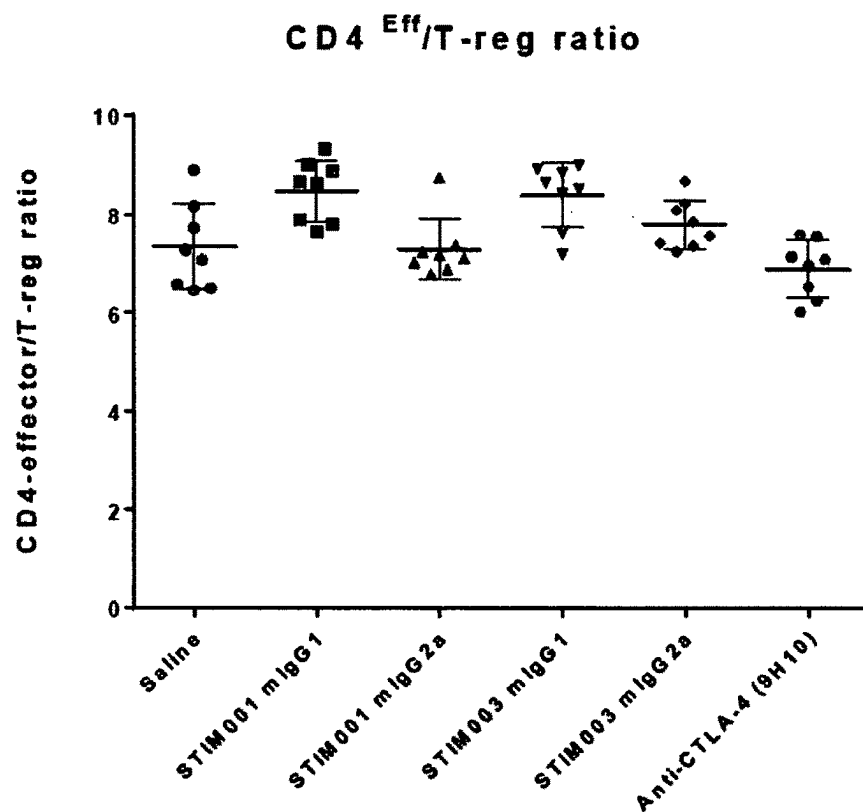
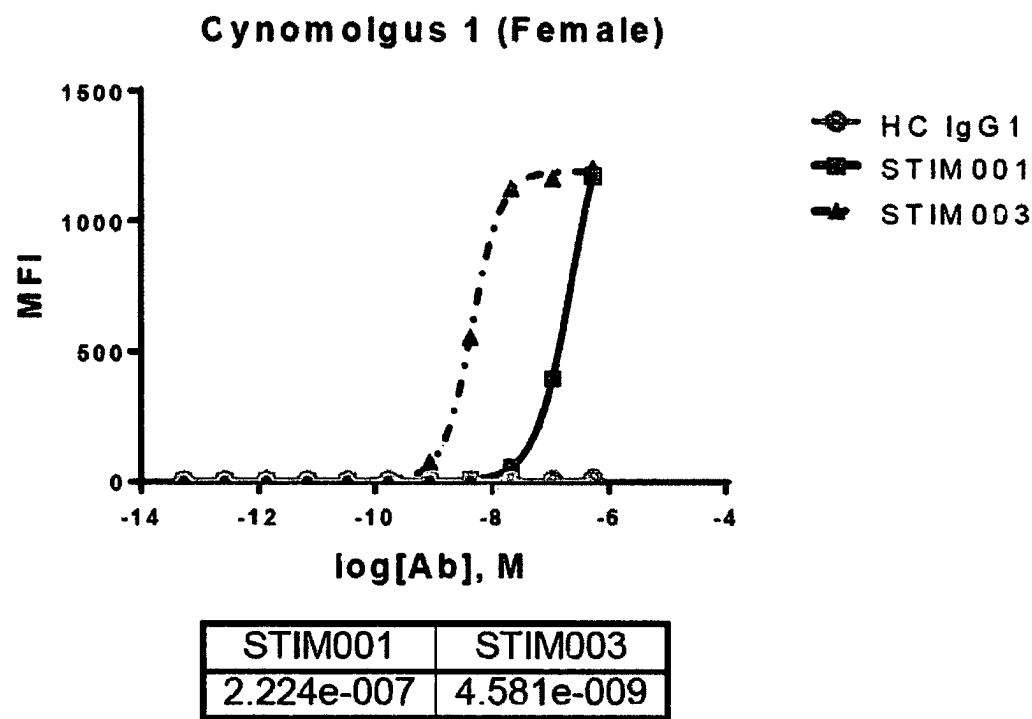


Figure 27

A.



B.

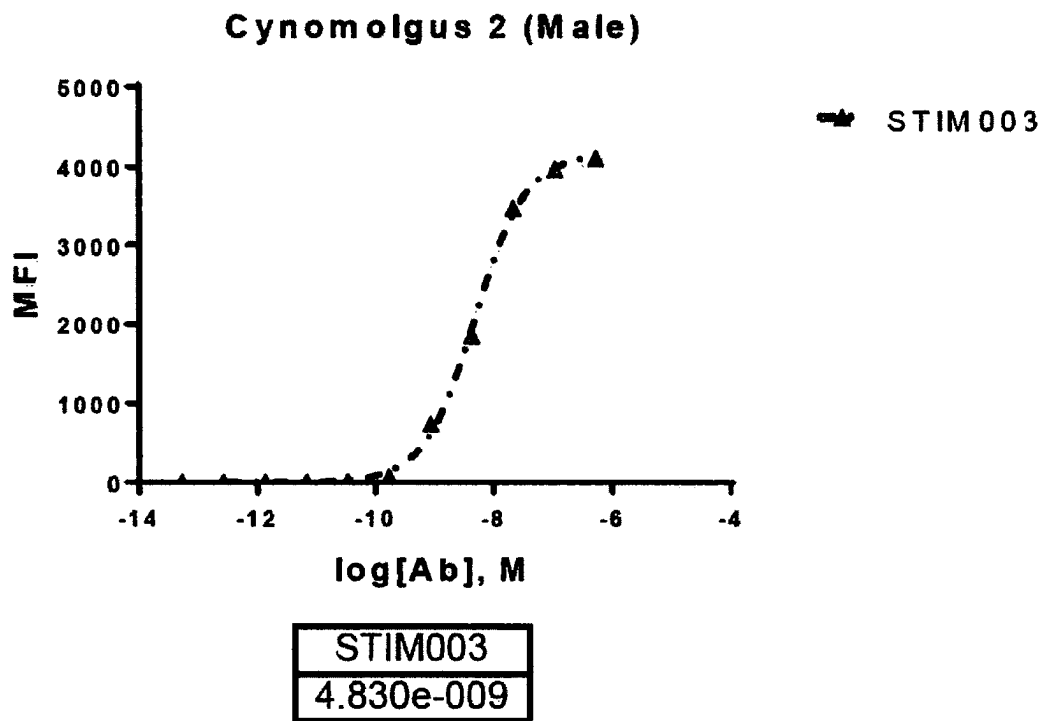


Figure 28

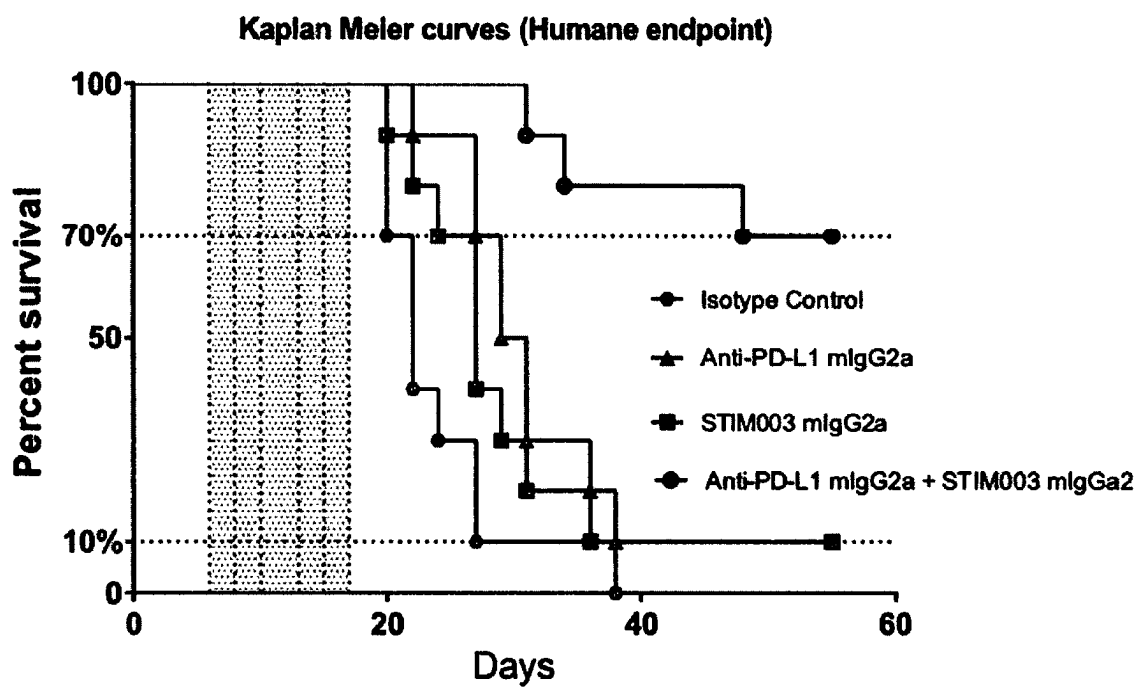


Figure 29

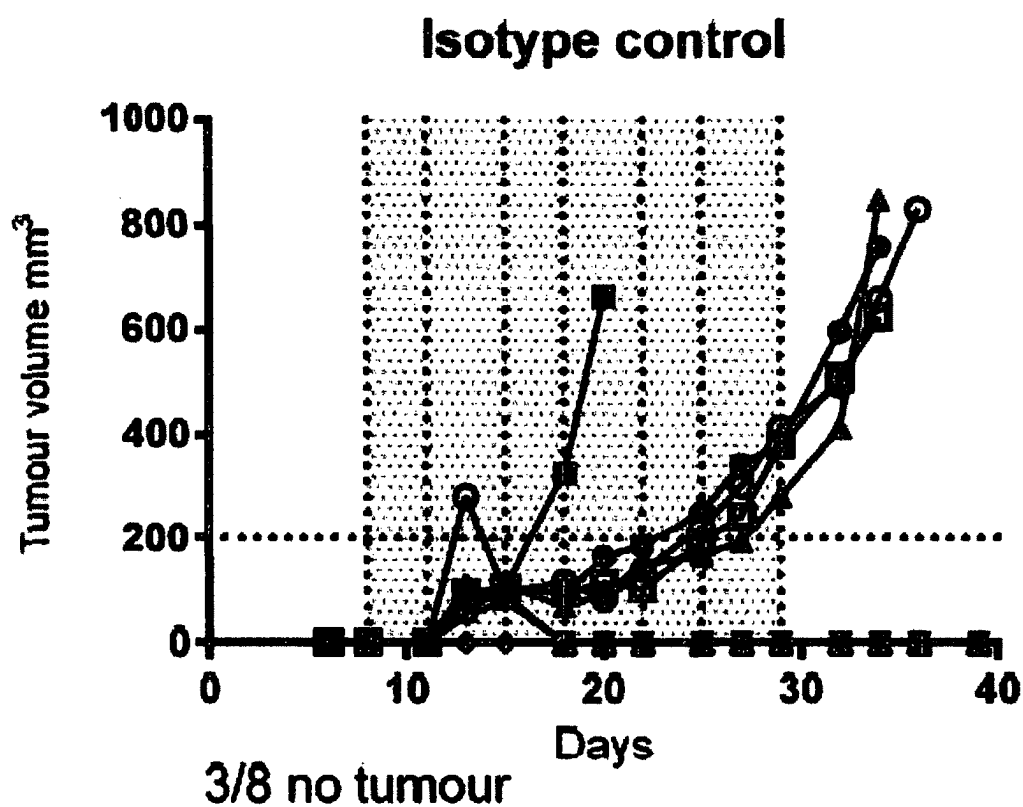


Figure 30

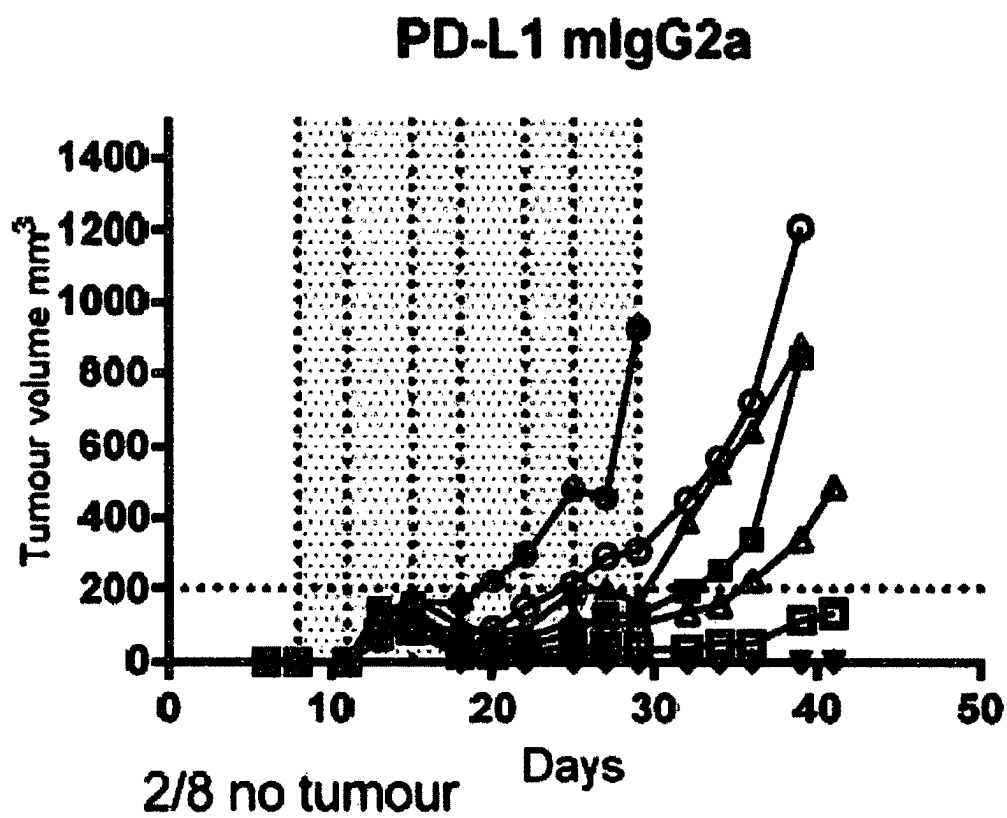


Figure 31

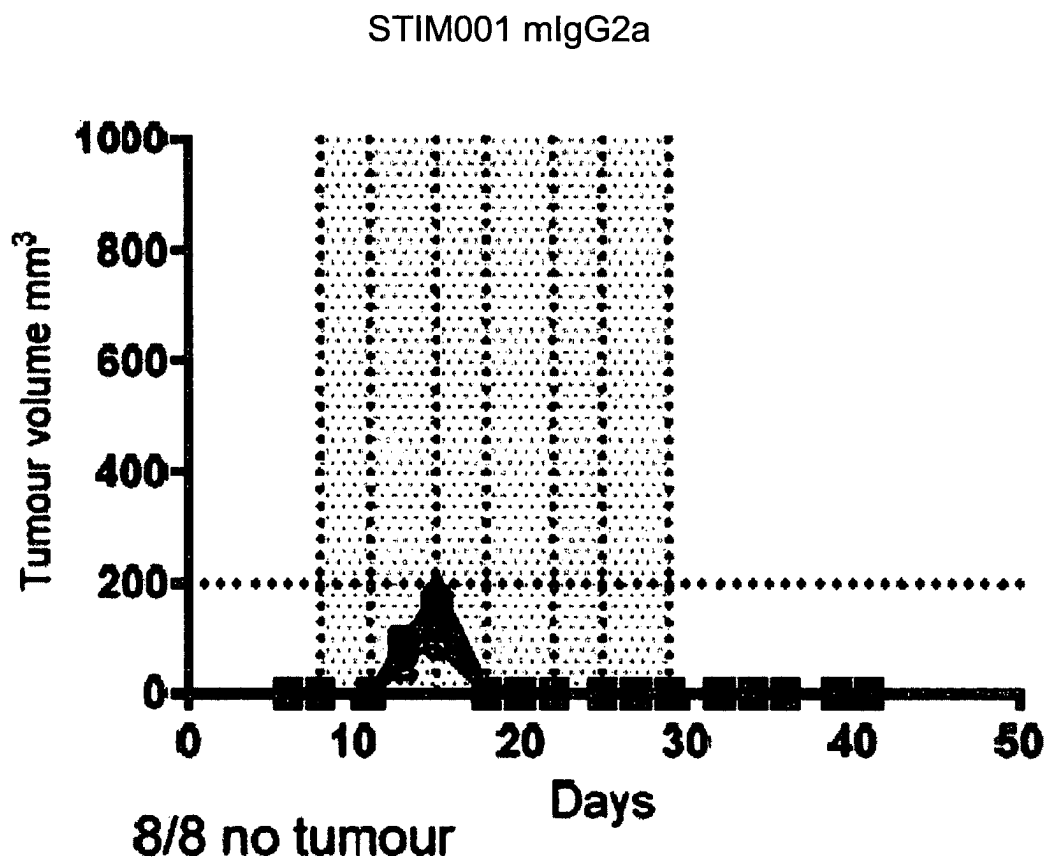


Figure 32

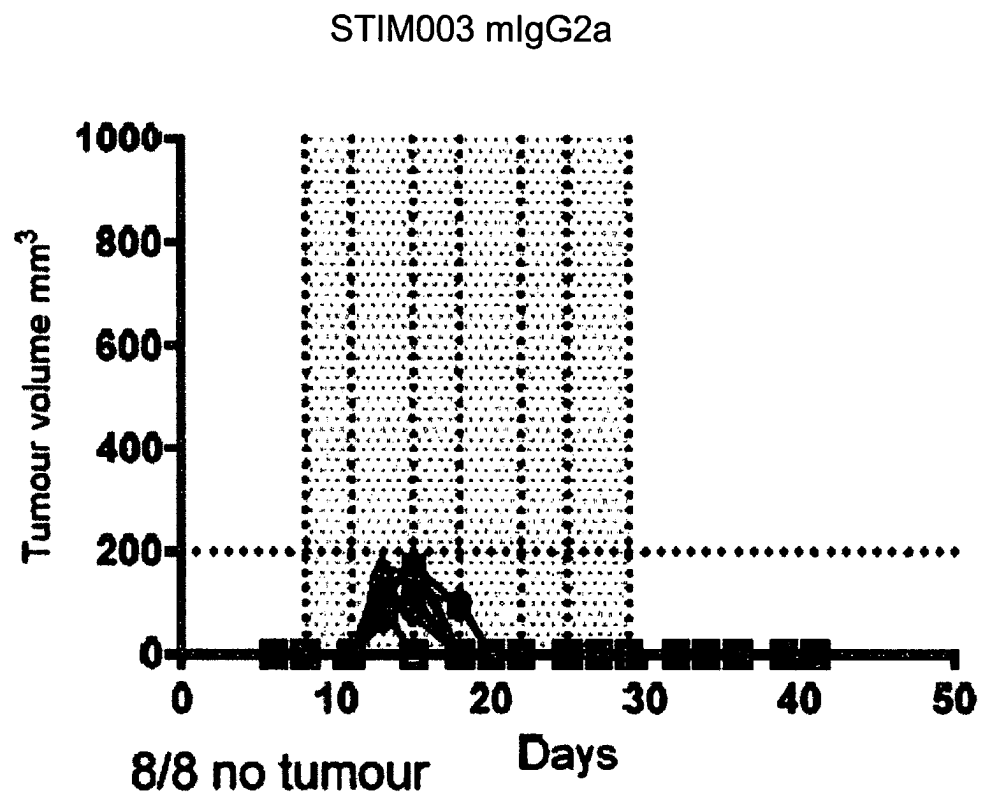


Figure 33

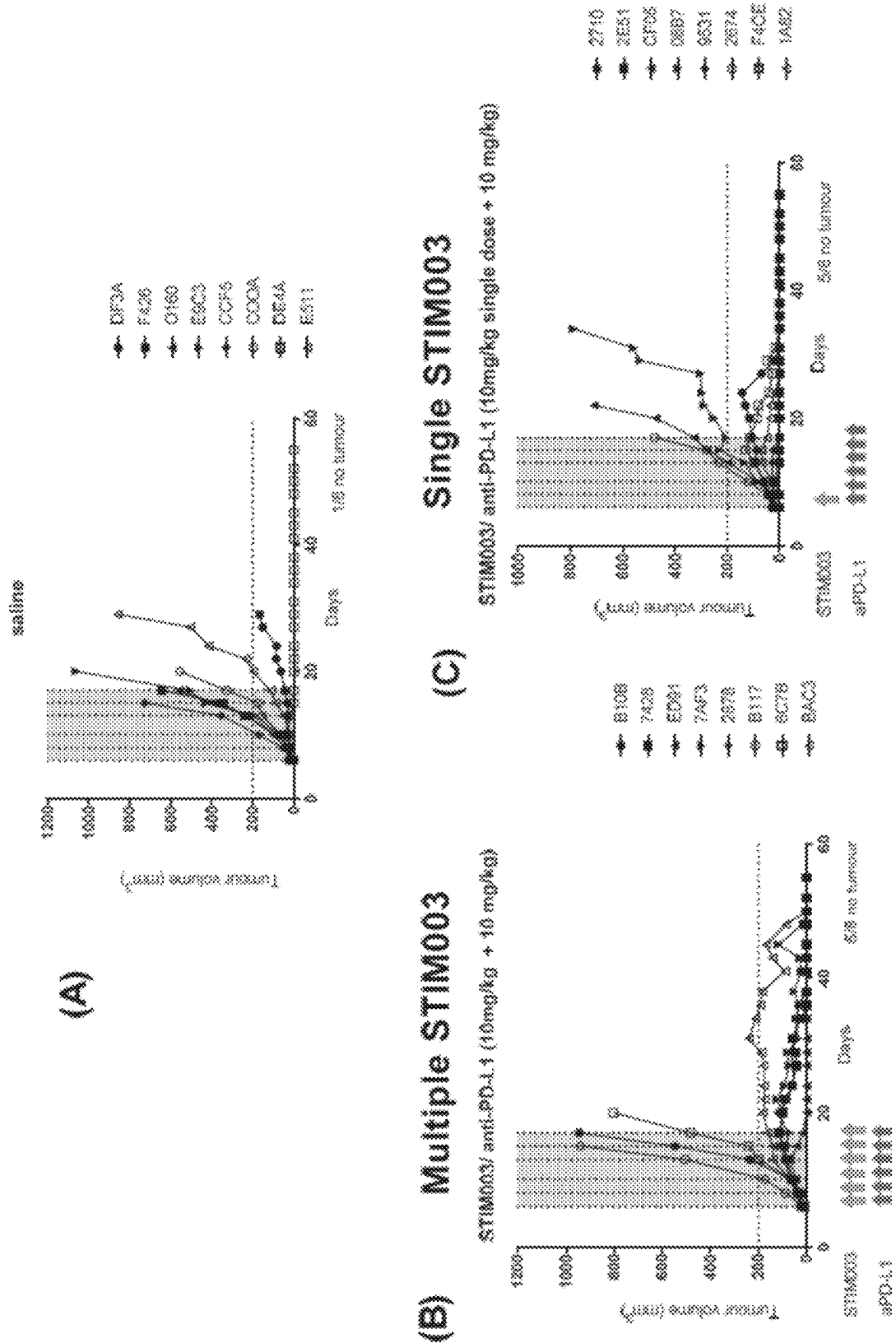


Figure 34

Heavy chain

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78

STIM002 QVQLVQSGGEVKKPGASVKVSKASGYTFTSYGFSWVRQAPGQGLEWMGWISAYNGNTNYAQKLQGRVTM
STIM002-B
STIM001 ...V...A.....STF.IT.....D.....N.....I.
CL-64912R.....V.....H.A.....G.....CG.S.
CL-64841S.
CL-64837S.
CL-64536S.
CL-61091 ...M...T.....T.....T.IT.....S.D.D...F...V
germlineA.....I.....

FW1 CDR1 FW2 CDR2 FW3

79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100 101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119 120 121 122 123 124 125 126 127 128

STIM002 TTDSTSTSTAYMELRSLRSDDTAVYYCARSTYFYGSGTLYGMDVWGQGTITVTVSS
STIM002-B
STIM001SGH.---YY.....
CL-64912 .A.....F...SY..A.....
CL-64841F...SY.....
CL-64837SY.....
CL-64536F...SY.....
CL-61091N.....K.....SGWP---HH.....
germline--.Y...NY.....

FW3 CDR3 FW4

Light chain

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79

STIM002-B DIVMTQSPLSLPVPTEPASISCRSSQSLHSDGYNCLDWYLQKPGQSPQLLIYLGSTRASGFPDRFSGS
STIM002Y.....
STIM001NE..Y.....F...N...V.....
CL-64912N.....
CL-64841N.....
CL-64837N.....
CL-64536N.....
CL-61091N.F.YF.....F.V.N...V.....
germlineN..Y.....N...V.....

FW1 CDR1 FW2 CDR2 FW3

80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100 101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119 120 121 122 123 124 125 126 127

STIM002-B GSGTDFTLKISRVEAEDVGYYCMQALQTPCSFGGQTKLEIK
STIM002L.....
STIM001T.....I.....S...LT..G...V...
CL-64912S.....
CL-64841
CL-64837
CL-64536
CL-61091I.....LT..G...V...
germlineDDSVS.L.....

FW3 CDR3 FW4

Figure 35

Heavy chain

	1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78
germline	EVQLVESGGGVVRRPGGSLRLSCAASGFTFDDYGMSWVRQAPGKGLEWVSGINWNGGSTGYADSVKGRFTI
STIM003	v...v.....d.d.s.....
CL-71642	i.....i.dn.d.....
CL-74570	i.....i.dn.d.....
	FW1 CDR1 FW2 CDR2 FW3
	79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100 101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119 120 121 122 123 124 125 126 127 128
germline	SRDNAKNSLYLQMNSLRAEDTALYYCARDYYGSGSYYN-YFDYWGQGTLVTVSS
STIM003	f.....hvp.....i.....
CL-71642	a.....vp.....
CL-74570	f.....vp.....
	FW3 CDR3 FW4

Light chain

	1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85
germline	EIVLTQSPGTLSSLSPGERATLSCRASQSVSSSYLAWYQQKPGQAPRLLIYGASSRATGIPDRFSGSGSGT
STIM003	r.....r.....d.....
CL-71642	
CL-74570	
	FW1 CDR1 FW2 CDR2 FW3
	86 87 88 89 90 91 92 93 94 95 96 97 98 99 100 101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119 120 121 122 123 124 125 126 127
germline	DFTLTISRLEPEDFAVYYCQQYGSSPFTFGPGTKVDIK
STIM003	s.....h..dm.....
CL-71642	h.....n.....
CL-74570	h.....n.....
	FW3 CDR3 FW4

Figure 36

Heavy chain

	1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77
germline	QITLKESGPTLVKPTQTLTLTCTFSGFSLSTSGVGVGWIRQPPGKALEWLALIIYWDDDKRYSPSLKSRLT
STIM007	t.....v.....
STIM008	v.....
	FW1 CDR1 FW2 CDR2 FW3
	78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100 101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119 120 121 122 123 124 125 126 127 128
germline	ITKDTSKNQVVLMTNMDPVDTATYYCAHRHGSESYYYYGMDVWGQGTTVTVSS
STIM007	f.t.gy..a...h.....
STIM008	f.t.gy..a...h.....
	FW3 CDR3 FW4

Light chain

	1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86
germline	EIVLTQSPATLSLSPGERATLSCRASQSVSSYLAWYQQKPGQAPRLIYDASNRTATGIPARFSGSGSGTD
STIM007	tn....h.....
STIM008	tn....h.....
	FW1 CDR1 FW2 CDR2 FW3
	87 88 89 90 91 92 93 94 95 96 97 98 99 100 101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119 120 121 122 123 124 125 126 127
germline	FTLTISSELPEDFAVYYCQQRSNWPLTFGGGKVEIK
STIM007	h.....
STIM008	
	FW3 CDR3 FW4

Figure 37

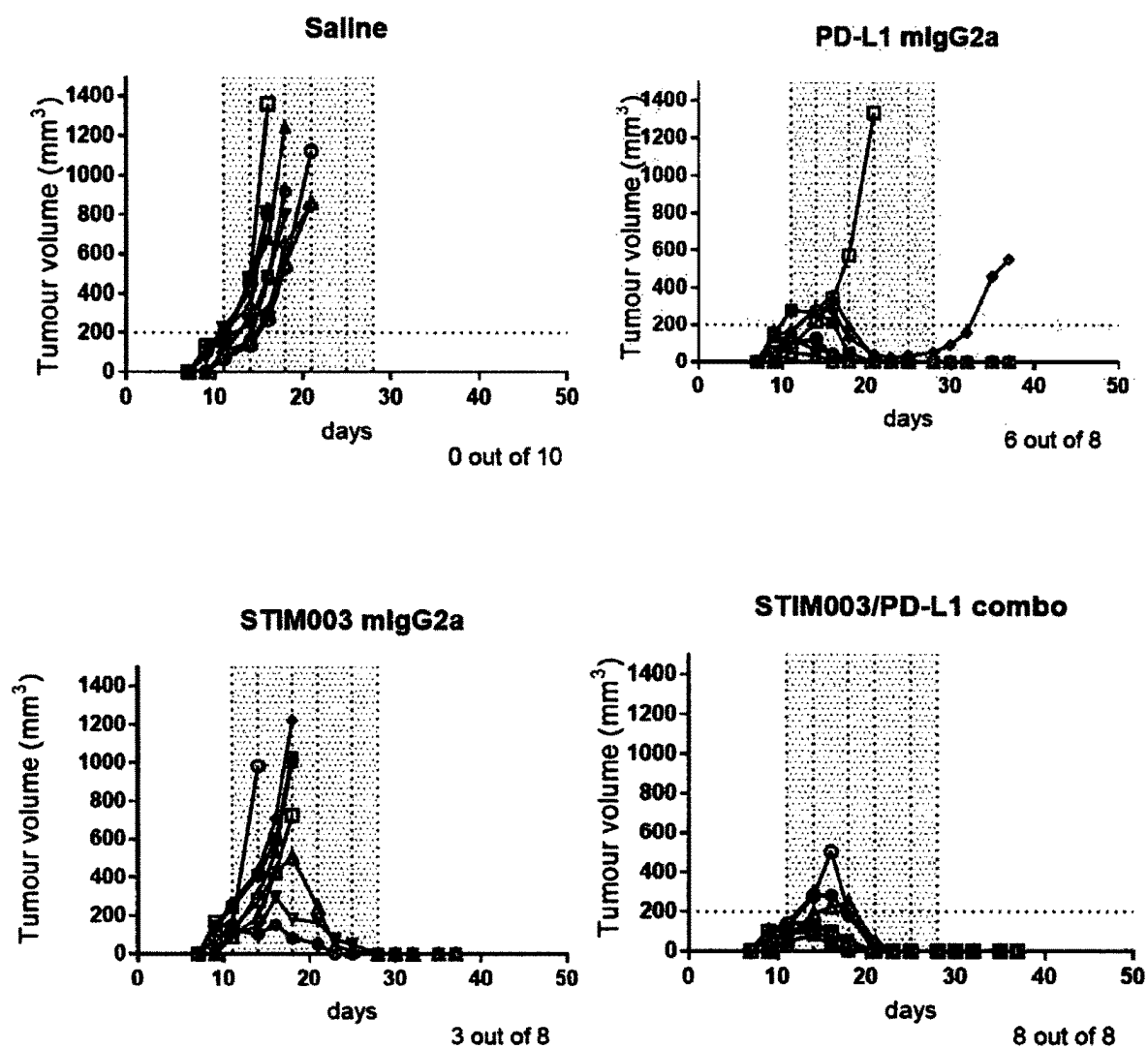


Figure 38

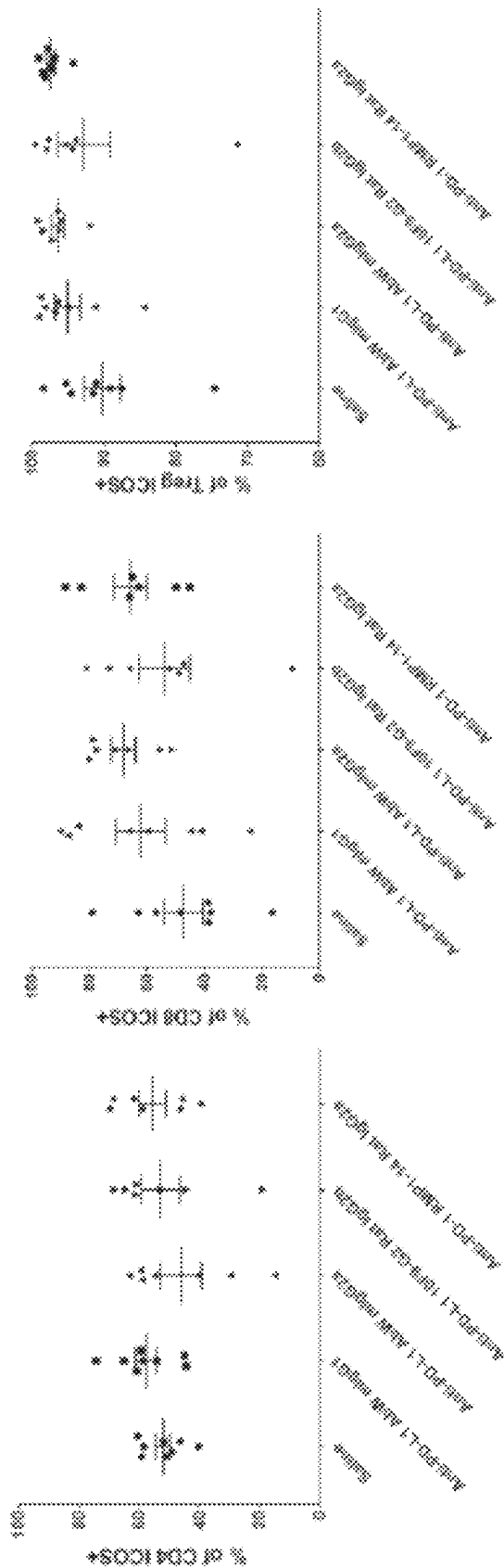


Figure 39A

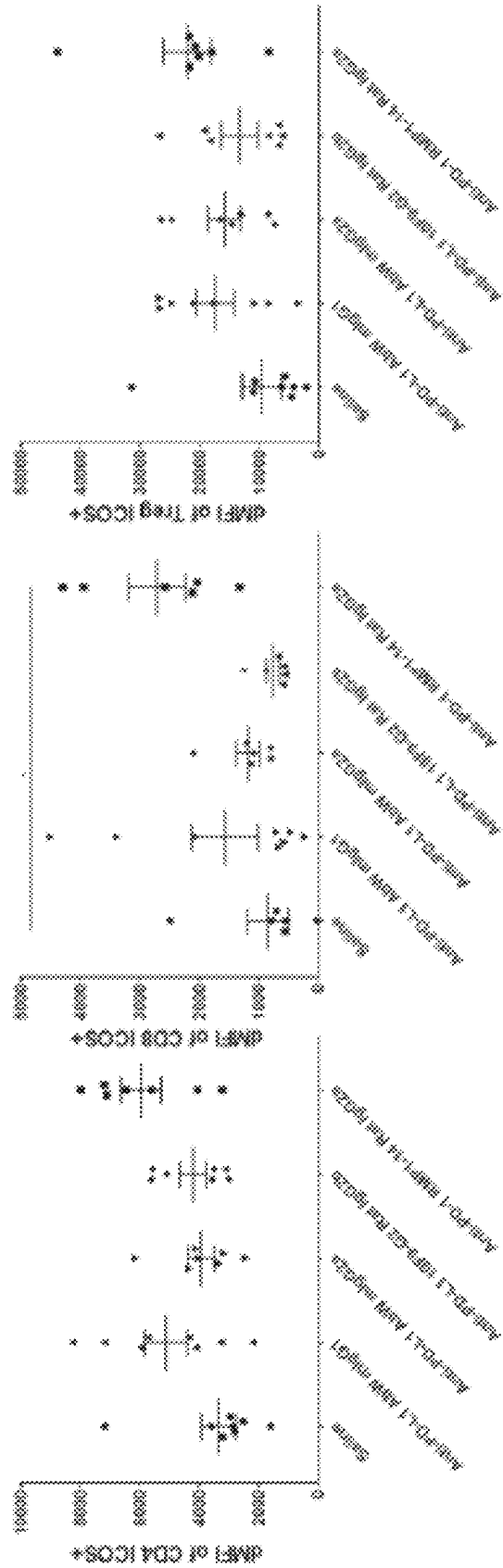
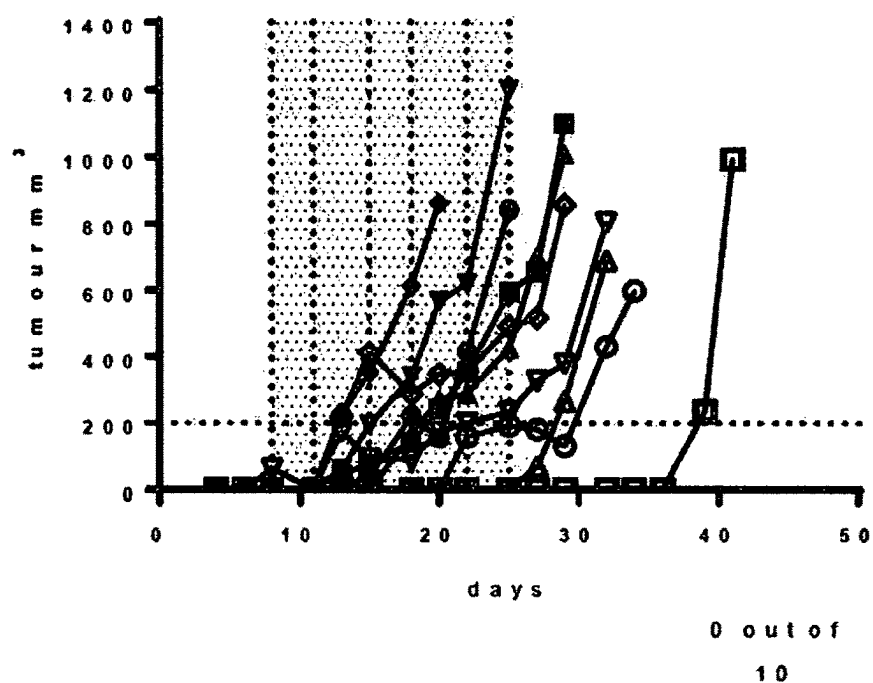


Figure 39B

A)



B)

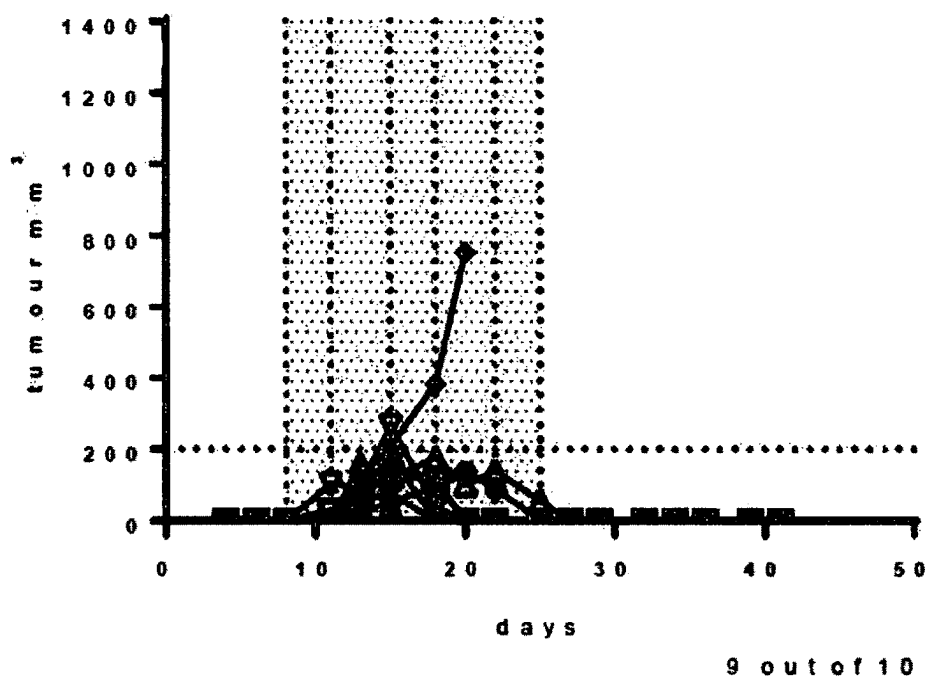


Figure 40 A and B

C)

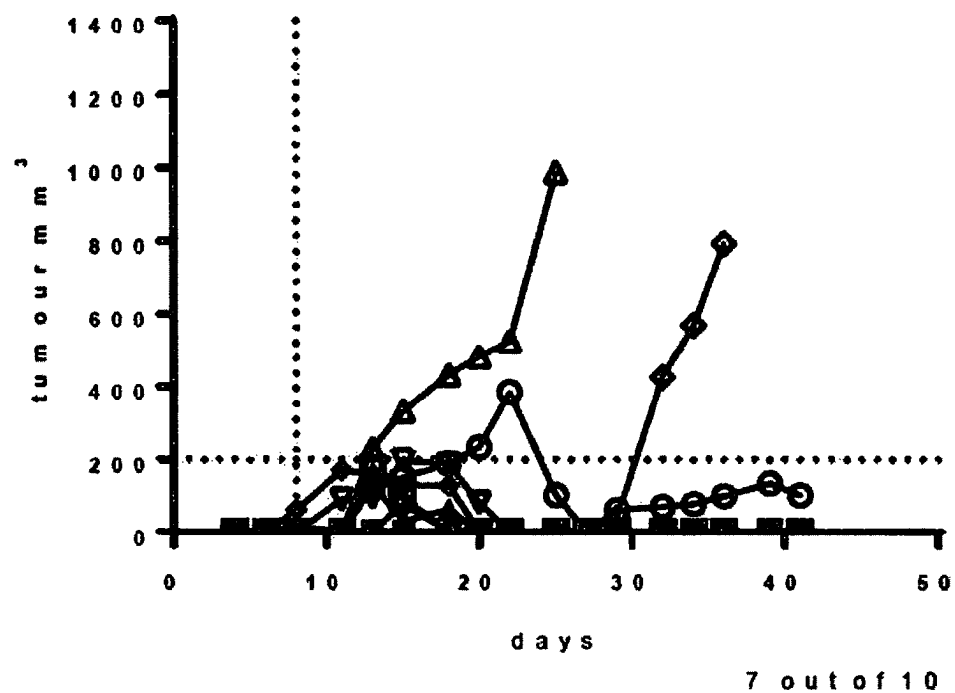


Figure 40C

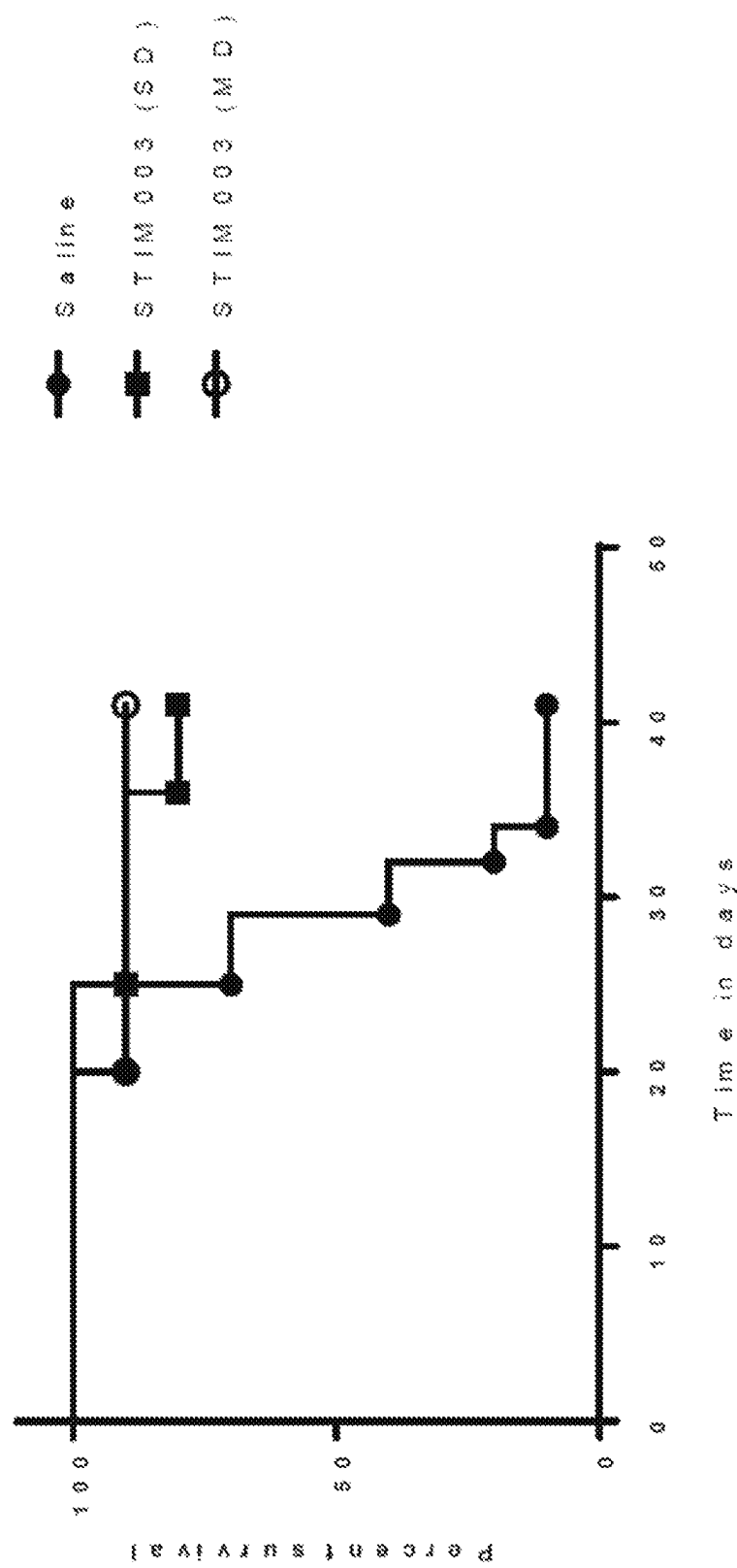


Figure 41

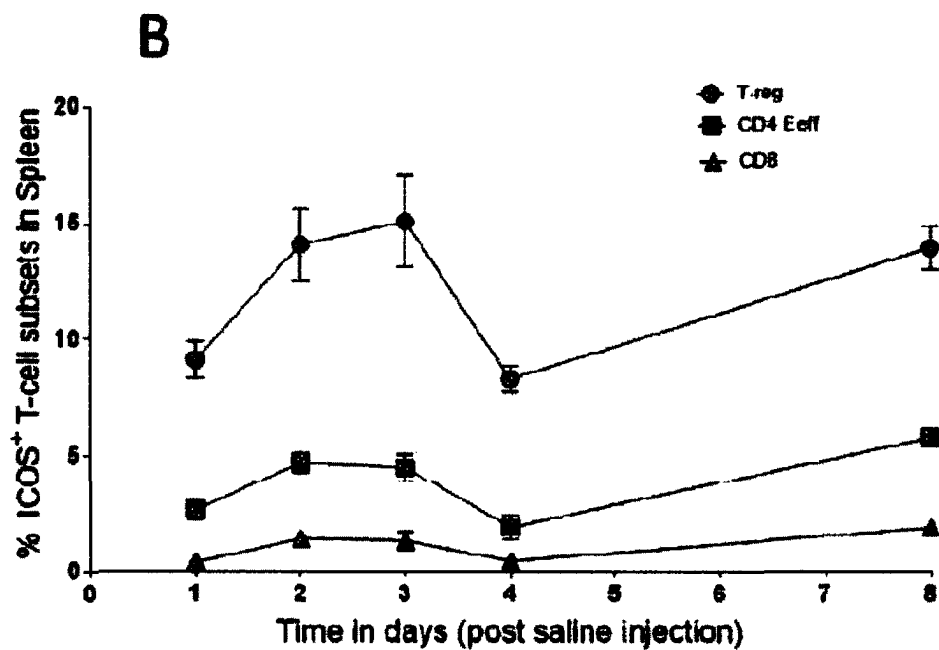
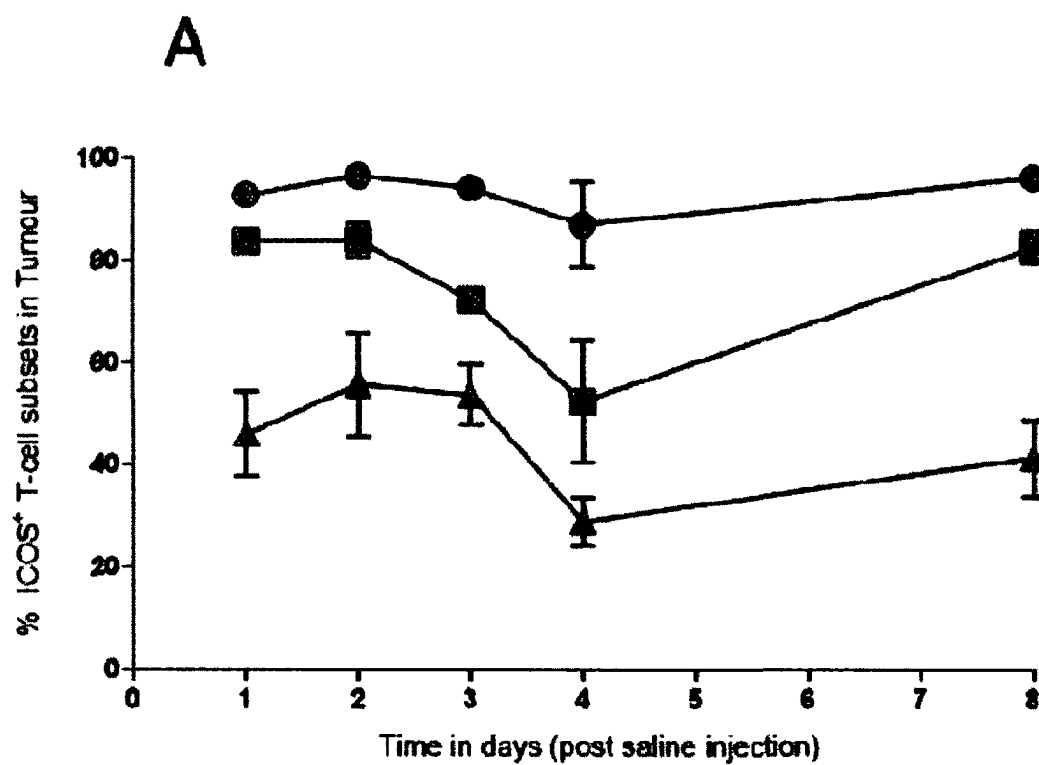
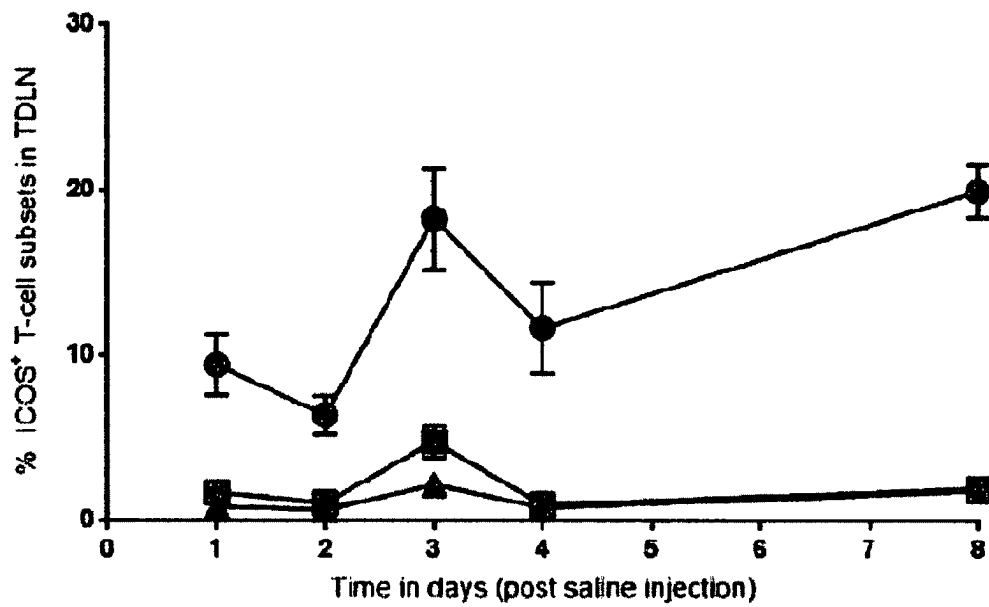


Figure 42 A and B

C



D

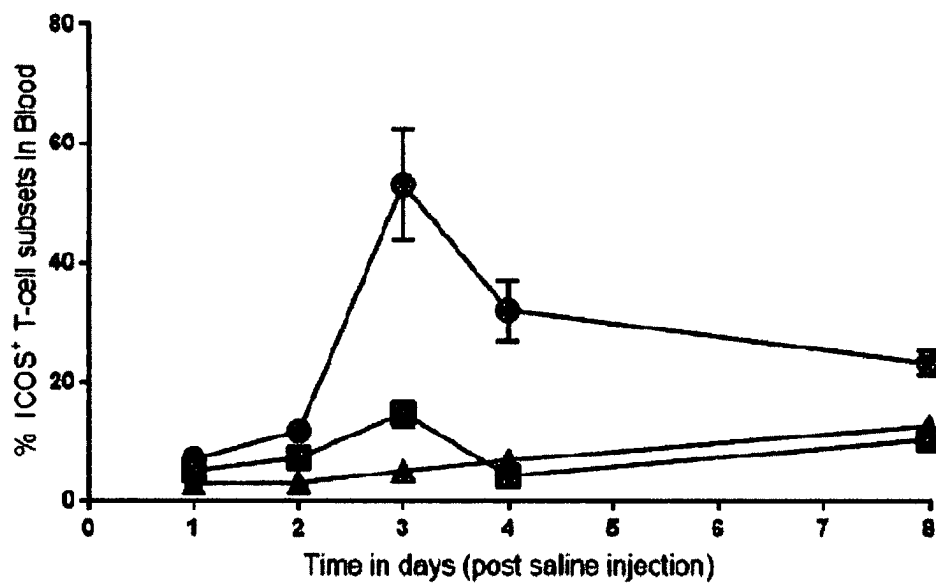


Figure 42 C and D

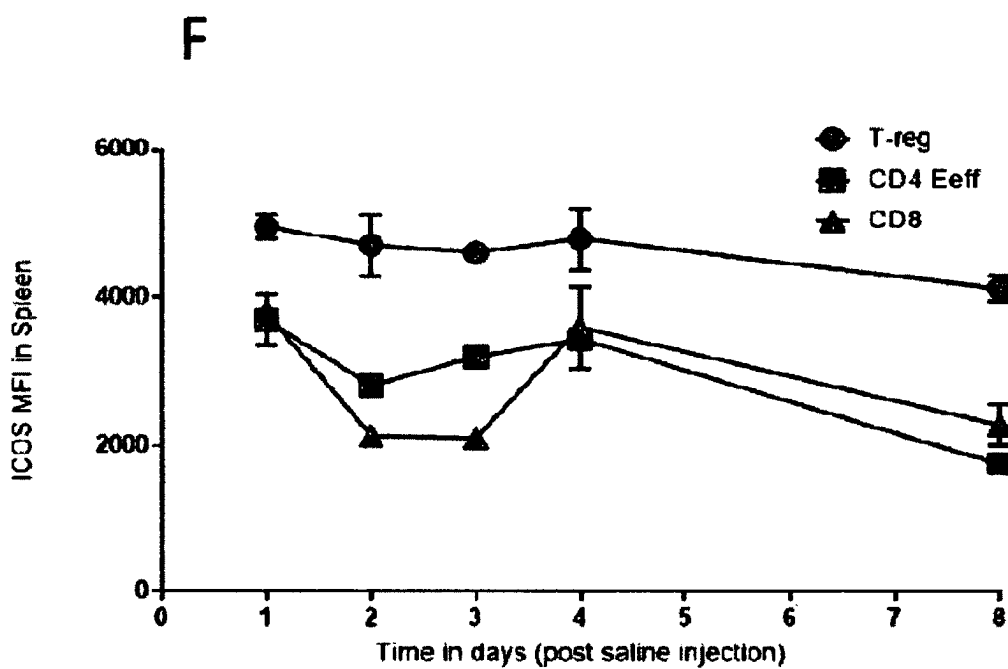
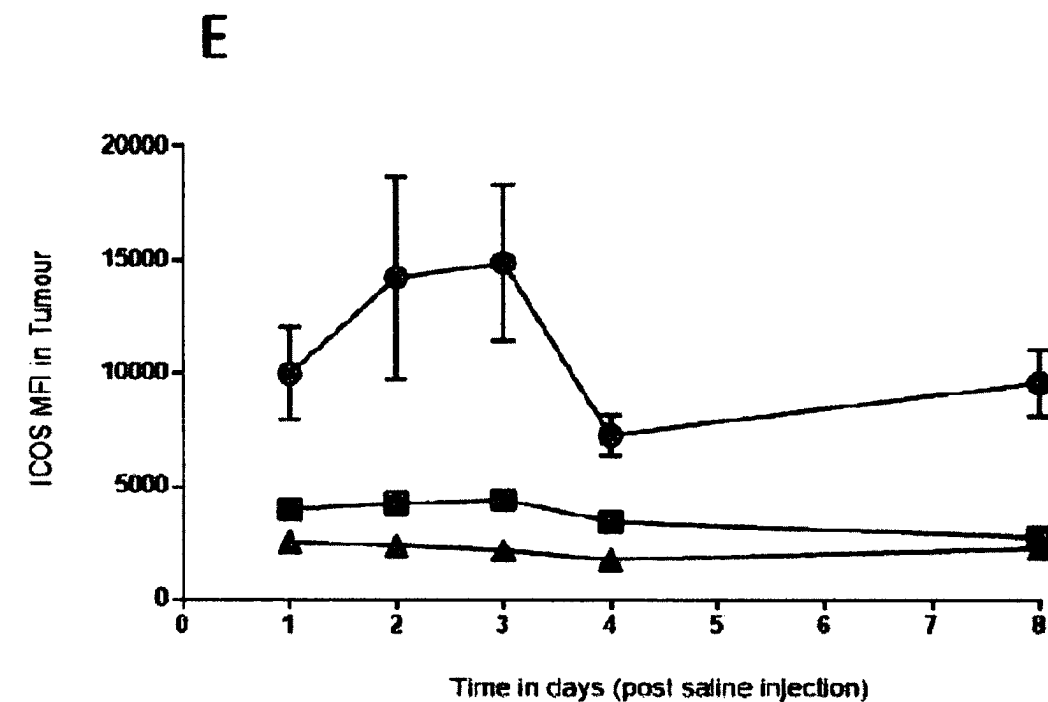


Figure 42 E and F

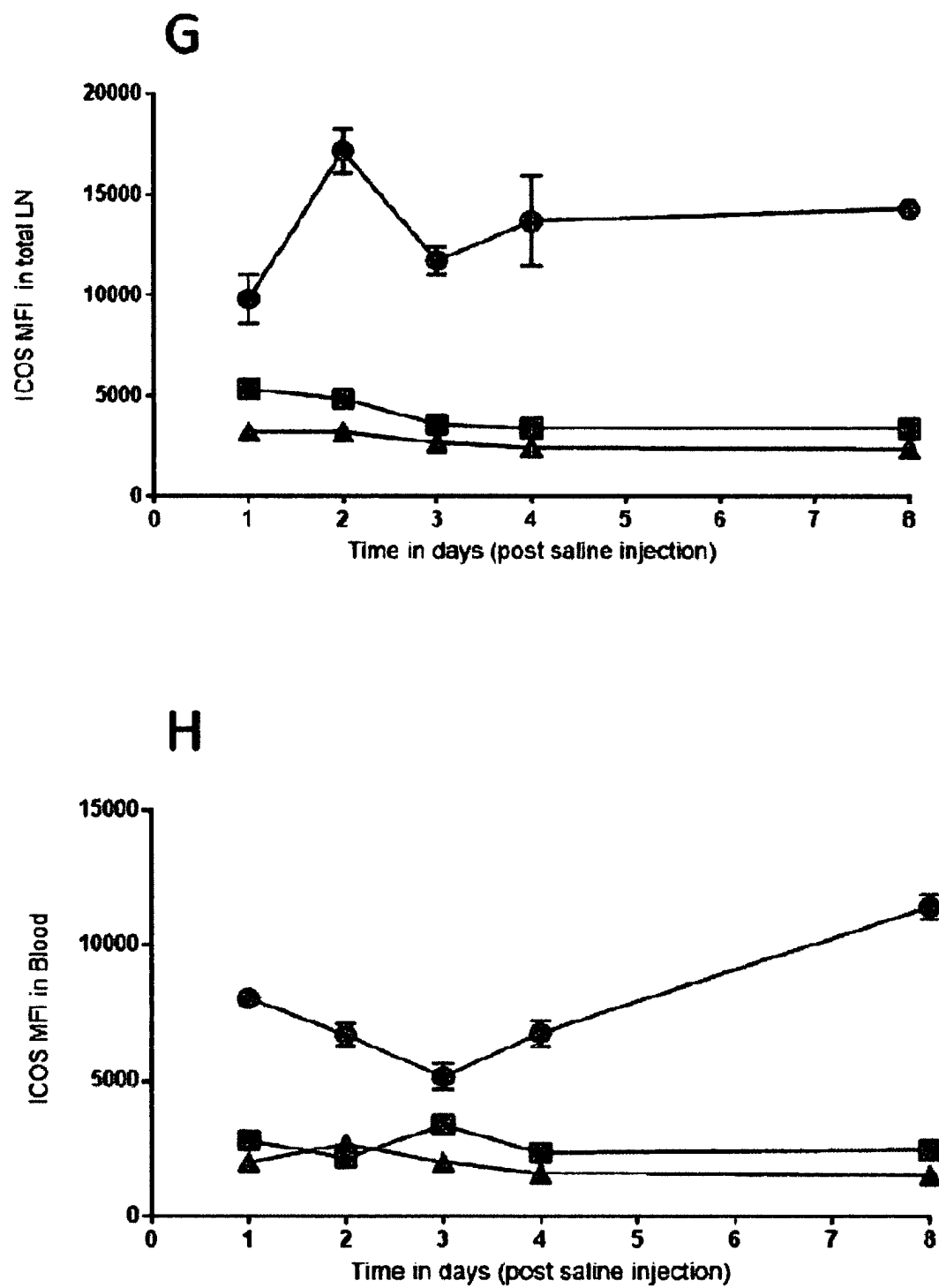


Figure 42 G and H

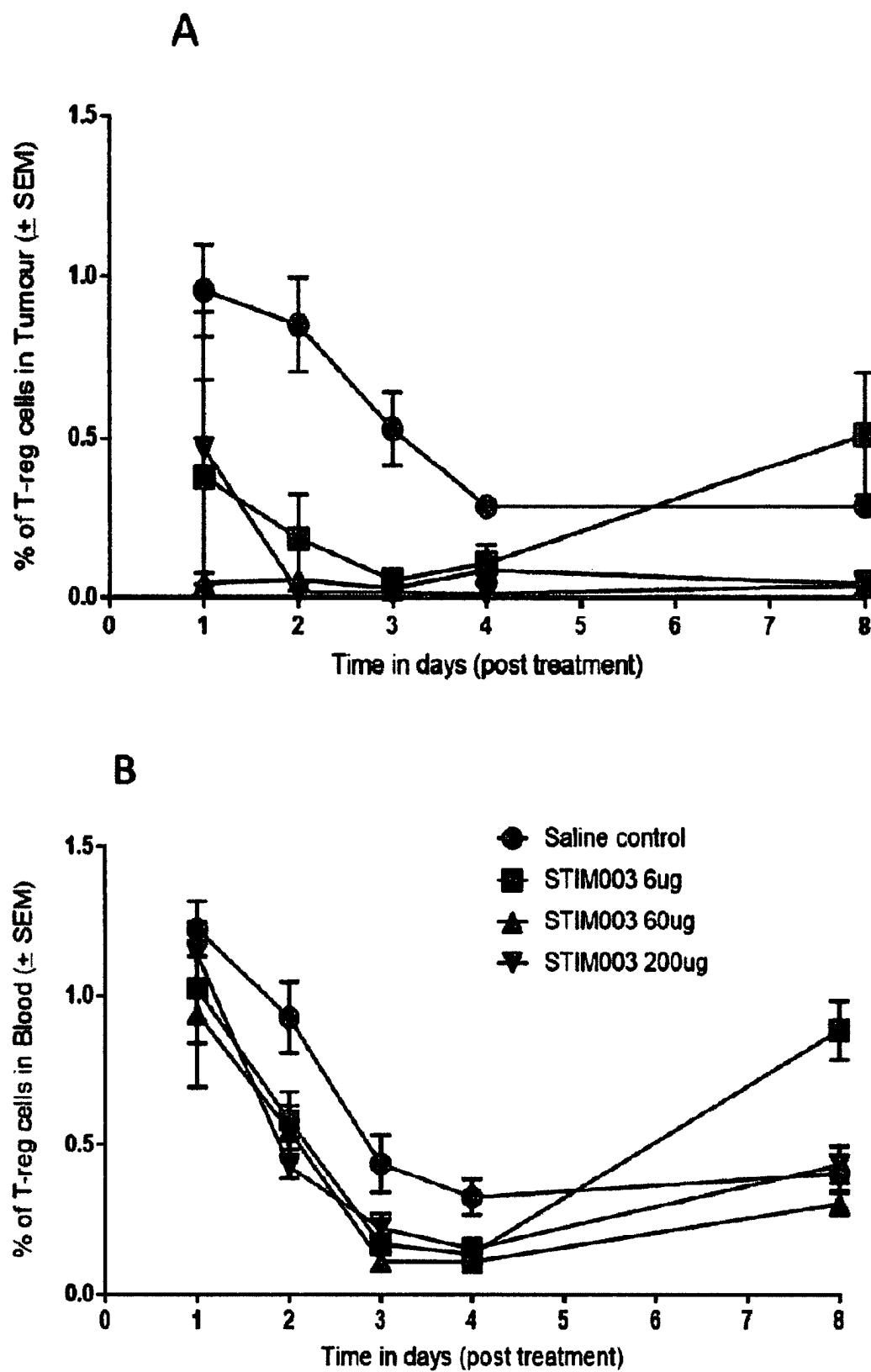


Figure 43

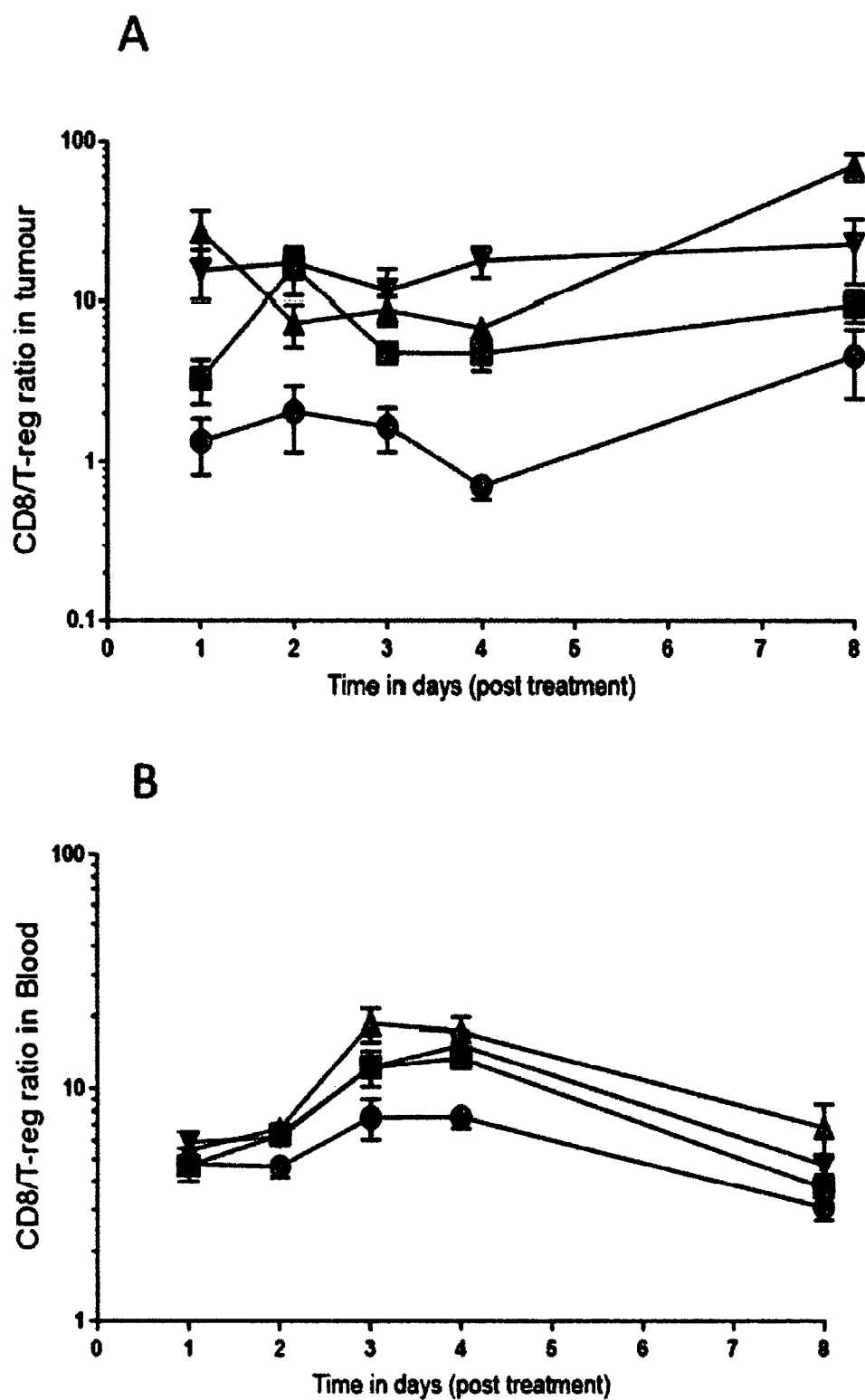


Figure 44 A and B

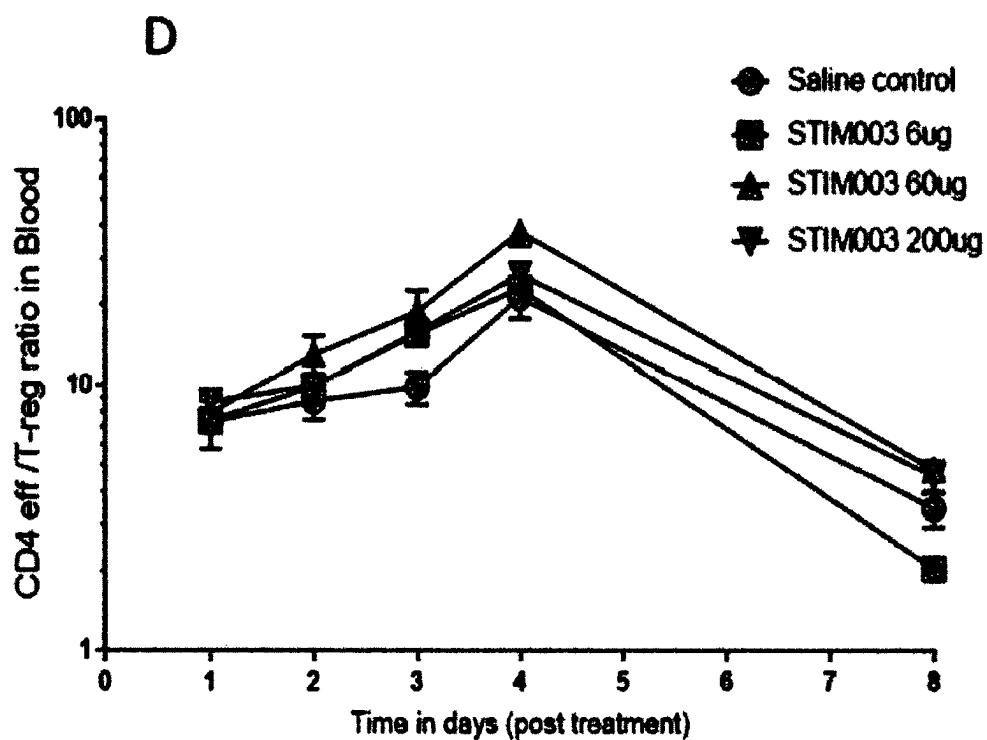
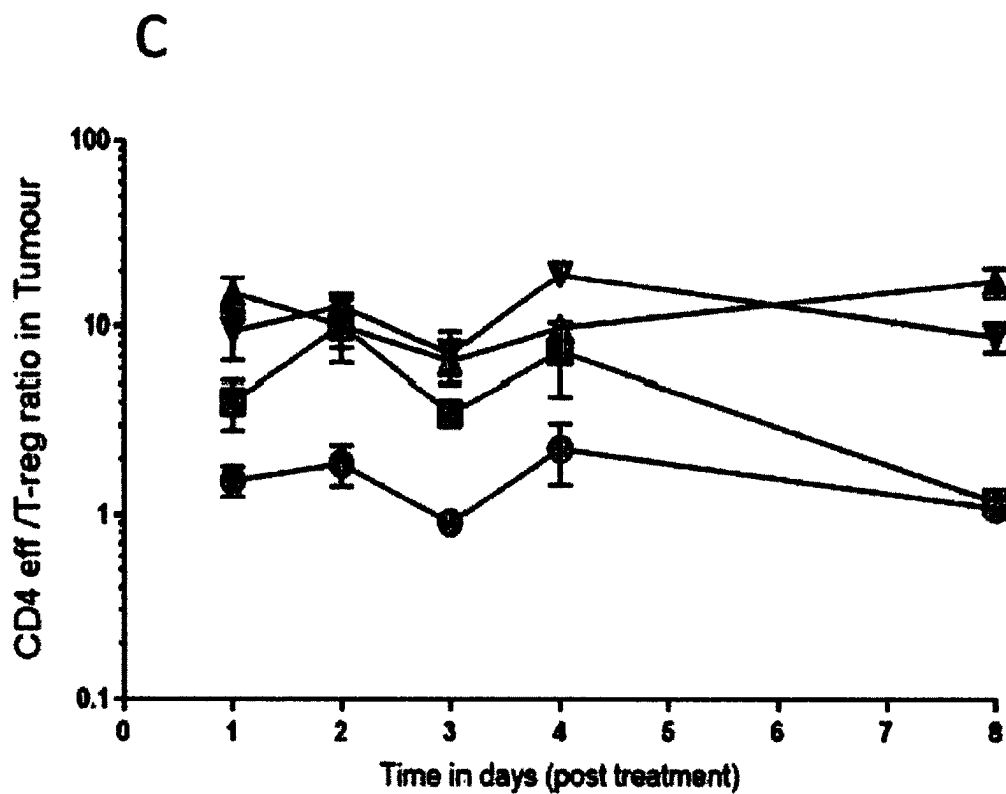


Figure 44 C and D

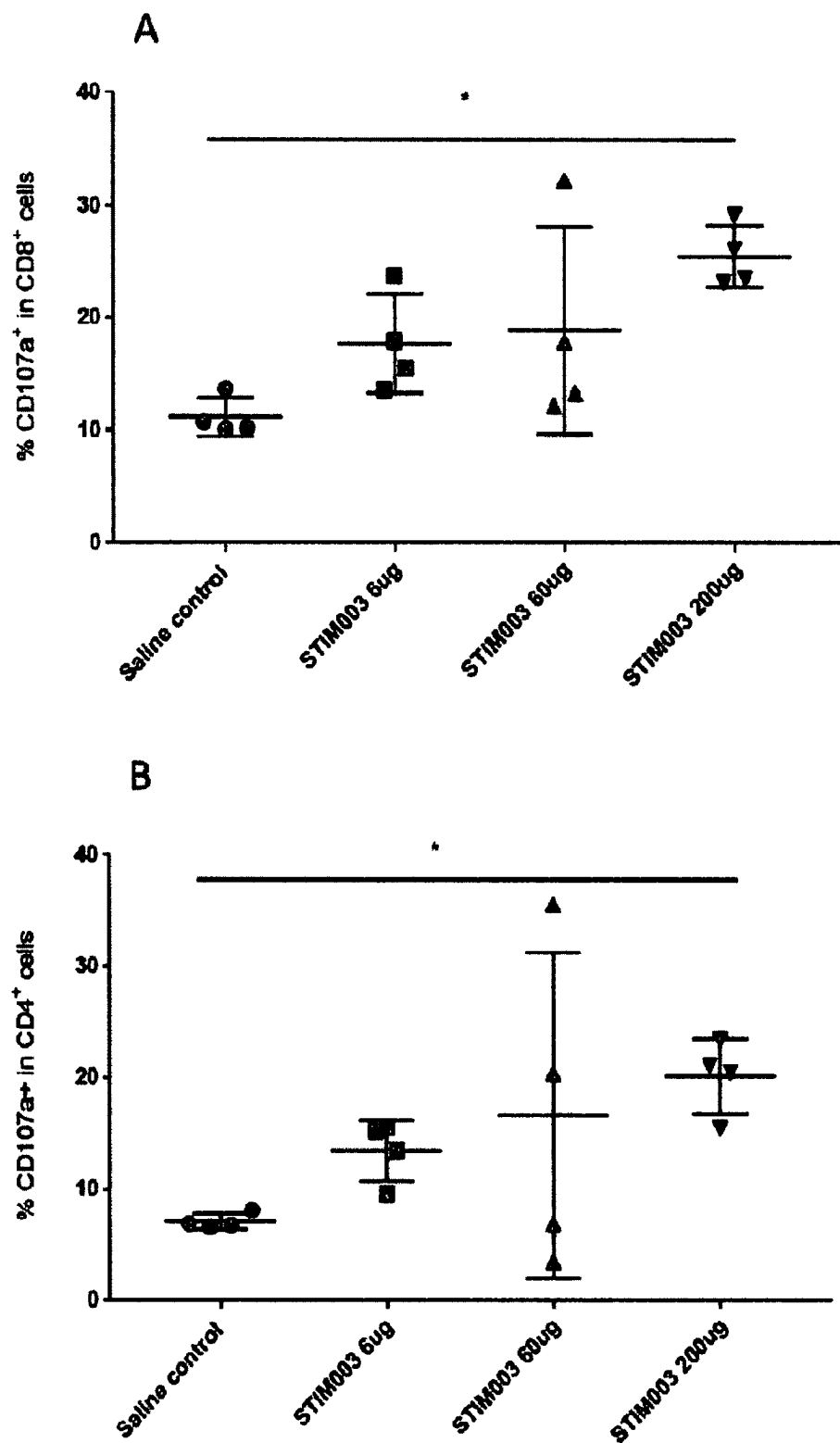


Figure 45 A and B

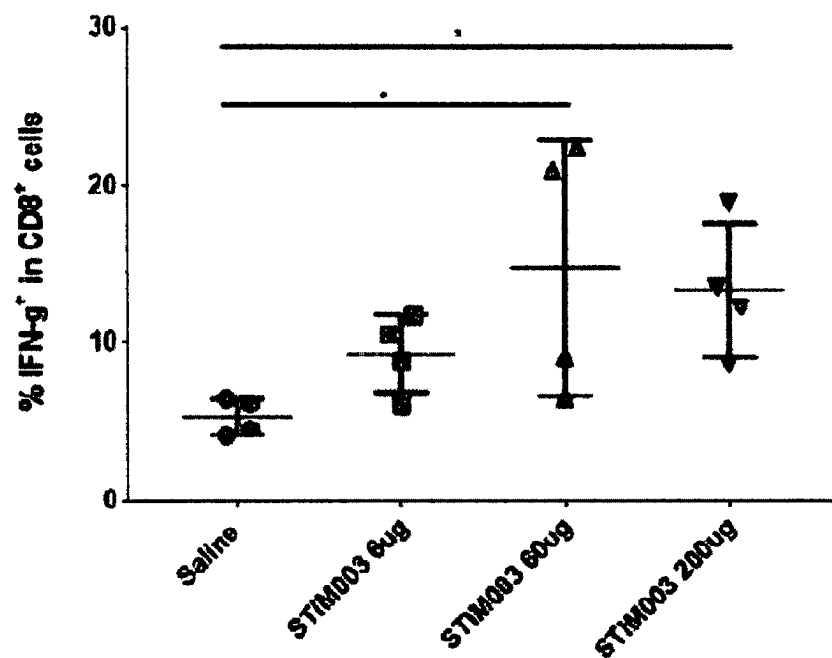
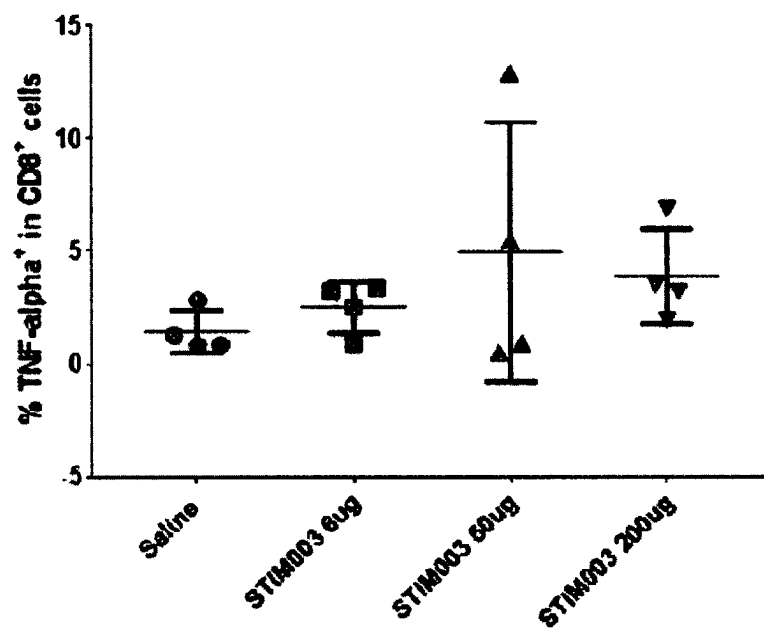
C**D**

Figure 45 C and D

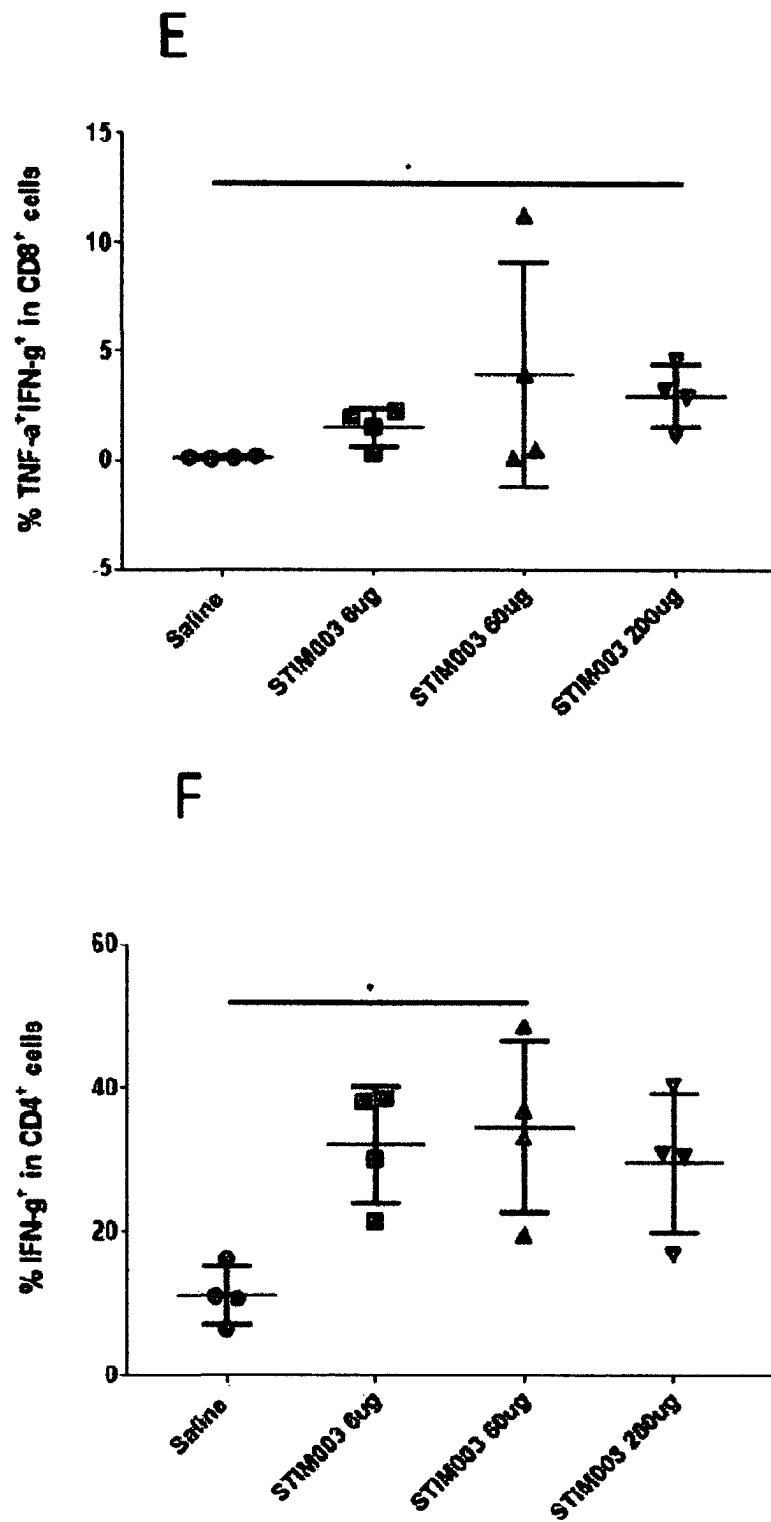


Figure 45 E and F

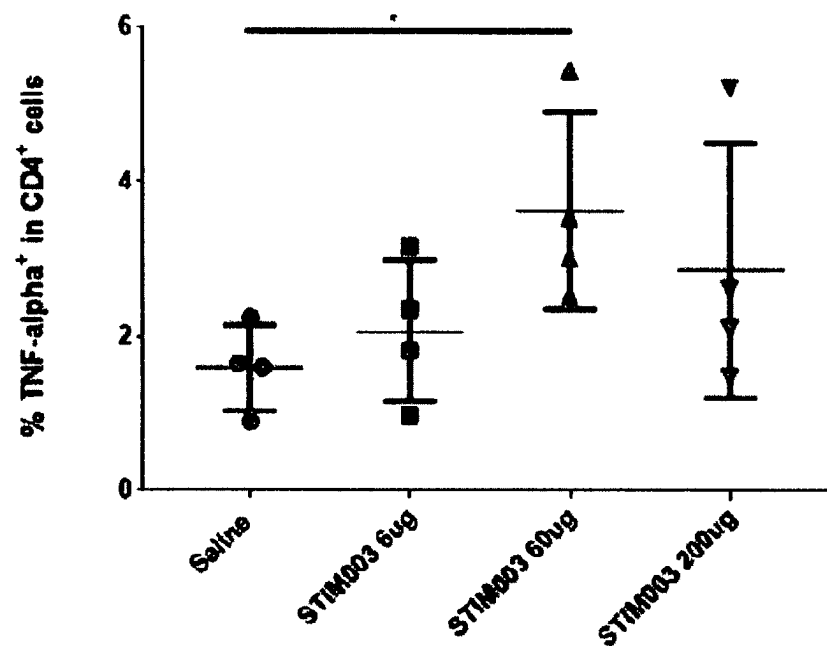
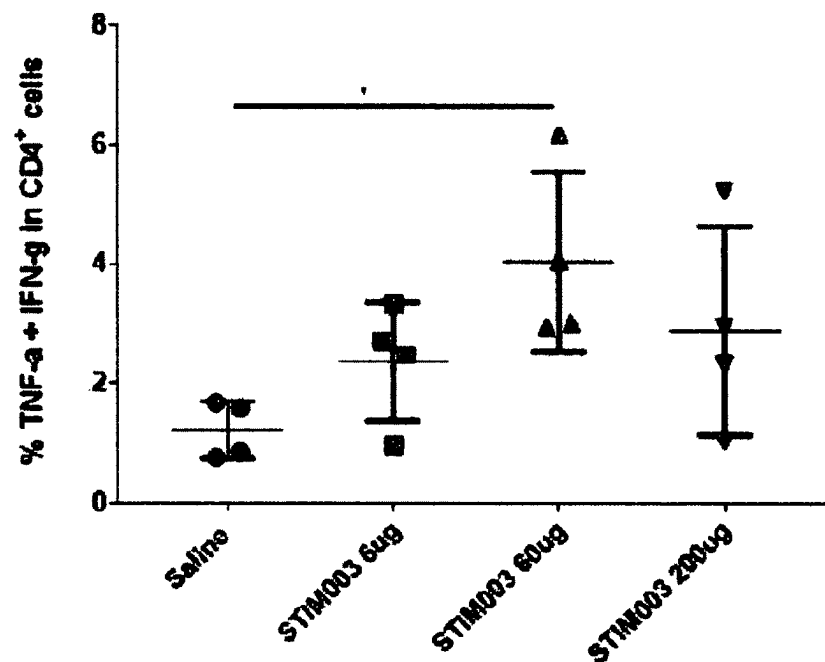
G**H**

Figure 45 G and H

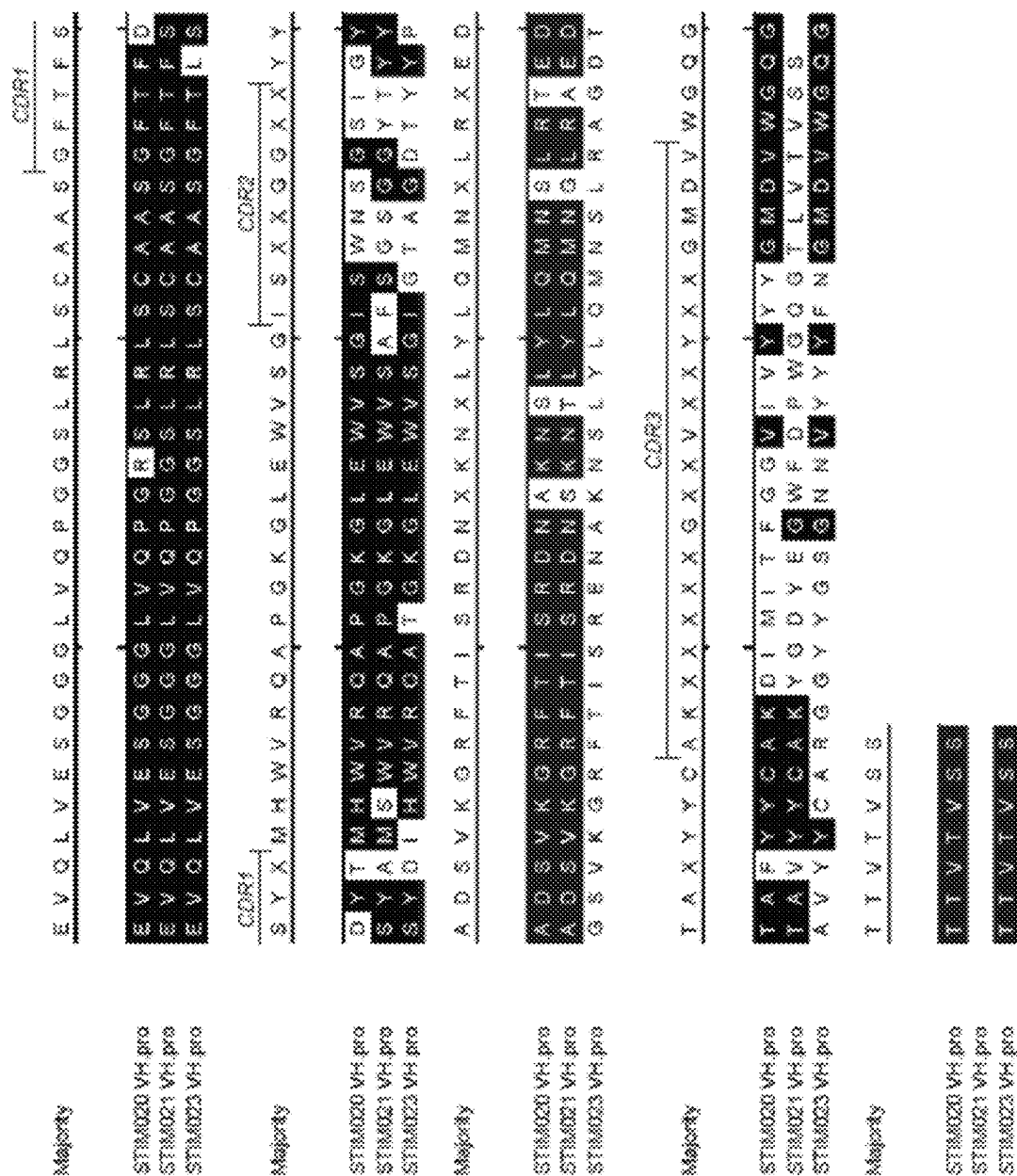


Figure 46a

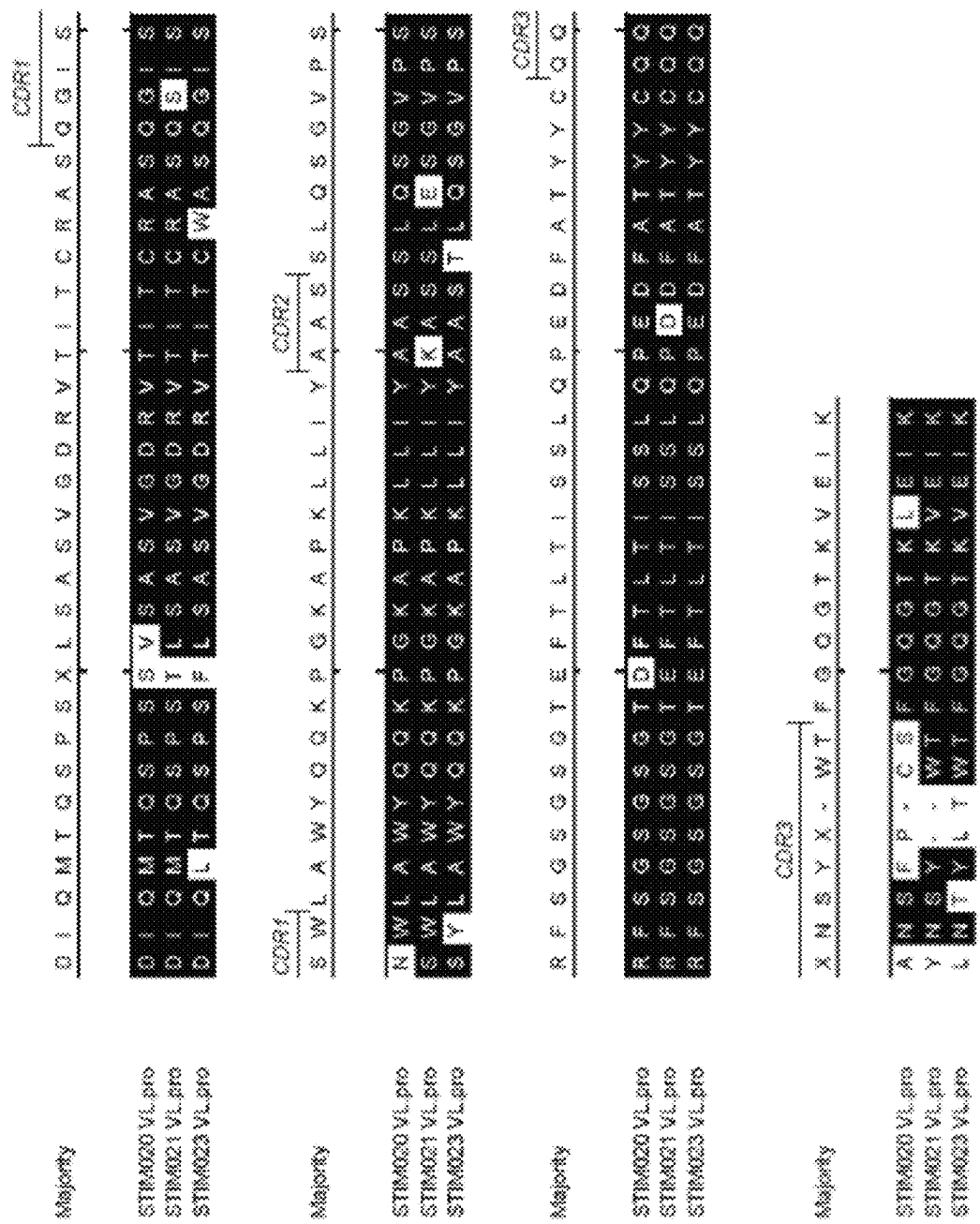


Figure 46b

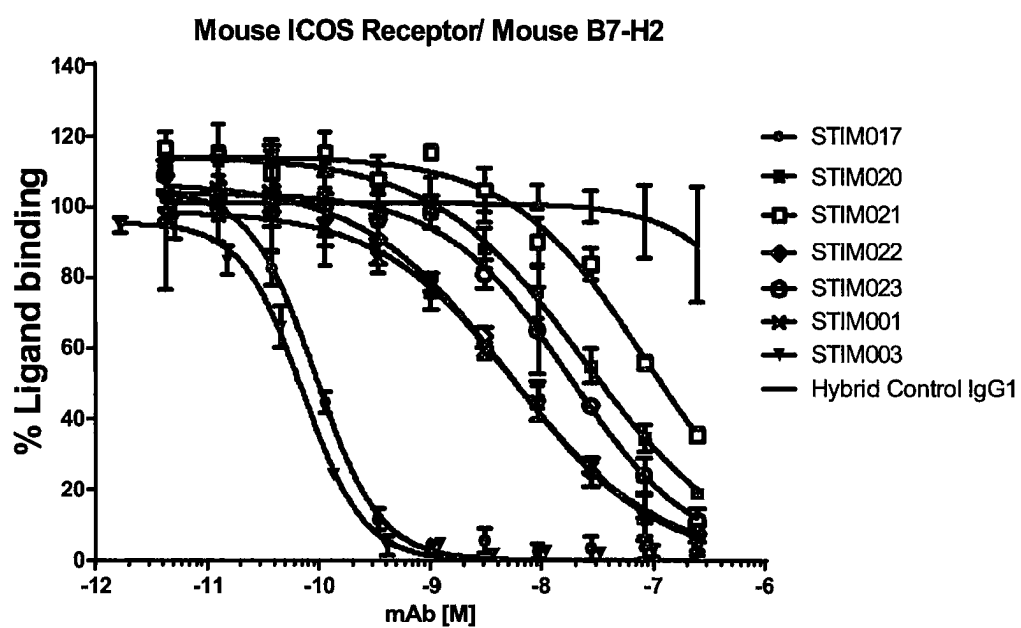


Figure 47

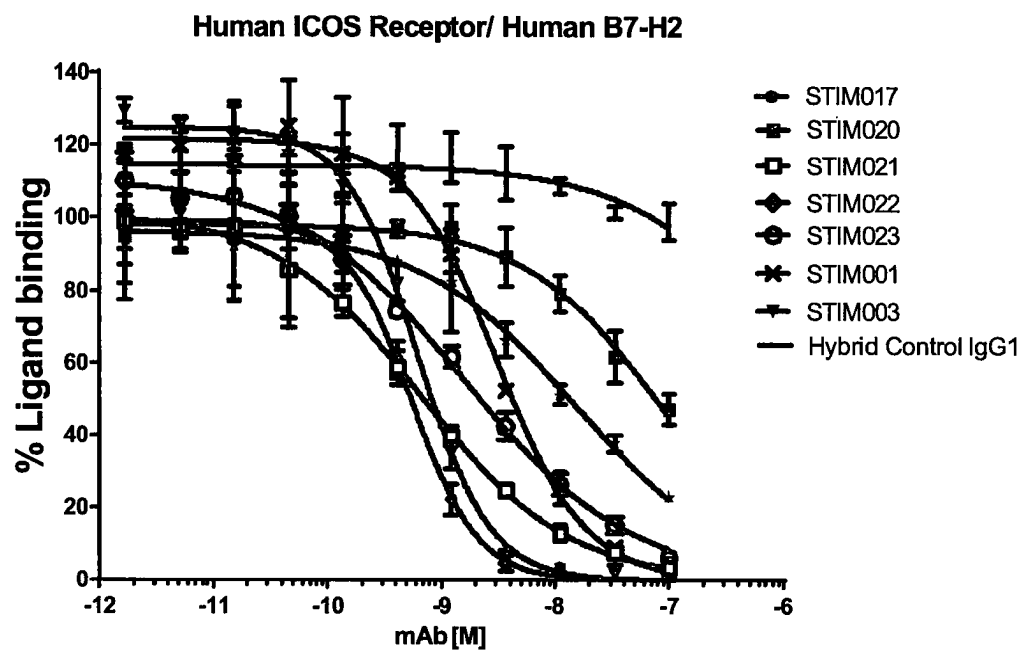


Figure 48

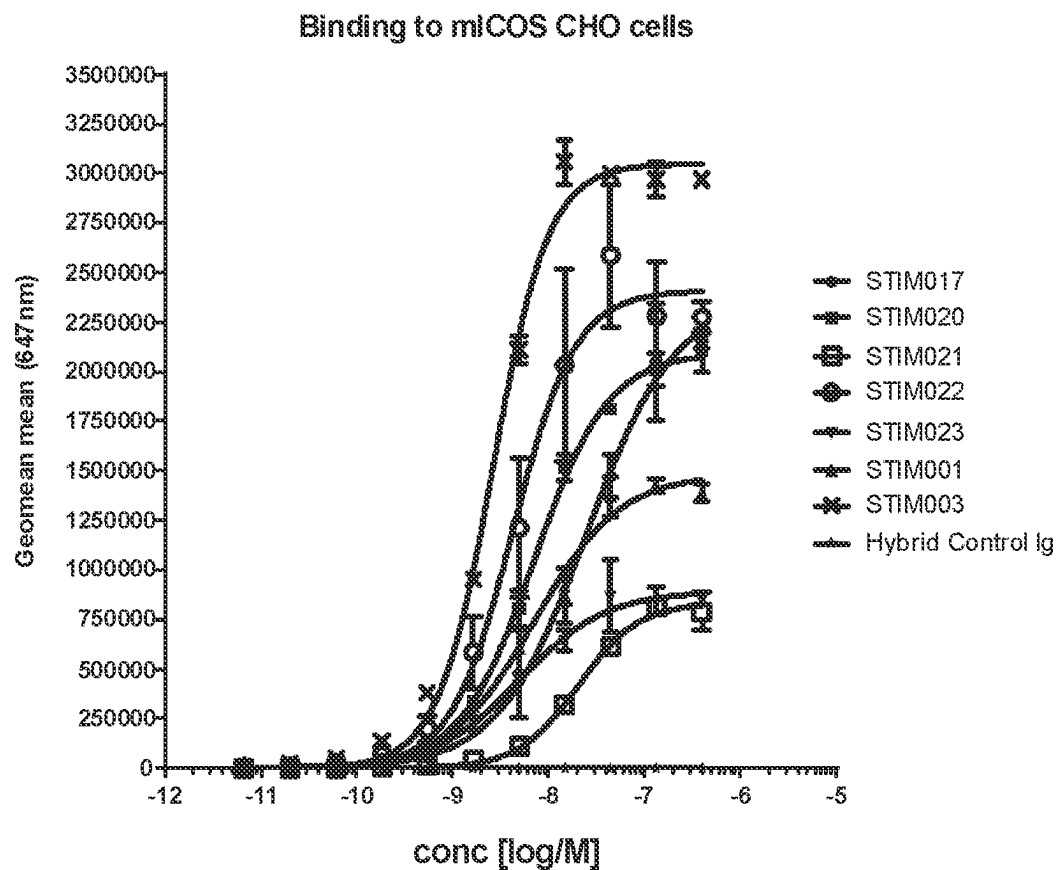


Figure 49

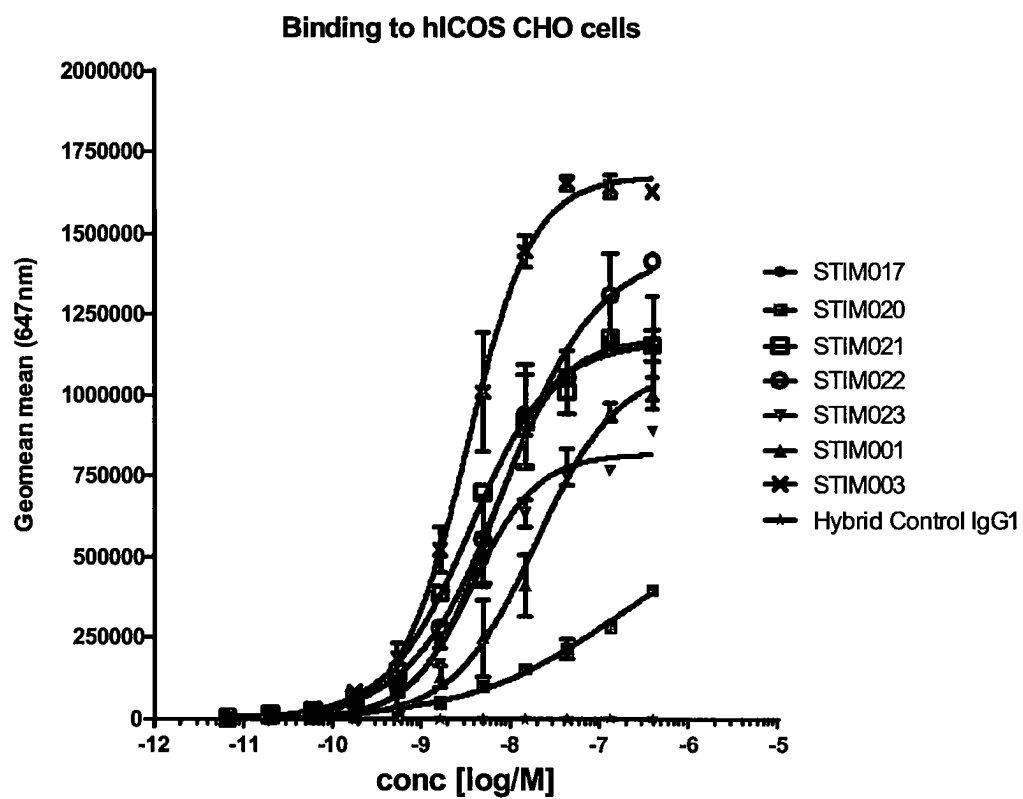


Figure 50

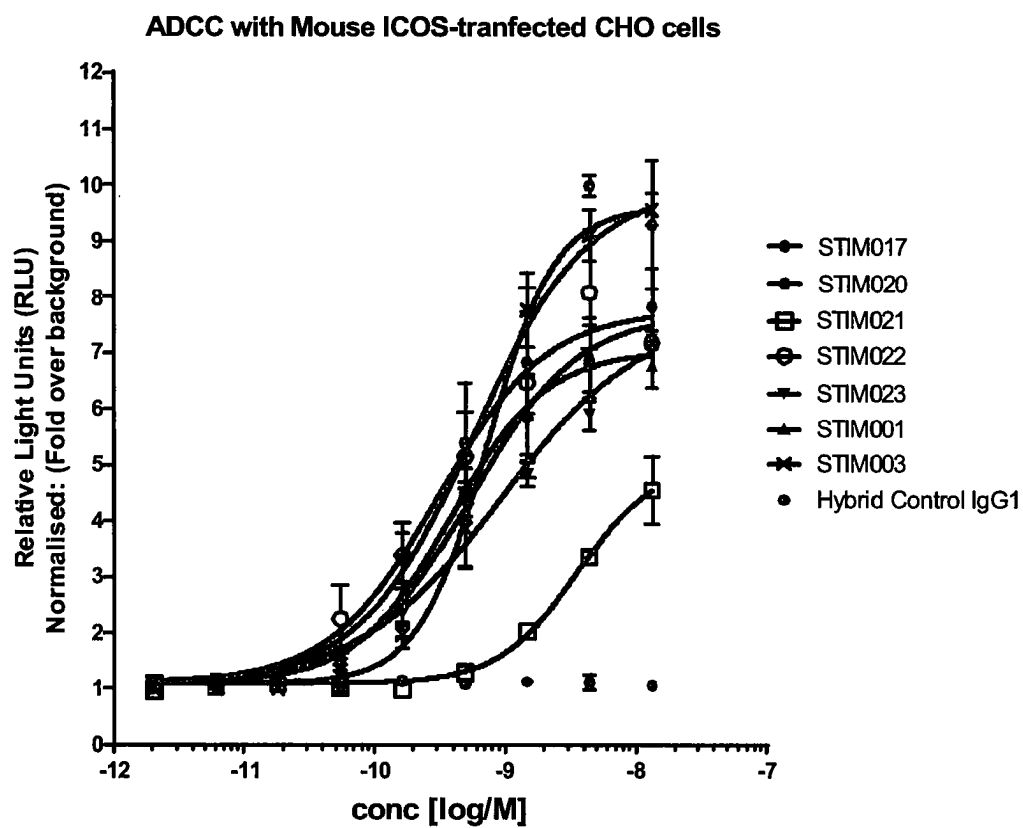


Figure 51

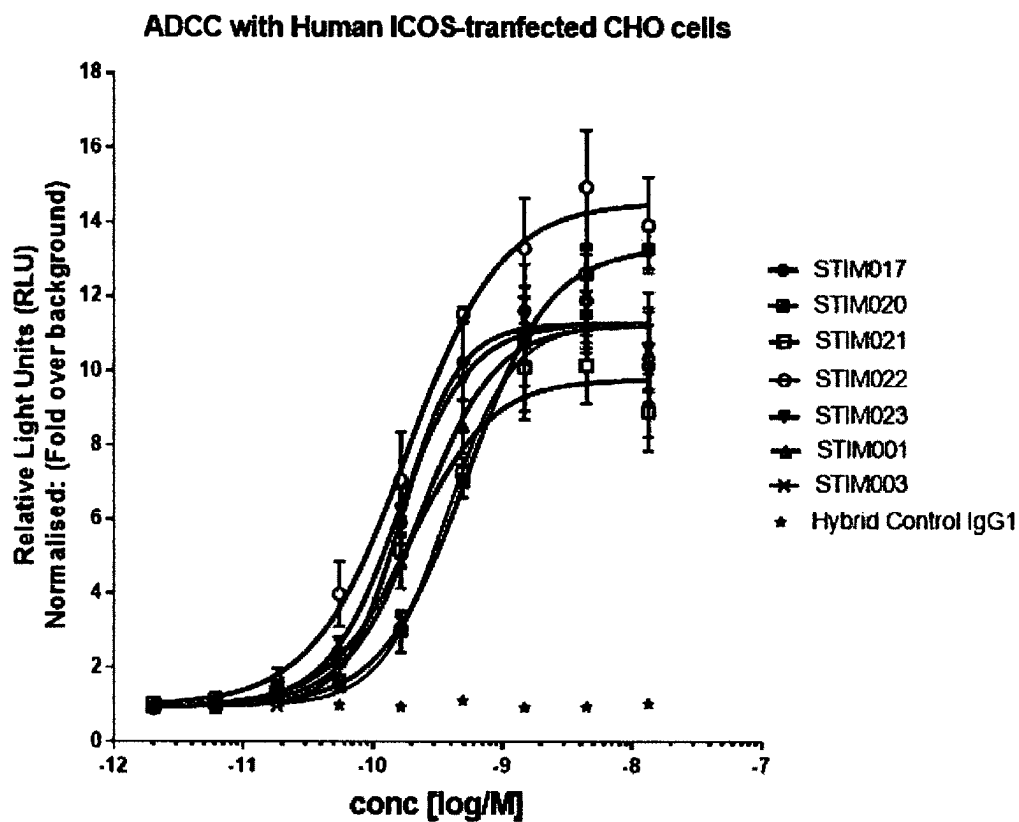


Figure 52

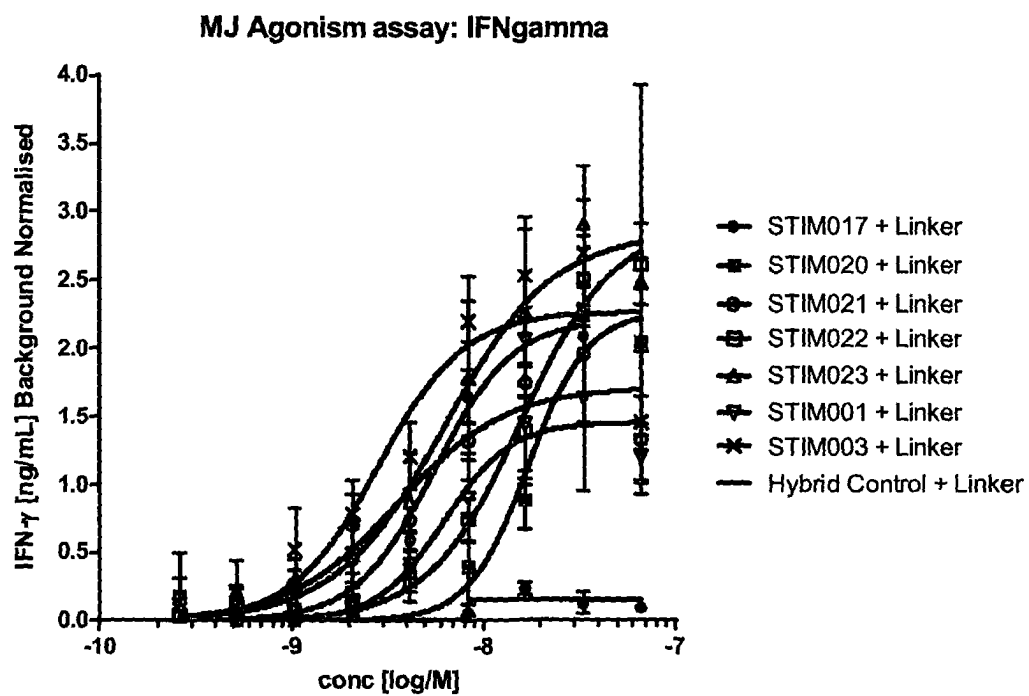


Figure 53

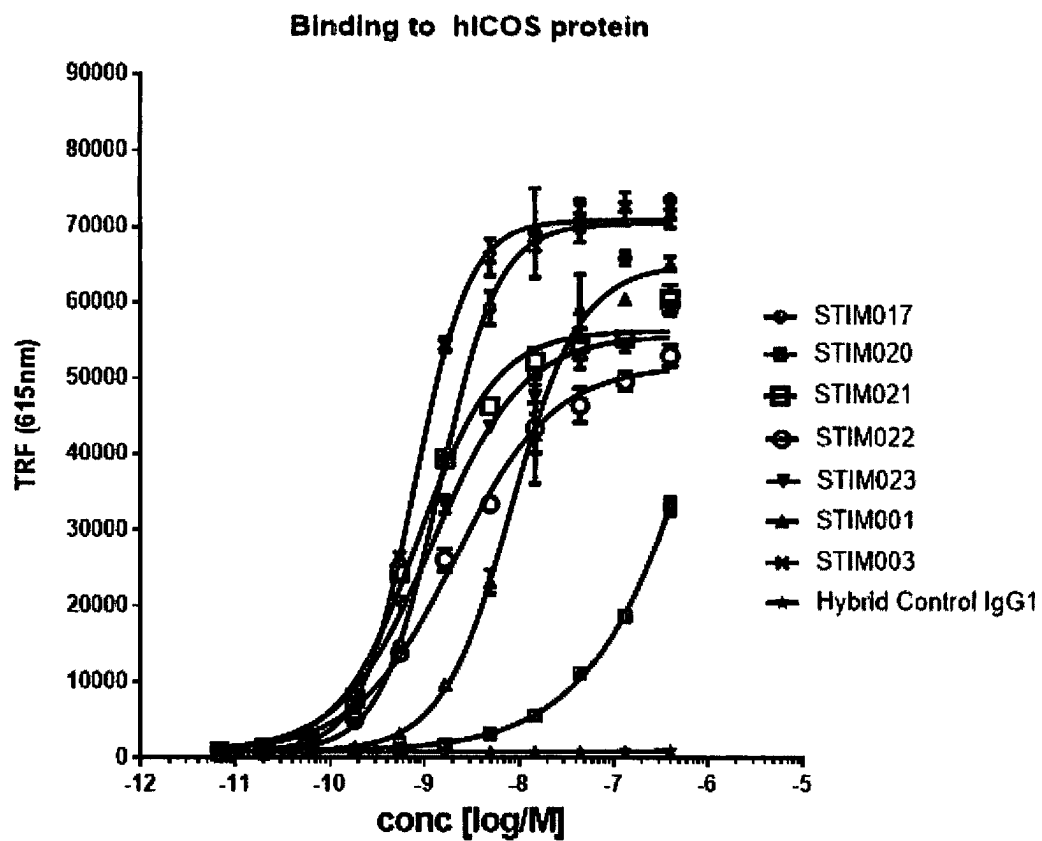


Figure 54

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ANTIBODIES TO ICOS**RELATED APPLICATIONS**

This application is a national stage filing under 35 U.S.C. § 371 of International Application No. PCT/GB2018/053701, filed on Dec. 19, 2018, which claims priority of British Patent Application No. 1721338.0, filed Dec. 19, 2017. The contents of these applications are each incorporated herein by reference.

SEQUENCE LISTING

The instant application contains a Sequence Listing which has been submitted electronically in ASCII format and is hereby incorporated by reference in its entirety. Said ASCII copy, created on Jun. 18, 2020, is named SequenceListing.txt and is 744,991 bytes in size.

FIELD OF THE INVENTION

This invention relates to compositions for stimulating the mammalian immune response, especially the T cell response. The invention also relates to medical use of such compositions in immuno-oncology, including anti-tumour therapy by promotion of anti-tumour T cell response in a patient, as well as to use of the compositions in other diseases and conditions where it is of therapeutic benefit to modulate the balance between effector T cells and regulatory T cells in favour of effector T cell activity, for example through stimulation of effector T cells and/or through depletion of regulatory T cells.

BACKGROUND

ICOS (Inducible T cell Co-Stimulator) is a member of the CD28 gene family involved in regulating immune responses, in particular humoral immune responses, first identified in 1999 [1]. It is a 55 kDa transmembrane protein, existing as a disulphide linked homodimer with two differentially glycosylated subunits. ICOS is exclusively expressed on T lymphocytes, and is found on a variety of T cell subsets. It is present at low levels on naïve T lymphocytes but its expression is rapidly induced upon immune activation, being upregulated in response to pro-inflammatory stimuli such as on engagement of TCR and co-stimulation with CD28 [2, 3]. ICOS plays a role in the late phase of T cell activation, memory T cell formation and importantly in the regulation of humoral responses through T cell dependent B cell responses [4, 5]. Intracellularly, ICOS binds PI3K and activates the kinases phosphoinositide-dependent kinase 1 (PDK1) and protein kinase B (PKB). Activation of ICOS prevents cell death and upregulates cellular metabolism. In the absence of ICOS (ICOS knockout) or in the presence of anti-ICOS neutralising antibodies there would be a suppression of pro-inflammatory responses.

ICOS binds to ICOS ligand (ICOSL) expressed on B-cells and antigen presenting cells (APC) [6, 7]. As a co-stimulatory molecule it serves to regulate TCR mediated immune responses and antibody responses to antigen. The expression of ICOS on T regulatory cells may be important, as it has been suggested that this cell type plays a negative role in immunosurveillance of cancer cells—there is emerging evidence for this in ovarian cancer [8]. Importantly, ICOS expression has been reported to be higher on intratumoural regulatory T cells (TRegs) compared with CD4+ and CD8+

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effector cells that are present in the tumour microenvironment. Depletion of TRegs using antibodies with Fc-mediated cellular effector function has demonstrated strong anti-tumour efficacy in a pre-clinical model [9]. Mounting evidence implicates ICOS in an anti-tumour effect in both animal models as well as patients treated with immune-checkpoint inhibitors. In mice deficient in ICOS or ICOSL the anti-tumor effect of anti-CTLA4 therapy is diminished while in normal mice ICOS ligand increases the effectiveness of anti-CTLA4 treatment in melanoma and prostate cancer [11]. Furthermore, in humans a retrospective study of advanced melanoma patients showed increased levels of ICOS following ipilimumab (anti-CTLA4) treatment [12]. In addition, ICOS expression is upregulated in bladder cancer patients treated with anti-CTLA4 [13]. It has also been observed that in cancer patients treated with anti-CTLA4 therapy the bulk of tumour specific IFN γ producing CD4 T-cells are ICOS positive while sustained elevation of ICOS positive CD4 T cells correlates with survival [12, 13, 14].

WO2016/120789 described anti-ICOS antibodies and proposed their use for activating T cells and for treating cancer, infectious disease and/or sepsis. A number of murine anti-ICOS antibodies were generated, of which a sub-set were reported to be agonists of the human ICOS receptor. The antibody “422.2” was selected as the lead anti-ICOS antibody and was humanised to produce a human “IgG4PE” antibody designated “H2L5”. H2L5 was reported to have an affinity of 1.34 nM for human ICOS and 0.95 nM for cynomolgus ICOS, to induce cytokine production in T cells, and to upregulate T cell activation markers in conjunction with CD3 stimulation. However, mice bearing implanted human melanoma cells were reported to show only minimal tumour growth delay or increase in survival when treated with H2L5 hIgG4PE, compared with control treated group. The antibody also failed to produce significant further inhibition of tumour growth in combination experiments with ipilimumab (anti-CTLA-4) or pembrolizumab (anti-PD-1), compared with ipilimumab or pembrolizumab monotherapy. Finally, In mice bearing implanted colon cancer cells (CT26), low doses of a mouse cross reactive surrogate of H2L5 in combination with a mouse surrogate of ipilimumab or pembrolizumab only mildly improved overall survival compared with anti-CTL4 and anti-PD1 therapy alone. A similar lack of strong therapeutic benefit was shown in mice bearing implanted EMT6 cells.

WO2016/154177 described further examples of anti-ICOS antibodies. These antibodies were reported to be agonists of CD4+ T cells, including effector CD8+ T cells (TEff), and to deplete T regulator cells (TRegs). Selective effects of the antibodies on TEff vs TReg cells were described, whereby the antibodies could preferentially deplete TRegs while having minimal effect on TEffs that express a lower level of ICOS. The anti-ICOS antibodies were proposed for use in treating cancer, and combination therapy with anti-PD-1 or anti-PD-L1 antibodies was described.

SUMMARY OF THE INVENTION

The present invention provides novel antibodies that bind human ICOS and their use in therapy.

There is a benefit to drug manufacturers and ultimately to patients in having a variety of antibody sequences from which to select optimal therapeutic molecules for use in the clinic. Antibodies can be characterised in terms of their antigen-binding properties and also in their biophysical and

chemical properties, as different sequences may exhibit different behaviour during formulation and solution handling, impacting on the ability and/or cost-effectiveness to generate pharmaceutical compositions containing these antibodies in a form suitable for administration to patients.

In various aspects the invention provides the antibodies, compositions comprising them, and methods for their administration to patients for treating diseases and conditions described herein, including combination therapies.

An antibody to ICOS that acts to increase effector T cell activity represents a therapeutic approach in immunooncology and in other medical contexts where a CD8⁺ T cell response is beneficial, including various diseases and conditions and in vaccination regimens. In many diseases and conditions involving an immune component, a balance exists between effector T cells (TEff) which exert the CD8⁺ T cell immune response, and regulatory T cells (TReg) which suppress that immune response by downregulating TEffs. The present invention relates to antibodies that modulate this TEff/TReg balance in favour of effector T cell activity. Antibodies that trigger the depletion of ICOS highly positive regulatory T cells would relieve the suppression of TEffs, and thus have a net effect of promoting the effector T cell response. An additional or complementary mechanism for an anti-ICOS antibody is via agonistic activity at the ICOS receptor level, to stimulate the effector T cell response.

The relative expression of ICOS on effector T cells (TEff) compared with regulatory T cells (TReg), and the relative activities of these cell populations, will influence the overall effect of an anti-ICOS antibody in vivo. An envisaged mode of action combines agonism of effector T cells with depletion of ICOS positive regulatory T cells. Differential and even opposing effects on these two different T cell populations may be achievable due to their different levels of ICOS expression. Dual-engineering of the variable and constant regions respectively of an anti-ICOS antibody can provide a molecule that exerts a net positive effect on effector T cell response by affecting the CD8/TReg ratio. An antigen-binding domain of an agonist antibody, which activates the ICOS receptor, may be combined with an antibody constant (Fc) region that promotes downregulation and/or clearance of highly expressing cells to which the antibody is bound. An effector positive constant region may be used to recruit cellular effector functions against the target cells (TRegs), e.g., to promote antibody-dependent cell-mediated cytotoxicity (ADCC) or antibody dependent cell phagocytosis (ADCP). The antibody may thus act both to promote effector T cell activation and to downregulate immunosuppressive T Regulatory cells. Since ICOS is more highly expressed on TRegs than on TEffs, a therapeutic balance may be achieved whereby TEff function is promoted while TRegs are depleted, resulting in a net increase in the T cell immune response (e.g. anti-tumour response or other therapeutically beneficial T cell response).

Several pre-clinical and clinical studies have shown a strong positive correlation between high effector T-cell to T-reg cell ratio in the tumour microenvironment (TME) and overall survival. In ovarian cancer patients the ratio of CD8⁺:T-reg cells has been reported to be an indicator of good clinical outcome [15]. A similar observation was made in metastatic melanoma patients after receiving ipilimumab [16]. In pre-clinical studies, it has also been shown that high effector cell:T-reg ratio in TME is associated with anti-tumour response [43].

This invention provides antibodies that bind human ICOS. The antibodies target the ICOS extracellular domain

and thereby bind to T cells expressing ICOS. Examples are provided of antibodies that have been designed to have an agonistic effect on ICOS, thus enhancing the function of effector T cells, as indicated by an ability to increase IFN γ expression and secretion. As noted, anti-ICOS antibodies may also be engineered to deplete cells to which they bind, which should have the effect of preferentially downregulating regulatory T cells, lifting the suppressive effect of these cells on the effector T cell response and thus promoting the effector T cell response overall. Through selection of appropriate antibody formats such as those including constant regions with a desired level of Fc effector function, or absence of such effector function where appropriate, the anti-ICOS antibodies may be tailored for use in a variety of medical contexts including treatment of diseases and conditions in which an effector T cell response is beneficial and/or where suppression of regulatory T cells is desired.

Exemplary antibodies include STIM017, STIM020, STIM021, STIM022, STIM023, STIM039, STIM040, STIM041, STIM042, STIM043, STIM044, STIM050, STIM051, STIM052, STIM053, STIM054, STIM055, STIM056, STIM057, STIM058, STIM059, STIM060, STIM061, STIM062, STIM063, STIM064, STIM065 and STIM066, sequences of which are set out herein.

STIM020, STIM021 and STIM023 exhibit species cross-reactivity, being able to bind human ICOS and mouse ICOS. The sequences of these antibodies can be considered to form a "cluster", and the sequence alignments show certain residues and motifs that are common to these antibodies. These sequence features are of particular interest in the antibody CDRs. In this group of cross-reactive antibodies it is observed that:

HCDR1 is G-F-T-X1-X2-X3-Y-X4 (SEQ ID NO: 611), wherein X1 is F or L; X2 is S and X3 are each independently selected from D and S; and X4 is any amino acid;

HCDR2 comprises I or F at IMGT position 56;

HCDR3 comprises A at IMGT position 105 and K or R at IMGT position 106;

LCDR1 is Q-X1-I-S-X2-X3 (SEQ ID NO: 612), wherein X1 is G or S, X2 is N or S, and X3 is W or Y;

LCDR2 is X1-A-S, wherein X1 is A or K; and

LCDR3 comprises Q-Q-X1-N-X2-X3-X4 (SEQ ID NO: 613), wherein X1 is A, L or Y, X2 is S or T, X3 is Y or F, and X4 is P, L or W.

An antibody according to the present invention may comprise an HCDR1, HCDR2, HCDR3, LCDR1, LCDR2 or LCDR3 as defined above. It may optionally comprise a set of HCDRs (HCDR1, HCDR2 and HCDR3) as defined, and/or optionally a set of LCDRs (LCDR1, LCDR2, LCDR3) as defined.

STIM039, STIM040, STIM041, STIM042, STIM043, STIM044, STIM050, STIM051, STIM052, STIM053, STIM054, STIM055, STIM056, STIM057, STIM058, STIM059, STIM060, STIM061, STIM062, STIM063, STIM064, STIM065 and STIM066 showed binding to human ICOS and, although lacking significant cross-reactivity with mouse ICOS, nevertheless represent diverse human anti-ICOS antibody sequences with therapeutic potential.

An antibody according to the invention may be one that competes for binding to human ICOS with an antibody (e.g., human IgG1, or an scFv) comprising the heavy and light chain complementarity determining regions (CDRs) of any of STIM017, STIM020, STIM021, STIM022, STIM023, STIM039, STIM040, STIM041, STIM042, STIM043, STIM044, STIM050, STIM051, STIM052, STIM053,

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STIM054, STIM055, STIM056, STIM057, STIM058, STIM059, STIM060, STIM061, STIM062, STIM063, STIM064, STIM065 and STIM066, optionally an antibody comprising the VH and VL domains of any of STIM017, STIM020, STIM021, STIM022, STIM023, STIM039, STIM040, STIM041, STIM042, STIM043, STIM044, STIM050, STIM051, STIM052, STIM053, STIM054, STIM055, STIM056, STIM057, STIM058, STIM059, STIM060, STIM061, STIM062, STIM063, STIM064, STIM065 and STIM066.

An antibody according to the present invention may comprise one or more CDRs of any of STIM017, STIM020, STIM021, STIM022, STIM023, STIM039, STIM040, STIM041, STIM042, STIM043, STIM044, STIM050, STIM051, STIM052, STIM053, STIM054, STIM055, STIM056, STIM057, STIM058, STIM059, STIM060, STIM061, STIM062, STIM063, STIM064, STIM065 and STIM066 (e.g., all 6 CDRs of any such antibody, or a set of HCDRs and/or LCDRs) or variants thereof as described herein.

The antibody may comprise an antibody VH domain comprising CDRs HCDR1, HCDR2 and HCDR3 and an antibody VL domain comprising CDRs LCDR1, LCDR2 and LCDR3, wherein the HCDR3 is an HCDR3 of an antibody selected from STIM017, STIM020, STIM021, STIM022, STIM023, STIM039, STIM040, STIM041, STIM042, STIM043, STIM044, STIM050, STIM051, STIM052, STIM053, STIM054, STIM055, STIM056, STIM057, STIM058, STIM059, STIM060, STIM061, STIM062, STIM063, STIM064, STIM065 and STIM066 or comprises that HCDR3 with 1, 2, 3, 4 or 5 amino acid alterations. The HCDR2 may be the HCDR2 of the selected antibody or it may comprise that HCDR2 with 1, 2, 3, 4 or 5 amino acid alterations. The HCDR1 may be the HCDR1 of the selected antibody or it may comprise that HCDR1 with 1, 2, 3, 4 or 5 amino acid alterations.

The antibody may comprise an antibody VL domain comprising CDRs HCDR1, HCDR2 and HCDR3 and an antibody VL domain comprising CDRs LCDR1, LCDR2 and LCDR3, wherein the LCDR3 is an LCDR3 of an antibody selected from STIM017, STIM020, STIM021, STIM022, STIM023, STIM039, STIM040, STIM041, STIM042, STIM043, STIM044, STIM050, STIM051, STIM052, STIM053, STIM054, STIM055, STIM056, STIM057, STIM058, STIM059, STIM060, STIM061, STIM062, STIM063, STIM064, STIM065 and STIM066 or comprises that LCDR3 with 1, 2, 3, 4 or 5 amino acid alterations. The LCDR2 may be the LCDR2 of the selected antibody or it may comprise that LCDR2 with 1, 2, 3, 4 or 5 amino acid alterations. The LCDR1 may be the LCDR1 of the selected antibody or it may comprise that LCDR1 with 1, 2, 3, 4 or 5 amino acid alterations.

An antibody may comprise:

an antibody VH domain comprising complementarity determining regions HCDR1, HCDR2 and HCDR3, and

an antibody VL domain comprising complementarity determining regions LCDR1, LCDR2 and LCDR3,

wherein the heavy chain complementarity determining regions are those of any of STIM017, STIM020, STIM021, STIM022, STIM023, STIM039, STIM040, STIM041, STIM042, STIM043, STIM044, STIM050, STIM051, STIM052, STIM053, STIM054, STIM055, STIM056, STIM057, STIM058, STIM059, STIM060, STIM061, STIM062, STIM063, STIM064, STIM065 and STIM066 or comprise the heavy chain comple-

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mentarity determining regions of said antibody with 1, 2, 3, 4 or 5 amino acid alterations; and/or wherein the light chain complementarity determining regions are those of any of STIM017, STIM020, STIM021, STIM022, STIM023, STIM039, STIM040, STIM041, STIM042, STIM043, STIM044, STIM050, STIM051, STIM052, STIM053, STIM054, STIM055, STIM056, STIM057, STIM058, STIM059, STIM060, STIM061, STIM062, STIM063, STIM064, STIM065 and STIM066, or comprise the light chain complementarity determining regions of said antibody with 1, 2, 3, 4 or 5 amino acid alterations.

An antibody may comprise a VH domain selected from the VH domain of any of STIM017, STIM020, STIM021, STIM022, STIM023, STIM039, STIM040, STIM041, STIM042, STIM043, STIM044, STIM050, STIM051, STIM052, STIM053, STIM054, STIM055, STIM056, STIM057, STIM058, STIM059, STIM060, STIM061, STIM062, STIM063, STIM064, STIM065 and STIM066 or it may comprise a VH domain having that VH domain amino acid sequence with 1, 2, 3, 4 or 5 conservative amino acid substitutions. The antibody may also comprise the VL domain of the selected antibody, or it may comprise a VL domain having that VL domain amino acid sequence with 1, 2, 3, 4 or 5 conservative amino acid substitutions.

Antibodies of the invention may comprise VH and/or VL domain framework regions corresponding to human germline gene segment sequences. For example, it may comprise one or more framework regions of any of STIM017, STIM020, STIM021, STIM022, STIM023, STIM039, STIM040, STIM041, STIM042, STIM043, STIM044, STIM050, STIM051, STIM052, STIM053, STIM054, STIM055, STIM056, STIM057, STIM058, STIM059, STIM060, STIM061, STIM062, STIM063, STIM064, STIM065 and STIM066. The framework region or framework regions may be a FR1, FR2, FR3 and/or FR4.

As described in Example 12 and Example 27, Table E12-1, Table E12-2 and Table E27-1 show the human germline gene segments that generated the VH and VL domains of example antibodies described herein through recombination. Antibody VH and VL domains of the present invention may be based on these V (D) J segments.

For example, it can be seen that some antibodies (e.g., STIM039 to STIM044) appear to have a VH domain derived from recombination of v segment IGHV3-13*01 and j segment IGJ6*02 with a d segment e.g., IGHD6-19*01 or IGHD3-22*01, and a VL domain derived from recombination of v segment IGKV3-20*01 or IGKV3-11*01 with j segment IGKJ5*01.

An antibody of the invention may comprise an antibody VH domain which

- (i) is derived from recombination of a human heavy chain V gene segment, a human heavy chain D gene segment and a human heavy chain J gene segment, wherein the V segment is IGHV3-13 (e.g., V3-13*01); the D gene segment is IGHD6-19 (e.g., IGHD6-19*01) or IGHD3-22 (e.g., IGHD3-22*01); and/or the J gene segment is IGJ6 (e.g., IGJ6*02), or
- (ii) comprises framework regions FR1, FR2, FR3 and FR4, wherein

FR1 aligns with human germline V gene segment IGHV3-13 (e.g., V3-13*01), optionally with 1, 2, 3, 4 or 5 amino acid alterations,

FR2 aligns with human germline V gene segment IGHV3-13 (e.g., V3-13*01), optionally with 1, 2, 3, 4 or 5 amino acid alterations,

FR3 aligns with human germline V gene segment IGHV3-13 (e.g., V3-13*01), optionally with 1, 2, 3, 4 or 5 amino acid alterations, and/or

FR4 aligns with human germline J gene segment IGJH6 (e.g., JH6*02), optionally with 1, 2, 3, 4 or 5 amino acid alterations.

FR1, FR2 and FR3 of the VH domain typically align with the same germline V gene segment. Thus, for example, the antibody may comprise a VH domain derived from recombination of human heavy chain V gene segment IGHV3-13 (e.g., VH3-13*01), a human heavy chain D gene segment and a human heavy chain J gene segment IGJH6 (e.g., JH6*02). An antibody may comprise VH domain framework regions FR1, FR2, FR3 and FR4, wherein FR1, FR2 and FR3 each align with human germline V gene segment IGHV3-13 (e.g., IGHV3-13*01) with up to 1, 2, 3, 4 or 5 amino acid alterations, and a FR4 that aligns with human germline J gene segment IGJH6 (e.g., IGJH6*02) with up to 1, 2, 3, 4 or 5 amino acid alterations. Alignment may be exact, but in some cases one or more residues can be mutated from germline, so there may be amino acid substitutions present, or in rarer cases deletions or insertions.

The antibody or VH domain may comprise the HCDRs of any of STIM039 to STIM044 or a variant thereof.

An antibody of the invention may comprise an antibody VL domain which

(i) is derived from recombination of a human light chain V gene segment and a human light chain J gene segment, wherein

the V segment is IGKV3-20 (e.g., IGKV3-20*01) or IGKV3-11 (e.g., IGKV3-11*01), and/or

the J gene segment is IGKJ5 (e.g., IGKJ5*01); or

(ii) comprises framework regions FR1, FR2, FR3 and FR4, wherein

FR1 aligns with human germline V gene segment IGKV3-20 (e.g., IGKV3-20*01) or IGKV3-11 (e.g., IGKV3-11*01), optionally with 1, 2, 3, 4 or 5 amino acid alterations,

FR2 aligns with human germline V gene segment IGKV3-20 (e.g., IGKV3-20*01) or IGKV3-11 (e.g., IGKV3-11*01), optionally with 1, 2, 3, 4 or 5 amino acid alterations,

FR3 aligns with human germline V gene segment IGKV3-20 (e.g., IGKV3-20*01) or IGKV3-11 (e.g., IGKV3-11*01), optionally with 1, 2, 3, 4 or 5 amino acid alterations, and/or

FR4 aligns with human germline J gene segment IGKJ5 (e.g., IGKJ5*01), optionally with 1, 2, 3, 4 or 5 amino acid alterations.

FR1, FR2 and FR3 of the VL domain typically align with the same germline V gene segment. Thus, for example, the antibody may comprise a VL domain derived from recombination of human light chain V gene segment IGKV3-20 (e.g., IGKV3-20*01) and human light chain J gene segment IGKJ5 (e.g., IGKJ5*01). An antibody may comprise VL domain framework regions FR1, FR2, FR3 and FR4, wherein FR1, FR2 and FR3 each align with human germline V gene segment IGKV3-20 (e.g., IGKV3-20*01) with up to 1, 2, 3, 4 or 5 amino acid alterations, and a FR4 that aligns with human germline J gene segment IGKJ5 (e.g., IGKJ5*01) with up to 1, 2, 3, 4 or 5 amino acid alterations. Alignment may be exact, but in some cases one or more residues can be mutated from germline, so there may be amino acid substitutions present, or in rarer cases deletions or insertions.

The antibody or VL domain may comprise the LCDRs of any of STIM039 to STIM044 or a variant thereof.

An antibody according to the invention may comprise an antibody VH domain which is the VH domain of any of STIM017, STIM020, STIM021, STIM022, STIM023, STIM039, STIM040, STIM041, STIM042, STIM043, STIM044, STIM050, STIM051, STIM052, STIM053, STIM054, STIM055, STIM056, STIM057, STIM058, STIM059, STIM060, STIM061, STIM062, STIM063, STIM064, STIM065 and STIM066, or which has an amino acid sequence at least 90% identical to the antibody VH domain sequence of any of STIM017, STIM020, STIM021, STIM022, STIM023, STIM039, STIM040, STIM041, STIM042, STIM043, STIM044, STIM050, STIM051, STIM052, STIM053, STIM054, STIM055, STIM056, STIM057, STIM058, STIM059, STIM060, STIM061, STIM062, STIM063, STIM064, STIM065 and STIM066. The amino acid sequence identity may be at least 95%, 96%, 97%, 98% or 99%.

The antibody may comprise an antibody VL domain which is the VL domain of any of STIM017, STIM020, STIM021, STIM022, STIM023, STIM039, STIM040, STIM041, STIM042, STIM043, STIM044, STIM050, STIM051, STIM052, STIM053, STIM054, STIM055, STIM056, STIM057, STIM058, STIM059, STIM060, STIM061, STIM062, STIM063, STIM064, STIM065 and STIM066, or which has an amino acid sequence at least 90% identical to the antibody VL domain sequence of any of STIM017, STIM020, STIM021, STIM022, STIM023, STIM039, STIM040, STIM041, STIM042, STIM043, STIM044, STIM050, STIM051, STIM052, STIM053, STIM054, STIM055, STIM056, STIM057, STIM058, STIM059, STIM060, STIM061, STIM062, STIM063, STIM064, STIM065 and STIM066. The amino acid sequence identity may be at least 95%, 96%, 97%, 98% or 99%.

An antibody VH domain having the HCDRs of any of STIM017, STIM020, STIM021, STIM022, STIM023, STIM039, STIM040, STIM041, STIM042, STIM043, STIM044, STIM050, STIM051, STIM052, STIM053, STIM054, STIM055, STIM056, STIM057, STIM058, STIM059, STIM060, STIM061, STIM062, STIM063, STIM064, STIM065 and STIM066, or having a variant of those CDRs, may be paired with an antibody VL domain having the LCDRs of the same antibody, or having a variant of those CDRs. Similarly, the VH domain of any of STIM017, STIM020, STIM021, STIM022, STIM023, STIM039, STIM040, STIM041, STIM042, STIM043, STIM044, STIM050, STIM051, STIM052, STIM053, STIM054, STIM055, STIM056, STIM057, STIM058, STIM059, STIM060, STIM061, STIM062, STIM063, STIM064, STIM065 and STIM066, or a variant of that VH domain, may be paired with a VL domain of the same antibody, or a VL domain variant of the same antibody.

Antibodies may include constant regions, optionally human heavy and/or light chain constant regions. An exemplary isotype is IgG, e.g., human IgG1.

Further aspects of the invention include nucleic acid molecules encoding sequences of the antibodies described herein, host cells containing such nucleic acids, and methods of producing the antibodies by culturing the host cells and expressing and optionally isolating or purifying the antibodies. The expressed antibody is thereby obtained. VH and VL domains of antibodies described herein may similarly be produced and are aspects of the present invention. Pharmaceutical compositions comprising the antibodies are also provided.

Also described herein are ICOS knock out non-human animals and their use for generating antibodies to human

ICOS. In an ICOS knock out animal, ICOS is not expressed, for example because the gene encoding ICOS has been inactivated or deleted from the animal's genome. Such animals are useful for generating species cross-reactive antibodies, which recognise both human ICOS and ICOS from the non-human species. The normal process of immune tolerance means that lymphocytes that recognise "self" antigens are deleted or inactivated to prevent autoimmune reactions in the body, whereas the absence of the endogenous ICOS antigen in the non-human knock out animal means that the animal's immune system should not be tolerised to that antigen and therefore can mount an immune response against ICOS when injected as recombinant protein or using cell lines or vesicles expressing ICOS. The immune repertoire of the knock out animal should contain lymphocytes able to recognise the ICOS protein from that animal species. A non-human test animal (e.g., a mouse) immunised with human ICOS may thus generate antibodies that bind both human ICOS and the test animal ICOS (e.g., mouse ICOS).

This has at least two advantages. First, a species cross-reactive antibody can be used for pre-clinical testing in the non-human test animal before being taken forward into development in human clinical trials. Second, a knock out animal's immune system may be able to recognise a greater number of possible epitopes on a human ICOS molecule compared with those recognised by an ICOS-expressing animal, so that the immune repertoire of the knock out animal may contain a greater functional diversity of antibodies. Since there is similarity between the sequences of homologous ICOS molecules from different species, the immune system of a non-human animal may ordinarily be tolerised to those regions of the human ICOS protein that match those of the non-human animal ICOS, whereas this tolerisation does not occur in a knock out animal. A number of human/mouse cross-reactive anti-ICOS antibodies are described herein, including STIM017, STIM020, STIM021, STIM022 and STIM023.

The ability to use an ICOS knock out animal, and its advantage for generating cross-reactive antibodies, is shown in the Examples. It is particularly surprising that an ICOS knock out animal could be successfully immunised to produce an antibody response, because ICOS itself is involved in the immune system biology such as formation and maintenance of the germinal centers and contributes to the generation of an immune response through its role on T follicular helper cells which are ICOS+ve cells [37]. With this in mind, an ICOS knock out animal might be predicted to generate a poor antibody response at best. Surprisingly, strong titres were obtained in ICOS knock out mice, and highly functional antibodies were isolated from among the antibody repertoire, including desirable cross-reactive antibodies.

Exemplary embodiments of the invention are set out in the appended claims.

BRIEF DESCRIPTION OF THE DRAWINGS

Certain aspects and embodiments of the invention will now be described in more detail with reference to the accompanying drawings.

FIG. 1: Determination of serum titres of ICOS KO and wild type Kymouse against both human and mouse ICOS expressed on CHO cells by flow cytometry. Data illustrate ability of immunoglobulin in sera of (a) ICOS KO mice (KO) or (b) wild type non-ICOS KO mice (HK or HL), each immunised with human ICOS expressing MEF cells and

human ICOS protein, to bind human ICOS (human ICOS binding) or mouse ICOS (mouse ICOS binding) expressed on CHO cells. Geometric mean is a measure of fluorescent intensity of immunoglobulin binding to cells as determined by flow cytometry.

FIG. 2: Human ICOS-ligand neutralisation HTRF with human ICOS receptor. Neutralisation profiles of STIM001 to STIM009 anti-ICOS mAbs in human IgG1 format compared to C398.4A and respective isotype controls. Data representative of four experiments.

FIG. 3: Mouse ICOS-Ligand neutralisation HTRF with mouse ICOS receptor. Neutralisation profiles of STIM001 to STIM009 anti-ICOS mAbs in human IgG1 format compared to C398.4A and respective isotype controls. Data representative of three experiments.

FIG. 4: Human ICOS-Ligand direct neutralisation HTRF with human ICOS receptor. Neutralisation profiles of STIM001 to STIM009 anti-ICOS mAbs in human IgG4.PE format compared to C398.4A and respective isotype controls. Data representative of four experiments.

FIG. 5: Mouse ICOS-Ligand neutralisation HTRF with mouse ICOS receptor. Neutralisation profiles of STIM001 to STIM009 anti-ICOS mAbs in human IgG4.PE format compared to C398.4A and respective isotype controls. Data representative of four experiments.

FIG. 6a: Concentration-dependent study of STIM001-mediated ADCC on MJ cells by using freshly isolated NK cells as effector cells. The effector cells and target cells (effector: target ratio of 5:1) were incubated together with antibody for 2 hours. BATDA releasing from lysed target cells was measured as described in the manufacturer kit instruction. HC is the hybrid isotype control.

FIG. 6b, c, d: Concentration-dependent study of STIM001 and STIM003-mediated ADCC on MJ cells with freshly isolated NK cells as effector cells. The effector cells and target cells (effector: target ratio of 5:1) were incubated together with antibody for 2 hours. BATDA releasing from lysed target cells was measured as described in the manufacturer kit instruction. HC is the hybrid isotype control.

FIG. 6e, f, g: Concentration-dependent study of STIM001 (hIgG1) and STIM003 (hIgG1)-mediated ADCC on ICOS-transfected CCRF-CEM cells with freshly isolated NK cells as effector cells. The effector cells and target cells (effector: target ratio of 5:1) were incubated together with antibody for 4 hours. BATDA releasing from lysed target cells was measured as described in the manufacturer kit instruction. HC is the hybrid isotype control.

FIG. 7, FIG. 8, FIG. 9: Anti-ICOS antibody inhibits CT26 tumour growth and improved survival when dosed as monotherapy or in combination with anti-PDL1. The STIM001 mIgG2a is more potent than the mIgG1 format. The number of animals cured or with stable disease is indicated on each graph.

FIG. 10: 2x2 combinations CT26 in vivo efficacy study. Each treatment groups is represented by a "spider plot" showing the tumour size of individual animals (n=10 per groups). When combined with anti-PDL1 antibodies, STIM001 delays tumour growth and improves the survival of treated animals. The efficacy observed in the presence of STIM001 mIgG2a is superior to that of STIM001 mIgG1. Finally, STIM001 mIgG2a in combination with anti-PDL1 mIgG2a was the most potent combination to trigger the anti-tumour response resulting in 60% of the animals cured of the disease. For each group, the number of animals cured of their disease is indicated on the top right of the respective graphs. Dosing was on days 6, 8, 10, 13, 15 and 17.

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FIG. 11: Graphs showing the CT26 tumour volumes over time of animals treated with anti-ICOS or anti-PDL1 mono-therapies or combination therapies. Each treatment group is represented by a "spider plot" showing the tumour size of individual animals (n=10 per group). For each group, the number of animals with tumour size below 100 mm³ (stable/cured of their disease) is indicated on the top right of the respective graphs. Dosing was performed on days 6, 8, 10, 13, 15 and 17. Dosing time is indicated by the shaded area. (a) Isotype control; (b) Anti-PDL1 mIgG2a AbW; (c) Anti ICOS STIM003 mIgG1; (d) Anti ICOS STIM003 mIgG2a; (e) Anti-PDL1 mIgG2a AbW+STIM003 mIgG1; (f) Anti-PDL1 mIgG2a AbW+STIM003 mIgG2a. STIM003 mIgG2 significantly inhibits CT26 tumour growth when combined with anti-PDL1 (AbW) mIgG2a.

FIG. 12: MJ cell in vitro activation assay-bead bound. Stimulation profiles of STIM001, STIM002 and STIM003 anti-ICOS mAbs bound to beads compared with anti-ICOS C398.4A and respective isotype controls. Data represent the average of two experiments (n=1 in the case of C398.4A isotype control beads).

FIG. 13: MJ cell in vitro activation assay-plate bound. Stimulation profiles of plate-bound STIM001, STIM002, STIM003 and STIM004 anti-ICOS mAbs compared with anti-ICOS C398.4A and respective isotype controls. Data represent the average of two experiments.

FIG. 14: FACS analysis of STIM001 and STIM003 hIgG1 binding to activated T cells. (a) shows a representative experiment of the dose response of pre-labelled antibodies binding to activated T cells, whereas (b) shows the binding following the dose response of naked antibodies followed by the detection with a secondary labelled antibody. Tables indicate relevant EC50 (M) as determined using GraphPad Prism.

FIG. 15: STIM001 and STIM003 showed isotype-dependent effects on the T cell compartment at the tumour site. A total of 1×10⁵ CT-26 tumour cells were implanted subcutaneously in Balb/c female mice. At day 13 and day 15 post implantation animals were dosed with antibodies or saline intraperitoneally (n=10/each group). On day 16 post implantation spleen and tumours were harvest from tumour bearing animals (n=8/each group), dissociated and stained for FACS analysis. A, percentage of CD3 cells that are positive for CD4 cells. B, percentage of CD3 cells that are positive for CD8 cells. C, percentage of CD4 cells that are Foxp3+ & CD25+. D, percentage of CD4 cells in spleen that are positive for Foxp3+ & CD25+. E, percentage of CD4 effector cells in total CD4 cells. F, ratio of CD8 effector to T-Reg cells. G, ratio of CD4 effector to T-Reg cells. Statistical analysis were performed using GraphPad Prism, all the Antibody treated groups were compared with saline treated group, P values were noted when significant (p<0.05). Values denote means±SD (n=8 mice/group). For F: Values denote mean±SEM.

FIG. 16: Example data from concentration-dependent study of STIM001 (hIgG1) and STIM003 (hIgG1) agonist effect on isolated human T-cells co-stimulated with anti-CD3/anti-CD28 dynabeads for 3-days in T cell activation assay 1 (see Example 9b). IFN- γ production was used as an indicator of the agonistic effect. STIM001 (hIgG1) and STIM003 (hIgG1) were tested in plate-bound, soluble or crosslinked soluble (Fc-linked Ab) formats and compared with a hybrid isotype control (HC hIgG1). Included for comparison in the plate-bound assay was hamster antibody C398.4A and its isotype control (hamster IgG). Upper panel shows data from plate-bound antibodies. Lower panel shows

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data from IgG1 antibodies in soluble and cross-linked forms. Left and right panels respectively use T cells from two independent human donors.

FIG. 17: Example data set for STIM001 in T cell activation assay 1 (see Example 9). Data indicate levels of IFN- γ induced by STIM001 (hIgG1) or its hybrid isotype control (HC IgG1) at one given dose for T cells from 8 independent human donors. Plate-bound antibody (FIG. 17a) was used at 5 μ g/ml. Soluble antibody (FIG. 17b) was used at 15 μ g/ml. Each dot represents one donor, identified by number (D214 for example). Significance was assessed using Wilcoxon statistic test: *, p<0.05 and **, p<0.01.

FIG. 18: Example data set for STIM003 in T cell activation assay 1 (see Example 9). Data indicate levels of IFN- γ induced by STIM003 (hIgG1) or its hybrid isotype control (HC hIgG1) at one given dose for T cells from 8 independent human healthy donors. Soluble antibody (FIG. 18a) was used at 15 μ g/ml. Plate-bound antibody (FIG. 18b) was used at 5 μ g/ml. Each dot represents one donor, identified by number (D214 for example). Significance was assessed using Wilcoxon statistic test: *, p<0.05 and **, p<0.01.

FIG. 19: Example data from T cell activation assay 2 (see Example 9c). Study of STIM001 (hIgG1) and STIM003 (hIgG1) agonist effect on isolated human T-cells stimulated with anti-CD3/anti-CD28 dynabeads for 3-days, then rested in medium for 3-days and finally re-stimulated with plate-bound STIM001, STIM003 or C398.4A Ab+/-CD3 Ab. Data comparing levels of IFN- γ (A, B), TNF- α (C, D) and IL-2 (E, F) induced by STIM001, STIM003 vs their hybrid control IgG1 (A, C, E) or C398.4A vs its hamster IgG control (B, D, F) at one given dose and in combination with CD3 Ab (TCR engagement). Each dot represents an independent donor identifiable by its number (D190 for example). Statistical significance between the Abs and their isotype control was assessed using Wilcoxon statistic test and p value indicated. Note that STIM003 concentration was slightly different to those of HC IgG1 (5.4 vs 5 μ g/ml).

FIG. 20: Graph showing the percentage of immune cells (CD8 T-Effector, CD4 T-Effector and CD4/FoxP3 TReg cells) in the CT26 tumours and in the spleen of tumour bearing animals that are expressing ICOS on their surface. Values denote mean±SD (n=8). P values were calculated using nonparametric Dunn's multiple comparisons test. NS=not significant; ***=p<0.001; ****=p<0.0001.

FIG. 21: Relative expression of ICOS on the surface of immune cells-CD8 T effector, CD4 T effector and CD4/FoxP3 TRegs—as determined by the mean fluorescence intensity (MFI). Values denote mean±SD (n=8). P values were calculated using nonparametric Dunn's multiple comparisons test. ****=p<0.0001, **=p<0.01. Note the difference in the fluorescence intensity between spleen (low) and tumours (high).

FIG. 22: Effect of STIM001 and STIM003 on the percentage of different immune cells in the microenvironment of CT26 tumours. *=p<0.05.

FIG. 23: Effect of antibodies STIM001 and STIM003 on the percentage of regulatory T cells (CD4+/FoxP3+ cells) in the microenvironment of CT26 tumours. **=p<0.05, ****=p<0.0001. Values denote mean±SD (n=8). P values were calculated using nonparametric Dunn's multiple comparisons test.

FIG. 24: STIM001 and STIM003 mIgG2 significantly increase the CD8 effector T cell to TReg ratio and the CD4 effector T cells to TReg ratio in CT26 tumours. The ratio was determined by dividing the percentage of effector cells in the tumour by the percentage of regulatory T cells in the tumour.

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FIG. 25: Effect of antibodies on percentage of immune cells in the spleen of CT26 tumour bearing animals.

FIG. 26: Effect of antibodies on percentage of regulatory T cells (CD4+/FoxP3+ cells) in the spleen of CT26 tumour bearing animals.

FIG. 27: (A) CD8 effector:Treg ratio and (B) CD4:TReg ratio in spleen of CT26 tumour bearing animals.

FIG. 28: Surface staining of AF647-conjugated STIM001, STIM003 and hIgG1 hybrid control (HC IgG1) on activated Mauritian cynomolgus pan T cells. Data from assays using different donor sources of T cells are shown in A and B respectively. EC50 values are indicated in the table.

FIG. 29: Kaplan Meier curves for CT26 Balb/C model. Shading shows dosing window. Log Rank $p < 0.0001$.

FIG. 30, FIG. 31, FIG. 32, FIG. 33: Graphs showing volumes of A20 tumours over time in mice for the study described in Example 20. Each treatment group is represented by a spider plot showing tumour size in individual animals, $n=8$ per group. For each group, the number of animals with no sign of tumour (indicating cured of disease) is indicated on the bottom left of the graph. Dosing was performed on days 8, 11, 15, 18, 22, 25 and 29 post tumour cell implantation and the dosing time is indicated by the grey shaded area. Compared with the control group (FIG. 30) and the anti-PD-L1 treatment group (FIG. 31), the STIM001 mIgG2a (FIG. 32) and STIM003 mIgG2a (FIG. 33) treatment groups showed significant inhibition of A20 tumour growth.

FIG. 34: Data from CT26 in vivo efficacy study described in Example 11c using combination of anti-PD-L1 mIgG2a antibody with single vs multiple doses of STIM003 mIgG2a. Each treatment group is represented by a "spider plot" showing the tumour size of individual animals ($n=8$ per group). For each group, the number of animals cured of their disease is indicated on the bottom right of the respective graph. Dosing days for each antibody are indicated by arrows below the respective graphs.

FIG. 35: STIM002 VH (top) (SEQ ID NO: 380) and VL (bottom) (SEQ ID NO: 387) domain amino acid sequences, showing residues that differ in the corresponding sequences of STIM001 VL (SEQ ID NO: 373) and VH (SEQ ID NO: 366), STIM002B VL (SEQ ID NO: 401) and VH (SEQ ID NO: 394) and related antibodies CL-61091 (STIM017) VL (SEQ ID NO: 590) and VH (SEQ ID NO: 564), CL-64536 VL (SEQ ID NO: 595) and VH (SEQ ID NO: 569), CL-64837 VL (SEQ ID NO: 598) and VH (SEQ ID NO: 572), CL-64841 VL (SEQ ID NO: 600) and VH (SEQ ID NO: 574), and CL-64912 VL (SEQ ID NO: 602) and VH (SEQ ID NO: 576) and/or in the human germline. Sequence numbering is according to IMGT.

FIG. 36: STIM003 VH (top) (SEQ ID NO: 408) and VL (bottom) (SEQ ID NO: 415) domain amino acid sequences, showing residues that differ in the corresponding sequences of related antibodies CL-71642 VL (SEQ ID NO: 604) and VH (SEQ ID NO: 581), and CL-74570 (STIM022) VL (SEQ ID NO: 607) and VH (SEQ ID NO: 586) and/or in the human germline. Sequence numbering is according to IMGT. The VL domain of antibody CL-71642 obtained from sequencing is shown here without the N terminal residue. From the alignment it can be seen that the full VH domain sequence would comprise an N terminal glutamic acid.

FIG. 37: STIM007 VH (top) (SEQ ID NO: 466) and VL (bottom) (SEQ ID NO: 473) domain amino acid sequences, showing residues that differ in the corresponding sequences of STIM008 VH (SEQ ID NO: 480) and VL (SEQ ID NO: 487) and/or in the human germline. Sequence numbering is according to IMGT.

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FIG. 38: Effect of STIM003 (anti-ICOS) and AbW (anti-PD-L1) mIgG2a antibodies in the J558 syngeneic model. Each treatment group is represented by a "spider plot" showing the tumour size of individual animals ($n=10$ or $n=8$ per group). STIM003 monotherapy demonstrated some efficacy with 3 of 8 animals cured from their disease. Similarly anti-PDL1 was effective in this model with 6 out of 8 animals cured from their disease by day 37. When combined with anti-PDL1 antibodies, STIM003 mIgG2 fully inhibited tumour growth and improved the survival of treated animals. For each group, the number of animals cured of their disease is indicated on the bottom right of the respective graph. Dosing days are indicated by dotted lines (day 11, 15, 18, 22, 25 and 29).

FIG. 39: Quantification of ICOS expression (percentage of positive cells and relative expression/dMFI) on the different TILS cell subtypes in the tumour tissue. (A) The % of immune cell subtypes that are positive for ICOS expression and (B) the ICOS dMFI (relative ICOS expression on ICOS positive cell) of immune cell subtypes of animals treated with saline or anti-PD-L1 or anti-PD-1 surrogate antibodies. The mice were implanted with $100 \mu\text{l}$ of 1×10^6 viable cells/ml on day 0 ($n=7$ or $n=8$). The animals were dosed i.p with $130 \mu\text{g}$ of antibody on day 13 and day 15. The tissue samples were isolated and analysed on day 16. CD4+/FOXP3+ cells were only included for the TReg population (right end side graphs) and were excluded from the "effector" CD4 cells (left end side graphs) which are all Foxp3 negative. See Example 22.

FIG. 40: Data from A20 in vivo efficacy study. Each treatment group is represented by a "spider plot" showing the tumour size of individual animals ($n=10$ per group). For each group, the number of animals cured of their disease is indicated on the respective graph. For the multiple dose, dosing was on days 8, 11, 15, 18, 22 and 25, indicated by dotted lines. For the single dose, animals received injection IP only on day 8. (A) Saline; (B) STIM003 mIgG2a multiple dose; (C) STIM003 mIgG2a single dose. See Example 23.

FIG. 41: Kaplan-Meier curves for study reported in Example 23 with STIM003 mIgG2a $60 \mu\text{g}$ fixed dose. SD=single dose, day 8. MD=multiple doses BIW from day 8.

FIG. 42: ICOS expression on major T cells subsets (T-reg [CD4+/FoxP3+], CD4 Eff [CD4+/FoxP3-] cells and CD8+) from CT26 tumour bearing animals ($n=4$ per time point) dosed with saline. Immune cells phenotyping were conducted on day 1, 2, 3, 4 and 8 post treatment and stained for ICOS expression in all the tissues at all time points. A-D showing the percentage of ICOS positive cells at all the time points in four different tissues. E-H show the ICOS dMFI (relative expression) all the time points in all the four different tissues. See Example 24.

FIG. 43: FACS analysis demonstrating T-reg depletion in the TME in response to STIM003 mIgG2a antibody. CT-26 tumour bearing animals were treated with a single dose (6, 60 or $200 \mu\text{g}$) of STIM003 on day 12 post tumour cell implantation. Tissues ($n=4$ per time point) were harvested for FACS analysis on day 1, 2, 3, 4 and 8 post treatment. The percentage of T-reg cells (CD4+CD25+Foxp3+) in total tumour (A) and the percentage of T-reg cells in the blood (B) are shown at the different time points. See Example 24.

FIG. 44: Increase in CD8:T-reg and CD4 eff:T-reg ratio in response to STIM003 mIgG2a. CT-26 tumour bearing animals received a single dose (6, 60 or $200 \mu\text{g}$) of STIM003 mIgG2a on day 12 post tumour cell implantation. Tissues ($n=4$ per time point) were harvested for FACS analysis on day 1, 2, 3, 4 and 8 post treatment and T eff to T-reg ratios

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were calculated. (A) & (B), CD8:T-reg ratio in tumour and blood, (C) & (D) CD4-eff:T-reg ratio in tumour and blood. See Example 24.

FIG. 45: STIM003 treatment correlates with increased degranulation and Th1 cytokine production by TILs. On day 8 post treatment TILs were isolated and FACS analysis were performed to detect CD107a expression on CD4 and CD8 T cells (A-B). In parallel, cells from dissociated tumours were rested for 4 hrs in the presence of Brefeldin-A, cells were stained for T cells markers and permeabilised for intracellular staining to detect IFN- γ and TNF- α (C—H). See Example 24.

FIG. 46: Alignment of (a) VH domain and (b) VL domain amino acid sequences of STIM020 VH (SEQ ID NO: 1013) and VL (SEQ ID NO: 1021), STIM021 VH (SEQ ID NO: 998) and VL (SEQ ID NO: 1005), STIM023 VH (SEQ ID NO: 982) and VL (SEQ ID NO: 990). Consensus sequence "majority" indicated above. Solid black shows residues that match the consensus sequence exactly.

FIG. 47: Mouse ICOS-Ligand neutralisation HTRF with mouse ICOS receptor. Neutralisation profiles of STIM017 to STIM023 anti-ICOS mAbs in human IgG1 format compared with benchmarks STIM001 and STIM003. Data representative of one experiment.

FIG. 48: Human ICOS-ligand neutralisation HTRF with human ICOS receptor. Neutralisation profiles of STIM017 to STIM023 anti-ICOS mAbs in human IgG1 format compared to benchmarks STIM001 and STIM003. Data representative of one experiment.

FIG. 49: Binding of selected antibodies to cell-expressed mouse ICOS. Antibodies were titrated on CHO cells expressed Human ICOS and bound antibody detected with an anti-human IgG AlexaFluor 647. Data is from a single experiment.

FIG. 50: Binding of selected antibodies to cell-expressed human ICOS. Antibodies were titrated on CHO cells expressed human ICOS and bound antibody detected with an anti-human IgG AlexaFluor 647. Data is from a single experiment.

FIG. 51: ADCC activity with mouse ICOS CHO cells. ADCC activity of selected mAbs STIM017 to STIM023 anti-ICOS mAbs in human IgG1 format compared to benchmarks (STIM001 and STIM003). Data representative of one experiment.

FIG. 52: ADCC activity with human ICOS CHO cells. ADCC activity of selected mAbs STIM017 to STIM023 anti-ICOS mAbs in human IgG1 format compared to benchmarks (STIM001 and STIM003). Data representative of one experiment.

FIG. 53: MJ cell in vitro activation assay-Soluble. Stimulation profiles of STIM017-STIM023 anti-ICOS mAbs with cross-linker compared to benchmarks STIM001 and STIM003 and respective isotype controls. Data average of one experiments (n=1).

FIG. 54: Binding activity to Human ICOS recombinant protein. Binding activity of selected mAbs STIM017 to STIM023 anti-ICOS mAbs in human IgG1 format compared to benchmarks (STIM001 and STIM003). Data representative of one experiment. EC50 values were obtained from the curve fitting the plotted mean data using the 4-parameter fit from Graphpad Prism Version 6.00.

DETAILED DESCRIPTION

ICOS

Antibodies according to the present invention bind the extracellular domain of human ICOS. Thus, the antibodies

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bind ICOS-expressing T lymphocytes. "ICOS" or "the ICOS receptor" referred to herein may be human ICOS, unless the context dictates otherwise. Sequences of human, cynomolgus and mouse ICOS are shown in the appended sequence listing, and are available from NCBI as human NCBI ID: NP_036224.1, mouse NCBI ID: NP_059508.2 and cynomolgus GenBank ID: EHH55098.1.

Cross-Reactivity

Antibodies according to the present invention are preferably cross-reactive, and may for example bind the extracellular domain of mouse ICOS as well as human ICOS. The antibodies may bind other non-human ICOS, including ICOS of primates such as cynomolgus. An anti-ICOS antibody intended for therapeutic use in humans must bind human ICOS, whereas binding to ICOS of other species would not have direct therapeutic relevance in the human clinical context. Nevertheless, the data herein indicate that antibodies that bind both human and mouse ICOS have properties that render them particularly suitable as agonist and depleting molecules. This may result from one or more particular epitopes being targeted by the cross-reactive antibodies. Regardless of the underlying theory, however, cross-reactive antibodies are of high value and are excellent candidates as therapeutic molecules for pre-clinical and clinical studies.

As explained in the experimental Examples, the STIM antibodies described here were generated using Kymouse™ technology where the mouse had been engineered to lack expression of mouse ICOS (an ICOS knock-out). ICOS knock-out transgenic animals and their use for generating cross-reactive antibodies are further aspects of the present invention.

One way to quantify the extent of species cross-reactivity of an antibody is as the fold-difference in its affinity for antigen or one species compared with antigen of another species, e.g., fold difference in affinity for human ICOS vs mouse ICOS. Affinity may be quantified as K_D , referring to the equilibrium dissociation constant of the antibody-antigen reaction as determined by SPR with the antibody in Fab format as described elsewhere herein. A species cross-reactive anti-ICOS antibody may have a fold-difference in affinity for binding human and mouse ICOS that is 30-fold or less, 25-fold or less, 20-fold or less, 15-fold or less, 10-fold or less or 5-fold or less. To put it another way, the K_D of binding the extracellular domain of human ICOS may be within 30-fold, 25-fold, 20-fold, 15-fold, 10-fold or 5-fold of the K_D of binding the extracellular domain of mouse ICOS. Antibodies can also be considered cross-reactive if the K_D for binding antigen of both species meets a threshold value, e.g., if the K_D of binding human ICOS and the K_D of binding mouse ICOS are both 10 nM or less, preferably 5 nM or less, more preferably 1 nM or less. The K_D may be 10 nM or less, 5 nM or less, 2 nM or less, or 1 nM or less. The K_D may be 0.9 nM or less, 0.8 nM or less, 0.7 nM or less, 0.6 nM or less, 0.5 nM or less, 0.4 nM or less, 0.3 nM or less, 0.2 nM or less, or 0.1 nM or less.

An alternative measure of cross-reactivity for binding human ICOS and mouse ICOS is the ability of an antibody to neutralise ICOS ligand binding to ICOS receptor, such as in an HTRF assay (see Example 8). Examples of species cross-reactive antibodies are provided herein, including STIM001, STIM002, STIM002-B, STIM003, STIM005 and STIM006, each of which was confirmed as neutralising binding of human B7-H2 (ICOS ligand) to human ICOS and neutralising binding of mouse B7-H2 to mouse ICOS in an HTRF assay. Any of these antibodies or their variants may be selected when an antibody cross-reactive for human and

mouse ICOS is desired. A species cross-reactive anti-ICOS antibody may have an IC50 for inhibiting binding of human ICOS to human ICOS receptor that is within 25-fold, 20-fold, 15-fold, 10-fold or 5-fold of the IC50 for inhibiting mouse ICOS to mouse ICOS receptor as determined in an HTRF assay. Antibodies can also be considered cross-reactive if the IC50 for inhibiting binding of human ICOS to human ICOS receptor and the IC50 for inhibiting binding of mouse ICOS to mouse ICOS receptor are both 1 nM or less, preferably 0.5 nM or less, e.g., 30 nM or less, 20 nM or less, 10 nM or less. The IC50s may be 5 nM or less, 4 nM or less, 3 nM or less or 2 nM or less. In some cases the IC50s will be at least 0.1 nM, at least 0.5 nM or at least 1 nM.

Specificity

Antibodies according to the present invention are preferably specific for ICOS. That is, the antibody binds its epitope on the target protein, ICOS (human ICOS, and preferably mouse and/or cynomolgus ICOS as noted above), but does not show significant binding to molecules that do not present that epitope, including other molecules in the CD28 gene family. An antibody according to the present invention preferably does not bind human CD28. The antibody preferably also does not bind mouse or cynomolgus CD28.

CD28 co-stimulates T cell responses when engaged by its ligands CD80 and CD86 on professional antigen presenting cells in the context of antigen recognition via the TCR. For various in vivo uses of the antibodies described herein, the avoidance of binding to CD28 is considered advantageous. Non-binding of the anti-ICOS antibody to CD28 should allow CD28 to interact with its native ligands and to generate appropriate co-stimulatory signal for T cell activation. Additionally, non-binding of the anti-ICOS antibody to CD28 avoids the risk of superagonism. Over-stimulation of CD28 can induce proliferation in resting T cells without the normal requirement for recognition of a cognate antigen via the TCR, potentially leading to runaway activation of T cells and consequent cytokine-release syndrome, especially in human subjects. The non-recognition of CD28 by antibodies according to the present invention therefore represents an advantage in terms of their safe clinical use in humans.

As discussed elsewhere herein, the present invention extends to multispecific antibodies (e.g., bispecifics). A multispecific (e.g., bispecific) antibody may comprise (i) an antibody antigen binding site for ICOS and (ii) a further antigen binding site (optionally an antibody antigen binding site, as described herein) which recognises another antigen (e.g., PD-L1). Specific binding of individual antigen binding sites may be determined. Thus, antibodies that specifically bind ICOS include antibodies comprising an antigen binding site that specifically binds ICOS, wherein optionally the antigen binding site for ICOS is comprised within an antigen-binding molecule that further includes one or more additional binding sites for one or more other antigens, e.g., a bispecific antibody that binds ICOS and PD-L1. Example multispecific and bispecific formats are described for example in U.S. 62/607,469 (filing date 19 Dec. 2017) and its corresponding international application filed on 19 Dec. 2018, the disclosure of which is incorporated by reference herein, including for example anti-ICOS antibodies including PD-L1 binding Fc region (Fcab). Further example multispecific formats are described in WO2017/220988, also incorporated by reference herein. The anti-ICOS VH and VL domain sequences disclosed herein may be incorporated into any such multispecific format.

Affinity

The affinity of binding of an antibody to ICOS may be determined. Affinity of an antibody for its antigen may be

quantified in terms of the equilibrium dissociation constant K_D , the ratio K_a/K_d of the association or on-rate (K_a) and the dissociation or off-rate (k_d) of the antibody-antigen interaction. K_d , K_a and k_d for antibody-antigen binding can be measured using surface plasmon resonance (SPR).

An antibody according to the present invention may bind the EC domain of human ICOS with a K_D of 10 nM or less, preferably 5 nM or less, more preferably 1 nM or less. The K_D may be 50 nM or less, 10 nM or less, 5 nM or less, 2 nM or less, or 1 nM or less. The K_D may be 0.9 nM or less, 0.8 nM or less, 0.7 nM or less, 0.6 nM or less, 0.5 nM or less, 0.4 nM or less, 0.3 nM or less, 0.2 nM or less, or 0.1 nM or less. The K_D may be at least 0.001 nM, for example at least 0.01 nM or at least 0.1 nM.

Quantification of affinity may be performed using SPR with the antibody in Fab format. A suitable protocol is as follows:

1. Coupling anti-human (or other antibody constant region species-matched) IgG to a biosensor chip (e.g., GLM chip) such as by primary amine coupling;
2. Exposing the anti-human IgG (or other matched species antibody) to a test antibody, e.g., in Fab format, to capture test antibody on the chip;
3. Passing the test antigen over the chip's capture surface at a range of concentrations, e.g., at 5000 nM, 1000 nM, 200 nM, 40 nM, 8 nM and 2 nM, and at 0 nM (i.e., buffer alone); and
4. Determining the affinity of binding of test antibody to test antigen using SPR at 25° C. Buffer may be at pH 7.6, 150 mM NaCl, 0.05% detergent (e.g., P20) and 3 mM EDTA. Buffer may optionally contain 10 mM HEPES. HBS-EP can be used as running buffer. HBS-EP is available from Teknova Inc (California; catalogue number H8022).

Regeneration of the capture surface can be carried out with 10 mM glycine at pH 1.7. This removes the captured antibody and allows the surface to be used for another interaction. The binding data can be fitted to 1:1 model inherent using standard techniques, e.g., using a model inherent to the ProteOn XPR36™ analysis software.

A variety of SPR instruments are known, such as Biacore™, ProteOn XPR36™ (Bio-Rad®), and KinExAR (Sapidyne Instruments, Inc). Worked examples of SPR are found in Example 7.

As described, affinity may be determined by SPR with the antibody in Fab format, with the antigen coupled to the chip surface and the test antibody passed over the chip in Fab format in solution, to determine affinity of the monomeric antibody-antigen interaction. Affinity can be determined at any desired pH, e.g., pH 5.5 or pH 7.6, and any desired temperature e.g., 25° C. or 37° C. As reported in Example 7, antibodies according to the present invention bound human ICOS with an apparent affinity of less than 2 nM, as determined by SPR using the antibody in monovalent (Fab) format.

Other ways to measure binding of an antibody to ICOS include fluorescence activated cell sorting (FACS), e.g., using cells (e.g., CHO cells) with exogenous surface expression of ICOS or activated primary T cells expressing endogenous levels of ICOS. Antibody binding to ICOS-expressing cells as measured by FACS indicates that the antibody is able to bind the extracellular (EC) domain of ICOS.

ICOS Receptor Agonism

The ICOS ligand (ICOSL, also known as B7-H2) is a cell surface expressed molecule that binds to the ICOS receptor [17]. This intercellular ligand-receptor interaction promotes multimerisation of ICOS on the T cell surface, activating the

receptor and stimulating downstream signalling in the T cell. In effector T cells, this receptor activation stimulates the effector T cell response.

Anti-ICOS antibodies may act as agonists of ICOS, mimicking and even surpassing this stimulatory effect of the native ICOS ligand on the receptor. Such agonism may result from ability of the antibody to promote multimerisation of ICOS on the T cell. One mechanism for this is where the antibodies form intercellular bridges between ICOS on the T cell surface and receptors on an adjacent cell (e.g., B cell, antigen-presenting cell, or other immune cell), such as Fc receptors. Another mechanism is where antibodies having multiple (e.g., two) antigen-binding sites (e.g., two VH-VL domain pairs) bridge multiple ICOS receptor molecules and so promote multimerisation. A combination of these mechanisms may occur.

Agonism can be tested for in *in vitro* T cell activation assays, using antibody in soluble form (e.g., in immunoglobulin format or other antibody format comprising two spatially separated antigen-binding sites, e.g., two VH-VL pairs), either including or excluding a cross-linking agent, or using antibody bound to a solid surface to provide a tethered array of antigen-binding sites. Agonism assays may use a human ICOS positive T lymphocyte cell line such as MJ cells (ATCC CRL-8294) as the target T cell for activation in such assays. One or more measures of T cell activation can be determined for a test antibody and compared with a reference molecule or a negative control to determine whether there is a statistically significant ($p < 0.05$) difference in T cell activation effected by the test antibody compared with the reference molecule or the control. One suitable measure of T cell activation is production of cytokines, e.g., IFN γ , TNF α or IL-2. The skilled person will include suitable controls as appropriate, standardising assay conditions between test antibody and control. A suitable negative control is an antibody in the same format (e.g., isotype control) that does not bind ICOS, e.g., an antibody specific for an antigen that is not present in the assay system. A significant difference is observed for test antibody relative to a cognate isotype control within the dynamic range of the assay is indicative that the antibody acts as an agonist of the ICOS receptor in that assay.

An agonist antibody may be defined as one which, when tested in a T cell activation assay:

- has a significantly lower EC50 for induction of IFN γ production compared with control antibody;
- induces significantly higher maximal IFN γ production compared with control antibody;
- has a significantly lower EC50 for induction of IFN γ production compared with ICOSL-Fc;
- induces significantly higher maximal IFN γ production compared with ICOSL-Fc;
- has a significantly lower EC50 for induction of IFN γ production compared with reference antibody C398.4A; and/or
- induces significantly higher maximal IFN γ production compared with reference antibody C398.4A.

In vitro T cell assays include the bead-bound assay of Example 13, the plate-bound assay of Example 14 and the soluble form assay of Example 15.

A significantly lower or significantly higher value may for example be up to 0.5-fold different, up to 0.75-fold different, up to 2-fold different, up to 3-fold different, up to 4-fold different or up to 5-fold different, compared with the reference or control value.

Thus, in one example, an antibody according to the present invention has a significantly lower, e.g., at least

2-fold lower, EC50 for induction of IFN γ in an MJ cell activation assay using the antibody in bead-bound format, compared with control.

The bead-bound assay uses the antibody (and, for control or reference experiments, the control antibody, reference antibody or ICOSL-Fc) bound to the surface of beads. Magnetic beads may be used, and various kinds are commercially available, e.g., Tosyl-activated DYNABEADS M-450 (DYNAL Inc, 5 Delaware Drive, Lake Success, N.Y. 11042 Prod No. 140.03, 140.04). Beads may be coated as described in Example 13, or generally by dissolving the coating material in carbonate buffer (pH 9.6, 0.2 M) or other method known in the art. Use of beads conveniently allows the quantity of protein bound to the bead surface to be determined with a good degree of accuracy. Standard Fc-protein quantification methods can be used for coupled protein quantification on beads. Any suitable method can be used, with reference to a relevant standard within the dynamic range of the assay. DELFIA is exemplified in Example 13, but ELISA or other methods could be used.

Agonism activity of an antibody can also be measured in primary human T lymphocytes *ex vivo*. The ability of an antibody to induce expression of IFN γ in such T cells is indicative of ICOS agonism. Described herein are two T cell activation assays using primary cells—see Example 2, T cell activation assay 1 and T cell activation assay 2. Preferably, an antibody will show significant ($p < 0.05$) induction of IFN γ at 5 $\mu\text{g/ml}$ compared with control antibody in T cell activation assay 1 and/or T cell activation assay 2. As noted above, an anti-ICOS antibody may stimulate T cell activation to a greater degree than ICOS-L or C398.4 in such an assay. Thus, the antibody may show significantly ($p < 0.05$) greater induction of IFN γ at 5 $\mu\text{g/ml}$ compared with the control or reference antibody in T cell activation assay 1 or 2. TNF α or IL-2 induction may be measured as an alternative assay readout.

Agonism of an anti-ICOS antibody may contribute to its ability to change the balance between populations of TReg and TEff cells *in vivo*, e.g., in a site of pathology such as a tumour microenvironment, in favour of TEff cells. The ability of an antibody to enhance tumour cell killing by activated ICOS-positive effector T cells may be determined, as discussed elsewhere herein.

T Cell Dependent Killing

Effector T cell function can be determined in a biologically relevant context using an *in vitro* co-culture assay where tumour cells are incubated with relevant immune cells to trigger immune cell-dependent killing, in which the effect of an anti-ICOS antibody on tumour cell killing by TEffs is observed.

The ability of an antibody to enhance tumour cell killing by activated ICOS-positive effector T cells may be determined. An anti-ICOS antibody may stimulate significantly greater ($p < 0.05$) tumour cell killing compared with a control antibody. An anti-ICOS antibody may stimulate similar or greater tumour cell killing in such an assay as compared with a reference molecule such as the ICOS ligand or the C398.4 antibody. A similar degree of tumour cell killing can be represented as the assay readout for the test antibody being less than two-fold different from that for the reference molecule.

ICOS Ligand-Receptor Neutralisation Potency

An antibody according to the present invention may be one which inhibits binding of ICOS to its ligand ICOSL.

The degree to which an antibody inhibits binding of the ICOS receptor to its ligand is referred to as its ligand-receptor neutralising potency. Potency is normally expressed

as an IC₅₀ value, in pM unless otherwise stated. In ligand-binding studies, IC₅₀ is the concentration that reduces receptor binding by 50% of maximal specific binding level. IC₅₀ may be calculated by plotting % specific receptor binding as a function of the log of the antibody concentration, and using a software program such as Prism (Graph-Pad) to fit a sigmoidal function to the data to generate IC₅₀ values. Neutralising potency may be determined in an HTRF assay. A detailed working example of an HTRF assay for ligand-receptor neutralising potency is set out in Example 8.

An IC₅₀ value may represent the mean of a plurality of measurements. Thus, for example, IC₅₀ values may be obtained from the results of triplicate experiments, and a mean IC₅₀ value can then be calculated.

An antibody may have an IC₅₀ of 1 mM or less in a ligand-receptor neutralisation assay, e.g., 0.5 mM or less. The IC₅₀ may be, 30 nM or less, 20 nM or less, 10 nM or less, 5 nM or less, 4 nM or less, 3 nM or less or 2 nM or less. The IC₅₀ may be at least 0.1 nM, at least 0.5 nM or at least 1 nM.

Antibodies

As described in more detail in the Examples, we isolated and characterised antibodies of particular interest, designated STIM001, STIM002, STIM002-B, STIM003, STIM004, STIM005, STIM006, STIM007, STIM008 and STIM009. Further antibodies were then isolated and characterised, including STIM017, STIM020, STIM021, STIM022, STIM023, STIM039, STIM040, STIM041, STIM042, STIM043, STIM044, STIM050, STIM051, STIM052, STIM053, STIM054, STIM055, STIM056, STIM057, STIM058, STIM059, STIM060, STIM061, STIM062, STIM063, STIM064, STIM065 and STIM066. In various aspects of the invention, unless context dictates otherwise, antibodies may be selected from any of these antibodies. Sequences of each of these antibodies are provided herein, wherein for each antibody the following sequences are shown: nucleotide sequence encoding VH domain; amino acid sequence of VH domain; VH CDR1 amino acid sequence; VH CDR2 amino acid sequence; VH CDR3 amino acid sequence; nucleotide sequence encoding VL domain; amino acid sequence of VL domain; VL CDR1 amino acid sequence; VL CDR2 amino acid sequence; and VL CDR3 amino acid sequence, respectively. The present invention encompasses anti-ICOS antibodies having the VH and/or VL domain sequences of all antibodies shown in the appended sequence listing and/or in the drawings, as well as antibodies comprising the HCDRs and/or LCDRs of those antibodies, and optionally having the full heavy chain and/or full light chain amino acid sequence.

STIM001 has a heavy chain variable region (V_H) amino acid sequence of Seq ID No: 366, comprising the CDRH1 amino acid sequence of Seq ID No:363, the CDRH2 amino acid sequence of Seq ID No:364, and the CDRH3 amino acid sequence of Seq ID No:365. The heavy chain nucleic acid sequence of the VH domain is Seq ID No:367. STIM001 has a light chain variable region (V_L) amino acid sequence of Seq ID No:373, comprising the CDRL1 amino acid sequence of Seq ID No:370, the CDRL2 amino acid sequence of Seq ID No:371, and the CDRL3 amino acid sequence of Seq ID No:372. The light chain nucleic acid sequence of the VL domain is Seq ID No:374. The VH domain may be combined with any of the heavy chain constant region sequences described herein, e.g. Seq ID No: 193, Seq ID No: 195, Seq ID No: 197, Seq ID No: 199, Seq ID No:201, Seq ID No:203, Seq ID No:205, Seq ID No:340, Seq ID No: 524, Seq ID No:526, Seq ID No:528, Seq ID No:530, Seq ID No:532 or Seq ID No:534. The VL domain

may be combined with any of the light chain constant region sequences described herein, e.g. Seq ID Nos: 207, 209, 211, 213, 215, 217, 219, 221, 223, 225, 227, 229, 231, 233, 235, 237, 536 and 538. A full length heavy chain amino acid sequence is Seq ID No:368 (heavy chain nucleic acid sequence Seq ID No:369). A full length light chain amino acid sequence is Seq ID No:375 (light chain nucleic acid sequence Seq ID No:376).

STIM002 has a heavy chain variable region (V_H) amino acid sequence of Seq ID No: 380, comprising the CDRH1 amino acid sequence of Seq ID No:377, the CDRH2 amino acid sequence of Seq ID No:378, and the CDRH3 amino acid sequence of Seq ID No:379. The heavy chain nucleic acid sequence of the VH domain is Seq ID No:381. STIM002 has a light chain variable region (V_L) amino acid sequence of Seq ID No:387, comprising the CDRL1 amino acid sequence of Seq ID No:384, the CDRL2 amino acid sequence of Seq ID No:385, and the CDRL3 amino acid sequence of Seq ID No:386. The light chain nucleic acid sequence of the VL domain is Seq ID No:388 or Seq ID No:519. The VH domain may be combined with any of the heavy chain constant region sequences described herein, e.g. Seq ID No:193, Seq ID No:195, Seq ID No: 197, Seq ID No: 199, Seq ID No:201, Seq ID No:203, Seq ID No:205, Seq ID No:340, Seq ID No:524, Seq ID No:526, Seq ID No:528, Seq ID No:530, Seq ID No:532 or Seq ID No: 534. The V_L domain may be combined with any of the light chain constant region sequences described herein, e.g. Seq ID Nos: 207, 209, 211, 213, 215, 217, 219, 221, 223, 225, 227, 229, 231, 233, 235, 237, 536 and 538. A full length heavy chain amino acid sequence is Seq ID No: 382 (heavy chain nucleic acid sequence Seq ID No:383). A full length light chain amino acid sequence is Seq ID No:389 (light chain nucleic acid sequence Seq ID No:390 or Seq ID No: 520).

STIM002-B has a heavy chain variable region (V_H) amino acid sequence of Seq ID No: 394, comprising the CDRH1 amino acid sequence of Seq ID No:391, the CDRH2 amino acid sequence of Seq ID No:392, and the CDRH3 amino acid sequence of Seq ID No:393. The heavy chain nucleic acid sequence of the VH domain is Seq ID No:395. STIM002-B has a light chain variable region (V_L) amino acid sequence of Seq ID No:401, comprising the CDRL1 amino acid sequence of Seq ID No:398, the CDRL2 amino acid sequence of Seq ID No:399, and the CDRL3 amino acid sequence of Seq ID No:400. The light chain nucleic acid sequence of the V_L domain is Seq ID No:402. The VH domain may be combined with any of the heavy chain constant region sequences described herein, e.g. Seq ID No:193, Seq ID No: 195, Seq ID No: 197, Seq ID No: 199, Seq ID No:201, Seq ID No:203, Seq ID No:205, Seq ID No:340, Seq ID No: 524, Seq ID No:526, Seq ID No:528, Seq ID No:530, Seq ID No:532 or Seq ID No:534. The V_L domain may be combined with any of the light chain constant region sequences described herein, e.g. Seq ID Nos: 207, 209, 211, 213, 215, 217, 219, 221, 223, 225, 227, 229, 231, 233, 235, 237, 536 and 538. A full length heavy chain amino acid sequence is Seq ID No:396 (heavy chain nucleic acid sequence Seq ID No:397). A full length light chain amino acid sequence is Seq ID No:403 (light chain nucleic acid sequence Seq ID No:404).

STIM003 has a heavy chain variable region (VH) amino acid sequence of Seq ID No: 408, comprising the CDRH1 amino acid sequence of Seq ID No:405, the CDRH2 amino acid sequence of Seq ID No:406, and the CDRH3 amino acid sequence of Seq ID No:407. The heavy chain nucleic acid sequence of the VH domain is Seq ID No:409 or Seq ID No:521. STIM003 has a light chain variable region (V_L)

STIM008 has a heavy chain variable region (VH) amino acid sequence of Seq ID No: 480, comprising the CDRH1 amino acid sequence of Seq ID No:477, the CDRH2 amino acid sequence of Seq ID No:478, and the CDRH3 amino acid sequence of Seq ID No:479. The heavy chain nucleic acid sequence of the VH domain is Seq ID No:481. STIM008 has a light chain variable region (VL) amino acid sequence of Seq ID No:487, comprising the CDRL1 amino acid sequence of Seq ID No:484, the CDRL2 amino acid sequence of Seq ID No:485, and the CDRL3 amino acid sequence of Seq ID No:486. The light chain nucleic acid

sequence of the V_L domain is Seq ID No:488. The VH domain may be combined with any of the heavy chain constant region sequences described herein, e.g. Seq ID No: 193, Seq ID No: 195, Seq ID No: 197, Seq ID No: 199, Seq ID No:201, Seq ID No:203, Seq ID No:205, Seq ID No:340, Seq ID No: 524, Seq ID No:526, Seq ID No:528, Seq ID No:530, Seq ID No:532 or Seq ID No:534. The V_L domain may be combined with any of the light chain constant region sequences described herein, e.g. Seq ID Nos: 207, 209, 211, 213, 215, 217, 219, 221, 223, 225, 227, 229, 231, 233, 235, 237, 536 and 538. A full length heavy chain amino acid sequence is Seq ID No:482 (heavy chain nucleic acid sequence Seq ID No:483). A full length light chain amino acid sequence is Seq ID No:489 (light chain nucleic acid sequence Seq ID No:490).

STIM009 has a heavy chain variable region (V_H) amino acid sequence of Seq ID No: 494, comprising the CDRH1 amino acid sequence of Seq ID No:491, the CDRH2 amino acid sequence of Seq ID No:492, and the CDRH3 amino acid sequence of Seq ID No:493. The heavy chain nucleic acid sequence of the VH domain is Seq ID No:495. STIM009 has a light chain variable region (V_L) amino acid sequence of Seq ID No:501, comprising the CDRL1 amino acid sequence of Seq ID No:498, the CDRL2 amino acid sequence of Seq ID No:499, and the CDRL3 amino acid sequence of Seq ID No:500. The light chain nucleic acid sequence of the V_L domain is Seq ID No:502. The VH domain may be combined with any of the heavy chain constant region sequences described herein, e.g. Seq ID No: 193, Seq ID No: 195, Seq ID No: 197, Seq ID No: 199, Seq ID No:201, Seq ID No:203, Seq ID No:205, Seq ID No:340, Seq ID No: 524, Seq ID No:526, Seq ID No:528, Seq ID No:530, Seq ID No:532 or Seq ID No:534. The V_L domain may be combined with any of the light chain constant region sequences described herein, e.g. Seq ID Nos: 207, 209, 211, 213, 215, 217, 219, 221, 223, 225, 227, 229, 231, 233, 235, 237, 536 and 538. A full length heavy chain amino acid sequence is Seq ID No:496 (heavy chain nucleic acid sequence Seq ID No:497). A full length light chain amino acid sequence is Seq ID No:503 (light chain nucleic acid sequence Seq ID No:504).

Antibodies according to the present invention are immunoglobulins or molecules comprising immunoglobulin domains, whether natural or partly or wholly synthetically produced. Antibodies may be IgG, IgM, IgA, IgD or IgE molecules or antigen-specific antibody fragments thereof (including, but not limited to, a Fab, $F(ab')_2$, Fv, disulphide linked Fv, scFv, single domain antibody, closed conformation multispecific antibody, disulphide-linked scfv, diabody), whether derived from any species that naturally produces an antibody, or created by recombinant DNA technology; whether isolated from serum, B-cells, hybridomas, transfectomas, yeast or bacteria. Antibodies can be humanised using routine technology. The term antibody covers any polypeptide or protein comprising an antibody antigen-binding site. An antigen-binding site (paratope) is the part of an antibody that binds to and is complementary to the epitope of its target antigen (ICOS).

The term "epitope" refers to a region of an antigen that is bound by an antibody. Epitopes may be defined as structural or functional. Functional epitopes are generally a subset of the structural epitopes and have those residues that directly contribute to the affinity of the interaction. Epitopes may also be conformational, that is, composed of non-linear amino acids. In certain embodiments, epitopes may include determinants that are chemically active surface groupings of molecules such as amino acids, sugar side chains, phospho-

ryl groups, or sulfonyl groups, and, in certain embodiments, may have specific three-dimensional structural characteristics, and/or specific charge characteristics.

The antigen binding site is a polypeptide or domain that comprises one or more CDRs of an antibody and is capable of binding the antigen. For example, the polypeptide comprises a CDR3 (e.g., HCDR3). For example the polypeptide comprises CDRs 1 and 2 (e.g., HCDR1 and 2) or CDRs 1-3 of a variable domain of an antibody (e.g., HCDRs1-3).

An antibody antigen-binding site may be provided by one or more antibody variable domains. In an example, the antibody binding site is provided by a single variable domain, e.g., a heavy chain variable domain (VH domain) or a light chain variable domain (VL domain). In another example, the binding site comprises a VH/VL pair or two or more of such pairs. Thus, an antibody antigen-binding site may comprise a VH and a VL.

The antibody may be a whole immunoglobulin, including constant regions, or may be an antibody fragment. An antibody fragment is a portion of an intact antibody, for example comprising the antigen binding and/or variable region of the intact antibody. Examples of antibody fragments include:

- (i) a Fab fragment, a monovalent fragment consisting of the VL, VH, CL and CH1 domains; (ii) a $F(ab')_2$ fragment, a bivalent fragment including two Fab fragments linked by a disulfide bridge at the hinge region; (iii) an Fd fragment consisting of the VH and CH1 domains;
- (iv) an Fv fragment consisting of the VL and VH domains of a single arm of an antibody,
- (v) a dAb fragment (Ward et al., (1989) Nature 341:544-546; which is incorporated by reference herein in its entirety), which consists of a VH or VL domain; and
- (vi) an isolated complementarity determining region (CDR) that retains specific antigen-binding functionality.

Further examples of antibodies are H2 antibodies that comprise a dimer of a heavy chain (5'-VH-(optional hinge)-CH2-CH3-3') and are devoid of a light chain.

Single-chain antibodies (e.g., scFv) are a commonly used fragment. Multispecific antibodies may be formed from antibody fragments. An antibody of the invention may employ any such format, as appropriate.

Optionally, the antibody immunoglobulin domains may be fused or conjugated to additional polypeptide sequences and/or to labels, tags, toxins or other molecules. Antibody immunoglobulin domains may be fused or conjugated to one or more different antigen binding regions, providing a molecule that is able to bind a second antigen in addition to ICOS. An antibody of the present invention may be a multispecific antibody, e.g., a bispecific antibody, comprising (i) an antibody antigen binding site for ICOS and (ii) a further antigen binding site (optionally an antibody antigen binding site, as described herein) which recognises another antigen (e.g., PD-L1).

An antibody normally comprises an antibody VH and/or VL domain. Isolated VH and VL domains of antibodies are also part of the invention. The antibody variable domains are the portions of the light and heavy chains of antibodies that include amino acid sequences of complementarity determining regions (CDRs; ie., CDR1, CDR2, and CDR3), and framework regions (FRs). Thus, within each of the VH and VL domains are CDRs and FRs. A VH domain comprises a set of HCDRs, and a VL domain comprises a set of LCDRs. VH refers to the variable domain of the heavy chain. VL refers to the variable domain of the light chain. Each VH and

VL is typically composed of three CDRs and four FRs, arranged from amino-terminus to carboxy-terminus in the following order: FR1, CDR1, FR2, CDR2, FR3, CDR3, FR4. According to the methods used in this invention, the amino acid positions assigned to CDRs and FRs may be defined according to Kabat (Sequences of Proteins of Immunological Interest (National Institutes of Health, Bethesda, Md., 1987 and 1991)) or according to IMGT nomenclature. An antibody may comprise an antibody VH domain comprising a VH CDR1, CDR2 and CDR3 and a framework. It may alternatively or also comprise an antibody VL domain comprising a VL CDR1, CDR2 and CDR3 and a framework. Examples of antibody VH and VL domains and CDRs according to the present invention are as listed in the appended sequence listing that forms part of the present disclosure. The CDRs shown in the sequence listing are defined according to the IMGT system [18]. All VH and VL sequences, CDR sequences, sets of CDRs and sets of HCDRs and sets of LCDRs disclosed herein represent aspects and embodiments of the invention. As described herein, a "set of CDRs" comprises CDR1, CDR2 and CDR3. Thus, a set of HCDRs refers to HCDR1, HCDR2 and HCDR3, and a set of LCDRs refers to LCDR1, LCDR2 and LCDR3. Unless otherwise stated, a "set of CDRs" includes HCDRs and LCDRs.

An antibody the invention may comprise one or more CDRs as described herein, e.g. a CDR3, and optionally also a CDR1 and CDR2 to form a set of CDRs. The CDR or set of CDRs may be a CDR or set of CDRs of any of STIM020, STIM021 and STIM023, or may be a variant thereof as described herein. The CDR or set of CDRs may be a CDR or set of CDRs of any of STIM039, STIM040, STIM041, STIM042, STIM043 and STIM044, or may be a variant thereof as described herein. The CDR or set of CDRs may be a CDR or set of CDRs of STIM017, or may be a variant thereof as described herein. The CDR or set of CDRs may be a CDR or set of CDRs of STIM022, or may be a variant thereof as described herein. The CDR or set of CDRs may be a CDR or set of CDRs of any of STIM050, STIM051, STIM052, STIM053, STIM054, STIM055, STIM056, STIM057, STIM058, STIM059, STIM060, STIM061, STIM062, STIM063, STIM064, STIM065 and STIM066, or may be a variant thereof as described herein.

The invention provides antibodies comprising an HCDR1, HCDR2 and/or HCDR3 of any of antibodies STIM017, STIM020, STIM021, STIM022, STIM023, STIM039, STIM040, STIM041, STIM042, STIM043, STIM044, STIM050, STIM051, STIM052, STIM053, STIM054, STIM055, STIM056, STIM057, STIM058, STIM059, STIM060, STIM061, STIM062, STIM063, STIM064, STIM065 and STIM066 and/or an LCDR1, LCDR2 and/or LCDR3 of any of these antibodies, e.g. a set of CDRs. The antibody may comprise a set of VH CDRs of one of these antibodies. Optionally it may also comprise a set of VL CDRs of one of these antibodies, and the VL CDRs may be from the same or a different antibody as the VH CDRs.

A VH domain comprising a disclosed set of HCDRs, and/or a VL domain comprising a disclosed set of LCDRs, are also provided by the invention.

Typically, a VH domain is paired with a VL domain to provide an antibody antigen-binding site, although as discussed further below a VH or VL domain alone may be used to bind antigen. The STIM020 VH domain may be paired with the STIM020 VL domain, so that an antibody antigen-binding site is formed comprising both the STIM020 VH and VL domains. Analogous embodiments are provided for

the other VH and VL domains disclosed herein. In other embodiments, the STIM020 VH is paired with a VL domain other than the STIM020 VL, e.g., it may be paired with the STIM021 or STIM023 VL domain. Light-chain promiscuity is well established in the art. Again, analogous embodiments are provided by the invention for the other VH and VL domains disclosed herein.

The VH of any STIM0xx antibody may be paired with the VL of any other STIM0xx antibody. For example, the VH of any of antibodies STIM020, STIM021 and STIM023 may be paired with the VL of any of antibodies STIM020, STIM021 and STIM023.

An antibody may comprise one or more CDRs, e.g. a set of CDRs, within an antibody framework. The framework regions may be of human germline gene segment sequences. Thus, the antibody may be a human antibody having a VH domain comprising a set of HCDRs in a human germline framework. Normally the antibody also has a VL domain comprising a set of LCDRs, e.g. in a human germline framework. An antibody "gene segment", e.g., a VH gene segment, D gene segment, or JH gene segment refers to oligonucleotide having a nucleic acid sequence from which that portion of an antibody is derived, e.g., a VH gene segment is an oligonucleotide comprising a nucleic acid sequence that corresponds to a polypeptide VH domain from FR1 to part of CDR3. Human V, D and J gene segments recombine to generate the VH domain, and human V and J segments recombine to generate the VL domain. The D domain or region refers to the diversity domain or region of an antibody chain. J domain or region refers to the joining domain or region of an antibody chain. Somatic hypermutation may result in an antibody VH or VL domain having framework regions that do not exactly match or align with the corresponding gene segments, but sequence alignment can be used to identify the closest gene segments and thus identify from which particular combination of gene segments a particular VH or VL domain is derived. When aligning antibody sequences with gene segments, the antibody amino acid sequence may be aligned with the amino acid sequence encoded by the gene segment, or the antibody nucleotide sequence may be aligned directly with the nucleotide sequence of the gene segment.

Alignments of STIM antibody VH and VL domain sequences against related antibodies and against human germline sequences are shown in FIG. 35, FIG. 36 and FIG. 37.

An antibody of the invention may be a human antibody or a chimaeric antibody comprising human variable regions and non-human (e.g., mouse) constant regions. The antibody of the invention for example has human variable regions, and optionally also has human constant regions.

Thus, antibodies optionally include constant regions or parts thereof, e.g., human antibody constant regions or parts thereof. For example, a VL domain may be attached at its C-terminal end to antibody light chain kappa or lambda constant domains. Similarly, an antibody VH domain may be attached at its C-terminal end to all or part (e.g. a CH1 domain or Fc region) of an immunoglobulin heavy chain constant region derived from any antibody isotype, e.g. IgG, IgA, IgE and IgM and any of the isotype sub-classes, such as IgG1 or IgG4.

Examples of human heavy chain constant regions are shown in Table S1.

Constant regions of antibodies of the invention may alternatively be non-human constant regions. For example, when antibodies are generated in transgenic animals (examples of which are described elsewhere herein), chimaeric

antibodies may be produced comprising human variable regions and non-human (host animal) constant regions. Some transgenic animals generate fully human antibodies. Others have been engineered to generate antibodies comprising chimaeric heavy chains and fully human light chains. Where antibodies comprise one or more non-human constant regions, these may be replaced with human constant regions to provide antibodies more suitable for administration to humans as therapeutic compositions, as their immunogenicity is thereby reduced.

Digestion of antibodies with the enzyme papain, results in two identical antigen-binding fragments, known also as “Fab” fragments, and a “Fc” fragment, having no antigen-binding activity but having the ability to crystallize. “Fab” when used herein refers to a fragment of an antibody that includes one constant and one variable domain of each of the heavy and light chains. The term “Fc region” herein is used to define a C-terminal region of an immunoglobulin heavy chain, including native-sequence Fc regions and variant Fc regions. The “Fc fragment” refers to the carboxy-terminal portions of both H chains held together by disulfides. The effector functions of antibodies are determined by sequences in the Fc region, the region which is also recognised by Fc receptors (FcR) found on certain types of cells. Digestion of antibodies with the enzyme pepsin, results in the a F(ab')₂ fragment in which the two arms of the antibody molecule remain linked and comprise two-antigen binding sites. The F(ab')₂ fragment has the ability to crosslink antigen.

“Fv” when used herein refers to the minimum fragment of an antibody that retains both antigen-recognition and antigen-binding sites. This region consists of a dimer of one heavy and one light chain variable domain in tight, non-covalent or covalent association. It is in this configuration that the three CDRs of each variable domain interact to define an antigen-binding site on the surface of the VH-VL dimer. Collectively, the six CDRs confer antigen-binding specificity to the antibody. However, even a single variable domain (or half of an Fv comprising only three CDRs specific for an antigen) has the ability to recognise and bind antigen, although at a lower affinity than the entire binding site.

Antibodies disclosed herein may be modified to increase or decrease serum half-life. In one embodiment, one or more of the following mutations: T252L, T254S or T256F are introduced to increase biological half-life of the antibody. Biological half-life can also be increased by altering the heavy chain constant region CH1 domain or CL region to contain a salvage receptor binding epitope taken from two loops of a CH2 domain of an Fc region of an IgG, as described in U.S. Pat. Nos. 5,869,046 and 6,121,022, the modifications described therein are incorporated herein by reference. In another embodiment, the Fc hinge region of an antibody or antigen-binding fragment of the invention is mutated to decrease the biological half-life of the antibody or fragment. One or more amino acid mutations are introduced into the CH2-CH3 domain interface region of the Fc-hinge fragment such that the antibody or fragment has impaired Staphylococcal protein A (SpA) binding relative to native Fc-hinge domain SpA binding. Other methods of increasing serum half-life are known to those skilled in the art. Thus, in one embodiment, the antibody or fragment is PEGylated. In another embodiment, the antibody or fragment is fused to an albumin-binding domain, e.g. an albumin binding single domain antibody (dAb). In another embodiment, the antibody or fragment is PASylated (i.e. genetic fusion of polypeptide sequences composed of PAS (XL-Protein GmbH) which forms uncharged random coil struc-

tures with large hydrodynamic volume). In another embodiment, the antibody or fragment is XTENylated®/rPEGylated (i.e. genetic fusion of non-exact repeat peptide sequence (Amunix, Versartis) to the therapeutic peptide). In another embodiment, the antibody or fragment is ELPylated (i.e. genetic fusion to ELP repeat sequence (PhaseBio)). These various half-life extending fusions are described in more detail in Strohl, BioDrugs (2015) 29:215-239, which fusions, e.g. in Tables 2 and 6, are incorporated herein by reference.

The antibody may have a modified constant region which increases stability. Thus, in one embodiment, the heavy chain constant region comprises a Ser228Pro mutation. In another embodiment, the antibodies and fragments disclosed herein comprise a heavy chain hinge region that has been modified to alter the number of cysteine residues. This modification can be used to facilitate assembly of the light and heavy chains or to increase or decrease the stability of the antibody.

Fc Effector Functions, ADCC, ADCP and CDC

As discussed above, anti-ICOS antibodies can be provided in various isotypes and with different constant regions. Examples of human IgG antibody heavy chain constant region sequences are shown in Table S1. The Fc region of the antibody primarily determines its effector function in terms of Fc binding, antibody-dependent cell-mediated cytotoxicity (ADCC) activity, complement dependent cytotoxicity (CDC) activity and antibody-dependent cell phagocytosis (ADCP) activity. These “cellular effector functions”, as distinct from effector T cell function, involve recruitment of cells bearing Fc receptors to the site of the target cells, resulting in killing of the antibody-bound cell. In addition to ADCC and CDC, the ADCP mechanism represents a means of depleting antibody-bound T cells, and thus targeting high ICOS expressing TRegs for deletion.

Cellular effector functions ADCC, ADCP and/or CDC may also be exhibited by antibodies lacking Fc regions. Antibodies may comprise multiple different antigen-binding sites, one directed to ICOS and another directed to a target molecule where engagement of that target molecule induces ADCC, ADCP and/or CDC, e.g., an antibody comprising two scFv regions joined by a linker, where one scFv can engage an effector cell.

An antibody according to the present invention may be one that exhibits ADCC, ADCP and/or CDC. Alternatively, an antibody according to the present invention may lack ADCC, ADCP and/or CDC activity. In either case, an antibody according to the present invention may comprise, or may optionally lack, an Fc region that binds to one or more types of Fc receptor. Use of different antibody formats, and the presence or absence of FcR binding and cellular effector functions, allow the antibody to be tailored for use in particular therapeutic purposes as discussed elsewhere herein.

A suitable antibody format for some therapeutic applications employs a wild-type human IgG1 constant region. A constant region may be an effector-enabled IgG1 constant region, optionally having ADCC and/or CDC and/or ADCP activity. A suitable wild type human IgG1 constant region sequence is SEQ ID NO: 340 (IGHG1*01). Further examples of human IgG1 constant regions are shown in Table S1.

For testing of candidate therapeutic antibodies in mouse models of human disease, an effector positive mouse constant region, such as mouse IgG2a (mIgG2a), may be included instead of an effector positive human constant region.

A constant region may be engineered for enhanced ADCC and/or CDC and/or ADCP.

The potency of Fc-mediated effects may be enhanced by engineering the Fc domain by various established techniques. Such methods increase the affinity for certain Fc-receptors, thus creating potential diverse profiles of activation enhancement. This can be achieved by modification of one or several amino acid residues [20]. Human IgG1 constant regions containing specific mutations or altered glycosylation on residue Asn297 (e.g., N297Q, EU index numbering) have been shown to enhance binding to Fc receptors. Example mutations are one or more of the residues selected from 239, 332 and 330 for human IgG1 constant regions (or the equivalent positions in other IgG isotypes). An antibody may thus comprise a human IgG1 constant region having one or more mutations independently selected from N297Q, S239D, I332E and A330L (EU index numbering). A triple mutation (M252Y/S254T/T256E) may be used to enhance binding to FcRn, and other mutations affecting FcRn binding are discussed in Table 2 of [21], any of which may be employed in the present invention.

Increased affinity for Fc receptors can also be achieved by altering the natural glycosylation profile of the Fc domain by, for example, generating under fucosylated or de-fucosylated variants [22]. Non-fucosylated antibodies harbour a tri-mannosyl core structure of complex-type N-glycans of Fc without fucose residue. These glycoengineered antibodies that lack core fucose residue from the Fc N-glycans may exhibit stronger ADCC than fucosylated equivalents due to enhancement of FcγRIIIa binding capacity. For example, to increase ADCC, residues in the hinge region can be altered to increase binding to Fc-gamma RIII [23]. Thus, an antibody may comprise a human IgG heavy chain constant region that is a variant of a wild-type human IgG heavy chain constant region, wherein the variant human IgG heavy chain constant region binds to human Fcγ receptors selected from the group consisting of FcγRIIB and FcγRIIA with higher affinity than the wild type human IgG heavy chain constant region binds to the human Fcγ receptors. The antibody may comprise a human IgG heavy chain constant region that is a variant of a wild type human IgG heavy chain constant region, wherein the variant human IgG heavy chain constant region binds to human FcγRIIB with higher affinity than the wild type human IgG heavy chain constant region binds to human FcγRIIB. The variant human IgG heavy chain constant region can be a variant human IgG1, a variant human IgG2, or a variant human IgG4 heavy chain constant region. In one embodiment, the variant human IgG heavy chain constant region comprises one or more amino acid mutations selected from G236D, P238D, S239D, S267E, L328F, and L328E (EU index numbering system). In another embodiment, the variant human IgG heavy chain constant region comprises a set of amino acid mutations selected from the group consisting of: S267E and L328F; P238D and L328E; P238D and one or more substitutions selected from the group consisting of E233D, G237D, H268D, P271G, and A330R; P238D, E233D, G237D, H268D, P271G, and A330R; G236D and S267E; S239D and S267E; V262E, S267E, and L328F; and V264E, S267E, and L328F (EU index numbering system). The enhancement of CDC may be achieved by amino acid changes that increase affinity for C1q, the first component of the classic complement activation cascade [24]. Another approach is to create a chimeric Fc domain created from human IgG1 and human IgG3 segments that exploit the higher affinity of IgG3 for C1q [25]. Antibodies of the present invention may comprise mutated amino acids at residues 329, 331 and/or 322 to alter

the C1q binding and/or reduced or abolished CDC activity. In another embodiment, the antibodies or antibody fragments disclosed herein may contain Fc regions with modifications at residues 231 and 239, whereby the amino acids are replaced to alter the ability of the antibody to fix complement. In one embodiment, the antibody or fragment has a constant region comprising one or more mutations selected from E345K, E430G, R344D and D356R, in particular a double mutation comprising R344D and D356R (EU index numbering system).

WO2008/137915 described anti-ICOS antibodies with modified Fc regions having enhanced effector function. The antibodies were reported to mediate enhanced ADCC activity as compared to the level of ADCC activity mediated by a parent antibody comprising the VH and VK domains and a wild type Fc region. Antibodies according to the present invention may employ such variant Fc regions having effector function as described therein.

ADCC activity of an antibody may be determined in an assay as described herein. ADCC activity of an anti-ICOS antibody may be determined in vitro using an ICOS positive T cell line as described in Example 10. ADCC activity of an anti-PD-L1 antibody may be determined in vitro in an ADCC assay using PD-L1 expressing cells.

For certain applications (such as in the context of vaccination) it may be preferred to use antibodies without Fc effector function. Antibodies may be provided without a constant region, or without an Fc region—examples of such antibody formats are described elsewhere herein. Alternatively, an antibody may have a constant region which is effector null. An antibody may have a heavy chain constant region that does not bind Fcγ receptors, for example the constant region may comprise a Leu235Glu mutation (i.e., where the wild type leucine residue is mutated to a glutamic acid residue). Another optional mutation for a heavy chain constant region is Ser228Pro, which increases stability. A heavy chain constant region may be an IgG4 comprising both the Leu235Glu mutation and the Ser228Pro mutation. This “IgG4-PE” heavy chain constant region is effector null.

An alternative effector null human constant region is a disabled IgG1. A disabled IgG1 heavy chain constant region may contain alanine at position 235 and/or 237 (EU index numbering), e.g., it may be a IgG1*01 sequence comprising the L235A and/or G237A mutations (“LAGA”).

A variant human IgG heavy chain constant region may comprise one or more amino acid mutations that reduce the affinity of the IgG for human FcγRIIIA, human FcγRIIA, or human FcγRI. In one embodiment, the FcγRIIB is expressed on a cell selected from the group consisting of macrophages, monocytes, B-cells, dendritic cells, endothelial cells, and activated T-cells. In one embodiment, the variant human IgG heavy chain constant region comprises one or more of the following amino acid mutations G236A, S239D, F243L, T256A, K290A, R292P, S298A, Y300L, V305I, A330L, I332E, E333A, K334A, A339T, and P396L (EU index numbering system). In one embodiment, the variant human IgG heavy chain constant region comprises a set of amino acid mutations selected from the group consisting of: S239D; T256A; K290A; S298A; I332E; E333A; K334A; A339T; S239D and I332E; S239D, A330L, and I332E; S298A, E333A, and K334A; G236A, S239D, and I332E; and F243L, R292P, Y300L, V305I, and P396L (EU index numbering system). In one embodiment, the variant human IgG heavy chain constant region comprises a S239D, A330L, or I332E amino acid mutations (EU index numbering system). In one embodiment, the variant human IgG heavy chain constant region comprises an S239D and I332E

amino acid mutations (EU index numbering system). In one embodiment, the variant human IgG heavy chain constant region is a variant human IgG1 heavy chain constant region comprising the S239D and I332E amino acid mutations (EU index numbering system). In one embodiment, the antibody or fragment comprises an afucosylated Fc region. In another embodiment, the antibody or fragment thereof is defucosylated. In another embodiment, the antibody or fragment is under fucosylated.

An antibody may have a heavy chain constant region that binds one or more types of Fc receptor but does not induce cellular effector functions, i.e., does not mediate ADCC, CDC or ADCC activity. Such a constant region may be unable to bind the particular Fc receptor(s) responsible for triggering ADCC, CDC or ADCC activity.

Generating and Modifying Antibodies

Methods for identifying and preparing antibodies are well known. Antibodies may be generated using transgenic mice (eg, the Kymouse™, Velocimouse®, Omnimouse®, Xenomouse®, HuMab Mouse® or MeMo Mouse®), rats (e.g., the Omnirat®), camelids, sharks, rabbits, chickens or other non-human animals immunised with ICOS or a fragment thereof or a synthetic peptide comprising an ICOS sequence motif of interest, followed optionally by humanisation of the constant regions and/or variable regions to produce human or humanised antibodies. In an example, display technologies can be used, such as yeast, phage or ribosome display, as will be apparent to the skilled person. Standard affinity maturation, e.g., using a display technology, can be performed in a further step after isolation of an antibody lead from a transgenic animal, phage display library or other library. Representative examples of suitable technologies are described in US20120093818 (Amgen, Inc), which is incorporated by reference herein in its entirety, eg, the methods set out in paragraphs to [0346].

Immunisation of an ICOS knock out non-human animal with human ICOS antigen facilitates the generation of antibodies that recognise both human and non-human ICOS. As described herein and illustrated in the Examples, an ICOS knock out mouse can be immunised with cells expressing human ICOS to stimulate production of antibodies to human and mouse ICOS in the mouse, which can be recovered and tested for binding to human ICOS and to mouse ICOS. Cross-reactive antibodies can thus be selected, which may be screened for other desirable properties as described herein. Methods of generating antibodies to an antigen (e.g., a human antigen), through immunisation of animals with the antigen where expression of the endogenous antigen (e.g. endogenous mouse antigen) has been knocked-out in the animal, may be performed in animals capable of generating antibodies comprising human variable domains. The genomes of such animals can be engineered to comprise a human or humanised immunoglobulin locus encoding human variable region gene segments, and optionally an endogenous constant region or a human constant region. Recombination of the human variable region gene segments generates human antibodies, which may have either a non-human or human constant region. Non-human constant regions may subsequently be replaced by human constant regions where the antibody is intended for in vivo use in humans. Such methods and knock-out transgenic animals are described in WO2013/061078.

Generally, a Kymouse™, VELOCIMMUNER or other mouse or rat (optionally an ICOS knock out mouse or rat, as noted) can be challenged with the antigen of interest, and lymphatic cells (such as B-cells) are recovered from the mice that express antibodies. The lymphatic cells may be

fused with a myeloma cell line to prepare immortal hybridoma cell lines, and such hybridoma cell lines are screened and selected to identify hybridoma cell lines that produce antibodies specific to the antigen of interest. DNA encoding the variable regions of the heavy chain and light chain may be isolated and linked to desirable isotypic constant regions of the heavy chain and light chain. Such an antibody protein may be produced in a cell, such as a CHO cell. Alternatively, DNA encoding the antigen-specific chimaeric antibodies or the variable domains of the light and heavy chains may be isolated directly from antigen-specific lymphocytes.

Initially, high affinity chimaeric antibodies are isolated having a human variable region and a mouse constant region. The antibodies are characterised and selected for desirable characteristics, including affinity, selectivity, agonism, T-cell dependent killing, neutralising potency, epitope, etc. The mouse constant regions are optionally replaced with a desired human constant region to generate the fully human antibody of the invention, for example wild-type or modified IgG1 or IgG4 (for example, SEQ ID NO: 751, 752, 753 in US2011/0065902 (which is incorporated by reference herein in its entirety). While the constant region selected may vary according to specific use, high affinity antigen-binding and target specificity characteristics reside in the variable region.

Thus, in a further aspect, the present invention provides a transgenic non-human mammal having a genome comprising a human or humanised immunoglobulin locus, wherein the mammal does not express ICOS. The mammal may for instance be a knock-out mouse or rat, or other laboratory animal species. Transgenic mice such as the Kymouse™ contain human heavy and light chain immunoglobulin loci inserted at the corresponding endogenous mouse immunoglobulin loci. A transgenic mammal according to the present invention may be one that contains such targeted insertions, or it may contain human heavy and light chain immunoglobulin loci or immunoglobulin genes that are randomly inserted in its genome, inserted at a locus other than the endogenous Ig locus, or provided on an additional chromosome or chromosomal fragment.

Further aspects of the invention are the use of such non-human mammals for producing antibodies to ICOS, and methods of producing antibodies or antibody heavy and/or light chain variable domains in such mammals.

A method of producing an antibody that binds the extracellular domain of human and non-human ICOS may comprise providing a transgenic non-human mammal having a genome comprising a human or humanised immunoglobulin locus, wherein the mammal does not express ICOS, and

- (a) immunising the mammal with human ICOS antigen (e.g., with cells expressing human ICOS or with purified recombinant ICOS protein);
- (b) isolating antibodies generated by the mammal;
- (c) testing the antibodies for ability to bind human ICOS and non-human ICOS; and
- (d) selecting one or more antibodies that binds both human and non-human ICOS.

Testing for ability to bind human ICOS and non-human ICOS may be done using surface plasmon resonance, HTFR, FACS or any other method described herein. Optionally, binding affinities for human and mouse ICOS are determined. The affinity, or fold-difference in affinity, of binding to human ICOS and mouse ICOS may be determined, and antibodies displaying species cross-reactivity may thus be selected (affinity thresholds and fold-differences that may be used as selection criteria are exemplified elsewhere herein). Neutralising potency, or fold difference in neutralising

potency, of the antibody for inhibiting human and mouse ICOS ligand binding to the human and mouse ICOS receptor respectively may also or alternatively be determined as a way to screen for cross-reactive antibodies, e.g., in an HTRF assay. Again, possible thresholds and fold-differences that may be used as selection criteria are exemplified elsewhere herein.

The method may comprise testing the antibodies for ability to bind non-human ICOS from the same species or from a different species as the immunised mammal. Thus, where the transgenic mammal is a mouse (e.g., a Kymouse™), antibodies may be tested for ability to bind mouse ICOS. Where the transgenic mammal is a rat, antibodies may be tested for ability to bind rat ICOS. However, it may be equally useful to determine cross-reactivity of an isolated antibody for non-human ICOS of another species. Thus, antibodies generated in goats may be tested for binding to rat or mouse ICOS. Optionally, binding to goat ICOS may be determined instead or additionally.

In other embodiments, the transgenic non-human mammal may be immunised with non-human ICOS, optionally ICOS of the same mammalian species (e.g., an ICOS knock-out mouse may be immunised with mouse ICOS) instead of human ICOS. Affinity of isolated antibodies for binding to human ICOS and non-human ICOS is then determined in the same way, and antibodies that bind both human and non-human ICOS are selected.

Nucleic acid encoding an antibody heavy chain variable domain and/or an antibody light chain variable domain of a selected antibody may be isolated. Such nucleic acid may encode the full antibody heavy chain and/or light chain, or the variable domain(s) without associated constant region (s). As noted, encoding nucleotide sequences may be obtained directly from antibody-producing cells of a mouse, or B cells may be immortalised or fused to generate hybridomas expressing the antibody, and encoding nucleic acid obtained from such cells. Optionally, nucleic acid encoding the variable domain(s) is then conjugated to a nucleotide sequence encoding a human heavy chain constant region and/or human light chain constant region, to provide nucleic acid encoding a human antibody heavy chain and/or human antibody light chain, e.g., encoding an antibody comprising both the heavy and light chain. As described elsewhere herein, this step is particularly useful where the immunised mammal produces chimaeric antibodies with non-human constant regions, which are preferably replaced with human constant regions to generate an antibody that will be less immunogenic when administered to humans as a medication. Provision of particular human isotype constant regions is also significant for determining the effector function of the antibody, and a number of suitable heavy chain constant regions are discussed herein.

Other alterations to nucleic acid encoding the antibody heavy and/or light chain variable domain may be performed, such as mutation of residues and generation of variants, as described herein.

The isolated (optionally mutated) nucleic acid may be introduced into host cells, e.g., CHO cells as discussed. Host cells are then cultured under conditions for expression of the antibody, or of the antibody heavy and/or light chain variable domain, in any desired antibody format. Some possible antibody formats are described herein, e.g., whole immunoglobulins, antigen-binding fragments, and other designs.

Variable domain amino acid sequence variants of any of the VH and VL domains or CDRs whose sequences are specifically disclosed herein may be employed in accordance with the present invention, as discussed.

There are many reasons why it may be desirable to create variants, which include optimising the antibody sequence for large-scale manufacturing, facilitating purification, enhancing stability or improving suitability for inclusion in a desired pharmaceutical formulation. Protein engineering work can be performed at one or more target residues in the antibody sequence, e.g., to substituting one amino acid with an alternative amino acid (optionally, generating variants containing all naturally occurring amino acids at this position, with the possible exception of Cys and Met), and monitoring the impact on function and expression to determine the best substitution. It is in some instances undesirable to substitute a residue with Cys or Met, or to introduce these residues into a sequence, as to do so may generate difficulties in manufacturing—for instance through the formation of new intramolecular or intermolecular cysteine-cysteine bonds. Where a lead candidate has been selected and is being optimised for manufacturing and clinical development, it will generally be desirable to change its antigen-binding properties as little as possible, or at least to retain the affinity and potency of the parent molecule. However, variants may also be generated in order to modulate key antibody characteristics such as affinity, cross-reactivity or neutralising potency.

An antibody may comprise a set of H and/or L CDRs of any of the disclosed antibodies with one or more amino acid mutations within the disclosed set of H and/or L CDRs. The mutation may be an amino acid substitution, deletion or insertion. Thus for example there may be one or more amino acid substitutions within the disclosed set of H and/or L CDRs. For example, there may be up to 12, 11, 10, 9, 8, 7, 6, 5, 4, 3 or 2 mutations e.g. substitutions, within the set of H and/or L CDRs. For example, there may be up to 6, 5, 4, 3 or 2 mutations, e.g. substitutions, in HCDR3 and/or there may be up to 6, 5, 4, 3, or 2 mutations, e.g. substitutions, in LCDR3. An antibody may comprise the set of HCDRs, LCDRs or a set of 6 (H and L) CDRs shown for any STIM antibody herein or may comprise that set of CDRs with one or two conservative substitutions.

One or more amino acid mutations may optionally be made in framework regions of an antibody VH or VL domain disclosed herein. For example, one or more residues that differ from the corresponding human germline segment sequence may be reverted to germline. Human germline gene segment sequences corresponding to VH and VL domains of example anti-ICOS antibodies are indicated in Table E12-1, Table E12-2 and Table E12-3, and alignments of antibody VH and VL domains to corresponding germline sequences are shown in the drawings.

An antibody may comprise a VH domain that has at least 60, 70, 80, 85, 90, 95, 96, 97, 98 or 99% amino acid sequence identity with a VH domain of any of the antibodies shown in the appended sequence listing, and/or comprising a VL domain that has at least 60, 70, 80, 85, 90, 95, 96, 97, 98 or 99% amino acid sequence identity with a VL domain of any of those antibodies. Algorithms that can be used to calculate % identity of two amino acid sequences include e.g. BLAST, FASTA, or the Smith-Waterman algorithm, e.g. employing default parameters. Particular variants may include one or more amino acid sequence alterations (addition, deletion, substitution and/or insertion of an amino acid residue).

Alterations may be made in one or more framework regions and/or one or more CDRs. Variants are optionally provided by CDR mutagenesis. The alterations normally do not result in loss of function, so an antibody comprising a thus-altered amino acid sequence may retain an ability to

bind ICOS. It may retain the same quantitative binding ability as an antibody in which the alteration is not made, e.g. as measured in an assay described herein. The antibody comprising a thus-altered amino acid sequence may have an improved ability to bind ICOS.

Alteration may comprise replacing one or more amino acid residue with a non-naturally occurring or non-standard amino acid, modifying one or more amino acid residue into a non-naturally occurring or non-standard form, or inserting one or more non-naturally occurring or non-standard amino acid into the sequence. Examples of numbers and locations of alterations in sequences of the invention are described elsewhere herein. Naturally occurring amino acids include the 20 "standard" L-amino acids identified as G, A, V, L, I, M, P, F, W, S, T, N, Q, Y, C, K, R, H, D, E by their standard single-letter codes. Non-standard amino acids include any other residue that may be incorporated into a polypeptide backbone or result from modification of an existing amino acid residue. Non-standard amino acids may be naturally occurring or non-naturally occurring.

The term "variant" as used herein refers to a peptide or nucleic acid that differs from a parent polypeptide or nucleic acid by one or more amino acid or nucleic acid deletions, substitutions or additions, yet retains one or more specific functions or biological activities of the parent molecule. Amino acid substitutions include alterations in which an amino acid is replaced with a different naturally-occurring amino acid residue. Such substitutions may be classified as "conservative", in which case an amino acid residue contained in a polypeptide is replaced with another naturally occurring amino acid of similar character either in relation to polarity, side chain functionality or size. Such conservative substitutions are well known in the art. Substitutions encompassed by the present invention may also be "non-conservative", in which an amino acid residue which is present in a peptide is substituted with an amino acid having different properties, such as naturally-occurring amino acid from a different group (e.g., substituting a charged or hydrophobic amino acid with alanine), or alternatively, in which a naturally-occurring amino acid is substituted with a non-conventional amino acid. In some embodiments amino acid substitutions are conservative. Also encompassed within the term variant when used with reference to a polynucleotide or polypeptide, refers to a polynucleotide or polypeptide that can vary in primary, secondary, or tertiary structure, as compared to a reference polynucleotide or polypeptide, respectively (e.g., as compared to a wild-type polynucleotide or polypeptide).

In some aspects, one can use "synthetic variants", "recombinant variants", or "chemically modified" polynucleotide variants or polypeptide variants isolated or generated using methods well known in the art. "Modified variants" can include conservative or non-conservative amino acid changes, as described below. Polynucleotide changes can result in amino acid substitutions, additions, deletions, fusions and truncations in the polypeptide encoded by the reference sequence. Some aspects use include insertion variants, deletion variants or substituted variants with substitutions of amino acids, including insertions and substitutions of amino acids and other molecules) that do not normally occur in the peptide sequence that is the basis of the variant, for example but not limited to insertion of ornithine which do not normally occur in human proteins. The term "conservative substitution," when describing a polypeptide, refers to a change in the amino acid composition of the polypeptide that does not substantially alter the polypeptide's activity. For example, a conservative substitution

refers to substituting an amino acid residue for a different amino acid residue that has similar chemical properties (e.g., acidic, basic, positively or negatively charged, polar or nonpolar, etc.). Conservative amino acid substitutions include replacement of a leucine with an isoleucine or valine, an aspartate with a glutamate, or a threonine with a serine. Conservative substitution tables providing functionally similar amino acids are well known in the art. For example, the following six groups each contain amino acids that are conservative substitutions for one another: 1) Alanine (A), Serine(S), Threonine (T); 2) Aspartic acid (D), Glutamic acid (E); 3) Asparagine (N), Glutamine (Q); 4) Arginine (R), Lysine (K); 5) Isoleucine (I), Leucine (L), Methionine (M), Valine (V); and 6) Phenylalanine (F), Tyrosine (Y), Tryptophan (W). (See also Creighton, Proteins, W. H. Freeman and Company (1984), incorporated by reference in its entirety.) In some embodiments, individual substitutions, deletions or additions that alter, add or delete a single amino acid or a small percentage of amino acids can also be considered "conservative substitutions" if the change does not reduce the activity of the peptide. Insertions or deletions are typically in the range of about 1 to 5 amino acids. The choice of conservative amino acids may be selected based on the location of the amino acid to be substituted in the peptide, for example if the amino acid is on the exterior of the peptide and expose to solvents, or on the interior and not exposed to solvents.

One can select the amino acid that will substitute an existing amino acid based on the location of the existing amino acid, including its exposure to solvents (i.e., if the amino acid is exposed to solvents or is present on the outer surface of the peptide or polypeptide as compared to internally localized amino acids not exposed to solvents). Selection of such conservative amino acid substitutions are well known in the art, for example as disclosed in Dordo et al, J. Mol Biol, 1999, 217, 721-739 and Taylor et al, J. Theor. Biol. 119 (1986); 205-218 and S. French and B. Robson, J. Mol. Evol. 19 (1983) 171. Accordingly, one can select conservative amino acid substitutions suitable for amino acids on the exterior of a protein or peptide (i.e. amino acids exposed to a solvent), for example, but not limited to, the following substitutions can be used: substitution of Y with F, T with S or K, P with A, E with D or Q, N with D or G, R with K, G with N or A, T with S or K, D with N or E, I with L or V, F with Y, S with T or A, R with K, G with N or A, K with R, A with S, K or P.

In alternative embodiments, one can also select conservative amino acid substitutions encompassed suitable for amino acids on the interior of a protein or peptide, for example one can use suitable conservative substitutions for amino acids is on the interior of a protein or peptide (i.e. the amino acids are not exposed to a solvent), for example but not limited to, one can use the following conservative substitutions: where Y is substituted with F, T with A or S, I with L or V, W with Y, M with L, N with D, G with A, T with A or S, D with N, I with L or V, F with Y or L, S with A or T and A with S, G, T or V. In some embodiments, non-conservative amino acid substitutions are also encompassed within the term of variants.

The invention includes methods of producing antibodies containing VH and/or VL domain variants of the antibody VH and/or VL domains shown in the appended sequence listing. Such antibodies may be produced by a method comprising

- (i) providing, by way of addition, deletion, substitution or insertion of one or more amino acids in the amino acid sequence of a parent antibody VH domain, an antibody

VH domain that is an amino acid sequence variant of the parent antibody VH domain,

wherein the parent antibody VH domain is the VH domain of any of antibodies STIM017, STIM020, STIM021, STIM022, STIM023, STIM039, STIM040, STIM041, STIM042, STIM043, STIM044, STIM050, STIM051, STIM052, STIM053, STIM054, STIM055, STIM056, STIM057, STIM058, STIM059, STIM060, STIM061, STIM062, STIM063, STIM064, STIM065 and STIM066 or a VH domain comprising the heavy chain complementarity determining regions of any of those antibodies,

- (ii) optionally combining the VH domain thus provided with a VL domain, to provide a VH/VL combination, and
- (iii) testing the VH domain or VH/VL domain combination thus provided to identify an antibody with one or more desired characteristics.

Desired characteristics include binding to human ICOS, binding to mouse ICOS, and binding to other non-human ICOS such as cynomolgus ICOS. Antibodies with comparable or higher affinity for human and/or mouse ICOS may be identified. Other desired characteristics include increasing effector T cell function indirectly, via depletion of immunosuppressive Tregs, or directly, via ICOS signalling activation on T effector cells. Identifying an antibody with a desired characteristic may comprise identifying an antibody with a functional attribute described herein, such as its affinity, cross-reactivity, specificity, ICOS receptor agonism, neutralising potency and/or promotion of T cell dependent killing, any of which may be determined in assays as described herein.

When VL domains are included in the method, the VL domain may be a VL domain of any of STIM017, STIM020, STIM021, STIM022, STIM023, STIM039, STIM040, STIM041, STIM042, STIM043, STIM044, STIM050, STIM051, STIM052, STIM053, STIM054, STIM055, STIM056, STIM057, STIM058, STIM059, STIM060, STIM061, STIM062, STIM063, STIM064, STIM065 and STIM066, or may be a variant provided by way of addition, deletion, substitution or insertion of one or more amino acids in the amino acid sequence of a parent VL domain, wherein the parent VL domain is the VL domain of any of STIM017, STIM020, STIM021, STIM022, STIM023, STIM039, STIM040, STIM041, STIM042, STIM043, STIM044, STIM050, STIM051, STIM052, STIM053, STIM054, STIM055, STIM056, STIM057, STIM058, STIM059, STIM060, STIM061, STIM062, STIM063, STIM064, STIM065 and STIM066 or a VL domain comprising the light chain complementarity determining regions of any of those antibodies.

Methods of generating variant antibodies may optionally comprise producing copies of the antibody or VH/VL domain combination. Methods may further comprise expressing the resultant antibody. It is possible to produce nucleotide sequences corresponding to a desired antibody VH and/or VL domain, optionally in one or more expression vectors. Suitable methods of expression, including recombinant expression in host cells, are set out in detail herein. Encoding Nucleic Acids and Methods of Expression

Isolated nucleic acid may be provided, encoding antibodies according to the present invention. Nucleic acid may be DNA and/or RNA. Genomic DNA, cDNA, mRNA or other RNA, of synthetic origin, or any combination thereof can encode an antibody.

The present invention provides constructs in the form of plasmids, vectors, transcription or expression cassettes which comprise at least one polynucleotide as above. Exemplary nucleotide sequences are included in the sequence listing. Reference to a nucleotide sequence as set out herein encompasses a DNA molecule with the specified sequence, and encompasses a RNA molecule with the specified sequence in which U is substituted for T, unless context requires otherwise.

The present invention also provides a recombinant host cell that comprises one or more nucleic acids encoding the antibody. Methods of producing the encoded antibody may comprise expression from the nucleic acid, e.g., by culturing recombinant host cells containing the nucleic acid. The antibody may thus be obtained, and may be isolated and/or purified using any suitable technique, then used as appropriate. A method of production may comprise formulating the product into a composition including at least one additional component, such as a pharmaceutically acceptable excipient.

Systems for cloning and expression of a polypeptide in a variety of different host cells are well known. Suitable host cells include bacteria, mammalian cells, plant cells, filamentous fungi, yeast and baculovirus systems and transgenic plants and animals.

The expression of antibodies and antibody fragments in prokaryotic cells is well established in the art. A common bacterial host is *E. coli*. Expression in eukaryotic cells in culture is also available to those skilled in the art as an option for production. Mammalian cell lines available in the art for expression of a heterologous polypeptide include Chinese hamster ovary (CHO) cells, HeLa cells, baby hamster kidney cells, NSO mouse melanoma cells, YB2/0 rat myeloma cells, human embryonic kidney cells, human embryonic retina cells and many others.

Vectors may contain appropriate regulatory sequences, including promoter sequences, terminator sequences, polyadenylation sequences, enhancer sequences, marker genes and other sequences as appropriate. Nucleic acid encoding an antibody can be introduced into a host cell. Nucleic acid can be introduced to eukaryotic cells by various methods, including calcium phosphate transfection, DEAE-Dextran, electroporation, liposome-mediated transfection and transduction using retrovirus or other virus, e.g. vaccinia or, for insect cells, baculovirus. Introducing nucleic acid in the host cell, in particular a eukaryotic cell may use a viral or a plasmid based system. The plasmid system may be maintained episomally or may be incorporated into the host cell or into an artificial chromosome. Incorporation may be either by random or targeted integration of one or more copies at single or multiple loci. For bacterial cells, suitable techniques include calcium chloride transformation, electroporation and transfection using bacteriophage. The introduction may be followed by expressing the nucleic acid, e.g., by culturing host cells under conditions for expression of the gene, then optionally isolating or purifying the antibody.

Nucleic acid of the invention may be integrated into the genome (e.g. chromosome) of the host cell. Integration may be promoted by inclusion of sequences that promote recombination with the genome, in accordance with standard techniques.

The present invention also provides a method that comprises using nucleic acid described herein in an expression system in order to express an antibody.

Therapeutic Use

An antibody described herein may be used in a method of treatment of the human or animal body by therapy. The antibodies find use in increasing effector T cell response, which is of benefit for a range of diseases or conditions, including treating cancers or solid tumours and in the context of vaccination. Increased Teff response may be achieved using an antibody that modulates the balance or ratio between Teffs and Tregs in favour of Teff activity.

Anti-ICOS antibodies may be used for depleting regulatory T cells and/or increasing effector T cell response in a patient, and may be administered to a patient to treat a disease or condition amenable to therapy by depleting regulatory T cells and/or increasing effector T cell response.

An antibody of the present invention, or a composition comprising such an antibody molecule or its encoding nucleic acid, may be used or provided for use in any such method. Use of the antibody, or of a composition comprising it or its encoding nucleic acid, for the manufacture of a medicament for use in any such method is also envisaged. The method typically comprises administering the antibody or composition to a mammal. Suitable formulations and methods of administration are described elsewhere herein.

One envisaged therapeutic use of the antibodies is treatment of cancer. The cancer may be a solid tumour, e.g., renal cell cancer (optionally renal cell carcinoma, e.g., clear cell renal cell carcinoma), head and neck cancer, melanoma (optionally malignant melanoma), non-small cell lung cancer (e.g., adenocarcinoma), bladder cancer, ovarian cancer, cervical cancer, gastric cancer, liver cancer, pancreatic cancer, breast cancer, testicular germ cell carcinoma, or the metastases of a solid tumour such as those listed, or it may be a liquid haematological tumour e.g., lymphoma (such as Hodgkin's lymphoma or Non-Hodgkin's lymphoma, e.g., diffuse large B-cell lymphoma, DLBCL) or leukaemia (e.g., acute myeloid leukaemia). An anti-ICOS antibody may enhance tumour clearance in melanoma, head and neck cancer and non-small cell lung cancer and other cancers with a moderate to high mutational load [26]. By enhancing patients' immune response to their neoplastic lesions, immunotherapy using an anti-ICOS antibody offers the prospect of durable cures or long-term remissions, potentially even in the context of late stage disease.

Cancers are a diverse group of diseases, but anti-ICOS antibodies offer the possibility of treating a range of different cancers by exploiting the patient's own immune system, which has the potential to kill any cancer cell through recognition of mutant or overexpressed epitopes that distinguish cancer cells from normal tissue. By modulating the Teff/Treg balance, anti-ICOS antibodies can enable and/or promote immune recognition and killing of cancer cells. While anti-ICOS antibodies are therefore useful therapeutic agents for a wide variety of cancers, there are particular categories of cancers for which anti-ICOS therapy is especially suited and/or where anti-ICOS therapy can be effective when other therapeutic agents are not.

One such group is cancer that is positive for expression of ICOS ligand. Cancer cells may acquire expression of ICOS ligand, as has been described for melanoma [27]. Expression of ICOS ligand may provide the cells with a selective advantage as the surface-expressed ligand binds ICOS on Tregs, promoting the expansion and activation of the Tregs and thereby suppressing the immune response against the cancer. Cancer cells expressing ICOS ligand may depend for their survival on this suppression of the immune system by Tregs, and would thus be vulnerable to treatment with anti-ICOS antibodies that target the Tregs. This applies also

to cancers derived from cells that naturally express ICOS ligand. Continued expression of ICOS ligand by these cells again provides a survival advantage through immune suppression. A cancer expressing ICOS ligand may be derived from antigen-presenting cells such as B cells, dendritic cells and monocytes and may be a liquid haematological tumour such as those mentioned herein. Interestingly it has been shown that these types of cancer are also high in ICOS and FOXP3 expression (TCGA data)—see Example 25. Example 20 herein demonstrates efficacy of exemplary anti-ICOS antibodies in treating tumours derived from cancerous B cells (A20 syngeneic cells) that express ICOS ligand.

Accordingly, anti-ICOS antibodies can be used in methods of treating cancers that are positive for expression of ICOS ligand. Further, a cancer to be treated with anti-ICOS antibody according to the present invention may be one that is positive for expression of ICOS and/or FOXP3, and optionally also expresses ICOS ligand.

Patients may undergo testing to determine whether their cancer is positive for expression of the protein of interest (e.g., ICOS ligand, ICOS and/or FOXP3), for example by taking a test sample (e.g., tumour biopsy) from the patient and determining expression of the protein of interest. Patients whose cancer has been characterised as positive for expression of one, two or all such proteins of interest are selected for treatment with anti-ICOS antibody. As discussed elsewhere herein, anti-ICOS antibody may be used as a monotherapy or in combination with one or more other therapeutic agents.

Anti-ICOS antibodies also offer hope to patients whose cancers are refractory to treatment with antibodies or other drugs directed to immune checkpoint molecules such as CTLA-4, PD-1, PD-L1, CD137, GITR or CD73. These immunotherapies are effective against some cancers but in some cases a cancer may not respond, or it may become unresponsive to continued treatment with the antibody. In common with antibodies to immune checkpoint inhibitors, anti-ICOS antibodies modulate the patient's immune system—nevertheless an anti-ICOS antibody may succeed where such other antibodies fail. It is shown herein that animals carrying A20 B cell lymphomas could be treated with anti-ICOS antibodies to reduce growth of the tumour, shrink the tumour and indeed clear the tumour from the body, whereas treatment with an anti-PD-L1 antibody was no better than control. The A20 cell line has also been reported to be resistant to anti-CTLA-4 [28].

Accordingly, anti-ICOS antibodies can be used in methods of treating cancers that are refractory to treatment with one or more immunotherapies, such as (any or all of) an anti-CTLA-4 antibody, anti-PD1 antibody, anti-PD-L1 antibody, anti-CD137 antibody, anti-GITR antibody, or anti-CD73 antibody. A cancer may be characterised as being refractory to treatment with an antibody or other drug if treatment with that antibody or drug does not significantly reduce growth of the cancer, e.g., if a tumour continues to grow or does not reduce in size or if after a response period the tumour re-initiates its growth. Non-response to a therapeutic agent may be determined ex vivo by testing a sample (e.g., tumour biopsy sample) for cancer cell killing or growth inhibition, and/or in the clinical setting by observing (e.g., using an imaging technology, including MRI) that a patient treated with the therapy is not responding to treatment. Patients whose cancer has been characterised as refractory to treatment with such an immunotherapy are selected for treatment with anti-ICOS antibody.

Further, anti-ICOS antibodies may be used to treat B-cell derived cancer that is resistant to treatment with an anti-CD20 antibody. Anti-ICOS antibodies represent a treatment for cancers that fail to respond to, or become resistant to, therapy with anti-CD20 antibodies like rituximab. Anti-ICOS antibody may be used as a second-line (or further, or additional) treatment for such cancers. The anti-CD20 antibody resistant cancer may be a B cell cancer, e.g., B cell lymphoma, such as diffuse large B cell lymphoma. Resistance of a cancer to anti-CD20 may be determined ex vivo by testing a sample (e.g., tumour biopsy sample) for cancer cell killing or growth inhibition by anti-CD20 antibody, and/or in the clinical setting by observing that a patient treated with the anti-CD20 antibody is not responding to treatment. Alternatively, or additionally, the cancer (e.g., a tumour biopsy sample) may be tested to assess expression of CD20, where an absence or low level of CD20 expression indicates loss of sensitivity to anti-CD20 antibody.

Samples obtained from patients may thus be tested to determine surface expression of a protein of interest, for example ICOS ligand, ICOS, FOXP3 and/or a target receptor to which another therapeutic agent (e.g., anti-receptor antibody) is directed. The target receptor may be CD20 (to which anti-CD20 antibody therapy such as rituximab is directed), or another receptor such as PD1, EGFR, HER2 or HER3. Surface expression of ICOS ligand, ICOS, FOXP3 and/or lack or loss of surface expression of the target receptor is an indication that the cancer is susceptible to anti-ICOS antibody therapy. Anti-ICOS antibodies can be provided for administration to a patient whose cancer is characterised by surface expression of ICOS ligand, ICOS, FOXP3 and/or lack or loss of surface expression of a target receptor, optionally where the patient has been previously treated with anti-CTLA4, anti-PD1, anti-PD-L1 or with an antibody to the target receptor and has not responded or has stopped responding to treatment with that antibody, as measured for example by continued or renewed cancer cell growth, e.g., increase in tumour size.

Any suitable method may be employed to determine whether cancer cells test positive for surface expression of a protein such as ICOS ligand, CD20 or other target receptors mentioned herein. A typical method is immunohistochemistry, where a sample of the cells (e.g., a tumour biopsy sample) is contacted with an antibody for the protein of interest, and binding of antibody is detected using a labelled reagent-typically a second antibody that recognises the Fc region of the first antibody and carries a detectable label such as a fluorescent marker. A sample may be declared to test positive where at least 5% of cells are labelled, as visualised by cell staining or other detection of the label. Optionally a higher cut-off such as 10% or 25% may be used. The antibody will generally be used in excess. Reagent antibodies to the molecules of interest are available or may be generated by straightforward methods. To test for ICOS ligand, the antibody MAB1651 is currently available from R&D systems as a mouse IgG that recognises human ICOS ligand. To test for CD20 expression, rituximab may be used. Detection of mRNA levels of the ICOS ligand or target receptor of interest is an alternative technique [27].

A further indication that a tumour will respond to treatment with anti-ICOS antibody is the presence of Tregs in the tumour microenvironment. Activated Tregs are characterised by ICOS-high and Foxp3-high surface expression. The presence of Tregs in a tumour, especially in elevated numbers, provides a further basis on which a patient may be selected for treatment with anti-ICOS antibody. Tregs may be detected in a tumour biopsy sample ex vivo, for example

by immunohistochemistry (assaying for co-expression of both Foxp3 and ICOS, using antibodies to the target protein followed by detection of labels, as described above) or by single cell dispersion of the sample for use in FACS with labelled antibodies to ICOS and Foxp3. FACS methods are exemplified in Example 17 and Example 18.

The anti-ICOS antibodies may be used for treating cancers associated with infectious agents, such as virally-induced cancers. In this category are head and neck squamous cell carcinoma, cervical cancer, Merkel cell carcinoma and many others. Viruses associated with cancer include HBV, HCV, HPV (cervical cancer, oropharyngeal cancer), and EBV (Burkitt's lymphomas, gastric cancer, Hodgkin's lymphoma, other EBV positive B cell lymphomas, nasopharyngeal carcinoma and post transplant lymphoproliferative disease). The International Agency for Research on Cancer (Monograph 100B) identified the following major cancer sites associated with infectious agents:

Stomach/Gastric: *Helicobacter pylori*

Liver: Hepatitis B virus, hepatitis C virus (HCV), *Opisthorchis viverrini*, *Clonorchis sinensis*

Cervix uteri: Human papillomavirus (HPV) with or without HIV

Anogenital (penile, vulva, vagina, anus): HPV with or without HIV

Nasopharynx: Epstein-Barr virus (EBV)

Oropharynx: HPV with or without tobacco or alcohol consumption

Kaposi's sarcoma: Human herpes virus type 8 with or without HIV

Non-Hodgkin lymphoma: *H. pylori*, EBV with or without HIV, HCV, human T-cell lymphotropic virus type 1

Hodgkin's lymphoma: EBV with or without HIV

Bladder: *Schistosoma haematobium*.

Antibodies according to the present invention may be used for treating cancer associated with or induced by any of these infectious agents, such as the cancers specified above.

Stimulation of effector T cell response can also contribute to immunity against infectious disease and/or to recovery from infectious disease in a patient. Thus, an anti-ICOS antibody may be used for treating infectious disease by administering the antibody to a patient.

Infectious diseases include those caused by pathogens, e.g., bacterial, fungal, viral or protozoal pathogens, and treatment may be to promote immune response in a patient against the pathogen infection. An example of a bacterial pathogen is tuberculosis. Examples of viral pathogens are hepatitis B and HIV. Examples of protozoal pathogens are *Plasmodium* species, which cause malaria, such as *P. falciparum*.

The antibody may be used for treating infections, e.g., infection by any pathogen mentioned herein. Infection may be persistent or chronic infection. Infection may be localised or systemic. Extended contact between a pathogen and the immune system may lead to exhaustion of the immune system or development of tolerance (manifested for example through increased levels of Tregs, and tipping of the Treg: T_H balance in favour of Tregs) and/or to immune evasion by the pathogen, through evolution and modification of displayed pathogen antigens. These features reflect similar processes that are believed to occur in cancer. Anti-ICOS antibodies present a therapeutic approach to treating infection by a pathogen, e.g., chronic infection, through modulation of the Treg:T_H ratio in favour of T_H and/or other effects described herein.

Treatment may be of patients who have been diagnosed as having an infectious disease or an infection. Alternatively,

treatment may be preventative, and administered to a patient to guard against contracting a disease, e.g., as a vaccine, as described elsewhere herein.

It has also been proposed that an immune response, particularly an IFN γ -dependent systemic immune response, could be beneficial for treatment of Alzheimer's disease and other CNS pathologies that share a neuroinflammatory component as part [29]. WO2015/136541 proposed treatment of Alzheimer's disease using an anti-PD-1 antibody. Anti-ICOS antibodies may be used in the treatment of Alzheimer's disease or other neurodegenerative diseases, optionally in combination with one or more other immunomodulators (e.g., antibody to PD-1).

Combination Therapy

Treatment with an immunomodulatory antibody such as anti-CTLA4, anti-PD1 or anti-PDL1, especially one with Fc effector function, may create an environment in which further depletion of ICOS highly expressing immune-suppressive cells is beneficial. It may be advantageous to combine an anti-ICOS antibody with such an immunomodulator to enhance its therapeutic effects.

A patient who has been treated with an immunomodulatory antibody (e.g., anti-PDL-1, anti-PD-1, anti-CTLA-4) may particularly benefit from treatment with an anti-ICOS antibody. One reason for this is that an immunomodulatory antibody may increase the number of ICOS-positive Tregs (e.g., intratumoural Tregs) in the patient. This effect is also observed with certain other therapeutic agents, such as recombinant IL-2. Anti-ICOS antibody may reduce and/or reverse a surge or rise in ICOS+ Tregs (e.g., intratumoural Tregs) resulting from treatment of the patient with another therapeutic agent. A patient selected for treatment with an anti-ICOS antibody may thus be one who has already received treatment with a first therapeutic agent, the first therapeutic agent being an antibody (e.g., immunomodulator antibody) or other agent (e.g., IL-2) that increases the number of ICOS+ Tregs in the patient.

Immunomodulators with which an anti-ICOS antibody may be combined include antibodies to any of: PDL1 (e.g., avelumab), PD-1 (e.g., pembrolizumab or nivolumab) or CTLA-4 (e.g., ipilimumab or tremelimumab). An anti-ICOS antibody may be combined with pidilizumab. In other embodiments, an anti-ICOS antibody is not administered in combination with anti-CTLA-4 antibody, and/or optionally is administered in combination with a therapeutic antibody that is not an anti-CTLA-4 antibody.

For example, an anti-ICOS antibody may be used in combination therapy with an anti-PDL1 antibody. Preferably, the anti-ICOS antibody is one that mediates ADCC, ADCP and/or CDC. Preferably, the anti-PDL1 antibody is one that mediates ADCC, ADCP and/or CDC. An example of such combination therapy is administration of an anti-ICOS antibody with an anti-PDL1 antibody wherein both antibodies have effector positive constant regions. Thus, the anti-ICOS antibody and the anti-PDL1 antibody may both be able to mediate ADCC, CDC and/or ADCP. Fc effector function and selection of constant regions is described in detail elsewhere herein, but as one example an anti-ICOS human IgG1 may be combined with an anti-PD-L1 human IgG1. The anti-ICOS antibody and/or the anti-PD-L1 antibody may comprise a wild type human IgG1 constant region. Alternatively, the effector positive constant region of an antibody may be one that is engineered for enhanced effector function, e.g., enhanced CDC, ADCC and/or ADCP. Example antibody constant regions, including wild type human IgG1 sequences and mutations that alter effector function, are discussed in detail elsewhere herein.

Anti-PDL1 antibodies with which an anti-ICOS antibody may be combined include:

Anti-PDL1 antibody that inhibits binding of PD-1 to PDL1 and/or inhibits PDL1, optionally as effector positive human IgG1;

Anti-PD-1 antibody that inhibits binding of PD-1 to PDL1 and/or PDL2;

Avelumab, a human IgG1 antibody which inhibits PD-1 binding to PDL-1. See WO2013/079174;

Durvalumab (or "MEDI4736"), a variant human IgG1 antibody having mutations L234A, L235A and 331. See WO2011/066389;

Atezolizumab, a variant human IgG1 antibody having mutations N297A, D356E and L358M. See US2010/0203056;

BMS-936559, a human IgG4 antibody comprising mutation S228P. See WO2007/005874.

Numerous further examples of anti-PD-L1 antibodies are disclosed herein and others are known in the art. Characterisation data for many of the anti-PD-L1 antibodies mentioned here has been published in U.S. Pat. Nos. 9,567,399 and 9,617,338, both incorporated by reference herein. Example anti-PD-L1 antibodies have VH and/or VL domains comprising the HCDRs and/or LCDRs of any of 1D05, 84G09, 1D05 HC mutant 1, 1D05 HC mutant 2, 1D05 HC mutant 3, 1D05 HC mutant 4, 1D05 LC mutant 1, 1D05 LC mutant 2, 1D05 LC mutant 3, 411B08, 411C04, 411D07, 385F01, 386H03, 389A03, 413D08, 413G05, 413F09, 414B06 or 416E01 as set out in U.S. Pat. No. 9,567,399 or U.S. Pat. No. 9,617,338. The antibody may comprise the VH and VL domain of any of these antibodies, and may optionally comprise a heavy and/or light chain having the heavy and/or light chain amino acid sequence of any of these antibodies. VH and VL domains of these anti-PD-L1 antibodies are further described elsewhere herein.

Further example anti-PD-L1 antibodies have VH and/or VL domains comprising the HCDRs and/or LCDRs of KN-035, CA-170, FAZ-053, M7824, ABBV-368, LY-3300054, GNS-1480, YW243.55.S70, REGN3504, or of an anti-PD-L1 antibody disclosed in any of WO2017/034916, WO2017/020291, WO2017/020858, WO2017/020801, WO2016/111645, WO2016/197367, WO2016/061142, WO2016/149201, WO2016/000619, WO2016/160792, WO2016/022630, WO2016/007235, WO2015/179654, WO2015/173267, WO2015/181342, WO2015/109124, WO2015/112805, WO2015/061668, WO2014/159562, WO2014/165082, WO2014/100079, WO2014/055897, WO2013/181634, WO2013/173223, WO2013/079174, WO2012/145493, WO2011/066389, WO2010/077634, WO2010/036959, WO2010/089411 and WO2007/005874. The antibody may comprise the VH and VL domain of any of these antibodies, and may optionally comprise a heavy and/or light chain having the heavy and/or light chain amino acid sequence of any of these antibodies. The anti-ICOS antibody which is used in combination therapy with anti-PD-L1 may be an antibody of the present invention as disclosed herein. Alternatively, the anti-ICOS antibody may comprise the CDRs of, or a VH and/or VL domain of, an anti-ICOS antibody disclosed in any of the following publications:

WO2016154177, US2016304610—for example any of antibodies 7F12, 37A10, 35A9, 36E10, 16G10, 37A10S713, 37A10S714, 37A10S715, 37A10S716, 37A10S717, 37A10S718, 16G10S71, 16G10S72, 16G10S73, 16G10S83, 35A9S79, 35A9S710, or 35A9S89; WO16120789, US2016215059—for example the antibody known as 422.2 and/or H2L5;

WO14033327, EP2892928, US2015239978—for example the antibody known as 314-8 and/or produced from hybridoma CNCM I-4180;
 WO12131004, EP2691419, U.S. Pat. No. 9,376,493, US20160264666—for example the antibody Icos145-1 and/or antibody produced by hybridoma CNCM I-4179;
 WO10056804—for example the antibody JMAb 136 or “136”;
 WO9915553, EP1017723B1, U.S. Pat. Nos. 7,259,247, 7,132,099, 7,125,551, 7,306,800, 7,722,872, WO05103086, EP1740617, U.S. Pat. Nos. 8,318,905, 8,916,155—for example the antibody MIC-944 or 9F3;
 WO983821, U.S. Pat. No. 7,932,358B2, US2002156242, EP0984023, EP1502920, U.S. Pat. Nos. 7,030,225, 7,045, 615, 7,279,560, 7,226,909, 7,196,175, 7,932,358, 8,389, 690, WO02070010, EP1286668, EP1374901, U.S. Pat. Nos. 7,438,905, 7,438,905, WO0187981, EP1158004, U.S. Pat. Nos. 6,803,039, 7,166,283, 7,988,965, WO0115732, EP1125585, U.S. Pat. Nos. 7,465,445, 7,998,478—for example any JMAb antibody, e.g., any of JMAb-124, JMAb-126, JMAb-127, JMAb-128, JMAb-135, JMAb-136, JMAb-137, JMAb-138, JMAb-139, JMAb-140, JMAb-141, e.g., JMAb136;
 WO2014/089113—for example antibody 17G9;
 WO12174338;
 US2016145344;
 WO11020024, EP2464661, US2016002336, US2016024211, U.S. Pat. Nos. 8,840,889; 8,497,244.

The anti-ICOS antibody optionally comprises the CDRs of 37A10S713 as disclosed in WO2016154177. It may comprise the VH and VL domains of 37A10S713, and may optionally have the antibody heavy and light chains of 37A10S713.

Combination of an anti-ICOS antibody with an immunomodulator may provide an increased therapeutic effect compared with monotherapy, and may allow therapeutic benefit to be achieved with a lower dose of the immunomodulator(s). Thus, for example, an antibody (e.g., anti-PD-L1 antibody, optionally ipilimumab) that is used in combination with anti-ICOS antibody may be dosed at 3 mg/kg rather than a more usual dose of 10 mg/kg. The administration regimen of the anti-PD-L1 or other antibody may involve intravenous administration over a 90 minute period every 3 weeks for a total of 4 doses.

An anti-ICOS antibody may be used to increase the sensitivity of a tumour to treatment with an anti-PD-L1 antibody, which may be recognised as a reduction in the dose at which the anti-PD-L1 antibody exerts a therapeutic benefit. Thus, anti-ICOS antibody may be administered to a patient to reduce the dose of anti-PD-L1 antibody effective to treat cancer or a tumour in the patient. Administration of anti-ICOS antibody may reduce the recommended or required dosage of anti-PD-L1 antibody administration to that patient to, for example, 75%, 50%, 25%, 20%, 10% or less, compared with the dosage when anti-PD-L1 antibody is administered without anti-ICOS. The patient may be treated by administration of anti-ICOS antibody and anti-PD-L1 antibody in a combination therapy as described herein.

The benefit of combining anti-PD-L1 with anti-ICOS may extend to a reduction in dosage of each agent when compared with its use as a monotherapy. Anti-PD-L1 antibody may be used to reduce the dose at which anti-ICOS antibody exerts a therapeutic benefit, and thus may be administered to a patient to reduce the dose of anti-ICOS antibody effective to treat cancer or a tumour in the patient. Thus, an anti-PD-L1 antibody may reduce the recommended or required

dosage of anti-ICOS antibody administration to that patient to, for example, 75%, 50%, 25%, 20%, 10% or less, compared with the dosage when anti-ICOS antibody is administered without anti-PD-L1. The patient may be treated by administration of anti-ICOS antibody and anti-PD-L1 antibody in a combination therapy as described herein.

As discussed in Example 22 herein, treatment with anti-PD-L1 antibody, especially antibody with effector positive Fc, appears not to increase the expression of ICOS on Teff cells. This is advantageous when administering such antibodies in combination with effector positive anti-ICOS antibodies, where an increase in ICOS expression on Teffs would undesirably render these cells more sensitive to depletion by the anti-ICOS antibody. In a combination with anti-PD-L1, anti-ICOS therapy may thus exploit a differential expression of ICOS on Teffs compared with Tregs, preferentially targeting the ICOS-high Tregs for depletion. This in turn relieves the suppression of TEffs and has a net effect of promoting the effector T cell response in a patient. The effect of targeting immune checkpoint molecules on expression of ICOS on T cells has also been studied previously—see Figure S6C in ref. (supplementary materials), where treatment with CTLA-4 antibody and/or anti-PD-1 antibody was reported to increase the percentage of CD4+ Tregs expressing ICOS. The effect of a therapeutic agent on ICOS expression in Tregs and Teffs may be a factor in selection of appropriate agents for use in combination with anti-ICOS antibodies, noting that effect of the anti-ICOS antibody may be enhanced under conditions where there is high differential expression of ICOS on Tregs versus Teffs.

As described herein, a single dose of anti-ICOS antibody may be sufficient to provide therapeutic effect, especially in combination with other therapeutic agents such as anti-PD-L1 antibody. In tumour therapy, the underlying rationale for this single dose benefit may be that the anti-ICOS antibody mediates its effect, at least in part, by resetting or altering the microenvironment of the tumour sufficiently to render the tumour more sensitive to immune attack and/or to the effects of other immunomodulators such as those mentioned. Tumour microenvironment resetting is triggered through for example depletion of ICOS positive tumour infiltrating Tregs. So, for example, a patient may be treated with a single dose of an anti-ICOS antibody followed by one or multiple doses of anti-PD-L1 antibody. Over a period of treatment, for example six months or a year, the anti-ICOS antibody may be administered in a single dose while other agents, e.g., anti-PD-L1 antibody, are optionally administered multiple times over that treatment period, preferably with at least one such dose being administered subsequent to treatment with the anti-ICOS antibody.

Further examples of combination therapy include combination of anti-ICOS antibody with:

- an antagonist of an adenosine A2A receptor (“A2AR inhibitor”);
- a CD137 agonist (e.g., agonist antibody);
- an antagonist of the enzyme indoleamine-2,3 dioxygenase, which catalyses the breakdown of tryptophan (“IDO inhibitor”). IDO is an immune checkpoint, activated in dendritic cells and macrophages, which contributes to immune suppression/tolerance.

Anti-ICOS antibodies may be used in combination therapy with IL-2 (e.g., recombinant IL-2 such as aldesleukin). The IL-2 may be administered at high dose (HD). Typical HD IL-2 therapy involves bolus infusion of over 500,000 IU/kg, e.g., bolus infusions of 600,000 or 720,000 IU/kg, per cycle of therapy, where 10-15 such bolus infusions are given at intervals of between 5-10 hours, e.g.,

up to 15 bolus infusions every 8 hours, and repeating the therapy cycle approximately every 14 to 21 days for up to 6 to 8 cycles. HD IL-2 therapy has been successful in treating tumours, especially melanoma (e.g., metastatic melanoma) and renal cell carcinoma, but its use is limited to the high toxicity of IL-2 which can cause severe adverse effects.

Treatment with high dose IL-2 has been shown to increase the population of ICOS-positive Tregs in cancer patients [31]. This increase in ICOS+ Tregs following the first cycle of HD IL-2 therapy was reported to correlate with worse clinical outcome—the higher the number of ICOS+Tregs, the worse the prognosis. An IL-2 variant F42K has been proposed as an alternative therapy to avoid this undesirable increase in ICOS+ Treg cells [32]. However, another approach would be to exploit the increase in ICOS+T regs by using an antibody in accordance with the present invention as a second-line therapeutic agent.

It may be beneficial to combine IL-2 therapy with anti-ICOS antibodies, capitalising on the ability of anti-ICOS antibodies to target Tregs that highly express ICOS, inhibiting these cells and improving the prognosis for patients undergoing IL-2 therapy. Concomitant administration of IL-2 and anti-ICOS antibody may increase the response rate while avoiding or reducing adverse events in the treated patient population. The combination may permit IL-2 to be used at lower dose compared with IL-2 monotherapy, reducing the risk or level of adverse events arising from the IL-2 therapy, while retaining or enhancing clinical benefit (e.g., reduction of tumour growth, clearance of solid tumour and/or reduction of metastasis). In this way, addition of anti-ICOS can improve treatment of patients who are receiving IL-2, whether high-dose (HD) or low-dose (LD) IL-2.

Accordingly, one aspect of the invention provides a method of treating a patient by administering an anti-ICOS antibody to the patient, wherein the patient is also treated with IL-2, e.g., HD IL-2. Another aspect of the invention is an anti-ICOS antibody for use in treating a patient, wherein the patient is also treated with IL-2, e.g., HD IL-2. The anti-ICOS antibody may be used as a second-line therapy. Thus, the patient may be one who has been treated with IL-2, e.g., having received at least one cycle of HD IL-2 therapy, and who has an increased level of ICOS+ Tregs. Assays may be performed on samples of cancer cells, e.g., tumour biopsy samples, using immunohistochemistry or FACS as described elsewhere herein to detect cells positive for ICOS, Foxp3, ICOSL and optionally one or more further markers of interest. Methods may comprise determining that the patient has an increased level of ICOS+ Tregs (e.g., in peripheral blood, or in a tumour biopsy) following IL-2 treatment, where an increased level is indicative that the patient would benefit from treatment with the anti-ICOS antibody. The increase in Tregs may be relative to control (untreated) individuals or to the patient prior to IL-2 therapy. Such patients with elevated Tregs represent a group who may not benefit from continued IL-2 treatment alone, but for whom a combination of anti-ICOS antibody and IL-2 therapy, or treatment with anti-ICOS antibody alone, offers therapeutic benefit. Thus, following a positive determination that the patient has an increased level of ICOS+ Tregs, anti-ICOS antibody and/or further IL-2 therapy may be administered. Treatment with the anti-ICOS antibody may selectively target and deplete the ICOS+ Tregs relative to other T cell populations in such patients. This provides a therapeutic effect by relieving the immunosuppression mediated by these cells and thereby enhancing activity of Tefs against the target cells, e.g., tumour cells or infected cells.

Combination therapy with anti-ICOS antibodies and IL-2 may be used for any therapeutic indication described herein, and particularly for treating a tumour, e.g., melanoma such as metastatic melanoma, or renal cell carcinoma. Thus, in one example, the patient treated with an anti-ICOS antibody is one who presents with metastatic melanoma and has been treated with IL-2, e.g., HD IL-2 therapy or LD IL-2 therapy.

In general, where an anti-ICOS antibody is administered to a patient who has received treatment with a first therapeutic agent (e.g., immunomodulator antibody) or other agent (e.g., IL-2), the anti-ICOS antibody may be administered after a minimum period of, for example, 24 hours, 48 hours, 72 hours, 1 week or 2 weeks following administration of the first therapeutic agent. The anti-ICOS antibody may be administered within 2, 3, 4 or 5 weeks after administration of the first therapeutic agent. This does not exclude additional administrations of either agent at any time, although it may be desirable to minimise the number of treatments administered, for ease of compliance for patients and to reduce costs. Rather, the relative timing of the administrations will be selected to optimise their combined effect, the first therapeutic agent creating an immunological environment (e.g., elevated ICOS+ Tregs, or antigen release as discussed below) in which the effect of the anti-ICOS antibody is especially advantageous. Thus, sequential administration of the first therapeutic agent and then the anti-ICOS antibody may allow time for the first agent to act, creating *in vivo* conditions in which the anti-ICOS antibody can exhibit its enhanced effect. Various administration regimens, including simultaneous or sequential combination treatments, are described herein and can be utilised as appropriate. Where the first therapeutic agent is one that increases the number of ICOS+ Tregs in the patient, the treatment regimen for the patient may comprise determining that the patient has an increased number of ICOS+ Tregs, and then administering the anti-ICOS antibody.

As noted, use of anti-ICOS antibodies in combination therapy may provide advantages of reducing the effective dose of the therapeutic agents and/or countering adverse effects of therapeutic agents that increase ICOS+ Tregs in patients. Yet further therapeutic benefits may be achieved through selecting a first therapeutic agent that causes release of antigens from target cells through “immunological cell death”, and administering the first therapeutic agent in combination with an anti-ICOS antibody. As noted, administration of the anti-ICOS antibody may sequentially follow administration of the first therapeutic agent, administration of the two agents being separated by a certain time window as discussed above.

Immunological cell death is a recognised mode of cell death, contrasting with apoptosis. It is characterised by release of ATP and HMGB1 from the cell and exposure of calreticulin on the plasma membrane [33, 34].

Immunological cell death in a target tissue or in target cells promotes engulfment of the cell by an antigen-presenting cell, resulting in display of antigens from the target cell, which in turn induces antigen-specific Tef cells. Anti-ICOS antibody may increase the magnitude and/or duration of the Tef response by acting as an agonist of ICOS on the Tef cells. In addition, where the anti-ICOS antibody is Fc effector function enabled (e.g., a human IgG1 antibody), the anti-ICOS antibody may cause depletion of antigen-specific Tregs. Thus, through a combination of either or both of these effects, the balance between Tef and Treg cells is modulated in favour of enhancing Tef activity. Combination of an anti-ICOS antibody with a treatment that induces immunological cell death in a target tissue or cell type, such as in a

tumour or in cancer cells, thereby promotes an immune response in the patient against the target tissue or cells, representing a form of vaccination in which the vaccine antigen is generated in vivo.

Accordingly, one aspect of the invention is a method of treating cancer in a patient by in vivo vaccination of the patient against their cancer cells. Another aspect of the invention is an anti-ICOS antibody for use in such a method. Anti-ICOS antibodies may be used in a method comprising:

treating the patient with a therapy that causes immunological cell death of the cancer cells, resulting in presentation of antigen to antigen-specific effector T cells, and administering an anti-ICOS antibody to the patient, wherein the anti-ICOS antibody enhances the antigen-specific effector T cell response against the cancer cells.

Treatments that induce immunological cell death include radiation (e.g., ionising irradiation of cells using UVC light or γ rays), chemotherapeutic agents (e.g., oxaliplatin, anthracyclines such as doxorubicin, idarubicin or mitoxantrone, BK channel agonists such as phloretin or pimarinic acid, bortezomib, cardiac glycosides, cyclophosphamide, GADD34/PP1 inhibitors with mitomycin, PDT with hypericin, polyinosinic-polycytidylic acid, 5-fluorouracil, gemcitabine, gefitinib, erlotinib, or thapsigargin with cisplatin) and antibodies to tumour-associated antigens. The tumour-associated antigen can be any antigen that is over-expressed by tumour cells relative to non-tumour cells of the same tissue, e.g., HER2, CD20, EGFR. Suitable antibodies include herceptin (anti-HER2), rituximab (anti-CD20), or cetuximab (anti-EGFR).

Thus, it is advantageous to combine an anti-ICOS antibody with one or more such treatments. Optionally, the anti-ICOS antibody is administered to a patient who has already received such treatment. The anti-ICOS antibody may be administered after a period of, for example, 24 hours, 48 hours, 72 hours, 1 week or 2 weeks following the treatment that induces immunological cell death, e.g., between 24 to 72 hours after the treatment. The anti-ICOS antibody may be administered within 2, 3, 4 or 5 weeks after the treatment. Other regimens for combination therapy are discussed elsewhere herein.

While "in vivo vaccination" has been described above, it is also possible to treat tumour cells to induce immunological cell death ex vivo, after which the cells may be reintroduced to the patient. Rather than administering the agent or treatment that induces immunological cell death directly to the patient, the treated tumour cells are administered to the patient. Treatment of the patient may be in accordance with administration regimens described above.

As already noted, a single dose of an anti-ICOS antibody may be sufficient to provide therapeutic benefit. Thus, in the methods of treatment described herein, the anti-ICOS antibody is optionally administered as a single dose. A single dose of anti-ICOS antibody may deplete Tregs in a patient, with consequent beneficial effects in diseases such as cancer. It has previously been reported that transient ablation of Tregs has anti-tumour effects, including reducing tumour progression, treating established tumours and metastases and extending survival, and that it can enhance the therapeutic effect of tumour irradiation [35]. Administration of a single dose of anti-ICOS may provide such Treg depletion, and may be used to enhance the effects of other therapeutic approaches used in combination, such as radiotherapy.

Antibodies to PD-L1

An antibody to PD-L1 for use in combination with an anti-ICOS antibody, whether as a separate therapeutic agent

or in a multispecific antibody as described herein, may comprise the antigen-binding site of any anti-PD-L1 antibody. Numerous examples of anti-PD-L1 antibodies are disclosed herein and others are known in the art. Characterisation data for many of the anti-PD-L1 antibodies mentioned here has been published in U.S. Pat. Nos. 9,567,399 and 9,617,338, both incorporated by reference herein.

1D05 has a heavy chain variable region (V_H) amino acid sequence of Seq ID No:33, comprising the CDRH1 amino acid sequence of Seq ID No:27 (IMGT) or Seq ID No:30 (Kabat), the CDRH2 amino acid sequence of Seq ID No:28 (IMGT) or Seq ID No:31 (Kabat), and the CDRH3 amino acid sequence of Seq ID No:29 (IMGT) or Seq ID No:32 (Kabat). The heavy chain nucleic acid sequence of the V_H domain is Seq ID No:34. 1D05 has a light chain variable region (V_L) amino acid sequence of Seq ID No:43, comprising the CDRL1 amino acid sequence of Seq ID No:37 (IMGT) or Seq ID No:40 (Kabat), the CDRL2 amino acid sequence of Seq ID No: 38 (IMGT) or Seq ID No:41 (Kabat), and the CDRL3 amino acid sequence of Seq ID No:39 (IMGT) or Seq ID No:42 (Kabat). The light chain nucleic acid sequence of the V_L domain is Seq ID No: 44. The V_H domain may be combined with any of the heavy chain constant region sequences described herein, e.g. Seq ID No: 193, Seq ID No: 195, Seq ID No:197, Seq ID No: 199, Seq ID No:201, Seq ID No:203, Seq ID No:205, Seq ID No:340, Seq ID No:524, Seq ID No: 526, Seq ID No:528, Seq ID No: 530, Seq ID No: 532 or Seq ID No: 534. The V_L domain may be combined with any of the light chain constant region sequences described herein, e.g. Seq ID Nos: 207, 209, 211, 213, 215, 217, 219, 221, 223, 225, 227, 229, 231, 233, 235, 237, 536 and 538. A full length heavy chain amino acid sequence is Seq ID No:35 (heavy chain nucleic acid sequence Seq ID No:36). A full length light chain amino acid sequence is Seq ID No: 45 (light chain nucleic acid sequence Seq ID No:46).

84G09 has a heavy chain variable (V_H) region amino acid sequence of Seq ID No:13, comprising the CDRH1 amino acid sequence of Seq ID No:7 (IMGT) or Seq ID No: 10 (Kabat), the CDRH2 amino acid sequence of Seq ID No:8 (IMGT) or Seq ID No:11 (Kabat), and the CDRH3 amino acid sequence of Seq ID No:9 (IMGT) or Seq ID No: 12 (Kabat). The heavy chain nucleic acid sequence of the V_H domain is Seq ID No:14. 84G09 has a light chain variable region (V_L) amino acid sequence of Seq ID No:23, comprising the CDRL1 amino acid sequence of Seq ID No:17 (IMGT) or Seq ID No:20 (Kabat), the CDRL2 amino acid sequence of Seq ID No: 18 (IMGT) or Seq ID No:21 (Kabat), and the CDRL3 amino acid sequence of Seq ID No:19 (IMGT) or Seq ID No:22 (Kabat). The light chain nucleic acid sequence of the V_L domain is Seq ID No: 24. The V_H domain may be combined with any of the heavy chain constant region sequences described herein, e.g. Seq ID No: 193, Seq ID No:195, Seq ID No: 197, Seq ID No: 199, Seq ID No:201, Seq ID No:203, Seq ID No:205, Seq ID No:340, Seq ID No:524, Seq ID No: 526, Seq ID No:528, Seq ID No:530, Seq ID No:532 or Seq ID No:534. The V_L domain may be combined with any of the light chain constant region sequences described herein, e.g. Seq ID Nos: 207, 209, 211, 213, 215, 217, 219, 221, 223, 225, 227, 229, 231, 233, 235, 237, 536 and 538. A full length heavy chain amino acid sequence is Seq ID No:15 (heavy chain nucleic acid sequence Seq ID No:16). A full length light chain amino acid sequence is Seq ID No:25 (light chain nucleic acid sequence Seq ID No:26).

1D05 HC mutant 1 has a heavy chain variable (V_H) region amino acid sequence of Seq ID No: 47, comprising the

ID05 LC mutant 2 has a heavy chain variable (V_H) region amino acid sequence of Seq ID No: 33, comprising the CDRH1 amino acid sequence of Seq ID No:27 (IMGT) or Seq ID No: 30 (Kabat), the CDRH2 amino acid sequence of Seq ID No:28 (IMGT) or Seq ID No:31 (Kabat), and the CDRH3 amino acid sequence of Seq ID No:29 (IMGT) or

Seq ID No:32 (Kabat). The heavy chain nucleic acid sequence of the V_H domain is Seq ID No:34. 1D05 LC mutant 2 has a light chain variable region (V_L) amino acid sequence of Seq ID No:51, comprising the CDRL1 amino acid sequence of Seq ID No:37 (IMGT) or Seq ID No:40 (Kabat), the CDRL2 amino acid sequence of Seq ID No:38 (IMGT) or Seq ID No:41 (Kabat), and the CDRL3 amino acid sequence of Seq ID No:39 (IMGT) or Seq ID No:42 (Kabat). The V_H domain may be combined with any of the heavy chain constant region sequences described herein, e.g. Seq ID No: 193, Seq ID No: 195, Seq ID No: 197, Seq ID No: 199, Seq ID No:201, Seq ID No:203, Seq ID No:205, Seq ID No:340, Seq ID No:524, Seq ID No:526, Seq ID No:528, Seq ID No:530, Seq ID No:532 or Seq ID No:534. The V_L domain may be combined with any of the light chain constant region sequences described herein, e.g. Seq ID Nos: 207, 209, 211, 213, 215, 217, 219, 221, 223, 225, 227, 229, 231, 233, 235, 237, 536 and 538. A full length heavy chain amino acid sequence is Seq ID No:35 (heavy chain nucleic acid sequence Seq ID No:36).

1D05 LC mutant 3 has a heavy chain variable (V_H) region amino acid sequence of Seq ID No: 33, comprising the CDRH1 amino acid sequence of Seq ID No:27 (IMGT) or Seq ID No: 30 (Kabat), the CDRH2 amino acid sequence of Seq ID No:28 (IMGT) or Seq ID No:31 (Kabat), and the CDRH3 amino acid sequence of Seq ID No:29 (IMGT) or Seq ID No:32 (Kabat). The heavy chain nucleic acid sequence of the V_H domain is Seq ID No:34. 1D05 LC mutant 3 has a light chain variable region (V_L) amino acid sequence of Seq ID No:298, comprising the CDRL1 amino acid sequence of Seq ID No:37 (IMGT) or Seq ID No:40 (Kabat), and the CDRL3 amino acid sequence of Seq ID No:39 (IMGT) or Seq ID No:42 (Kabat). The CDRL2 sequence of 1D05 LC Mutant 3 is as defined by the Kabat or IMGT systems from the V_L sequence of Seq ID No:298. The light chain nucleic acid sequence of the V_L domain is Seq ID No: 44. The V_H domain may be combined with any of the heavy chain constant region sequences described herein, e.g. Seq ID No: 193, Seq ID No: 195, Seq ID No: 197, Seq ID No: 199, Seq ID No:201, Seq ID No:203, Seq ID No:205 or Seq ID No:340, Seq ID No:524, Seq ID No: 526, Seq ID No:528, Seq ID No:530, Seq ID No:532 or Seq ID No:534. The V_L domain may be combined with any of the light chain constant region sequences described herein, e.g. Seq ID Nos: 207, 209, 211, 213, 215, 217, 219, 221, 223, 225, 227, 229, 231, 233, 235, 237, 536 and 538. A full length heavy chain amino acid sequence is Seq ID No:35 (heavy chain nucleic acid sequence Seq ID No:36). A full length light chain amino acid sequence is Seq ID No: 45 (light chain nucleic acid sequence Seq ID No:46).

411B08 has a heavy chain variable (V_H) region amino acid sequence of Seq ID No:58, comprising the CDRH1 amino acid sequence of Seq ID No:52 (IMGT) or Seq ID No:55 (Kabat), the CDRH2 amino acid sequence of Seq ID No:53 (IMGT) or Seq ID No:56 (Kabat), and the CDRH3 amino acid sequence of Seq ID No:54 (IMGT) or Seq ID No:57 (Kabat). The heavy chain nucleic acid sequence of the V_H domain is Seq ID No:59. 411B08 has a light chain variable region (V_L) amino acid sequence of Seq ID No:68, comprising the CDRL1 amino acid sequence of Seq ID No:62 (IMGT) or Seq ID No:65 (Kabat), the CDRL2 amino acid sequence of Seq ID No:63 (IMGT) or Seq ID No:66 (Kabat), and the CDRL3 amino acid sequence of Seq ID No: 64 (IMGT) or Seq ID No:67 (Kabat). The light chain nucleic acid sequence of the V_L domain is Seq ID No:69. The V_H domain may be combined with any of the heavy chain constant region sequences described herein, e.g. Seq ID No:

193, Seq ID No: 195, Seq ID No:197, Seq ID No: 199, Seq ID No:201, Seq ID No:203, Seq ID No:205, Seq ID No:340, Seq ID No:524, Seq ID No: 526, Seq ID No:528, Seq ID No:530, Seq ID No:532 or Seq ID No:534. The V_L domain may be combined with any of the light chain constant region sequences described herein, e.g. Seq ID Nos: 207, 209, 211, 213, 215, 217, 219, 221, 223, 225, 227, 229, 231, 233, 235, 237, 536 and 538. A full length heavy chain amino acid sequence is Seq ID No:60 (heavy chain nucleic acid sequence Seq ID No:61). A full length light chain amino acid sequence is Seq ID No:70 (light chain nucleic acid sequence Seq ID No:71).

411C04 has a heavy chain variable (V_H) region amino acid sequence of Seq ID No:78, comprising the CDRH1 amino acid sequence of Seq ID No:72 (IMGT) or Seq ID No:75 (Kabat), the CDRH2 amino acid sequence of Seq ID No:73 (IMGT) or Seq ID No:76 (Kabat), and the CDRH3 amino acid sequence of Seq ID No:74 (IMGT) or Seq ID No:77 (Kabat). The heavy chain nucleic acid sequence of the V_H domain is Seq ID No:79. 411C04 has a light chain variable region (V_L) amino acid sequence of Seq ID No:88, comprising the CDRL1 amino acid sequence of Seq ID No:82 (IMGT) or Seq ID No:85 (Kabat), the CDRL2 amino acid sequence of Seq ID No:83 (IMGT) or Seq ID No:86 (Kabat), and the CDRL3 amino acid sequence of Seq ID No: 84 (IMGT) or Seq ID No:87 (Kabat). The light chain nucleic acid sequence of the V_L domain is Seq ID No:89. The V_H domain may be combined with any of the heavy chain constant region sequences described herein, e.g. Seq ID No:193, Seq ID No: 195, Seq ID No:197, Seq ID No: 199, Seq ID No:201, Seq ID No:203, Seq ID No:205, Seq ID No:340, Seq ID No:524, Seq ID No: 526, Seq ID No:528, Seq ID No:530, Seq ID No:532 or Seq ID No:534. The V_L domain may be combined with any of the light chain constant region sequences described herein, e.g. Seq ID Nos: 207, 209, 211, 213, 215, 217, 219, 221, 223, 225, 227, 229, 231, 233, 235, 237, 536 and 538. A full length heavy chain amino acid sequence is Seq ID No:80 (heavy chain nucleic acid sequence Seq ID No:81). A full length light chain amino acid sequence is Seq ID No:90 (light chain nucleic acid sequence Seq ID No:91).

411D07 has a heavy chain variable (V_H) region amino acid sequence of Seq ID No:98, comprising the CDRH1 amino acid sequence of Seq ID No:92 (IMGT) or Seq ID No:95 (Kabat), the CDRH2 amino acid sequence of Seq ID No:93 (IMGT) or Seq ID No:96 (Kabat), and the CDRH3 amino acid sequence of Seq ID No:94 (IMGT) or Seq ID No:97 (Kabat). The heavy chain nucleic acid sequence of the V_H domain is Seq ID No:99. 411D07 has a light chain variable region (V_L) amino acid sequence of Seq ID No:108, comprising the CDRL1 amino acid sequence of Seq ID No: 102 (IMGT) or Seq ID No: 105 (Kabat), the CDRL2 amino acid sequence of Seq ID No:103 (IMGT) or Seq ID No:106 (Kabat), and the CDRL3 amino acid sequence of Seq ID No:104 (IMGT) or Seq ID No: 107 (Kabat). The light chain nucleic acid sequence of the V_L domain is Seq ID No: 109. The V_H domain may be combined with any of the heavy chain constant region sequences described herein, e.g. Seq ID No:193, Seq ID No: 195, Seq ID No: 197, Seq ID No:199, Seq ID No:201, Seq ID No:203, Seq ID No:205, Seq ID No:340, Seq ID No:524, Seq ID No:526, Seq ID No:528, Seq ID No:530, Seq ID No:532 or Seq ID No: 534. The V_L domain may be combined with any of the light chain constant region sequences described herein, e.g. Seq ID Nos: 207, 209, 211, 213, 215, 217, 219, 221, 223, 225, 227, 229, No: 100 (heavy chain nucleic acid sequence Seq ID

No:101). A full length light chain amino acid sequence is Seq ID No: 110 (light chain nucleic acid sequence Seq ID No:111).

385F01 has a heavy chain variable (V_H) region amino acid sequence of Seq ID No:118, comprising the CDRH1 amino acid sequence of Seq ID No:112 (IMGT) or Seq ID No:115 (Kabat), the CDRH2 amino acid sequence of Seq ID No:113 (IMGT) or Seq ID No:116 (Kabat), and the CDRH3 amino acid sequence of Seq ID No:114 (IMGT) or Seq ID No:117 (Kabat). The heavy chain nucleic acid sequence of the V_H domain is Seq ID No:119. 385F01 has a light chain variable region (V_L) amino acid sequence of Seq ID No:128, comprising the CDRL1 amino acid sequence of Seq ID No:122 (IMGT) or Seq ID No:125 (Kabat), the CDRL2 amino acid sequence of Seq ID No:123 (IMGT) or Seq ID No:126 (Kabat), and the CDRL3 amino acid sequence of Seq ID No:124 (IMGT) or Seq ID No:127 (Kabat). The light chain nucleic acid sequence of the V_L domain is Seq ID No:129. The V_H domain may be combined with any of the heavy chain constant region sequences described herein, e.g. Seq ID No: 193, Seq ID No: 195, Seq ID No:197, Seq ID No: 199, Seq ID No:201, Seq ID No:203, Seq ID No:205, Seq ID No:340, Seq ID No:524, Seq ID No:526, Seq ID No:528, Seq ID No:530, Seq ID No:532 or Seq ID No: 534. The V_L domain may be combined with any of the light chain constant region sequences described herein, e.g. Seq ID Nos: 207, 209, 211, 213, 215, 217, 219, 221, 223, 225, 227, 229, 231, 233, 235, 237, 536 and 538. A full length heavy chain amino acid sequence is Seq ID No: 120 (heavy chain nucleic acid sequence Seq ID No:121). A full length light chain amino acid sequence is Seq ID No:130 (light chain nucleic acid sequence Seq ID No:131).

386H03 has a heavy chain variable (V_H) region amino acid sequence of Seq ID No:158, comprising the CDRH1 amino acid sequence of Seq ID No:152 (IMGT) or Seq ID No:155 (Kabat), the CDRH2 amino acid sequence of Seq ID No:153 (IMGT) or Seq ID No:156 (Kabat), and the CDRH3 amino acid sequence of Seq ID No: 154 (IMGT) or Seq ID No: 157 (Kabat). The heavy chain nucleic acid sequence of the V_H domain is Seq ID No: 159. 386H03 has a light chain variable region (V_L) amino acid sequence of Seq ID No:168, comprising the CDRL1 amino acid sequence of Seq ID No: 162 (IMGT) or Seq ID No: 165 (Kabat), the CDRL2 amino acid sequence of Seq ID No: 163 (IMGT) or Seq ID No: 166 (Kabat), and the CDRL3 amino acid sequence of Seq ID No:164 (IMGT) or Seq ID No: 167 (Kabat). The light chain nucleic acid sequence of the V_L domain is Seq ID No: 169. The V_H domain may be combined with any of the heavy chain constant region sequences described herein, e.g. Seq ID No:193, Seq ID No: 195, Seq ID No: 197, Seq ID No: 199, Seq ID No:201, Seq ID No:203, Seq ID No:205, Seq ID No:340, Seq ID No:524, Seq ID No:526, Seq ID No:528, Seq ID No:530, Seq ID No:532 or Seq ID No: 534. The V_L domain may be combined with any of the light chain constant region sequences described herein, e.g. Seq ID Nos: 207, 209, 211, 213, 215, 217, 219, 221, 223, 225, 227, 229, No: 160 (heavy chain nucleic acid sequence Seq ID No:161). A full length light chain amino acid sequence is Seq ID No: 170 (light chain nucleic acid sequence Seq ID No:171).

389A03 has a heavy chain variable (V_H) region amino acid sequence of Seq ID No:178, comprising the CDRH1 amino acid sequence of Seq ID No:172 (IMGT) or Seq ID No:175 (Kabat), the CDRH2 amino acid sequence of Seq ID No:173 (IMGT) or Seq ID No:176 (Kabat), and the CDRH3 amino acid sequence of Seq ID No:174 (IMGT) or Seq ID No:177 (Kabat). The heavy chain nucleic acid sequence of

the V_H domain is Seq ID No:179. 389A03 has a light chain variable region (V_L) amino acid sequence of Seq ID No:188, comprising the CDRL1 amino acid sequence of Seq ID No: 182 (IMGT) or Seq ID No:185 (Kabat), the CDRL2 amino acid sequence of Seq ID No: 183 (IMGT) or Seq ID No:186 (Kabat), and the CDRL3 amino acid sequence of Seq ID No: 184 (IMGT) or Seq ID No: 187 (Kabat). The light chain nucleic acid sequence of the V_L domain is Seq ID No:189. The V_H domain may be combined with any of the heavy chain constant region sequences described herein, e.g. Seq ID No:193, Seq ID No:195, Seq ID No:197, Seq ID No: 199, Seq ID No:201, Seq ID No:203, Seq ID No:205, Seq ID No:340, Seq ID No:524, Seq ID No:526, Seq ID No:528, Seq ID No:530, Seq ID No:532 or Seq ID No: 534. The V_L domain may be combined with any of the light chain constant region sequences described herein, e.g. Seq ID Nos: 207, 209, 211, 213, 215, 217, 219, 221, 223, 225, 227, 229, 231, 233, 235, 237, 536 and 538. A full length heavy chain amino acid sequence is Seq ID No: 180 (heavy chain nucleic acid sequence Seq ID No: 181). A full length light chain amino acid sequence is Seq ID No:190 (light chain nucleic acid sequence Seq ID No:191).

413D08 has a heavy chain variable (V_H) region amino acid sequence of Seq ID No:138, comprising the CDRH1 amino acid sequence of Seq ID No:132 (IMGT) or Seq ID No:135 (Kabat), the CDRH2 amino acid sequence of Seq ID No:133 (IMGT) or Seq ID No: 136 (Kabat), and the CDRH3 amino acid sequence of Seq ID No:134 (IMGT) or Seq ID No:137 (Kabat). The heavy chain nucleic acid sequence of the V_H domain is Seq ID No:139. 413D08 has a light chain variable region (V_L) amino acid sequence of Seq ID No:148, comprising the CDRL1 amino acid sequence of Seq ID No:142 (IMGT) or Seq ID No: 145 (Kabat), the CDRL2 amino acid sequence of Seq ID No:143 (IMGT) or Seq ID No:146 (Kabat), and the CDRL3 amino acid sequence of Seq ID No:144 (IMGT) or Seq ID No: 147 (Kabat). The light chain nucleic acid sequence of the V_L domain is Seq ID No: 149. The V_H domain may be combined with any of the heavy chain constant region sequences described herein, e.g. Seq ID No: 193, Seq ID No: 195, Seq ID No: 197, Seq ID No:199, Seq ID No:201, Seq ID No:203, Seq ID No:205, Seq ID No:340, Seq ID No:524, Seq ID No:526, Seq ID No:528, Seq ID No:530, Seq ID No:532 or Seq ID No: 534. The V_L domain may be combined with any of the light chain constant region sequences described herein, e.g. Seq ID Nos: 207, 209, 211, 213, 215, 217, 219, 221, 223, 225, 227, 229, 231, 233, 235, 237, 536 and 538. A full length heavy chain amino acid sequence is Seq ID No: 140 (heavy chain nucleic acid sequence Seq ID No:141). A full length light chain amino acid sequence is Seq ID No: 150 (light chain nucleic acid sequence Seq ID No:151).

413G05 has a heavy chain variable (V_H) region amino acid sequence of Seq ID No:244, comprising the CDRH1 amino acid sequence of Seq ID No:238 (IMGT) or Seq ID No:241 (Kabat), the CDRH2 amino acid sequence of Seq ID No:239 (IMGT) or Seq ID No:242 (Kabat), and the CDRH3 amino acid sequence of Seq ID No:240 (IMGT) or Seq ID No:243 (Kabat). The heavy chain nucleic acid sequence of the V_H domain is Seq ID No:245. 413G05 has a light chain variable region (V_L) amino acid sequence of Seq ID No:254, comprising the CDRL1 amino acid sequence of Seq ID No:248 (IMGT) or Seq ID No:251 (Kabat), the CDRL2 amino acid sequence of Seq ID No:249 (IMGT) or Seq ID No:252 (Kabat), and the CDRL3 amino acid sequence of Seq ID No:250 (IMGT) or Seq ID No:253 (Kabat). The light chain nucleic acid sequence of the V_L domain is Seq ID No:255. The V_H domain may be combined with any of the

heavy chain constant region sequences described herein, e.g. Seq ID No:193, Seq ID No:195, Seq ID No:197, Seq ID No:199, Seq ID No:201, Seq ID No:203, Seq ID No:205, Seq ID No:340, Seq ID No:524, Seq ID No:526, Seq ID No:528, Seq ID No:530, Seq ID No:532 or Seq ID No: 534. The V_L domain may be combined with any of the light chain constant region sequences described herein, e.g. Seq ID Nos: 207, 209, 211, 213, 215, 217, 219, 221, 223, 225, 227, 229, 231, 233, 235, 237, 536 and 538. A full length heavy chain amino acid sequence is Seq ID No: 246 (heavy chain nucleic acid sequence Seq ID No:247). A full length light chain amino acid sequence is Seq ID No:256 (light chain nucleic acid sequence Seq ID No:257).

413F09 has a heavy chain variable (V_H) region amino acid sequence of Seq ID No:264, comprising the CDRH1 amino acid sequence of Seq ID No:258 (IMGT) or Seq ID No:261 (Kabat), the CDRH2 amino acid sequence of Seq ID No:259 (IMGT) or Seq ID No:262 (Kabat), and the CDRH3 amino acid sequence of Seq ID No:260 (IMGT) or Seq ID No:263 (Kabat). The heavy chain nucleic acid sequence of the V_H domain is Seq ID No:265. 413F09 has a light chain variable region (V_L) amino acid sequence of Seq ID No:274, comprising the CDRL1 amino acid sequence of Seq ID No:268 (IMGT) or Seq ID No:271 (Kabat), the CDRL2 amino acid sequence of Seq ID No:269 (IMGT) or Seq ID No:272 (Kabat), and the CDRL3 amino acid sequence of Seq ID No:270 (IMGT) or Seq ID No:273 (Kabat). The light chain nucleic acid sequence of the V_L domain is Seq ID No:275. The V_H domain may be combined with any of the heavy chain constant region sequences described herein, e.g. Seq ID No:193, Seq ID No: 195, Seq ID No: 197, Seq ID No: 199, Seq ID No:201, Seq ID No:203, Seq ID No:205, Seq ID No:340, Seq ID No:524, Seq ID No:526, Seq ID No:528, Seq ID No:530, Seq ID No:532 or Seq ID No: 534. The V_L domain may be combined with any of the light chain constant region sequences described herein, e.g. Seq ID Nos: 207, 209, 211, 213, 215, 217, 219, 221, 223, 225, 227, 229, No: 266 (heavy chain nucleic acid sequence Seq ID No:267). A full length light chain amino acid sequence is Seq ID No:276 (light chain nucleic acid sequence Seq ID No:277).

414B06 has a heavy chain variable (V_H) region amino acid sequence of Seq ID No:284, comprising the CDRH1 amino acid sequence of Seq ID No:278 (IMGT) or Seq ID No:281 (Kabat), the CDRH2 amino acid sequence of Seq ID No:279 (IMGT) or Seq ID No:282 (Kabat), and the CDRH3 amino acid sequence of Seq ID No:280 (IMGT) or Seq ID No:283 (Kabat). The heavy chain nucleic acid sequence of the V_H domain is Seq ID No:285. 414B06 has a light chain variable region (V_L) amino acid sequence of Seq ID No:294, comprising the CDRL1 amino acid sequence of Seq ID No:288 (IMGT) or Seq ID No:291 (Kabat), the CDRL2 amino acid sequence of Seq ID No:289 (IMGT) or Seq ID No:292 (Kabat), and the CDRL3 amino acid sequence of Seq ID No:290 (IMGT) or Seq ID No:293 (Kabat). The light chain nucleic acid sequence of the V_L domain is Seq ID No:295. The V_H domain may be combined with any of the heavy chain constant region sequences described herein, e.g. Seq ID No: 193, Seq ID No:195, Seq ID No:197, Seq ID No: 199, Seq ID No:201, Seq ID No:203, Seq ID No:205, Seq ID No:340, Seq ID No:524, Seq ID No:526, Seq ID No:528, Seq ID No:530, Seq ID No:532 or Seq ID No: 534. The V_L domain may be combined with any of the light chain constant region sequences described herein, e.g. Seq ID Nos: 207, 209, 211, 213, 215, 217, 219, 221, 223, 225, 227, 229, 231, 233, 235, 237, 536 and 538. A full length heavy chain amino acid sequence is Seq ID No: 286 (heavy chain

nucleic acid sequence Seq ID No:287). A full length light chain amino acid sequence is Seq ID No:296 (light chain nucleic acid sequence Seq ID No:297).

416E01 has a heavy chain variable region (V_H) amino acid sequence of Seq ID No:349, comprising the CDRH1 amino acid sequence of Seq ID No:343 (IMGT) or Seq ID No:346 (Kabat), the CDRH2 amino acid sequence of Seq ID No:344 (IMGT) or Seq ID No:347 (Kabat), and the CDRH3 amino acid sequence of Seq ID No:345 (IMGT) or Seq ID No:348 (Kabat). The heavy chain nucleic acid sequence of the V_H domain is Seq ID No:350. 416E01 has a light chain variable region (V_L) amino acid sequence of Seq ID No:359, comprising the CDRL1 amino acid sequence of Seq ID No:353 (IMGT) or Seq ID No:356 (Kabat), the CDRL2 amino acid sequence of Seq ID No:354 (IMGT) or Seq ID No:357 (Kabat), and the CDRL3 amino acid sequence of Seq ID No:355 (IMGT) or Seq ID No:358 (Kabat). The light chain nucleic acid sequence of the V_L domain is Seq ID No:360. The V_H domain may be combined with any of the heavy chain constant region sequences described herein, e.g. Seq ID No:193, Seq ID No: 195, Seq ID No: 197, Seq ID No:199, Seq ID No:201, Seq ID No:203, Seq ID No:205, Seq ID No:340, Seq ID No:524, Seq ID No:526, Seq ID No:528, Seq ID No:530, Seq ID No:532 or Seq ID No: 534. The V_L domain may be combined with any of the light chain constant region sequences described herein, e.g. Seq ID Nos: 207, 209, 211, 213, 215, 217, 219, 221, 223, 225, 227, 229, No: 351 (heavy chain nucleic acid sequence Seq ID No:352). A full length light chain amino acid sequence is Seq ID No:361 (light chain nucleic acid sequence Seq ID No:362).

Antibody-Drug Conjugates

Anti-ICOS antibodies can be used as carriers of cytotoxic agents, to target Tregs. As reported in Example 18, Tregs located in the tumour microenvironment (TME) strongly express ICOS. ICOS is more strongly expressed on intratumoural Tregs than on intratumoural Tefs or peripheral Tregs. Thus, anti-ICOS antibodies labelled with a toxic drug or pro-drug may preferentially target Tregs in the TME to deliver the toxic payload, selectively inhibiting those cells. Such targeting of cytotoxic agents provides an additional route to removing the immune suppressive effect of Tregs, thereby altering the Treg:Teff balance in favour of Teff activity and may be used as an alternative to, or in combination with, any one or more of the other therapeutic approaches discussed herein (e.g., Fc effector-mediated inhibition of Tregs, agonism of effector T cells).

Accordingly, the invention provides an anti-ICOS antibody that is conjugated to a cytotoxic drug or pro-drug. In the case of a pro-drug, the pro-drug is activatable in the TME or other target site of therapeutic activity to generate the cytotoxic agent. Activation may be in response to a trigger such as photoactivation, e.g., using near-infrared light to activate a photoabsorber conjugate [36]. Spatially-selective activation of a pro-drug further enhances the cytotoxic effect of the antibody-drug conjugate, combining with the high ICOS expression on intratumoural Tregs to provide a cytotoxic effect that is highly selective for these cells.

For use in an antibody-drug conjugate, the cytotoxic drug or pro-drug is preferably non-immunogenic and non-toxic (dormant or inactive) during circulation of the antibody-drug conjugate in the blood. Preferably the cytotoxic drug (or the pro-drug, when activated) is potent—e.g., two to four molecules of the drug may be sufficient to kill the target cell. A photoactivatable pro-drug is silicaphthalocyanine dye (IRDye 700 DX), which induces lethal damage to the cell membrane after near-infrared light exposure. Cytotoxic drugs include

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anti-mitotic agents such as monomethyl auristatin E and microtubule inhibitors such as maytansine derivatives, e.g., mertansine, DM1, emtansine.

Conjugation of the drug (or pro-drug) to the antibody will usually be via a linker. The linker may be a cleavable linker, e.g., disulphide, hydrazone or peptide link. Cathepsin-cleavable linkers may be used, so that the drug is released by cathepsin in tumour cells. Alternatively, non-cleavable linkers can be used, e.g., thioether linkage. Additional attachment groups and/or spacers may also be included.

The antibody in the antibody-drug conjugate may be an antibody fragment, such as Fab'2 or other antigen-binding fragment as described herein, as the small size of such fragments may assist penetration to the tissue site (e.g., solid tumour).

An anti-ICOS antibody according to the present invention may be provided as an immunocytokine. Anti-ICOS antibodies may also be administered with immunocytokines in combination therapy. A number of examples of antibodies are described herein for use in combination therapy with anti-ICOS, and any of these (e.g., an anti-PD-L1 antibody) may be provided as immunocytokines for use in the present invention. An immunocytokine comprises an antibody molecule conjugated to a cytokine, such as IL-2. Anti-ICOS: IL-2 conjugates and anti-PD-L1: IL-2 conjugates are thus further aspects of the present invention.

An IL-2 cytokine may have activity at the high ($\alpha\beta\gamma$) affinity IL-2 receptor and/or the intermediate affinity ($\alpha\beta$) IL-2 receptor. IL-2 as used in an immunocytokine may be human wild type IL-2 or a variant IL-2 cytokine having one or more amino acid deletions, substitutions or additions, e.g., IL-2 having a 1 to 10 amino acid deletion at the N-terminus. Other IL-2 variants include mutations R38A or R38Q.

An example anti-PD-L1 immunocytokine comprises an immunoglobulin heavy chain and an immunoglobulin light chain, wherein the heavy chain comprises in N- to C-terminal direction:

- a) A V_H domain comprising CDRH1, CDRH2 and CDRH3; and
 - b) A heavy chain constant region;
- and wherein the light chain comprises in N- to C-terminal direction:
- c) A V_L domain comprising CDRL1, CDRL2 and CDRL3;
 - d) A light chain constant region, (CL);
 - e) Optionally, a linker, (L); and
 - f) An IL-2 cytokine;

wherein the V_H domain and V_L domain are comprised by an antigen-binding site that specifically binds to human PD-L1; and

wherein the immunocytokine comprises a V_H domain which comprises a CDRH3 comprising the motif $X_1\text{GSGX}_2\text{YGX}_3\text{X}_4\text{FD}$ (SEQ ID NO: 609), wherein X_1 , X_2 and X_3 are independently any amino acid, and X_4 is either present or absent, and if present, may be any amino acid.

The V_H and V_L domain may be the V_H and V_L domain of any anti-PD-L1 antibody mentioned herein, e.g., the 1D05 V_H and V_L domains.

The IL-2 may be human wild type or variant IL-2.

Vaccination

Anti-ICOS antibodies may be provided in vaccine compositions or co-administered with vaccines preparations. ICOS is involved in T follicular helper cell formation and the germinal centre reaction [37]. Agonist ICOS antibodies thus have potential clinical utility as molecular adjuvants to

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enhance vaccine efficacy. The antibodies may be used to increase protective efficacy of numerous vaccines, such as those against hepatitis B, malaria, HIV.

In the context of vaccination, the anti-ICOS antibody will generally be one that lacks Fc effector function, and thus does not mediate ADCC, CDC or ADCP. The antibody may be provided in a format lacking an Fc region, or having an effector null constant region. Optionally, an anti-ICOS antibody may have a heavy chain constant region that binds one or more types of Fc receptor but does not induce ADCC, CDC or ADCP activity, or that exhibits lower ADCC, CDC and ADCP activity compared with wild type human IgG1. Such a constant region may be unable to bind, or may bind with lower affinity, the particular Fc receptor(s) responsible for triggering ADCC, CDC or ADCP activity. Alternatively, where cellular effector functions are acceptable or desirable in the context of the vaccination, the anti-ICOS antibody may comprise a heavy chain constant region that is Fc effector function positive. Any of IgG1, IgG4 and IgG4.PE formats may for instance be used for anti-ICOS antibodies in vaccination regimens, and other examples of suitable isotypes and antibody constant regions are set out in more detail elsewhere herein.

Formulations and Administration

Antibodies may be monoclonal or polyclonal, but are preferably provided as monoclonal antibodies for therapeutic use. They may be provided as part of a mixture of other antibodies, optionally including antibodies of different binding specificity.

Antibodies according to the invention, and encoding nucleic acid, will usually be provided in isolated form. Thus, the antibodies, V_H and/or V_L domains, and nucleic acids may be provided purified from their natural environment or their production environment. Isolated antibodies and isolated nucleic acid will be free or substantially free of material with which they are naturally associated, such as other polypeptides or nucleic acids with which they are found in vivo, or the environment in which they are prepared (e.g., cell culture) when such preparation is by recombinant DNA technology in vitro. Optionally an isolated antibody or nucleic acid (1) is free of at least some other proteins with which it would normally be found, (2) is essentially free of other proteins from the same source, e.g., from the same species, (3) is expressed by a cell from a different species, (4) has been separated from at least about 50 percent of polynucleotides, lipids, carbohydrates, or other materials with which it is associated in nature, (5) is operably associated (by covalent or noncovalent interaction) with a polypeptide with which it is not associated in nature, or (6) does not occur in nature.

Antibodies or nucleic acids may be formulated with diluents or adjuvants and still for practical purposes be isolated—for example they may be mixed with carriers if used to coat microtitre plates for use in immunoassays, and may be mixed with pharmaceutically acceptable carriers or diluents when used in therapy. As described elsewhere herein, other active ingredients may also be included in therapeutic preparations. Antibodies may be glycosylated, either naturally in vivo or by systems of heterologous eukaryotic cells such as CHO cells, or they may be (for example if produced by expression in a prokaryotic cell) unglycosylated. The invention encompasses antibodies having a modified glycosylation pattern. In some applications, modification to remove undesirable glycosylation sites may be useful, or e.g., removal of a fucose moiety to increase ADCC function [38]. In other applications, modification of galactosylation can be made in order to modify CDC.

Typically, an isolated product constitutes at least about 5%, at least about 10%, at least about 25%, or at least about 50% of a given sample. An antibody may be substantially free from proteins or polypeptides or other contaminants that are found in its natural or production environment that would interfere with its therapeutic, diagnostic, prophylactic, research or other use.

An antibody may have been identified, separated and/or recovered from a component of its production environment (eg, naturally or recombinantly). The isolated antibody may be free of association with all other components from its production environment, eg, so that the antibody has been isolated to an FDA-approvable or approved standard. Contaminant components of its production environment, such as that resulting from recombinant transfected cells, are materials that would typically interfere with research, diagnostic or therapeutic uses for the antibody, and may include enzymes, hormones, and other proteinaceous or non-proteinaceous solutes. In some embodiments, the antibody will be purified: (1) to greater than 95% by weight of antibody as determined by, for example, the Lowry method, and in some embodiments, to greater than 99% by weight; (2) to a degree sufficient to obtain at least 15 residues of N-terminal or internal amino acid sequence by use of a spinning cup sequenator, or (3) to homogeneity by SDS-PAGE under non-reducing or reducing conditions using Coomassie blue or silver stain. Isolated antibody includes the antibody in situ within recombinant cells since at least one component of the antibody's natural environment will not be present. Ordinarily, however, an isolated antibody or its encoding nucleic acid will be prepared by at least one purification step.

The invention provides therapeutic compositions comprising the antibodies described herein. Therapeutic compositions comprising nucleic acid encoding such antibodies are also provided. Encoding nucleic acids are described in more detail elsewhere herein and include DNA and RNA, e.g., mRNA. In therapeutic methods described herein, use of nucleic acid encoding the antibody, and/or of cells containing such nucleic acid, may be used as alternatives (or in addition) to compositions comprising the antibody itself. Cells containing nucleic acid encoding the antibody, optionally wherein the nucleic acid is stably integrated into the genome, thus represent medicaments for therapeutic use in a patient. Nucleic acid encoding the anti-ICOS antibody may be introduced into human B lymphocytes, optionally B lymphocytes derived from the intended patient and modified ex vivo. Optionally, memory B cells are used. Administration of cells containing the encoding nucleic acid to the patient provides a reservoir of cells capable of expressing the anti-ICOS antibody, which may provide therapeutic benefit over a longer term compared with administration of isolated nucleic acid or isolated antibody.

Compositions may contain suitable carriers, excipients, and other agents that are incorporated into formulations to provide improved transfer, delivery, tolerance, and the like. A multitude of appropriate formulations can be found in the formulary known to all pharmaceutical chemists: Remington's Pharmaceutical Sciences, Mack Publishing Company, Easton, Pa. These formulations include, for example, powders, pastes, ointments, jellies, waxes, oils, lipids, lipid (cationic or anionic) containing vesicles (such as LIPO-FECTINT™), DNA conjugates, anhydrous absorption pastes, oil-in-water and water-in-oil emulsions, emulsions carbowax (polyethylene glycols of various molecular weights), semi-solid gels, and semi-solid mixtures containing carbowax. See also Powell et al. "Compendium of excipients for parenteral formulations" PDA (1998) J Pharm

Sci Technol 52:238-311. Compositions may comprise the antibody or nucleic acid in combination with medical injection buffer and/or with adjuvant.

Antibodies, or their encoding nucleic acids, may be formulated for the desired route of administration to a patient, e.g., in liquid (optionally aqueous solution) for injection. Various delivery systems are known and can be used to administer the pharmaceutical composition of the invention. Methods of introduction include, but are not limited to, intradermal, intramuscular, intraperitoneal, intravenous, subcutaneous, intranasal, epidural, and oral routes. Formulating antibodies for subcutaneous administration typically requires concentrating them into a smaller volume compared with intravenous preparations. The high potency of antibodies according to the present invention may lend them to use at sufficiently low doses to make subcutaneous formulation practical, representing an advantage compared with less potent anti-ICOS antibodies.

The composition may be administered by any convenient route, for example by infusion or bolus injection, by absorption through epithelial or mucocutaneous linings (e.g., oral mucosa, rectal and intestinal mucosa, etc.) and may be administered together with other biologically active agents. Administration can be systemic or local.

The pharmaceutical composition can be also delivered in a vesicle, in particular a liposome (see Langer (1990) Science 249:1527-1533; Treat et al. (1989) in Liposomes in the Therapy of Infectious Disease and Cancer, Lopez Berestein and Fidler (eds.), Liss, New York, pp. 353-365; Lopez-Berestein, *ibid.*, pp. 317-327; see generally *ibid.*).

In certain situations, the pharmaceutical composition can be delivered in a controlled release system. In one embodiment, a pump may be used (see Langer, *supra*; Sefton (1987) CRC Crit. Ref. Biomed. Eng. 14:201). In another embodiment, polymeric materials can be used; see, Medical Applications of Controlled Release, Langer and Wise (eds.), CRC Pres., Boca Raton, Fla. (1974). In yet another embodiment, a controlled release system can be placed in proximity of the composition's target, thus requiring only a fraction of the systemic dose (see, e.g., Goodson, in Medical Applications of Controlled Release, *supra*, vol. 2, pp. 115-138, 1984).

The injectable preparations may include dosage forms for intravenous, subcutaneous, intracutaneous and intramuscular injections, drip infusions, etc. These injectable preparations may be prepared by methods publicly known. For example, the injectable preparations may be prepared, e.g., by dissolving, suspending or emulsifying the antibody or its salt described above in a sterile aqueous medium or an oily medium conventionally used for injections. As the aqueous medium for injections, there are, for example, physiological saline, an isotonic solution containing glucose and other auxiliary agents, etc., which may be used in combination with an appropriate solubilizing agent such as an alcohol (e.g., ethanol), a polyalcohol (e.g., propylene glycol, polyethylene glycol), a nonionic surfactant [e.g., polysorbate 80, HCO-50 (polyoxyethylene (50 mol) adduct of hydrogenated castor oil)], etc. As the oily medium, there are employed, e.g., sesame oil, soybean oil, etc., which may be used in combination with a solubilizing agent such as benzyl benzoate, benzyl alcohol, etc. The injection thus prepared can be filled in an appropriate ampoule. A pharmaceutical composition of the present invention can be delivered subcutaneously or intravenously with a standard needle and syringe. It is envisaged that treatment will not be restricted to use in the clinic. Therefore, subcutaneous injection using a needle-free device is also advantageous. With respect to subcutaneous delivery, a pen delivery device readily has applications in

delivering a pharmaceutical composition of the present invention. Such a pen delivery device can be reusable or disposable. A reusable pen delivery device generally utilizes a replaceable cartridge that contains a pharmaceutical composition. Once all of the pharmaceutical composition within the cartridge has been administered and the cartridge is empty, the empty cartridge can readily be discarded and replaced with a new cartridge that contains the pharmaceutical composition. The pen delivery device can then be reused. In a disposable pen delivery device, there is no replaceable cartridge. Rather, the disposable pen delivery device comes prefilled with the pharmaceutical composition held in a reservoir within the device. Once the reservoir is emptied of the pharmaceutical composition, the entire device is discarded. Numerous reusable pen and autoinjector delivery devices have applications in the subcutaneous delivery of a pharmaceutical composition of the present invention. Examples include, but certainly are not limited to AUTOPEN™ (Owen Mumford, Inc., Woodstock, UK), DISETRONIC™ pen (Disetronic Medical Systems, Burghdorf, Switzerland), HUMALOG MIX 75/25™ pen, HUMALOG™ pen, HUMALIN 70/30™ pen (Eli Lilly and Co., Indianapolis, Ind.), NOVOPEN™ I, II and III (Novo Nordisk, Copenhagen, Denmark), NOVOPEN JUNIOR™ (Novo Nordisk, Copenhagen, Denmark), BD™ pen (Becton Dickinson, Franklin Lakes, N.J.), OPTIPEN™, OPTIPEN PRO™, OPTIPEN STARLET™, and OPTICLIK™ (Sanofi-Aventis, Frankfurt, Germany), to name only a few. Examples of disposable pen delivery devices having applications in subcutaneous delivery of a pharmaceutical composition of the present invention include, but certainly are not limited to the SOLOSTAR™ pen (Sanofi-Aventis), the FLEXPEN™ (Novo Nordisk), and the KWIKPEN™ (Eli Lilly).

Advantageously, the pharmaceutical compositions for oral or parenteral use described above are prepared into dosage forms in a unit dose suited to fit a dose of the active ingredients. Such dosage forms in a unit dose include, for example, tablets, pills, capsules, injections (ampoules), suppositories, etc. The amount of the aforesaid antibody contained is generally about 5 to about 500 mg per dosage form in a unit dose; especially in the form of injection, the aforesaid antibody may be contained in about 5 to about 100 mg and in about 10 to about 250 mg for the other dosage forms.

The antibody, nucleic acid, or composition comprising it, may be contained in a medical container such as a vial, syringe, IV container or an injection device. In an example, the antibody, nucleic acid or composition is in vitro, and may be in a sterile container. In an example, a kit is provided comprising the antibody, packaging and instructions for use in a therapeutic method as described herein.

One aspect of the invention is a composition comprising an antibody or nucleic acid of the invention and one or more pharmaceutically acceptable excipients, examples of which are listed above. "Pharmaceutically acceptable" refers to approved or approvable by a regulatory agency of the USA Federal or a state government or listed in the U.S. Pharmacopeia or other generally recognized pharmacopeia for use in animals, including humans. A pharmaceutically acceptable carrier, excipient, or adjuvant can be administered to a patient, together with an agent, e.g., any antibody or antibody chain described herein, and does not destroy the pharmacological activity thereof and is nontoxic when administered in doses sufficient to deliver a therapeutic amount of the agent.

In some embodiments, an anti-ICOS antibody will be the sole active ingredient in a composition according to the present invention. Thus, a composition may consist of the antibody or it may consist of the antibody with one or more pharmaceutically acceptable excipients. However, compositions according to the present invention optionally include one or more additional active ingredients. Detailed description of agents with which the anti-ICOS antibodies may be combined is provided elsewhere herein. Optionally, compositions contain multiple antibodies (or encoding nucleic acids) in a combined preparation, e.g., a single formulation comprising the anti-ICOS antibody and one or more other antibodies. Other therapeutic agents that it may be desirable to administer with antibodies or nucleic acids according to the present invention include analgesic agents. Any such agent or combination of agents may be administered in combination with, or provided in compositions with antibodies or nucleic acids according to the present invention, whether as a combined or separate preparation. The antibody or nucleic acid according to the present invention may be administered separately and sequentially, or concurrently and optionally as a combined preparation, with another therapeutic agent or agents such as those mentioned.

Anti-ICOS antibodies for use in a particular therapeutic indication may be combined with the accepted standard of care. Thus, for anti-cancer treatment, the antibody therapy may be employed in a treatment regimen that also includes chemotherapy, surgery and/or radiation therapy for example. Radiotherapy may be single dose or in fractionated doses, either delivered to affected tissues directly or to the whole body.

Multiple compositions can be administered separately or simultaneously. Separate administration refers to the two compositions being administered at different times, e.g. at least 10, 20, 30, or 10-60 minutes apart, or 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12 hours apart. One can also administer compositions at 24 hours apart, or even longer apart. Alternatively, two or more compositions can be administered simultaneously, e.g. less than 10 or less than 5 minutes apart. Compositions administered simultaneously can, in some aspects, be administered as a mixture, with or without similar or different time release mechanism for each of the components.

Antibodies, and their encoding nucleic acids, can be used as therapeutic agents. Patients herein are generally mammals, typically humans. An antibody or nucleic acid may be administered to a mammal, e.g., by any route of administration mentioned herein.

Administration is normally in a "therapeutically effective amount", this being an amount that produces the desired effect for which it is administered, sufficient to show benefit to a patient. The exact amount will depend on the purpose of the treatment, and will be ascertainable by one skilled in the art using known techniques (see, for example, Lloyd (1999) The Art, Science and Technology of Pharmaceutical Compounding). Prescription of treatment, e.g. decisions on dosage etc, is within the responsibility of general practitioners and other medical doctors and may depend on the severity of the symptoms and/or progression of a disease being treated. A therapeutically effective amount or suitable dose of antibody or nucleic acid can be determined by comparing its in vitro activity and in vivo activity in an animal model. Methods for extrapolation of effective dosages in mice and other test animals to humans are known.

As indicated by the *in vivo* studies described in the Examples herein, anti-ICOS antibody may be effective at a range of doses. Pharmacodynamic studies are reported in Example 24.

Anti-ICOS antibodies may be administered in an amount in one of the following ranges per dose:

- about 10 µg/kg body weight to about 100 mg/kg body weight,
- about 50 µg/kg body weight to about 5 mg/kg body weight,
- about 100 µg/kg body weight to about 10 mg/kg body weight,
- about 100 µg/kg body weight to about 20 mg/kg body weight,
- about 0.5 mg/kg body weight to about 20 mg/kg body weight, or
- about 5 mg/kg body weight or lower, for example less than 4, less than 3, less than 2, or less than 1 mg/kg of the antibody.

An optimal therapeutic dose may be between 0.1 and 0.5 mg/kg in a human, for example about 0.1 mg/kg, 0.15 mg/kg, 0.2 mg/kg, 0.25 mg/kg, 0.3 mg/kg, 0.35 mg/kg, 0.4 mg/kg, 0.45 mg/kg or 0.5 mg/kg. For fixed dosing in adult humans, a suitable dose may be between 8 and 50 mg, or between 8 and 25 mg, e.g., 15 mg or 20 mg.

In methods of treatment described herein, one or more doses may be administered. In some cases, a single dose may be effective to achieve a long-term benefit. Thus, the method may comprise administering a single dose of the antibody, its encoding nucleic acid, or the composition. Alternatively, multiple doses may be administered, usually sequentially and separated by a period of days, weeks or months. Anti-ICOS antibody may be repeatedly administered to a patient at intervals of 4 to 6 weeks, e.g., every 4 weeks, every 5 weeks, or every 6 weeks. Optionally, the anti-ICOS antibody may be administered to a patient once a month, or less frequently, e.g., every two months or every three months. Accordingly, a method of treating a patient may comprise administering a single dose of the anti-ICOS antibody to the patient, and not repeating the administration for at least one month, at least two months, at least three months, and optionally not repeating the administration for at least 12 months.

As discussed in Example 11c, comparable therapeutic effects may be obtained using either one or multiple doses of anti-ICOS antibody, which may be a result of a single dose of antibody being effective to reset the tumour microenvironment. Physicians can tailor the administration regimen of the anti-ICOS antibody to the disease and the patient undergoing therapy, taking into account the disease status and any other therapeutic agents or therapeutic measures (e.g., surgery, radiotherapy etc) with which the anti-ICOS antibody is being combined. In some embodiments, an effective dose of an anti-ICOS antibody is administered more frequently than once a month, such as, for example, once every three weeks, once every two weeks, or once every week. Treatment with anti-ICOS antibody may include multiple doses administered over a period of at least a month, at least six months, or at least a year.

As used herein, the terms “treat,” “treatment,” “treating,” or “amelioration” refer to therapeutic treatments, wherein the object is to reverse, alleviate, ameliorate, inhibit, slow down or stop the progression or severity of a condition associated with a disease or disorder. The term “treating” includes reducing or alleviating at least one adverse effect or symptom of a condition, disease or disorder. Treatment is generally “effective” if one or more symptoms or clinical

markers are reduced. Alternatively, treatment is “effective” if the progression of a disease is reduced or halted. That is, “treatment” includes not just the improvement of symptoms or markers, but also a cessation of, or at least slowing of, progress or worsening of symptoms compared to what would be expected in the absence of treatment. Beneficial or desired clinical results include, but are not limited to, alleviation of one or more symptom(s), diminishment of extent of disease, stabilised (i.e., not worsening) state of disease, delay or slowing of disease progression, amelioration or palliation of the disease state, remission (whether partial or total), and/or decreased mortality, whether detectable or undetectable. The term “treatment” of a disease also includes providing relief from the symptoms or side-effects of the disease (including palliative treatment). For treatment to be effective a complete cure is not contemplated. The method can in certain aspects include cure as well. In the context of the invention, treatment may be preventative treatment.

T Cell Therapy

WO2011/097477 described use of anti-ICOS antibodies for generating and expanding T cells, by contacting a population of T cells with a first agent that provides a primary activation signal (e.g., an anti-CD3 antibody) and a second agent that activates ICOS (e.g., an anti-ICOS antibody), optionally in the presence of a Th17 polarising agent such as IL-1B, IL-6, neutralising anti-IFN γ and/or anti-IL-4. Anti-ICOS antibodies described herein may be used in such methods to provide T cell populations. Populations of cultured expanded T cells having therapeutic activity (e.g., anti-tumour activity) may be generated. As described in WO2011/097477, such T cells may be used therapeutically in methods of treating patients by immunotherapy.

Morphological Assay for Anti-ICOS Antibodies as Therapeutic Candidates

It was observed that when candidate therapeutic anti-ICOS antibodies were coupled to a solid surface and brought into contact with ICOS-expressing T cells, they were able to induce morphological change in the cells. On addition of ICOS+ T cells to wells that were internally coated with anti-ICOS antibodies, cells were seen to change from their initial rounded shape, adopting a spindle-shape, spreading and adhering to the antibody-coated surface. This morphological change was not observed with control antibody. Moreover, the effect was found to be dose-dependent, with faster and/or more pronounced shape change occurring as the concentration of antibody on the surface increased. The shape change provides a surrogate indicator of T cell binding to ICOS, and/or of agonism by anti-ICOS antibody. The assay may be used to identify an antibody that promotes multimerisation of ICOS on the T cell surface. Such antibodies represent therapeutic candidate agonist antibodies. Conveniently, the visual indicator provided by this assay is a simple method of screening antibodies or cells, particularly in large numbers. The assay may be automated to run in a high-throughput system.

Accordingly, one aspect of the invention is an assay for selecting an antibody that binds ICOS, optionally for selecting an ICOS agonist antibody, the assay comprising:

- providing an array of antibodies immobilised (attached or adhered) to a substrate in a test well;
- adding ICOS-expressing cells (e.g., activated primary T cells, or MJ cells) to the test well;
- observing morphology of the cells;
- detecting shape change in the cells from rounded to flattened against the substrate within the well; wherein

the shape change indicates that the antibody is an antibody that binds ICOS, optionally an ICOS agonist antibody, and

selecting the antibody from the test well.

The assay may be run with multiple test wells, each containing a different antibody for testing, optionally in parallel, e.g., in a 96 well plate format. The substrate is preferably an inner surface of the well. Thus, a two-dimensional surface is provided against which flattening of the cells may be observed. For example, the bottom and/or wall of a well may be coated with antibody. Tethering of antibody to the substrate may be via a constant region of the antibody.

A negative control may be included, such as an antibody known not to bind ICOS, preferably an antibody that does not bind an antigen on the surface of the ICOS-expressing cells to be used. The assay may comprise quantifying the degree of morphological change and, where multiple antibodies are tested, selecting an antibody that induces greater morphological change than one or more other test antibodies.

Selection of antibody may comprise expressing nucleic acid encoding the antibody present in the test well of interest, or expressing an antibody comprising the CDRs or antigen binding domain of that antibody. The antibody may optionally be reformatted, for example to provide an antibody comprising the antigen binding domain of the selected antibody, e.g., an antibody fragment, or an antibody comprising a different constant region. A selected antibody is preferably provided with a human IgG1 constant region or other constant region as described herein. A selected antibody may further be formulated in a composition comprising one or more additional ingredients-suitable pharmaceutical formations are discussed elsewhere herein.

Clauses

Embodiments of the invention are set out in the following numbered clauses, which are part of the description.

Clause 1. An isolated antibody that binds the extracellular domain of human and/or mouse ICOS, wherein the antibody comprises a VH domain comprising an amino acid sequence having at least 95% sequence identity to the STIM003 VH domain SEQ ID NO: 408 and a VL domain comprising an amino acid sequence having at least 95% sequence identity to the STIM003 VL domain SEQ ID NO: 415.

Clause 2. An antibody according to clause 1, wherein the VH domain comprises a set of heavy chain complementarity determining regions (HCDRs) HCDR1, HCDR2 and HCDR3, wherein
HCDR1 is the STIM003 HCDR1 having amino acid sequence SEQ ID NO: 405,
HCDR2 is the STIM003 HCDR2 having amino acid sequence SEQ ID NO: 406,
HCDR3 is the STIM003 HCDR3 having amino acid sequence SEQ ID NO: 407.

Clause 3. An antibody according to clause 1 or clause 2, wherein the V_L domain comprises a set of light chain complementarity determining regions (LCDRs) LCDR1, LCDR2 and LCDR3, wherein
LCDR1 is the STIM003 LCDR1 having amino acid sequence SEQ ID NO: 412,
LCDR2 is the STIM003 LCDR2 having amino acid sequence SEQ ID NO: 413,
LCDR3 is the STIM003 LCDR3 having amino acid sequence SEQ ID NO: 414.

Clause 4. An antibody according to clause 1, wherein the V_H domain amino acid sequence is SEQ ID NO: 408 and/or wherein the V_L domain amino acid sequence is SEQ ID NO: 415.

Clause 5. An isolated antibody that binds the extracellular domain of human and/or mouse ICOS, comprising an antibody V_H domain comprising complementarity determining regions (CDRs) HCDR1, HCDR2 and HCDR3, and

an antibody V_L domain comprising complementarity determining regions LCDR1, LCDR2 and LCDR3, wherein

HCDR1 is the HCDR1 of STIM001, STIM002, STIM002-B, STIM003, STIM004, STIM005, STIM006, STIM007, STIM008 or STIM009, or comprises that HCDR1 with 1, 2, 3, 4 or 5 amino acid alterations,

HCDR2 is the HCDR2 of STIM001, STIM002, STIM002-B, STIM003, STIM004, STIM005, STIM006, STIM007, STIM008 or STIM009, or comprises that HCDR2 with 1, 2, 3, 4 or 5 amino acid alterations, and/or

HCDR3 is the HCDR3 of STIM001, STIM002, STIM002-B, STIM003, STIM004, STIM005, STIM006, STIM007, STIM008 or STIM009 or comprises that HCDR3 with 1, 2, 3, 4 or 5 amino acid alterations.

Clause 6. An antibody according to clause 5, wherein the antibody heavy chain CDRs are those of STIM001, STIM002, STIM002-B, STIM003, STIM004, STIM005, STIM006, STIM007, STIM008 or STIM009 or comprise the STIM001, STIM002, STIM002-B, STIM003, STIM004, STIM005, STIM006, STIM007, STIM008 or STIM009 heavy chain CDRs with 1, 2, 3, 4 or 5 amino acid alterations.

Clause 7. An antibody according to clause 6, wherein the antibody VH domain has the heavy chain CDRs of STIM003.

Clause 8. An isolated antibody that binds the extracellular domain of human and/or mouse ICOS, comprising an antibody VH domain comprising complementarity determining regions HCDR1, HCDR2 and HCDR3, and

an antibody VL domain comprising complementarity determining regions LCDR1, LCDR2 and LCDR3, wherein LCDR1 is the LCDR1 of STIM001, STIM002, STIM002-B, STIM003, STIM004, STIM005, STIM006, STIM007, STIM008 or STIM009, or comprises that LCDR1 with 1, 2, 3, 4 or 5 amino acid alterations,

LCDR2 is the LCDR2 of STIM001, STIM002, STIM002-B, STIM003, STIM004, STIM005, STIM006, STIM007, STIM008 or STIM009, or comprises that LCDR2 with 1, 2, 3, 4 or 5 amino acid alterations, and/or

LCDR3 is the LCDR3 of STIM001, STIM002, STIM002-B, STIM003, STIM004, STIM005, STIM006, STIM007, STIM008 or STIM009 or comprises that LCDR3 with 1, 2, 3, 4 or 5 amino acid alterations.

Clause 9. An antibody according to any of clauses 5 to 8, wherein the antibody light chain CDRs are those of STIM001, STIM002, STIM002-B, STIM003, STIM004, STIM005, STIM006, STIM007, STIM008 or STIM009, or comprise the STIM001, STIM002, STIM002-B, STIM003, STIM004, STIM005, STIM006, STIM007, STIM008 or STIM009 light chain CDRs with 1, 2, 3, 4 or 5 amino acid alterations.

Clause 10. An antibody according to clause 9, wherein the antibody VL domain has the light chain CDRs of STIM003.

Clause 11. An antibody according to any of clauses 5 to 10, comprising VH and/or VL domain framework regions of human germline gene segment sequences. 5

Clause 12. An antibody according to any of clauses 5 to 11, comprising a VH domain which

(i) is derived from recombination of a human heavy chain V gene segment, a human heavy chain D gene segment and a human heavy chain J gene segment, wherein the V segment is IGHV1-18 (e.g., V1-18*01), IGVH3-20 (e.g. V3-20*d01), IGVH3-11 (e.g. V3-11*01) or IGVH2-5 (e.g., V2-5*10); 10

the D gene segment is IGHD6-19 (e.g., IGHD6-19*01), IGHD3-10 (e.g., IGHD3-10*01) or IGHD3-9 (e.g., IGHD3-9*01); and/or

the J gene segment is IGHJ6 (e.g., IGHJ6*02), IGHJ4 (e.g., IGHJ4*02) or IGHJ3 (e.g., IGHJ3*02), or

(ii) comprises framework regions FR1, FR2, FR3 and FR4, wherein 20

FR1 aligns with human germline V gene segment IGHV1-18 (e.g., V1-18*01), IGVH3-20 (e.g. V3-20*d01), IGVH3-11 (e.g. V3-11*01) or IGVH2-5 (e.g., V2-5*10), optionally with 1, 2, 3, 4 or 5 amino acid alterations, 25

FR2 aligns with human germline V gene segment IGHV1-18 (e.g., V1-18*01), IGVH3-20 (e.g. V3-20*d01), IGVH3-11 (e.g. V3-11*01) or IGVH2-5 (e.g., V2-5*10), optionally with 1, 2, 3, 4 or 5 amino acid alterations, 30

FR3 aligns with human germline V gene segment IGHV1-18 (e.g., V1-18*01), IGVH3-20 (e.g. V3-20*d01), IGVH3-11 (e.g. V3-11*01) or IGVH2-5 (e.g., V2-5*10), optionally with 1, 2, 3, 4 or 5 amino acid alterations, and/or 35

FR4 aligns with human germline J gene segment IGJH6 (e.g., JH6*02), IGJH4 (e.g., JH4*02) or IGJH3 (e.g., JH3*02), optionally with 1, 2, 3, 4 or 5 amino acid alterations. 40

Clause 13. An antibody according to any of clauses 5 to 12, comprising an antibody V_L domain which

(i) is derived from recombination of a human light chain V gene segment and a human light chain J gene segment, wherein 45

the V segment is IGKV2-28 (e.g., IGKV2-28*01), IGKV3-20 (e.g., IGKV3-20*01), IGKV1D-39 (e.g., IGKV1D-39*01) or IGKV3-11 (e.g., IGKV3-11*01), and/or

the J gene segment is IGKJ4 (e.g., IGKJ4*01), IGKJ2 (e.g., IGKJ2*04), IGLJ3 (e.g., IGKJ3*01) or IGKJ1 (e.g., IGKJ1*01); or 50

(ii) comprises framework regions FR1, FR2, FR3 and FR4, wherein

FR1 aligns with human germline V gene segment 55

IGKV2-28 (e.g., IGKV2-28*01), IGKV3-20 (e.g., IGKV3-20*01), IGKV1D-39 (e.g., IGKV1D-39*01) or IGKV3-11 (e.g., IGKV3-11*01), optionally with 1, 2, 3, 4 or 5 amino acid alterations,

FR2 aligns with human germline V gene segment 60

IGKV2-28 (e.g., IGKV2-28*01), IGKV3-20 (e.g., IGKV3-20*01), IGKV1D-39 (e.g., IGKV1D-39*01) or IGKV3-11 (e.g., IGKV3-11*01), optionally with 1, 2, 3, 4 or 5 amino acid alterations,

FR3 aligns with human germline V gene segment 65

IGKV2-28 (e.g., IGKV2-28*01), IGKV3-20 (e.g., IGKV3-20*01), IGKV1D-39 (e.g., IGKV1D-39*01)

or IGKV3-11 (e.g., IGKV3-11*01), optionally with 1, 2, 3, 4 or 5 amino acid alterations, and/or

FR4 aligns with human germline J gene segment IGKJ4 (e.g., IGKJ4*01), IGKJ2 (e.g., IGKJ2*04), IGKJ3 (e.g., IGKJ3*01) or IGKJ1 (e.g., IGKJ1*01), optionally with 1, 2, 3, 4 or 5 amino acid alterations.

Clause 14. An antibody according to any of clauses 5 to 13, comprising an antibody VH domain which is the VH domain of STIM001, STIM002, STIM002-B, STIM003, STIM004, STIM005, STIM006, STIM007, STIM008 or STIM009, or which has an amino acid sequence at least 90% identical to the antibody VH domain sequence of STIM001, STIM002, STIM002-B, STIM003, STIM004, STIM005, STIM006, STIM007, STIM008 or STIM009.

Clause 15. An antibody according to any of clauses 5 to 14, comprising an antibody VL domain which is the VL domain of STIM001, STIM002, STIM002-B, STIM003, STIM004, STIM005, STIM006, STIM007, STIM008 or STIM009, or which has an amino acid sequence at least 90% identical to the antibody VL domain sequence of STIM001, STIM002, STIM002-B, STIM003, STIM004, STIM005, STIM006, STIM007, STIM008 or STIM009.

Clause 16. An antibody according to clause 15, comprising

an antibody VH domain which is selected from the VH domain of STIM001, STIM002, STIM002-B, STIM003, STIM004, STIM005, STIM006, STIM007, STIM008 or STIM009, or which has an amino acid sequence at least 90% identical to the antibody VH domain sequence of STIM001, STIM002, STIM002-B, STIM003, STIM004, STIM005, STIM006, STIM007, STIM008 or STIM009, and

an antibody VL domain which is the VL domain of said selected antibody, or which has an amino acid sequence at least 90% identical to the antibody VL domain sequence of said selected antibody.

Clause 17. An antibody according to clause 16, comprising the STIM003 VH domain and the STIM003 VL domain.

Clause 18. An antibody according to any of the preceding clauses, comprising an antibody constant region.

Clause 19. An antibody according to clause 18, wherein the constant region comprises a human heavy and/or light chain constant region.

Clause 20. An antibody according to clause 18 or clause 19, wherein the constant region is Fc effector positive.

Clause 21. An antibody according to clause 20, comprising an Fc region that has enhanced ADCC, ADCP and/or CDC function compared with a native human Fc region.

Clause 22. An antibody according to any of clauses 18 to 21, wherein the antibody is an IgG1.

Clause 23. An antibody according to clause 21 or clause 22, wherein the antibody is afucosylated.

Clause 24. An antibody according to any of the preceding clauses which is conjugated to a cytotoxic drug or pro-drug.

Clause 25. An antibody according to any of the preceding clauses, which is a multispecific antibody.

Clause 26. An isolated antibody that binds the extracellular domain of human and mouse ICOS with an affinity (K_D) of less than 50 nM as determined by surface plasmon resonance.

Clause 27. An antibody according to clause 26, wherein the antibody binds the extracellular domain of human and mouse ICOS with an affinity (K_D) of less than 5 nM as determined by surface plasmon resonance.

Clause 28. An antibody according to clause 26 or clause 27, wherein the K_D of binding the extracellular domain of human ICOS is within 10-fold of the K_D of binding the extracellular domain of mouse ICOS.

Clause 29. A composition comprising an isolated antibody according to any of the preceding clauses and a pharmaceutically acceptable excipient.

Clause 30. A composition comprising isolated nucleic acid encoding an antibody according to any of clauses 1 to 28 and a pharmaceutically acceptable excipient.

Clause 31. A method of modulating the balance of regulatory T cells (Tregs) to effector T cells (Teffs) to increase Teff response in a patient, comprising administering an antibody according to any of clauses 1 to 28 or composition according to clause 29 to the patient.

Clause 32. A method of treating a disease or condition amenable to therapy by depleting regulatory T cells (Tregs) and/or increasing effector T cell (Teff) response in a patient, the method comprising administering an antibody according to any of clauses 1 to 28 or a composition according to clause 29 to the patient.

Clause 33. An antibody according to any of clauses 1 to 28, or a composition according to clause 29, for use in a method of treatment of the human body by therapy.

Clause 34. An antibody or composition for use according to clause 33, for use in modulating the balance of regulatory T cells (Tregs) to effector T cells (Teffs) to increase effector T cell response in a patient.

Clause 35. An antibody or composition for use according to clause 33, for use in treating a disease or condition amenable to therapy by depleting regulatory T cells (Tregs) and/or increasing effector T cell (Teff) response in a patient.

Clause 36. A method according to clause 32, or an antibody or a composition for use according to clause 35, wherein the disease is a cancer or a solid tumour.

Clause 37. An antibody according to any of clauses 1 to 28 or a composition according to clause 29, for use in a method of treating cancer in a human patient.

Clause 38. A method of treating cancer in a human patient, comprising administering an antibody according to any of clauses 1 to 28 or a composition according to clause 29 to the patient.

Clause 39. A method or an antibody or composition for use according to any of clauses 36 to 38, wherein the cancer is renal cell cancer, head and neck cancer, melanoma, non small cell lung cancer or diffuse large B-cell lymphoma.

Clause 40. A method or an antibody or composition for use according to any of clauses 31 to 39, wherein the method comprises administering the antibody and another therapeutic agent and/or radiation therapy to the patient.

Clause 41. A method or an antibody or composition for use according to clause 40, wherein the therapeutic agent is an anti-PD-L1 antibody.

Clause 42. A method or an antibody or composition for use according to clause 41, wherein the anti-PD-L1 antibody comprises a VH domain having amino acid sequence SEQ ID NO: 299 and a VL domain having amino acid sequence SEQ ID NO: 300.

Clause 43. A method or an antibody or composition for use according to clause 41 or clause 42, wherein the therapeutic agent is an anti-PD-L1 IL-2 immunocytokine.

Clause 44. A method or an antibody or composition for use according to clause 43, wherein the anti-PD-L1 antibody is an immunocytokine comprising human wild type or variant IL-2.

Clause 45. A method or an antibody or composition for use according to clause 44, wherein the anti-ICOS antibody and the anti-PDL1 antibody are each able to mediate ADCC, ADCP and/or CDC.

Clause 46. A method or an antibody or composition for use according to any of clauses 41 to 45, wherein the anti-ICOS antibody is a human IgG1 antibody and the anti-PDL1 antibody is a human IgG1 antibody.

Clause 47. A method or an antibody or composition for use according to clause 40, wherein the therapeutic agent is an anti-PD-1 antibody.

Clause 48. A method or an antibody or composition for use according to clause 40, wherein the other therapeutic agent is IL-2.

Clause 49. A method or an antibody or composition for use according to any of clauses 40 to 48, wherein the method comprises administering the anti-ICOS antibody after administering the other therapeutic agent and/or radiation therapy.

Clause 50. A method or an antibody or composition for use according to any of clauses 31 to 49, wherein the anti-ICOS antibody is conjugated to a pro-drug, and wherein the method or use comprises administering the anti-ICOS antibody to a patient and selectively activating the pro-drug at a target tissue site.

Clause 51. A method or an antibody or composition for use according to clause 50, wherein the patient has a solid tumour and the method comprises selectively activating the pro-drug in the tumour.

Clause 52. A method or an antibody or composition for use according to clause 50 or clause 51, comprising selectively activating the pro-drug through photoactivation.

Clause 53. Combination of anti-ICOS human IgG1 antibody and anti-PDL1 human IgG1 antibody for use in a method of treating cancer in a patient.

Clause 54. A method of treating cancer in a patient, comprising administering an anti-ICOS human IgG1 antibody and an anti-PD-L1 human IgG1 antibody to the patient.

Clause 55. Anti-ICOS antibody for use in a method of treating cancer in a patient, the method comprising administering the anti-ICOS antibody and the anti-PD-L1 antibody to the patient, wherein a single dose of the anti-ICOS antibody is administered.

Clause 56. Anti-ICOS antibody for use according to clause 55, wherein the anti-ICOS antibody is a human IgG1 antibody and the anti-PD-L1 antibody is a human IgG1 antibody.

Clause 57. Combination according to clause 53, method according to clause 54 or anti-ICOS antibody for use according to clause 55 or clause 56, wherein the cancer is renal cell cancer, head and neck cancer, melanoma, non small cell lung cancer or diffuse large B-cell lymphoma.

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- Clause 58. A method or an antibody, composition or combination for use according to any of clauses 41 to 46 or 53 to 54, the method comprising administering the anti-ICOS antibody and the anti-PD-L1 antibody to the patient, wherein a single dose of the anti-ICOS antibody is administered. 5
- Clause 59. A method or an antibody, composition or combination for use according to clause 58, wherein the method comprises administering a single dose of the anti-ICOS antibody followed by multiple doses of the anti-PD-L1 antibody. 10
- Clause 60. A method or an antibody, composition or combination for use according to any of clauses 41 to 46 or 53 to 54, wherein the anti-ICOS antibody and the anti-PDL1 antibody are provided in separate compositions for administration. 15
- Clause 61. A method or an antibody, composition or combination for use according to any of clauses 41 to 46 or 53 to 60, wherein the anti-ICOS antibody and/or the anti-PD-L1 antibody comprises a human IgG1 constant region comprising amino acid sequence SEQ ID NO: 340. 20
- Clause 62. Anti-ICOS antibody for use in a method of treating a patient, the method comprising administering the anti-ICOS antibody to a patient who has an increased level of ICOS-positive regulatory T cells following treatment with another therapeutic agent. 25
- Clause 63. A method of treating a patient, the method comprising administering an anti-ICOS antibody to a patient who has an increased level of ICOS-positive regulatory T cells following treatment with another therapeutic agent. 30
- Clause 64. An anti-ICOS antibody for use according to clause 62, or a method according to clause 63, wherein the method comprises administering a therapeutic agent to the patient, determining that the patient has an increased level of ICOS-positive regulatory T cells following the treatment with said agent, and administering an anti-ICOS antibody to the patient to reduce the level of regulatory T cells. 40
- Clause 65. An anti-ICOS antibody for use or a method according to any of clauses 62 to 64, wherein the therapeutic agent is IL-2 or an immunomodulatory antibody (e.g., anti-PDL-1, anti-PD-1 or anti-CTLA-4). 45
- Clause 66. An anti-ICOS antibody for use or a method according to any of clauses 62 to 65, wherein the method comprises treating a tumour, e.g., melanoma, such as metastatic melanoma.
- Clause 67. Anti-ICOS antibody for use in a method of treating cancer in a patient by in vivo vaccination of the patient against their cancer cells, the method comprising treating the patient with a therapy that causes immunological cell death of the cancer cells, resulting in presentation of antigen to antigen-specific effector T cells, and administering an anti-ICOS antibody to the patient, wherein the anti-ICOS antibody enhances the antigen-specific effector T cell response. 60
- Clause 68. A method of treating cancer in a patient by in vivo vaccination of the patient against their cancer cells, the method comprising treating the patient with a therapy that causes immunological cell death of the cancer cells, resulting in presentation of antigen to antigen-specific effector T cells, and 65

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- administering an anti-ICOS antibody to the patient, wherein the anti-ICOS antibody enhances the antigen-specific effector T cell response.
- Clause 69. A method of treating cancer in a patient by in vivo vaccination of the patient against their cancer cells, the method comprising administering an anti-ICOS antibody to the patient, wherein the patient is one who has been previously treated with a therapy that causes immunological cell death of the cancer cells, resulting in presentation of antigen to antigen-specific effector T cells, and wherein the anti-ICOS antibody enhances the antigen-specific effector T cell response.
- Clause 70. Anti-ICOS antibody for use or a method according to any of clauses 67 to 69, wherein the therapy that causes immunological cell death is radiation of the cancer cells, administration of a chemotherapeutic agent and/or administration of an antibody directed to a tumour-associated antigen.
- Clause 71. Anti-ICOS antibody for use or a method according to clause 70, wherein the chemotherapeutic agent is oxaliplatin.
- Clause 72. Anti-ICOS antibody for use or a method according to clause 70, wherein the tumour-associated antigen is HER2 or CD20.
- Clause 73. Anti-ICOS antibody for use in a method of treating a cancer in a patient, wherein the cancer is or has been characterised as being positive for expression of ICOS ligand and/or FOXP3.
- Clause 74. A method of treating a cancer in a patient, wherein the cancer is or has been characterised as being positive for expression of ICOS ligand and/or FOXP3, the method comprising administering an anti-ICOS antibody to the patient.
- Clause 75. Anti-ICOS antibody for use according to clause 73, or a method according to clause 74, wherein the method comprises: testing a sample from a patient to determine that the cancer expresses ICOS ligand and/or FOXP3; selecting the patient for treatment with the anti-ICOS antibody; and administering the anti-ICOS antibody to the patient.
- Clause 76. Anti-ICOS antibody for use according to clause 73, or a method according to clause 74, wherein the method comprises administering an anti-ICOS antibody to a patient from whom a test sample has indicated that the cancer is positive for expression of ICOS ligand and/or FOXP3.
- Clause 77. Anti-ICOS antibody for use or a method according to clause 75 or clause 76, wherein the sample is biopsy sample of a solid tumour.
- Clause 78. Anti-ICOS antibody for use in a method of treating a cancer in a patient, wherein the cancer is or has been characterised as being refractory to treatment with an immunooncology drug, e.g., anti-CTLA-4 antibody, anti-PD1 antibody, anti-PD-L1 antibody, anti-CD137 antibody or anti-GITR antibody.
- Clause 79. A method of treating a cancer in a patient, wherein the cancer is or has been characterised as being refractory to treatment with an immunooncology drug, e.g., anti-CTLA-4 antibody, anti-PD1 antibody, anti-PD-L1 antibody, anti-CD137 antibody or anti-GITR antibody, the method comprising administering an anti-ICOS antibody to the patient.
- Clause 80. Anti-ICOS antibody for use according to clause 78 or a method according to clause 79, wherein the method comprises:

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treating the patient with the immunooncology drug;
determining that the cancer is not responsive to the
drug;
selecting the patient for treatment with the anti-ICOS
antibody; and
administering the anti-ICOS antibody to the patient.

Clause 81. Anti-ICOS antibody for use according to
clause 78, or a method according to clause 79, wherein
the method comprises administering an anti-ICOS anti-
body to a patient whose cancer was not responsive to
prior treatment with the immunooncology drug.

Clause 82. Anti-ICOS antibody for use or a method
according to any of clauses 73 to 81, wherein the cancer
is a tumour derived from cells that have acquired ability
to express ICOS ligand.

Clause 83. Anti-ICOS antibody for use or a method
according to clause 82, wherein the cancer is mela-
noma.

Clause 84. Anti-ICOS antibody for use or a method
according to any of clauses 73 to 81, wherein the cancer
is derived from an antigen-presenting cell, such as a B
lymphocyte (e.g., B cell lymphoma, such as diffuse
large B cell lymphoma) or a T lymphocyte.

Clause 85. Anti-ICOS antibody for use or a method
according to any of clauses 73 to 81, wherein the cancer
is resistant to treatment with an anti-CD20 antibody.

Clause 86. Anti-ICOS antibody for use or a method
according to clause 85, wherein the cancer is B cell
lymphoma.

Clause 87. Anti-ICOS antibody for use or a method
according to clause 86, wherein the anti-CD20 anti-
body is rituximab.

Clause 88. Anti-ICOS antibody for use or a method
according to any of clauses 85 to 87, wherein the
method comprises treating the patient with the anti-
CD20 antibody;
determining that the cancer is not responsive to the
anti-CD20 antibody;
testing a sample from a patient to determine that the
cancer expresses ICOS ligand;
selecting the patient for treatment with the anti-ICOS
antibody; and
administering the anti-ICOS antibody to the patient.

Clause 89. Anti-ICOS antibody for use or a method
according to any of clauses 85 to 87, wherein the
method comprises administering an anti-ICOS anti-
body to a patient whose cancer was not responsive to
prior treatment with anti-CD20 antibody.

Clause 90. Anti-ICOS antibody for use or a method
according to any of clauses 67 to 89, wherein the cancer
is a solid tumour.

Clause 91. Anti-ICOS antibody for use or a method
according to any of clauses 67 to 89, wherein the cancer
is a haematological liquid tumour.

Clause 92. Anti-ICOS antibody for use or a method
according to clause 90 or 91, wherein the tumour is
high in regulatory T cells.

Clause 93. Anti-ICOS antibody for use or a method
according to any of clauses 53 to 92, wherein the
anti-ICOS antibody is as defined in any of clauses 1 to
28 or is provided in a composition according to clause
29.

Clause 94. A transgenic non-human mammal having a
genome comprising a human or humanised immuno-
globulin locus encoding human variable region gene
segments, wherein the mammal does not express ICOS.

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Clause 95. A method of producing an antibody that binds
the extracellular domain of human and non-human
ICOS, comprising
(a) immunising a mammal according to clause 94 with
human ICOS antigen;
(b) isolating antibodies generated by the mammal;
(c) testing the antibodies for ability to bind human
ICOS and non-human ICOS; and
(d) selecting one or more antibodies that binds both
human and non-human ICOS.

Clause 96. A method according to clause 95, comprising
immunising the mammal with cells expressing human
ICOS.

Clause 97. A method according to clause 95 or clause 96,
comprising
(c) testing the antibodies for ability to bind human
ICOS and non-human ICOS using surface plasmon
resonance and determining binding affinities; and
(d) selecting one or more antibodies for which the K_D
of binding to human ICOS is less than 50 nM and the
 K_D of binding to non-human ICOS is less than 500
nM.

Clause 98. A method according to clause 97, comprising
(d) selecting one or more antibodies for which the K_D
of binding to human ICOS is less than 10 nM and the
 K_D of binding to non-human ICOS is less than 100
nM.

Clause 99. A method according to any of clauses 95 to 98,
comprising
(c) testing the antibodies for ability to bind human
ICOS and non-human ICOS using surface plasmon
resonance and determining binding affinities; and
(d) selecting one or more antibodies for which the K_D
of binding to human ICOS is within 10-fold of the
 K_D of binding to non-human ICOS.

Clause 100. A method according to clause 99, comprising
(d) selecting one or more antibodies for which the K_D
of binding to human ICOS is within 5-fold of the K_D
of binding to non-human ICOS.

Clause 101. A method according to any of clauses 95 to
100, comprising testing the antibodies for ability to
bind non-human ICOS from the same species as the
mammal.

Clause 102. A method according to any of clauses 95 to
101, comprising testing the antibodies for ability to
bind non-human ICOS from a different species as the
mammal.

Clause 103. A method according to any of clauses 95 to
102, wherein the mammal is a mouse or a rat.

Clause 104. A method according to any of clauses 95 to
103, wherein the non-human ICOS is mouse ICOS or
rat ICOS.

Clause 105. A method according to any of clauses 95 to
104, wherein the human or humanised immunoglobulin
locus comprises human variable region gene segments
upstream of an endogenous constant region.

Clause 106. A method according to clause 105, compris-
ing
(a) immunising a mammal according to clause 94 with
human ICOS antigen, wherein the mammal is a
mouse;
(b) isolating antibodies generated by the mouse;
(c) testing the antibodies for ability to bind human
ICOS and mouse ICOS; and
(d) selecting one or more antibodies that binds both
human and mouse ICOS.

Clause 107. A method according to any of clauses 95 to 106, comprising isolating nucleic acid encoding an antibody heavy chain variable domain and/or an antibody light chain variable domain.

Clause 108. A method according to any of clauses 95 to 107, wherein the mammal generates antibodies through recombination of human variable region gene segments and an endogenous constant region.

Clause 109. A method according to clause 107 or clause 108, comprising conjugating the nucleic acid encoding the heavy and/or light chain variable domain to a nucleotide sequence encoding a human heavy chain constant region and/or human light chain constant region respectively.

Clause 110. A method according to any of clauses 107 to 109, comprising introducing the nucleic acid into a host cell.

Clause 111. A method according to clause 110, comprising culturing the host cell under conditions for expression of the antibody, or of the antibody heavy and/or light chain variable domain.

Clause 112. An antibody, or antibody heavy and/or light chain variable domain, produced by the method according to any of clauses 95 to 111.

Clause 113. A method of selecting an antibody that binds ICOS, optionally for selecting an ICOS agonist antibody, the assay comprising:

providing an array of antibodies immobilised (attached or adhered) to a substrate in a test well;

adding ICOS-expressing cells (e.g., activated primary T cells, or MJ cells) to the test well;

observing morphology of the cells;

detecting shape change in the cells from rounded to flattened against the substrate within the well; wherein the shape change indicates that the antibody is an antibody that binds ICOS, optionally an ICOS agonist antibody;

selecting the antibody from the test well;

expressing nucleic acid encoding the CDRs of the selected antibody; and

formulating the antibody into a composition comprising one or more additional components.

Various further aspects and embodiments of the present invention will be apparent to those skilled in the art in view of the present disclosure. All documents mentioned in this specification, including published US counterparts of any patents or patent applications referred to, are incorporated herein by reference in their entirety.

EXPERIMENTAL EXAMPLES

The following Examples describe the generation, characterisation and performance of anti-ICOS antibodies. Antibodies were generated using the Kymouse™, a transgenic mouse platform capable of generating antibodies with human variable domains. Antibodies from the Kymouse™ have human variable domains, generated from human V (D) and J segments, and mouse constant domains. The endogenous mouse variable genes have been silenced and make up a very small portion of the repertoire (less than 0.5% of all heavy chain variable regions are of mouse origin). The Kymouse™ system is described in Lee et al 2014 [39], WO2011/004192, WO2011/158009 and WO2013/061098. This project employed the Kymouse™ HK strain, in which the heavy chain locus and light chain kappa locus are humanised.

ICOS knock-out Kymouse™ were immunised with either ICOS protein or a combination of alternating boosts of protein and cells expressing human and mouse ICOS.

Hits which bound to human ICOS were identified. The primary selection criteria for the screen was binding to human cell expressed ICOS (CHO cells) and binding to ICOS protein (HTRF). Binding to mouse ICOS protein and mouse cell expressed ICOS (CHO cells) was also assessed and taken into consideration when selecting primary screen hits. Using these criteria hits were progressed to a secondary screen. In the secondary screen hits were confirmed by determining binding to human and mouse ICOS expressed on CHO cells by flow cytometry.

From a large number of antibodies screened, a small panel were identified which bind to human/cynomolgus and mouse ICOS as determined by surface plasmon resonance and flow cytometry. These antibodies included STIM001, STIM002 and its variant STIM002-B, STIM003, STIM004 and STIM005. An additional four antibodies STIM006, STIM007, STIM008 and STIM009 were also selected, showing less cross-reactivity with mouse ICOS but demonstrating agonism of the human ICOS receptor.

A further screening campaign was then undertaken to identify additional anti-ICOS antibodies from antigen-selected B cells obtained from the immunised mice. Approximately 570 additional clones were selected and sequenced. Using bioinformatics analysis, 467 of these clones were selected for synthesis and expression in HEK cells. Following functional screening of cell supernatants, 8 human/mouse cross-reactive anti-ICOS antibodies and 38 human-specific anti-ICOS antibodies were selected for further study. Additionally, based on analysis of sequences of the human-specific anti-ICOS antibodies, an additional further 20 sequenced clones from the immunised mice were synthesised, purified and tested for binding to human and mouse ICOS. This further screening campaign resulted in selection of cross-reactive anti-ICOS antibodies STIM017, STIM020, STIM021, STIM022, STIM023 and anti-human-ICOS antibodies STIM039, STIM040, STIM041, STIM042, STIM043, STIM044, STIM050, STIM051, STIM052, STIM053, STIM054, STIM055, STIM056, STIM057, STIM058, STIM059, STIM060, STIM061, STIM062, STIM063, STIM064, STIM065 and STIM066.

The data presented here indicate the ability of anti-ICOS antibodies to act as agonists of the ICOS receptor in an ICOS positive CD4+ cell line and also in a primary T cell-based assay, show cell-killing ability in an ADCC assay and an ability to promote an anti-tumour immune response in vivo.

Example 1: Generation of ICOS Knock-Out Mouse

An ICOS knock-out Kymouse™ line was generated by homologous recombination in Kymouse™ HK ES cells. In brief, a 3.5 kb targeting vector encoding a puromycin selection was targeted into ES cells. Successful targeting resulted in the replacement of a small region (72 bp) of the mouse ICOS locus with the puromycin cassette, disrupting the signal peptide/start codon of the gene. Positive ES clones were expanded and microinjected into mouse blastocysts and resulting chimaeras bred in order to ultimately generate animals homozygous for both the humanised heavy and kappa immunoglobulin loci and the modified, functionally-null, ICOS locus.

Example 2: Antigen and Cell-Line Preparation

Generation of Stably Transfected MEF and CHO-S Cells Expressing Human or Mouse ICOS

Full length DNA sequences encoding human and mouse ICOS were codon optimised for mammalian expression, ordered as synthetic string DNA and cloned into an expression vector under the control of the CMV promoter and flanked by 3' and 5' piggyBac specific terminal repeat sequences facilitating stable integration into the cell genome (see [40]). The expression vector contained a puromycin selection cassette to facilitate stable cell line generation. For generation of human ICOS expressing and mouse ICOS expressing cell lines respectively, the human or mouse ICOS expression plasmid was co-transfected with a plasmid

mouse Fc or a FLAG His peptide motif to aid purification and detection. These were added to the DNA constructs by overlap extension. All constructs were sequenced prior to expression to ensure their correct sequence composition.

Example 3: Immunisation

ICOS knock out HK Kymice™ (see Example 1), Kymouse™ wild type HK strain and Kymouse™ wild type HL strain were immunised according to the regimens shown in Table E3. Kymouse™ wild type HK and HL strains express wild type mouse ICOS. In the HK strain the immunoglobulin heavy chain locus and light chain kappa locus are humanised, and in the HL strain the immunoglobulin heavy chain locus and light chain lambda locus are humanised.

TABLE E3

Immunisation regimens for Kymouse™ strains						
Regime	Mouse	Prime	Boost 1	Boost 2	Boost 3	Final Boost
KM103	ICOS KO	mICOS Fc	hICOS MEF	mICOS Fc	hICOS MEF	mICOS Fc
KM103	ICOS KO	mICOS Fc	hICOS Fc	mICOS Fc	hICOS Fc	N/A
KM111	ICOS KO	mICOS Fc + hICOS Fc	mICOS MEF + hICOS MEF	mICOS Fc + hICOS Fc	mICOS MEF + hICOS MEF	mICOS Fc + hICOS Fc
KM111	ICOS KO	hICOS Fc	hICOS MEF	hICOS Fc	hICOS MEF	hICOS Fc
KM111	ICOS KO	hICOS Fc	hICOS MEF	hICOS Fc	hICOS MEF	hICOS Fc
KM111	ICOS KO	mICOS Fc	mICOS MEF	mICOS Fc	mICOS MEF	mICOS Fc
KM111	HK and HL	hICOS Fc	hICOS MEF	hICOS Fc	hICOS MEF	hICOS Fc
KM135	ICOS KO		mICOS Fc 1 prime and 6 boosts (RIMMS)			
KM135	ICOS KO		hICOS Fc 1 prime and 6 boosts (RIMMS)			

encoding piggyBac transposase into a mouse embryonic fibroblast (MEF) cell line and CHO-S cells using the Free-Style Max transfection reagent (Invitrogen) according to manufacturer instructions. MEF cells were generated from embryos obtained from a 129S5 crossed to C57BL6 female mouse. Twenty four hours after transfection, the media was supplemented with puromycin and grown for at least two weeks to select stable cell lines. Cell culture medium was replaced every 3-4 days. Expression of human or mouse ICOS protein was assessed by flow cytometry using anti-human or anti-mouse ICOS-PE conjugated antibodies (eBioscience) respectively. Complete MEF media was made up of Dulbecco's Modified Eagle's Medium (Gibco) supplemented with 10% v/v fetal bovine serum (Gibco). Complete CHO-S media was made up of CD-CHO media supplemented with 8 mM Glutamax (Gibco). CHO-S cells are the CHO-3E7 cell line included with the pTT5 system available from the National Research Council of Canada, but other CHO cell lines could be employed.

Preparation of MEF Cells for Mouse Immunisations

Cell culture medium was removed and cells washed once with 1xPBS. Cells were treated for 5 minutes with trypsin to loosen cells from tissue culture surface. Cells were collected and the trypsin neutralized by the addition of complete media containing 10% v/v fetal bovine serum (FCS). Cells were then centrifuged at 300 g for 10 minutes and washed with 25 ml of 1xPBS. Cells were counted and resuspended at the appropriate concentration in 1xPBS.

Cloning and Expression of Recombinant Proteins

Synthetic DNA encoding the extracellular domains of human ICOS (NCBI ID: NP_036224.1), mouse ICOS (NCBI ID: NP_059508.2) and cynomolgus ICOS (GenBank ID: EHH55098.1) were cloned into either a pREP4 (Invitrogen) or a pTT5 (National Research Council Of Canada) expression plasmid using standard molecular biology techniques. The constructs also contained either a human Fc, a

Key to Table:

mICOS Fc=mouse ICOS protein with human Fc

hICOS Fc=human ICOS protein with human Fc

mICOS MEF=mouse ICOS expressed on MEF cells

hICOS MEF=human ICOS expressed on MEF cells

mICOS Fc+ hICOS Fc=mouse ICOS protein with human Fc+human ICOS protein with human Fc administered simultaneously

mICOS MEF+ hICOS MEF=mouse ICOS expressed on MEF cells+human ICOS expressed on MEF cells administered simultaneously

ICOS KO=ICOS knockout HK Kymouse

HK and HL=wild type Kymouse HK and HL genotype

RIMMS is a modified sub-cutaneous immunisation procedure (rapid immunisation at multiple sites); modified after Kilpatrick et al. [41]). Immunisation regimens KM103 and KM111 were prime-rest-boost by intraperitoneal (i.p.) administration. Sigma Adjuvant System was used for all immunisations and rest intervals were usually between 2 and 3 weeks. Final boosts were administered by intravenously in absence of adjuvant.

Sera from serial or terminal blood samples were analysed for the presence of specific antibodies by flow cytometry and the titre data was used (where possible) to select mice to be used for B cell sorting.

Example 4: Comparison of Serum Titres Between ICOS KO and Wild Type Mice

Serum titres of immunised ICOS KO and immunised wild type Kymouse were determined using flow cytometry. In ICOS KO mice, immunisation with human ICOS antigen induced a serum immunoglobulin response with Ig binding to both human and mouse ICOS expressed on CHO cells (FIG. 1a). Conversely, in the wild type Kymouse (expressing mouse ICOS), immunisation with the same human ICOS

antigen produced sera that showed markedly reduced binding to mouse ICOS compared with binding of the same serum to human ICOS (FIG. 1b).

Method

CHO-S cells expressing human ICOS or mouse ICOS (see Example 2) or untransfected CHO-S cells (referred to as wild type (WT)), suspended in FACS buffer (PBS+1% w/v BSA+0.1% w/v sodium azide) were distributed to a 96-well, V-bottom plate (Greiner) at a density of 10^5 cells per well. A titration of mouse serum was prepared, diluting samples in FACS buffer. 50 μ L/well of this titration was then added to the cell plate. To determine the change in activity level due to immunisation, serum from each animal prior to immunisation was diluted to 1/100 in FACS buffer and 50 μ L/well added to the cells. Cells were incubated at 4° C. for 1 hour. Cells were washed twice with 150 μ L PBS, centrifuging after each wash step and aspirating supernatant (centrifuged at 300 $\times$ g for 3 minutes). To detect antibody binding, APC goat-anti-mouse IgG (Jackson ImmunoResearch) was diluted 1/500 in FACS buffer and 50 μ L was added to the cells. In some instances AF647 goat-anti-mouse IgG (Jackson ImmunoResearch) was used. Cells were incubated 1 hour at 4° C. in the dark, then washed twice with 150 μ L PBS as above. To fix cells, 100 μ L 2% v/v paraformaldehyde was added and cells incubated for 30 minutes at 4° C. Cells were then pelleted by centrifugation at 300 $\times$ g and the plates resuspended in 50 μ L of FACS buffer. Fluorescent signal intensity (geometric mean) was measured by flow cytometry using a BD FACS Array instrument.

Example 5: Sorting of Antigen-Specific B Cells by FACS

B-cells expressing anti-ICOS antibodies were recovered from immunised mice, using techniques substantially as described in Example 1 of WO2015/040401. In brief, splenocytes and/or lymph node cells isolated from the immunisation regimes were stained with an antibody cocktail containing markers for the selection of cells of interest (CD19), whereas unwanted cells were excluded from the final sorted population (IgM, IgD, 7AAD). CD19⁺ B-cells were further labelled with fluorescently-tagged human ICOS ECD-Fc dimers and fluorescently-tagged mouse ICOS ECD-Fc to detect B-cells producing anti-ICOS antibodies. Fluorescent labelling of human and mouse ICOS was with AlexaFluor647 and AlexaFluor488, respectively-see Example 6. Cells binding human ICOS, or both human and mouse ICOS were selected. These cells were single cell sorted by FACS into lysis buffer. V-region sequences were recovered using RT-PCR and two further rounds of PCR, then bridged to mouse IgG1 constant region and expressed in HEK293 cells. Supernatants from HEK293 cells were screened for the presence of ICOS binding and functional antibodies. This method is hereafter referred to as BCT.

Example 6: Screening of Antibodies from BCT

HTRF Screening of BCT Supernatants for Binding to Recombinant Human and Mouse ICOS-Fc

Supernatants collected from BCT in Example 5 were screened for the ability of secreted antibodies to bind to human ICOS Fc and mouse ICOS Fc expressed as recombinant proteins. Binding of secreted antibodies to recombinant human and mouse ICOS were identified by HTRF® (Homogeneous Time-Resolved Fluorescence, Cisbio) assay format using FluoProbes®647H (Innova Biosciences) labelled ICOS (referred to herein as 647 hICOS or 647

mICOS for human ICOS and mouse ICOS labelled with FluoProbes®647H respectively). 5 μ L BCT supernatant was transferred to a white 384-well, low-volume, non-binding surface polystyrene plate (Greiner). 5 μ L of 20 nM 647 hICOS or 647 mICOS diluted in HTRF assay buffer was added to all wells. For human ICOS binding assay the reference antibody was diluted in BCT media (Gibco #A14351-01) to 120 nM and 5 μ L added to plate. For negative control wells for human ICOS binding assay, 5 μ L of mouse IgG1 (Sigma M9269 in some instances referred to as CM7) diluted to 120 nM in BCT media. In the case of mouse ICOS binding assay the reference antibody was diluted in BCT media (Gibco #A14351-01) to 120 nM and 5 μ L added to plate. A rat IgG2b isotype control (R&D systems) was added to negative control wells (R&D Systems) diluted in BCT media to 120 nM and 5 μ L added to plate. Binding of secreted antibodies to human ICOS was detected by addition of 10 μ L of goat anti-mouse IgG (Southern Biotech) directly labelled with Europium cryptate (Cisbio) diluted 1/2000 in HTRF assay buffer. In the case of the mouse ICOS binding assay 5 μ L of mouse anti-Rat IgG2B-UBLB (Southern Biotech) was added to positive and negative control wells, and 5 μ L of HTRF assay buffer added to all other wells of plate. Then 5 μ L of goat anti-mouse IgG (Southern Biotech) directly labelled with Europium cryptate (Cisbio) diluted 1/1000 in HTRF assay buffer was added to detect binding. The plate was left to incubate in the dark for 2 hours prior to reading time resolved fluorescence at 620 nm and 665 nm emission wavelengths, 100 flashes, using an EnVision plate reader (Perkin Elmer).

Data were analysed by calculating 665/620 ratio and percent effect for each sample according to equation 2 and equation 1 respectively.

For KM103 and KM11-B1, primary hits were selected based on greater than or equal to 5 percent effect for binding to human and mouse ICOS. For KM135 primary hits were selected based on greater than or equal to 10 percent effect for binding to human and mouse ICOS. For KM111-B2 primary hits were defined as greater than or equal to 4 percent effect for binding to human and greater than or equal to 3 percent effect for binding to mouse ICOS.

Equation 1: Calculation of Percentage Effect from Primary Screen Envision Cell Binding and HTRF
Using Well Ratio Value (Equation 3) or 665/620 nm Ratio (See Equation 2) (HTRF)

$$\text{Percent effect} = \frac{(\text{sample well} - \text{non-specific binding}) \times 100}{(\text{total binding} - \text{non-specific binding})}$$

Non-specific binding =

values from wells containing isotype control mouse IgG1

Total Binding = values from wells containing reference antibody

Equation 2: Calculation of 665/620 Ratio

$$665/620 \text{ ratio} = (\text{sample } 665/620 \text{ nm value}) \times 10,000$$

Equation 3: Calculation of 647/FITC Ratio

Data were first normalised for cell number by dividing mAb channel (647) by FITC (cell stain) channel to give "well ratio value":

$$\text{Well Ratio Value} = \frac{647 \text{ Channel}}{\text{FITC Channel}}$$

Screening of BCT Supernatants for Binding to Cell-Expressed Human and Mouse ICOS

Supernatants collected from BCT in Example 5 were screened for the ability of secreted antibodies to bind to human or mouse ICOS expressed on the surface of CHO-S cells. To determine CHO-S human and mouse ICOS binding, cells were plated in black-walled, clear-bottom tissue culture treated 384-well plates (Perkin Elmer) at 4×10^4 /well in F12 media (Gibco) supplemented with 10% FBS (Gibco) and cultured overnight. Culture media was removed from 384-well assay plates. At least 50 μ L of BCT supernatant or 50 μ L reference antibody at 2 μ g/mL in BCT media or isotype IgG1 control antibody (referred to in some instances as Cm7, Sigma M9269), at a final concentration of 2 μ g/mL) diluted in BCT media were added to each well. Plates were incubated for 1 hour at 4° C. Supernatant was aspirated and 50 μ L of goat anti-Mouse 647 (Jackson immunoresearch) at 5 μ g/ml with vibrant green DNA stain (Life Technologies) diluted 1 in 500 in secondary antibody buffer (1xPBS+1% BSA+0.1% Sodium Azide) was added to detect antibody binding and visualise cells. Plates were incubated for 1 hr at 4 degrees. Supernatant was aspirated and 25 μ L of 4% v/v paraformaldehyde added and plates were incubated for 15 minutes at room temperature. Plates were washed twice with 100 μ L PBS and then the wash buffer was completely removed. Fluorescence intensity was measured using Envision plate reader (Perkin Elmer) measuring FITC (excitation 494 nm, emission 520 nm) and alexafluor 647 (excitation 650 nm, emission 668 nm). Assay signal was determined as described in equation 3 and percent effect as in equation 1. Total binding was defined using reference antibody at a final assay concentration of 2 μ g/mL. Non-specific binding was defined using mouse IgG1 isotype control (Sigma) at a final assay concentration of 2 μ g/mL. Criteria for hit selection were based on assay signal and percent effect.

For KM103, KM111-B1 and KM135, primary hits were selected based on greater than or equal to 10 percent effect. For KM111-B2, primary hits were selected based on greater than or equal to 4 percent effect.

Summary of Primary Screening Results

TABLE E6

Summary of number of BCT supernatants screened from immunisations, and number of supernatants meeting primary screening selection criteria for binding to human and mouse ICOS.		
Experiment ID	Supernatants screened	Primary hits selected
KM103	1232	40
KM111-B1	1056	198
KM111-B2	1056	136
KM135	704	31

FACS Screening for Binding to Cell Expressed Human and Mouse ICOS

BCT supernatants and HEK293 expressed antibodies from Example 5 were tested for ability to bind to CHO-S cells expressing human or mouse ICOS.

CHO-S cells expressing human or mouse ICOS (see Example 2), were diluted in FACS buffer (PBS 1% BSA 0.1% sodium azide) and were distributed to a 96-well, V-bottom plate (Greiner) at a density of 1×10^5 cells per well. Cells were washed with 150 μ L PBS and centrifuged at 300 g for 3 minutes. For supernatant screening, supernatant was aspirated and 150 μ L PBS added. This wash step was repeated. 30 μ L BCT undiluted supernatant or 50 μ L of reference antibody or control antibody diluted to 5 μ g/ml in

BCT media was added to the washed cells. Cells were incubated at 4° C. for 60 minutes. 150 μ L FACS buffer was added and cells washed as described above. To detect antibody binding, 50 μ L of goat anti-mouse APC (Jackson ImmunoResearch) diluted to 2 μ g/ml in FACS buffer was added to cells. Cells were incubated 4° C. for 60 minutes. Cells were washed twice with 150 μ L FACS buffer, centrifuging at 300 g for 3 minutes after each wash step and aspirating supernatant. Cells were fixed by addition of 25 μ L 4% paraformaldehyde for 20 minutes at room temperature. Cells were washed once as above and resuspended in FACS buffer for analysis. APC signal intensity (geometric mean) was measured by flow cytometry using a BD FACS Array instrument. Data were plotted as geometric mean values without further calculation.

A small sub-set of antibodies were selected as meeting more stringent species cross-reactivity criteria in this further screening compared with the primary screening. In brief:

From KM103, 4 antibodies were selected by taking the average geomean of the hybrid control binding to hICOS, mICOS and WT CHO cells and identifying mouse and human binders that were >4 fold above. These 4 antibodies were designated STIM001, STIM002-B, STIM007 and STIM009.

From KM111-B1, 4 antibodies were selected by taking the average of geomean of the negative control (Armenian hamster: clone HTK888) binding to hICOS, mICOS and WT CHO cells and identifying mouse and human binders that were >10 fold above.

From KM111-B2, 4 antibodies were selected by taking the average of geomean of the negative control (Armenian hamster: clone HTK888) binding to hICOS, mICOS and WT CHO cells and identifying mouse and human binders that were >4 fold above. These 4 antibodies included STIM003, STIM004 and STIM005.

From KM135, no cross-reactive antibodies were identified. Due to a technical failure of the FACS secondary screening method, screening was also carried out using SPR and HTRE, but no antibodies were found to meet the desired cross-reactivity level.

In conclusion, from the various multiple immunisation regimens described in Example 3, upward of 4000 BCT supernatants (from the ICOS KO mice only) were screened for binding to human ICOS and mouse ICOS, and a small panel of candidates, including STIM001, STIM002-B, STIM003, STIM004, STIM005, STIM007 and STIM009, were identified as having the most promising characteristics for further development. These were taken forward for more detailed characterisation.

Separately, two antibodies STIM006 and STIM008, which did not meet the species cross-reactivity criteria, were also chosen for further characterisation on the basis of their ability to bind human ICOS.

Example 7: Affinity Determination by Surface Plasmon Resonance (SPR)

Fab affinities of the ICOS leads were generated by SPR using the ProteOn XPR3 6 (BioRad). An anti-human IgG capture surface was created on a GLC biosensor chip by primary amine coupling, immobilising three anti-human IgG antibodies (Jackson Labs 109-005-008, 109-006-008 and 309-006-008). The human Fc tagged human ICOS (hICOS) and mouse ICOS (mICOS) were captured individually on the anti-human IgG surface and the purified Fabs were used as analytes at 5000 nM, 1000 nM, 200 nM, 40 nM and 8 nM, except for STIM003 which was used at 1000 nM, 200 nM,

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40 nM, 8 nM and 2 nM. Binding sensorgrams were double referenced using a buffer injection (i.e. 0 nM), and the data was fitted to the 1:1 model inherent to the ProteOn XPR36 analysis software. The assay was run at 25° C. and using HBS-EP as running buffer.

TABLE E7-1

Affinity and kinetic data for selected antibodies as measured by SPR.				
Sample Ab	Ligand	ka	kd	K _D (nM)
STIM006	hICOS	6.67E+05	9.20E-03	13.8
STIM003	hICOS	6.56E+05	8.62E-04	1.3
STIM001	hICOS	2.54E+04	1.12E-03	44.0
STIM002	hICOS	3.20E+04	3.43E-02	1070.0
STIM006	mICOS	1.57E+03	5.00E-04	318.0
STIM003	mICOS	1.29E+06	5.03E-04	0.4
STIM001	mICOS	5.66E+04	2.30E-02	407.0
STIM002	mICOS	weak	weak	weak

In addition, a comparison was performed of antibody: antigen binding affinity at different pH values. As before, the dimeric human ICOS protein, presented as the extracellular domain of ICOS fused to a human Fc region, was captured on the anti-human Fc capture surface created using the 3 antibody cocktail, immobilised on the GLC biosensor chip by primary amine coupling. SPR analysis of recombinantly expressed anti-ICOS Fabs was carried out on the ProteOn XPR36 Array system (Biorad). The Fab fragments were used as analyte to generate binding sensorgrams, which were double referenced with a buffer injection (i.e., 0 nM). The subsequent referenced sensorgrams were fitted to the 1:1 model inherent to the ProteOn analysis software. Table E7-2 presents affinity and kinetic data for the antibodies, all run at 37° C. unless stated, using either HBS-EP at pH 7.4/7.6 or pH 5.5 as indicated. Data were fitted to the 1:1 model. Note that data for STIM002 fitted poorly to the 1:1 model at both pH 7.4 and 5.5—the affinity for this antibody may therefore be lower than indicated in the table.

TABLE E7-2

Relative affinity of STIM001, STIM002, STIM002-B and STIM003 Fabs against recombinant human ICOS, at 37° C. except where stated.				
Antibody (Fab)	pH	ka	kd	KD (nM)
STIM001	7.4	5.08E+04	3.23E-03	63.5
	5.5	4.90E+04	3.58E-03	73.1
	7.6	8.29E+04	3.54E-03	42.6
	5.5	6.77E+04	5.41E-03	80.3
	7.6 (25° C.)	2.54E+04	1.12E-03	44
STIM002	7.4	3.72E+04	8.31E-03	223
	5.5	8.79E+04	3.67E-03	4.17
	7.4	8.28E+04	3.46E-03	41.8
STIM002-B	5.5	8.64E+04	2.30E-03	26.6
	7.4	1.49E+06	2.54E-03	1.71
STIM003	5.5	1.55E+06	1.58E-03	1.02
	7.6	1.87E+06	3.70E-03	1.98
	5.5	1.71E+06	1.94E-03	1.15
	7.6 (25° C.)	6.65E+06	0.862E-03	1.31

Comparison of the affinity data at different pH values indicated that the antibodies retain binding to their target across a physiological pH range. The tumour microenvironment may be relatively acidic compared with blood, thus maintenance of affinity at low pH is a potential advantage in vivo to improve intra-tumoural T-reg depletion.

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Example 8: Neutralisation of ICOS Ligand Binding to ICOS Receptor Assayed by HTRF

Selected anti-ICOS antibodies were further assessed for their ability to neutralise ICOS ligand (B7-H2) binding to ICOS, using homogenous time resolved fluorescence (HTRF). Human IgG1 and human IgG4.PE (null-effector) isotypes of the mAbs were assessed in:

HTRF assay for neutralisation of human B7-H2 binding to human ICOS; and

HTRF assay for neutralisation of mouse B7-H2 binding to mouse ICOS. Anti-ICOS antibody C398.4A (hamster IgG in each case) was included for comparison.

A number of antibodies were found to have high neutralising potency for human and/or mouse ICOS receptor-ligand binding, and the results indicated that some of these antibodies showed good cross-reactivity. The antibody isotype had no significant effect, differences in results between the IgG1 and IgG4.PE assays being within experimental error. IgG1

In the human IgG1 assays, antibody C398.4A produced an IC50 of 1.2±0.30 nM for the neutralisation of human ICOS ligand and an IC50 of 0.14±0.01 nM for the neutralisation of mouse ICOS ligand.

IgG1 mAbs STIM001, STIM002, STIM003 and STIM005 produced similar IC50 to C398.4A using the human ICOS ligand neutralisation system and were also cross-reactive, neutralising binding of mouse ICOS ligand to the mouse ICOS receptor.

Two additional cross-reactive mAbs, STIM002-B and STIM004, showed weaker human and mouse ICOS ligand neutralisation.

STIM006, STIM007, STIM008 and STIM009 showed neutralisation of human ICOS ligand but did not demonstrate significant cross-reactivity in the mouse ICOS ligand neutralisation system. Neutralising IC50 values for mouse B7-H2 ligand could not be calculated for these antibodies

TABLE E8-1

IC50 values for human IgG1 isotype mAb for neutralisation of human ICOS Receptor binding to human B7-H2. See also FIG. 2.		
	Mean IC50 (nM)	SD (nM) (n = 4)
STIM001	2.2	1.3
STIM002	1.9	0.8
STIM002-B	3.6	3.5
STIM003	1.3	0.5
STIM004	233	123
STIM005	2.5	0.8
STIM006	2.2	1.5
STIM007	1.1	0.5
STIM008	1.6	1.4
STIM009	30.5	53
C398.4A	1.2	0.3

TABLE E8-2

IC50 values for human IgG1 isotype mAb for neutralisation of mouse ICOS Receptor binding to mouse B7-H2. See also FIG. 3.		
	Mean IC50 (nM)	SD (nM) (n = 3)
STIM001	6.5	2.5
STIM002	6.9	2.1
STIM002-B	30	11.4

TABLE E8-2-continued

IC50 values for human IgG1 isotype mAb for neutralisation of mouse ICOS Receptor binding to mouse B7-H2. See also FIG. 3.		
	Mean IC50 (nM)	SD (nM) (n = 3)
STIM003	0.1	0
STIM004	22.1	15.4
STIM005	0.3	0.2
C398.4A	0.1	0

IgG4.PE

As expected, IgG4.PE mAbs produced similar results to the IgG1 isotypes. STIM001, STIM003 and STIM005 showed similar IC50 values to C398.4A using the human ICOS ligand neutralisation system. These mAbs were also cross-reactive at neutralising mouse ICOS ligand. STIM002-B and STIM004 produced weaker IC50 values for human ICOS B7-H2 neutralisation and mouse B7-H2 ligand. STIM007, STIM008 and STIM009 showed neutralisation of human ICOS ligand binding to human ICOS receptor but neutralising IC50 values for mouse B7-H2 ligand could not be calculated in these assays. IgG4.PE isotypes of STIM006 and STIM002 were not assayed.

TABLE E8-3

IC50 values for human IgG4.PE isotype mAb for neutralisation of human ICOS Receptor binding to human B7-H2. See also FIG. 4.		
	Mean IC50 (nM)	SD (nM) (n = 4 unless otherwise stated)
STIM001	1.3	0.2
STIM002-B	3.4	1.8
STIM003	1.2	0.3
STIM004	161	152 (n = 3)
STIM005	1.6	0.2
STIM006	0.8	(n = 1)
STIM007	0.8	0.1
STIM008	0.8	0.1
STIM009	4.6	2.2
C398.4A	2.8	3.8

TABLE E8-4

IC50 values for human IgG4.PE isotype mAb for neutralisation of mouse ICOS Receptor binding to mouse B7-H2. See also FIG. 5.		
	Mean IC50 (nM)	SD (nM) (n = 3)
STIM001	4.7	2.1
STIM002-B	43.9	25.7
STIM003	0.2	0.1
STIM004	30	14
STIM005	0.3	0.1
C398.4A	0.2	0.1

Materials and Methods

Test antibodies and isotype controls were diluted in assay buffer (0.53 M Potassium Fluoride (KF), 0.1% Bovine Serum Albumin (BSA) in 1xPBS) from a starting working concentration of up to 4 μ M, 1 μ M final to 0.002 nM, 5.64e-4 nM final over 11 point titration, 1 in 3 dilutions. Titrations of 5 μ l of antibody were added to 384w white walled assay plate (Greiner Bio-One). Positive control wells received 5 μ l of assay buffer only.

5 μ l of ICOS receptor (human ICOS-mFc, 20 nM, 5 nM final or mouse ICOS-mFc 4 nM, 1 nM final (Chimerigen))

was added to required wells. Plate was incubated for 1 hr at room temperature (RT). After incubation, 5 μ l of either mouse or human ICOS ligand, (B7-H2, R&D Systems) conjugated to Alexa 647 (Innova Bioscience) was diluted to either 32 nM (8 nM final) for human B7-H2 or 30 nM, 7.5 nM final for mouse B7-H2 and added to all wells of assay plate except negative control wells which instead received 5 μ l of assay buffer.

Finally, 5 μ l of anti-mouse IgG donor mAb (Southern Biotech) labelled with europium cryptate (Cis Bio), 40 nM, 10 nM final was added to each well and the assay was left in the dark at RT to incubate for a further 2 hours. After incubation, assay was read on Envision plate reader (Perkin Elmer) using a standard HTRF protocol. 620 nm and 665 nm channel values were exported to Microsoft Excel (Microsoft) and % Delta-F and % Neutralisation calculations performed. Titration curves and IC50 values [M] were plotted using Graphpad (Prism). IC50 values were calculated by first transforming the data using equation $X = \text{Log}(X)$. The transformed data was then fitted using nonlinear regression, using fitting algorithm, log (inhibitor) vs. response-variable slope (four parameters).

% Delta F =

$$\frac{(665/620 \text{ nm Well Signal Ratio} - \text{Signal Negative Control})}{(\text{Signal Negative Control})} * 100$$

Signal Negative control = average of minimum signal ratio.

% Neutralisation:

% Max (neutralisation) =

$$\frac{(\% \text{ Delta} - F \text{ of sample well} - \text{Negative Control})}{(\text{Positive Control} - \text{Negative Control})} * 100$$

Example 9a: T-Cell Activation

STIM001 and STIM003 agonistic potential on cytokine production was tested in plate-bound and soluble format in a human primary T-cell activation assay where anti-CD3 and anti-CD28 Abs were added concurrently to the anti-ICOS Ab to induce ICOS expression on effector T-cells. Effect of the ICOS co-stimulation on the level of IFN γ produced by these activated T-cells were assessed using ELISA at 72 hrs post-activation.

Materials and Methods

T-Cell Activation Assay 1:

Isolation of Mononuclear Cells from Human Peripheral Blood (PBMC):

Leukocyte cones were collected from healthy donors and their content was diluted up to 50 ml with phosphate buffered saline (PBS, from Gibco) and layered into 2 centrifuge tubes on top of 15 mL Ficoll-Paque (from GE Healthcare). PBMC were separated by density gradient centrifugation (400 g for 40 min without brake), transferred to a clean centrifuge tube and then washed with 50 mL PBS, twice by centrifuging at 300 g for 5 min and twice by centrifuging at 200 g for 5 min. PBMC were then resuspended in R10 media (RPMI+10% heat-inactivated Fetal

Bovine Serum, both from Gibco) and their cell count and viability assess with EVE™ Automated Cell Counter (from NanoEnTek).

ICOS Antibodies (Abs) Preparation and Dilutions:

STIM001 and STIM003 were tested in 3 formats: plate-bound, soluble or soluble plus F(ab')₂ Fragments (109-006-170 from Jackson Immuno Research) which crosslink the anti-ICOS Abs.

For plate-bound format: the anti-ICOS Abs and their isotype control were serially diluted 1:3 in PBS to give final antibody concentrations ranging from 45 to 0.19 µg/mL. 100 µL of diluted antibodies were coated in duplicate into a 96-well, high-binding, flat-bottom plate (Corning EIA/RIA plate) overnight at 4° C. Plate was then washed with PBS and 125 µL of R10 were added to each well.

For soluble format: The anti-ICOS Abs and their isotype control were serially diluted 1:3 in R10 media to give an 2×Ab stock concentrations ranging from 90 to 0.38 µg/mL. 125 µL of diluted Abs were pipetted in duplicate into a 96-well, flat-bottom plate.

For crosslinked soluble format: The anti-ICOS Abs and their isotype control were mixed with F(ab')₂ Fragments at 1M to 1M ratio. Abs/F(ab')₂ Fragments mixes were then 1:3 serially diluted in R10 media to give an 2×Ab concentrations ranging from 90 to 0.38 µg/mL for ICOS and from 60 to 0.24 µg/mL for F(ab')₂ Fragments. 125 µL of diluted Abs were pipetted in duplicate into a 96-well, flat-bottom plate.

T-Cell Isolation, Cultures and IFN-γ Quantification:

T-cell were negatively isolated from PBMC using the EasySep Human T Cell Isolation Kit (from Stemcell Technologies) and resuspended at 2×10⁶/mL in R10 media supplemented with 40 µL/ml of Dynabeads Human T-Activator CD3/CD28 (from Life Technologies).

125 µL of T-cell suspension were added to Ab-containing plates to give a final cell concentration of 1×10⁶ cells/mL and cultured for 72 hrs at 37° C. and 5% CO₂. Cell free supernatants were then collected and kept at -20° C. until analysis of secreted IFNγ by ELISA (duoset kit from R&D).

This experiment was repeated on T-cells isolated from 6 independent donors and 2 technical replicates were included for each assay condition.

T-Cell Activation Assay 2 (STIM-REST-STIM Assay):

STIM001 and STIM003 agonist potential on cytokine release were also tested plate-bound in a human T-cell assay where T-cells were prestimulated by anti-CD3 and anti-CD28 Abs for 3-days to induce ICOS expression before being rested for 3-days to reduce their activation levels. ICOS expression was confirmed by FACS staining after stimulation (Day 3) and resting (Day 6). These stimulated rested T-cells were then cultured with STIM001 or STIM003 in presence or absence of CD3 Ab to assess the requirement of TCR engagement. Effects of the ICOS co-stimulation were assessed after 72 hrs on the levels of IFNγ, TNFα and IL-2 present in the culture.

ICOS Abs Dilutions and Coating:

Anti-human CD3 (clone UCHT1 from eBioscience) was diluted in PBS to a 2×Ab concentration of 10 µg/mL. 50 µL of PBS or 50 µL of diluted CD3 Ab were pipetted into a 96-well, high-binding, flat-bottom plate. STIM001, STIM003 and their isotype control were 1:2 serially diluted in PBS to give final 2× antibody concentrations ranging from 20 to 0.62 µg/mL. 50 µL of diluted anti-ICOS Ab were added to wells containing either PBS (no TCR engagement) or diluted CD3 Ab (TCR engagement). Plates were coated overnight at 4° C.

T-Cell Isolation, Cultures and IFN-γ Quantification:

PBMC from leukocyte cones were obtained as described in T-cell activation assay 1. T-cell were negatively isolated from this PBMC using the EasySep Human T Cell Isolation Kit (from Stemcell Technologies). T-cells were resuspended at 1×10⁶/mL in R10 media supplemented with 20 µL/mL of Dynabeads Human T-Activator CD3/CD28 (from Life Technologies) and cultured for 3-days at 37° C. and 5% CO₂ (Stimulation). At day 3 dynabeads were removed from the culture. T-cells were then washed (300 g for 5 min), counted and resuspended at 1.5×10⁶/mL in R10 media and culture at 37° C. and 5% CO₂ for 3-more days (Resting phase).

At day 6 stimulated rested T-cells were then washed (300 g for 5 min), counted and resuspended at 1×10⁶/mL in R10 media and 250 µL of T-cell suspension were added to ICOS Ab-coated plates and cultured for 72 hrs at 37° C. and 5% CO₂. Cell free supernatants were then collected and kept at -20° C. until analysis of secreted cytokines on the MSD platform. This experiment was repeated with T-cells isolated from 5 independent donors and 3 technical replicates were included for each assay condition.

Results

Both STIM001 and STIM003 tested positive for inducing IFNγ expression therefore demonstrating agonism in both assays.

Example 9b: T Cell Activation Assay 1 Data

T cell activation assay 1 was performed as described in Example 9a, using T cells isolated from 8 independent donors, testing each of STIM001 and STIM003 in human IgG1 format. Hamster anti-ICOS antibody C398.4A and a hamster antibody isotype control were included for comparison. 2 technical replicates were included for each assay condition.

Results are shown in FIG. 16, FIG. 17 and FIG. 18. As noted before, both STIM001 and STIM003 tested positive for inducing IFNγ expression therefore demonstrating an agonistic effect on human primary T cells.

Cross-linked antibodies acted as agonists of T cell activation, as indicated by the strong enhancement of IFNγ induction in the presence of the Fc-linking F(ab')₂ fragments, compared with either soluble antibody or with control. IFNγ expression in the T cells increased with increasing concentration of cross-linked STIM001 or STIM003 (FIG. 16, lower panels). Agonism was also observed for both STIM001 and STIM003 in plate-bound form and, more weakly, for the hamster antibody C398.4A, as indicated by the increase in IFNγ expression observed in the T cells with increasing concentration of antibody (FIG. 16, top panels).

Magnitude of the IFNγ response varied between T cells obtained from different donors, but STIM001 consistently produced an increase in IFNγ expression in T cells compared with IFNγ expression observed with control antibody (HC IgG1). When considering data from assays with T cells from all 8 donors, it is seen that treatment of T cells with STIM001 significantly increased IFNγ expression compared with treatment with isotype control antibody, in plate-bound form, soluble form and cross-linked form (FIG. 17). STIM001 thus behaved as an agonist of T cell activation in all three formats.

Similar effects were observed with STIM003 (FIG. 18). Levels of IFNγ induced by STIM003 hIgG1 were compared with levels of IFNγ induced by its isotype control (HC IgG1) at a given dose of antibody in the assay, for 8 independent donors. Despite the variability between donors, the mean increase in IFNγ level induced by STIM003 was significant

when compared against HC IgG1. It is proposed that STIM001, and the other STIM antibodies described here, have the potential to similarly promote T cell activation in vivo. As discussed previously, agonism of activated ICOS-expressing T cells may be mediated by the anti-ICOS antibody binding to and inducing multimerisation of the ICOS receptor on the T cell surface. Example 9c: T cell activation assay 2 data

T cell activation assay 2 was performed as described in Example 9a.

In the absence of TCR engagement (no anti-CD3 antibody), levels of cytokines produced from the primary T cells were low and no increase was induced by STIM001 (hIgG1), STIM003 (hIgG1) or antibody C398.4A even at the highest concentration of 10 µg/ml. In contrast, when the anti-ICOS antibodies were added to T cells in combination with the anti-CD3 antibody, each of STIM001 (hIgG1), STIM003 (hIgG1) and C398.4A showed a dose-dependent trend to increase expression of IFN γ , TNF α and, to a lesser degree, IL-2.

Data from primary T cells treated with anti-ICOS antibodies under conditions of TCR engagement are shown in FIG. 19. Although marked increases in cytokine expression were observed for each of STIM001, STIM003 and C398.4A relative to their respective isotype controls, the difference did not reach statistical significance in this assay. Further replicates of the assay with responsive primary T cells from more donors would be expected to generate statistically significant results.

Example 10a: ADCC Assay

STIM001 and STIM003 potential to kill via ADCC was tested in the Delfia BATDA cytotoxicity assay (Perkin Elmer) using human primary NK cells as effector and ICOS high MJ cell line (ATCC, CRL-8294) as target cells. MJ cells are human CD4 T-lymphocyte cells that express high levels of ICOS protein.

This method is based on loading target cells with an acetoxymethyl ester of fluorescence enhancing ligand (BATDA) which quickly penetrates the cell membrane. Within the cell the ester bonds are hydrolysed to form a hydrophilic ligand (TDA) which no longer passes the membrane. After cytolysis the ligand is released and can be detected by addition of Europium which forms with the BATDA a highly fluorescent and stable chelate (EuTDA). The measured signal correlates directly with the amount of lysed cells.

Materials and Methods

Target Cell Labelling:

According to the manufacturer's instructions, MJ cells were resuspended at 1×10^6 /mL in assay media (RPMI+10% ultra-low IgG FBS, from Gibco) and loaded with 5 µL/mL of Delfia BATDA reagent (Perkin Elmer) for 30 min at 37° C. MJ cells were then washed 3 times with 50 mL PBS (300 g for 5 min) and resuspended at 8×10^5 /mL in assay media supplemented with 2 mM Probenecid (from Life technologies) to reduce BATDA spontaneous release from the cells. ICOS Ab Dilution:

STIM001, STIM003 and their isotype control were 1:4 serially diluted in assay media +2 mM Probenecid to give final 4 $\times$ antibody concentrations across a range down to 80 pg/mL.

NK-Cell Isolation and Culture:

PBMC from leukocyte cones were obtained as described in T-cell activation assay 1. NK-cell were negatively isolated from this PBMC using the EasySep Human NK Cell Isolation Kit (from Stemcell Technologies) and resuspended at 4×10^6 /mL in R10 media +2 mM Probenecid. NK cell purity was checked to be above 90% by staining for CD3-/CD56+.

50 µL of diluted Ab, 50 µL of BATDA loaded MJ cells, 50 µL of NK cells and 50 µL of assay media +2 mM Probenecid (final volume of 200 µL/well) were added in each well to give a final Ab concentration across a range down to 20 µg/mL and an effector: target ratio of 5:1. Wells containing MJ cells only or MJ cells+delfia lysis buffer (Perkin Elmer) are used to determine spontaneous and 100% BATDA release.

The assay was run at 37° C., 5% CO $_2$ for 2 hrs before transferring 50 µL of cell-free supernatant into a DELFIA Microtitration Plates (Perkin Elmer). 200 µL of Delfia Europium solution (Perkin Elmer) was added to the supernatants and incubated for 15 min at Room Temperature. Fluorescent signal was then quantified with EnVision Multilabel Reader (PerkinElmer).

Specific release induced by STIM001 and STIM003 was calculated according to the kit instructions. This experiment was repeated with NK-cells from independent donors and 3 technical replicates were included for each assay condition. Results

Anti-ICOS antibodies STIM001 (hIgG1) and STIM003 (hIgG1) kill ICOS positive human MJ cells in a primary NK dependent ADCC assay (2 hour time point). See also FIG. 6a. Sub-Nanomolar EC50 were achieved in this assay for both molecules tested.

TABLE E10-1

EC50 (Molar unit) of STIM001 in the NK primary cells ADCC assay from 2 donors (2 hour time point).		
EC50	Donor 1	Donor 2
STIM001	1.21e-10	5.29e-10

Example 10b: ADCC Assay with MJ Target Cells

The experiment was performed according to the Materials and Methods set out in Example 10a. STIM001, STIM003 and isotype control were 1:4 serially diluted in assay media+2 mM Probenecid to give final 4 $\times$ antibody concentrations ranging from 40 µg/mL to 80 µg/mL. 50 µL of diluted Ab, 50 µL of BATDA loaded MJ cells, 50 µL of NK cells and 50 µL of assay media +2 mM Probenecid (final volume of 200 µL/well) were added in each well to give a final Ab concentration ranging from 10 µg/mL to 20 µg/mL and an effector: target ratio of 5:1.

Results are shown in FIG. 6 (b-d) and in the table below. STIM001 (hIgG1) and STIM003 (hIgG1) killed ICOS positive human MJ cells in the primary NK dependent ADCC assay, measured at the two hour time point.

TABLE E10-2

EC50 (Molar unit) of STIM001 and STIM003 in the NK primary cell ADCC assay from 3 donors (2 hour time point).			
EC50	Donor 1	Donor 2	Donor 3
STIM001	1.21e-10 (0.121 nM)	5.29e-10 (0.529 nM)	2.92e-09 (2.92 nM)

TABLE E10-2-continued

EC50 (Molar unit) of STIM001 and STIM003 in the NK primary cell ADCC assay from 3 donors (2 hour time point).			
EC50	Donor 1	Donor 2	Donor 3
STIM003	2.33e-12 (2.33 pM)	3.58-e-11 (35.8 pM)	1.01e-10 (0.101 nM)

Example 10c: ADCC Assay with ICOS-Transfected CCRF-CEM Target Cells

STIM001 and STIM003 hIgG1 potential to kill via ADCC was further tested in the Delfia BATDA cytotoxicity assay (Perkin Elmer) using human primary NK cells as effector and ICOS-transfected CCRF-CEM cells (ATCC, CRL-119) as target cells. CCRF-CEM is a human T lymphoblast line, originating from peripheral blood from a patient with acute lymphoblastic leukaemia. Antibody-mediated killing of CCRF-CEM cells was confirmed for both STIM001 and STIM003 in this assay.

Materials and Methods

Materials and Methods were as set out in Example 10a, but using CCRF-CEM cells obtained from ATCC (ATCC CCL-119) rather than MJ cells as the target cells, and using an incubation time of 4 hours.

CCRF-CEM cells were transfected with ICOS. A synthetic string DNA encoding full length human ICOS (with signal peptide, as shown in the appended sequence listing), codon-optimised for mammalian expression, was cloned into an expression vector under control of the CMV promoter and flanked by 3' and 5' piggyBac specific terminal repeat sequences facilitating stable integration into the cell genome (see [40]). The expression vector contained a puromycin selection cassette to facilitate stable cell line generation. The human ICOS expression plasmid was co-transfected with a plasmid encoding piggyBac transposase into CEM CCRF cells by electroporation. 24 hours after transfection, the media was supplemented with puromycin and grown for at least two weeks to select stable cell lines, with media being exchanged every 3-4 days. The expression of human ICOS was assessed by flow cytometry using an anti-human ICOS-PE conjugated antibody (eBioscience). Complete CEM media was made up of Advanced RPMI Medium containing 10% (v/v) FBS and 2 mM Glutamax.

STIM001 (hIgG1), STIM003 (hIgG1) and an isotype control antibody (HC IgG1) were serially diluted in assay media to give final 4x antibody concentrations ranging from 20 µg/mL to 80 µg/mL.

50 µl of diluted Ab, 50 µl of BATDA loaded ICOS-transfected CEM cells, 50 µl of NK cells and 50 µl of assay media (final volume of 200 µl/well) were added in each well to give a final Ab concentration ranging from 5 µg/mL to 20 µg/mL and an effector: target ratio of 5:1.

Results

STIM001 (hIgG1) and STIM003 (hIgG1) killed ICOS-transfected CCRF-CEM cells in the primary NK dependent ADCC assay, measured at the four hour time point. Results are shown in FIG. 6 (e-g) and in the table below.

TABLE E10-3

EC50 (Molar unit) of STIM001 and STIM003 in the NK primary cell ADCC assay from 3 donors (4 hour time point).			
EC50	Donor 4	Donor 5	Donor 6
STIM001	3.92e-12 (3.92 pM)	3.95e-12 (3.95 pM)	3.75e-12 (3.75 pM)
STIM003	Approx 3 pM*	8.95e-13 (0.895 pM)	1.03e-12 (1.03 pM)

*Value estimated from incomplete curve.

Example 11a: CT26 Syngeneic Model

Improved anti-tumour in vivo efficacy was shown in a CT26 syngeneic model by combining anti-ICOS (STIM001 mIgG2a, effector enable) with anti-PDL1 (10F9G2).

Materials and Methods

Efficacy studies were performed in Balb/c mice using the sub-cutaneous CT26 colon carcinoma model (ATCC, CRL-2638). This model is poorly sensitive to PD1/PDL1 blockade and only tumour growth delay (no stable disease or cure) is usually observed in response to 10F9.G2 (anti-PDL1) and RMT1-14 (anti-PD1) monotherapies. Therefore this model constitutes a relevant model for looking at anti-PD1, anti-PDL1 intrinsic resistance for combination studies. All in vivo experiments were performed in accordance with the UK Animal (Scientific Procedures) Act 1986 and the EU Directive 86/609, under a UK Home Office Project Licence and approved by the Babraham Institute Animal Welfare and Ethical Review Body.

Balb/c mice were supplied by Charles River UK at 6-8 weeks of age and >18 g and housed under specific pathogen-free conditions. A total of 1x10⁵ CT26 cells (passage number below P20) were subcutaneously injected into the left flanks of mice. Unless stated otherwise, treatment were initiated at day 6 post tumour cells injection. The CT26 cells were passaged in vitro by using Accutase (Sigma), washed twice in PBS and resuspended in RPMI supplemented with 10% fetal calf serum. Cell viability was confirmed to be above 90% at the time of tumour cell injection.

For in vivo studies STIM001 anti-ICOS agonist (cross reactive to mouse ICOS protein) was reformatted as mouse IgG1 and mouse IgG2a to test the as effector function null and as effector function enable, respectively. The Anti-PDL1 was sourced from Biolegend (Cat. no: 124325). The hybrid controls were generated in Kymab (mIgG2a isotype) or from commercial source (hamster isotype HTK888, Biolegend (Part No92257, Lot B215504)). All antibodies were dosed intraperitoneal (IP) at 10 mg/kg (1 mg/ml in 0.9% saline) three times a week from day 6 (dosing for 2 weeks day 6-18) as monotherapies or by combining anti-PDL1 with anti-ICOS antibodies. Animal weight and tumour volume were measured 3 times per week from the day of tumour cell injection. Tumour volume was calculated by use of the modified ellipsoid formula $\frac{1}{2}(\text{Length} \times \text{Width}^2)$. Mice were kept on studies until their tumour reached an average diameter of 12 mm³ or, in rare case, when incidence of tumour ulceration was observed (welfare). The experiment was stopped at day 50. The human endpoint survival statistics were calculated using the Kaplan-Meier method with Prism. This approach was used to determine if specific treatments were associated with improved survival.

TABLE E11-1

Treatment groups		
Groups	Number of animals	Treatments (T.I.W, IP from day 6)
1	10	10 mg/kg mIgG2a and 10 mg/kg IgG isotype Control (HTK888)
2	10	10 mg/kg Anti-ICOS STIM001 mIgG1
3	10	10 mg/kg Anti-ICOS STIM001 mIgG2a
4	10	10 mg/kg Anti-PD-L1 (10F9.G2)
5	10	10 mg/kg anti-PD-L1 plus 10 mg/kg Anti-ICOS STIM001 mIgG1
6	10	10 mg/kg anti-PD-L1 plus 10 mg/kg Anti-ICOS STIM001 mIgG2a

Results

As shown in FIG. 7, FIG. 8 and FIG. 9, ICOS agonists can delay disease progression and cure a proportion of animals from the CT-26 subcutaneous tumours either as a monotherapy or in combination with anti-PDL1. Anti-PDL1 monotherapy induced tumour growth delay but no stable disease or curative potential was observed. The combination was more effective at treating the tumours than the anti-ICOS monotherapies. This study also highlighted that STIM001 in the mouse IgG2a format (effector function enable) was more potent than the mouse IgG1 (effector null) format at triggering an anti-tumour response in this model.

Example 11b: Strong Anti-Tumour Efficacy In Vivo in CT26 Syngeneic Model for Combination of Anti-ICOS mIgG2a with Anti-PDL1 mIgG2a

An in vivo combination study was performed with STIM001 with a mouse cross reactive anti-human PDL1 antibody designated AbW. For this in vivo work, STIM001 was reformatted as mouse IgG1 and mouse IgG2a to compare its efficacy with low effector function or as effector function enabled molecule, respectively. The anti-PDL1 AbW was generated in the same formats (mouse IgG1 and mouse IgG2A).

The efficacy studies were performed in Balb/c mice using the sub-cutaneous CT26 colon carcinoma model (ATCC, CRL-2638). Balb/c mice were supplied by Charles River UK at 6-8 weeks of age and >18 g and housed under specific pathogen-free conditions. A total of 1×10E5 CT26 cells (passage number below P20) were subcutaneously injected into the right flanks of mice. Unless stated otherwise, treatment was initiated at day 6 post tumour cells injection. The CT26 cells were passaged in vitro by using TrypLET Express Enzyme (Thermofisher), washed twice in PBS and resuspended in RPMI supplemented with 10% foetal calf serum. Cell viability was confirmed to be above 90% at the time of tumour cell injection.

STIM001 and anti-PDL1 antibodies were dosed concurrently in combinations intraperitoneal (IP) at 200 µg each (1 mg/ml in 0.9% saline) three times a week from day 6 (dosing for 2 weeks between day 6-17) post tumour cell implantation. Tumour growth was monitored and compared to tumours of animals treated with a mixture of isotype control antibodies (mIgG1 and mIgG2A). Animal weight and tumour volume were measured 3 times a week from the day of tumour cell injection. Tumour volume was calculated by use of the modified ellipsoid formula $\frac{1}{2}(\text{Length} \times \text{Width}^2)$. Mice were kept on studies until their tumour reached an average diameter of 12 mm³ or, in rare case, when incidence of tumour ulceration was observed (welfare). The experiment was stopped at day 60. The human endpoint survival

statistics were calculated using the Kaplan-Meier method with Prism. This approach was used to determine if specific treatments were associated with improved survival.

TABLE E11-2

Treatment groups for STIM001 2 × 2 combinations		
Group	Number of animals	Treatment regimen (3 time a week for 2 weeks)
1	10	mIgG2a + mIgG1 isotypes 200 µg each
2	10	Anti-ICOS mIgG1 STIM001 + Anti-PD-L1 mIgG1 (AbW) 200 µg each
3	10	Anti-ICOS mIgG2a STIM001 + Anti-PD-L1 mIgG2a (AbW) 200 µg each
4	10	Anti-ICOS mIgG2a STIM001 + Anti-PD-L1 mIgG1 (AbW) 200 µg each
5	10	Anti-ICOS mIgG1 STIM001 + Anti-PD-L1 mIgG2a (AbW) 200 µg each

Results are shown in FIG. 10. All antibody combinations delayed tumour growth and extended the survival (time to reach human endpoint) of treated animals when compared to isotype control-treated animals. Interestingly, when combined with anti-PDL1 (independently of its format, mIgG1 or mIgG2a), STIM001 mIgG2a antibody was more effective at inhibiting tumour growth than STIM001 in the mIgG1 format. These data suggest the advantage of an anti-ICOS antibody having effector function to maximize anti-tumour efficacy. Notably, STIM001 mIgG2a in combination with aPD-L1 mIgG2a demonstrated the strongest anti-tumour efficacy and improved survival (90% of animals to show response and 60% were cured from the disease at day 60).

Similarly, STIM003 mIgG1 and mIgG2a were tested as monotherapy or in combination with anti-PDL1 (AbW) mIgG2a in the same CT26 tumour models. STIM003 and anti-PDL1 antibodies were dosed in animals as monotherapy or in combination by intraperitoneal injection (IP) of 200 µg of antibodies each (1 mg/ml in 0.9% saline) three times a week from day 6 (dosing for 2 weeks between day 6-17) post tumour cell implantation. In this experiment tumour sizes were monitored for 41 days. The human endpoint survival statistics were calculated using the Kaplan-Meier method with Prism. This approach was used to determine if specific treatments were associated with improved survival.

TABLE E11-3

Treatment groups for STIM003 combination with anti-PDL1 AbW IgG2a		
Group	Number of animals	Treatment regimen (3 times a week for 2 weeks from day 6)
1	10	mIgG2a + mIgG1 isotypes control 200 µg each
2	10	Anti-PD-L1 mIgG2a (AbW) 200 µg
3	10	STIM003 mIgG1 200 µg
4	10	STIM003 mIgG2a 200 µg
5	10	STIM003 mIgG1 + Anti-PD-L1 mIgG2a (AbW) 200 µg each
6	10	STIM003 mIgG2a + Anti-PD-L1 mIgG2a (AbW) 200 µg each

Results are shown in FIG. 11. Monotherapies using aPD-L1 (AbW) and STIM003 mIgG2a demonstrated mild anti-tumour activity (one animal was cured of the disease in each group). Combinations of STIM003 mIgG1 or mIgG2a with aPD-L1 (AbW) mIgG2a showed strong anti-tumour efficacy. Interestingly, by day 41, when combined with aPD-L1 mIgG2a, STIM003 mIgG2a was more potent at inhibiting tumour growth than STIM003 mIgG1 (60% vs

30% of animals cured of the disease, respectively). The data further highlighted the advantage of an effector format for anti-ICOS antibodies to maximize anti-tumour efficacy.

Altogether, these data demonstrate that combination of an anti-ICOS antibody STIM001 or STIM003 with anti-PDL1 results in the strongest anti-tumour response when both antibodies have an effector enabled function. Suitable corresponding human antibody isotypes would include human IgG1, optionally with further enhanced effector function e.g., afucosylated IgG1.

Kaplan Meier curves for mice treated with the combination of anti-PDL1 mIgG2a and STIM003 mIgG2a and for each agent individually are shown in FIG. 29.

Example 11c: Single Dose of STIM003 Antibody Resets the Tumour Microenvironment (TME) and Results in Strong Anti-Tumour Efficacy in Combination with Continuous Anti-PD-L1 Dosing

This study compared single vs multiple dosing of STIM003 mIgG2A together with multiple dosing of anti-PDL1 antibody (AbW). The data indicate that a single dose of anti-ICOS antibody could alter the tumour microenvironment so as to allow an anti-PD-L1 antibody to exert a greater effect. This can be envisaged as a “resetting” of the TME by the anti-ICOS antibody.

As before, these efficacy studies were performed in Balb/c mice using the sub-cutaneous CT26 colon carcinoma model (ATCC, CRL-2638). Balb/c mice were supplied by Charles River UK at 6-8 weeks of age and >18 g and housed under specific pathogen-free conditions. A total of 1×10^5 CT26 cells (passage number below P20) were subcutaneously injected into the right flanks of mice. Unless stated otherwise, treatments were initiated at day 6 post tumour cells injection. The CT26 cells were passaged in vitro by using TrypLE™ Express Enzyme (Thermofisher), washed twice in PBS and resuspended in RPMI supplemented with 10% foetal calf serum. Cell viability was confirmed to be above 90% at the time of tumour cell injection.

Treatment groups are shown in Table E11-4. STIM003 and anti-PDL1 antibodies were dosed intraperitoneal (IP) at 10 mg/kg (1 mg/ml in 0.9% saline). Treatments were initiated from day 6 post tumour cell implantation. Tumour growth was monitored and compared with tumours of animals treated with saline. Animal weight and tumour volume were measured 3 times a week from the day of tumour cell injection. Tumour volume was calculated by use of the modified ellipsoid formula $\frac{1}{2}(\text{Length} \times \text{Width}^2)$. Mice were kept on studies until their tumour reached an average diameter of 12 mm³ or, in rare case, when incidence of tumour ulceration was observed (welfare). The experiment was stopped at day 55.

Data are shown in FIG. 34. Concurrent dosing of STIM003 and anti-PDL1 for 6 doses from day 6 resulted in a strong anti-tumour efficacy in the CT26 model with 5/8 animals being tumour free at the end of the study (day 55). Interestingly, similar anti-tumour efficacy was achieved with a single dose of STIM003 followed by multiple dose of anti-PDL1 as monotherapy. When combined with anti-PD-L1 mIgG2a, similar overall efficacy was observed between dosing STIM003 once (C) vs dosing 6 times (B). When compared with saline treated group (A) where only one animal had a spontaneous tumour rejection (rare in this model), the groups treated with combined drugs had full tumour rejection in 62.5% of the animals by the end of the experiment (day 55). The data suggest that the STIM003 antibody could be used to reset the tumour microenviron-

ment and that the antibody allows immune-checkpoint resistant tumours to become sensitive to anti-PDL1. As previously shown (Example 11b), the CT26 tumour cell line is not strongly responsive to anti-PDL1 monotherapy. It appears that STIM003 causes changes that facilitate anti-tumour activity of the anti-PDL1 therapy.

Example 12: Antibody Sequence Analysis

Framework regions of antibodies STIM001, STIM002, STIM002-B, STIM003, STIM004, STIM005, STIM006, STIM007, STIM008 and STIM009 were compared with human germline gene segments to identify the closest match. See Table E12-1 and Table E12-2.

TABLE E12-1

Heavy chain germline gene segments of anti-ICOS Abs			
Heavy chain	V	D	J
STIM001	IGHV1-18*01	IGHD6-19*01	IGHJ6*02
STIM002	IGHV1-18*01	IGHD3-10*01	IGHJ6*02
STIM002-B	IGHV1-18*01	IGHD3-10*01	IGHJ6*02
STIM003	IGHV3-20*d01	IGHD3-10*01	IGHJ4*02
STIM004	IGHV3-20*d01	IGHD3-10*01	IGHJ4*02
STIM005	IGHV1-18*01	IGHD3-9*01	IGHJ3*02
STIM006	IGHV3-11*01	IGHD3-10*01	IGHJ6*02
STIM007	IGHV2-5*10	IGHD3-10*01	IGHJ6*02
STIM008	IGHV2-5*10	IGHD3-10*01	IGHJ6*02
STIM009	IGHV3-11*01	IGHD3-9*01	IGHJ6*02

TABLE E12-2

Kappa light chain germline gene segments of anti-ICOS Abs		
Light chain	V	J
STIM001	IGKV2-28*01	IGKJ4*01
STIM002	IGKV2-28*01	IGKJ2*04
STIM002-B	IGKV2-28*01	IGKJ2*04
STIM003	IGKV3-20*01	IGKJ3*01
STIM004	IGKV3-20*01	IGKJ3*01
STIM005	IGKV1D-39*01	IGKJ1*01
STIM006	IGKV2-28*01	IGKJ2*04
STIM007	IGKV3-11*01	IGKJ4*01
STIM008	IGKV3-11*01	IGKJ4*01
STIM009	IGKV2-28*01	IGKJ1*01

Additional antibody sequences were obtained by next generation sequencing of PCR-amplified antibody DNA from further ICOS-specific cells that were sorted from the immunised mice as described in Example 3. This identified a number of antibodies that could be grouped into clusters with STIM001, STIM002 or STIM003 based their heavy and light chain v and j gene segments and CDR3 length. CL-61091 (STIM017) clustered with STIM001; CL-64536, CL-64837, CL-64841 and CL-64912 clustered with STIM002; and CL-71642 and CL-74570 (STIM022) clustered with STIM003. Sequence alignments of the antibody VH and VL domains are shown in FIGS. 35 to 37.

TABLE E12-3

Antibodies clustered by sequence.						
ANTIBODIES	VH_V_GENE	VH_J_GENE	VH_CDR3_NT_LENGTH	VL_V_GENE	VL_J_GENE	VL_CDR3_NT_LENGTH
STIM001, CL-61091 (STIM017)	1-18	6	42	2-28	4	27
STIM002, CL-64536, CL-64837, CL-64841, CL-64912	1-18	6	51	2-28	2	27
STIM003, CL-71642, CL-74570 (STIM022)	3-20	4	51	3-20	3	27
STIM004	3-20	4	51	3-20	3	24
STIM005	1-18	3	51	1D-39	1	24
STIM006	3-11	6	63	2-28	2	30
STIM007, STIM008	2-5	6	48	3-11	4	27
STIM009	3-11	6	60	2-28	1	27

Example 13: Agonism of ICOS-Expressing MJ Cells by Bead-Bound Antibody

Antibodies STIM001, STIM002 and STIM003, the anti-ICOS antibody C398.4A, and ICOS ligand (ICOSL-Fc), were each covalently coupled to beads and assessed for their ability to induce expression of the cytokine IFN- γ from MJ cells grown in culture. Human IgG1 and Clone C398.4A isotype controls coupled to beads were assessed in parallel.

Data are shown in FIG. 12 and Table E13 below.

Each of the anti-ICOS antibodies demonstrated agonism in this assay, stimulating MJ cells as determined by IFN- γ quantification significantly above that observed by their cognate isotype controls within the dynamic range of the assay.

STIM003 and Clone C398.4A produced lower top asymptote values (95% CI: 3.79 to 5.13 and 3.07 to 4.22, respectively) but more potent Log EC50 values (95% CI: -9.40 to -9.11 and -9.56 to -9.23, respectively) compared with STIM001 (Top 95% CI: 7.21 to 8.88 and Log EC50 95% CI: -8.82 to -8.63) and STIM002 (Top 95% CI: 5.38 to 6.95 and Log EC50 95% CI: -9.00 to -8.74). Because incomplete curves (Top out of dynamic range of assay) were produced for ICOSL-Fc and Clone C398.4A isotype control, the fitted Top and Log EC50 values were not treated as reliable. Human IgG1 Hybrid Control produced a complete curve, however the area under the curve was not significantly different from 0 and it was therefore not deemed to be an agonist.

MJ Cell Activation Assay Materials and Methods-Bead-Bound

25 Coupling Proteins of Interest to Magnetic Particles

Anti-ICOS antibodies, control antibodies, and ICOSL-Fc, were coupled to beads as follows.

Dynabeads M-450 Tosylactivated (Invitrogen; approximately 2×10^8 beads/sample) were incubated with 100 μ g of each protein sample overnight at room temperature with agitation. Beads were washed three times with DPBS (Gibco) and incubated with 1M Tris-HCl, pH 8.0 (Ultra-Pure™, Gibco) for 1 hr at room temperature with agitation to block the uncoupled reactive sites. Beads were washed again three times with DPBS and finally resuspended in 0.5 ml of DPBS/sample.

The quantity of each protein of interest on the beads was then determined as follows. Black flat bottom, high binding ELISA plates (Greiner) were coated with Anti-human IgG (Southern Biotech) or Anti-Armenian Hamster IgG (Jackson ImmunoResearch) capture antibody at 4 μ g/ml in DPBS, 50 μ l/well, overnight at 4° C. Wells were then washed three times with DPBS+0.1% Tween, 200 μ l/well and blocked with 200 μ l/well of DPBS+1% BSA for 1 hr at RT. Wells were washed again three times with DPBS+0.1% Tween. Stock protein samples were quantified spectrometrically and beads were counted on a cell counter. Dilution series of protein samples and beads were then incubated in the plates at 50 μ l/well for 1 hr at RT before washing again three times with DPBS+0.1% Tween. 50 μ l/well of either biotinylated anti-armenian hamster antibody or anti-human IgG-euro-

TABLE E13

Summary table of bead bound MJ cell in vitro activation assay.							
Best-fit values	STIM001-beads	STIM002-beads	STIM003-beads	Human IgG1 Hybrid Control-beads	Clone C398.4A-beads	Clone C398.4A IC-beads	Clone C398.4A ICOSL-Fc-beads
Bottom	-0.24 to 0.36	-0.38 to 0.56	-0.39 to 0.34	-0.26 to 0.06	-0.37 to 0.42	-0.20 to -0.09	-0.62 to 0.49
Top	7.21 to 8.88	5.38 to 6.95	3.79 to 5.13	-0.04 to 0.20	3.07 to 4.22	NA	-45.66 to 65.37
LogEC50	-8.82 to -8.63	-9.00 to -8.74	-9.40 to -9.11	-10.43 to -8.60	-9.56 to -9.23	NA	-12.80 to -2.37
HillSlope	0.89 to 1.38	0.82 to 1.85	0.69 to 1.56	-3.47 to 6.61	0.64 to 1.90	NA	-0.38 to 1.88

NA—not applicable.

pium in DPBS+0.1% BSA were added and incubated for 1 hr at RT. In the case of addition of biotinylated anti-Armenian hamster antibody C398.4A another incubation step with 50 μ l/well of streptavidin-europium (Perkin Elmer) diluted 1:500 in Assay buffer (Perkin Elmer) were added and incubated for 1 hr at RT. The wells were washed three times with 200 μ l/well of TBS+0.1% Tween before developing the assay by adding 50 μ l/well of Delfia enhancement solution (Perkin Elmer), incubating for 10 mins at RT and measuring the fluorescence emitted at 615 nm on the EnVision Multilabel Plate Reader. The quantity of protein on the beads was determined by extrapolating values from the signals obtained from known concentrations of uncoupled protein samples.

MJ Cell In Vitro Activation Assay-Bead Bound

MJ [G11] cell line (ATCC CRL-8294) was grown in IMDM (Gibco or ATCC) supplemented with 20% heat inactivated FBS. Cells were counted and 15000 cells/well (50 μ l/well) of cell suspension was added to 96-well clear flat bottom polystyrene sterile TC-treated microplates. Beads were counted and serial 1:2 dilutions ranging from 1.5×10^6 beads/well to approximately 5860 beads/well (50 μ l/well) were added to the cells in duplicate or in triplicate. To account for background several wells of the plate contained MJ cells only (100 μ l/well). The cells and beads were co-cultured in the plates for 3 days at 37° C. and 5% CO₂ after which supernatants were harvested by centrifugation and collected for IFN- γ content determination.

Measuring IFN- γ Levels

The IFN- γ content in each well was determined using a modification of the Human IFN-gamma DuoSet ELISA kit (R&D systems). Capture antibody (50 μ l/well) was coated overnight at 4 μ g/ml in DPBS on black flat bottom, high binding plates (Greiner). The wells were washed three times with 200 μ l/well of DPBS+0.1% Tween. The wells were blocked with 200 μ l/well of 1% BSA in DPBS (w/v), washed three times with 200 μ l/well of DPBS+0.1% Tween and then 50 μ l/well of either the IFN- γ standard solutions in RPMI or neat cell supernatant were added to each well and incubated for 1 hr at RT. The wells were washed three times with 200 μ l/well of DPBS+0.1% Tween before adding 50 μ l/well of biotinylated detection antibody at 200 ng/ml in DPBS+0.1% BSA and incubated for 1 hr at RT. The wells were washed three times with 200 μ l/well of DPBS+0.1% Tween before adding 50 μ l/well of streptavidin-europium (Perkin Elmer) diluted 1:500 in Assay buffer (Perkin Elmer) and incubated for 1 hr at RT. The wells were washed three times with 200 μ l/well of TBS+0.1% Tween before developing the assay by adding 50 μ l/well of Delfia enhancement solution (Perkin

Elmer) and incubating for 10 mins at RT and measuring the fluorescence emitted at 615 nm on the EnVision Multilabel Plate Reader.

Data Analysis

IFN- γ values for each well were interpolated from the standard curve and the average background levels from cell-only wells were subtracted. The background corrected values were then used in GraphPad prism to fit a 4-parameter log-logistic concentration response curve.

Example 14: Agonism of ICOS-Expressing MJ Cells by Plate-Bound Antibody

An alternative assay for agonism of ICOS-expressing T cells uses antibodies in a plate-bound format.

MJ Cell Activation Assay Materials and Methods—Plate-Bound

Antibody coating: 96-well, sterile, flat, high binding plates (Costar) were coated overnight at 4° C. with 100 μ l/well of serial 1:2 dilutions of proteins of interest (anti-ICOS antibodies, control antibodies, and ICOSL-Fc) in DPBS (Gibco) ranging from 10 μ g/ml to 0.02 μ g/ml in duplicate or in triplicate. To account for background several wells of the plate were coated with DPBS only. Plates were then washed three times with 200 μ l/well of DPBS before the addition of cells.

Cell stimulation: MJ [G11] cell line (ATCC CRL-8294) was grown in IMDM (Gibco or ATCC) supplemented with 20% heat inactivated FBS. The cells were counted and 15000 cells/well (100 μ l/well) of cell suspension were added to the protein coated plates. Cells were cultured in the plates for 3 days at 37° C. and 5% CO₂. Cells were separated from the media by centrifugation and the supernatants collected for IFN- γ content determination.

Measurement of IFN γ levels and data analysis was as described in Example 13.

Results

Results are shown in FIG. 13 and in Table E14-1 below. In summary, STIM001, STIM002 and STIM003 all showed potent agonism as measured by IFN- γ secretion with similar Log EC50 values (Log EC50 95% CI: -7.76 to -7.64, -7.79 to -7.70 and -7.82 to -7.73, respectively) and Top values (Top 95% CI: 2.06 to 2.54, 2.44 to 2.93 and 2.01 to 2.41, respectively). Clone C398.4A exhibited a similar Log EC50 value (Log EC50 95% CI: -7.78 to -7.60) but lower Top value (Top 95% CI: 1.22 to 1.63) than STIM001 to STIM003. STIM004 also showed agonism in this assay, but was less potent, reaching a moderate Top value (Top 95% CI: 0.16 to 0.82) with a similar Log EC50 value (Log EC50 95% CI: -7.91 to -7.21). STIM001, STIM002 and STIM003 were stronger agonists than ICOSL-Fc (Log EC50 95% CI: -7.85 to -7.31 and Top 95% CI: 0.87 to 2.45).

TABLE E14-1

Summary of plate-bound MJ cell in vitro activation assay.									
Best-fit values									
95% CI	STIM001	STIM002	STIM003	STIM004	IgG1	Clone C398.4A	ICOSL-Fc	Clone C398.4AIC	
Bottom	-0.03 to 0.13	-0.08 to 0.11	-0.10 to 0.07	-0.06 to 0.07	NA	-0.03 to 0.11	-0.16 to 0.09	-0.07 to 0.04	
Top	2.06 to 2.54	2.44 to 2.93	2.01 to 2.41	0.16 to 0.82	NA	1.22 to 1.63	0.87 to 2.45	0.05 to 0.29	
LogEC50	-7.76 to -7.64	-7.79 to -7.70	-7.82 to -7.73	-7.91 to -7.21	NA	-7.78 to -7.60	-7.85 to -7.31	NA	
HillSlope	2.06 to 5.38	0.16 to 10.88	1.77 to 6.5	-1.46 to 6.77	NA	1.24 to 8.20	0.26 to 3.97	NA	

IgG1 = Human IgG1 hybrid control antibody.

Example 15: Agonism of ICOS-Expressing MJ Cells by Antibody in Soluble Form

In contrast to the assays described in Example 13 and Example 14, which used antibody arrayed on a solid surface, this assay determines whether antibody in soluble form acts as agonist of ICOS-expressing T cells.

MJ Cell Activation Assay Materials and Methods-Soluble

MJ [G11] cell line (ATCC CRL-8294) was grown in IMDM (Gibco or ATCC) supplemented with 20% heat inactivated FBS. Cells were counted and 15000 cells/well (50 μ l/well) of cell suspension was added to 96-well clear flat bottom polystyrene sterile TC-treated microplates. Serial 1:2 dilutions of proteins of interest ranging from 10 μ g/ml to 0.01953125 μ g/ml either alone or with the addition of a cross-linking reagent (AffiniPure F(ab')₂ Fragment Goat Anti-Human IgG, Fc Fragment Specific; Jackson ImmunoResearch) were added to the cells in duplicate or in triplicate (50 μ l/well). To account for background several wells of the plate contained MJ cells only (100 μ l/well). The cells and beads were co-cultured in the plates for 3 days at 37° C. and 5% CO₂ after which supernatants were harvested by centrifugation and collected for IFN- γ content determination.

Measurement of IFN γ levels and data analysis was as described in Example 13.

Results

STIM001 and STIM002 both showed significant soluble agonism as measured by IFN- γ secretion compared to Human IgG4. PE hybrid control. MAb cross-linking via Goat Anti-Human IgG Fc F(ab')₂ Fragment increased secreted IFN- γ levels even more.

Example 16: Binding of Antibody to Activated T Cells

A. Human ICOS

Ability of anti-ICOS antibodies to recognise the ICOS extracellular domain in its native context on the surface of activated primary human T cells is confirmed in this assay.

Pan T-cells (CD3 cells) were isolated and cultured for 3 days with CD3/CD28 dynabeads (ThermoFisher) to induce ICOS expression on their surface. Surface staining of STIM001, STIM003 and the hIgG1 hybrid control (HC IgG1) was determined by two methods, namely detection following direct binding of pre-labelled antibodies (antibodies directly conjugated with AF647) or indirectly via the use of a secondary AF647-Goat anti-human Fc antibody. Stained cells were ran on the Attune and staining intensity was presented as Mean of fluorescence intensity (MFI). EC50 was determined using GraphPad Prism.

Results are shown in FIG. 14. Once activated, pan CD3 T cells were clearly stained by both STIM001 and STIM003 hIgG1. Notably, the saturation of STIM003 binding to activated T cells occurred at a lower concentration than that of STIM001, suggesting higher affinity of STIM003 to human ICOS. The EC50 of STIM003 was roughly 100 $\times$ lower than that of STIM001 (0.148 nM vs 17 nM for the indirect binding assay).

B. ICOS from Non-Human Primates

Ability of anti-ICOS antibodies to recognise the ICOS extracellular domain in its native context on the surface of activated primary T cells from non-human primates (NHP) is confirmed in this assay.

PBMC from whole blood of 2 Mauritian cynomolgus macaques (Wickham Laboratories) were isolated by gradient centrifugation and cultured for 3 days with CD2/CD3/

CD28 MACSiBeads (Miltenyi) to induce ICOS expression on their surface. Surface staining of STIM001, STIM003 and the hIgG1 hybrid control (HC IgG1) was determined following direct binding of AF647 pre-labelled antibodies (from 80 μ g to 8 μ g/ml). Cells were also labelled with V450-CD3 to assess staining on T-cell subsets. Stained cells were run on Attune (ThermoFisher) and staining intensity was presented as mean fluorescence intensity (MFI). EC50 was determined using GraphPad Prism.

Results are shown in FIG. 28. Once activated, T cells were clearly stained by both STIM001 and STIM003 hIgG1. As was observed with binding to human T cells, saturation of STIM003 binding to activated NHP T cells occurred at lower concentration than that of STIM001, indicating that STIM0003 has the higher affinity of these two antibodies ICOS. EC50 values for binding to NHP ICOS were similar to those obtained for binding to human ICOS.

TABLE E16

EC50 (Molar) calculated for antibody binding to ICOS on activated NHP T cells			
Pan T-cells	EC50	Cynomolgus donor 1	Cynomolgus donor 2
NHP	STIM001	2.224e-7	not tested
	STIM003	4.581e-9	4.830e-9
Human	STIM001	2.209e-7	1.207e-7
	STIM003	2.293e-9	8.953e-10

Example 17: Analysis of T Cell Sub-Populations Among Tumour Infiltrating Lymphocytes and Peripheral T Cells

A pharmacodynamics study revealed that anti-ICOS antibodies STIM001 and STIM003 in mIgG2a isotype significantly deplete TRegs, increase the percentage of CD4+ effector cells and increase the CD4+ effector/TReg ratio as well as the CD8+/TReg ratio within the tumour microenvironment (TME).

The increased CD8+/TReg ratio and increased number of CD4+ effector cells within the TME may collectively contribute to the CT26 tumour clearance observed when these anti-ICOS antibodies were co-injected with anti-PDL1 antibody in the STIM001 & STIM003 efficacy study (Example 11).

Method

The pharmacodynamics study was performed in female Balb/c mice bearing CT-26 mouse colon carcinoma cells (ATCC, CRL-2638). Balb/c mice were supplied by Charles River UK at 6-8 weeks of age and >18 g and housed under specific pathogen-free conditions. A total of 1 $\times$ 10⁵ CT-26 tumour cells (passage number: P8) were subcutaneously injected in the right flank. All CT-26 tumour bearing animals were assigned to 6 groups (Table E17-1) and individual mice were dosed twice (on Day 13 & Day 15 post tumour cell implantation) with 200 μ g of antibody or saline. CD3+ T-cells from the CT-26 tumour bearing animals were analysed by FACS on day 16 post tumour cell implantation.

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TABLE E17-1

Treatment groups		
Group	Number of animals	Treatment regimen (Day 13 and Day 15 post tumour cell implantation)
1	10	Saline
2	10	Anti-ICOS (STIM001) mIgG1 200 µg each
3	10	Anti-ICOS (STIM001) mIgG2a 200 µg each
4	10	Anti-ICOS (STIM003) mIgG1 200 µg each
5	10	Anti-ICOS (STIM003) mIgG2a 200 µg each
6	10	Anti-CTLA-4 (9H10) 200 µg each

Results

Animals treated with STIM001 & STIM003 in the mIgG2a isotype showed a lower percentage of CD4+CD3+ CD45+ cells at the tumour site when compared with saline treated group (FIG. 15A), whereas STIM001 or STIM003 treatments had very marginal effect of the percentage of CD8+CD3+CD45+ cells at the tumour site (FIG. 15B). The decrease in CD4+ T cells could be attributed to a profound decrease in the percentages of T-Regulatory cells in all the groups treated with STIM001 and STIM003 antibodies. Notably, animals treated with STIM001 and STIM003 in the mIgG2a isotype showed a dramatic reduction in T-Regs (CD4+ Foxp3+CD25+) within the TME, whereas STIM001 & STIM003 in the mIgG1 isotype had only a modest effect on T-Reg content in TME. In addition, animals treated with STIM001 & STIM003 in the mIgG2a isotype had reduced T-Reg in the TME when compared with the animals treated with a commercial anti-CTLA-4 (9H10, Biolegend Cat #106208) antibody which is known to deplete T-Reg [42], but this result did not reach statistical significance (FIG. 15C). The effect of STIM001 and STIM003 either in mIgG1 or mIgG2a isotypes on T-Reg compartment was more specific with tumour infiltrating lymphocytes (TILs). T-reg depletion was not observed in the periphery (as previously described for anti-CTLA4 [43]) (FIG. 15D). The changes in T-Reg contents also resulted in a significant increase in the percentage of intra-tumoural CD4-effector cells (CD4+ Foxp3- CD25-) (FIG. 15E), similarly the ratio of CD4 effector/T-Reg and CD8/T-Reg ratio in the animals treated with STIM001 & STIM003 in the mIgG2a was also significantly increased within TME (FIG. 15F & FIG. 15G).

Example 18: Effect of Anti-ICOS Antibody on
Level of ICOS-Expressing T Cells in CT26
Tumour and Spleen

Analysis was performed to quantify the percentage of immune cells within the tumour compared with the spleen, by analysis of total immune cells in the tumour and spleen tissues, following treatment with anti-ICOS antibody STIM001 or STIM003. STIM001 and STIM003 mIgG2a each caused a significant reduction in Treg within the tumour, but not in the spleen, indicating a tumour-selective effect. This depletion was selective for Tregs compared with other T cell subtypes. The results presented here assist in understanding the effects of the STIM antibodies on the immune contexture, and confirm that anti-ICOS antibodies with effector-function-enabled Fc regions can strongly deplete TRegs.

Materials and Methods

Mice bearing CT26 tumours were dosed twice with STIM001, STIM003, or anti-CTLA4 antibody (9H10). The anti-CTLA4 antibody was included as a positive control for

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Treg depletion, as anti-CTLA4 antibody had been previously shown to selectively reduce Tregs in tumours [43].

The immune contexture within the tumours and the spleen of treated animals was analysed by FACS following tissue disaggregation.

Details of FACS antibodies used in this study are shown in Table E18. All FACS antibodies were used at a concentration recommended by the supplier. FACS data were acquired using Attune N×T flow cytometer and data were analysed using FlowJo software.

TABLE E18

FACS antibodies.				
Marker	Supplier	Cat. number	Lot number	Fluorophore
Live/dead	Life technologies	L-34959	1784156	Fixable Yellow
CD45	E-bioscience	45-0451-82	E08336-1636	PerCp-Cy5.5
CD3	E-bioscience	48-0032-82	4278794	eFlour 450
CD4	E-bioscience	11-0042-86	E0084-1633	FITC
CD8	E-bioscience	12-0081-85	E01039-1635	PE
Foxp3	E-bioscience	17-5773-82	4291991	APC
CD25	E-bioscience	47-0251-82	4277960	APC eF 780
ICOS	E-bioscience	25-9942-82	E17665-103	PE-CY7
Fc/Block	E-bioscience	16-0161-86	E06357-1633	—

Results

ICOS expression was determined in the CT26 tumours and in the spleen of tumour-bearing animals. We observed an increased percentage of tumour infiltrating immune cells expressing ICOS protein (FIG. 20), indicating that immune cells in the tumours are more often positive for ICOS expression than immune cells in the periphery. TRegs in the tumour of untreated animals were nearly all (>90%) positive for ICOS expression, whereas CD8+ effector T cells in the tumour were not (approx. 60%). Comparing T cell subpopulations (again in untreated mice) in tumour with those in spleen, a significantly higher (p<0.0001) percentage of intratumoural Tregs were positive for ICOS compared with Tregs in spleen, and a significantly higher (p<0.001) percentage of intratumoural CD4+ Teff cells were positive for ICOS compared with CD4+ Teff cells in spleen.

Also in the mice before treatment, the level of ICOS expression was much higher on immune cells in the microenvironment of CT26 tumours, when compared with immune cells in the spleen (FIG. 21). ICOS expression was increased on the surface of all immune cell subsets analysed (CD8 T-Effector, CD4 T-Effector and CD4/FoxP3 TReg cells) in the tumour microenvironment. Note that although immune cells in the tumours and the spleen are both expressing ICOS, immune cells in the tumour are expressing significantly more ICOS (indicated by higher MFI, FIG. 21) than cells in the spleen (indicated by lower MFI, FIG. 21). Importantly, TRegs in the tumour are expressing the highest levels of ICOS, as previously reported [11].

CT26 tumour bearing animals were treated with 2 doses of antibody STIM001 or STIM003 and with an anti-CTLA-4 antibody. The STIM antibodies did not affect the overall percentage of CD45 positive cells (a marker for immune cells) in the tumours, when used in either mIgG1 or mIgG2 format. Nor did treatment with these antibodies significantly affect the percentage of CD8 effector T cells in CT26 tumours (FIG. 22). Treatment with STIM001 in mIgG2a isotype led to a significant (p<0.05) depletion of CD4+ effector T cells, but none of STIM001 mIgG1, STIM003 mIgG1 and STIM003 mIgG2a affected the percentage of CD4+ effector T cells.

Anti-CTLA-4 treatment produced a notable (albeit not statistically significant) increase in CD45+ cells and CD8+ effector T cells in the TME, but did not affect CD4+ effector T cells (FIG. 22).

The STIM antibodies significantly affected regulatory T cells in the tumour. As shown in FIG. 23, STIM001 mIgG2a and STIM003 mIgG2a significantly and selectively depleted TRegs (which are high for ICOS expression) in the tumour microenvironment. Interestingly, the anti-CTLA4 antibody which, despite being included as a positive control for TReg depletion in this experiment, was less effective than the STIM mIgG2a antibodies at depleting TRegs.

This selective depletion of TRegs resulted in an increase in the ratio CD8 effector T cells to TRegs in the tumour, and an increase in the ratio of CD4 effector T cells to TRegs in the tumour, both of which should favour an anti-tumour immune response. Ratio data are shown in FIG. 24.

In contrast to the depletion of intratumoural Tregs by STIM001 mIgG2a and STIM003 mIgG2a, no such effect was observed on Tregs in spleen (FIG. 25, FIG. 26, FIG. 27). This indicates that the effects of the anti-ICOS antibodies on depletion of Tregs depletion was not systemic in all the tissues. Such selectivity could be advantageous for therapeutic anti-ICOS antibodies in treating tumours in patients, as preferential depletion of Tregs in the tumour microenvironment could selectively relieve suppression of anti-tumour effector T cells, while minimising side effects at other sites in the body. The anti-ICOS antibodies may thus promote an anti-tumour response in the immune system with a low risk of undesirable activation of a wider T cell response that could cause treatment-limiting autoimmune adverse events.

Example 19: Antibody Stability

STIM003 human IgG1 was tested for stability during storage, freeze/thawing and purification, and was found to maintain its stability under all tested conditions. % aggregation was determined by HPLC.

There was no significant change in the percentage of monomer (>99%) after 3 months storage at 4° C. in buffer (10 mM sodium phosphate, 40 mM sodium chloride, pH 7.0).

On thermal denaturation testing, all samples (n=15) had the same Tm (no significant difference between aliquots) and had comparable thermal denaturation curves.

There was no significant change in Tm (~70.3° C.), the percentage of monomer or the profile on SDS-PAGE after 3 cycles of freeze and thaw.

There was no significant change in Tm (~70.3° C.), the percentage of monomer or the profile on SDS-PAGE after 7 days' storage at room temperature.

There was 90% recovery post protein A purification.

Example 20: Monotherapeutic Efficacy of Anti-ICOS Ab Against A20 Tumour Growth in Mouse

Anti-ICOS antibodies STIM001 mIgG2a and STIM003 mIgG2a each showed strong anti-tumour efficacy when used as monotherapies in vivo in a mouse A20 syngeneic model.

Materials and Methods

The efficacy study was performed in BALB/c mice using the sub-cutaneous A20 reticulum cell sarcoma model (ATCC, TIB-208). The A20 cell line is a BALB/c B cell

lymphoma line derived from a spontaneous reticulum cell neoplasm found in an old BALB/cAnN mouse. This cell line has been reported to be positive for ICOSL.

BALB/c mice were supplied by Charles River UK>18 gram and housed under specific pathogen-free conditions. A total of 5×10⁵ A20 cells (passage number below P20) were subcutaneously injected into the right flanks of mice. The A20 cells were passaged in vitro washed twice in PBS and re-suspended in RPMI supplemented with 10% foetal calf serum. Cell viability was confirmed to be above 85% at the time of tumour cell injection. Unless stated otherwise, antibody or isotype administration was initiated from day 8 post tumour cells injection.

STIM001 and STIM003 anti-ICOS antibodies were generated in mouse IgG2a isotype format. The mouse cross reactive anti-PD-L1 antibody (AbW) was also generated in the same isotype format (mouse IgG2a). STIM001, STIM003 and anti-PD-L1 antibodies were dosed intraperitoneally (IP) at 200 µg of each antibody twice a week starting from day 8 (dosing for 3 weeks between day 8-29) post tumour cell implantation. Animal weights and tumour volume were measured 3 times a week from the day of tumour cell injection. Tumour volume was calculated by use of the modified ellipsoid formula $\frac{1}{2}(\text{Length} \times \text{Width}^2)$. Mice were kept on study until their tumour reached an average diameter of 12 mm. The experiment was stopped at day 43 post tumour cell implantation. Tumour growth was monitored and compared with tumours of animals treated with isotype control (mIgG2a) antibody. Treatment groups are shown in Table E20 below.

TABLE E20

Treatment groups for A20 study.		
Group	Number of animals	Treatment regimen (twice per week for 3 weeks 7 doses)
1	8	mIgG2a isotype control 200 µg/mouse/each dose
2	8	Anti-PD-L1 mIgG2a (AbW) 200 µg/mouse/each dose
3	8	Anti-ICOS mIgG2a STIM001 200 µg/mouse each dose
4	8	Anti-ICOS mIgG2a STIM003 200 µg/mouse/each dose

Results

Monotherapy administration of either STIM001 or STIM003 (mIgG2a) in the A20 tumour model produced a complete anti-tumour response (FIG. 32, FIG. 33). All the animals administered with either STIM001 or STIM003 were cured of the disease. This contrasts with the results in the isotype control and PD-L1 mIgG2a groups (FIG. 30, FIG. 31). In rare cases, regression of tumours was observed for some animals in the isotype control (spontaneous regression) and anti-PDL-1 groups, but treatment with anti-ICOS antibody produced significantly greater efficacy. At the end of the study, 3 of 8 control animals and 2 of 8 anti-PDL-1 treated animals had no tumour. However, all animals treated with either STIM001 or STIM003 were tumour free at the end of the study (8 of 8 mice in both groups), representing 100% cure using the anti-ICOS antibodies.

Example 21: Strong Anti-Tumour Efficacy In Vivo in the J558 Myeloma Syngeneic Model for Combination of Anti-ICOS Antibody and Anti-PD-L1 Antibody

Anti-ICOS antibody STIM003 mIgG2a and anti-PD-L1 antibody AbW mIgG2a were administered individually and

in combination in the J558 tumour model. This is a syngeneic mouse model of myeloma. The anti-ICOS antibody was found to inhibit tumour growth when dosed as monotherapy or in combination with anti-PD-L1.

Materials & Methods

Anti-tumour efficacy studies were performed in Balb/c mice using the sub-cutaneous J558 plasmacytoma: myeloma cell line (ATCC, TIB-6). Balb/c mice were supplied by Charles River UK at 6-8 weeks of age and >18 g and housed under specific pathogen-free conditions. A total of 5×10^6 cells (passage number below P15) were subcutaneously injected (in 100 μ l) into the right flanks of mice. Unless stated otherwise, on day 11 post tumour cells injection, the animals were randomised based on tumour size and treatments were initiated. The J558 cells were passaged in vitro by using TrypLET Express Enzyme (ThermoFisher), washed twice in PBS and resuspended in DMEM supplemented with 10% foetal calf serum. Cell viability was confirmed to be above 90% at the time of tumour cell injection.

Treatment was initiated when the tumours reached an average volume of $\sim 140 \text{ mm}^3$. Animals were then allocated to 4 groups with similar average tumour size (see Table E-21 for the dosing groups). Both antibodies, which are mouse cross-reactive, were dosed IP from day 11 (post tumour cell implantation) twice a week for 3 weeks (FIG. 38) unless the animals had to be removed from study due to welfare (rare) or tumour size. As a control, a group of animals (n=10) was dosed at the same time using a saline solution. For the combination group, both STIM003 and anti-PDL1 antibodies were dosed concurrently IP at 60 μ g and 200 μ g respectively (in 0.9% saline). Tumour growth was monitored over 37 days and compared to tumours of animals treated with saline. Animal weight and tumour volume were measured 3 times a week from the day of tumour cell injection. Tumour volume was calculated by use of the modified ellipsoid formula $\frac{1}{2}(\text{Length} \times \text{Width}^2)$. Mice were kept on studies until their tumour reached an average diameter of 12 mm³ or, in rare cases, when incidence of tumour ulceration was observed (welfare).

TABLE E21

Treatment groups for J558 efficacy study.		
Groups	Number of animals	Treatment regimen twice per week from day 11
1	10	Saline
2	8	Anti-PD-L1 mIgG2a 200 μ g (AbW)
3	8	Anti-ICOS STIM003 mIgG2a/anti-PD-L1 mIgG2a (AbW) combination 60 μ g/200 μ g (respectively)
4	8	Anti-ICOS STIM003 mIgG2a 60 μ g

Results

J558 syngeneic tumours were highly aggressive and all the animals in the saline control group (n=10) had to be removed from studies by day 21 due to tumour size. The anti-STIM003 mIgG2a and the anti-PDL1 mIgG2a both demonstrated good efficacy as monotherapies in this model with 37.5% and 75% of the animals cured of disease, respectively. Importantly, combination of the two antibodies resulted in 100% of the animals having rejected the plasmacytoma tumours by day 37. Data are shown in FIG. 38.

Example 22: Administration of Anti-PD1 Increases ICOS Expression on TILs Significantly More than Anti-PD-L1 Antibody

A pharmacodynamic study was performed in animals harbouring established CT26 tumours to evaluate the effect of treatment with anti-PD-L1 or anti-PD-1 antibodies on ICOS expression on subsets of tumour infiltrating lymphocytes (TILs). The following antibodies were compared:

- anti-PD-L1 AbW mIgG1 [limited effector function]
- anti-PD-L1 AbW mIgG2a [with effector function]
- anti-PD-L1 10F9.G2 rat IgG2b [with effector function]
- anti-PD1 antibody RMT1-14 rat IgG2a [effector null].

Tumours of treated mice were isolated, dissociated to single cells and stained for CD45, CD3, CD4, CD8, FOXP3 and ICOS.

Materials & Methods

Rat anti-PD-1 RMP1-14 IgG2a (BioXCell; Catalog number: BE0146), rat anti-PD-L1 10F9.G2 IgG2b (Bio-Legend; Catalog number: 124325) and anti-PD-L1 AbW mIgG1 and mIgG2a were tested in the CT26 tumour model by dosing i.p. with 130 μ g on days 13 and 15 post tumour cell implantation. On day 16, animals were culled and the mouse tumours were harvested for FACS analysis. Tumours were dissociated using a mouse tumour dissociation kit (Miltenyi Biotec) and homogenised. The resulting cell suspensions were clarified through 70 μ m filters, pelleted and resuspended in FACS buffer at 2 million cells/well in a 96 well plate. The cell suspensions were incubated with anti-16/32 mAb (eBioscience) and stained with FACS antibodies specific for CD3 (17A2), CD45 (30-F11), CD4 (RM4-5), CD8 (53-6.7) and ICOS (7E.17G9) all obtained from eBioscience Ltd. Cells were also stained with LiveDead Yellow fixable viability dye (Life technologies). For the Foxp3 intracellular staining, samples were fixed, permeabilised, and stained with antibody specific for Foxp3 (eBioscience, FJK-16s). The samples were resuspended in PBS and data acquired on the Attune flow cytometer (Invitrogen) and analysed using FlowJo V10 software (Treestar).

Results

Treatment with anti-PD1 and anti-PD-L1 antibodies only resulted in a marginal increase in the percentage on CD8 cells and T Regs expressing ICOS at the measured time-point. However, in response to anti-PD1 rat IgG2a, a clear and significant (over the saline treated group) increase in ICOS expression (increased dMFI) was observed on the surface of ICOS+ve CD8 cells. ICOS expression was also noted to be upregulated on CD4 effector and CD4 T Reg cells although this did not reach statistical significance. This anti-PD1 antibody induced a marked increase in ICOS expression on CD8 effector cells that was barely seen with the anti-PD-L1 mIgG2a. Similarly, when comparing the different formats of anti-PD-L1 antibodies, in some of the animals treated it was observed that the antibody having the lowest effector function (mIgG1) was associated with higher ICOS expression on effector CD8 and CD4 cells when compared with antibody having effector function (mIgG2a and ratIgG2b), which rarely showed this. See FIG. 39.

An increase in ICOS expression on effector CD8/CD4 T cells may have the effect of rendering these cells more sensitive to depletion by anti-ICOS antibody (e.g., on treatment of mice with STIM003 mIgG2a). An antibody that exhibits lower ICOS induction in effector CD8 and CD4 T cells may be preferable for use in combination with anti-ICOS antibody. The data from this study indicate that

anti-PD-L1 effector positive antibody may be especially suitable for combination with anti-ICOS effector positive antibody, reflecting the anti-tumour efficacy observed when combining anti-PDL1 mIgG2a with STIM003 mIgG2a reported in other Examples herein.

Example 23: Strong Anti-Tumour Efficacy of
Single Dose Anti-ICOS Antibody Monotherapy In
Vivo in a B Cell Lymphoma Syngeneic Model

This experiment confirms the anti-tumour efficacy of STIM003 mIgG2a as monotherapy. Strong anti-tumour efficacy was demonstrated after short exposure of STIM003 mIgG2a.

Materials & Methods

Efficacy studies were performed in BALB/c mice using the sub-cutaneous A20 Reticulum Cell Sarcoma model (ATCC number CRL-TIB-208). BALB/c mice were supplied by Charles River UK at 6-8 weeks of age and >18 g and housed under specific pathogen-free conditions. A total of 5×10E5 A20 cells (passage number below P20) were subcutaneously injected into the right flanks of mice. Treatments were initiated at day 8 post tumour cells injection as shown in the table below. The A20 cells were passaged in vitro by using TryPLET Express Enzyme (Thermofisher), washed twice in PBS and resuspended in RPMI supplemented with 10% foetal calf serum. Cell viability was confirmed to be above 85% at the time of tumour cell injection. STIM003 mIgG2a was used either as a single dose (SD) of 60 µg (equivalent to 3 mg/kg for a 20 g animal) or as multiple doses (MD, twice a week for 3 weeks) of 60 µg. Anti-tumour efficacy observed in response to the two schedules was compared to that of animals "treated" with saline (MD, twice a week for 3 weeks). The antibodies were dosed intraperitoneal (IP) as 1 mg/ml in 0.9% saline. Animal weight and tumour volume were measured 3 times a week from the day of tumour cell injection. Tumour volume was calculated by use of the modified ellipsoid formula $\frac{1}{2}(\text{Length} \times \text{Width}^2)$. Mice were kept on study until their tumour reached an average diameter of 12 mm or, rarely, when incidence of tumour ulceration was observed (welfare).

TABLE E23-1

Treatment groups.		
Group	Number of animals	Treatment regimen (IP injection)
1	10	Saline (multiple dose from day 8, twice a week for 3 weeks)
2	10	STIM003 mIgG2 A (multiple dose from day 8, twice a week for 3 weeks)
3	10	STIM003 mIgG2 A (Single dose on day 8)

Results

Both multiple and single dose of STIM003 mIgG2a resulted in strong and significant monotherapy anti-tumour efficacy as shown by the number of animals with no signs of tumour growth at endpoint (Day 41). SD resulted in 7 out of 10 animals cured from the disease whereas the multiple dose cured 9 out of 10 animals injected with A20 B cell lymphoblast. All animals in the saline treated group had to be removed from the study by day 40 due to tumour size. See FIG. 40.

Humane endpoint survival statistics were calculated from the Kaplan-Meier curves (FIG. 41) using GraphPad Prism V7.0. This approach was used to determine if the treatments were associated with improved survival. The Hazard Ratio (Mantel-Haenszel) values and their associated P values (Log-Rank Mantel-Cox) are shown in the table below.

TABLE E23-1

Hazard Ratio (Mantel-Haenszel) values and their associated P values (Log-Rank Mantel-Cox) corresponding to FIG. 41 Kaplan-Meier curves.			
Hazard Ratio (Mantel-Haenszel)	MD vs Saline	SD vs Saline	MD vs SD
Ratio (and its reciprocal)	0.09995	0.1076	0.5314
95% CI of ratio	0.02604 to 0.3837	0.02856 to 0.4052	0.05522 to 5.115
P Value	0.0008	0.001	0.5842

Example 24: Time and Dose Dependent Effects of
Anti-ICOS Antibody in CT-26 Tumour Bearing
Animals

This Example presents the results of a pharmacodynamic study evaluating the effects of anti-ICOS antibody on immune cells in mice bearing CT-26 tumours. T and B cell subtypes from different tissues were analysed by FACS after a single dose of STIM003 mIgG2a.

Methods

CT-26 tumour bearing animals were dosed i.p. with either saline or STIM003 at 200 µg, 60 µg or 6 µg on day 12 post tumour cell implantation. Tumour tissues, blood, tumour draining lymph node (TDLN) and spleen were harvested on day 1, 2, 3, 4, and day 8 post treatment. The tumours were dissociated to make single cell suspension using mouse tumour dissociation kit (Miltenyi Biotec). Spleen tissue was dissociated using gentle MACS dissociation, red blood cells were lysed using RBC lysis buffer. Tumour draining lymph nodes were mechanically disaggregated to make single cells suspensions. The resulting cell suspensions were clarified through either 70 UM or 40 µM filters depending on the tissue, cells were then washed twice in RMPI complete media and finally resuspended in ice cold FACS buffer. Total blood was collected into plasma tubes and red blood cells were lysed using RBC lysis buffer, cells were washed twice in RMPI complete media and finally resuspended in ice cold FACS buffer. The single cell suspension from all the tissues were distributed into 96 deep well plates for FACS analysis. Cells were stained with Live Dead Fixable Yellow viability dye (Life technologies). The cell suspensions were incubated with anti-CD16/CD32 mAb (eBioscience) and stained with FACS antibodies specific for CD3 (17A2), CD45 (30-F11), CD4 (RM4-5), CD8 (53-6.7), CD25 (PC61.5), ICOSL (HK5.3), B220 (RA3-6B2), Ki-67 (SolA15), CD107a (eBio1D4B), IFN-γ (XMG1.2), TNF-α (MP6-XT22), Foxp3 (FJK-16s) and ICOS (7E.17G9) all obtained from eBioscience Ltd. For cytokine readout by FACS, single cells suspensions from the tumours were plated in 24 well plate for 4 hours in the presence of Brefeldin-A. For the intracellular staining, samples were fixed, permeabilised, and stained with specific antibodies. The samples were finally resuspended in PBS and data acquired on the Attune flow cytometer (Invitrogen) and analysed using FlowJo V10 software (Treestar).

Results are presented and discussed below.
 ICOS Expression is High on Intra-Tumoral T-Regs in the CT26 Model

When the percentage of tumour infiltrating lymphocytes (TILs) expressing ICOS was compared to the percentage of immune cells in the spleen, blood, and TDLN, we demonstrated that more immune cells in the microenvironment of CT-26 tumours expressed ICOS vs other tissues. More importantly, the percentage of ICOS positive T-reg cells in all the tissues and at all the time points was higher than the percentage of CD4 or CD8 effector T cells positive for ICOS. Importantly, the dMFI (relative expression) for ICOS also followed the similar ranking in expression with intra-tumoural T-reg being highly positive for ICOS expression vs other TILs subtypes. Interestingly, there was no striking change in the percentage of ICOS⁺ TILs within the time frame of this experiment. Similar results were also seen in spleen and TDLN. On the other hand, in the blood, ICOS expression is relatively stable on T effector cells but increased on T-regs during the course of the experiment. Altogether the data demonstrated that more cells expressed ICOS in the tumour microenvironment and these positive cells also expressed more ICOS molecules on their surface. More importantly, T regs in TILs are highly positive for ICOS. See FIG. 42.

Strong Depletion of Intra-Tumoural T-Reg Cells in Response to STIM003 Administration

In response to the STIM003 mIgG2a antibody, there was strong and rapid depletion of T-reg cells (CD4+CD25+ Foxp3) in TME. As T-regs have high ICOS expression compared with the other T cells subsets, it is expected that an anti-ICOS antibody with effector function would preferentially deplete these cells. At the lower dose of STIM003 (6 µg corresponding to a 0.3 mg/kg for a 20 g animal) there was a continuous depletion of T-reg and by day 3 most of the T-reg were depleted from TME. Interestingly, by day 8, T-reg cells repopulate the TME then reach a level slightly above that observed in the saline treated animals. The repopulation of T-reg cells at lower dose can be attributed to the increase in the proliferating CD4 T cells in TME as evidenced by an observed increase in Ki-67+CD4 T-cells. At a dose higher than 6 µg there was a long-term depletion of T-reg cells in TME as shown by full T Reg depletion until the last time point analysed in this study (day 8). Whereas in the blood there was a transient depletion of T-reg cells at all doses. Importantly, by day 8, all the treated animals had similar (or higher for the 6 µg dose) level of T-reg cells in the blood when compared to the saline treated animals. Data are shown in FIG. 43. Notably, and similarly to data previously published for depleting CTLA-4 antibodies, there was no significant change in the percentage of T-reg cells in the spleen or TDLN tissues, suggesting that T-reg cells may be protected from depletion in these organs.

In summary, strong depletion of T-reg cells in TME was achieved in CT-26 model at a dose as low as 6 µg per animal. However, a dose of 60 µg resulted in long term depletion up to 8 days post STIM003 mIgG2a injection. This was not improved by using higher dose (200 µg). STIM003 mIgG2a Increased CD8:T Reg and CD4:T Reg Ratios

Effects of STIM003 on T-eff:T-reg ratios are shown in FIG. 44.

STIM003 mIgG2a increased the CD8:T-reg ratio as well as the CD4 eff:T-reg ratio. Although all the treatment doses were associated with an increase in T-eff to T-reg ratio, the

intermediate dose of 60 µg (the equivalent of 3 mg/kg for a 20 g animals) was associated with the highest ratio by day 8 post treatment.

Interestingly, at the 6 µg dose, the ratios were high until day 4 but by day 8 post treatment they were matching that of the saline treated animals. This can be explained by the repopulation of TRegs observed for this dose by day 8 post treatment. On the other hand, at a dose of 60 or 200 µg, the T-eff to T-reg ratios remained high at all time points. This is explained by a long term depletion of Tregs at these doses. Notably, at higher dose (200 µg), despite the long term Treg depletion there was only a moderate improvement in the ratio by day 8. This can be explained by some depletion of ICOSINT effector cells at high concentration of STIM003. Altogether, the data demonstrated TReg depletion and increased Effector:T reg ratio at all doses tested. However, an optimal dose of 60 µg (~3 mg/kg) achieved both a long-term depletion of T-reg, as well as the highest T-eff to T-reg ratios which would be associated with the most favourable immune context to initiate an anti-tumour immune response. Interestingly a similar pattern was observed in the blood, with the intermediate dose of 60 µg associated with the highest T-eff to T-reg ratio. Importantly, in the blood, improvement of the ratio was observed at an earlier time point (between day 3 and day 4).

Activation of Effector Cells in Response to STIM003

Surface expression of CD107a on the tumour infiltrating T effector cells was previously identified as a reliable marker for cells that have been activated and exert cytotoxic activity [44]. In the present study employed this marker to confirm that STIM003, in addition to depleting T-regs, can stimulate the cytotoxic activity of effector T cells in the TME. Interestingly, on day 8 post treatment, there was an increase in surface expression of CD107a on both the CD4 and CD8 effector T cell compartments at all doses of STIM003. Furthermore, this upregulation of CD107a expression on the surface on both CD4 and CD8 T cells appeared to plateau when animals were dosed at 60 µg as no improvement was seen at 200 µg dosing.

To further demonstrate activation of effector cells in the TME, the cytokine release by CD4 and CD8 TILs was analysed by FACS. As expected and consistent with the in-vitro agonism data presented in earlier Examples herein, STIM003 mIgG2a at all doses promoted pro-inflammatory cytokine IFN-γ and TNF-α production by effector CD4 and CD8 T cells. The induction of pro-inflammatory cytokine production appeared to be high at the optimum dose of 60 µg. Indeed, 60 µg of STIM003 significantly increased cytokine production by CD4 T cells. A similar trend was seen for the proinflammatory cytokine IFN-γ and TNF-α production by effector CD8 T cells in TME. Data are shown in FIG. 45.

In summary, STIM003 at all the doses resulted in T cells activation in the TME as shown by (1) the presence of the degranulation marker CD107a on their surface and (2) by the production of Th1 cytokines (IFNγ and TNFα) by T cells. This indicates that STIM003 strongly affects the immune context in the TME and plays the dual role of depleting Treg cells and stimulate the killing activity of T effector cells.

Human Dose Estimations

Based on the pre-clinical efficacy data seen in mice, initial predictions can be made of the clinical dose appropriate for human patients, based on corresponding biological surface area (BSA) [45].

For example, taking the optimal anti-ICOS IgG dose in mouse to be 3 mg/kg (60 µg), and following the methods of ref. [45], the corresponding dose for a human is 0.25 mg/kg.

Using the Mosteller formulae, for an individual of 60 kg and 1.70 m the BSA 1.68 m². Multiplying the dose in mg/kg by a factor of 35.7 (60/1.68) gives a fixed dose of 15 mg. For an individual of 80 kg the corresponding fixed dose would be 20 mg.

Doses may be optimised for human therapy in clinical trials to determine safe and effective treatment regimens.

Example 25: Bioinformatic Analysis of Data from Tumour Samples

One target group of cancers according to the present invention is those cancers that are associated with a relatively high level of ICOS+ immunosuppressive Tregs.

To identify cancer types associated with a high content of Tregs, transcriptome data was obtained from The Cancer Genome Atlas (TCGA) public dataset and analysed for ICOS and FOXP3 expression levels. TCGA is a large-scale study that has catalogued genomic and transcriptomic data accumulated for many different types of cancers, and includes mutations, copy number variation, mRNA and miRNA gene expression, and DNA methylation along with substantial sample metadata.

Gene Set enrichment analysis (GSEA) was conducted as follows. Gene expression RNA seq data collected as part of the TCGA consortium was downloaded from the UCSC Xena Functional Genomics Browser as log 2 (normalized_count+1). Non-tumour tissue samples were removed from the dataset, leaving data for 20530 genes from 9732 samples. An algorithm from and its implementation in that calculates enrichment scores for genes within a specified gene set was used to transpose gene level counts to gene set scores for each sample. The gene set of interest was defined as containing both ICOS and FOXP3. Samples were grouped by primary disease and the ssGSEA scores for each group were compared across the 33 primary disease groups. The disease groups that showed the highest median scores were found to be lymphoid neoplasm diffuse large b-cell lymphoma, thymoma, head and neck squamous cell carcinoma, although diffuse large b-cell lymphoma showed a multimodal distribution of scores with a subset scoring highly and the rest scoring below the group median.

In rank order of highest to lowest ssGSEA score for ICOS and FOXP3 expression, the top 15 cancer types were:

- DLBC (n=48) lymphoid neoplasm diffuse large b-cell lymphoma
- THYM (n=120) thymoma
- HNSC (n=522) head and neck squamous cell carcinoma
- TGCT (n=156) testicular germ cell tumour
- STAD (n=415) stomach adenocarcinoma
- SKCM (n=473) skin cutaneous melanoma
- CESC (n=305) cervical squamous cell carcinoma and endocervical adenocarcinoma
- LUAD (n=517) lung adenocarcinoma
- LAML (n=173) acute myeloid leukemia
- ESCA (n=185) esophageal carcinoma
- LUSC (n=502) lung squamous cell carcinoma
- READ (n=95) rectum adenocarcinoma
- COAD (n=288) colon adenocarcinoma
- BRCA (n=1104) breast invasive carcinoma
- LIHC (n=373) liver hepatocellular carcinoma

In which n is the number of patient samples for that cancer type in TCGA dataset. Anti-ICOS antibodies described herein may be used for treatment of these and other cancers.

Cancers that are associated with a relatively high level of ICOS+ immunosuppressive Tregs and which further express PD-L1 may respond especially well to treatment with a

combination of anti-ICOS antibody and anti-PD-L1 antibody. Appropriate treatment regimens and antibodies for this purpose have already been detailed in the foregoing description.

Using the TCGA dataset as before, enrichment scores for ICOS and FOXP3 were correlated with expression levels of PD-L1 using Spearman's rank correlation and grouped by primary disease indication. P-values were calculated for each group and a p-value of 0.05 (with Bonferroni's multiple comparison correction) was taken as statistically significant. The disease groups with the highest correlations between ICOS/FOXP3 and PD-L1 expression were:

- TGCT (n=156) testicular germ cell tumour
- COAD (n=288) colon adenocarcinoma
- READ (n=95) rectum adenocarcinoma
- BLCA (n=407) bladder urothelial carcinoma
- OV (n=308) ovarian serous cystadenocarcinoma
- BRCA (n=1104) breast invasive carcinoma
- SKCM (n=473) skin cutaneous melanoma
- CESC (n=305) cervical squamous cell carcinoma and endocervical adenocarcinoma
- STAD (n=415) stomach adenocarcinoma
- LUAD (n=517) lung adenocarcinoma

Patients may be selected for treatment following an assay determining that their cancer is associated with ICOS+ immunosuppressive Tregs and expression of PD-L1. For cancer types in which, as above, there is a high correlation score, it may suffice to determine that one of ICOS+ immunosuppressive Tregs and expression of PD-L1 is present (e.g., above a threshold value). PD-L1 immunohistochemistry assays may be used in this context.

Example 26: Assessment of Further Anti-ICOS Antibodies

CL-74570 (STIM022) and CL-61091 (STIM017) antibody sequences identified in Example 12 were synthesised and expressed in IgG1 format in HEK cells.

Functional characterisation of these antibodies was performed using an HTRF assay similar to that described in Example 6, with modifications to adapt the assay to use of purified IgG1 rather than BCT supernatant. 5 µL of supernatant containing human IgG1 antibodies expressed from HEK cells was used in place of the BCT supernatant, and the total volume made up to 20 µl per well using HTRF buffer as before. A human IgG1 antibody was used as a negative control. Both antibodies exhibited greater than 5% effect for binding to human and mouse ICOS as calculated using Equation 1 and were therefore confirmed to test positive in this assay.

Ability of these antibodies to bind human and mouse ICOS expressed on the surface of CHO-S cells was further confirmed using a Mirrorball assay. In this assay, 5 µl supernatant containing the anti-ICOS IgG1 was transferred to each well of 384 mirrorball black plates (Corning). Binding of anti-ICOS antibodies was detected by adding 10 µl of goat anti-human 488 (Jackson ImmunoResearch) diluted in assay buffer (PBS+1% BSA+0.1% Sodium Azide) at a concentration of 0.8 mg/ml to all wells.

For positive control wells, 5 µL reference antibody diluted in assay media to 2.2 µg/ml was added to the plates. For negative control wells, 5 µl of Hybrid control IgG1 diluted in assay media to 2.2 µg/mL was added to the plates. 10 U_M of DRAQ5 (Thermoscientific) was added to 0.4×10⁶/ml cells resuspended in assay buffer and 5 µl was added to all wells. Plates were incubated for 2 hr at 4 degrees.

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Fluorescence intensity was measured using Mirrorball plate reader (TTP Labtech), measuring Alexafluor 488 (excitation 493 nm, emission 519 nm) from a population of 500-700 single cells. Assay signal was measured as Median (FL2) Mean Intensity.

Total binding was defined using reference antibody at an assay concentration of 2.2 µg/mL. Non-specific binding was defined using Hybrid control hIgG1 at an assay concentration of 2.22 µg/mL. Both antibodies exhibited greater than 1 percent effect and were therefore confirmed to test positive in this assay.

$$\text{Percent effect} = \frac{(\text{sample well} - \text{non-specific binding})}{(\text{total binding} - \text{non-specific binding})} \times 100$$

Each of CL-74570 (STIM022) and CL-61091 (STIM017) also demonstrated binding to human and mouse ICOS expressed on CHO-S cells as determined by flow cytometry. FACS screening was performed using a method similar to that described in Example 6, with modifications to adapt the assay to use of purified IgG1 rather than BCT supernatant. Both antibodies exhibited binding >10 fold above the average of geomean of the negative control binding to hICOS, mICOS and WT CHO cells.

TABLE E26-1

Functional characterisation of CL-74570 (STIM022) and CL-61091 (STIM017).						
Primary Screen						
HTRF (Protein)		Mirrorball (ICOS CHO Cell)		Secondary screen		
Human	Mouse	Human	Mouse	FACS		Clone ID
1:100 dil Percent Effect [%]	1:100 dil Percent Effect [%]	1:100 dil Percent Effect [%]	1:100 dil Percent Effect [%]	Human ICOS CHO (1:10 dil) % Binding-APC	Mouse ICOS CHO (1:10 dil) % Binding-APC	
94.42	60.86	107.02	127.03	122.97	96.41	CL-74570 STIM022
83.43	76.65	54.14	113.10	19.08	62.94	CL-61091 STIM017

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Example 27: Additional Screening Campaign for Anti-ICOS Antibodies

As mentioned in Example 12, additional antibody sequences were obtained by next generation sequencing of PCR-amplified antibody DNA from further ICOS-specific

Example 28: Analysis of Further Antibody Sequences

Table E28-1 and Table E28-2 identify the human germline gene segments that are believed to have recombined to generate the additional anti-ICOS antibodies.

TABLE E28-1

Gene segment and mutation characteristics of identified STIM antibodies									
Antibody ID	IGH V gene	IGH D gene	IGH J gene	CDR3 length	Non-germline CDR3 amino acid	IGH L gene	IGL J gene	CDR3 length	Non-germline CDR3 amino acid
STIM044	IGHV3-13*01	IGHD6-19*01	IGHJ6*02	19	7	IGKV3-20*01	IGKJ5*01	11	7
STIM043	IGHV3-13*01	IGHD6-19*01	IGHJ6*02	19	7	IGKV3-20*01	IGKJ5*01	10	2
STIM042	IGHV3-13*01	IGHD6-19*01	IGHJ6*02	19	7	IGKV3-20*01	IGKJ5*01	10	2
STIM041	IGHV3-13*01	IGHD3-22*01	IGHJ6*02	19	9	IGKV3-11*01	IGKJ5*01	10	3
STIM040	IGHV3-13*01	IGHD3-22*01	IGHJ6*02	19	10	IGKV3-11*01	IGKJ5*01	10	2
STIM039	IGHV3-13*01	IGHD3-22*01	IGHJ6*02	19	9	IGKV3-11*01	IGKJ5*01	10	3
STIM023	IGHV3-13*01	IGHD3-10*01	IGHJ6*02	21	7	IGKV1-9*d01	IGKJ1*01	10	3

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cells that were sorted from immunised mice. Approximately 570 additional clones were selected and sequenced. Using bioinformatics analysis, 467 of these clones were selected for synthesis and expression in HEK cells. Functional screening of cell supernatants was performed essentially as described in Example 6 and Example 26. Antibodies that passed the screening criteria included:

8 human/mouse cross-reactive anti-ICOS antibodies, of which the best were STIM017, STIM020, STIM021, STIM022 and STIM023; and

38 human-specific anti-ICOS antibodies, of which the best were STIM050, STIM051, STIM052, STIM053, STIM054, STIM055, STIM056, STIM057, STIM058, STIM059, STIM060, STIM061, STIM062, STIM063, STIM064, STIM065 and STIM066.

An additional further 20 sequenced clones from the immunised mice were also synthesised, purified and tested for binding to human and mouse ICOS. Of these, the best binders to human ICOS were STIM039, STIM040, STIM041, STIM042, STIM043 and STIM044, STIM050, STIM051, STIM052, STIM053, STIM054, STIM055, STIM056, STIM057, STIM058, STIM059, STIM060, STIM061, STIM062, STIM063, STIM064, STIM065 and STIM066.

TABLE E28-1-continued

Gene segment and mutation characteristics of identified STIM antibodies									
Antibody ID	IGH V gene	IGH D gene	IGH J gene	CDR3 length	Non-germline CDR3 amino acid	IGH L gene	IGL J gene	CDR3 length	Non-germline CDR3 amino acid
STIM022	IGHV3-20*d01	IGHD3-10*01	IGHJ4*02	17	4	IGKV3-20*01	IGKJ3*01	9	2
STIM021	IGHV3-23*04	IGHD4-17*01	IGHJ5*02	12	2	IGKV1-5*03	IGKJ1*01	8	0
STIM020	IGHV3-9*01	IGHD3-16*02	IGHJ6*02	20	0	IGKV1D-12*02	IGKJ2*04	9	0
STIM017	IGHV1-18*01	IGHD6-19*01	IGHJ6*02	14	3	IGKV2-28*01	IGKJ4*01	9	0

TABLE E28-2

Gene segment and mutation characteristics of identified STIM antibodies									
Antibody ID	IGH V gene	IGH D gene	IGH J gene	CDR3 length	Non-germline CDR3 amino acid	IGH L gene	IGL J gene	CDR3 length	Non-germline CDR3 amino acid
STIM064	IGHV3-13*01	IGHD6-19*01	IGHJ6*02	19	5	IGKV3-20*01	IGKJ5*01	10	2
STIM063	IGHV3-11*01	IGHD3-10*01	IGHJ6*02	15	6	IGKV1-5*03	IGKJ1*01	8	1
STIM062	IGHV3-11*01	IGHD3-10*01	IGHJ3*02	15	6	IGKV1-5*03	IGKJ1*01	8	1
STIM061	IGHV3-7*01	IGHD3-10*01	IGHJ6*02	15	4	IGKV1-5*03	IGKJ1*01	8	2
STIM060	IGHV3-11*01	IGHD3-10*01	IGHJ4*02	13	6	IGKV1-5*03	IGKJ1*01	8	0
STIM059	IGHV3-13*01	IGHD3-10*01	IGHJ6*02	19	8	IGKV3-11*01	IGKJ2*04	9	3
STIM058	IGHV3-11*01	IGHD4-17*01	IGHJ6*02	17	4	IGKV1-9*d01	IGKJ4*01	9	1
STIM057	IGHV3-13*01	IGHD3-10*01	IGHJ4*02	19	8	IGKV1-12*01	IGKJ4*01	9	0
STIM066	IGHV3-13*01	IGHD6-19*01	IGHJ6*02	19	7	IGKV3-20*01	IGKJ5*01	10	2
STIM065	IGHV3-11*01	IGHD3-10*01	IGHJ6*02	19	6	IGKV1-9*d01	IGKJ1*01	9	0
STIM056	IGHV3-13*01	IGHD3-22*01	IGHJ6*02	19	9	IGKV3-11*01	IGKJ5*01	10	3
STIM055	IGHV3-13*01	IGHD3-10*01	IGHJ6*02	19	6	IGKV3-11*01	IGKJ5*01	10	2
STIM054	IGHV3-9*01	IGHD3-10*01	IGHJ4*02	17	6	IGKV1D-12*02	IGKJ1*01	9	0
STIM053	IGHV3-11*01	IGHD3-10*01	IGHJ6*02	15	5	IGKV1-5*03	IGKJ1*01	8	1
STIM052	IGHV3-13*01	IGHD4-23*01	IGHJ6*02	19	7	IGKV3-20*01	IGKJ2*04	10	1
STIM051	IGHV3-11*01	IGHD1-20*01	IGHJ6*02	13	4	IGKV1-5*03	IGKJ1*01	8	1
STIM050	IGHV4-4*02	IGHD3-9*01	IGHJ5*02	14	3	IGKV2-28*01	IGKJ4*01	9	1

FIG. 46a shows an alignment of the V_H domain sequences of STIM020, STIM021 and STIM023.

FIG. 46b shows an alignment of the V_L domain sequences of STIM020, STIM021 and STIM023.

Example 29: Functional Characterisation of Additional Cross-Reactive Anti-ICOS Antibodies in HTRF Neutralisation Assay

Materials & Methods—Neutralisation of ICOS Ligand Binding to ICOS Receptor by HTRF

Test antibodies and isotype controls were diluted in assay buffer (0.53 M Potassium Fluoride (KF), 0.1% Bovine Serum Albumin (BSA) in 1xPBS) from a starting working concentration of up to 1 μ M, 400 nM to 0.0169 nM final and 0.0068 nM final respectively, over 11 point titration, 1 in 3 dilutions. Titrations of 5 μ l of antibody were added to 384w white walled assay plate (Greiner Bio-One). Positive control wells received 5 μ l of assay buffer only. 5 μ l of ICOS receptor (human ICOS-mFc, 20 nM, 5 nM final (Chimerigen) or mouse ICOS-mFc 4 nM, 1 nM final (Chimerigen)) was added to required wells. Plate was incubated for 1 hour (hr) at room temperature (RT). After incubation, 5 μ l of either mouse or human ICOS ligand, (B7-H2, R&D Systems) conjugated to Alexa 647 (Innova Bioscience) was diluted to either 25.72 nM (6.43 nM final) for human B7-H2 or 27.32 nM, 6.83 nM final for mouse B7-H2 and added to all wells of assay plate except negative control wells which instead received 5 μ l of assay buffer. Finally, 5 μ l of

anti-mouse IgG donor mAb (Southern Biotech) labelled with europium cryptate (Cis Bio), 4.32 μ M, 4.32 nM final was added to each well and the assay was left in the dark at RT to incubate for a further 2 hours. After incubation, assay was read on Envision plate reader (Perkin Elmer) using a standard HTRF protocol. 620 nm and 665 nm channel values were exported to Microsoft Excel (Microsoft) and % Delta-F and % Neutralisation calculations performed. Titration curves and IC50 values [M] were plotted using Graphpad (Prism). IC50 values were calculated by first transforming the data using equation $X = \text{Log}(X)$. The transformed data was then fitted using nonlinear regression, using fitting algorithm, log (inhibitor) vs. response-variable slope (four parameters).

% Delta-F Calculation

665/620 nm ratio for ratio metric data reduction.

% Delta $F =$

$$\frac{(665/620 \text{ nm Well Signal Ratio} - \text{Signal Negative Control})}{(\text{Signal Negative Control})} * 100$$

Signal Negative control = average of minimum signal ratio.

% Neutralisation

% Max (neutralisation) =

$$\frac{(\% \text{ Delta} - F \text{ of sample well} - \text{Negative Control})}{(\text{Positive Control} - \text{Negative Control})} * 100$$

Results

Five panel antibodies (mAbs) and two benchmarks (STIM001 and STIM003) were assessed for ICOS ligand (B7-H2) neutralisation using HTRF. mAbs capable of neutralising both human and mouse ICOS B7-H2 ligand (cross-reactive mAbs) were preferred over mAbs only capable of neutralising human or mouse ICOS ligand only. Human IgG1 isotypes of panel mAbs were assessed in both assays.

In the human IgG1 assays, benchmark antibody STIM001 produced an IC50 of 3.0 nM for the neutralisation of human ICOS ligand and an IC50 of 6.0 nM for the neutralisation of mouse ICOS ligand.

In the human IgG1 assays, benchmark antibody STIM003 produced an IC50 of 0.61 nM for the neutralisation of human ICOS ligand and an IC50 of 0.07 nM for the neutralisation of mouse ICOS ligand.

Cross-reactive mAb STIM017 produced similar IC50 to the STIM003 benchmark (IC50, 0.54 nM) using the human ICOS ligand neutralisation system. Similar to the benchmark, this mAb was also cross-reactive at neutralising mouse ICOS ligand (IC50, 0.09 nM).

Cross-reactive mAbs STIM021 and STIM023 produced similar IC50 to the STIM003 benchmark (IC50, 0.66 nM and 1.56 nM, respectively), using the human ICOS ligand neutralisation system. However, these mAbs produced weaker values for mouse ICOS ligand neutralisation (IC50, 84 nM and 17 nM, respectively).

Cross-reactive mAb STIM020 is a weak human (IC50, 80 nM) and mouse (IC50, 23 nM) ICOS ligand neutraliser.

Cross-reactive mAb STIM022 is a weak human (IC50, 15 nM) and strong mouse (IC50, 4.9 nM) ICOS ligand neutraliser.

TABLE E29-1

Summary of neutralisation data with human ICOS receptor with human B7-H2 ligand or mouse ICOS receptor with mouse B7-H2 ligand. IC50 (nM), n = 1 each assay.		
	Human ICOS Receptor/ Human B7-H2 IC50 (M)	Mouse ICOS Receptor/ Mouse B7-H2 IC50 (M)
STIM022	1.50795E-08	4.90853E-09
STIM021	6.68335E-10	8.47637E-08
STIM020	8.04167E-08	2.36411E-08
STIM017	5.49957E-10	9.31593E-11
STIM023	1.56931E-09	1.78505E-08
STIM003	6.19913E-10	7.18607E-11
STIM001	3.06618E-09	6.0775E-09
Hybrid	1.60491E-06	1.83055E-06
Control IgG1		

Data are illustrated in FIG. 47 and FIG. 48.

Example 30: Functional Characterisation of Additional Cross-Reactive Anti-ICOS Antibodies in FACS Assay

Binding to Cell-Expressed Mouse ICOS

Five selected antibodies (STIM017, STIM020, STIM021, STIM022, STIM023) and two benchmarks (STIM003 and

STIM001), and one isotype control were assessed for mouse ICOS binding using flow cytometry. These were characterised using anti-human IgG AlexaFluor 647 detection.

CHO-S cells transfected with mouse ICOS were diluted in FACS buffer (PBS+1% w/v BSA+0.1% w/v sodium azide) and were distributed to a 96-well V-bottom plate (Greiner) at a density of 1×10^5 cells per well. Antibody titrations were prepared from 399 nM working concentration as a 1:3 dilution series in FACS buffer over a 11 point titration. Plates were centrifuged at 300xg for 3 minutes and supernatant aspirated. 50 μ L per well of antibody titrations were added to cells and incubated at 4° C. for 1 hour. Cells were washed with 150 μ L of PBS and centrifuged at 300 g for 3 minutes. Supernatant was aspirated and 150 μ L PBS added per well. This wash step was repeated. Presence of bound antibody was detected by addition of 50 μ L per well of Anti-Human IgG AlexaFluor 647 (Jackson ImmunoResearch) diluted to 3 μ g/ml in FACS buffer. Cells were incubated for 1 hour at 4° C. in the dark. Cells were washed as previously described. To fix cells, 50 μ L per well of 4% v/v paraformaldehyde was added and cells incubated for 20 minutes at 4° C., cells were pelleted by centrifugation at 300xg and the plates resuspended in 110 μ L PBS. AlexaFluor 647 signal intensity (geometric mean) was measured by flow cytometry using a Beckman Coulter CytoFLEX instrument.

Antibodies STIM020, STIM022, STIM023, produced EC50 values within 3-fold to benchmark STIM003 (7.1 nM, 4.3 nM, 5.0 nM and 2.66 nM respectively).

Antibodies STIM017, STIM021, STIM023, produced EC50 values within 3-fold to benchmark STIM001 (25 nM, 20.5 nM and 7.14 nM respectively).

The isotype control did not show binding therefore its EC50 value cannot be calculated.

TABLE E30-1

Binding to mouse ICOS in FACS assay.	
	mICOS EC50 (M)
STIM022	4.3E-09
STIM021	2.05E-08
STIM020	7.1E-09
STIM017	2.51E-08
STIM023	5.04E-09
STIM003	2.66E-09
STIM001	7.14E-09

Data on binding of selected antibodies to the cell-expressed mouse ICOS are illustrated in FIG. 49.

Binding to Cell-Expressed Human ICOS

Five selected antibodies (STIM017, STIM020, STIM021, STIM022, STIM023) and two benchmarks (STIM003, STIM001) and one isotype control were assessed for human ICOS binding using flow cytometry. These were characterised using anti-human IgG AlexaFluor 647 detection.

CHO-S cells transfected with Human ICOS were diluted in FACS buffer (PBS+1% w/v BSA+0.1% w/v sodium azide) and were distributed to a 96-well V-bottom plate (Greiner) at a density of 1×10^5 cells per well. Antibody titrations were prepared from 399 nM working concentration as a 1:3 dilution series in FACS buffer over a 11 point titration. Plates were centrifuged at 300xg for 3 minutes and supernatant aspirated. 50 μ L per well of antibody titrations were added to cells and incubated at 4° C. for 1 hour. Cells were washed with 150 μ L of PBS and centrifuged at 300 g for 3 minutes. Supernatant was aspirated and 150 μ L PBS added per well. This wash step was repeated. Presence of

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bound antibody was detected by addition of 50 μ L per well of Anti-Human IgG AlexaFluor 647 (Jackson ImmunoResearch) diluted to 3 μ g/ml in FACS buffer. Cells were incubated for 1 hour at 4° C. in the dark. Cells were washed as previously described. To fix cells, 50 μ L per well of 4% v/v paraformaldehyde was added and cells incubated for 20 minutes at 4° C., cells were pelleted by centrifugation at 300 $\times$ g and the plates resuspended in 110 μ L PBS. AlexaFluor 647 signal intensity (geometric mean) was measured by flow cytometry using a Beckman Coulter CytoFLEX instrument.

Antibodies STIM017, STIM021, STIM022, STIM023 produced EC50 values within 3-fold to benchmark STIM003 (5.86 nM, 3.53 nM, 8.89 nM, 3.75 nM and 3.27 nM respectively). STIM020 showed binding but its EC50 could not be accurately ascertained, due to incomplete curve fit. The isotype control did not show binding therefore its EC50 value cannot be calculated.

TABLE E30-2

Binding to human ICOS in FACS assay.	
	hICOS EC50 (M)
STIM022	8.88746E-09
STIM021	3.52932E-09
STIM020	Not calculated
STIM017	5.85594E-09
STIM023	3.7483E-09
STIM003	3.27374E-09
STIM001	1.97189E-08

Data on binding of selected antibodies to the cell-expressed human ICOS are illustrated in FIG. 50.

Example 31: Functional Characterisation of Additional Cross-Reactive Anti-ICOS Antibodies in ADCC Assay

The antibody-dependent cell-mediated cytotoxicity (ADCC) activity of the selected antibodies was evaluated using an ADCC Reporter Bioassay. CHO cells expressing mouse or human ICOS were coincubated with effector cells (engineered Jurkat cells stably expressing human Fc γ RIIIa receptor-V158, Promega) that in the presence of an ADCC-enabled antibody produce luciferase in a concentration dependent manner. The soluble luciferase activity was assessed by measuring the luminescence produced as the luciferase transformed a luminogenic substrate into a luminescent product.

Results are shown in Table E31-1 and in FIG. 51 and FIG. 52.

TABLE E31-1

EC50 (Molar unit) of selected mAbs and benchmarks STIM001 and STIM003 in the mICOS CHO ADCC assay from n = 1 and in the hICOS CHO ADCC assay from n = 1.		
	Mouse ICOS EC50 (M)	Human ICOS EC50 (M)
STIM022	3.1E-10	1.82E-10
STIM021	3.56E-09	1.93E-10
STIM020	5.79E-10	5.26E-10
STIM017	5.45E-10	1.68E-10

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TABLE E31-1-continued

EC50 (Molar unit) of selected mAbs and benchmarks STIM001 and STIM003 in the mICOS CHO ADCC assay from n = 1 and in the hICOS CHO ADCC assay from n = 1.		
	Mouse ICOS EC50 (M)	Human ICOS EC50 (M)
STIM023	9.32E-10	1.57E-10
STIM003	7.31E-10	3.72E-10
STIM001	3.86E-10	2.32E-10

Method:

Target cells, CHO cells expressing mouse or human ICOS receptor were diluted in assay media (RPMI 1640+L-Glutamine (Gibco)+4% Low IgG FBS (Hyclone)) to 2000,000 cells/ml and plated into 96-well solid white plates at 25 μ L/well.

mAb panel of antibodies including Benchmarks (STIM001 and STIM003) were diluted in assay media to give 3 $\times$ antibody concentration and then serially diluted 9 $\times$ over a 1:3 titration in assay media, concentrations ranging from 6 μ g/ml to 914 μ g/ml.

25 μ L of Jurkat NFAT luciferase reporter effector cells diluted in assay media to 400,000 cells/ml and 25 μ L of diluted antibody (final volume 75 μ L/well) were added to each well to give a final antibody concentration ranging from 2 μ g/ml to 300 μ g/ml and a target:effector ratio of 5:1. After overnight incubation at 37° C., 5% CO₂, luciferase activity was measured by adding 75 μ L/well of luminogenic BioGlo substrate (Promega) and read on an Envision (Perlin Elmer) plate reader.

For data analysis, relative light unit (RLU) values from the raw data (Envision reads) were first normalised to 'Fold of induction' using the following equation:

$$\text{Fold of induction} = \frac{RLU(\text{induced}) - RLU(\text{background})}{RLU(\text{background})}$$

Mean and standard deviation "Fold of induction" values for each antibody concentration were then plotted and curves were fitted using GraphPad Prism 4-parameter log-logistic curve. A "Fold of induction" value above 2 was considered significant.

Example 32: Functional Characterisation of Additional Cross-Reactive Anti-ICOS Antibodies in MJ Soluble Agonism Assay

Five panel antibodies (mAbs), two benchmarks (STIM001 and STIM003) in the presence of a F(ab')₂ cross-linker, were assessed for their ability to induce expression of the cytokine IFN- γ from MJ cells grown in culture. Human IgG1 isotype controls cross-linked to F(ab')₂ were assessed in parallel.

Agonist antibodies were determined to be those antibodies that stimulate MJ cells as determined by IFN- γ quantification significantly above that observed by their cognate isotype controls within the dynamic range of the assay

STIM003 produced more potent EC50 values compared to STIM001 (2.84 nM and 6.24 nM, respectively). STIM017, STIM021 and STIM023 produced EC50 values that are within 3-fold of STIM003 (5.56 nM, 3.57 nM, 6.28 nM and 2.84 nM, respectively). STIM020 and STIM022 produced EC50 values that are within 3-fold of STIM001 (17.3 nM, 15.8 nM and 6.24 nM, respectively).

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Data from the soluble MJ cell in vitro activation assay with F(ab')₂ cross-linker are shown in Table E32-1 and in FIG. 53.

TABLE E32-1

Summary table of soluble MJ cell in vitro activation assay with cross-linker.	
	EC50 (M)
STIM022 + Linker	1.58E-08
STIM021 + Linker	3.57E-09
STIM020 + Linker	1.73E-08
STIM017 + Linker	5.56E-09
STIM023 + Linker	6.28E-09
STIM003 + Linker	2.84E-09
STIM001 + Linker	6.24E-09

Method:

MJ [G11] cell line (ATCC® CRL-8294™) was grown in IMDM (Gibco or ATCC) supplemented with 20% heat inactivated FBS. Cells were counted and 15000 cells/well (50 µl/well) of cell suspension was added to 96-well clear flat bottom polystyrene sterile TC-treated microplates. Serial 1:2 dilutions of antibodies of interest including benchmarks STIM003 and STIM001, ranging from 10 µg/ml to 0.01953125 µg/ml with the addition of a cross-linking reagent (AffiniPure F(ab')₂ Fragment Goat Anti-Human IgG, Fc Fragment Specific; Jackson ImmunoResearch) were added to the cells in triplicate (50 µl/well). The cells were co-cultured in the plates for 3 days at 37° C. and 5% CO₂ after which supernatants were harvested by centrifugation and collected for IFN-γ content determination.

The IFN-γ content in each well was determined using a modification of the Human IFN-γ DuoSet ELISA kit (R&D systems). Capture antibody (50 µl/well) was coated overnight at 4 µg/ml in DPBS on black flat bottom, high binding plates (Greiner). The wells were washed three times with 200 µl/well of DPBS+0.1% Tween. The wells were blocked with 200 µl/well of 1% BSA in DPBS (w/v), washed three times with 200 µl/well of DPBS+0.1% Tween and then 50 µl/well of either the IFN-γ standard solutions in RPMI or neat cell supernatant were added to each well. The wells were washed three times with 200 µl/well of DPBS+0.1% Tween before adding 50 µl/well of the detection antibody at 200 ng/ml in DPBS+0.1% BSA. The wells were washed three times with 200 µl/well of DPBS+0.1% Tween before adding 50 µl/well of streptavidin-europium (Perkin Elmer) diluted 1:500 in Assay buffer (Perkin Elmer). The wells were washed three times with 200 µl/well of TBS+0.1% Tween before developing the assay by adding 50 µl/well of DELFIA enhancement solution (Perkin Elmer) and measuring the fluorescence emitted at 615 nm on the EnVision Multilabel Plate Reader.

IFN-γ values for each well were interpolated from the standard curve and the average background levels from Hybrid Control-only treated wells were subtracted. The background corrected values were then used in GraphPad prism to fit a 4-parameter log-logistic dose response curve.

Example 33: Functional Characterisation of Additional Cross-Reactive Anti-ICOS Antibodies in DELFIA Assay

The binding activity of five selected monoclonal antibodies and benchmarks STIM001 and STIM003 to human ICOS was determined using dissociation-enhanced lanthanide fluorescence immunoassay (DELFLIA).

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In-vitro binding of antibodies to recombinant soluble human ICOS extracellular domain Fc-fusion proteins captured on solid support was measured by detection with anti-hIgG antibody Eu-labelled conjugate. The Eu is dissociated from the antibody and measured by time-resolved fluorescence (TRF) at 615 nm. The final reported values were EC50 values derived from a curve fitting process.

STIM017, STIM021, STIM022 and STIM023 showed binding to human ICOS protein and gave EC50 values within 3-fold of the benchmark STIM003 (1.43 nM, 0.84 nM, 2.08 nM, 1.21 nM and 0.78 nM, respectively). STIM020 showed binding to human ICOS protein but EC50 could not be determined accurately due to an in-complete curve-fit. See Table E33-1 and FIG. 54.

TABLE E33-1

EC50 (Molar unit) of selected mAbs and benchmarks STIM001 and STIM003 in the hICOS DELFLIA assay from n = 1.	
	EC50 (M)
STIM022	2.089E-09
STIM021	8.43622E-10
STIM020	Not Calculated
STIM017	1.43347E-09
STIM023	1.21023E-09
STIM003	7.861E-10
STIM001	8.29438E-09

Method:

Human ICOS-mouseFc (chimerigen) protein was diluted in 1xPBS and added to Hi-Bind solid black plates (Griener) at a concentration of 4 µg/ml, 50 µl/well. The plates were left to incubate overnight at 4° C. The next day plates were washed 3x with 300 µl/well of 1xPBS+0.1% Tween followed by the addition of 200 µl/well of 1% w/v BSA/PBS blocking buffer. Wells were blocked for 1 hr at room temperature (RT) on a plate shaker. Plates were washed again 3x as above.

Five anti-ICOS antibodies and benchmarks STIM001 and STIM003 or isotype control antibodies were diluted in 0.1% w/v BSA/PBS to 60 µg/ml, then serially diluted 1 in 3 to generate 11 point set of concentrations. 50 µl/well of these dilutions were added to the blocked wells of the Hi-Bind plates and left to incubate for 1 hr at RT on a plate shaker. Plates were washed again 3x as above.

DELFLIA® Eu-labelled anti-human IgG (Perkin-Elmer) was diluted 1:500 in DELFLIA Assay buffer (Perkin-Elmer) and added to wells at 50 µl/well and left to incubate for 1 hr at RT on a plate shaker. Plates were then washed 3x with 300 µl/well 1xDELFLIA wash buffer (Perkin-Elmer) before the addition of 50 µl/well of DELFLIA Enhancement Solution (Perkin-Elmer), incubated for a minimum of 5 minutes in the dark before reading plate on an Envision (Perkin Elmer) plate reader at 615 nm.

Example 34a: Functional Characterisation of Additional Anti-Human ICOS Antibodies STIM039 to STIM044 in FACS Assay

Antibodies STIM039 to STIM044, two benchmarks (STIM003, STIM001) and one isotype control were assessed for human ICOS and mouse ICOS binding using flow cytometry. These were characterised using anti-human IgG AlexaFluor 647 detection.

Antibodies STIM039, STIM040, STIM041, STIM042, STIM043, STIM044, produced EC50 values within 3-fold

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to benchmark STIM003 (5.52 nM, 5.89 nM, 5.79 nM, 8.01 nM, 7.6 nM, 6.57 nM and 6.69 nM respectively).

None of the selected antibodies showed any binding to mouse ICOS-expressing CHO cells, confirming them to be human-specific binders. The isotype control did not show binding to human ICOS or mouse ICOS in this assay.

TABLE E34-1

	EC50 (M)
STIM039	5.52045E-09
STIM040	5.89514E-09
STIM041	5.79131E-09
STIM042	8.01378E-09
STIM043	7.60866E-09
STIM044	6.57914E-09
STIM003	6.69899E-09
STIM001	1.45395E-08

Method: Binding of Selected Antibodies to Human or Mouse ICOS Expressing CHO Cells

CHO-S cells transfected with either human ICOS or mouse ICOS were diluted in FACS buffer (PBS+1% w/v BSA+0.1% w/v sodium azide) and were distributed to a 96-well V-bottom plate (Greiner) at a density of 1×10^5 cells per well. Antibody titrations were prepared from 399 nM or 66.6 nM working concentration as a 1:3 dilution series in FACS buffer over a 11 point titration. Plates were centrifuged at $300 \times g$ for 3 minutes and supernatant aspirated. 50 μ L per well of antibody titrations were added to cells and incubated at 4° C. for 1 hour. Cells were washed with 150 μ L of PBS and centrifuged at $300 \times g$ for 3 minutes. Supernatant was aspirated and 150 μ L PBS added per well. This wash step was repeated. Presence of bound antibody was detected by addition of 50 μ L per well of Anti-Human IgG AlexaFluor 647 (Jackson ImmunoResearch) diluted to 3 μ g/ml in FACS buffer. Cells were incubated for 1 hour at 4° C. in the dark. Cells were washed as previously described. To fix cells, 50 μ L per well of 4% v/v paraformaldehyde was added and cells incubated for 20 minutes at 4° C., cells were pelleted by centrifugation at $300 \times g$ and the plates resuspended in 110 μ L PBS. AlexaFluor 647 signal intensity (geometric mean) was measured by flow cytometry using a Beckman Coulter CytoFLEX instrument.

Example 34b: Functional Characterisation of Additional Anti-Human ICOS Antibodies STIM050 to STIM066 in FACS Assay

Seventeen antibodies ranging from STIM050 to STIM066, two benchmarks (STIM003, STIM001) and one isotype control were assessed for human ICOS and mouse ICOS binding using flow cytometry. These were characterised using anti-human IgG AlexaFluor 647 detection.

Antibodies STIM050 to STIM066 produced Geomean values within 4.3-fold to benchmark STIM003 and 2.3 fold to benchmark STIM001 for human ICOS binding.

None of the selected antibodies showed any binding to mouse ICOS-expressing CHO cells, confirming them to be human-specific binders. The isotype control did not show binding to human ICOS or mouse ICOS in this assay.

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TABLE E34-2

Binding of selected antibodies to CHO cells expressing Human ICOS or Mouse ICOS. Antibody binding was detected with an anti-human IgG AlexaFluor 647 and Geomean was compared to benchmarks STIM001 and STIM003. Data is from a single experiment. Antibody concentration was 6.66 μ g/ml in all cases.

	STIM ID	Geomean (647)	
		human ICOS CHO cells	mouse ICOS CHO cells
10	STIM050	607000	442
	STIM051	996000	400
	STIM052	631000	346
15	STIM053	1120000	373
	STIM054	1630000	424
	STIM055	1410000	420
20	STIM056	1440000	383
	STIM057	675000	323
	STIM058	642000	319
25	STIM059	1080000	321
	STIM060	1400000	690
	STIM061	962000	371
30	STIM062	502000	346
	STIM063	863000	382
	STIM064	1310000	436
35	STIM065	1440000	344
	STIM066	1590000	1145
	STIM003	2160000	940000
40	STIM001	1190000	300010
	Hybrid	810	1508
	Control IgG1		

Method: Binding of Selected Antibodies to Human or Mouse ICOS Expressing CHO Cells

CHO-S cells transfected with either Human ICOS or Mouse ICOS (generated in-house) were diluted in FACS buffer (PBS+1% w/v BSA+0.1% w/v sodium azide) and were distributed to a 96-well V-bottom plate (Greiner) at a density of 1×10^5 cells per well. Antibodies were diluted to 6.66 μ g/ml in FACS buffer. Plates were centrifuged at $300 \times g$ for 3 minutes and supernatant aspirated. 50 μ L per well of antibody titrations were added to cells and incubated at 4° C. for 1 hour. Cells were washed with 150 μ L of PBS and centrifuged at $300 \times g$ for 3 minutes. Supernatant was aspirated and 150 μ L PBS added per well. This wash step was repeated. Presence of bound antibody was detected by addition of 50 μ L per well of Anti-Human IgG AlexaFluor 647 (Jackson ImmunoResearch) diluted to 2 μ g/ml in FACS buffer. Cells were incubated for 1 hour at 4° C. in the dark. Cells were washed as previously described. To fix cells, 50 μ L per well of 4% v/v paraformaldehyde was added and cells incubated for 20 minutes at 4° C., cells were pelleted by centrifugation at $300 \times g$ and the plates resuspended in 110 μ L PBS. AlexaFluor 647 signal intensity (geometric mean) was measured by flow cytometry using a Beckman Coulter CytoFLEX instrument.

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- Sequences
- Sequences of human/mouse cross-reactive anti-ICOS antibodies STIM020, STIM021 and STIM023, and sequences of anti-human-ICOS antibodies STIM039 to STIM044 and STIM050 to STIM066, are set out below. For each antibody, sequences are shown in the following order:
 - Nucleotide sequence encoding VL domain
 - VL domain amino acid sequence
 - Nucleotide sequence encoding VH domain
 - VH domain amino acid sequence
- Table of CDRs defined according to Kabat and IMGT systems respectively.

STIM066

Nucleotide sequence encoding VL domain

(SEQ ID NO: 629)

GAAATTGTGTTGACGCAGTCTCCAGGCACCCCTGTCTTGTCTCCAGGGGAAAGAGCCACCCTCTCCTGCAGGGCCAG
 TCAGAGTGTAGCAGCAGCTCCTTAGCTGGTACCAGCAGAAACCTGGCCAGGCTCCCAGGCTCCTCATCTATGGTG
 CGTCCAGCAGGGCCACTGGCATCCAGACAGGTTAGTGGCAGTGGGTCTGGGACAGACTTCACTCTCACCATCAGC
 AGACTGGAGCCTGAAGATTTTGAGTGTATTACTGTCAGCAGTTTGGTAGCTCACCTCCGATCACCTTCGGCCAAGG
 GACACGACTGGAGATTAAAC

VL domain amino acid sequence

(SEQ ID NO: 628)

EIVLTQSPGTLSPGERATLSCRASQSVSSSLAWYQQKPGQAPRLLIYGASSRATGIP
 DRFSGSGSGTDFTLTISRLEPEDFAVYYCQQFGSPFITFGQGRLEIK

Nucleotide sequence encoding VH domain

(SEQ ID NO: 621)

GAGGTGCAGCTGGTGGAGTCTGGGGGAGGCTTGGTACTGCCCGGGGGTCCCTGAGACTCTCCTGTGCAGCCTCTGG
 ATTCACCTTCAATATGTACGACATGCATGGGTCCGCCAAGTTACAGGAAAAGTCTGGAGTGGGTCGCAGCTATAG
 GTACTGCTGGTGACACATACTATACAGGCTCCGTGAAGGTCGACTCACCATCTCCAGAGAAAATGCCAAGAAGTCC
 TTGTATCTTCAAATGAACAGCCTGAGAGCCGGTGACACGGCTGTGTATTACTGTGTAAGAGATCGTAGAGCAGTGGC
 TGGTCACTACTACTACTTCTACGGTTTGGACGCTCTGGGGCCAAGGGACCACGGTCACCGTCTCCTCAG

VH domain amino acid sequence

(SEQ ID NO: 620)

EVQLVESGGGLVLPGGSRLRLSCAASGFTFNMYDMHWVRQVTGKGLEWVAIAGTAGDITYT
 GSVKGRLTISRNAKSLYLQMNSLRAGDTAVYYCVRDRRAVAGHYFFYGLDVWGQGT

VTVSS

Scheme	Locus	CDR1	CDR2	CDR3	Locus	CDR1	CDR2	CDR3
kabat	H	MYDMH (SEQ ID NO: 617)	AIGTAGDITYT GSVKG (SEQ ID NO: 618)	DRRAVAGHYFFY GLDV (SEQ ID NO: 619)	L	RASQSVSS SSLA (SEQ ID NO: 625)	GASS RAT (SEQ ID NO: 626)	QQFGSSP PIT (SEQ ID NO: 627)
imgt	H	GFTFN MYD (SEQ ID NO: 614)	IGTAGDT (SEQ ID NO: 615)	VRDRRAVAGHYFFY FYGLDV (SEQ ID NO: 616)	L	QSVSSSS (SEQ ID NO: 622)	GAS (SEQ ID NO: 624)	QQFGSSP PIT (SEQ ID NO: 624)

STIM065

Nucleotide sequence encoding VL domain

(SEQ ID NO: 645)

GACATCCAGTTGACCCAGTCTCCATCCTTCCTGTCTGCATCTGTAGGAGACAGAGTCACCATCACTTGCTGGGCCAG
 TCAGGGCATTAGCAGTTATTTAGCCTGGTATCAGCAAAACCAGGGAAAGTCCCTAAGCTCCAGATCTATGCTGCAT
 CCACCTTGCAAAGCGGGTCCCATCAAGTTACAGCGCAGTGGATCTGGGACAGAATCACTCTCACAATCAGCAGC
 CTGCAGCCTGAAGATTTTGCAACTTATTACTGTCAACAGCTTAATAGTTACCCGTGGACGTTTCGGCCAAGGGACCAA
 GGTGGAATCAAAC

VL domain amino acid sequence

(SEQ ID NO: 644)

DIQLTQSPSFLSASVGDRVTITCWASQGISSYLAWYQQKPKVPLQIYAASLTQSGVPS
 RFSGSGSGTEFTLTISLQPEDFATYYCQQLNSYPWTFGQGTKVEIK

Nucleotide sequence encoding VH domain

(SEQ ID NO: 637)

CAGGTGCAGCTGGTGGAGTCTGGGGGAGGCTTGGTCAAGCCTGGAGGGTCCCTGAGACTCTCCTGTGCAGCCTCTGG
 ATTCACCTTCAGTGACTACTACATGAGCTGGATCCGCCAGGCTCCAGGGAAGGGCTGGAGTGGGTTTCATACATTG
 GTGGTAGTGGTACTACCATATACTACGCAGACTCTGTGAAGGGCCGATTACCATCTCCAGGGACAACGCCAAGAAC
 TCACTGTATCTGCAAATGAACAGCCTGAGAGCCGAGGACACGGCCGTGTATTACTGTGCGAGAGGGGATATTACTA
 TGTTTCGGGACTTATACTACTTCGTATGGGCGTCTGGGGCCAAGGGACCACGGTCACCGTCTCCTCAG

-continued

VH domain amino acid sequence

(SEQ ID NO: 636)

QVQLVESGGGLVKPGGSLRLSCAASGFTFSDYYMSWIRQAPGKGLEWVS YIGSGTTIYY

ADSVKGRFTISRDNKNSLYLQMNSLRAEDTAVYYCARGGYYYSYFGMGVWGQGT

TVTVSS

Scheme	Locus	CDR1	CDR2	CDR3	Locus	CDR1	CDR2	CDR3
kabat	H	DYMS (SEQ ID NO: 633)	YIGSGGT- TIYYA DSVKG (SEQ ID NO: 634)	GGYYYSGTYNYF GMGV (SEQ ID NO: 635)	L	WASQGIS SYLA (SEQ ID NO: 641)	AAS LQS (SEQ ID NO: 642)	QQLNSY PWT (SEQ ID NO: 643)
imgt	H	GFTFS DYY (SEQ ID NO: 630)	IGSGGTI (SEQ ID NO: 631)	ARGGYYSGTYN YFGMGV (SEQ ID NO: 632)	L	QGISSY (SEQ ID NO: 638)	AAS (SEQ ID NO: 640)	QQLNSY PWT (SEQ ID NO: 640)

STIM064

Nucleotide sequence encoding VL domain

(SEQ ID NO: 661)

GAAATGTGTTGACGAGTCTCCAGGACCCCTGTCTTTGTCTCCAGGGGAAAGAGCCACCCTCTCCTGCAGGGCCAG

TCAGAGTGTAGCAGCAGCTCCTTAGCCTGGTACCAGCAGAAACCTGGCCAGGCTCCAGGCTCCTCATCTCTGGTG

CATCCAGCAGGGCCACTGGCATCCAGACAGGTTAGTGGCAGTGGGTCTGGACAGACTTCACTCTCACCATCAGC

AGACTGGAGCCTGAAGATTTTACAATGTATTACTGTGACAGTGGGTAGCTCACCTCCGATCACCTTCGGCCAAGG

GACACGACTGGAGATTAAAC

VL domain amino acid sequence

(SEQ ID NO: 660)

EIVLTQSPGTLSPGERATLSCRASQSVSSSSSLAWYQQKPGQAPRLLISGASSRATGIP

DRFSGSGSGTDFTLTISRLEPEDFTMYCQQFGSSPPI TFGQGTREIK

Nucleotide sequence encoding VH domain

(SEQ ID NO: 653)

GAGGTGCAGCTGGTGGAGTCTGGGGGAGGCTTGGTACAGCCGGGGGGTCCCTGAGACTCTCCTGTGCAGCCTCTGG

ATTACCTTCAGTAGCTACGACATGCAGTGGTCCGCCAAGCTACAGGAAAAGTCTGGAGTGGGTCTCAGTTATTG

GTACTGCTGGTGACACATACTATCCAGGCTCCGTGAAGGGCCGATTACCATCTCCAGAGAGAATGCCAAGAACTCC

TTGTATCTTCAAATGAACAGCTGAGAGCCGGGACACGGCTGTGTATTACTGTGTAAGAGATCGTAGAGCAGTGGC

TGGTCATTACTACTATTACTACGGTATGGAGCTCTGGGGCCAAGGGACCACGGTCACCGTCTCCTCAG

VH domain amino acid sequence

(SEQ ID NO: 652)

EVQLVESGGGLVQPGGSLRLSCAASGFIFSSYDMHWVRQATGKGLEWVS VIGTAGDTYYP

GSVKGRFTISRDNKNSLYLQMNSLRAGDTAVYYCVRDRRAVAGHYYYYYGMDVWGQGT

VTVSS

Scheme	Locus	CDR1	CDR2	CDR3	Locus	CDR1	CDR2	CDR3
kabat	H	SYDMH (SEQ ID NO: 649)	VIGTAGDTYYP GSVKG (SEQ ID NO: 655)	DRRAVAGHYYYYY GMDV (SEQ ID NO: 651)	L	RASQSVSS SSLA (SEQ ID NO: 657)	GASS RAT (SEQ ID NO: 658)	QQFGSSP PIT (SEQ ID NO: 659)
imgt	H	GFIFS SYD (SEQ ID NO: 646)	IGTAGDT (SEQ ID NO: 647)	VRDRRAVAGHYYY YYGMDV (SEQ ID NO: 648)	L	QSVSSSS (SEQ ID NO: 654)	GAS (SEQ ID NO: 656)	QQFGSSP PIT (SEQ ID NO: 656)

STIM063

Nucleotide sequence encoding VL domain

(SEQ ID NO: 676)

GACATCCAGATGACCCAGTCTCCTTCCACCTGTCTGCATCTGTAGGAGACAGAGTACCATCACTTCCCGGCCAG

TCAGAGTATTAGTAGTTGGTTGGCCTGGTATCAGCAGAAACAGGGAAAGCCCTAAACTCCTGATCTATAAGGCGT

- continued

CTAGTTTAGAAAGTGGGGCCCCATCAAGGTTTCAGCGGCAGTGGATCGGGGACGCAGTTCACCTCTCACCATCAGCAGT
 TTGCAGCCTGATGATTTTGCCACTTATTACTGCCAACAGTATTATAGTTATTGGACGTTTCGGCCAAGGGACCAAGGT
 GGAAATCAAAC

VL domain amino acid sequence

(SEQ ID NO: 675)

DIQMTQSPSTLSASVGDRTITCRASQSISSWLANWYQQKPKAPKLLIYKASSLESGLAPS
 RFSGSGSGTQFTLTISLQPDFFATYYCQQYYSYWTFGQGTKVEIK

Nucleotide sequence encoding VH domain

(SEQ ID NO: 669)

CAGGTGCAGCTGGTGGAGTCTGGGGGAGACTTGGTCAAGCCTGGAGGGTCCCTGAGACTCTCCTGTACAGCCTCTGG
 ATTCACCTTCAGTGACTACTACATGGACTGGATCCGCCAGGCTCCAGGGAAGGGCTGGAGTGGGTTTCATACATTA
 GTAGTAGTGGTGATACCATATACTACGCAGACTCTGTGAAGGGCCGATTACCATCTCCAGGGACAACGCCAAGAAC
 TCACTGTTTCTGCAAATGGACAGCCTGAGAGCCGAGGACACGGCCGTATATTACTGTGCGAGAGATGCCCACTATGT
 TTCGGGGAGCAAGGGTGTGGACGCTCTGGGGCCAAGGGACCACGGTCACCGTCTCCTCAG

VH domain amino acid sequence

(SEQ ID NO: 668)

QVQLVESGGDLVKPGGSLRLSCTASGFTFSDDYMDWIRQAPGKLEWVSYISSSGDTIYY
 ADSVKGRFTISRDNKNSLFLQMDSLRAEDTAVYYCARDAHYVSGSKGVDVWGQGTITVTV

SS

Scheme	Locus	CDR1	CDR2	CDR3	Locus	CDR1	CDR2	CDR3
kabat	H	DYYMD	YIS- SSGDTIYYA	DAHVVS	L	RASQSISS	KASSL	
		(SEQ ID NO: 665)	DSVKG (SEQ ID NO: 666)	DV (SEQ ID NO: 667)		WLA (SEQ ID NO: 673)	ES (SEQ ID NO: 647)	
imgt	H	GFTFSD YY (SEQ ID NO: 662)	ISSSGDTI (SEQ ID NO: 663)	ARDAHYVSGSK GVDV (SEQ ID NO: 664)	L	QSISSW (SEQ ID NO: 670)	KAS	QQYYSY WT (SEQ ID NO: 672)

STIM062

Nucleotide sequence encoding VL domain

(SEQ ID NO: 691)

GACATCCAGATGACCCAGTCTCCTTCCACCCGTGTGTCATCTGTAGGAGACAGAGTACCACCTACTTGC CGGCCAG
 TCAGAGTATTAGTAGCTGGTTGGCCTGGTATCAGCAGAAACAGGGAAAGCCCTAAGCTCCTGATCTATAAGGCGT
 CTAGTTTAGAAAGTGGGGTCCCATCAAGGTTTCAGCGGCAGTGGATCTGGGACAGAATTCACCTCTCACCATCAGCAGC
 CTGCAGCCTGATGATTTTGCAACTTATTACTGCCAACATATCTTAGTTATTGGACGTTTCGGCCAAGGGACCAAGGT
 GGAAATCAAAC

VL domain amino acid sequence

(SEQ ID NO: 690)

DIQMTQSPSTLSASVGDRTITCRASQSISSWLANWYQQKPKAPKLLIYKASSLESGLVPS
 RFSGSGSGTEFTLTISLQPDFFATYYCQQYLSYWTFGQGTKVEIK

Nucleotide sequence encoding VH domain

(SEQ ID NO: 684)

CAGGTGCAGCTGGTGGAGTCTGGGGGAGGCTGGTCAAGCCTGGAGGGTCCCTGAGACTCTCCTGTGAAGCCTCTGG
 ATTCACCTTCAGTGACTACTACATGAGCTGGATTCGCCAGGCTCCAGGGAAGGGCTGGAGTGGGTTTCATACATTA
 GTAGTAGTGGTAGTGCATATACTACGCAGACTCTGTGAAGGGCCGATTACCATCTCCAGGGACAACGCCAAGAAC
 TCACTGTATCTGCAAATGAACAGCCTGAGAGTCGAGGACACGGCCGTGTATTACTGTGCGAGAGATGAGGATTACCA
 TGTTTCGGGGAGTCCCCGGGATATCTGGGGCCAAGGGACAATGGTCACCGTCTCTTCAG

-continued

VH domain amino acid sequence

(SEQ ID NO: 683)

QVQLVESGGGLVPGGSLRLSCEASGFTFSDIYMSWIRQAPGKGLEWVS_YISSSGSAIYYADSVKGRFTISRDN_{AK}NSLYLQMNSLRVEDTAVYYCARDEDYHGSGSPRDIWGQGTMTVTV

SS

Scheme	Locus	CDR1	CDR2	CDR3	Locus	CDR1	CDR2	CDR3
kabat	H	DYMS (SEQ ID NO: 680)	YISSSG- SAIYYA DSVKG (SEQ ID NO: 681)	DEDYHGSGSPR DI (SEQ ID NO: 682)	L	RASQSISS WLA (SEQ ID NO: 688)	KASSL ES (SEQ ID NO: 689)	
imgt	H	GFTFSD YY (SEQ ID NO: 677)	ISSSGSAI (SEQ ID NO: 678)	ARDEDYHGSGS PRDI (SEQ ID NO: 679)	L	QSISSW (SEQ ID NO: 685)	KAS WT (SEQ ID NO: 687)	QQYLSY

STIM061

Nucleotide sequence encoding VL domain

(SEQ ID NO: 706)

GACATCCAGATGACCCAGTCTCCTTCCACCCGTGTCTGCATCTGTGGGAGACAGAGTCACCATCACTTCCGGGCCAG

TCAGAGTATTAGTAGCTGGTTGGCCTGGTATCAGCAGAAACCAGGGAAGCCCTAACCTCCTGATCTATAAGGCGT

CTAGTTTAGAAAGTGGGGTCCCATCAAGTTCAGCGGCAGTGGATCTGGGACAGAATCACTCTCACCATCAGCAGC

CTGCAGCCTGATGATTTTGCAACTTATTACTGCCAACAGTATTATTTATTGGACGTCGGCCAAGGACCAAGGT

GGAAATCAAAC

VL domain amino acid sequence

(SEQ ID NO: 705)

DIQMTQSPSTLSASVGD_{RV}TI_{TC}RASQSISSWLA_{WY}QKPKAPNLLIYKASSLES_{GV}PSRFSGSGSGTEFTLT_{IS}SLQ_PDDFATYYCQQYYIY_{WT}FGQGT_KVEIK

Nucleotide sequence encoding VH domain

(SEQ ID NO: 699)

GAGGTGCAGCTGGTGGAGTCTGGGGGAGGCTTGGCCAGCCTGGGGGTCCTGAGACTCTCCTGTGCAGCCTCTGG

ATTCTCCTTTAGTAATTATTGGATGA_{ACT}GGGTCCGCCAGGCTCCAGGGAAGGGCTGGAGTGGGTGGCCAACATAA

AGCGAGATGGAAGTGAAAAATATTATGTGGACTCTGTGAAGGGCCGATTACCATCTCCAGAGACAACGCCAAGAAC

TCACTATATCTGCAATGAACAGCCTGAGAGCCGAGGACACGGCTGTATATTACTGTGCGAGAGATGATTACTATGG

TTCGGA_{CT}ACTCCGGTATGGACGTCTGGGGCCAAGGACACGGTCACCGTCTCCTCAG

VH domain amino acid sequence

(SEQ ID NO: 698)

EVQLVESGGGLAQPGGSLRLS_{CA}ASGFSFSNYWMN_{WVR}QAPGKGLEWVANI_{KRD}GSEKYYVDSVKGRFTISRDN_{AK}NSLYLQMNSLRAEDTAVYYCARD_{DY}YGS_{DY}SGMDVWGQGT_{TV}TV

SS

Scheme	Locus	CDR1	CDR2	CDR3	Locus	CDR1	CDR2	CDR3
kabat	H	NYWMN (SEQ ID NO: 695)	NIKRDGSEKYYDDY _Y GS _{DY} SGM DSVKG (SEQ ID NO: 696)	DV (SEQ ID NO: 697)	L	RASQSISS WLA (SEQ ID NO: 703)	KASSL ES (SEQ ID NO: 704)	
imgt	H	GFSFSN YW (SEQ ID NO: 692)	IKRDGSEK (SEQ ID NO: 693)	ARDDY _Y GS _{DY} S GMDV (SEQ ID NO: 694)	L	QSISSW (SEQ ID NO: 700)	KAS WT (SEQ ID NO: 702)	QQYYIY

STIM060

Nucleotide sequence encoding VL domain

(SEQ ID NO: 721)

GACATCCAGATGACCCAGTCTCCTTCCACCCGTGTCTGCATCTGTAGGAGACAGAGTCACCATCACTTCCGGGCCAG

TCAGAGTATTAGTGGCTGGTTGGCCTGGTATCAGCAGAAACCAGGGAAGCCCTAAGCTCCTGATCTATAAGGCGT

- continued

CTAGTTTAGAAAGTGGGGTCCCATCAAGGTTACGCGGCAGTGGATCTGGGACAGAATCACTCTCACCATCAGCAGC
CTGCAGCCTGATGATTTTGCAACTTATTACTGCCAACAGTATAATAGTTATTGGACGTTCCGCCAAGGGACCAAGGT
GGAAATCAAAC

VL domain amino acid sequence

(SEQ ID NO: 720)

DIQMTQSPSTLSASVGDRTITCRASQISGNLAWYQQKPKAPKLLIYKASSLESQVPS
RFSGSGSGTEFTLTISSLQPDFATYYCQYNSYWTFGQGTKVEIK

Nucleotide sequence encoding VH domain

(SEQ ID NO: 714)

CAGGTGCAGCTGGTGGAGTCTGGGGGAGGCTTGGTCAAGCCTGGAGGGTCCCTGAGACTCTCCTGTGCAGCCTCTGG
ATTACCTTCAGTGACTACTACATGAGCTGGGTCCGCCAGGCTCCAGGGAAGGGGCTCGAGTGGGTTCATACATTA
GTGGTAGTGGTAGTACCATATACTACGCAGACTCTGTGAAGGGCCGATTACCATCTCCAGGACAACGCCAAGAAT
TCACTGTATCTGCAAATGAACAGCCTGAGAGCCGAGGATACGGCCGTGTATTACTGTGTGAGAGATCAAGACTATGG
TTTGGGGAGTTATGAATACTGGGGCCAGGGAACCTGGTCACCGTCTCCTCAG

VH domain amino acid sequence

(SEQ ID NO: 713)

QVQLVESGGGLVKPGSSLRSCAASGFTFSDDYMSWVRQAPGKLEWVSYISGSGSTIYY
ADSVKGRFTISRDNKNSLYLQMNSLRAEDTAVYYCVRDQDYGLGSYEWGQGLTLVTSS

Scheme	Locus	CDR1	CDR2	CDR3	Locus	CDR1	CDR2	CDR3
kabat	H	DYMS (SEQ ID NO: 710)	YISGSGSTIYY ADSVKG (SEQ ID NO: 711)	DQDYGLGSYE Y (SEQ ID NO: 712)	L	RASQSIG WLA (SEQ ID NO: 718)	KASSL ES (SEQ ID NO: 719)	
imgt	H	GFTFSD YY (SEQ ID NO: 707)	ISGSGSTI (SEQ ID NO: 708)	VRDQDYGLGS YEY (SEQ ID NO: 709)	L	QSISGW (SEQ ID NO: 715)	KAS (SEQ ID NO: 717)	QQYNSY WT (SEQ ID NO: 717)

STIM059

Nucleotide sequence encoding VL domain

(SEQ ID NO: 737)

GAAATTGTGTTGACACAGTCTCCAGCCACCCTGTCTTTGTCTCCAGGGAAAGAGCCACCCTCTCCTGCAGGGCCAG
TCAGAGTGTTAGCAGTACTTAGCCTGGTACCAACAGAAACCTGGCCAGGCTCCAGGCTCCTCATCTATGATGCAT
CCAACAGGGCCACTGGCATCCAGCCAGGTTCACTGGCACTGGGTCTGGGACAGACTTCACTCTCACCATCAGCAGC
CTAGAGTCTGAAGATTTTGCAGTTTATTCTGTCTAGCAACGTTACAACCTGGCCTCCGGGTTTGGCCAGGGGACCAA
GCTGGAGATCAAAC

VL domain amino acid sequence

(SEQ ID NO: 736)

EIVLTQSPATLSLSPGERATLSCRASQSVSSYLAWYQQKPGQAPRLLIYDASNRTGIPA
RFSGTGSGTDFTLTISLSEDFAVYFCQQRYNWPPGFGQGTKLEIK

Nucleotide sequence encoding VH domain

(SEQ ID NO: 729)

GAGGTACAGCTGGTGAATCTGGGGGAGGCTTGGTACAGCCTGGGGGCTCCCTGAGACTCTCCTGTACAGCCTCTGG
ATTACCTTCAGTAATTACGACATGCATTGGGTCCGCCAAGTTACAGGAAAAGTCTGGAATGGGTCTCAATTATTG
GTACTGCTGGTGACATACTATTAGGCTCCGTGAAGGGCCGATTACCATCTCCAGAGATAATGCCAAGAACTCC
GTGTATCTTCAAATGAACAGCCTGAGTCCGGGGACACGGCTGTGTATTATTGTGTAAGAGAGTTATAGTGGTTCG
GGGAGGTAATTACTACAACACGGTATGGACGCTGGGGCCAAAGGGACCACGGTCACCGTCTCCTCAG

-continued

VH domain amino acid sequence

(SEQ ID NO: 728)

EVQLVESGGGLVQPGGSLRLSCTASGFTFSNYDMHWVRQVTGKGLEWVSIIGTAGDTYYS

GSVKGRFTISRDNAKNSVYLQMNSLSAGDTAVYYCVREIVVRGGNYNYGMDVWGQGT

VTSS

Scheme	Locus	CDR1	CDR2	CDR3	Locus	CDR1	CDR2	CDR3
kabat	H	NYDMH (SEQ ID NO: 725)	IIGTAGDTYYS (SEQ ID NO: 726)	EVIVVRGGNYNY (SEQ ID NO: 727)	L	RASQSVS (SEQ ID NO: 733)	DASNR (SEQ ID NO: 734)	QQRYNW (SEQ ID NO: 735)
imgt	H	GFTFS NYD (SEQ ID NO: 722)	IGTAGDT (SEQ ID NO: 723)	VREVIVVRGGNY NYGMDV (SEQ ID NO: 724)	L	QSVSSY (SEQ ID NO: 730)	DAS	QQRYNW PPG (SEQ ID NO: 732)

STIM058

Nucleotide sequence encoding VL domain

(SEQ ID NO: 753)

GACATCCAGTTGACCCAGTCTCCATCCTTCCTGTCTGCATCTGTAGGAGACAGAGTCACCATCACTTGCTGGGCCAG

TCAGGGCATTAGCAGTTATTTAGCCTGGTATCAGCAAAACAGGGAAAGCCCTCAGCTCCTGATCTATGATGCAT

CCACTTTGCAAAAGTGGGGTCCCATCAAGGTTGAGCGGCAGTGGATCTGGAACAGACTTCACTCTCACAATCAGCAGC

CTGCAGCCTGAAGATTTGTCAGCTTATTACTGTCAACAGGTTAATAGTTACCCGCTCACTTTCGGCGGAGGGACCAA

GGTGGAGAGCAAAAC

VL domain amino acid sequence

(SEQ ID NO: 752)

DIQLTQSPSFLSASVGRVITITCWASQGISSYLAWYQKPKAPQLLIYDASTLQSGVPS

RFSGSGSGTDFTLTISSLQPEDFAAAYCQQVNSYPLTFGGGTVKVESK

Nucleotide sequence encoding VH domain

(SEQ ID NO: 745)

CAGGTGCTGGTGGTGGAGTCTGGGGGAGGCTTGGTCAAGCCTGGAGGGTCCCTGAGACTCTCCTGTGCAGCCTCTGG

ATTACCTTCAGTGACTACTACATGAGCTGGATCCGCCAGGCTCCAGGGAAGGGTCTGGAGTGGGTTTCATACATTA

GTAGTAGTGGTAGTACCACATACTGCGCAGACTCTGTGAAGGGCCGATTACCATCTCCAGGACAACGACAGGAAC

TCATTTTATCTGCAATGAACAGCCTGAGAGCCGAGGACACGGCCGTGTATTATTGTGCGAGGGGACCTACGGTGA

CTACCGCTACTATTACTTCGGTATGGACGTCTGGGGCCAAGGGACCACGGTCACCGTCTCCTCAG

VH domain amino acid sequence

(SEQ ID NO: 744)

QVLVVESGGGLVKPGGSLRLSCAASGFTFSDYYSWIRQAPGKGLEWVSYISSSGSTTYC

ADSVKGRFTISRDNDRNSFYLMNSLRAEDTAVYYCARGTYGDYRYYYFGMDVWGQGT

TVSS

Scheme	Locus	CDR1	CDR2	CDR3	Locus	CDR1	CDR2	CDR3
kabat	H	DYYMS (SEQ ID NO: 741)	YISSSGST- TYCA DSVKG (SEQ ID NO: 742)	GTYGDYRYYYFG MDV (SEQ ID NO: 743)	L	WASQGIS (SEQ ID NO: 749)	DASTL QS (SEQ ID NO: 750)	QQVNSY PLT (SEQ ID NO: 751)
imgt	H	GFTFS DYY (SEQ ID NO: 738)	ISSSGSTT (SEQ ID NO: 739)	ARGTYGDYRYYY FGMDV (SEQ ID NO: 740)	L	QGISSY (SEQ ID NO: 746)	DAS	QQVNSY PLT (SEQ ID NO: 748)

STIM057

Nucleotide sequence encoding VL domain

(SEQ ID NO: 769)

GACATCCAGATGACCCAGTCTCCATCTTCCGTGTCTGCATCTGTAGGAGACAGAGTCACCATCACTTGTCGGGCGAG

TCAGGTATTAGCAGCTGGTTAGCCTGGTATCAGCAGAAACAGGGAAAGCCCTAAGCTCCTGATCTATGCTGCAT

- continued

CCAGTTTGCAAAGTGGGGTCCCATCAAGGTTTCAGCGGCAGTGGATCTGGGACAGATTCTACTCTCACCATCACCAGC
CTGCAGCCTGAAGATTTTGCAACTTACTATTGTCAACAGGCTAACAGTTTCCCTCTCACTTTCGGCGGAGGGACCAA
GGTGGAGATCAAAC

VL domain amino acid sequence

(SEQ ID NO: 768)

DIQMTQSPSSVSASVGDRTITCRASQGISSWLAWYQQKPKAPKLLIYAASSLQSGVPS

RFSGSGSGTDFTLTITSLQPEDFATYYCQQANSFPLTFGGGTKEIK

Nucleotide sequence encoding VH domain

(SEQ ID NO: 761)

GAGGTGCAGCTGGTGGAGTCTGGGGGAGGCTTGGTACAGCCTGGGGGTCCTGAGACTCTCCTGTGCAGCCTCTGG
ATTCACTTCAGTACCTACGACATGACTGGATCCGCCAAGCTCCAGGAAAAGTCTGGAGTGGGTCTCAGCTATTG
GTACTGCTGGTGACACATACTATCCAGACTCCGTGAAGGGCCGATTACCATCTCCAGAGAAAATGCCAAGAACTCC
TTGTATCTTCAGATGAACAGTCTGAGAGCCGGGACACGGCTGTGTATTACTGTGCAAGGGCTCGTTCTTACTATGT
TTCGGGGACTTATTCCTGACTACTTTGACTACTGGGGCCAGGGAACCTGGTCATTGTTTCTCTAG

VH domain amino acid sequence

(SEQ ID NO: 760)

EVQLVESGGGLVQPGGSLRLSCAASGFTFSTYDMYWIRQAPGKLEWVSAIGTAGDTYYP

DSVKGRFTISRENAKNSLYLQMNSLRAGDTAVYYCARARSYYVSGTYSPIYFDYWGQGTL

VIVSS

Scheme	Locus	CDR1	CDR2	CDR3	Locus	CDR1	CDR2	CDR3
kabat	H	TYDMY (SEQ ID NO: 757)	AIGTAGDTYYP DSVKG (SEQ ID NO: 758)	ARSYYVSGTYSY YFDY (SEQ ID NO: 759)	L	RASQGIS SWLA (SEQ ID NO: 765)	AASSL QS (SEQ ID NO: 766)	QQANSF PLT (SEQ ID NO: 767)
imgt	H	GFTFS TYD (SEQ ID NO: 754)	IGTAGDT (SEQ ID NO: 755)	ARARSYYVSGTYS PYFDY (SEQ ID NO: 756)	L	QGISSW (SEQ ID NO: 762)	AAS	QQANSF PLT (SEQ ID NO: 764)

STIM056

Nucleotide sequence encoding VL domain

(SEQ ID NO: 785)

GAAAGTTGTTTTGACACAGTTTCCAGCCACCCCTTTCTTTGTCTCCAGGGGAGAGAGCCACCCTCTCCTGCAGGGCCAG
TCAGAATATTGCGAGTTACTTAGCTGGTTCCAACAGAACTGGCCAGGCTCCAGGCTCCTCATCTATGGTGCAT
CCAATAGGGCCACTGGCATCCAGCCAGGTTCACTGGCAGTGGGTCTGGGACAGACTTCACTCTCACCATCAGCAGC
CTAGAGCCTGAAGATTTTGAGTTTATTACTGTGACGAACGAATCAGCTGGCCTCCGATCACCTTCGGCCCAGGGAC
ACGACTGGAGATTAAAC

VL domain amino acid sequence

(SEQ ID NO: 784)

EVVLTQFPATLSLSPGERATLSCRASQNIIRSYLAWFQQKPGQAPRLLIYGASNRATGIPA

RFSGSGSGTDFTLTITSLLEPEDFAVYYCQQRISWPPITFGPGTRLEIK

Nucleotide sequence encoding VH domain

(SEQ ID NO: 777)

GAGGTGCGGCTGGTGGAGTCTGGGGGAGGCTTGGTACAGCCTGGGGGTCCTGAGACTCTCCTGTGCAACCTCTGA
ATTCACTTTCAATTCTACGACATGCACTGGGTCCGCCAAGTTACAGGAAAAGTCTGGAGTGGGTCTCAGCTATTG
GCACTGCTGGTGACACATACTATTAGACTCCGTGAAGGGCCGATTACCATCTCCAGAGAAGATGCCAGGAATCC
CTGTATCTTCAAATGAACAGCCTGCGAGCCGGTGACACGGCTGTTTATTACTGTACAGGAGATAGGACTGTGGTTAG
AGGAGTCAATTATTACTACTACGGTATGGACGTCTGGGGCCAAGGGACCACGGTCACCGTCGCCTCAG

-continued

VH domain amino acid sequence

(SEQ ID NO: 776)

EVRLVESGGGLVQPGGSLRLSCATSEFIFNSYDMHWVRQVTGKGLEWVSAIGTAGDTYYS

DSVKGRFTISRREDARNSLYLQMNSLRAGDTAVYYCTGDRTVVRGVNYYYYGMDVWGQGT

VTVAS

Scheme	Locus	CDR1	CDR2	CDR3	Locus	CDR1	CDR2	CDR3
kabat	H	SYDMH (SEQ ID NO: 773)	AIGTAGDTYYS (SEQ ID NO: 774)	DRTVVVRGVNYYYY L (SEQ ID NO: 775)	RASQNI (SEQ ID NO: 781)	RASQNI (SEQ ID NO: 781)	GASN (SEQ ID NO: 782)	QQRISWP (SEQ ID NO: 783)
imgt	H	EFIFN (SEQ ID NO: 770)	IGTAGDT (SEQ ID NO: 771)	TGDRTVVVRGVNYY L (SEQ ID NO: 772)	QNIRSY (SEQ ID NO: 778)	QNIRSY (SEQ ID NO: 778)	GAS (SEQ ID NO: 780)	QQRISWP (SEQ ID NO: 780)

STIM055

Nucleotide sequence encoding VL domain

(SEQ ID NO: 801)

GAAGTTGTTTTGACACAGTCTCCAGCCACCCTGTCTTTGTCTCCAGGGAGAGAGCCACCCTCTCCTGCAGGGCCAG

TCAGAATATTTCGAACTACTTACGCTGGTTCCAACAGAAACCTGGCCAGGCTCCAGGCTCCTCATCTATGGTGCAT

CCAACAGGGCCACTGGCATCCAGCCAGGTTCACTGGCAGTGGGTCTGGGACAGACTTCACTCTCACCATCAGCAGC

CTAGAGCCTGAAGATTTTCAGTTTATTACTGTCTAGCAACGTATCAACTGGCCTCCGATCACCTTCGGCCAAGGGAC

ACGACTGGAGATTAAAC

VL domain amino acid sequence

(SEQ ID NO: 800)

EVVLTQSPATLSLSPGERATLSCRASQNIIRNYLAWFQQKPGQAPRLLIYGASNRATGIPA

RFGSGSGTDFTLTISSLEPEDFAVYYCQQRINWPPITFGQGRLEIK

Nucleotide sequence encoding VH domain

(SEQ ID NO: 793)

GAGGTGCAGTTGGTGGAGTCTGGGGGAGGCTTGGTACAGCCGGGGGGTCCCTGAGACTCTCCTGTGCAGCCTCTGA

ATTATCTTCAGTACCTACGACATGCACTGGGTCCGCCAAGTTACAGGAAAAGGTCTGGAGTGGGTCTCAGCTATTG

GCACTGTCTGGTGACACATACTATTAGACTCCGTGAAGGGCCGATTACCATCTCCAGAGAAAATGCCAAAACTCC

TTGTATCTTCAAATGAACAGCCTGAGAGACGGTGACACGGCTGTATTTCTGTACGGGGGATAGGACTGTGGTTCG

GGGAGTCAATTATTACTACTACGGTATGGACGTCTGGGGCCAAGGGACCACGGTCACCGTCTCCTCAG

VH domain amino acid sequence

(SEQ ID NO: 792)

EVQLVESGGGLVQPGGSLRLSCAASEFIFSTYDMHWVRQVTGKGLEWVSAIGTAGDTYYS

DSVKGRFTISRRENAKNSLYLQMNSLRDGDVAVYFCTGDRTVVRGVNYYYYGMDVWGQGT

VTVSS

Scheme	Locus	CDR1	CDR2	CDR3	Locus	CDR1	CDR2	CDR3
kabat	H	TYDMH (SEQ ID NO: 789)	AIGTAGDTYYS (SEQ ID NO: 790)	DRTVVVRGVNYYYY L (SEQ ID NO: 791)	RASQNI (SEQ ID NO: 797)	RASQNI (SEQ ID NO: 797)	GASN (SEQ ID NO: 798)	QQRINWP (SEQ ID NO: 799)
imgt	H	EFIFS (SEQ ID NO: 786)	IGTAGDT (SEQ ID NO: 787)	TGDRTVVVRGVNYY L (SEQ ID NO: 788)	QNIRNY (SEQ ID NO: 794)	QNIRNY (SEQ ID NO: 794)	GAS (SEQ ID NO: 796)	QQRINWP (SEQ ID NO: 796)

STIM054

Nucleotide sequence encoding VL domain

(SEQ ID NO: 817)

GACATCCAGATGACCCAGTCTCCATCTTCCGTGTCTGCATCTGTAGGAGACAGAGTCAACATCACTTGTCTGGGCGAG

TCAGGGTATTAGCAGCTGGTTAGCCTGGTATCAGCAGAAACAGGGAAAGCCCTAAGTCTCTGATCTATGCTGCAT

- continued

CCAGTTTGCAAAGTGGGGTCCCATCAAGGTTACAGCGCAGTGGATCTGGGACAGATTCTACTCTCACCATCAGCAGC
CTGCAGCCTGAAGATTTTGCAACTTACTATTGTCAACAGGCTAACAGTTTCCCGTGGACGTTTCGGCCAAGGGACCAA
GGTGAAATCAAAC

VL domain amino acid sequence

(SEQ ID NO: 816)

DIQMTQSPSSVSASVGDRTITCRASQGISSWLANWYQQKPKAPFLIYAASSLQSGVPS
RFSGSGSGTDFTLTISSLQPEDFATYYCQQANSFPWTFGQGTKVEIK

Nucleotide sequence encoding VH domain

(SEQ ID NO: 809)

GAAGTGCTTCTGGTGGAGTCTGGGGGAGGCTTGGTACAGCCTGGCAGGTCCTGAGACTCTCCTGTGAAGCCTCTGG
ATTCACCTTTGATGATTATGCCATGCACCTGGGTCCGGCAAGCTCCAGGGAAGGGCTGGAGTGGGTCTCAGGTATTA
CTTGGAATAGTGGTAGCATAGACTATACGGACTCTGTGAAGGGCCGATTACCATCTCCAGAGACAACGCCAAGAAC
TCCCTGTATCTGCAATGAACAGTCTGAGAGCTGAGGACACGGCCTTGATTACTGTGCAAAGAATCTACTTCGGG
GAGTTATTATAACGTTCCCTGGTTTGACTACTGGGGCCAGGGAACCTGGTCACCGTCTCCTCAG

VH domain amino acid sequence

(SEQ ID NO: 808)

EVLVLESGGGLVQPGRLRLSCEASGFTFDDYAMHWVRQAPGKGLEWVSGITWNSGSDY
TDSVKGRTISRDNKNSLYLQMNSLRAEDTALYYCAKESTSGSYYNVPWFYWGQGLV
TVSS

Scheme	Locus	CDR1	CDR2	CDR3	Locus	CDR1	CDR2	CDR3
kabat	H	DYAMH (SEQ ID NO: 805)	GITWNSGSI- DYT DSVKG (SEQ ID NO: 806)	ESTSGSYYNVPW FDY (SEQ ID NO: 807)	L	RASQGIS SWLA (SEQ ID NO: 813)	AASSL QS (SEQ ID NO: 814)	QQANSF PWT (SEQ ID NO: 815)
imgt	H	GFTFD DYA (SEQ ID NO: 802)	ITWNSGSI (SEQ ID NO: 803)	AKESTSGSYYNV PWFYD (SEQ ID NO: 804)	L	QGISSW (SEQ ID NO: 810)	AAS (SEQ ID NO: 812)	QQANSF PWT (SEQ ID NO: 812)

STIM053

Nucleotide sequence encoding VL domain

(SEQ ID NO: 832)

GACATCCAGATGACCCAGTCTCCTTCCACCCGTGTGCATCTGTAGGAGACAGAGTACCATCACCTGCCGGGCCAG
TCAGAGTATTAGTAGCTGGTTGGCCTGGTATCAGCAGAAACAGGGAAGCCCTAAGCTCCTTATCTATAAGGCGT
CTAGTTTAGAAGTGGGGTCCCATCAAGGTTACAGCGCAGTGGATCTGGGACAGAATTCTACTCTCACCATCAGCAGC
CTGCAGCCTGATGATTTTGCAACTTATTACTGCCAACAGTATTATAGTTATTGGACGTTTCGGCCAAGGGACCAAGGT
GGAAATCAAAC

VL domain amino acid sequence

(SEQ ID NO: 831)

DIQMTQSPSTLSASVGDRTITCRASQSISSWLANWYQQKPKAPKLLIYKASSLEGGVPS
RFSGSGSGTEFTLTISSLQPDFATYYCQQYYSYWTFGQGTKVEIK

Nucleotide sequence encoding VH domain

(SEQ ID NO: 825)

CAGGTGCAACTGCTGGAGTCTGGGGGAGGCTTGGTCAAGCCTGGAGGTCCTGAGACTCTCCTGTGCAGCCTCTGG
ATTCACCTTCAGTGACTACTACATGAACCTGGATCCGCCAGGCTCCAGGGAAGAGACTGGAGTGGGTTTCATACATTA
GTAATAGTGGTACTACCATATACTACGTAGACTCTGTGAGGGCCGATTACCGTCTCCAGGGACAACGCCAAGAAC
TCACTGTATCTGCAATGAACAGCCTGAGAGCCGAGGACACGGCCGTATATTACTGTGCGAGAGATGCCCACTATGT
TTCGGGGAGCGAGGGTATGGACGCTCTGGGGCCAAGGACCACGGTCACCGTCTCCTCAG

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VH domain amino acid sequence

(SEQ ID NO: 824)

QVQLLES^{GGGLVKPGGSLRL}SCAASGFTFS^{DIYM}NWIRQAPGK^{RL}EWVS^{YIS}NSGTTI^{YY}VDSV^{RG}RF^{TVSR}DN^{AKNS}LYLQ^{MNS}LR^{AE}DTAV^{YY}CARD^{AH}YVSG^{SE}GMDV^{WG}QGT^{TV}TV

SS

Scheme	Locus	CDR1	CDR2	CDR3	Locus	CDR1	CDR2	CDR3
kabat	H	DYYMN (SEQ ID NO: 821)	YISNSGT- TIYYV DSVRG (SEQ ID NO: 822)	DAHYVSGSEGM DV (SEQ ID NO: 823)	L	RASQSISS WLA (SEQ ID NO: 829)	KASSL EG (SEQ ID NO: 830)	
imgt	H	GFTFSD YY (SEQ ID NO: 818)	ISNSGTTI (SEQ ID NO: 819)	ARDAHYVSGSE GMDV (SEQ ID NO: 820)	L	QSISSW (SEQ ID NO: 826)	KAS WT (SEQ ID NO: 828)	QQYYSY

STIM052

Nucleotide sequence encoding VL domain

(SEQ ID NO: 848)

GAAATGTGTTGACGAGTCTCCAGGCACCCGTGTCTTTGTCTCCAGGGGAAAGAGCCACCCCTCTCCTGCAGGGCCAG

TCAGAGTGTTAGCAGTATTTTCTTAGCCTGGTACCAGCAGAAACCTGGCCAGGCTCCAGGCTCCTCATCTATGGTG

CCTCCGGCAGGGCCACTGGCATCCAGCCAGGTTCACTGGCAGTGGGTCTGGGACAGACTTCACTCTCACCATCAGC

AGACTGGAACCTGAAGATTTTGCAGTGTATTACTGTGACAGTATGGTAGCTCACCCCTTGTGCAGTTTTGGCCAGGG

GACCAAGCTGGAGATCAAAC

VL domain amino acid sequence

(SEQ ID NO: 847)

EIVLTQSPGTL^{SLSP}GERATL^{SCR}ASQSV^{SSI}FLAWYQ^{QKPGQ}APRLLIYGASGRATGIPARFSGSGSGTDFTLTISRLEPEDFAV^{YY}CQY^{GSS}PLCSFGQGT^{KLEIK}

Nucleotide sequence encoding VH domain

(SEQ ID NO: 840)

GAGGTGCAGCTGGTGGAGTCTGGGGGAGGCTTGGTACAGCCTGGGGGTCCTGAGACTCTCCTGTGCAGCCTCTGG

ATTACCTTCAGTACCTACGAAATGCAGTGGGTCCGCCAAGCTACAGGAAAAGTCTGGAGTGGGTCTCAACTATTG

GTACTGCTGGTGACACATACTATCCAGGTCCTGGAAGGGCCGATTACCATCTCCAGAGAAAATGCCAAGAACTCC

TTGTATCTTCAAATGAACAGCCTGAGAGCCGGGACACGGCTATATATTACTGTGCAAGGTCTCTACGGTGGTATC

CGGCTATCGCTATGACTACTACGGCATGGAGTCTGGGGCCAAGGGACCACGGTCACCGTCTCCTCAG

VH domain amino acid sequence

(SEQ ID NO: 839)

EVQLVESGGGLV^{QPGGSLRL}SCAASGFTFS^{TYEM}HWVRQATGK^{LEWV}STIGTAGDT^{YY}PGSVK^{GRFT}ISRENAKNSLYLQ^{MNS}LRAGDTAI^{YY}CARSSTV^{VS}GYRYDY^{GMDV}WGQGT

VTVSS

Scheme	Locus	CDR1	CDR2	CDR3	Locus	CDR1	CDR2	CDR3
kabat	H	TYEMH (SEQ ID NO: 836)	TIGTAGDTYYP GSVKG (SEQ ID NO: 837)	SSTVVS ^{GYRYDY} GMDV (SEQ ID NO: 838)	L	RASQSVSS IFLA (SEQ ID NO: 844)	GASG RAT (SEQ ID NO: 845)	QQYGSSP LCS (SEQ ID NO: 846)
imgt	H	GFTFS TYE (SEQ ID NO: 833)	IGTAGDT (SEQ ID NO: 834)	ARSSTVVS ^{GYRYD} YYGMDV (SEQ ID NO: 835)	L	QSVSSIF (SEQ ID NO: 841)	GAS (SEQ ID NO: 843)	QQYGSSP LCS (SEQ ID NO: 843)

STIM051

Nucleotide sequence encoding VL domain

(SEQ ID NO: 863)

GACATCCAGATGACCCAGTCTCCTTCCACCCGTGTCTGCATCTGTAGGAGACAGAGTCACCATCACTTGCCGGGCCAG

TCAGAGTATTAGTAGTTGGTGGCCTGGTATCAGCAGAAACAGGGAAAGCCCTAATCTCCTGATCTATAAGGCGT

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CTAGTTTAGAAAGTGGGGTCCCATCAAGGTTACGCGCAGTGGATCTGGGACAGAATCACTCTCACCATCAGCAGC
CTGCAGCCTGATGATTTTGCAACTTATTACTGCCAACAGTATTATAGTTATTGGACGTTTCGGCCAAGGGACCAAGGT
GGAAATCAAAC

VL domain amino acid sequence

(SEQ ID NO: 862)

DIQMTQSPSTLSASVGDRTITCRASQSISSWLANVYQKPGKAPNLLIYKASSLESQVPS

RFSGSGSGTEFTLTISSLQPDFFATYYCQQYYSYWTFGQGTKVEIK

Nucleotide sequence encoding VH domain

(SEQ ID NO: 856)

CAGGTGCAGCTGGTGGAGTCTGGGGGAGGCTTGGTCAAGCCTGGAGGGTCCCTGAGACTCTCCTGTGCAGCCTCTGG

ATTCACCTTCAGTGACTACTACATGAACCTGGTTCGCCAGGCTCCAGGGAAGGGCTGGAGTGGGTTTCATACATTA

GTAGTGGTGTGATACCATATACTACACAGACTCTGTGAAGGCCGATTACCATCTCCAGGGACAGCGCAAGAAC

TCACTATATCTGCAAAATGAACGCCTGAGAGCCGAGGACACGGCCGTGTATTACTGTGCGAGAGATCGGTTAACTGG

AACGGGAGGTATGGACGTCTGGGGCCAAGGGACCACGGTCACCGTCTCCTCAG

VH domain amino acid sequence

(SEQ ID NO: 855)

QVQLVESGGGLVKPGSSLRSCAASGFTFSDDYIMNWERQAPGKLEWVSYISSGDTIYY

TDSVKGRFTISRDSAKNSLYLQMNSLRAEDTAVYYCARDRLTGTGGMDVWGQGTITVTVSS

Scheme	Locus	CDR1	CDR2	CDR3	Locus	CDR1	CDR2	CDR3
kabat	H	DYYMN (SEQ ID NO: 852)	YISSGDTIYY TDSVKG (SEQ ID NO: 853)	DRLTGTGGMD V (SEQ ID NO: 854)	L	RASQSISS WLA (SEQ ID NO: 860)	KASSL ES (SEQ ID NO: 861)	
imgt	H	GFTFSD YY (SEQ ID NO: 849)	ISSGDTI (SEQ ID NO: 850)	ARDRLTGTGG MDV (SEQ ID NO: 851)	L	QSISSW (SEQ ID NO: 857)	KAS WT (SEQ ID NO: 859)	QQYYSY

STIM050

Nucleotide sequence encoding VL domain

(SEQ ID NO: 879)

GATATTGTGATGACTCAGTCTCCACTCTCCCTGCCCGTCACCCCTGGAGAGCCGGCCTCCATCTCCTGCAGGTCTAG

TCAGAGCCTCCTGCATAGTAATGGATACAACATATTGGATTGGTACCTGCAGAAGCCAGGGCAGTCTCCACAGCTCC

TGATCTATTGGGTTCTAATCGGGCCTCCGGGTCCTGACAGGTTTCTAGTGGCAGTGGATCAGGCACAGATTTTACA

CTGAAAATCAGCAGAGTGGAGGCTGAGGATGTTGGGGTTTATTACTGCATGCAAACTCTGCAAACTCCGCTCACTTT

CGGCGGAGGGACCAAGGTGGAGATCAAAC

VL domain amino acid sequence

(SEQ ID NO: 878)

DIVMTQSPLSLPVTGPGEPAISCRSSQSLHNSNGYNYLDWYLQKPGQSPQLLIYLGSNRA

SGVPDRFSGSGSDFTLTKISRVEAEDVGVYYCMQTLQTPLTFGGGTKVEIK

Nucleotide sequence encoding VH domain

(SEQ ID NO: 871)

CAGTTGCAGCTGCAGAGTCGGGCCAGGACTGCTGAAGCCTTCGGGGACCCTGTCCCTCACCTGCCTGTCTCTGG

TGGCTCCATCAGCAGTAGTAGTGTGGAGTTGGGTCCGCCAGCCCCAGGGAAGGGCTGGAGTGGATTGGGGAAA

TCTATCACTAGGGAGCACCAACTACAGCCCGTCCCTCAAGAGTCGAGTCACCATATCAGTAGACAAGTCCAAGAAC

CAGTTCTCCCTGAAGCTGAGCTCTGTGACCGCCGCGGACACGGCCGTGTATTACTGTGCGAGAGGTCATTACGATAT

TTGACTGGTTCCTTCGACCCCTGGGGCCAGGGAACCTGGTCACCGTCTCCTCAG

VH domain amino acid sequence

(SEQ ID NO: 870)

QLQLQESGPGLLKPSGTLSTCAVSGGSISSSSWWSWVRQPPGKLEWIGEIYHTGSTNY

SPSLKSRVTISVDKSKNQFSLKLSSVTAADTAVYYCARGHYDILTGSDPDWGQGTLVTVS

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Scheme	Locus	CDR1	CDR2	CDR3	Locus	CDR1	CDR2	CDR3
kabat	H	SSSWWS (SEQ ID NO: 867)	EIYHTGSTNYS PSLKS (SEQ ID NO: 868)	GHYDILTGS FDP (SEQ ID NO: 869)	L	RSSQSLLS NGYNYLD (SEQ ID NO: 875)	LGSN RAS (SEQ ID NO: 876)	MQTLQT PLT (SEQ ID NO: 877)
imgt	H	GGSISS SSW (SEQ ID NO: 864)	IYHTGST (SEQ ID NO: 865)	ARGHYDILT GSFDP (SEQ ID NO: 866)	L	QSLLSNGY NY (SEQ ID NO: 872)	LGS	MQTLQT PLT (SEQ ID NO: 874)

STIM044

Nucleotide sequence encoding VL domain

(SEQ ID NO: 895)

GAAATGTGTTGACGCAGTCTCCAGGCACCCGTGCTTTGTCTCCAGGGGAAAGAGCCACC
CTCTCCTGCAGGGCCAGTCAGAGTGTTAGCACCAGCAGAACCTTAGCCTGGTACCAGCAG
AAACCTGACCAGGCTCCCAGGCTCCTCATCTATGGTGCGTCTACCAGGGCCACTGGCATC
CCAGACAGGTTTACGTGGCAGTGGGTCTGGGACAGACTTCACTCTCACCATCAGCAGACTG
GAGCCTGAAGATTTTGCAGTGTATTACTGTGAGCTATTTGGTAGCTCACCTCCGATCACC
TTCGGCCAAGGGACACGACTGGAGATTAAAC

VL domain amino acid sequence

(SEQ ID NO: 894)

EIVLTQSPGTL~~SL~~SPGERATL~~SCR~~ASQSVTSRT~~LAWYQQKPDQAPRLLI~~YGA~~STRATGI~~
PDRFSGSGSGTDFTLTISRLEPEDFAVYYCQLFGSSPPITFGQGTRLEIK

Nucleotide sequence encoding VH domain

(SEQ ID NO: 887)

GAGGTGCAGCTGGTGGAGTCTGGGGGAGGCTTGGTACAGCCCGGGGGTCCCTGAGACTC
TCCTGTGCAGCCTCTGGATTACCTTTAATGGGTACGACATGCACTGGGTCCGCCAACCT
ACAGGAAAAGGCTCGAATGGGTGCGACTTATAGTACTGCTGGAGACACATACTATTCA
GACTCCGTGAAGGGCCGACTCACCATCTCCAGAGAAAATGCCAAGAAGTCCCTGTATCTT
CAATGAACAGCCTGAGAGTCGGGACACGGCTGTGTATTACTGTGTAAGAGATCGTAGA
GCAGTGGCTGGTCATTACTACTACTTCTACGGTTTGGACGCTCTGGGGCCAAGGGACCAG
GTCACCGTCTCCTCAG

VH domain amino acid sequence

(SEQ ID NO: 886)

EVQLVESGGGLVQP~~GS~~SLRLSCAASGFTFNGYDMHWVRQPTGKGLEWVALIGTAGDTYY~~S~~
DSVKGR~~L~~ITISRENAKSLYLQMNSLRVGD~~T~~AVYYCVRDRRAVAGHY~~Y~~FYGLDVWGQGT~~T~~
VTVSS

Scheme	Locus	CDR1	CDR2	CDR3	Locus	CDR1	CDR2	CDR3
kabat	H	GYDMH (SEQ ID NO: 883)	LIGTAGDTYY S DSVKG (SEQ ID NO: 884)	DRRAVAGHY Y FY GLDV (SEQ ID NO: 885)	L	RASQSVST S RTLA (SEQ ID NO: 891)	GAST RAT (SEQ ID NO: 892)	QLFGSS PPIT (SEQ ID NO: 893)
imgt	H	GFTFN GYD (SEQ ID NO: 880)	IGTAGDT (SEQ ID NO: 881)	VRDRRAVAGHY Y FYGLDV (SEQ ID NO: 882)	L	QSVSTSR T (SEQ ID NO: 888)	GAS	QLFGSS PPIT (SEQ ID NO: 890)

STIM043

Nucleotide sequence encoding VL domain

(SEQ ID NO: 911)

GAAATGTGTTGACGCAGTCTCCAGGCACCCGTGCTTTGTCTCCAGGGGAAAGAGCCACC
CTCTCCTGCAGGGCCAGTCAGAGTGTCAGCACCAGCTCCTTAGTCTGGTACCAGCAGAAA
CCTGGCCAGGCTCCCAGGCTCCTCATCTATGGTGCGTCTACCAGGGCCACTGGCATCCCA

- continued

GACAGGTTTCAGTGGCAGTGGGTCTGGGACAGACTTCACTCTCACCATCAGCAGACTGGAG
CCTGAAGATTTTGCAGTGTATTACTGTCAGCAGTTTGGTAGCTCACCTCCGATTACCTTC
GGCCAAGGACACGACTGGAGATTAAAC

VL domain amino acid sequence

EIVLTQSPGTLSPGERATLSCRASQSVSTSSLVWYQQKPGQAPRLLIYGASTRATGIP
DRFSGSGSGTDFTLTISRLEPEDFAVYYCQQFGSSPPIITFGQGRLEIK

(SEQ ID NO: 910)

Nucleotide sequence encoding VH domain

GAGGTGCAGTTGGTGGAGTCTGGGGGAGGCTTGGTACAGCCCGGGGGTCCCTGAGACTC
TCCTGTGCAGCCTCTGGATTACCTTTAATGGGTACGATATGCACTGGGTCCGCCAAGCT
ACAGGAAAAGGCTCGAATGGGTGCGCACTTATAGGTACTGCTGGTGACACATACTATTCA
GACTCCGTAAAGGGCCGACTCACCATCTCCAGAGAAAATGCCAAGAAGTCTTGTATCTT
CAATGAATAGCCTGAGAGTCGGGGACACGGCTGTGTATTACTGTGTAAGAGATCGTAGA
GCAGTGGCTGGTCATTACTACTTCTACGGTTTGGACGCTCTGGGGCCAAGGGACCACG
GTCACCGTCTCCTCAG

(SEQ ID NO: 903)

VH domain amino acid sequence

EVQLVESGGGLVQPGGSLRLSCAASGFTFNGYDMHWVRQATGKGLEWVALIGTAGDTYY
DSVKGRITISRENAKSLYLQMNSLRVGDYAVYYCVRDRRAVAGHYFFYGLDVWGQGT
VTVSS

(SEQ ID NO: 902)

Scheme	Locus	CDR1	CDR2	CDR3	Locus	CDR1	CDR2	CDR3
kabat	H	GYDMH (SEQ ID NO: 899)	LIGTAGDTYY DSVKG (SEQ ID NO: 900)	DRRAVAGHYFFY GLDV (SEQ ID NO: 901)	L	RASQSVST SSLV (SEQ ID NO: 907)	GAST RAT (SEQ ID NO: 908)	QQFGSSP PIT (SEQ ID NO: 909)
imgt	H	GFTFN GYD (SEQ ID NO: 896)	IGTAGDT (SEQ ID NO: 897)	VRDRRAVAGHYFFY FYGLDV (SEQ ID NO: 898)	L	QSVSTSS (SEQ ID NO: 904)	GAS (SEQ ID NO: 905)	QQFGSSP PIT (SEQ ID NO: 906)

STIM042

Nucleotide sequence encoding VL domain

GACATTGTGTTGACGAGTCTCCAGGCACCTGTCTTTGTCTCCAGGGGAAAGAGCCACC
CTCTCCTGCAGGGCCAGTCAGAGTGTAGCAGCAGCTCCTTAGCCTGGTACCAGCAGAAA
CCTGGCCAGGCTCCAGGCTCCTCATCTATGGTGCCTCCAGCAGGGCCACTGGCATCCCA
GACAGGTTTCAGTGGCAGTGGGTCTGGGACAGACTTCACTCTCACCATCAGCAGACTGGAG
CCTGAAGATTTTGCAGTGTATTACTGTCAGCAGTTTGGTAGCTCACCTCCGATCACCTTC
GGCCAAGGACACGACTGGAGATTAAAC

(SEQ ID NO: 927)

VL domain amino acid sequence

DIVLTQSPGTLSPGERATLSCRASQSVSSSLAWYQQKPGQAPRLLIYGASSRATGIP
DRFSGSGSGTDFTLTISRLEPEDFAVYYCQQFGSSPPIITFGQGRLEIK

(SEQ ID NO: 926)

Nucleotide sequence encoding VH domain

GAGGTGCAGCTGGTGGAGTCTGGGGGAGGCTTGGTACTGCCCGGGGGTCCCTGAGACTC
TCCTGTGCAGCCTCTGGATTACCTTCAATATGTACGACATGCACTGGGTCCGCCAAGTT
ACAGGAAAAGGCTCGAGTGGGTGCGCAGCTATAGGTACTGCTGGTGACACATACTATACA
GGCTCCGTGAAGGTCGACTCACCATCTCCAGAGAAAATGCCAAGAAGTCTTGTATCTT

(SEQ ID NO: 919)

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CAAATGAACAGCCTGAGAGCCGGGACACGGCTGTGTATTACTGTGTAAGAGATCGTAGA
 GCAGTGGCTGGTCATTACTACTTCTACGGTTTGGACGTCTGGGGCCAAGGGACCACG
 GTCACCGTCTCCTCAG

VH domain amino acid sequence

(SEQ ID NO: 918)

EVQLVESGGGLVLPGGSLRLSCAASGFTFNMYDMHWVRQVTGKGLEWVAIAGTAGDTYYT
 GSVKGRITISRENAKISLYLQMNSLRAGDTAVVYCVRRRAVAGHYIYFYGLDVWGQGT
 VTVSS

Scheme	Locus	CDR1	CDR2	CDR3	Locus	CDR1	CDR2	CDR3
kabat	H	MYDMH (SEQ ID NO: 915)	AIGTAGDTYYT (SEQ ID NO: 916)	DRRAVAGHYIYFY (SEQ ID NO: 917)	L	RASQSVSS SSLA (SEQ ID NO: 923)	GASS RAT (SEQ ID NO: 924)	QQFGSSP PIT (SEQ ID NO: 925)
imgt	H	GFTFN MYD (SEQ ID NO: 912)	IGTAGDT (SEQ ID NO: 913)	VRDRRAVAGHYIY FYGLDV (SEQ ID NO: 914)	L	QSVSSSS (SEQ ID NO: 920)	GAS	QQFGSSP PIT (SEQ ID NO: 922)

STIM041

Nucleotide sequence encoding VL domain

(SEQ ID NO: 943)

GAAGTTGTTTTGACACAGTTTCCAGCCACCTGTCTTTGTCTCCAGGGAGAGAGCCACC
 CTCTCCTGCAGGGCCAGTCAGAATATTCGCGATTACTTAGCCTGGTTCCAACAGAAACCT
 GGCCAGGCTCCCAGGCTCCTCATCTATGGTGCATCCAATAGGGCCACTGGCATCCCAGCC
 AGGTTCACTGGCAGTGGGTCTGGGACAGACTTCACTCTACCATCAGCAGCCTAGAGCCT
 GAAGATTTGCAGTTTATTACTGTCTAGCAACGAATCAGCTGGCCTCCGATCACCTTCGGC
 CCAGGGACACGACTGGAGATTAAAC

VL domain amino acid sequence

(SEQ ID NO: 942)

EVVLTPFPATLSLSPGERATLSRASQNIIRSYLAWFQQKPGQAPRLLIYGASNRATGIPA
 RFSGSGSGTDFTLTISLLEPEDFAVIYCCQRIISWPPITFGPGTRLEIK

Nucleotide sequence encoding VH domain

(SEQ ID NO: 935)

GAGGTGCGGCTGGTGGAGTCTGGGGAGTCTTGGTACAGCCTGGGGGTCCCTGAGACTC
 TCCTGTGCAGCCTCTGAATTCATCTTCAATTCCTACGACATGCACTGGGTCCGCCAAGTT
 ACAGGAAAAGGTCTGGAGTGGGTCTCAGCGATTGGTACTGCTGGTGACACATACTATTCA
 GACTCCGTGAAGGGCCGATTACCATCTCCAGAGAAGATGCCAGGAACCTCCCTGTATCTT
 CAAATGAACAGCCTGCGAGCCGGTGACACGGCTGTTTATTACTGTACAGGAGATAGGACT
 GTGGTTAGAGGAGTCAATTATTACTACTACGGTATGGACGTCTGGGGCCAAGGGACCACG
 GTCACCGTCGCCTCAG

VH domain amino acid sequence

(SEQ ID NO: 934)

EVRLVESGGVLVQPGGSLRLSCAASEFIFNSYDMHWVRQVTGKGLEWVSAIAGTAGDTYYT
 DSVKGRFTISREDARNSLYLQMNSLRAGDTAVVYCTGDRTVVRGVNYYYYGMDVWGQGT
 VTVAS

Scheme	Locus	CDR1	CDR2	CDR3	Locus	CDR1	CDR2	CDR3
kabat	H	SYDMH (SEQ ID NO: 931)	AIGTAGDTYYT DSVKG (SEQ ID NO: 932)	DRTVVRGVNYYYY GMDV (SEQ ID NO: 933)	L	RASQNIIR SYLA (SEQ ID NO: 939)	GASN RAT (SEQ ID NO: 940)	QQRIISW PIT (SEQ ID NO: 941)

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imgt	H	EFIFN SYD (SEQ ID NO: 928)	IGTAGDT (SEQ ID NO: 929)	TGDRTVVRGVNYY L YGM DV (SEQ ID NO: 930)	QNIRSY (SEQ ID NO: 936)	GAS	QQRISWP PIT (SEQ ID NO: 938)
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STIM040

Nucleotide sequence encoding VL domain

(SEQ ID NO: 959)

GAAGTTGTTTTGACACAGTTTCCAGCCACCCCTGTCTTTGTCTCCAGGGGAGAGAGCCACC
 CTCTCCTGCAGGGCCAGTCAGAATATTCGCAGTTACTTAGCCTGGTTCCAACAGAAGCCT
 GGCCAGGCTCCCAGGCTCCTCATCTATGGTGCATCCAATAGGGCCACTGGCATCCCAGCC
 AGGTTTCAGTGGCAGTGGGTCTGGGACAGACTTCACTCTCACCATCAGCAGCCTAGAGCCT
 GAAGATTTTGAGTTTATTACTGTGACGAACGAATCAACTGGCCTCCGATCACCTTCGGC
 CCAGGGACACGACTGGAGATTAAAC

VL domain amino acid sequence

(SEQ ID NO: 958)

EVVLTQFPATLSLSPGERATLSCRASQNIIRSYLAWFQQKPGQAPRLLIYGASNRATGIPA
 RFSGSGSGTDFTLTISLLEPEDFAVYYCQQRINWPPITFGPGTRLEIK

Nucleotide sequence encoding VH domain

(SEQ ID NO: 951)

GAGGTGCGGCTGGTGGAGTCTGGGGGAGGCTTGGTACAGCCTGGGGGTCCCTGAGACTC
 TCCTGTGCAGCCTCTGAATTCATCTTCAATCCTACGACATGCACTGGGTCCGCCAAGTT
 ACAGGAAAAGGTCTGGAGTGGGTCTCAGCTATTGGTACTGCTGGTGACACATACTATTCA
 GACTCCGTGAAGGGCCGGTTCACCATCTCCAGAGAAGATGCCAGGAACCTCCCTGTATCTT
 CAAATGAACAGCCTGCGAGCCGGTGACACGGCTGTTTATTACTGTACAGGAGATAGGACT
 GTGGTTAGAGGAGTCAATTATTACTACTTCGGTATGGACGTCTGGGGCCAAGGGACCACG
 GTCACCGTCGCCTCAG

VH domain amino acid sequence

(SEQ ID NO: 950)

EVRLVESGGGLVQPGGSLRLSCAASEFIFNSYDMHWVRQVTGKGLEWVSAIGTAGDTYYG
 DSVKGRFTISREDARNSLYLQMNSLRAGDTAVYYCTGDRTVVRGVNYYYFGMDVWGQGT

VTVAS

Scheme	Locus	CDR1	CDR2	CDR3	Locus	CDR1	CDR2	CDR3
kabat	H	SYDMH (SEQ ID NO: 947)	AIGTAGDTYYG (SEQ ID NO: 948)	DRTVVRGVNYYYF (SEQ ID NO: 949)	L	RASQNIIR (SEQ ID NO: 955)	GASN RAT (SEQ ID NO: 956)	QQRINWP PIT (SEQ ID NO: 957)
imgt	H	EFIFN SYD (SEQ ID NO: 944)	IGTAGDT (SEQ ID NO: 945)	TGDRTVVRGVNYY YFGMDV (SEQ ID NO: 946)	L	QNIRSY (SEQ ID NO: 952)	GAS	QQRINWP PIT (SEQ ID NO: 954)

STIM039

Nucleotide sequence encoding VL domain

(SEQ ID NO: 975)

GAAGTTGTTTTGACACAGTTTCCAGCCACCCCTGTCTTTGTCTCCAGGGGAGAGAGCCACC
 CTCTCCTGCAGGGCCAGTCAGAATATTCGCAGTTACTTAGCCTGGTTCCAACAGAAACCT
 GGCCAGGCTCCCAGGCTCCTCATCTATGGTGCATCCAATAGGGCCACTGGCATCCCAGCC
 AGGTTTCAGTGGCAGTGGGTCTGGGACAGACTTCACTCTCACCATCAGCAGCCTAGAGCCT
 GAAGATTTTGAGTTTATTACTGTGACGAACGAATCAGCTGGCCTCCGATCACCTTCGGC
 CCAGGGACACGACTGGAGATTAAAC

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VL domain amino acid sequence

(SEQ ID NO: 974)

EVVLTQFPATLSLSPGERATLSCRASQNIIRSYLANFQQKPGQAPRLLIYGASNRATGIPA
 RFSGSGSGTDFTLTISLLEPEDFAVIYCQQRISWPPITFGPGTRLEIK

Nucleotide sequence encoding VH domain

(SEQ ID NO: 967)

GAGGTGCGGCTGGTGGAGTCTGGGGGTGGCTTGGTACAGTCTGGGGGGTCCCTGAGACTC
 TCCTGTGCAGCCTCTGAATTCATCTTCAATTCCTACGACATGCACTGGGTCCGCCAAGTT
 ACAGGAAAAGGTCTGGAGTGGGTCTCAGCTATTGGTACTGCTGGTGACACATACTATTCTG
 GACTCCGTGAAGGGCCGATTACCATCTCCAGAGAAGATGCCAGGAACTCCCTGTATCTT
 CAAATGAGCAGCCTGCGAGCCGGTGACACGGCTGTTTATTATTGTACAGGAGATAGGACT
 GTGGTTAGAGGAGTCAATTATTATTATTACGGTATGGACGTCTGGGGCCAAGGGACCACG
 GTCACCGTCGCCTCAG

VH domain amino acid sequence

(SEQ ID NO: 966)

EVRLVESGGGLVQSGGSLRLSCAASEFIFNSYDMHWVRQVTGKGLEWVSAIGTAGDTYYIS
 DSVKGRFTISREDARNSLYLQMSSLRAGDTAVYYCTGDRTVVRGVNYYYYGMDVWGQGT
 TVTAS

Scheme	Locus	CDR1	CDR2	CDR3	Locus	CDR1	CDR2	CDR3
kabat	H	SYDMH (SEQ ID NO: 963)	AIGTAGDTYYIS DSVKG (SEQ ID NO: 964)	DRTVVRGVNYYYY GMDV (SEQ ID NO: 965)	L	RASQNIIR SYLA (SEQ ID NO: 971)	GASN RAT (SEQ ID NO: 972)	QQRISWP PIT (SEQ ID NO: 973)
imgt	H	EFIFN SYD (SEQ ID NO: 960)	IGTAGDT (SEQ ID NO: 961)	TGDRTVVRGVNYY YYGMDV (SEQ ID NO: 962)	L	QNIRSY (SEQ ID NO: 968)	GAS (SEQ ID NO: 970)	QQRISWP PIT (SEQ ID NO: 970)

STIM023

Nucleotide sequence encoding VL domain

(SEQ ID NO: 991)

GACATCCAGTTGACCCAGTCTCCATCCTTCCTGTCTGCATCTGTAGGAGACAGAGTCAACATCACTTGCTGGGCCAG
 TCAGGGCATTAGCAGTTATTTAGCCTGGTATCAGCAAAAACAGGAAAGCCCTAACTCCTGATCTATGCTGCAT
 CCACTTTGCAAAGTGGGGTCCCATCAAGGTTACGCGCAGTGGATCTGGGACAGAATCACTCTCACAAATCAGCAGC
 CTGCAGCCTGAAGATTTTGCAACTTATTACTGTCAACAGCTTAATACTTACCTCACGTGGACGTTTCGGCCAAGGGAC
 CAAGGTGGAATCAAA

VL domain amino acid sequence

(SEQ ID NO: 990)

DIQLTQSPSFLSASVGDRTVITCWAQQGISSYLAWYQKPGKAPKLLIYAASLTQSGVPS
 RFSGSGSGTEFTLTISLQPEDFATYYCQQLNTYLTWTFGQGTKVEIK

Nucleotide sequence encoding VH domain

(SEQ ID NO: 983)

GAGGTGCAGCTGGTGGAGTCTGGGGGAGGCTTGGTTCAGCCGGGGGGTCCCTGAGACTCTCCTGTGCAGCCTCTGG
 ATTCACCCCTCAGTAGCTACGACATACACTGGGTCCGCCAAGCTACAGGAAAAGTCTGGAGTGGGTCTCAGGTATTG
 GAACTGCTGGTGACACATACTATCCGGGCTCCGTGAAGGGCCGATTACCATCTCCAGAGAAAATGCCAAGAACTCC
 TTGTATCTTCAAATGAACAGCCTGAGAGCCGGGGACACGGCTGTGTATTACTGTGCAAGAGGGGTACTATGGTTC
 GGGGAATAACGTTTACTACTACTTTAACGGTATGGACGTCTGGGGCCAAGGGACCAGGTACCGTCTCCTCA

-continued

VH domain amino acid sequence

(SEQ ID NO: 982)

EVQLVESGGGLVQPGGSLRLSCAASGFTLSSYDIHWVRQATGKGLEWVSGICTAGDTYYP

GSVKGRFTISRENAKNSLYLQMNSLRAGDTAVYYCARGGYGSGNNVYYYFNGMDVWGQG

TTVTVSS

Scheme	Locus	CDR1	CDR2	CDR3	Locus	CDR1	CDR2	CDR3
kabat	H	SYDIH (SEQ ID NO: 979)	GIGTAGDTYYP (SEQ ID NO: 980)	GGYYGSGNNVYYYF L (SEQ ID NO: 981)	L	WASQGIS (SEQ ID NO: 987)	AAS (SEQ ID NO: 988)	QQLNTY (SEQ ID NO: 989)
imgt	H	GFTLS SYD (SEQ ID NO: 976)	IGTAGDT (SEQ ID NO: 977)	ARGGYGSGNNVYY L YFNGMDV (SEQ ID NO: 978)	L	QGISSY (SEQ ID NO: 984)	AAS	QQLNTY LTWT (SEQ ID NO: 986)

STIM021

Nucleotide sequence encoding VL domain

(SEQ ID NO: 1006)

GACATCCAGATGACCCAGTCTCCTTCCACCTGTCTGCATCTGTAGGAGACAGAGTCACCATCACTTGCCGGGCCAG

TCAGAGTATTAGTAGCTGGTTGGCCTGTATCAGCAGAAACAGGAAAGCCCTAAGCTCCTGATCTATAAGGCGT

CTAGTTTAGAAAGTGGGGTCCCATCAAGGTTACGCGCAGTGGATCTGGGACAGAATTCACCTCTACCATCAGCAGC

CTGCAGCCTGATGATTTTGCACTTATTACTGCCAACAGTATAATAGTTATTGGACGTTTCGGCCAAGGGACCAAGGT

GGAAATCAAA

VL domain amino acid sequence

(SEQ ID NO: 1005)

DIQMTQSPSTLSASVGRVITICRASQSISSWLAWYQQKPKAPKLLIYKASSLESQVPS

RFSGSGSGTEFTLTITISLQPDFATYYCQQYNSYWTFGQGTKVEIK

Nucleotide sequence encoding VH domain

(SEQ ID NO: 999)

GAGGTGCAGCTGGTGGAGTCTGGGGGAGGCTTGTTTCAGCCGGGGGGTCCCTGAGACTCTCCTGTGCAGCCTCTGG

ATTACACCTTTAGCAGCTATGCCATGAGCTGGGTCCGCCAGGCTCCAGGGAAGGGCTGGAGTGGGTCTCAGCTTTTA

GTGGTAGTGGTGGTTACACATACTACGCAGACTCCGTGAAGGGCCGGTTCACCATCTCCAGAGACAATTCAGAAGAC

ACGCTGTATCTGCAATGAACGCCTGAGAGCCGAGGACACGGCCGTATATTACTGTGCGAAATACGGTGACTATGA

GGGTTGGTTCGACCCCTGGGGCCAGGGAACCTGGTCACCGTCTCTCTCA

VH domain amino acid sequence

(SEQ ID NO: 998)

EVQLVESGGGLVQPGGSLRLSCAASGFTFSSYAMSWVRQAPGKLEWVSAFSGSGGYTY

ADSVKGRFTISRDNKNTLYLQMNLRAEDTAVYYCAKYGDYEGWFDPWGQGLTVTVSS

Scheme	Locus	CDR1	CDR2	CDR3	Locus	CDR1	CDR2	CDR3
kabat	H	SYAMS (SEQ ID NO: 995)	AFSGSGGYTY ADSVKG (SEQ ID NO: 996)	YGDYEGWFD P (SEQ ID NO: 997)	L	RASQSISS WLA (SEQ ID NO: 1003)	KASSL ES (SEQ ID NO: 1004)	
imgt	H	GFTFSS YA (SEQ ID NO: 992)	FSGSGGYT (SEQ ID NO: 993)	AKYGDYEGW FDP (SEQ ID NO: 994)	L	QSISSW (SEQ ID NO: 1000)	KAS	QQYNSY WT (SEQ ID NO: 1002)

STIM020

Nucleotide sequence encoding VL domain

(SEQ ID NO: 1022)

GACATCCAGATGACCCAGTCTCCTTCCACCTGTCTGCATCTGTAGGAGACAGAGTCACCATCACTTGCCGGGCCAG

TCAGGTTATTAGCAACTGGTTAGCCTGTATCAGCAGAAACAGGGAAGCCCTAAGCTCCTGATCTATGCTGCAT

CCAGTTTGCAAAGTGGGGTCCCATCAAGGTTACGCGCAGTGGATCTGGGACAGATTTCACTCTACCATCAGCAGC

- continued

 CTGCAGCCTGAAGATTTTGCAACTTACTATTGTCAACAGGCTAACAGTTTCCCGTCAGTTTGGCCAGGGGACCAA

GCTGGAGATCAAA

VL domain amino acid sequence

(SEQ ID NO: 1021)

DIQMTQSPSSVSASVGDRTITCRASQGISNWLAWYQKPGKAPKLLIYAASSLQSGVPS

RFSGSGSGTDFTLTISSLQPEDFATYYCQQANSFPCSFQGTGLEIK

Nucleotide sequence encoding VH domain

(SEQ ID NO: 1014)

GAAGTGCAGCTGGTGGAGTCTGGGGAGGCTTGGTACAGCCTGGCAGGTCCTGAGACTCTCCTGTGCAGCCTCTGG

ATTACACCTTTGATGATTATACCATGCACCTGGGTCGGCAGGCTCCAGGAAGGGCCTGGAATGGGTCTCAGGTATTA

GTTGGAATAGTGGTAGCATAGGCTATGCGGACTCTGTGAAGGGCCGATTACCATCTCCAGAGACAACGCCAAGAAC

TCCCTGTATCTGCAAATGAACAGTCTGAGAACTGAGGACACGGCCTTTTATTACTGTGCAAAAGATATTATGATTAC

GTTTGGGGAGTTATCGTCTACTATTATGGTATGGACGCTCTGGGGCCAAGGGACCACGGTCACCGTCTCCTCA

VH domain amino acid sequence

(SEQ ID NO: 1013)

EVQLVESGGGLVQPGRSLRLSCAASGFTFDDYTMHWVRQAPGKGLEWVSGISWNSGSIQY

ADSVKGRFTISRDNKNSLYLQMNSLRTEDTAFYYCAKDIMITFGGVIVYYYGMDVWGQG

TTVTVSS

Scheme	Locus	CDR1	CDR2	CDR3	Locus	CDR1	CDR2	CDR3
kabat	H	DYTMH (SEQ ID NO: 1010)	GISWNSGSIQY ADSVKG (SEQ ID NO: 1011)	DIMITFGGVIVYYY L GMDV (SEQ ID NO: 1012)	RASQGIS NWLA (SEQ ID NO: 1018)	AASS LQS (SEQ ID NO: 1019)	QQANSF PCS (SEQ ID NO: 1020)	
imgt	H	GFTFD DYT (SEQ ID NO: 1007)	ISWNSGSI (SEQ ID NO: 1008)	AKDIMITFGGVIVY L YYGMDV (SEQ ID NO: 1009)	QGISNW (SEQ ID NO: 1015)	AAS (SEQ ID NO: 1017)	QQANSF PCS (SEQ ID NO: 1017)	

Anti-ICOS antibodies STIM001 to STIM009

Antibody STIM001

VH domain nucleotide sequence:

SEQ ID NO: 367

CAGGTTCAAGTGGTGCAGTCTGGAGCTGAGGTGAAGAAGCCTGGGGCCTCAGTGAAGGTCTCCTGCAAGGCTTCTGG

TTACACCTTTTCCACCTTTGGTATCACCTGGGTGCGACAGGCCCTGGACAAGGGCTTGAATGGATGGGATGGATCA

GCGCTTACAATGGTGACACAACTATGCACAGAATCTCCAGGGCAGAGTCATCATGACCACAGACACATCCACGAGC

ACAGCCTACATGGAGCTGAGGAGCCTGAGATCTGACGACACGGCCGTTTATTACTGTGCGAGGAGCAGTGGCCACTA

CTACTACTACGGTATGGACGCTCTGGGGCCAAGGGACCACGGTCACCGTCTCCTCA

VH domain amino acid sequence:

SEQ ID NO: 366

QVQVQSGAEVKKPGASVKVSCKASGYTFSTFGITWVRQAPGQGLEWMGWISAYNGDTNYAQNLRVIMTTDTSTS

TAYMELRSLRSDDTAVYYCARSSGHYYYGMDVWGQGTITVTSS

VH CDR1 amino acid sequence:

SEQ ID NO: 363

GYTFSTFG

VH CDR2 amino acid sequence:

SEQ ID NO: 364

ISAYNGDT

VH CDR3 amino acid sequence:

SEQ ID NO: 365

ARSSGHYYYGMDV

-continued

VL domain nucleotide sequence:

SEQ ID NO: 374

GATATTGTGATGACTCAGTCTCCACTCTCCCTGCCCGTCACCCCTGGAGAGCCGGCCTCCATCTCCTGCAGGTCTAG
TCAGAGCCTCCTGCATAGTAATGAATACAACATATTTGGATTGGTACCTGCAGAAGCCAGGGCAGTCTCCACAGCTCC
TGATCTTTTTGGGTTCTAATCGGGCCTCCGGGTTCCCTGACAGGTTCACTGGCAGTGGATCAGGCACAGATTTTACA
CTGAAAATCACCAGAGTGGAGGCTGAGGATGTTGGAATTTATTACTGCATGCAATCTCTACAACTCCGCTCACTTT
CGGCGGAGGGACCAAGGTGGAGATCAAA

VL domain amino acid sequence:

SEQ ID NO: 373

DIVMTQSPLSLPVTPGEPASISCRSSQSLHLSNEYNYLDWYLQKPGQSPQLLI FLGSNRRASGVPDRFSGSGSDTFT
LKITRVEAEDVGIYYCMQSLQTPLTFGGGTKVEIK

VL CDR1 amino acid sequence:

SEQ ID NO: 370

QSLHLSNEYNY

VL CDR2 amino acid sequence:

SEQ ID NO: 371

LGS

VL CDR3 amino acid sequence:

SEQ ID NO: 372

MQSLQTPLT

Antibody STIM002

VH domain nucleotide sequence:

SEQ ID NO: 381

CAGGTTCAACTGGTGCAGTCTGGAGGTGAGGTGAAGAAGCCTGGGGCCTCAGTGAAGGTCTCCTGCAAGGCTTCTGG
TTACACCTTTACCAGCTATGGTTTCAGCTGGGTGCGACAGGCCCTGGACAAGGACTAGAGTGGATGGGATGGATCA
GCGCTTACAATGGTAACACAACTATGCACAGAAGCTCCAGGGCAGAGTCACCATGACCACAGACACATCCACGAGC
ACAGCCTACATGGAGCTGAGGAGCTTGAGATCTGACGACACGGCCGTGTATTACTGTGCGAGATCTACGTATTTCTA
TGTTTCGGGGACCTCTACGGTATGGACGTCTGGGGCCAAGGGACCACGGTCACCGTCTCCTCA

VH domain amino acid sequence:

SEQ ID NO: 380

QVQLVQSGGEVKKPGASVKVSKASGYTFTSYGFSWVRQAPGQGLEWMGWISAYNGNTNYAQKLQGRVTMTTDTSTS
TAYMELRSLRSDDTAVYYCARSTYFYGSGTLYGMDVWGQGTITVTVSS

VH CDR1 amino acid sequence:

SEQ ID NO: 377

GYTFTSYG

VH CDR2 amino acid sequence:

SEQ ID NO: 378

ISAYNGNT

VH CDR3 amino acid sequence:

SEQ ID NO: 379

ARSTYFYGSGTLYGMDV

VL domain nucleotide sequence: 388

GATATTGTGATGACTCAGTCTCCACTCTCCCTGCCCGTCACCCCTGGAGAGCCGGCCTCCATCTCCTGCAGGTCTAG
TCAGAGCCTCCTGCATAGTGATGGATACAACATGTTTGGATTGGTACCTGCAGAAGCCAGGGCAGTCTCCACAGCTCC
TGATCTATTTGGGTTCTACTCGGGCCTCCGGGTTCCCTGACAGGTTCACTGGCAGTGGATCAGGCACAGATTTTACA
CTGAAAATCAGCAGAGTGGAGGCTGAGGATGTTGGGGTTTATTACTGCATGCAAGCTCTACAACTCCGTGCAGTTT
TGGCCAGGGGACCAAGCTGGAGATCAAA

Corrected STIM002 VL domain nucleotide sequence:

SEQ ID NO: 519

GATATTGTGATGACTCAGTCTCCACTCTCCCTGCCCGTCACCCCTGGAGAGCCGGCCTCCATCTCCTGCAGGTCTAG
TCAGAGCCTCCTGCATAGTGATGGATACAACATATTTGGATTGGTACCTGCAGAAGCCAGGGCAGTCTCCACAGCTCC

-continued

TGATCTATTGGGTTCTACTCGGGCTCCGGGTTCCCTGACAGGTTCTAGTGGCAGTGGATCAGGCACAGATTTTACA
CTGAAATCAGCAGAGTGGAGGCTGAGGATGTTGGGGTTTATTACTGCATGCAAGCTCTACAACTCCGCTCAGTTT
TGGCCAGGGGACCAAGCTGGAGATCAAA

VL domain amino acid sequence:

SEQ ID NO: 387

DIVMTQSPLSLPVTGPASPISCRSSQSLHSDGYNLDWYLQKPGQSPQLLIYLGSTRASGFPDRFSGSGSDFT
LKISRVEAEDVGVYYCMQALQTPLSFGQGTKLEIK

VL CDR1 amino acid sequence:

SEQ ID NO: 384

QSLHSDGYN

VL CDR2 amino acid sequence:

SEQ ID NO: 385

LGS

VL CDR3 amino acid sequence:

SEQ ID NO: 386

MQALQTPLS

Antibody STIM002-B

VH domain nucleotide sequence:

SEQ ID NO: 395

CAGGTTCAACTGGTGCAGTCTGGAGGTGAGGTGAAGAAGCCTGGGGCTCAGTGAAGTCTCCTGCAAGGCTTCTGG
TTACACCTTTACCAGCTATGGTTTCAGCTGGGTGCGACAGGCCCTGGACAAGGACTAGAGTGGATGGGATGGATCA
GCGCTTACAATGGTAACACAACTATGCACAGAAGCTCCAGGGCAGAGTACCATGACCACAGACATCCACGAGC
ACAGCCTACATGGAGCTGAGGAGCTTGAGATCTGACGACACGGCCGTGTATTACTGTGCGAGATCTACGTATTCTA
TGGTTCGGGGACCTCTACGGTATGGACGTCTGGGGCCAAGGACACGGTCACCGTCTCCTCA

VH domain amino acid sequence:

SEQ ID NO: 394

QVQLVQSGGEVKKPGASVKVSKASGYTFTSYGFSWVRQAPGQGLEWMGWI SAYNGNTNYAQLQGRVTMTDTST
TAYMELRSLRSDDTAVYYCARSTYFYSGTLYGMDVWGQGTITVTVSS

VH CDR1 amino acid sequence:

SEQ ID NO: 391

GYTFTSYG

VH CDR2 amino acid sequence:

SEQ ID NO: 392

ISAYNGNT

VH CDR3 amino acid sequence:

SEQ ID NO: 393

ARSTYFYSGTLYGMDV

VL domain nucleotide sequence:

SEQ ID NO: 402

GATATTGTGATGACTCAGTCTCCACTCTCCCTGCCCGTCACCCCTGGAGAGCCGGCTCCATCTCCTGCAGGTCTAG
TCAGAGCCTCCTGCATAGTGATGGATACAACTGTTGGATTGGTACCTGCAGAAGCCAGGGCAGTCTCCACAGCTCC
TGATCTATTGGGTTCTACTCGGGCTCCGGGTTCCCTGACAGGTTCTAGTGGCAGTGGATCAGGCACAGATTTTACA
CTGAAATCAGCAGAGTGGAGGCTGAGGATGTTGGGGTTTATTACTGCATGCAAGCTCTACAACTCCGTGCAGTTT
TGGCCAGGGGACCAAGCTGGAGATCAAA

VL domain amino acid sequence:

SEQ ID NO: 401

DIVMTQSPLSLPVTGPASPISCRSSQSLHSDGYNCLDWYLQKPGQSPQLLIYLGSTRASGFPDRFSGSGSDFT
LKISRVEAEDVGVYYCMQALQTPLSFGQGTKLEIK

-continued

VL CDR1 amino acid sequence:

SEQ ID NO: 398

QSLHSDGYNC

VL CDR2 amino acid sequence:

SEQ ID NO: 399

LGS

VL CDR3 amino acid sequence:

SEQ ID NO: 400

MQALQTPCS

Antibody STIM003

VH domain nucleotide sequence:

SEQ ID NO: 409

GAGGTGCAGCTGGTGGAGTCTGGGGGAGGTGTGGTACGGCCTGGGGGGTCCCTGAGACTCTCCTGTGTAGCCTCTGG

AGTCACCTTTGATGATTATGGCATGAGCTGGGTCCGCCAAGCTCCAGGGAAGGGGCTGGARTGGGTCTCTGGTATTA

ATTGGAATGGTGGCGACACAGATTATTCAGACTCTGTGAAGGGCCGATTACCATCTCCAGAGACAACGCCAAGAAC

TCCCTGTATCTACAAATGAATAGTCTGAGAGCCGAGGACACGGCCTTGTATTACTGTGCGAGGGATTTCATGGTTC

GGGGAGTTATTATCACGTTCTCTTTTGGTACTACTGGGGCCAGGGAATCCTGGTCACCGTCTCCTCA

Corrected STIM003 VH domain nucleotide sequence:

SEQ ID NO: 521

GAGGTGCAGCTGGTGGAGTCTGGGGGAGGTGTGGTACGGCCTGGGGGGTCCCTGAGACTCTCCTGTGTAGCCTCTGG

AGTCACCTTTGATGATTATGGCATGAGCTGGGTCCGCCAAGCTCCAGGGAAGGGGCTGGAGTGGGTCTCTGGTATTA

ATTGGAATGGTGGCGACACAGATTATTCAGACTCTGTGAAGGGCCGATTACCATCTCCAGAGACAACGCCAAGAAC

TCCCTGTATCTACAAATGAATAGTCTGAGAGCCGAGGACACGGCCTTGTATTACTGTGCGAGGGATTTCATGGTTC

GGGGAGTTATTATCACGTTCTCTTTTGGTACTACTGGGGCCAGGGAATCCTGGTCACCGTCTCCTCA

VH domain amino acid sequence:

SEQ ID NO: 408

EVQLVESGGGVVRPGGSLRLSCVAGVTFDDYGMSSWRQAPGKLEWVSGINWNGGDTDYSDSVKGRPTISRDNANK

SLYLQMNSLRLEDTALYYCARDFYGSGSYHVPFDYWGQILVTVSS

VH CDR1 amino acid sequence:

SEQ ID NO: 405

GVTFDDYG

VH CDR2 amino acid sequence:

SEQ ID NO: 406

INWNGGDT

VH CDR3 amino acid sequence:

SEQ ID NO: 407

ARDFYGSGSYHVPFDY

VL domain nucleotide sequence:

SEQ ID NO: 416

GAAATTGTGTTGACGAGTCTCCAGGGACCTGTCTTTGTCTCCAGGGGAAAGACCACCTCTCCTGCAGGGCCAG

TCAGAGTGTTAGCAGAAGCTACTTAGCCTGTACCAGCAGAAACGTGGCCAGGCTCCCAGGCTCCTCATCTATGGTG

CATCCAGCAGGGCCACTGGCATCCAGACAGGTTAGTGGCGATGGGTCTGGGACAGACTTCACTCTCTCCATCAGC

AGACTGGAGCCTGAAGATTTTGCAGTGTATTACTGTACCAGTATGATATGTACCATTCACTTTTCGGCCCTGGGAC

CAAAGTGGATATCAAA

VL domain amino acid sequence:

SEQ ID NO: 415

EIVLTQSPGTLTSLSPGERATLSCRASQSVRSYLAWYQQKRGQAPRLLIYGASSRATGIPDRFSGDGSQDFTLSIS

RLEPEDFAVYYCHQYDMSFPFTFGPGTKVDIK

-continued

VL CDR1 amino acid sequence:

SEQ ID NO: 412

QSVRSY

VL CDR2 amino acid sequence:

SEQ ID NO: 413

GAS

VL CDR3 amino acid sequence:

SEQ ID NO: 414

HQYDMSPT

Antibody STIM004

VH domain nucleotide sequence:

SEQ ID NO: 423

GAGGTGCAGCTGGTGGAGTCTGGGGGAGGTGTGGTACGGCCTGGGGGGTCCCTGAGACTCTCCTGTGCAGCCTCTGG

ACTCACCTTTGATGATTATGGCATGAGCTGGGTCCGCCAAGTTCAGGGAAGGGGCTGGAGTGGGTCTCTGGTATTA

ATTGGAATGGTGATAACACAGATTATGCAGACTCTGTGAAGGGCCGATTACCATCTCCAGAGACAACGCCAAGAAC

TCCCTGTATCTGCAAATGAACAGTCTGAGAGCCGAGGACACGGCCTTGTATTACTGTGCGAGGGATTACTATGGTTC

GGGGAGTTATTATAACGTTCTTTTGTACTACTGGGGCCAGGGAACCTGGTCACCGTCTCCTCA

VH domain amino acid sequence:

SEQ ID NO: 422

EVQLVESGGGVVRPGGSLRLSCAASGLTEDDYGMSWVRQVPGKGLEWVSGINWNGDNTDYADSVKGRFTISRDNANK

SLYLQMNSLR AEDTALYYCARDYYGSGSYYNVFPDYWGQGLVTVSS

VH CDR1 amino acid sequence:

SEQ ID NO: 419

GLTFDDYG

VH CDR2 amino acid sequence:

SEQ ID NO: 420

INWNGDNT

VH CDR3 amino acid sequence:

SEQ ID NO: 421

ARDYYGSGSYYNVFPDY

VL domain nucleotide sequence:

SEQ ID NO: 431

GAAATTGTGTTGACGAGTCTCCAGGCACCTGTCTTTGTCTCCAGGGGAAAGAGCCACCCTCTCCTGCAGGGCCAG

TCAGAGTGTTAGCAGCAGCTACTTAGCCTGGTACCAGCAGAAACCTGGCCAGGCTCCAGGCTCCTCATATATGGTG

CATCCAGCAGGGCCACTGGCATCCAGACAGGTTAGTGGCAGTGGGTCTGGGACAGACTTCACTCTCACCATCAGA

AGACTGGAGCCTGAAGATTTTGCAGTGATTACTGTGACAGTATGGTAGTTCACCATTCATTTCGGCCCTGGGACC

AAAGTGGATATCAAA

VL domain amino acid sequence as encoded by the above VL domain nucleotide sequence.

Corrected VL domain nucleotide sequence:

SEQ ID NO: 430

GAAATTGTGTTGACGAGTCTCCAGGCACCTGTCTTTGTCTCCAGGGGAAAGAGCCACCCTCTCCTGCAGGGCCAG

TCAGAGTGTTAGCAGCAGCTACTTAGCCTGGTACCAGCAGAAACCTGGCCAGGCTCCAGGCTCCTCATATATGGTG

CATCCAGCAGGGCCACTGGCATCCAGACAGGTTAGTGGCAGTGGGTCTGGGACAGACTTCACTCTCACCATCAGA

AGACTGGAGCCTGAAGATTTTGCAGTGATTACTGTGACAGTATGGTAGTTCACCATTCATTTCGGCCCTGGGACCA

AGTGGATATCAAA

Corrected VL domain amino acid sequence:

SEQ ID NO: 432

EIVLTQSPGTLSPGERATLSCRASQSVSSSYLA WYQQKPGQAPRLLIYGASSRATGIPDRFSGSGSGTDFTLTIR

RLEPEDFAVYYCQQYGSSPFFGPGTKVDIK

-continued

VL CDR1 amino acid sequence:

SEQ ID NO: 426

QSVSSSY

VL CDR2 amino acid sequence:

SEQ ID NO: 427

GAS

VL CDR3 amino acid sequence:

SEQ ID NO: 428

QQYGSSPF

Antibody STIM005

VH domain nucleotide sequence:

SEQ ID NO: 439

CAGGTTTCAGTTGGTGACAGTCTGGAGCTGAGGTGAAGAAGCCTGGGGCCTCAGTGAAGGTCCTCTGCAAGGCTTCTGG

TTACACCTTTAATAGTTATGGTATCATCTGGGTGCGACAGGCCCTGGACAAGGGCTTGAGTGGATGGGATGGATCA

GCGTTTCACAATGGTAACACAACTGTGCACAGAAGCTCCAGGGTAGAGTCACCATGACCACAGACACATCCACGAGC

ACAGCCTACATGGAGCTGAGGAGCCTGAGAACTGACGACACGGCCGTGTATTACTGTGCGAGAGCGGGTTACGATAT

TTTGACTGATTTTCCGATGCTTTTGATATCTGGGGCCACGGGACAATGGTCACCGTCTCTTCA

VH domain amino acid sequence:

SEQ ID NO: 438

QVQLVQSGAEVKKPGASVKVSKASGYTFNSYGIWVRQAPGQGLEWMGWI SVHNGNTNCAQKLQGRVTMTDTSTTS

TAYMELRSLRTDDTAVYYCARAGYDILTDFSDAFDIWGHGTMVTVSS

VH CDR1 amino acid sequence:

SEQ ID NO: 435

GYTFNSYG

VH CDR2 amino acid sequence:

SEQ ID NO: 436

ISVHNGNT

VH CDR3 amino acid sequence:

SEQ ID NO: 437

ARAGYDILTDFSDAFDI

VL domain nucleotide sequence:

SEQ ID NO: 446

GACATCCAGATGACCCAGTCTCCATCCTCCCTGTCTGCATCTGTAGGAGACAGAGTCACCATCACTTGCCGGGCAAG

TCAGAACATTAATAACTTTTAAATTGGTATCAGCAGAAAGAAGGAAAGGCCCTAAGCTCCTGATCTATGCAGCAT

CCAGTTTGCAAAGAGGGATACCATCAACGTTTCAGTGGCAGTGGATCTGGGACAGACTTCACTCTCACCATCAGCAGT

CTGCAACCTGAAGATTTTGCAACTTACATCTGTCAACAGAGCTACGGTATCCCGTGGGTCGGCCAAGGGACCAAGGT

GGAAATCAAA

VL domain amino acid sequence:

SEQ ID NO: 445

DIQMTQSPSSLSASVGDRTITCRASQNNINFLNWIYQQKEGKPKLLIYAASSLQRGIPSTFSGSGSGTDFTLTIS

LQPEDFATYICQQSYGIPWVGQGTKVEIK

VL CDR1 amino acid sequence:

SEQ ID NO: 442

QNINNF

VL CDR2 amino acid sequence:

SEQ ID NO: 443

AAS

VL CDR3 amino acid sequence:

SEQ ID NO: 444

QQSYGIPW

-continued

Antibody STIM006

VH domain nucleotide sequence:

SEQ ID NO: 453

CAGGTGCAGCTGGTGGAGTCTGGGGGAGGCTTGGTCAAGCCTGGAGGGTCCCTGAGACTCTCCTGTGCAGCCTCTGG
ATTCACCTTCAGTGACTACTTTCATGAGCTGGATCCGCCAGGCAGGGAAGGGGCTGGAGTGGATTTACATACATTA
GTTCTAGTGGTAGTACCATATACTACGCAGACTCTGTGAGGGGCCGATTACCATCTCCAGGGACAACGCCAAGTAC
TCACTGTATCTGCAAAATGAACAGCCTGAGATCCGAGGACACGCCGTGTATTACTGTGCGAGAGATCACTACGATGG
TTCGGGGATTATCCCTCTACTACTATTACGGTTTGGACGTCTGGGGCCAGGGGACCACGGTCACCGTCTCCTCA

VH domain amino acid sequence:

SEQ ID NO: 454

QVQLVESGGGLVKPGGSLRLSCAASGFTFSDFMSWIRQAPGKLEWISYISSSGSTIYYADSVRGRFTISRDNKY
SLYLQMNSLRSEDTAVYYCARDHYDGSIGYPLYYYGLDVWGQGTITVTVSS

VH CDR1 amino acid sequence:

SEQ ID NO: 449

GFTFSDF

VH CDR2 amino acid sequence:

SEQ ID NO: 450

ISSSGSTI

VH CDR3 amino acid sequence:

SEQ ID NO: 451

ARDHYDGSIGYPLYYYGLDV

VL domain nucleotide sequence:

SEQ ID NO: 460

ATTGTGATGACTCAGTCTCCACTCTCCCTACCCGTCACCCCTGGAGAGCCGGCTCCATCTCCTGCAGGTCTAGTCA
GAGCCTCCTGCATAGTAATGGATACAATATTGGATTATTACCTGCAGAAGCCAGGGCAGTCTCCACAGCTCCTGA
TCTATTGGGTTCTTATCGGGCTCCGGGGTCCCTGACAGGTTCACTGGCAGTGGATCAGGCACAGATTTTACACTG
AAAATCAGCAGAGTGGAGGCTGAGGATGTTGGGGTTTATTACTGCATGCAAGCTCTACAACTCCTCGCAGTTTGG
CCAGGGGACCACGCTGGAGATCAAA

VL domain amino acid sequence:

SEQ ID NO: 459

IVMTQSPSLPVTGPGEPAISCRSSQSLHNSGYNLYDYLLQKPGQSPQLLIYLGSRASGVDPDRFSGSGSGDTFTL
KISRVEAEDVGVYYCMQALQTPRSPGQGTTLLEIK

VL CDR1 amino acid sequence:

SEQ ID NO: 456

QSLHNSGYN

VL CDR2 amino acid sequence:

SEQ ID NO: 457

LGS

VL CDR3 amino acid sequence:

SEQ ID NO: 458

MQALQTPRS

Antibody STIM007

VH domain nucleotide sequence:

SEQ ID NO: 467

CAGATCACCTGAAGGAGTCTGGTCTACGTGGTGAACCCACACAGACCCTCACGCTGACCTGCACCTTCTCTGG
GTTCTCACTCAGCACTACTGGAGTGGGTGTGGGCTGGATCCGTGAGCCCCAGGAAAGGCCCTGGAGTGGCTTGCAG
TCATTTATTGGGATGATGATAAGCGCTACAGCCCATCTCTGAAGAGCAGACTACCATCACCAAGGACACCTCCAAA
AACCAGGTGGTCTTACAATGACCAACATGGACCCTGTGGACACAGCCACATATTTCTGTACACACGGATATGGTTC
GGCGAGTTATTACCACTACGGTATGGACGTCTGGGGCCAAGGGACCACGGTCACCGTCTCCTCA

-continued

VH domain amino acid sequence:

SEQ ID NO: 466

QITLKESGPTLVKPTQTLTLTCTFSGFSLSTTGVGVGWIRQPPGKALEWLAVIYWDDDKRYSPSLKSRLTITKDTSK

NQVVLMTNMDPVDATATYFCTHGYGSASYHYGMDVWGQGTITVTVSS

VH CDR1 amino acid sequence:

SEQ ID NO: 463

GFSLSSTTGVG

VH CDR2 amino acid sequence:

SEQ ID NO: 464

IYWDDDK

VH CDR3 amino acid sequence:

SEQ ID NO: 465

THGYGSASYHYGMDV

VL domain nucleotide sequence:

SEQ ID NO: 474

GAAATTGTATTGACACAGTCTCCAGCCACCCTGTCTTTGTCTCCAGGGAAAGAGCCACCCTCTCCTGCAGGGCCAG

TCAGAGTGTTACCAACTACTTAGCCTGGCACCAACAGAAACCTGGCCAGGCTCCAGGCTCCTCATCTATGATGCAT

CCAACAGGGCCACTGGCATCCAGCCAGGTTCACTGGCAGTGGGTCTGGGACAGACTTCACTCTCACCATCAGCAGC

CTAGAGCCTGAAGATTTTGCAGTTTATTACTGTCTAGCACCCTAGCAACTGGCCTCTCACTTTCGGCGGAGGGACCAA

GGTGGAGATCAAAAC

VL domain amino acid sequence:

SEQ ID NO: 473

EIVLTQSPATLSLSPGERATLSCRASQSVTNYLAWHQKPGQAPRLLIYDASNRATGIPARFSGSGSGTDFTLTSS

LEPEDFAVYYCQHRSNWPLTFGGGKVEIK

VL CDR1 amino acid sequence:

SEQ ID NO: 470

QSVTNY

VL CDR2 amino acid sequence:

SEQ ID NO: 471

DAS

VL CDR3 amino acid sequence:

SEQ ID NO: 472

QHRSNWPLT

Antibody STIM008

VH domain nucleotide sequence:

SEQ ID NO: 481

CAGATCACCTGAAGGAGTCTGGTCTACGCTGGTGAAACCCACACAGACCCTCACGCTGACCTGCACCTTCTCTGG

GTTCTCACTCAGCACTAGTGGAGTGGGTGTGGGCTGGATCCGTACGCCCCAGGAAAGGCCCTGGAGTGGCTTGCAG

TCATTATTGGGATGATGATAAGCGCTACAGCCCATCTCTGAAGAGCAGGCTCACCATCACCAAGGACACCTCCAAA

AACCAGGTGGTCTTACAATGACCAACATGGACCCTGTGGACACAGCCACATATTTCTGTACACACGGATATGGTTC

GGCGAGTTATTACCACTACGGTATGGACGTCTGGGGCCAAGGGACCACGGTCACCGTCTCCTCA

VH domain amino acid sequence:

SEQ ID NO: 480

QITLKESGPTLVKPTQTLTLTCTFSGFSLSTSGVGVGWIRQPPGKALEWLAVIYWDDDKRYSPSLKSRLTITKDTSK

NQVVLMTNMDPVDATATYFCTHGYGSASYHYGMDVWGQGTITVTVSS

VH CDR1 amino acid sequence:

SEQ ID NO: 477

GFSLSSTSGVG

VH CDR2 amino acid sequence:

SEQ ID NO: 478

IYWDDDK

VH CDR3 amino acid sequence:

SEQ ID NO: 479

THGYGSASYHYGMDV

-continued

VL domain nucleotide sequence:

SEQ ID NO: 488

GAAATTGTTGTTGACACAGTCTCCAGCCACCCTGTCTTTGTCTCCAGGGGAAAGAGCCACCCTCTCCTGCAGGGCCAG
TCAGAGTGTTACCAACTACTTAGCCTGGCACCAACAGAAACCTGGCCAGGCTCCAGGCTCCTCATCTATGATGCAT
CCAACAGGGCCACTGGCATCCAGCCAGGTTTCAGTGGCAGTGGGTCTGGGACAGACTTCACTCTCACCATCAGCAGC
CTAGAGCCTGAAGATTTTGCAGTTTATTACTGTGACGAGCGTAGCAACTGGCCTCTCACTTTCGGCGGAGGGACCAA
GGTGGAGATCAAA

VL domain amino acid sequence:

SEQ ID NO: 489

EIVLTQSPATLSLSPGERATLSCRASQSVTNYLAWHQKPGQAPRLLIYDASNRTGIPARFSGSGSGTDFTLTSS
LEPEDFAVYYCQQRSNWPLTFGGGKVEIK

VL CDR1 amino acid sequence:

SEQ ID NO: 484

QSVTNY

VL CDR2 amino acid sequence:

SEQ ID NO: 485

DAS

VL CDR3 amino acid sequence:

SEQ ID NO: 486

QQRSNWPLT

Antibody STIM009

VH domain nucleotide sequence:

SEQ ID NO: 495

CAGGTGCAGCTGGTGGAGTCTGGGGAGGCTTGGTCAAGCCTGGAGGTCCTTGAGACTCTCCTGTGCAGCCTCTGG
ATTCACCTTCAGTGACTACTACATGAGCTGGATCCGCCAGGCTCCAGGAAGGGGCTGGAGTGGGTTTCATACATTA
GTAGTAGTGGTAGTACCATATACTACGCACTCTGTGAAGGGCCGATTACCATCTCCAGGGACAACGCCAAGAAC
TCACTGTATCTGCAAATTAACAGCCTGAGAGCCGAGGACACGGCCGTGTATTACTGTGCGAGAGATTTTACGATAT
TTTGACTGATAGTCCGTACTTCTACTACGGTGTGGACGTCTGGGGCCAAGGACCACGGTCACCGTCTCCTCA

VH domain amino acid sequence:

SEQ ID NO: 494

QVQLVESGGGLVKPGGSLRLSCAASGFTFSFYMSWIRQAPGKGLEWVSYYSSSGSTIYYADSVKGRFTISRDAKN
SLYLQINSLRAEDTAVYYCARDFYDILTDSPIFYGVVDVWGQGTITVTVSS

VH CDR1 amino acid sequence:

SEQ ID NO: 491

GFTFSYY

VH CDR2 amino acid sequence:

SEQ ID NO: 492

ISSSGSTI

VH CDR3 amino acid sequence:

SEQ ID NO: 493

ARDFYDILTDSPIFYGVVDV

VL domain nucleotide sequence:

SEQ ID NO: 502

GATATTGTGATGACTCAGTCTCCACTCTCCCTGCCCGTCACCCCTGGAGAGCCGGCTCCATCTCCTGCAGGTCTAG
TCAGAGCCTCCTGCATAGTAATGGATACAACTATTTGGATTGGTACCTGCAGAAGCCAGGGCAGTCTCCACAGCTCC
TGATCTATTTGGGTTCTAATCGGGCTCCGGGTCCTGACAGGTTTCAGTGGCAGTGGATCAGGCACAGATTTTACA
CTGAAATCAGCAGAGTGGAGGCTGAGGATGTTGGGGTTTATTACTGCATGCAAGCTCTACAACTCCTCGGACGTT
CGGCCAAGGGACCAAGGTGAAATCAAA

VL domain amino acid sequence:

SEQ ID NO: 501

DIVMTQSPLSLPVTPGEPASISCRSSQSLHNSGYNLYLDWYLGKPGQSPQLLIYLGSNRAGVDPDRFSGSGSGTDFT
LKISRVEAEDVGVVYCMQALQTPRTFGQGTKVEIK

-continued

VL CDR1 amino acid sequence:

QSLHNSGYNV

SEQ ID NO: 498

VL CDR2 amino acid sequence:

LGS

SEQ ID NO: 499

VL CDR3 amino acid sequence:

MQALQTPRT

SEQ ID NO: 500

TABLE S1

SEQ ID NOS: 1-342			
SEQ ID NO:	Name	Description	Sequence
1	Human PD-L1	NCBI number: NP_054862.1 (ECD highlighted in BOLD, cytoplasmic domain underlined)	MRIFAVFIFMTYWHLLNAFTVTVPKDLVVEYGSNMTIECKFPVEKQL DLAALIVYWEMEDKNIIQFVHGEEDLVQHSYRQARLLKQQLSLGN AALQITDVKLQDAGVYRCMISYGGADYKRITVKVNAPYNKINQRLV DPVTSEHELTCQAEGYPKAEVIWTSDDHQLVSGKTTTNSKREEKLFN VTSTLRINTTTNEIFYCTFRRLDPEENHTAELVIPLEPLAHPPNERTH LVILGAILLCLGVALTFIFRLRKGRMDVKKCGIQDTNSKKQSDTHLE ET
2	Cyno PD-L1	NCBI number: XP_014973154.1 (ECD highlighted in BOLD)	MGWSCIIILFLVATATGVHSMFTVTVPKDLVVEYGSNMTIECKFPVEK QLDLTSLIVYWEMEDKNIIQFVHGEEDLVQHSNRYRQRAQLKQQLSL GNAALRITDVKLQDAGVYRCMISYGGADYKRITVKVNAPYNKINQRL VVDPTSEHELTCQAEGYPKAEVIWTSDDHQLVSGKTTTNSKREEKLFN LNVSTLRINTTANEIFYCIFRRLDPEENHTAELVIPLEPLALPPNER T
3	Human PD-L1 His	Human PD-L1 ECD with C-terminal His tag	MRIFAVFIFMTYWHLLNAFTVTVPKDLVVEYGSNMTIECKFPVEKQL DLAALIVYWEMEDKNIIQFVHGEEDLVQHSYRQARLLKQQLSLGN AALQITDVKLQDAGVYRCMISYGGADYKRITVKVNAPYNKINQRLV DPVTSEHELTCQAEGYPKAEVIWTSDDHQLVSGKTTTNSKREEKLFN VTSTLRINTTTNEIFYCTFRRLDPEENHTAELVIPLEPLAHPPNERTH HHHHH
4	Human PD-L1 Fc	Human PD-L1 ECD with C-term Fc fusion (in bold)	MRIFAVFIFMTYWHLLNAFTVTVPKDLVVEYGSNMTIECKFPVEKQL DLAALIVYWEMEDKNIIQFVHGEEDLVQHSYRQARLLKQQLSLGN AALQITDVKLQDAGVYRCMISYGGADYKRITVKVNAPYNKINQRLV DPVTSEHELTCQAEGYPKAEVIWTSDDHQLVSGKTTTNSKREEKLFN VTSTLRINTTTNEIFYCTFRRLDPEENHTAELVIPLEPLAHPPNERTI EGREPKSCDKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEV KCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLT VTLHQLDNLNGKEYCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSRD ELTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSGF FLYSKLTVDKSRWQQGNVFCFSVMHEALHNHYTQKSLSLSPGK
5	Cyno PD-L1 FLAG	Cynomolgus PD-L1 ECD with N-term FLAG tag	MGWSCIIILFLVATATGVHSLDSPDRPWNPTFSPALLVVTGEDNATFT CSFSNTSESFVLNWMSPSNQTDKLAAPPEDRSQPGQDCRFVVTQLP NGRDFHMSVVRARRNDSTYLCGAISLAPKAQIKESLRAELRVTERRA EVPTAHPSPSPRPAGQKLENLYFQGLEGRMDEPKSCDKTHTCPPCPAP ELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVD GVEVHNAKTKPREEQYNSTYRVVSVLTVTLHQLDNLNGKEYCKVSNKAL PAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGFYPSD IAVEWESNGQPENNYKTTTPVLDSDGSGFFLYSKLTVDKSRWQQGNVFC FSVMHEALHNHYTQKSLSLSP
6	Human PD-1 Fc	Human PD-1 full length sequence derived from cDNA as human Fc fusion	MGWSCIIILFLVATATGVHSLDSPDRPWNPTFSPALLVVTGEDNATFT CSFSNTSESFVLNWMSPSNQTDKLAAPPEDRSQPGQDCRFVVTQLP NGRDFHMSVVRARRNDSTYLCGAISLAPKAQIKESLRAELRVTERRA EVPTAHPSPSPRPAGQKLENLYFQGLEGRMDEPKSCDKTHTCPPCPAP ELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVD GVEVHNAKTKPREEQYNSTYRVVSVLTVTLHQLDNLNGKEYCKVSNKAL PAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGFYPSD IAVEWESNGQPENNYKTTTPVLDSDGSGFFLYSKLTVDKSRWQQGNVFC FSVMHEALHNHYTQKSLSLSP
7	84G09-CDRH1 (IMGT)	Amino acid sequence of CDRH1 of 84G09 using IMGT	GFTFDYYA

TABLE S1-continued

SEQ ID NOS: 1-342			
SEQ ID NO:	Name	Description	Sequence
8	84G09-CDRH2 (IMGT)	Amino acid sequence of CDRH2 of 84G09 using IMGT	ISWKSNI
9	84G09-CDRH3 (IMGT)	Amino acid sequence of CDRH3 of 84G09 using IMGT	ARDITGSGSYGWFPD
10	84G09-CDRH1 (Kabat)	Amino acid sequence of CDRH1 of 84G09 using Kabat	DYAMH
11	84G09-CDRH2 (Kabat)	Amino acid sequence of CDRH2 of 84G09 using Kabat	GISWKSNIIGYADSVKG
12	84G09-CDRH3 (Kabat)	Amino acid sequence of CDRH3 of 84G09 using Kabat	DITGSGSYGWFPD
13	84G09-Heavy chain variable region	Amino acid sequence of V_H of 84G09 (mutations from germline are shown in bold letters)	EVQLVESGGGLVQPGRSLRLSCAASGFTFDDYAMHWVRQTPGKGLEWVSGISWKSNIIGYADSVKGRFTISRDNAKNSLYLQMNSLRAEDTALYYCAR ARDITGSGSYGWFPDPWGQGLVTVSS
14	84G09-Heavy chain variable region	Nucleic acid sequence of V_H of 84G09	CAaGAAAAAGCTTGCCGCCACCATGGAGTTTGGGCTGAGCTGGATTTT CCTTTTGGCTATTTTAAAGGTGTCCAGTGTGAAGTACAATTGGTGGA GTCCGGGGGAGGCTTGGTACAGCCTGGCAGGTCCCTGAGACTCTCTG TGCAGCCTCTGGATTACCTTTGATGATTATGCCATGCACTGGGTCCG ACAAACTCCAGGAAGGGCCTGGAGTGGGTCTCAGGTATAAGTTGGAA GAGTAATATCATAGGCTATGCGGACTCTGTGAAGGGCCGATTACCAT CTCCAGAGACAACGCCAAGAACTCCCTGTATCTGCAAATGAACAGTCT GAGAGCTGAGGACACGGCCTTGTATTATTGTGCAAGAGATATAACGGG TTCGGGGAGTTATGGCTGGTTCGACCCCTGGGGCCAGGGAACCTGGT CACCGTCTCCTCAGCCAAACGACACCCCATCTGTCTATCCACTGGC CCCTGAATCTGCTAAACTCAGCCTCCG
15	84G09-full heavy chain sequence	Amino acid sequence of 84G09 heavy chain (mutations from germline are shown in bold letters)	EVQLVESGGGLVQPGRSLRLSCAASGFTFDDYAMHWVRQTPGKGLEWVSGISWKSNIIGYADSVKGRFTISRDNAKNSLYLQMNSLRAEDTALYYCAR ARDITGSGSYGWFPDPWGQGLVTVSSASTKGPSVFPLAPCSRSTSEST AALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVT VPSSSLGKTYTCNVDHKPSNTKVDKRVESKYGPPCPCPAPEFEGGP SVFLPPPKPDLMISRTPEVTCVVVDVSQEDPEVQFNWYVDGVVHN AKTKPREEQFNSTYRVVSVLTVLHQDWLNGKEYCKVSNKGLPSSIEK TISKAKGPPEPQVYTLPPSQEEMTKNQVSLTCLVKGFYPSDIAVEWE SNGQPENNYKTPPVLDSGDSFFLYSRLTVDKSRWQEGNVFSCSVMHE ALHNHYTQKSLSLSLGLK
16	84G09-full heavy chain sequence	Nucleic acid sequence of 84G09 heavy chain	GAAAGTGACGCTGGTGAATCTGGCGCGGACTGGTGCAGCCTGGCAGATCCCTGAGACTGTCTGTGCGCCTCCGGCTTCACCTTCGACGACTAC GCTATGCACTGGGTGCGACAGACCCCTGGCAAGGGCCTGGAATGGGTG TCCGCATCTCTGGAAGTCCAACATCATCGGCTACGCCGACTCCGTG AAGGGCCGGTTACCATCTCCCGGACCAACGCCAAGAACTCCCTGTAC CTGCAGATGAACAGCCTGCGGGCCGAGGACACCGCCCTGTACTACTG GCCAGAGACATCACCGGCTCCGGCTCCTACGGATGGTTCGATCCTGG GGCCAGGGCACCTCGTGACCGTGTCTCTGCCAGCACCAAGGGCCCC TCTGTGTTCCCTCTGGCCCTTCCAGCAAGTCCACCTCTGGCGGAACA GCCGCTCTGGGCTGCCTCGTGAAGGACTACTTCCCGAGCCTGTGACC GTGTCTGGAATCTGGCGCTCTGACAGCGGAGTGACACCTTCCT GCTGTGCTGACGCTCTCCGGCTGTACTCCCTGTCTCTCCGTGCTGAC GTGCTTCCAGCTCTCTGGGCACCCAGACCTACATCTGCAACGTGAAC CACAAGCCTTCAACACCAAGGTGGACAAGAGGTGGAACCCCAAGTCC TGCGACAAGACCCACACCTGTCCCTTGTCTGCCCCGAACTGCTG GGCGGACCTTCCGTGTTCTGTTCCTTCCCCCAAGCCCAAGGACACCTG ATGATCTCCCGGACCCCGAAGTGACCTGCGTGGTGGTGGATGTGTCC CACGAGGACCCCTGAAGTGAAGTTCAATTGGTACGTGGACGGCGTGGAA GTGCACAACGCCAAGACCAAGCCTAGAGAGGAACAGTACAACCTCACC TACCGGGTGGTGTCCGTGCTGACCGTGTGCACAGGATTGGCTGAAC GGCAAAGAGTACAAGTGAAGGTGTCACAAGGCCCTGCTGCCCCC ATCGAAAAGACCATCTCCAAGGCCAAGGCCAGCCCCGGGAACCCAG GTGTACACACTGCCCCCTAGCAGGGACGAGCTGACCAAGAACCGGTG

TABLE S1-continued

SEQ ID NOS: 1-342			
SEQ ID NO:	Name	Description	Sequence
			TCCCTGACCTGTCTCGTGAAAGGCTTCTACCCCTCCGATATCGCCGTG GAATGGGAGTCCAACGGCCAGCCTGAGAACAACTACAAGACCACCCCC CCTGTGCTGGACTCCGACGGCTCATTCTTCTGTACAGCAAGCTGACA GTGGACAAGTCCCGGTGGCAGCAGGGCAACGTGTTCTCTGTCTCCGTG ATGCACGAGGCCCTGCACAACCACTACACCCAGAAGTCCCTGTCCCTG AGCCCCGGCAAG
17	84G09- CDRL1 (IMGT)	Amino acid sequence of CDRL1 of 84G09 using IMGT	QSISSY
18	84G09- CDRL2 (IMGT)	Amino acid sequence of CDRL2 of 84G09 using IMGT	VAS
19	84G09- CDRL3 (IMGT)	Amino acid sequence of CDRL3 of 84G09 using IMGT	QQSYSNPIT
20	84G09- CDRL1 (Kabat)	Amino acid sequence of CDRL1 of 84G09 using Kabat	RASQSISSYLN
21	84G09- CDRL2 (Kabat)	Amino acid sequence of CDRL2 of 84G09 using Kabat	VASSLQS
22	84G09- CDRL3 (Kabat)	Amino acid sequence of CDRL3 of 84G09 using Kabat	QQSYSNPIT
23	84G09- Light chain variable region	Amino acid sequence of V _L of 84G09	DIQMTQSPSSLSASVGDRTITCRASQSISSYLNWYQQKPKGKAPKPLI YVASSLQSGVPSFSGSGSDFTLTISLQPEDFATYYCQQSYSNPI TFGQGRLEIK
24	84G09- Light chain variable region	Nucleic acid sequence of V _L of 84G09	GACATCCAGATGACCCAGTCTCCATCCTCCCTGTCTGCATCTGTAGGA GACAGAGTCACCATCACTTGCCGGGCAAGTCAGAGCATTAGCAGCTAT TTAAATTGGTATCAGCAGAAACAGGGAAAGCCCTAAGCCCCTGATC TATGTTGCATCCAGTTTGCAAAGTGGGGTCCCATCAAGTTTCAGTGGC AGTGGATCTGGGACAGATTTCACTCTCACCATCAGCAGTCTGCAACCT GAAGATTTTGCAACTTACTACTGTCAACAGAGTTACAGTAATCCGATC ACCTTCGGCCAAGGGACACGACTGGAGATCAAA
25	84G09- full light chain sequence	Amino acid sequence of 84G09 light chain	DIQMTQSPSSLSASVGDRTITCRASQSISSYLNWYQQKPKGKAPKPLI YVASSLQSGVPSFSGSGSDFTLTISLQPEDFATYYCQQSYSNPI TFGQGRLEIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREA KVQWKVDNALQSGNSQESVTEQDSKDSYLSSTLTLSKADYEKHKVY ACEVTHQGLSPVTKSFNRGEC
26	84G09- full light chain sequence	Nucleic acid sequence of 84G09 light chain	GACATCCAGATGACCCAGTCTCCATCCTCCCTGTCTGCATCTGTAGGA GACAGAGTCACCATCACTTGCCGGGCAAGTCAGAGCATTAGCAGCTAT TTAAATTGGTATCAGCAGAAACAGGGAAAGCCCTAAGCCCCTGATC TATGTTGCATCCAGTTTGCAAAGTGGGGTCCCATCAAGTTTCAGTGGC AGTGGATCTGGGACAGATTTCACTCTCACCATCAGCAGTCTGCAACCT GAAGATTTTGCAACTTACTACTGTCAACAGAGTTACAGTAATCCGATC ACCTTCGGCCAAGGGACACGACTGGAGATCAAAACGTACGGTGGCCGCT CCCTCCGTGTTTCTTCCACCTTCCGACGAGCAGCTGAAGTCCGGC ACCGCTTCTGTCTGTGCTGCTGAACAACCTTCTACCCCGCGAGGGCC AAGGTGCAGTGAAGGTGGACAACGCCCTGCAGTCCGGCAACTCCAG GAATCCGTGACCGAGCAGGACTCCAAGGACAGCACCTACTCCCTGTCC TCCACCCTGACCCTGTCCAAGGCCGACTACGAGAAGCACAAGGTGTAC GCCTGCGAAGTGACCCACAGGGCCTGTCTAGCCCCGTGACCAAGTCT TTCAACCGGGGCGAGTGT
27	1D05- CDRH1 (IMGT)	Amino acid sequence of CDRH1 of 1D05 using IMGT	GFTFDDYA
28	1D05- CDRH2 (IMGT)	Amino acid sequence of CDRH2 of 1D05 using IMGT	ISWIRTGI

TABLE S1-continued

SEQ ID NOS: 1-342			
SEQ ID NO:	Name	Description	Sequence
29	1D05-CDRH3 (IMGT)	Amino acid sequence of CDRH3 of 1D05 using IMGT	AKDMKSGTYGGWFDT
30	1D05-CDRH1 (Kabat)	Amino acid sequence of CDRH1 of 1D05 using Kabat	DYAMH
31	1D05-CDRH2 (Kabat)	Amino acid sequence of CDRH2 of 1D05 using Kabat	GISWIRTGIGYADSVKG
32	1D05-CDRH3 (Kabat)	Amino acid sequence of CDRH3 of 1D05 using Kabat	DMKSGTYGGWFDT
33	1D05-Heavy chain variable region	Amino acid sequence of V_H of 1D05 (mutations from germline are shown in bold letters)	EVQLVESGGGLVQPGRSLRLSCAASGFTFDDYAMHWVRQVPKGLEWVSGISW IRT GIGYADSVKGRFTI FR DNNAKNSLYLQMNSLRAEDTALYYCAKDMKSGTYGGWFDTWGQGLTVTVSS
34	1D05-Heavy chain variable region	Nucleic acid sequence of V_H of 1D05	AAGCTTGCCGCCACCATGGAGTTTGGGCTGAGCTGGATTTTCCTTTTGCTATTTTAAAAGGTGTCCAGTGTGAAGTGCAGCTGGTGGAGTCTGGGGGAGGCTTGGTGCAGCCTGGCAGGTCCTGAGACTCTCCTGTGCAGCCTCTGGATTACCTTTGATGATTATGCCATGCACCTGGGTCCGGCAAGTCCAGGGAAGGGCCTGGAATGGGTCTCAGGCATTAGTTGGATTCTGACTGGCATAGGCTATGCGGACTCTGTGAAGGGCCGATTACCATTTTCAGAGACAACGCCAAGAAATCCCTGTATCTGCAAAATGAACAGTCTGAGAGCTGAGGACACGGCCTTGATTACTGTGCAAAAGATATGAAGGGTTCGGGGACTTATGGGGGTGGTTCGACACCTGGGGCCAGGGAACCTGGTCACCGTCTCTCAGCCAAACAACAGCCCCATCGGTCTATCCACTGGCCCCCTGC
35	1D05-full heavy chain sequence	Amino acid sequence of 1D05 heavy chain	EVQLVESGGGLVQPGRSLRLSCAASGFTFDDYAMHWVRQVPKGLEWVSGISWIRTGIGYADSVKGRFTI FR DNNAKNSLYLQMNSLRAEDTALYYCAKDMKSGTYGGWFDTWGQGLTVTVSSASTKGPSVFPPLAPCSRSTSETAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQKTYTCNVDPKPKNTKVDKRVESKYGPCCPAPPEFEGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSQEDPEVQFNWYVDGVEVHNAKTKPREEQFNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKGLPSSIEKTISKAKGQPREPQVYTLPPSQEEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLYSRLTVDKSRWQEGNVPFSCVMHEALHNYHTQKSLSLSLGK
36	1D05-full heavy chain sequence	Nucleic acid sequence of 1D05 heavy chain	GAGTGCAGCTGGTGAATCTGGCGGCGGACTGGTGCAGCCTGGCAGATCCCTGAGACTGTCTGTGCCGCCTCCGGCTTCACCTTCGACGACTACGCTATGCAGCTGGGTGCGCAGGTGCCAGGCAAGGGCCTGGAATGGGTGTCGGCATCTCTTGGATCCGGACCGGCATCGGCTACGCCGACTCTGTGAAGGGCCGGTTACCATCTCCGGGACACGCCAAGAAGTCCCTGTACTGTCAGATGAACAGCCTGCGGGCCGAGGACACCGCCCTGTACTACTGCGCAAGGACATGAAGGGCTCCGGCACCTACGGCGGATGGTTGCATACTGGGGCCAGGGCACCTCGTGACCGTGTCTCTGCCAGCACCAAGGGCCCTCTGTGTTCCCTCTGGCCCTTCCAGCAAGTCCACCTCTGGCGGAACAGCCGCTCTGGGCTGCCTCGTGAAGGACTACTTCCCCGAGCCTGTGACCGTGTCTTGAACTCTGGCGCTCTGACCAGCGGAGTGACACCTTCCCTGTGTGCTGCAGTCTCCGGCCTGTACTCCCTGTCTCCGTCTGTACCGTGCCCTTCCAGCTCTCTGGGCACCCAGACCTACATCTGCAACGTGAACCAAGCCCTCCAAACACCAAGGTGGACAAGAAGGTGGAACCCAAGTCCCTGCGACAAGACCCACACCTGTCCCCCTTGTCTGCCCCCTGAAGTCTGGGCGGACCTTCCGTGTTCTGTTCCTGTTCCCCCAAAGCCCAAGGACACCTGATGATCTCCCGGACCCCGAAGTGACCTGCGTGGTGGTGGATGTGTCCACGAGGACCCCTGAAGTGAAGTTCAATTGGTACGTGGACGGCGTGAAGTGACACAACGCCAAGACCAAGCCTAGAGAGGAACAGTACCACTCCACCTACCGGGTGGTGTCCGTGTGACCGTGTGCAACAGGATTGGCTGAACGGCAAGAGTACAAGTGCAAGGTGTCCAACAGGCCCTGCCTGCCCTCCATCGAAAAGACCATCTCCAAGGCCAAGGGCCAGCCCCGGGAACCCAGGTGTACACACTGCCCCCTAGCAGGGACGAGCTGACCAAGAACCCAGGTGTCCTGACCTGTCTCGTGAAAGGCTTCTACCCCTCCGATATCGCGTGGAATGGGAGTCCAACGGCCAGCCTGAGAACAACACAGACCAACCCCCCTGTGCTGGACTCCGACGGCTCATCTCTGTACAGCAAGCTGACAGTGGACAAGTCCCGTGGCAGCAGGGCAACGTGTTCTCTGTCTCCGTGATGACAGGGCCCTGCACAACCACTACACCCAGAAGTCCCTGTCTCTGAGCCCCGCAAG

TABLE S1-continued

SEQ ID NOS: 1-342			
SEQ ID NO:	Name	Description	Sequence
37	1D05-CDRL1 (IMGT)	Amino acid sequence of CDRL1 of 1D05 using IMGT	QSISSY
38	1D05-CDRL2 (IMGT)	Amino acid sequence of CDRL2 of 1D05 using IMGT	VAS
39	1D05-CDRL3 (IMGT)	Amino acid sequence of CDRL3 of 1D05 using IMGT	QQSYSTPIT
40	1D05-CDRL1 (Kabat)	Amino acid sequence of CDRL1 of 1D05 using Kabat	RASQSISSYLN
41	1D05-CDRL2 (Kabat)	Amino acid sequence of CDRL2 of 1D05 using Kabat	VASSLQS
42	1D05-CDRL3 (Kabat)	Amino acid sequence of CDRL3 of 1D05 using Kabat	QQSYSTPIT
43	1D05-Light chain variable region	Amino acid sequence of V_L of 1D05 (mutations from germline are shown in bold letters)	DIQMTQSPSSLSASVGDRTITCRASQSISSYLNWYQQKPGKAPKLLIYVASSLQSGVPSRFRSGSGSDFTLTISLQPEDFATYYCQQSYSTPI TFGQGTREIK
44	1D05-Light chain variable region	Nucleic acid sequence of V_L of 1D05	AAAGCTTGCCGCCACCATGAGGCTCCCTGCTCAGCTTCTGGGGCTCCTGCTACTCTGGCTCCGAGGTGCCAGATGTGACATCCAGATGACCCAGTCTCCATCCTCCCTGTCTGCATCTGTAGGAGACAGAGTCACCATCACTTGCCGGCAAGTCAGAGCATTAGCAGCTATTTAAATTGGTATCAGCAGAAACCAGGGAAAGCCCCCTAAACTCCTGATCTATGTTGCATCCAGTTTGCA AAGTGGGGTCCCATCAAGGTTCAAGTGGCAGTGGATCTGGGACAGATTTCACTCTCACTATCAGCAGTCTGCAACCTGAAGATTTGCAACTTACTA CTGTCAACAGAGTTACAGTACCCCGATCACCTTCGGCCAAGGGACACG TCTGGAGATCAAACGTACGGATGCTGCACCAACT
45	1D05-full light chain sequence	Amino acid sequence of 1D05 light chain	DIQMTQSPSSLSASVGDRTITCRASQSISSYLNWYQQKPGKAPKLLIYVASSLQSGVPSRFRSGSGSDFTLTISLQPEDFATYYCQQSYSTPI TFGQGTREIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREA KVQWKVDNALQSGNSQESVTEQDSKDSITYLSLSTLTLSKADYEKHKVY ACEVTHQGLSSPVTKSFNRGEC
46	1D05-full light chain sequence	Nucleic acid sequence of 1D05 light chain	GACATCCAGATGACCCAGTCCCCCTCCAGCCTGTCTGCTTCCGTGGGC GACAGAGTGACCATCACCTGTGCGGCCCTCCAGTCCATCTCCTCTAC CTGAATCGGTATCAGCAGAAGCCCGCAAGGCCCAAGCTGCTGATC TACGTGGCCAGCTCTCTGCAGTCCGGCGTGCCCTCTAGATTCTCCGGC TCTGGCTCTGGCACCGACTTTACCTGACCATCAGCTCCCTGCAGCCC GAGGACTTCGCCACCTACTACTGCCAGCAGTCTTACTCCACCCCTATC ACCTTCGGCCAGGGCACCCGGCTGGAATCAAACGTACGGTGGCCGCT CCCTCCGTGTTTCATCTCCACCTTCCGACGAGCAGCTGAAGTCCGGC ACCGCTTCTGTGCTGTGCTGCTGAACAACCTTCTACCCCGCGAGGGCC AAGGTGAGTGAAGGTGGACAACGCCCTGCAGTCCGGCAACTCCAG GAATCCGTGACCGAGCAGGACTCCAAGGACAGACCTACTCCCTGTCC TCCACCTGACCTGTCCAAGGCCGACTACGAGAAGCACAAGGTGTAC GCCTGCGAAGTGACCCACAGGGCCTGTCTAGCCCCGTGACCAAGTCT TCAACCGGGGCGAGTGT
47	Mutated 1D05-HC mutant 1	Amino acid sequence of 1D05 heavy chain with V to A back-mutation in framework region to germline highlighted with IgG1 disabled (LAGA) constant region	EVQLVESGGGLVQPGRSLRLSCAASGFTFDDYAMHWVRQAPGKGLEWVSGISWIRTGIGYADSVKGRFTIFRDNAKNSLYLQMNSLRÆDTALYYCAKDMKSGTYGGWFDTWGQGLTVTVSSASTKGPSVFPLAPCSRSTSES TAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYTCNVDHKPSNTKVDKRVESKYGPCCPAPCPELAGA PSVFLFPPPKPKDTLMISRPEVTCTVVDVSQEDPEVQFNWYVDGVEVH NAKTKPREQFNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKGLPSSIE KTISKAKGQPREPQVYTLPPSQEEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLYSRLTVDKSRWQEGNVFSCVMHEALHNHYTQKSLSLSLGK

TABLE S1-continued

SEQ ID NOS: 1-342			
SEQ ID NO:	Name	Description	Sequence
48	Mutated 1D05-HC mutant 2	Amino acid sequence of 1D05 heavy chain with F to S back-mutation in framework region to germline highlighted with IgG1 disabled (LAGA) constant region	EVQLVESGGGLVQPGRSLRLSCAASGFTFDDYAMHWVRQVPKGLEWVSGISWIRTGIGYADSVKGRFTISRDNAKNSLYLQMNSLRAEDTALYYCAKDMKSGSGTYGGWFDTWGQGLTVTVSSASTKGPSVFPLAPCSRSTSESTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTCTYTCNVDHKPSNTKVDKRVESKYGPCCPPCPAPELAGAPSVFLFPPKPKDTLMISRTPEVTCVVVDVSQEDPEVQFNWYVDGVEVHNAKTKPREEQFNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKGLPSSIEKTISKAKGQPREPQVYTLPPSQEEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSLRTVDKSRWQEGNVFSCVMHEALHNHYTQKSLSLSLGK
49	Mutated 1D05-HC mutant 3	Amino acid sequence of 1D05 heavy chain with ELLG to -PVA back-mutation in constant region to germline highlighted	EVQLVESGGGLVQPGRSLRLSCAASGFTFDDYAMHWVRQVPKGLEWVSGISWIRTGIGYADSVKGRFTIFRDNAKNSLYLQMNSLRAEDTALYYCAKDMKSGSGTYGGWFDTWGQGLTVTVSSASTKGPSVFPLAPCSRSTSESTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTCTYTCNVDHKPSNTKVDKRVESKYGPCCPPCPAP-PVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSQEDPEVQFNWYVDGVEVHNAKTKPREEQFNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKGLPSSIEKTIISKAKGQPREPQVYTLPPSQEEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSLRTVDKSRWQEGNVFSCVMHEALHNHYTQKSLSLSLGK
50	Mutated 1D05-LC mutant 1	Amino acid sequence of 1D05 kappa light chain with V to A back-mutation in CDR2 to germline highlighted	DIQMTQSPSSLSASVGRVTITCRASQSISSYLNWYQQKPKAPKLLIYAASSLQSGVPSRFSGSGSGTDFTLTISLQPEDFATYYCQQSYSTPI TFGQGTRLLEIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREA KVQWKVDNALQSGNSQESVTEQDSKDSSTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC
51	Mutated 1D05-LC mutant 2	Amino acid sequence of 1D05 kappa light chain with L to F back-mutation in framework to germline highlighted	DIQMTQSPSSLSASVGRVTITCRASQSISSYLNWYQQKPKAPKLFIVASSLQSGVPSRFSGSGSGTDFTLTISLQPEDFATYYCQQSYSTPI TFGQGTRLLEIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREA KVQWKVDNALQSGNSQESVTEQDSKDSSTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC
52	411B08-CDRH1 (IMGT)	Amino acid sequence of CDRH1 of 411B08 using IMGT	GFTFSSYW
53	411B08-CDRH2 (IMGT)	Amino acid sequence of CDRH2 of 411B08 using IMGT	IKEDGSEK
54	411B08-CDRH3 (IMGT)	Amino acid sequence of CDRH3 of 411B08 using IMGT	ARNRLYSDFLDN
55	411B08-CDRH1 (Kabat)	Amino acid sequence of CDRH1 of 411B08 using Kabat	SYWMS
56	411B08-CDRH2 (Kabat)	Amino acid sequence of CDRH2 of 411B08 using Kabat	NIKEDGSEKYYVDSVKG
57	411B08-CDRH3 (Kabat)	Amino acid sequence of CDRH3 of 411B08 using Kabat	NRLYSDFLDN
58	411B08-Heavy chain variable region	Amino acid sequence of V_H of 411B08	EVQLVESGGGLVQPGGSLRLSCAASGFTFSSYWMSWVRQAPKGLEWV ANIKEDGSEKYYVDSVKGFRFTISRDNAKNSLYLQMNSLRAEDTSVYYC ARNRLYSDFLDNWWGQGLTVTVSS
59	411B08-Heavy chain variable region	Nucleic acid sequence of V_H of 411B08	GAGGTGCAGCTGGTGGAGTCTGGGGGAGGCTTGGTCCAGCCTGGGGG TCCTTGAGACTCTCCTGTGCAGCCTCTGGATCACGTTTAGTAGCTAT TGGATGAGTTGGGTCCGCCAGGCTCCAGGGAAGGGGCTGGAGTGGGTG GCCAACATCAAGAAGATGGAAGTGAATACTATGTCTGACTCTGTG AAGGGCCGATTACCATCTCCAGAGACACGCAAGAAGTCACTGTAT CTGCAATGAACAGCCTGAGAGCCGAGGACAGCTCTGTGTATTACTGT

TABLE S1-continued

SEQ ID NOS: 1-342			
SEQ ID NO:	Name	Description	Sequence
			GCGAGAAATCGACTCTACAGTGACTTCTTGACAACTGGGGCCAGGGA ACCCTGGTCACCGTCTCCTCAG
60	411B08- full heavy chain sequence	Amino acid sequence of 411B08 heavy chain	EVQLVESGGGLVQPGGSLRLSCAASGFTFSSYWMSEWRQAPGKGLEWV ANIKEDGSEKYVDSVKGRFTISRDNAKNSLYQMNSLRAEDTSVYYC ARNRLYSDFLDNWQGQTLVTVSSASTKGPSVFPLAPSSKSTSGGTAAL GCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPS SSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKHTHTCPPCPAPPELLGGP SVFLFPPPKPDLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHN AKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEK TISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWE SNGQPENNYKTPPVLDSDGSFFLYSLKLTVDKSRWQQGNVFSQSVMHE ALHNHYTQKSLSLSPGK
61	411B08- full heavy chain sequence	Nucleic acid sequence of 411B08 heavy chain	GAGGTGCAGCTGGTGGAGTCTGGGGGAGGCTTGGTCCAGCCTGGGGGG TCCCTGAGACTCTCCTGTGCAGCCTCTGGATTCACGTTAGTAGCTAT TGGATGAGTTGGGTCCGCCAGGCTCCAGGGAAGGGGCTGGAGTGGGTG GCCAACATCAAAGAAGTGAAGTGAGAAATACTATGTCGACTCTGTG AAGGGCCGATTACCATCTCCAGAGACAACGCCAAGAACTCACTGTAT CTGCAAATGAACAGCTGAGAGCCGAGGACACGTCTGTGTATTACTGT GCGAGAAATCGACTCTACAGTGACTTCTTGACAACTGGGGCCAGGGA ACCCTGGTCACCGTCTCCTCAGCCAGCACCAAGGGCCCTCTGTGTTT CCTCTGGCCCCCTCCAGCAAGTCCACCTCTGGCGGAACAGCCGCTCTG GGCTGCCTCGTGAAGGACTACTTCCCCGAGCCTGTGACCGTGTCTCTGG AACTCTGGCGCTCTGACCAGCGGAGTGACACCTTCCCTGCTGTGCTG CAGTCTCTCCGGCTGTACTCCCTGTCTCCGTCGTGACCGTGCCTTCC AGCTCTCTGGGCACCCAGACCTACATCTGCAACGTGAACCACAAGCCC TCCAACACCAAGGTGGACAAGAAGGTGGAACCAAGTCTTGCGACAAG ACCCACACCTGTCCCCCTGTCTCTGCCCTGAACGTGGGCGGACCT TCCGTGTTCTCTGTTCCCCCAAGGCCAAGGACACCCCTGATGATCTCC CGGACCCCCGAAGTGACCTGCGTGGTGGTGGATGTGTCCACGAGGAC CCTGAAGTGAAGTTCAATTGGTACGTGGACGGCTGGAAGTGACACAAC GCCAAGACCAAGCCTAGAGAGGAACAGTACAACCTCACCTACCGGGTG GTGTCCTGTGCTGACCGTGTGCTGACCAGGATTGGCTGAACGGCAAGAG TACAAGTGCAAGGTGTCCAACAAGGCCCTGCCTGCCCCCATCGAAAAG ACCATCTCCAAGGCCAAGGGCCAGCCCCGGGAACCCAGGTGTACACA CTGCCCCCTAGCAGGACGAGCTGACCAAGAACCAGGTGTCCCTGACC TGTCTCGTGAAAGGCTTCTACCCCTCCGATATCGCCGTGGAATGGGAG TCCAACGGCCAGCCTGAGAACAACTACAAGACCACCCCTGTGCTG GACTCCGACGGCTCATTCTTCTGTACAGCAAGCTGACAGTGGACAAG TCCCGTGGCAGCAGGGCAACGTGTTCTCCTGCTCCGTGATGCACGAG GCCCTGCACAACCACTACACCCAGAAGTCCCTGTCCCTGAGCCCCGGC AAG
62	411B08- CDRL1 (IMGT)	Amino acid sequence of CDRL1 of 411B08 using IMGT	QGVSSW
63	411B08- CDRL2 (IMGT)	Amino acid sequence of CDRL2 of 411B08 using IMGT	GAS
64	411B08- CDRL3 (IMGT)	Amino acid sequence of CDRL3 of 411B08 using IMGT	QQANSIPFT
65	411B08- CDRL1 (Kabat)	Amino acid sequence of CDRL1 of 411B08 using Kabat	RASQGVSSWLA
66	411B08- CDRL2 (Kabat)	Amino acid sequence of CDRL2 of 411B08 using Kabat	GASSLQS
67	411B08- CDRL3 (Kabat)	Amino acid sequence of CDRL3 of 411B08 using Kabat	QQANSIPFT
68	411B08- Light chain variable region	Amino acid sequence of V _L of 411B08	DIQMTQSPSSVSASVGDVITITCRASQGVSSWLAWYQQKSGKAPKLLI YGASSLQSGVPSRFSGSGTEFILTISLQPEDFATYYCQQANSIPF TFGPGTKVDIK

TABLE S1-continued

SEQ ID NOS: 1-342			
SEQ ID NO:	Name	Description	Sequence
69	411B08-Light chain variable region	Nucleic acid sequence of V_L of 411B08	GACATCCAGATGACCCAGTCTCCATCTTCCGTGTCTGCATCTGTGCGGA GACAGAGTCACCATCACTTGTCTGGGCGAGTCAGGGTGTAGCAGCTGG TTAGCCTGGTATCAGCAGAAATCAGGGAAAGCCCCTAAGCTCCTGATC TATGGTGCATCCAGTTTGCAAAGTGGGGTCCCATCAAGATTCAGCGGC AGTGGATCTGGGACAGAGTTCATTCTCACCATCAGCAGCCTGCAGCCT GAAGATTTTGCAACTTACTATTGTCAACAGGCTAACAGTATCCCATT ACTTTCGGCCCTGGGACCAAAGTGGATATCAAAAC
70	411B08-full light chain sequence	Amino acid sequence of 411B08 light chain	DIQMTQSPSSVSASVGDRTITCRASQGVSSWLAWYQQKSGKAPKLLI YGASSLQSGVPSRFRSGSGSGTEFILTISLQPEDFATYYCQANSIPF TFPGPTKVDIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREA KVQWKVDNALQSGNSQESVTEQDSKDSYSTLSSTLTLSKADYEKHKVY ACEVTHQGLSPVTKSFNRGEC
71	411B08-full light chain sequence	Nucleic acid sequence of 411B08 light chain	GACATCCAGATGACCCAGTCTCCATCTTCCGTGTCTGCATCTGTGCGGA GACAGAGTCACCATCACTTGTCTGGGCGAGTCAGGGTGTAGCAGCTGG TTAGCCTGGTATCAGCAGAAATCAGGGAAAGCCCCTAAGCTCCTGATC TATGGTGCATCCAGTTTGCAAAGTGGGGTCCCATCAAGATTCAGCGGC AGTGGATCTGGGACAGAGTTCATTCTCACCATCAGCAGCCTGCAGCCT GAAGATTTTGCAACTTACTATTGTCAACAGGCTAACAGTATCCCATT ACTTTCGGCCCTGGGACCAAAGTGGATATCAAAACGTACGGTGGCCGCT CCCTCCGTGTTTCACTTCCACCTTCCGACGAGCAGCTGAAGTCCGGC ACCGCTTCTGTCTGTGCTGCTGTAACAACTTCTACCCCGCGAGGCC AAGGTGCAGTGAAGGTGGACAACGCCCTGCAGTCCGGCAACTCCCAG GAATCCGTGACCGAGCAGGACTCCAAGGACAGCACCTACTCCTGTCC TCCACCTGACCCTGTCCAAGGCCGACTACGAGAAGCACAGGTGTAC GCCTGCGAAGTGACCCACAGGGCCTGTCTAGCCCCGTGACCAAGTCT TTCAACCGGGGGGAGTGT
72	411C04-CDRH1 (IMGT)	Amino acid sequence of CDRH1 of 411C04 using IMGT	GFTFSSYW
73	411C04-CDRH2 (IMGT)	Amino acid sequence of CDRH2 of 411C04 using IMGT	IKEDGSEK
74	411C04-CDRH3 (IMGT)	Amino acid sequence of CDRH3 of 411C04 using IMGT	ARVRLYSDFLDY
75	411C04-CDRH1 (Kabat)	Amino acid sequence of CDRH1 of 411C04 using Kabat	SYWMS
76	411C04-CDRH2 (Kabat)	Amino acid sequence of CDRH2 of 411C04 using Kabat	NIKEDGSEKYYVDSLKG
77	411C04-CDRH3 (Kabat)	Amino acid sequence of CDRH3 of 411C04 using Kabat	VRLYSDFLDY
78	411C04-Heavy chain variable region	Amino acid sequence of V_H of 411C04	EVQLVDSGGGLVQPGGSLRLSCAASGFTFSSYWMSWVRQAPGKGLEWV ANIKEDGSEKYYVDSLKGRFTISRDNAKNSLYLQMNSLRAEDTSVYYC ARVRLYSDFLDYWGQGLTVTVSS
79	411C04-Heavy chain variable region	Nucleic acid sequence of V_H of 411C04	GAGGTGCAGCTGGTGGACTCTGGGGGAGGCTTGGTCCAGCCTGGGGGG TCCTGAGACTCTCCTGTGCAGCCTCTGGATCACGTTTAGTAGCTAT TGGATGAGTTGGGTCCGCCAGGCTCCAGGAAGGGGCTGGAGTGGGTG GCCAACATAAAGAAGATGGAAGTGAGAAATACTATGTAGACTCTTTG AAGGGCCGATTACCATCTCCAGAGACACGCCAAGAAGTCACTGTAT CTGCAATGAACAGCCTGAGAGCCGAGGACAGCTCTGTGTATTACTGT GCGAGAGTTCGACTCTACAGTGACTTCCTTGACTACTGGGGCCAGGGA ACCTGGTCACTGCTCTCTCAG
80	411C04-full heavy chain sequence	Amino acid sequence of 411C04 heavy chain	EVQLVDSGGGLVQPGGSLRLSCAASGFTFSSYWMSWVRQAPGKGLEWV ANIKEDGSEKYYVDSLKGRFTISRDNAKNSLYLQMNSLRAEDTSVYYC ARVRLYSDFLDYWGQGLTVTVSSASTKGPSVFPLAPSSKSTSGGTAAL GCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPS SSLGTQTYICNVNHKPSNTKVDKVEPKSCDKHTHTCPPCPAPELGGP SVFLFPPKPKDTLMISRTPEVTCVVDVSHEDPEVKFNWYVDGVEVHN

SEO ID NOS: 1-342

SEQ ID NOS: 1-342			
SEQ ID			
NO:	Name	Description	Sequence
			AKTKPREEQNSTYRVVSVLTVLHQDNLNGKEYKCKVSNKALPAPIEK TISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWE SNQGPENNYKTTTPVLDSDGSFPLYSKLTVDKSRWQQGNVFSQSVMHE ALHNHYTKQSLSLSPGK
81	411C04- full heavy chain sequence	Nucleic acid sequence of 411C04 heavy chain	GAGGTGCAGCTGGTGGACTCTGGGGGAGGCTTGGTCCAGCCTGGGGGG TCCTTGAGACTCTCCTGTGCAGCCTCTGGATTACGTTTAGTAGCTAT TGGATGAGTTGGGTCCGCCAGGCTCCAGGAAGGGGCTGGAGTGGGTG GCCAACATAAAAGAAGATGGAAGTGAGAAATACTATGTAGACTCTTTG AAGGGCCGATTCACCATCTCCAGAGACAACGCCAAGAACTCACTGTAT CTGCAATGAACAGCCTGAGAGCCGAGGACACGCTCTGTGTATTACTGT GCGAGAGTTCGACTCTACAGTGACTTCCTTGACTACTGGGGCCAGGGA ACCTTGGTCACCGTCTCCTCAGCCAGCACCAAGGGCCCTCTGTGTTC CCTCTGGCCCTTCCAGCAAGTCCACCTCTGGCGGAACAGCCGCTCTG GGCTGCCTCGTGAAGGACTACTTCCCCGAGCCTGTGACCGTGTCCCTGG AACTCTGGCGCTCTGACCAGCGAGTGCACACCTTCCCTGCTGTGCTG CAGTCTCTCCGGCCTGTACTCCTGTCTCCTGCTGTGACCGTGCCTTCC AGCTCTCTGGGCACCCAGACCTACATCTGCAACGTGAACCACAAGCCC TCCAACACCAAGGTGGACAAGAAGGTGGAACCCAAGTCTTGCAGACAAG ACCCACACCTGTCCCCCTTGCTCTGCCCTGAACTGCTGGGCGGACCT TCCGTGTCTCTGTTCCCCCACAAGCCCAAGGACACCTGATGATCTCC CGGACCCCCGAAGTGACCTGCGTGGTGGTGGATGTGTCCACAGGAGAC CCTGAAGTGAAGTTCAATTGGTACGTGGACGGCGTGAAGTGCACAAAC GCCAAGACCAAGCCTAGAGAGGAACAGTACAACCTCCACCTACCGGGTG GTGTCGTGCTGACCGTGTCTGCACAGGATTGGCTGAACGGCAAAAGAG TACAAGTGAAGGTGTCCAACAAGGCCCTGCCTGCCCCCATCGAAAAG ACCATCTCCAAGGCCAAGGGCCAGCCCCGGGAACCCAGGTGTACACA TGCCCCCTAGCAGGGACGAGCTGACCAAGAACCAGGTGTCCTTGACC TGCTCTCGTGAAGGCTTCTACCCCTCCGATATCGCCGTGGAATGGGAG TCCAACGGCCAGCCTGAGAACAACTACAAGACCACCCCTCTGTGTG GACTCCGACGGCTCATTCTTCTGTACAGCAAGCTGACAGTGGACAAG TCCCGGTGGCAGCAGGGCAACGTGTTCTCCTGCTCCGTGATGCACAG GCCTGCACAACCACTACACCCAGAAGTCCCTGTCCCTGAGCCCCGGC AAG
82	411C04- CDRL1 (IMGT)	Amino acid sequence of CDRL1 of 411C04 using IMGT	QGVSSW
83	411C04- CDRL2 (IMGT)	Amino acid sequence of CDRL2 of 411C04 using IMGT	GAS
84	411C04- CDRL3 (IMGT)	Amino acid sequence of CDRL3 of 411C04 using IMGT	QQANSIPFT
85	411C04- CDRL1 (Kabat)	Amino acid sequence of CDRL1 of 411C04 using Kabat	RASQGVSSWLA
86	411C04- CDRL2 (Kabat)	Amino acid sequence of CDRL2 of 411C04 using Kabat	GASSLQS
87	411C04- CDRL3 (Kabat)	Amino acid sequence of CDRL3 of 411C04 using Kabat	QQANSIPFT
88	411C04- Light chain variable region	Amino acid sequence of V_L of 411C04	DIQMTQSPSSVSASVGDRTVITCRASQGVSSWLAWYQQKSGKAPKLLI YGASSLQSGVPSRFSFGSGSGTEFILSISSLPQEDFATYYCQQANSIPF TFPGTKVDIK
89	411C04- Light chain variable region	Nucleic acid sequence of V_L of 411C04	GACATCCAGATGACCCAGTCTCCATCTTCCGTGTCTGCATCTGTCCGA GACAGAGTCACCATCACTTGTCTGGGCGAGTCAGGGTGTTAGCAGTTGG TTAGCCTGGTATCAGCAGAAATCAGGGAAGGCCCTTAAGCTCCTGATC TATGGTGCCTCCAGTTTGCAAAGTGGGGTCCCATCAAGATTGAGCGGC AGTGGATCTGGGACAGAGTTCATTCTCAGCATCAGCAGCCTGCAGCCT GAAGATTTTGCAACTTACTATTGTCAACAGGCTAACAGTATCCCATTC ACTTTCCGGCCCTGGGACCAAGTGGATATCAAAC

TABLE S1-continued

SEQ ID NOS: 1-342			
SEQ ID NO:	Name	Description	Sequence
90	411C04-full light chain sequence	Amino acid sequence of 411C04 light chain	DIQMTQSPSSVSASVGDRTITCRASQGVSSWLAWYQQKSGKAPKLLI YGASSLQSGVPSRFSGSGSGTEFILSISSLQPEDFATYYCQQANSIPF TFGPGTKVDIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREA KVQWKVDNALQSGNSQESVTEQDSKSTYLSLSTLTLSKADYEKHKVY ACEVTHQGLSSPVTKSFNRGEC
91	411C04-full light chain sequence	Nucleic acid sequence of 411C04 light chain	GACATCCAGATGACCCAGTCTCCATCTTCCGTGTCTGCATCTGTCCGA GACAGAGTCACCATCACTTGTCTGGGCGAGTCAGGGTGTAGCAGTTGG TTAGCCTGGTATCAGCAGAAATCAGGGAAGCCCTAAGCTCCTGATC TATGGTGCCCTCCAGTTTGCAAAGTGGGGTCCCATCAAGATTCAGCGGC AGTGGATCTGGGACAGAGTTCATTCTCAGCATCAGCAGCCTGCAGCCT GAAGATTTTGCAACTTACTATTGTCAACAGGCTAACAGTATCCCATTC ACTTTCGGCCCTGGGACCAAGTGATATCAACGTACGGTGGCCGCT CCCTCCGTGTTTCATCTTCCCACCTTCCGACGAGCAGCTGAAGTCCGGC ACCGCTTCTGTCTGTCTGTCTGTGAACAACCTTCTACCCCGCGAGGCC AAGGTGCAGTGGAAGGTGGACAACGCCCTGCAGTCCGGCACTCCAG GAATCCGTGACCGAGCAGGACTCCAAGGACAGCACCTACTCCCTGTCC TCCACCTTGACCTGTCTCAAGGCCGACTACGAGAAGCACAAAGGTGTAC GCCTGCGAAGTGACCCACAGGGCCTGTCTAGCCCGTGACCAAGTCT TTCAACCGGGCGAGTGT
92	411D07-CDRH1 (IMGT)	Amino acid sequence of CDRH1 of 411D07 using IMGT	GGSIISDW
93	411D07-CDRH2 (IMGT)	Amino acid sequence of CDRH2 of 411D07 using IMGT	IFHSGRT
94	411D07-CDRH3 (IMGT)	Amino acid sequence of CDRH3 of 411D07 using IMGT	ARDGSGSY
95	411D07-CDRH1 (Kabat)	Amino acid sequence of CDRH1 of 411D07 using Kabat	SSDWWN
96	411D07-CDRH2 (Kabat)	Amino acid sequence of CDRH2 of 411D07 using Kabat	EIFHSGRTNYPNLSKS
97	411D07-CDRH3 (Kabat)	Amino acid sequence of CDRH3 of 411D07 using Kabat	DGSGSY
98	411D07-Heavy chain variable region	Amino acid sequence of V_H of 411D07	QVQLQESGPGLVKPSGTLSTLCIVSGGSIISDWWNWRQPPGKLEW IGEIFHSGRTNYPNLSKSRVTISIDKSKNQFSLRLSSVTAADTAVYYC ARDGSGSYWGQGLTVTVSS
99	411D07-Heavy chain variable region	Nucleic acid sequence of V_H of 411D07	CAGGTGCAGCTGCAGGAGTCGGGCCAGGACTGGTGAAGCCTTCGGGG ACCCTGTCCCTCACCTGCATTGTCTCTGGTGGCTCCATCATCAGTAGT GACTGGTGAATTGGGTCCGCCAGCCCCAGGGAAGGGCTGGAGTGG ATTGGAGAAATCTTTCATAGTGGGAGGACCAACTACAACCCGTCCCTC AAGAGTCGAGTCACCATATCAATAGACAAGTCCAAGAATCAGTTCTCC CTGAGGCTGAGCTCTGTGACCGCCGCGGACACGGCCGTGATTACTGT GCGAGAGATGGTTCGGGGAGTTACTGGGGCCAGGGAACCTGGTCACC GTCTCCTCAG
100	411D07-full heavy chain sequence	Amino acid sequence of 411D07 heavy chain	QVQLQESGPGLVKPSGTLSTLCIVSGGSIISDWWNWRQPPGKLEW IGEIFHSGRTNYPNLSKSRVTISIDKSKNQFSLRLSSVTAADTAVYYC ARDGSGSYWGQGLTVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLV KDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLG TQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCPPCPAPELLGGPSVFL FPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTK PREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISK AKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESNGQ PENNYKTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFCSCVMHEALHN HYTQKSLSLSPGK

TABLE S1-continued

SEQ ID NOS: 1-342			
SEQ ID NO:	Name	Description	Sequence
101	411D07-full heavy chain sequence	Nucleic acid sequence of 411D07 heavy chain	CAGGTGCAGCTGCAGGAGTCGGGCCAGGACTGGTGAAGCCTTCGGGG ACCCTGTCCCTCACCTGCATTGTCTCTGGTGGCTCCATCATCAGTAGT GACTGGTGGAAATTGGGTCCGCCAGCCCCAGGGAAGGGCTGGAGTGG ATTGGAGAAATCTTTCATAGTGGGAGGCCAACTACAACCCGTCCCTC AAGAGTCGAGTCACCATATCAATAGACAAGTCCAAGAATCAGTTCTCC CTGAGGCTGAGCTCTGTGACCGCCGCGACACGGCCGTGATTACTGT GCGAGAGATGGTTCGGGGAGTTACTGGGGCCAGGGAAACCTGGTCACC GTCTCCTCAGCCAGCACCAAGGGCCCCCTCTGTGTTCCCTCTGGCCCCCT TCCAGCAAGTCCACCTCTGGCGGAACAGCCGCTCTGGGCTGCCTCGTG AAGGACTACTTCCCGAGCCTGTGACCGTGTCTTGAACTCTGGCGCT CTGACCAGCGGAGTGCACACCTTCCCTGCTGTGCTGCAGTCCCTCCGGC CTGTACTCCCTGTCTCCGTCTGTACCGTGCCTTCCAGCTCTCTGGGC ACCCAGACCTACATCTGCAACGTGAACCAAGCCCTCCAACACCAAG GTGGACAAGAAGGTGGAACCCAAAGTCTGCGACAAGACCCACACCTGT CCCCCTTGTCTGCCCTGAAGTGTGGGCGGACCTTCCGTGTTCTGT TTCCCCCAAGCCCAAGGACACCTGATGATCTCCCGACCCCGAA GTGACCTGCGTGGTGGTGGATGTGTCCACGAGGACCTGAAGTGAAG TTCAATTGGTACGTGGACGGCGTGGAAAGTGACACAACGCCAAGACCAAG CCTAGAGAGGAACAGTACAATCCACCTACCGGGTGGTGTCCGTGCTG ACCGTGCTGCACCAAGATTGGGTGAACGGCAAGAGTACAAGTGAAG GTGTCCAACAAGGCCCTGCCTGCCCCATCGAAAAGACCATCTCCAAG GCCAAGGGCCAGCCCCGGGAACCCAGGTGTACACACTGCCCCCTAGC AGGGACGAGCTGACCAAGAACCAGGTGTCCCTGACCTGTCTCGTGAAA GGCTTCTACCCCTCCGATATCGCCGTGGAATGGGAGTCCAACGGCCAG CCTGAGAACAACTACAAGACCACCCCCCTGTGTGCTGGACTCCGACGGC TCATTCTTCTGTACAGCAAGCTGACAGTGGACAAGTCCCGGTGGCAG CAGGGCAACGTGTTCTCTGCTCCGTGATGCACGAGGCCCTGCACAAC CACTACACCCAGAAGTCCCTGTCCCTGAGCCCCGGCAAG
102	411D07-CDRL1 (IMGT)	Amino acid sequence of CDRL1 of 411D07 using IMGT	QSVLYSSNNKNY
103	411D07-CDRL2 (IMGT)	Amino acid sequence of CDRL2 of 411D07 using IMGT	WAS
104	411D07-CDRL3 (IMGT)	Amino acid sequence of CDRL3 of 411D07 using IMGT	QQYYSNRS
105	411D07-CDRL1 (Kabat)	Amino acid sequence of CDRL1 of 411D07 using Kabat	KSSQSVLYSSNNKNYLA
106	411D07-CDRL2 (Kabat)	Amino acid sequence of CDRL2 of 411D07 using Kabat	WASTRES
107	411D07-CDRL3 (Kabat)	Amino acid sequence of CDRL3 of 411D07 using Kabat	QQYYSNRS
108	411D07-Light chain variable region	Amino acid sequence of V_L of 411D07	DIVMTQSPDSLAVSLGERATINCKSSQSVLYSSNNKNYLA ^{WY} QQKSGQ PPKLLIYWASTRESGVPDRFSGSGSGTDFTLT ^{ISS} LQTEDVAVYYCQQ YYSNRSFQGQGT ^{KLEI} K
109	411D07-Light chain variable region	Nucleic acid sequence of V_L of 411D07	GACATCGTGATGACCCAGTCTCCAGACTCCCTGGCTGTGTCTCTGGGC GAGAGGGCCACCATCAACTGCAAGTCCAGCCAGAGTGT ^{TTT} TATACAGC TCCAACAATAAGAATTACTTAGCTTGGTACCAGCAGAAATCAGGACAC CCTCCTAAGTTGCTCATTACTGGGCATCTACCCGGAATCCGGGGTC CCTGACCGATTAGTGGCAGCGGTCTGGGACAGATTTCACTCTCACC ATCAGCAGCCTGCAGACTGAAGATGTGGCAGTTTATTACTGTACGAA TATTATAGTAATCGCAGTTT ^{TG} GCCAGGGGACCAAGCTGGAGATCAAA C
110	411D07-full light chain sequence	Amino acid sequence of 411D07 light chain	DIVMTQSPDSLAVSLGERATINCKSSQSVLYSSNNKNYLA ^{WY} QQKSGQ PPKLLIYWASTRESGVPDRFSGSGSGTDFTLT ^{ISS} LQTEDVAVYYCQQ YYSNRSFQGQGT ^{KLEI} KRTVAAPSVFIFPPSDEQLKSGTASV ^{VCL} LN ^{NF} YPREAKVQWKVDNALQSGNSQESVTEQDSK ^{STY} LSSTLTLSKADYE KHKVYACEVTHQGLSPVTKSFNRGEC

TABLE S1-continued

SEQ ID NOS: 1-342			
SEQ ID NO:	Name	Description	Sequence
111	411D07-full light chain sequence	Nucleic acid sequence of 411D07 light chain	GACATCGTGATGACCCAGTCTCCAGACTCCCTGGCTGTGTCTCTGGGC GAGAGGGCCACCATCAACTGCAAGTCCAGCCAGAGTGTCTTATACAGC TCCAACAATAAGAATTACTTAGCTTGGTACCAGCAGAAATCAGGACAG CCTCCTAAGTTGCTCATTACTGGGCATCTACCCGGGAATCCGGGGTC CCTGACCGATTGAGTGGCAGCGGGTCTGGGACAGATTTCACTCTCACC ATCAGCAGCCTGCAGACTGAAGATGTGGCAGTTTATTACTGTCAGCAA TATTATAGTAATCGCAGTTTGGCCAGGGGACCAAGCTGGAGATCAAA CGTACGGTGGCCGCTCCCTCCGTGTTTCATCTTCCACCTTCCGACGAG CAGCTGAAGTCCGGCACCGCTTCTGTCTGTGCTGCTGAACAACTTC TACCCCGCGAGGCCAAGGTGCAGTGGAAGGTGGACAACGCCCTGCAG TCCGGCAACTCCAGGAATCCGTGACCGAGCAGGACTCCAAGGACAGC ACCTACTCCCTGTCTCCACCTGACCTGTCCAAGGCCGACTACGAG AAGCAAGGTGTACGCTGCGAAGTGACCCACCAGGGCCTGTCTAGC CCCCTGACCAAGTCTTTCAACCGGGGCGAGTGT
112	385F01-CDRH1 (IMGT)	Amino acid sequence of CDRH1 of 385F01 using IMGT	GPTFSSYW
113	385F01-CDRH2 (IMGT)	Amino acid sequence of CDRH2 of 385F01 using IMGT	IKEDGSEK
114	385F01-CDRH3 (IMGT)	Amino acid sequence of CDRH3 of 385F01 using IMGT	ARNRLYSDFLDN
115	385F01-CDRH1 (Kabat)	Amino acid sequence of CDRH1 of 385F01 using Kabat	SYWMS
116	385F01-CDRH2 (Kabat)	Amino acid sequence of CDRH2 of 385F01 using Kabat	NIKEDGSEKYYVDSVKG
117	385F01-CDRH3 (Kabat)	Amino acid sequence of CDRH3 of 385F01 using Kabat	NRLYSDFLDN
118	385F01-Heavy chain variable region	Amino acid sequence of V_H of 385F01	EVQLVESGGGLVQPGGSLRLSCAASGFTFSSYWMSWVRQAPGKGLEWV ANIKEDGSEKYYVDSVKGRFTISRDNAKNSLYLQMNSLRAEDTSVYYC ARNRLYSDFLDNHWGQGLTVTVSS
119	385F01-Heavy chain variable region	Nucleic acid sequence of V_H of 385F01	GAGGTGCAGCTGGTGGAGTCTGGGGGAGGCTTGGTCCAGCCTGGGGG TCCCTGAGACTCTCCTGTGCAGCCTCTGGATTACGTTTAGTAGCTAT TGGATGAGTTGGGTCCGCCAGGCTCCAGGGAAGGGGCTGGAGTGGGTG GCCAACATCAAAGAAGATGGAAGTGAGAAATACTATGTCGACTCTGTG AAGGGCCGATTACCATCTCCAGAGACAACGCCAAGAACTCACTGTAT CTGCAAAATGAACAGCCTGAGAGCCGAGGACACGCTCTGTGTATTACTGT GCGAGAAATCGACTCTACAGTGACTTCCTTGACAACCTGGGGCCAGGGA ACCCTGGTCACCGTCTCCTCAG
120	385F01-full heavy chain sequence	Amino acid sequence of 385F01 heavy chain	EVQLVESGGGLVQPGGSLRLSCAASGFTFSSYWMSWVRQAPGKGLEWV ANIKEDGSEKYYVDSVKGRFTISRDNAKNSLYLQMNSLRAEDTSVYYC ARNRLYSDFLDNHWGQGLTVTVSSASTKGPSVFPLAPSSKSTSGGTAAL GCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPS SSLTGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHSTCPPCPAPELLGGP SVFLPPPKPKDLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHN AKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYCKVSNKALPAPIEK TISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWE SNGQPENNYKTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFPSCSVHME ALHNHYTQKSLSLSPGK
121	385F01-full heavy chain sequence	Nucleic acid sequence of 385F01 heavy chain	GAGGTGCAGCTGGTGGAGTCTGGGGGAGGCTTGGTCCAGCCTGGGGG TCCCTGAGACTCTCCTGTGCAGCCTCTGGATTACGTTTAGTAGCTAT TGGATGAGTTGGGTCCGCCAGGCTCCAGGGAAGGGGCTGGAGTGGGTG GCCAACATCAAAGAAGATGGAAGTGAGAAATACTATGTCGACTCTGTG AAGGGCCGATTACCATCTCCAGAGACAACGCCAAGAACTCACTGTAT CTGCAAAATGAACAGCCTGAGAGCCGAGGACACGCTCTGTGTATTACTGT GCGAGAAATCGACTCTACAGTGACTTCCTTGACAACCTGGGGCCAGGGA ACCCTGGTCACCGTCTCCTCAGCCAGCACCAAGGCCCTCTGTGTTT

TABLE S1-continued

SEQ ID NOS: 1-342			
SEQ ID NO: Name	Description	Sequence	
		CCTCTGGCCCCCTTCCAGCAAGTCCACCTCTGGCGGAACAGCCGCTCTG GGCTGCCTCGTGAAGGACTACTTCCCCGAGCCTGTGACCGTGTCTCTGG AACTCTGGCGCTCTGACCAGCGGAGTGCACACCTTCCCTGTGTGCTG CAGTCCTCCGGCCTGTACTTCCCTGTCTCCGTGACCGTGCCCTTCC AGCTCTCTGGGCACCCAGACCTACATCTGCAACGTGAACCACAAGCCC TCCAACACCAAGGTGGACAAGAAGGTGGAACCCAAGTCTTGCACAAAG ACCCACACCTGTCCCCCTGTCTGCTGCCCCCTGAACTGCTGGGCGGACCT TCCGTGTTCTGTCTCCCCCAAGCCCAAGGACACCTGATGATCTCC CGGACCCCCGAAGTGACCTGCGTGGTGGTGGATGTGTCCACGAGGAC CCTGAAGTGAAGTTCATTTGGTACGTGGACGGCGTGGAAGTGCACAAC GCCAAGACCAAGCCTAGAGAGGAACAGTACAACCTCCACCTACCGGGTG GTGTCCGTGCTGACCGTGTGCTGACACAGGATTGGCTGAACGGCAAGAG TACAAGTGAAGGTGTCCAACAAGGCCCTGCCTGCCCCATCGAAAAG ACCATCTCCAAGGCCAAGGGCCAGCCCGGGAAACCCAGGTGTACACA CTGCCCCCTAGCAGGGACGAGCTGACCAAGAACCAGGTGTCCCTGACC TGTCTCGTGAAGGCTTCTACCCCTCCGATATCGCCGTGGAATGGGAG TCCAACGGCCAGCCTGAGAACAACTACAAGACCAACCCCTGTGTCTG GACTCCGACGGCTCATTCTTCTGTACAGCAAGCTGACAGTGGACAAG TCCCGGTGGCAGCAGGGCAACGTGTTCTCTGCTCCGTGATGCACGAG GCCCTGCACAACCACTACACCCAGAAGTCCCTGTCCCTGAGCCCCGGC AAG	
122 385F01-CDRL1 (IMGT)	Amino acid sequence of CDRL1 of 385F01 using IMGT	QGVSSW	
123 385F01-CDRL2 (IMGT)	Amino acid sequence of CDRL2 of 385F01 using IMGT	GAS	
124 385F01-CDRL3 (IMGT)	Amino acid sequence of CDRL3 of 385F01 using IMGT	QQANSIPFT	
125 385F01-CDRL1 (Kabat)	Amino acid sequence of CDRL1 of 385F01 using Kabat	RASQGVSSWLA	
126 385F01-CDRL2 (Kabat)	Amino acid sequence of CDRL2 of 385F01 using Kabat	GASSLQS	
127 385F01-CDRL3 (Kabat)	Amino acid sequence of CDRL3 of 385F01 using Kabat	QQANSIPFT	
128 385F01-Light chain variable region	Amino acid sequence of V_L of 385F01	DIQMTQSPSSVSASVGDRTITCRASQGVSSWLAWYQQKSGKAPKLLI YGASSLQSGVPSRFSGSGSGTEFILTISLQPEDFATYYCQQANSIPF TFGPGTKVDIK	
129 385F01-Light chain variable region	Nucleic acid sequence of V_L of 385F01	GACATCCAGATGACCCAGTCTCCATCTTCCGTGTCTGCATCTGTGGA GACAGAGTCACCATCACTTGTGCGGCGAGTCAGGGTGTAGCAGCTGG TTAGCCTGGTATCAGCAGAAATCAGGGAAGCCCTAAGCTCCTGATC TATGGTGATCCAGTTTGCAAAGTGGGGTCCCATCAAGATTGACGGC AGTGATCTGGGACAGAGTTCATTCTCACCATCAGCAGCCTGCAGCCT GAAGATTTTGCAACTTACTATTGTCAACAGGCTAACAGTATCCCATTC ACTTTCGGCCCTGGGACCAAGTGGATATCAAAC	
130 385F01-full light chain sequence	Amino acid sequence of 385F01 light chain	DIQMTQSPSSVSASVGDRTITCRASQGVSSWLAWYQQKSGKAPKLLI YGASSLQSGVPSRFSGSGSGTEFILTISLQPEDFATYYCQQANSIPF TFGPGTKVDIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREA KVQWKVDNALQSGNSQESVTEQDSKDSYLSSTLTLSKADYEKHKVY ACEVTHQGLSPVTKSFNRGEC	
131 385F01-full light chain sequence	Nucleic acid sequence of 385F01 light chain	GACATCCAGATGACCCAGTCTCCATCTTCCGTGTCTGCATCTGTGGA GACAGAGTCACCATCACTTGTGCGGCGAGTCAGGGTGTAGCAGCTGG TTAGCCTGGTATCAGCAGAAATCAGGGAAGCCCTAAGCTCCTGATC TATGGTGATCCAGTTTGCAAAGTGGGGTCCCATCAAGATTGACGGC AGTGATCTGGGACAGAGTTCATTCTCACCATCAGCAGCCTGCAGCCT GAAGATTTTGCAACTTACTATTGTCAACAGGCTAACAGTATCCCATTC ACTTTCGGCCCTGGGACCAAGTGGATATCAAACGTAACGTTGGCCGT CCGTCCGTGTTATCTTCCACCTTCCGACGAGCAGCTGAAGTCCGGC	

TABLE S1-continued

SEQ ID NOS: 1-342			
SEQ ID NO:	Name	Description	Sequence
			ACCGCTTCTGTCGTGTCCTGCTGAACAACCTCTACCCCCGCGAGGCC AAGGTGCAGTGGGAAGGTGGACAACGCCCTGCAGTCCGGCAACTCCCAG GAATCCGTGACCGAGCAGGACTCCAAGGACAGCACCTACTCCCTGTCC TCCACCTTGACCTGTCCAAGGCCGACTACGAGAAGCACAAGGTGTAC GCCTGCGAAGTGACCCACCAGGGCCTGTCTAGCCCCGTGACCAAGTCT TTCAACCGGGGCGAGTGT
132	413D08- CDRH1 (IMGT)	Amino acid sequence of CDRH1 of 413D08 using IMGT	GFTFRIYG
133	413D08- CDRH2 (IMGT)	Amino acid sequence of CDRH2 of 413D08 using IMGT	IWYDGSNK
134	413D08- CDRH3 (IMGT)	Amino acid sequence of CDRH3 of 413D08 using IMGT	ARDMDYFGMDV
135	413D08- CDRH1 (Kabat)	Amino acid sequence of CDRH1 of 413D08 using Kabat	IYGMH
136	413D08- CDRH2 (Kabat)	Amino acid sequence of CDRH2 of 413D08 using Kabat	VIWYDGSNKYYADSVKG
137	413D08- CDRH3 (Kabat)	Amino acid sequence of CDRH3 of 413D08 using Kabat	DMDYFGMDV
138	413D08- Heavy chain variable region	Amino acid sequence of V_H of 413D08	QVQLVESGGGVVQPGRSRLRLSCAASGFTFRIYGMHWVRQAPGKGLEWV AVIWYDGSNKYYADSVKGRFTISRDNNDNTLYLQMNSLRAEDTAVYYC ARDMDYFGMDVWGQGTITVSS
139	413D08- Heavy chain variable region	Nucleic acid sequence of V_H of 413D08	CAGGTGCAGCTGGTGGAGTCTGGGGGAGGCGTGGTCCAGCCTGGGAGG TCCCTGAGACTCTCCTGTGCAGCGTCTGGATTACCTTCCGTATTTAT GGCATGCAGTGGGTCCGCCAGGCTCCAGGCAAGGGGCTGGAGTGGGTG GCAGTTATATGGTATGATGGAAGTAATAAATACTATGCTGACTCCGTG AAGGGCCGATTACCATCTCCAGAGACAATTCCGACAACACGCTGTAT CTGCAAAATGAACAGCCTGAGAGCCGAGGACACGGCTGTGTATTACTGT GCGAGAGATATGGACTACTTCGGTATGGACGCTCTGGGGCCAAGGGACC ACGGTCACCGTCTCCTCAG
140	413D08- full heavy chain sequence	Amino acid sequence of 413D08 heavy chain	QVQLVESGGGVVQPGRSRLRLSCAASGFTFRIYGMHWVRQAPGKGLEWV AVIWYDGSNKYYADSVKGRFTISRDNNDNTLYLQMNSLRAEDTAVYYC ARDMDYFGMDVWGQGTITVSSASTKGPSVFPLAPSSKSTSGGTAAALG CLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSS SLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHSTCPPCPAPPELLGGPS VFLFPPKPKDTLMISRTPEVTCVVDVSHEDPEVKFNWYVDGVEVHNA KTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKT ISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWES NGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFSQSVSMHEA LHNHYTQKSLSLSPGK
141	413D08- full heavy chain sequence	Nucleic acid sequence of 413D08 heavy chain	CAGGTGCAGCTGGTGGAGTCTGGGGGAGGCGTGGTCCAGCCTGGGAGG TCCCTGAGACTCTCCTGTGCAGCGTCTGGATTACCTTCCGTATTTAT GGCATGCAGTGGGTCCGCCAGGCTCCAGGCAAGGGGCTGGAGTGGGTG GCAGTTATATGGTATGATGGAAGTAATAAATACTATGCTGACTCCGTG AAGGGCCGATTACCATCTCCAGAGACAATTCCGACAACACGCTGTAT CTGCAAAATGAACAGCCTGAGAGCCGAGGACACGGCTGTGTATTACTGT GCGAGAGATATGGACTACTTCGGTATGGACGCTCTGGGGCCAAGGGACC ACGGTCACCGTCTCCTCAGCAGCACCAGGGGCCCTCTGTGTTCCCT CTGGCCCCCTTCCAGCAAGTCCACCTCTGGCGGAACAGCCGCTCTGGG TGCCCTCGTGAAGGACTACTTCCCGAGCCTGTGACCGTGTCTTGGAAC TCTGGCGCTCTGACCAGCGAGTGACACCTTCCCTGTGTGCTGCAG TCCTCCGGCCTGTACTCCCTGTCTCCGTCGTGACCGTGCCTTCCAGC TCTCTGGGCACCCAGACCTACATCTGCAACGTGAACCAAGCCCTCC AACACCAAGGTGGACAAGAAGGTGGAACCAAGTCTCTGCGACAAGACC CACACCTGTCCCCCTGTCTGCTGCCCTGAACGTGTGGGCGGACCTTCC GTGTTCCCTGTCCCCCAAGCCCAAGGACACCTGATGATCTCCCGG ACCCCCGAAGTGACCTGCGTGGTGGTGGATGTGTCCACGAGGACCTT

TABLE S1-continued

SEQ ID NOS: 1-342			
SEQ ID NO: Name	Description	Sequence	
		GAAGTGAAGTTC AATTGGTACGTGGACGGCGTGGAAGTGCAACAACGCC AAGACCAAGCCTAGAGAGGAACAGTACAACTCCACTACCGGGTGGTG TCCGTGCTGACCGTGCTGCACCAGGATTGGCTGAACGGCAAGAGTAC AAGTGCAAGGTGTCCAACAAGGCCCTGCCTGCCCCATCGAAAAGACC ATCTCCAAGGCCAAGGGCCAGCCCCGGGAACCCAGGTGTACACACTG CCCCCTAGCAGGGACGAGCTGACCAAGAACCAGGTGTCCCTGACCTGT CTCGTGAAGGCTTCTACCCCTCCGATATCGCCGTGGAATGGGAGTCC AACGGCCAGCCTGAGAACAACTACAAGACCACCCCCCTGTGCTGGAC TCCGACGGCTCATTTCTCCTGTACAGCAAGCTGACAGTGGACAAGTCC CGGTGGCAGCAGGGCAACGTGTTCTCCTGCTCCGTGATGCACGAGGCC CTGCACAACCACTACACCAGAAGTCCCTGTCCCTGAGCCCCGGCAAG	
142 413D08-CDRL1 (IMGT)	Amino acid sequence of CDRL1 of 413D08 using IMGT	QGIRND	
143 413D08-CDRL2 (IMGT)	Amino acid sequence of CDRL2 of 413D08 using IMGT	AAS	
144 413D08-CDRL3 (IMGT)	Amino acid sequence of CDRL3 of 413D08 using IMGT	LQHNSYPRT	
145 413D08-CDRL1 (Kabat)	Amino acid sequence of CDRL1 of 413D08 using Kabat	RASQGIRNDLG	
146 413D08-CDRL2 (Kabat)	Amino acid sequence of CDRL2 of 413D08 using Kabat	AASSLQS	
147 413D08-CDRL3 (Kabat)	Amino acid sequence of CDRL3 of 413D08 using Kabat	LQHNSYPRT	
148 413D08-Light chain variable region	Amino acid sequence of V_L of 413D08	DLQMTQSPSSLSASVGDRTITCRASQGIRNDLGWYQQKPGKAPKRLI YAASSLQSGVPSRFSGSGSGTEFTLTISSLQPEDFATYYCLQHNSYPRTFGQGTKVEIK	
149 413D08-Light chain variable region	Nucleic acid sequence of V_L of 413D08	GACCTCCAGATGACCCAGTCTCCATCCTCCCTGTCTGCATCTGTAGGA GACAGAGTCACCATCACTTGCCGGGCAAGTCAGGGCATTAGAAATGAT TTAGGCTGGTATCAGCAGAAACAGGGAAAGCCCCAAGCGCCTGATC TATGCTGCATCCAGTTTGCAAAGTGGGGTCCCATCAAGGTTCAGCGGC AGTGGATCTGGGACAGAATTCACCTCTACAATCAGCAGCCTGCAGCCT GAAGATTTTGCAACTTATTACTGTCTACAGCATAATAGTTACCCCTCGG ACGTTCCGCCAAGGGACCAAGGTGGAATCAAAC	
150 413D08-full light chain sequence	Amino acid sequence of 413D08 light chain	DLQMTQSPSSLSASVGDRTITCRASQGIRNDLGWYQQKPGKAPKRLI YAASSLQSGVPSRFSGSGSGTEFTLTISSLQPEDFATYYCLQHNSYPRTFGQGTKVEIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREA KVQWKVDNALQSGNSQESVTEQDSKDSSTYSLSSTLTLSKADYEKHKVYACEVTHQGLSPVTKSFNRGEC	
151 413D08-full light chain sequence	Nucleic acid sequence of 413D08 light chain	GACCTCCAGATGACCCAGTCTCCATCCTCCCTGTCTGCATCTGTAGGA GACAGAGTCACCATCACTTGCCGGGCAAGTCAGGGCATTAGAAATGAT TTAGGCTGGTATCAGCAGAAACAGGGAAAGCCCCAAGCGCCTGATC TATGCTGCATCCAGTTTGCAAAGTGGGGTCCCATCAAGGTTTACGCGGC AGTGGATCTGGGACAGAATTCACCTCTACAATCAGCAGCCTGCAGCCT GAAGATTTTGCAACTTATTACTGTCTACAGCATAATAGTTACCCCTCGG ACGTTCCGCCAAGGGACCAAGGTGGAATCAAACGTACGGTGGCCGCT CCCTCCGTGTTTCATCTCCCACTTCCGACGAGCAGCTGAAGTCCGGC ACCGCTTCTGTCGTGTGCCTGCTGAACAACTTCTACCCCCGCGAGGCC AAGGTGCAAGTGAAGGTGGACAACGCCCTGCAGTCCGGCAACTCCCAG GAATCCGTGACCGAGCAGGACTCCAAGGACAGCACCTACTCCCTGTCC TCCACCCTGACCCTGTCCAAGGCCGACTACGAGAAGCACAAGGTGTAC GCCTGCGAAGTGACCCACAGGGCCTGTCTAGCCCCGTGACCAAGTCT TTCAACCGGGCGAGTGT	

TABLE S1-continued

SEQ ID NOS: 1-342			
SEQ ID NO:	Name	Description	Sequence
152	386H03-CDRH1 (IMGT)	Amino acid sequence of CDRH1 of 386H03 using IMGT	GGSISSSDW
153	386H03-CDRH2 (IMGT)	Amino acid sequence of CDRH2 of 386H03 using IMGT	IFHSGNT
154	386H03-CDRH3 (IMGT)	Amino acid sequence of CDRH3 of 386H03 using IMGT	VRDGSGSY
155	386H03-CDRH1 (Kabat)	Amino acid sequence of CDRH1 of 386H03 using Kabat	SSDWWS
156	386H03-CDRH2 (Kabat)	Amino acid sequence of CDRH2 of 386H03 using Kabat	EIFHSGNTNYPNPSLKS
157	386H03-CDRH3 (Kabat)	Amino acid sequence of CDRH3 of 386H03 using Kabat	DGSGSY
158	386H03-Heavy chain variable region	Amino acid sequence of V_H of 386H03	QVQLQESGPGLVKPSGTLSTCAVSGGSISSSDWWSWVRQPPGKGLEW IGEIPFHSGHTNYPNPSLKSRTVISVDKSKNQISLRNLNSVTAADTAVYYC VRDGSGSYWGQGLVTVSS
159	386H03-Heavy chain variable region	Nucleic acid sequence of V_H of 386H03	CAGGTGCAGCTGCAGGAGTCGGGCCAGGACTGGTGAAGCCTTCGGGG ACCCGTGTCCTCACCTGCGCTGTCTCTGGTGGCTCCATCAGCAGTAGT GACTGGTGGAGTTGGGTCCGCCAGCCCCAGGGAAGGGGCTGGAGTGG ATTGGGGAAATCTTTCATAGTGGGAACCACTACAACCCGTCCCTC AAGAGTCGAGTCACCATATCAGTAGACAAGTCCAAGAACCAGATCTCC CTGAGGCTGAACCTCTGTGACCGCCGCGACACGGCCGTGTATTACTGT GTGAGAGATGGTTCGGGGAGTTACTGGGGCCAGGGAACCTGGTCACC GTCTCCTCAG
160	386H03-full heavy chain sequence	Amino acid sequence of 386H03 heavy chain	QVQLQESGPGLVKPSGTLSTCAVSGGSISSSDWWSWVRQPPGKGLEW IGEIPFHSGHTNYPNPSLKSRTVISVDKSKNQISLRNLNSVTAADTAVYYC VRDGSGSYWGQGLVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLV KDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLG TQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCPPCPAPELLGGPSVFL FPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTK PREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISK AKGQPREPQVYITLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESNGQ PENNYKTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFSQSVMHREALHN HYTQKSLSLSPGK
161	386H03-full heavy chain sequence	Nucleic acid sequence of 386H03 heavy chain	CAGGTGCAGCTGCAGGAGTCGGGCCAGGACTGGTGAAGCCTTCGGGG ACCCGTGTCCTCACCTGCGCTGTCTCTGGTGGCTCCATCAGCAGTAGT GACTGGTGGAGTTGGGTCCGCCAGCCCCAGGGAAGGGGCTGGAGTGG ATTGGGGAAATCTTTCATAGTGGGAACCACTACAACCCGTCCCTC AAGAGTCGAGTCACCATATCAGTAGACAAGTCCAAGAACCAGATCTCC CTGAGGCTGAACCTCTGTGACCGCCGCGACACGGCCGTGTATTACTGT GTGAGAGATGGTTCGGGGAGTTACTGGGGCCAGGGAACCTGGTCACC GTCTCCTCAGCCAGACCAAGGGCCCTCTGTGTTCCCTCTGGCCCT TCCAGCAAGTCCACCTCTGGCGGAACAGCCGCTCTGGGCTGCCTCGTG AAGGACTACTTCCCGAGCCTGTGACCGTGTCTGGAACCTGGCGCT CTGACCAGCGGAGTGACACCTTCCCTGCTGTGCTGCAGTCTCTCGGC CTGTACTCCCTGTCTCCGTGCTGACCGTGCTTCCAGCTCTCTGGG ACCCAGACCTACATCTGCAACGTGAACACACAGCCCTCCAACACCAAG GTGGACAAGAAGGTGGAACCAAGTCTCTGCGACAAGACCAACCTGT CCCCCTTGCTCTGCCCTGAACTGCTGGGCGGACCTTCCGTGTTCTGT TCCCCCCTAAAGCCCAAGGACACCTGATGATCTCCCGGACCCCGAA GTGACCTGCTGGTGGTGGATGTGTCCACGAGGACCTGAAGTGAAG TTCAATTGGTACGTGGACGGCGTGAAGTGCACAACGCCAAGACCAAG CCTAGAGAGGAACAGTACAACCTCCACCTACCGGGTGGTGTCCGTGTG ACCGTGCTGCACAGGATTGGCTGAACGGCAAGAGTACAAGTGAAG GTGTCCAACAAGGCCCTGCCTGCCCTCATCGAAAAGACCATCTCCAAG GCCAAGGGCCAGCCCCGGGAACCCAGGTGTACACACTGCCCTTAGC AGGGACGAGCTGACCAAGAACAGGTGTCCCTGACCTGTCTCGTGAAA

TABLE S1-continued

SEQ ID NOS: 1-342			
SEQ ID NO: Name	Description	Sequence	
		GGCTTCTACCCCTCCGATATCGCCGTGGAATGGGAGTCCAACGGCCAG CCTGAGAACAACTACAAGACCACCCCTGTGCTGGACTCCGACGGC TCATTCTTCTGTACAGCAAGCTGACAGTGGACAAGTCCCGGTGGCAG CAGGGCAACGTGTTCTCTGCTCCGTGATGCACGAGGCCCTGCACAAC CACTACACCCAGAAGTCCCTGTCCCTGAGCCCCGCAAG	
162 386H03- CDRL1 (IMGT)	Amino acid sequence of CDRL1 of 386H03 using IMGT	QSVLYSSNNKNY	
163 386H03- CDRL2 (IMGT)	Amino acid sequence of CDRL2 of 386H03 using IMGT	WAS	
164 386H03- CDRL3 (IMGT)	Amino acid sequence of CDRL3 of 386H03 using IMGT	QQYYSTRS	
165 386H03- CDRL1 (Kabat)	Amino acid sequence of CDRL1 of 386H03 using Kabat	KSSQSVLYSSNNKNYLA	
166 386H03- CDRL2 (Kabat)	Amino acid sequence of CDRL2 of 386H03 using Kabat	WASTRES	
167 386H03- CDRL3 (Kabat)	Amino acid sequence of CDRL3 of 386H03 using Kabat	QQYYSTRS	
168 386H03- Light chain variable region	Amino acid sequence of V_L of 386H03	DIVMTQSPDSLAVSLGERATINCKSSQSVLYSSNNKNYLAWYQQKPGQ PPKLLIYWASTRESGVDRFSGSGSGTDFTLTISLQAEDVAVYYCQQ YYSTRSFGQGTGLEIK	
169 386H03- Light chain variable region	Nucleic acid sequence of V_L of 386H03	GACATCGTGATGACCCAGTCTCCAGACTCCCTGGCTGTGTCTCTGGGC GAGAGGGCCACCATCAACTGCAAGTCCAGCCAGAGTGTCTTATACAGC TCCAACAATAAGAACTACTTAGCTTGGTACCAGCAGAAACAGGACAG CCTCCTAAACTGCTCATTACTGGGCATCTACCCGGGAATCCGGGGTC CCTGACCGATTAGTGGCAGCGGGTCTGGGACAGATTTCACTCTCACC ATCAGCAGCCTGCAGGCTGAAGATGTGGCAGTTTATTACTGTACAGAA TATTATAGTACTCGCAGTTTGGCCAGGGGACCAAGCTGGAGATCAAA C	
170 386H03- full light chain sequence	Amino acid sequence of 386H03 light chain	DIVMTQSPDSLAVSLGERATINCKSSQSVLYSSNNKNYLAWYQQKPGQ PPKLLIYWASTRESGVDRFSGSGSGTDFTLTISLQAEDVAVYYCQQ YYSTRSFGQGTGLEIKRTVAAPSVFIFPPSDEQLKSGTASVCLLNNF YPREAKVQWKVDNALQSGNSQESVTEQDSKDSYLSSTLTLSKADYE KHKVYACEVTHQGLSPVTKSFNRGEC	
171 386H03- full light chain sequence	Nucleic acid sequence of 386H03 light chain	GACATCGTGATGACCCAGTCTCCAGACTCCCTGGCTGTGTCTCTGGGC GAGAGGGCCACCATCAACTGCAAGTCCAGCCAGAGTGTCTTATACAGC TCCAACAATAAGAACTACTTAGCTTGGTACCAGCAGAAACAGGACAG CCTCCTAAACTGCTCATTACTGGGCATCTACCCGGGAATCCGGGGTC CCTGACCGATTAGTGGCAGCGGGTCTGGGACAGATTTCACTCTCACC ATCAGCAGCCTGCAGGCTGAAGATGTGGCAGTTTATTACTGTACAGAA TATTATAGTACTCGCAGTTTGGCCAGGGGACCAAGCTGGAGATCAAA CGTACGGTGGCCGCTCCCTCCGTGTTATCTTCCACCTTCCGACGAG CAGCTGAAGTCCGGCACCGCTTCTGTCTGTGCTGCTGAACAACCTTC TACCCCGCGAGGCCAAGGTGCAAGTGGAAAGTGGACAACGCCCTGCAG TCCGGCAACTCCCAGGAATCCGTGACCGAGCAGGACTCCAAGGACAGC ACCTACTCCCTGTCTCCACCTGACCTGTCCAAGGCCGACTACGAG AAGCACAAGGTGACGCTGCGAAGTGACCCACAGGGCCTGTCTAGC CCCGTGACCAAGTCTTTCAACCGGGCGAGTGT	
172 389A03- CDRH1 (IMGT)	Amino acid sequence of CDRH1 of 389A03 using IMGT	GGSISSSSYY	
173 389A03- CDRH2 (IMGT)	Amino acid sequence of CDRH2 of 389A03 using IMGT	IYSTGYT	

TABLE S1-continued

SEQ ID NOS: 1-342			
SEQ ID NO:	Name	Description	Sequence
174	389A03-CDRH3 (IMGT)	Amino acid sequence of CDRH3 of 389A03 using IMGT	AISTAAGPEYFHR
175	389A03-CDRH1 (Kabat)	Amino acid sequence of CDRH1 of 389A03 using Kabat	SSSYCYG
176	389A03-CDRH2 (Kabat)	Amino acid sequence of CDRH2 of 389A03 using Kabat	SIYSTGYTYYNPSLKS
177	389A03-CDRH3 (Kabat)	Amino acid sequence of CDRH3 of 389A03 using Kabat	STAAGPEYFHR
178	389A03-Heavy chain variable region	Amino acid sequence of V _H of 389A03	QLQESGPGLVKPKSETLSLTCTVSGGSISSSYCGWIRQPPGKGLDWI GSIYSTGYTYYNPSLKSRTVISIDTSKNQFSCLILTSVTAADTAVYYC AISTAAGPEYFHRWGQGLVTVSS
179	389A03-Heavy chain variable region	Nucleic acid sequence of V _H of 389A03	CAGCTGCAGGAGTCGGGCCAGGCCTGGTGAAGCCTTCGGAGACCCCTG TCCCTCACCTGCACGTGTCTCTGGTGGCTCCATCAGCAGTAGTAGTTAT TACTGCGGCTGGATCCGCCAGCCCCCTGGGAAGGGCTGGACTGGATT GGGAGTATCTATTCTACTGGGTACACCTACTACAACCCGTCCTCAAG AGTCGAGTCACCATTTCCATAGACACGTCCAAGAACCAGTTCTCATGC CTGATACTGACCTCTGTGACCGCCGACAGACCGCTGTGTATTACTGT GCGATAAGTACAGCAGCTGGCCCTGAATACTTCCATCGCTGGGGCCAG GGCACCCCTGGTCACCGTCTCCTCAGCCAGCACCAAGGGCCCTCTGTG TTCCCTCTGGCCCCCTCCAGCAAGTCCACCTCTGGCGGAACAGCCGCT CTGGGCTGCCTCGTGAAGGACTACTTCCCCGAGCCTGTGACCGTGTCC TGGAACCTCTGGCGCTCTGACCAGCGGAGTGCACACCTTCCCTGCTGTG CTGCAGTCTCTCGGCTGTACTCCCTGTCTCCGTCTGACCGTGCCT TCCAGCTCTCTGGGCACCCAGACCTACATCTGCAACGTGAACCAAG CCCTCCAACACCAAGGTGGACAAGAAGGTGGAACCAAGTCTTGCAC AAGACCCACACCTGTCCCCCTTGTCTGCCCCCTGAAGTGTGGGCGGA CCTTCCGTGTTCTGTCCCCCAAGGCCAAGGACACCTGATGATC TCCCCGACCCCGAAGTGACCTGCGTGGTGGATGTGTCCACGAG GACCTGAAGTGAAGTTCAATTGGTACGTGGACGGCGTGAAGTGCAC AAGCGCAAGACCAAGCTAGAGAGGAACAGTACAACCTCCACCTACCGG GTGGTGTCCGTGCTGACCGTGTGCACACAGGATTGGCTGAACGGCAAA GAGTACAAGTGCAAGGTGTCCAACAAGGCCCTGCCTGCCCTCATCGAA AAGACCATCTCCAAGGCCAAGGGCCAGCCCCGGGAACCCAGGTGTAC ACACTGCCCCCTAGCAGGGACGAGCTGACCAAGAACCAGGTGTCCCTG ACCTGTCTCGTGAAGGCTTCTACCCCTCCGATATCGCCGTGGAATGG GAGTCCAACGGCCAGCCTGAGAACAATAACAAGACACCCCCCTGTG CTGGACTCCGACGGCTCATTCTTCTGTACAGCAAGCTGACAGTGGAC AAGTCCCGGTGGCAGCAGGGCAACGTGTTCTCTGCTCCGTGATGCAC GAGGCCCTGCACAACCACTACACCCAGAAGTCCCTGTCCCTGAGCCCC GGCAAG
180	389A03-full heavy chain sequence	Amino acid sequence of 389A03 heavy chain	QLQESGPGLVKPKSETLSLTCTVSGGSISSSYCGWIRQPPGKGLDWI GSIYSTGYTYYNPSLKSRTVISIDTSKNQFSCLILTSVTAADTAVYYC AISTAAGPEYFHRWGQGLVTVSSASTKGPSVFPLAPSSKSTSGGTAA LGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVP SSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKHTCTPPCPAPELLGG PSVFLFPPKPKDTLMISRTPEVTCVVDVSHEDPEVKFNWYVDGVEVH NAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIE KTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGFYPSDIAVEW ESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVPSCSVMH EALHNHYTQKSLSLSPGK
181	389A03-full heavy chain sequence	Nucleic acid sequence of 389A03 heavy chain	CAGCTGCAGGAGTCGGGCCAGGCCTGGTGAAGCCTTCGGAGACCCCTG TCCCTCACCTGCACGTGTCTCTGGTGGCTCCATCAGCAGTAGTAGTTAT TACTGCGGCTGGATCCGCCAGCCCCCTGGGAAGGGCTGGACTGGATT GGGAGTATCTATTCTACTGGGTACACCTACTACAACCCGTCCTCAAG AGTCGAGTCACCATTTCCATAGACACGTCCAAGAACCAGTTCTCATGC CTGATACTGACCTCTGTGACCGCCGACAGACCGCTGTGTATTACTGT GCGATAAGTACAGCAGCTGGCCCTGAATACTTCCATCGCTGGGGCCAG GGCACCCCTGGTCACCGTCTCCTCAGCCAGCACCAAGGGCCCTCTGTG TTCCCTCTGGCCCCCTCCAGCAAGTCCACCTCTGGCGGAACAGCCGCT CTGGGCTGCCTCGTGAAGGACTACTTCCCCGAGCCTGTGACCGTGTCC TGGAACCTCTGGCGCTCTGACCAGCGGAGTGCACACCTTCCCTGCTGTG CTGCAGTCTCTCGGCTGTACTCCCTGTCTCCGTCTGACCGTGCCT TCCAGCTCTCTGGGCACCCAGACCTACATCTGCAACGTGAACCAAG CCCTCCAACACCAAGGTGGACAAGAAGGTGGAACCAAGTCTTGCAC AAGACCCACACCTGTCCCCCTTGTCTGCCCCCTGAAGTGTGGGCGGA CCTTCCGTGTTCTGTCCCCCAAGGCCAAGGACACCTGATGATC TCCCCGACCCCGAAGTGACCTGCGTGGTGGATGTGTCCACGAG GACCTGAAGTGAAGTTCAATTGGTACGTGGACGGCGTGAAGTGCAC AAGCGCAAGACCAAGCTAGAGAGGAACAGTACAACCTCCACCTACCGG GTGGTGTCCGTGCTGACCGTGTGCACACAGGATTGGCTGAACGGCAAA GAGTACAAGTGCAAGGTGTCCAACAAGGCCCTGCCTGCCCTCATCGAA AAGACCATCTCCAAGGCCAAGGGCCAGCCCCGGGAACCCAGGTGTAC ACACTGCCCCCTAGCAGGGACGAGCTGACCAAGAACCAGGTGTCCCTG ACCTGTCTCGTGAAGGCTTCTACCCCTCCGATATCGCCGTGGAATGG GAGTCCAACGGCCAGCCTGAGAACAATAACAAGACACCCCCCTGTG CTGGACTCCGACGGCTCATTCTTCTGTACAGCAAGCTGACAGTGGAC AAGTCCCGGTGGCAGCAGGGCAACGTGTTCTCTGCTCCGTGATGCAC GAGGCCCTGCACAACCACTACACCCAGAAGTCCCTGTCCCTGAGCCCC GGCAAG

TABLE S1-continued

SEQ ID NOS: 1-342			
SEQ ID NO:	Name	Description	Sequence
182	389A03-CDRL1 (IMGT)	Amino acid sequence of CDRL1 of 389A03 using IMGT	QSVLYSSNSKNF
183	389A03-CDRL2 (IMGT)	Amino acid sequence of CDRL2 of 389A03 using IMGT	WAS
184	389A03-CDRL3 (IMGT)	Amino acid sequence of CDRL3 of 389A03 using IMGT	QQYYSTPRT
185	389A03-CDRL1 (Kabat)	Amino acid sequence of CDRL1 of 389A03 using Kabat	KSSQSVLYSSNSKNFLA
186	389A03-CDRL2 (Kabat)	Amino acid sequence of CDRL2 of 389A03 using Kabat	WASTRGS
187	389A03-CDRL3 (Kabat)	Amino acid sequence of CDRL3 of 389A03 using Kabat	QQYYSTPRT
188	389A03-Light chain variable region	Amino acid sequence of V_L of 389A03	DIVMTQSPDLSAVSLGERATINCKSSQSVLYSSNSKNFLAWYQQKPGQYYPKLFYIWASTRGSVGPDRISGSGSGTDFNLTISLQAEDVAVYYCQQYYSTPRTFGQGTKVEIK
189	389A03-Light chain variable region	Nucleic acid sequence of V_L of 389A03	GACATCGTGATGACCCAGTCTCCAGACTCCCTGGCTGTGTCTCTGGGC GAGAGGGCCACCATCAACTGCAAGTCCAGCCAGAGTGTCTTATACAG TCCAACAGTAAGAAGTCTTAGCTTGGTACCAGCAGAAACCGGGACAG CCTCTAAGCTGTTTACTTACTGGGCATCTACCCGGGGATCCGGGGT CCGTACCGAATCAGTGGCAGCGGGTCTGGGACAGATTTCAATCTCACC ATCAGCAGCTGCAGGCTGAAGATGTGGCAGTTTATTACTGTCAACAA TATTATAGTACTCCTCGGACGTTTCGCCAAGGGACCAAGGTGGAGATC AAAC
190	389A03-full light chain sequence	Amino acid sequence of 389A03 light chain	DIVMTQSPDLSAVSLGERATINCKSSQSVLYSSNSKNFLAWYQQKPGQYYPKLFYIWASTRGSVGPDRISGSGSGTDFNLTISLQAEDVAVYYCQQYYSTPRTFGQGTKVEIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNN FYPREAKVQWKVDNALQSGNSQESVTEQDSKDSYSLSTLTLSKADY EKHKVYACEVTHQGLSSPVTKSFNRGEC
191	389A03-full light chain sequence	Nucleic acid sequence of 389A03 light chain	GACATCGTGATGACCCAGTCTCCAGACTCCCTGGCTGTGTCTCTGGGC GAGAGGGCCACCATCAACTGCAAGTCCAGCCAGAGTGTCTTATACAG TCCAACAGTAAGAAGTCTTAGCTTGGTACCAGCAGAAACCGGGACAG CCTCTAAGCTGTTTACTTACTGGGCATCTACCCGGGGATCCGGGGT CCGTACCGAATCAGTGGCAGCGGGTCTGGGACAGATTTCAATCTCACC ATCAGCAGCTGCAGGCTGAAGATGTGGCAGTTTATTACTGTCAACAA TATTATAGTACTCCTCGGACGTTTCGCCAAGGGACCAAGGTGGAGATC AAACGTACGGTGGCCGCTCCCTCCGTGTTCTCTCCACCTTCCGAC GAGCAGCTGAAGTCCGGCACCGCTTCTGTCTGTGCCTGCTGAACAAC TTCTACCCCGCGAGGCCAAGGTGCAGTGGAGGTGGACAACGCCCTG CAGTCCGGCACTCCAGGAATCCGTGACCGAGCAGGACTCCAAGGAC AGCACCTACTCCTGTCTCCACCTGACCTGTCCAAGGCCGACTAC GAGAAGCACAAAGGTGTACGCTGCGAAGTGACCCACAGGGCCTGTCT AGCCCCGTGACCAAGTCTTTCAACCGGGCGAGTGT
192	Human IgG4 heavy chain constant region #1	IGHG*01 & IGHG4*04 Heavy Chain Constant Region Nucleotide Sequence	gcttcaccaagggccatccgtcttccccctggcgccctgctccagg agcacctccgagagcacagccgctgggtgctggtcaaggactac tccccgaaccgggtgacgggtgctggtggaactcaggcgccctgaccagc ggcggtgcacaccttcccggtgtcctacagtcctcaggactctactcc ctccagcagcgtggtgacggtgccctccagcagcttgggcacgaagacc tacacctgcaacgtatgacacaagccagcaacaccaaggtggacaag agagttgagtcacaaatagggtcccccatgcccatcatgccagcacct gagttcctggggggaccatcagttcttctgttcccccaaaaaccaag gacactctcatgatctccggacccctgaggtcaggtgcgtggtggtg gacgtgagccagggaagaccccgaggtccagttcaactggtacgtggat ggcggtgaggtgcataatgccaagacaagccgcgaggagagcagttc aacagcacgtaccgtgtggtcagcgtcctcacgctcctgacccaggac tgggtgaacggcaaggagtacaagtgcaaggtctccaacaaaggcctc

TABLE S1-continued

SEQ ID NOS: 1-342			
SEQ ID NO: Name	Description	Sequence	
193	Heavy Chain Constant Region Amino Acid Sequence	ccgtctccatcgagaaaaccatctccaaagccaaagggcagccccga gagccacaggtgtacacctgcccccatcccaggaggagatgaccaag aaccaggtcagcctgacctgctggtcaaaggcttctaccccagcgac atcgccgtggagtgggagagcaatgggcagccggagagaacaactacaag accacgcctcccgtgctggactccgacggctccttctctctacagc aggtcaaccgtggacaagagcaggtggcaggagggaatgtcttctca tgctccgtgatgcatagaggctctgcacaaccactacacacagaagagc ctctccctgtctctgggtaaa ASTKGPSVFPLAPCSRSTSESTAALGCLVKDYFPEPVTVSWNSGALTS GVHTFPAVLQSSGLYSLSSVTVPSSSLGKTYTCNVDHKPSNTKVDK RVESKYGPPCPSCPAPFLGGPSVFLPPPKDITLMI SRTPEVTCVVV DVSQEDPEVQFNWYVDGVEVHNAKTKPREEQFNSTYRVVSVLTVLHQD WLNGKEYKCKVSNKGLPSSIEKTI SKAKGQPREPQVYTLPPSQEEMTK NQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLYS RLTVDKSRWQEGNVFSCSVMEALHNHYTQKSLSLSLGK	
194 Human IgG4 heavy chain constant region #2	IGHG*02 Heavy Chain Constant Region Nucleotide Sequence	gcttcaccaagggcccatccgtcttccccctggcgccctgctccagg agcacctccgagagcacagccgcccctgggtgctcaggactac tccccgaaccggtagcgggtgctggaactcaggcgccctgaccagc ggcgtgcacaccttcccggctgtctacagtcctcaggactctactcc ctccagcagcgtggtagcctgcccctccagcagcttgggcacgaagacc tacacctgcaacgtatgcataagcccagcaacaccaaggtggacaag agagttgagtcctcaaatatgggtccccctgcccctcatgcccagcacct gaggctcctggggggaccatcagttcttctgttccccccaaaaccgaag gacactctcatgatctccggaccctcagggtcagctgctggtggtg gacgtgagccaggaagaccccagaggtccagttcaactggtagctggat ggcgtggaggtgcataatgccaaagacaagccgcccggaggagcagttc aacagcacgtaccgtgtgggtcagcgtcctcaccgtcgtgcaccagga tgggtgaacggcaaggagtagcaagtgcaaggtctccaacaaaggcctc ccgtcctccatcgagaaaaccatctccaaagccaaagggcagccccga gagccacaggtgtacacctgcccccatcccaggaggagatgaccaag aaccaggtcagcctgacctgctggtcaaaggcttctaccccagcgac atcgccgtggagtgggagagcaatgggcagccggagagaacaactacaag accacgcctcccgtgctggactccgacggctccttctctctacagc aggtcaaccgtggacaagagcaggtggcaggagggaatgtcttctca tgctccgtgatgcatagaggctctgcacaaccactacacgcagaagagc ctctccctgtctctgggtaaa ASTKGPSVFPLAPCSRSTSESTAALGCLVKDYFPEPVTVSWNSGALTS GVHTFPAVLQSSGLYSLSSVTVPSSSLGKTYTCNVDHKPSNTKVDK RVESKYGPPCPSCPAPFLGGPSVFLPPPKDITLMI SRTPEVTCVVV DVSQEDPEVQFNWYVDGVEVHNAKTKPREEQFNSTYRVVSVLTVVHQD WLNGKEYKCKVSNKGLPSSIEKTI SKAKGQPREPQVYTLPPSQEEMTK NQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLYS RLTVDKSRWQEGNVFSCSVMEALHNHYTQKSLSLSLGK	
195	Heavy Chain Constant Region Amino Acid Sequence	gcttcaccaagggcccatccgtcttccccctggcgccctgctccagg agcacctccgagagcacagccgcccctgggtgctcaggactac tccccgaaccggtagcgggtgctggaactcaggcgccctgaccagc ggcgtgcacaccttcccggctgtctacagtcctcaggactctactcc ctccagcagcgtggtagcctgcccctccagcagcttgggcacgaagacc tacacctgcaacgtatgcataagcccagcaacaccaaggtggacaag agagttgagtcctcaaatatgggtccccctgcccctcatgcccagcacct gaggctcctggggggaccatcagttcttctgttccccccaaaaccgaag gacactctcatgatctccggaccctcagggtcagctgctggtggtg gacgtgagccaggaagaccccagaggtccagttcaactggtagctggat ggcgtggaggtgcataatgccaaagacaagccgcccggaggagcagttc aacagcacgtaccgtgtgggtcagcgtcctcaccgtcctgcaccaggac tggctgaacggcaaggagtagcaagtgcaaggtctccaacaaaggcctc ccgtcctccatcgagaaaaccatctccaaagccaaagggcagccccga gagccacaggtgtacacctgcccccatcccaggaggagatgaccaag aaccaggtcagcctgacctgctggtcaaaggcttctaccccagcgac atcgccgtggagtgggagagcaatgggcagccggagagaacaactacaag accacgcctcccgtgctggactccgacggctccttctctctacagc aggtcaaccgtggacaagagcaggtggcaggagggaatgtcttctca tgctccgtgatgcatagaggctctgcacaaccactacacgcagaagagc ctctccctgtctctgggtaaa ASTKGPSVFPLAPCSRSTSESTAALGCLVKDYFPEPVTVSWNSGALTS GVHTFPAVLQSSGLYSLSSVTVPSSSLGKTYTCNVDHKPSNTKVDK RVESKYGPPCPSCPAPFLGGPSVFLPPPKDITLMI SRTPEVTCVVV DVSQEDPEVQFNWYVDGVEVHNAKTKPREEQFNSTYRVVSVLTVVHQD WLNGKEYKCKVSNKGLPSSIEKTI SKAKGQPREPQVYTLPPSQEEMTK NQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLYS RLTVDKSRWQEGNVFSCSVMEALHNHYTQKSLSLSLGK	
196 Human IgG4 heavy chain constant region #3	IGHG*03 Heavy Chain Constant Region Nucleotide Sequence	gcttcaccaagggcccatccgtcttccccctggcgccctgctccagg agcacctccgagagcacagccgcccctgggtgctcaggactac tccccgaaccggtagcgggtgctggaactcaggcgccctgaccagc ggcgtgcacaccttcccggctgtctacagtcctcaggactctactcc ctccagcagcgtggtagcctgcccctccagcagcttgggcacgaagacc tacacctgcaacgtatgcataagcccagcaacaccaaggtggacaag agagttgagtcctcaaatatgggtccccctgcccctcatgcccagcacct gaggctcctggggggaccatcagttcttctgttccccccaaaaccgaag gacactctcatgatctccggaccctcagggtcagctgctggtggtg gacgtgagccaggaagaccccagaggtccagttcaactggtagctggat ggcgtggaggtgcataatgccaaagacaagccgcccggaggagcagttc aacagcacgtaccgtgtgggtcagcgtcctcaccgtcctgcaccaggac tggctgaacggcaaggagtagcaagtgcaaggtctccaacaaaggcctc ccgtcctccatcgagaaaaccatctccaaagccaaagggcagccccga gagccacaggtgtacacctgcccccatcccaggaggagatgaccaag aaccaggtcagcctgacctgctggtcaaaggcttctaccccagcgac atcgccgtggagtgggagagcaatgggcagccggagagaacaactacaag accacgcctcccgtgctggactccgacggctccttctctctacagc aagctcaccgtggacaagagcaggtggcaggagggaacgtcttctca tgctccgtgatgcatagaggctctgcacaaccactacacgcagaagagc ctctccctgtctctgggtaaa ASTKGPSVFPLAPCSRSTSESTAALGCLVKDYFPEPVTVSWNSGALTS GVHTFPAVLQSSGLYSLSSVTVPSSSLGKTYTCNVDHKPSNTKVDK RVESKYGPPCPSCPAPFLGGPSVFLPPPKDITLMI SRTPEVTCVVV DVSQEDPEVQFNWYVDGVEVHNAKTKPREEQFNSTYRVVSVLTVLHQD WLNGKEYKCKVSNKGLPSSIEKTI SKAKGQPREPQVYTLPPSQEEMTK NQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLYS KLTVDKSRWQEGNVFSCSVMEALHNHYTQKSLSLSLGK	
197	Heavy Chain Constant Region Amino Acid Sequence	gcttcaccaagggcccatccgtcttccccctggcgccctgctccagg agcacctccgagagcacagccgcccctgggtgctcaggactac tccccgaaccggtagcgggtgctggaactcaggcgccctgaccagc ggcgtgcacaccttcccggctgtctacagtcctcaggactctactcc ctccagcagcgtggtagcctgcccctccagcagcttgggcacgaagacc tacacctgcaacgtatgcataagcccagcaacaccaaggtggacaag agagttgagtcctcaaatatgggtccccctgcccctcatgcccagcacct gaggctcctggggggaccatcagttcttctgttccccccaaaaccgaag gacactctcatgatctccggaccctcagggtcagctgctggtggtg gacgtgagccaggaagaccccagaggtccagttcaactggtagctggat ggcgtggaggtgcataatgccaaagacaagccgcccggaggagcagttc aacagcacgtaccgtgtgggtcagcgtcctcaccgtcctgcaccaggac tggctgaacggcaaggagtagcaagtgcaaggtctccaacaaaggcctc ccgtcctccatcgagaaaaccatctccaaagccaaagggcagccccga gagccacaggtgtacacctgcccccatcccaggaggagatgaccaag aaccaggtcagcctgacctgctggtcaaaggcttctaccccagcgac atcgccgtggagtgggagagcaatgggcagccggagagaacaactacaag accacgcctcccgtgctggactccgacggctccttctctctacagc aagctcaccgtggacaagagcaggtggcaggagggaacgtcttctca tgctccgtgatgcatagaggctctgcacaaccactacacgcagaagagc ctctccctgtctctgggtaaa ASTKGPSVFPLAPCSRSTSESTAALGCLVKDYFPEPVTVSWNSGALTS GVHTFPAVLQSSGLYSLSSVTVPSSSLGKTYTCNVDHKPSNTKVDK RVESKYGPPCPSCPAPFLGGPSVFLPPPKDITLMI SRTPEVTCVVV DVSQEDPEVQFNWYVDGVEVHNAKTKPREEQFNSTYRVVSVLTVLHQD WLNGKEYKCKVSNKGLPSSIEKTI SKAKGQPREPQVYTLPPSQEEMTK NQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLYS KLTVDKSRWQEGNVFSCSVMEALHNHYTQKSLSLSLGK	

TABLE S1-continued

SEQ ID NOS: 1-342			
SEQ ID NO:	Name	Description	Sequence
198	IgG4 heavy chain constant region-IgG4-PE	Heavy Chain Constant Region Nucleotide Sequence-Synthetic Version A	gctccaccacaggcccatccgtcttccccctggcgccctgctccagg agcacctccgagagcacggccgcccctgggtgcctggtaaggactac tccccgaaccagtgacgggtgctgctggaactcaggcgccctgaccagc ggcggtgcacaccttccggctgtcctacagtcctcaggactctactcc ctccagcagcgtggtgaccgtgccctccagcagcttgggcacgaagacc tacacctgcaacgtagatcacaagcccagcaacaccaaggtggacaag agagttgagtgccaaatatgggtcccccatgccaccatgccagcgccct gaatttgaggggggaccatcagttctcctgttcccccaaaacccaag gacactctcatgatctcccgaccctcagggtcagctgcgtgggtgggtg gacgtgagccaggaagaccccgaggtccagttcaactggtacgtggat ggcggtgaggtgcataatgccaagacaaagccgaggaggagcagttc aacagcagctaccgtgtgggtcagcgtcctcaccgtcctgaccaggac tgggtgaacggcaaggagtacaagtgaaggtctccaacaaggcctc ccgtcatcgatcgagaaaaccatctccaagccaaagggcagccccga gagccacaggtgtacacctgcccccatcccaggaggagatgaccaag aaccaggtcagcctgacctgcctgggtcaaggccttctacccagcgac atcgccgtggagtgggagagcaatgggcagccggagaacaactacaag accacgctcccgctgctggactccgacggatccttcttctctacagc aggtgaacctggacaagagcaggtggcaggagggaatgtcttctca tgctccgtgatgcagaggtctgcacacccactacacacagaagagc ctctccctgtctctgggtaaa ASTKGPSVFPLAPCSRSTSESTAALGCLVKDYFPEPVTVSWNSGALTS GVHTFPAVLQSSGLYSLSSVTVPSSSLGKTYTCNVDHKPSNTKVDK RVESKYGPPCPCPAPEFEGGSPVFLPPPKPKDTLMISRTPEVTCVVV DVSQEDPEVQFNWYVDGVEVHNAKTKPREEQFNSTYRVVSVLTVHQD WLNGKEYKCKVSNKGLPSSIEKTIISKAKGQPREPQVYTLPPSQEEMTK NQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTPPVLDSDGSFFLYS RLTVDKSRWQEGRVFSQSVMEALHNHYTQKSLSLSLGK
199	IgG4 heavy chain constant region-IgG4-PE	Heavy Chain Constant Region Amino Acid Sequence-Encoded by Synthetic Version A, B & C(Two residues that differ from the wild-type sequence are identified in bold)	Gcctccaccacaggacactagcgtgttccctctcgccccctgttccagg tcacacagcgagtcacccgctgccctcggtgtctggtgaaagactac ttccccgagcccgtagcgtctcctggaatagcggagccctgacctcc ggcggtgcacacatttcccgccgtgctgcagagcagcggactgtatagc ctgagcagcgtggtgaccgtgccagctccagcctcgccaccacaaacc tacacctgcaacgtggaccacaagccctccaacaccaaggtggacaag cgggtggagagcaagtagcggcccccttgccctccttgctcctgccct gagttcgaggaggaccctccgtgttctgtttcccccaaaacccaag gacaccctgatgatctcccgacaccccgaggtgacctgtgtgggtcggtg gacgtcagccagggaggaccccgaggtgcagttcaactggtatgtggac ggctggaggtgcacaatgcaaaaccaaagccagggaggagcagttc aattccacctacagggtggtgagcgtgctgaccgtcctgcatcaggat tgggtgaacggcaaggagtacaagtgaaggtgtccaacaaggagctg ccagctccatcgagaagaccatcagcaaggctaaaggccagccgagg gagccccaggtgtatacctgcctcctagccaggaagagatgaccaag aaccaagtgtccctgacctgcctgggtgaagggtattctacctccgac atcgccgtggagtgggagagcaatggccagcccgagaacaactacaaa acaacccctcccgctgctgatagcgagcggcagcttcttctctacagc cggctgacagtggaacaagagcaggtggcaggagggaacgtgttctcc tgttccgtgatgcacgagggccctgcacaatcactacaccagaagagc ctctccctgtccctgggcaag
200	IgG4 heavy chain constant region-IgG4-PE	Heavy Chain Constant Region Nucleotide Sequence-Synthetic Version B	Gcctccaccacaggacactagcgtgttccctctcgccccctgttccagg tcacacagcgagtcacccgctgccctcggtgtctggtgaaagactac ttccccgagcccgtagcgtctcctggaatagcggagccctgacctcc ggcggtgcacacatttcccgccgtgctgcagagcagcggactgtatagc ctgagcagcgtggtgaccgtgccagctccagcctcgccaccacaaacc tacacctgcaacgtggaccacaagccctccaacaccaaggtggacaag cgggtggagagcaagtagcggcccccttgccctccttgctcctgccct gagttcgaggaggaccctccgtgttctgtttcccccaaaacccaag gacaccctgatgatctcccgacaccccgaggtgacctgtgtgggtcggtg gacgtcagccagggaggaccccgaggtgcagttcaactggtatgtggac ggctggaggtgcacaatgcaaaaccaaagccagggaggagcagttc aattccacctacagggtggtgagcgtgctgaccgtcctgcatcaggat tgggtgaacggcaaggagtacaagtgaaggtgtccaacaaggagctg ccagctccatcgagaagaccatcagcaaggctaaaggccagccgagg gagccccaggtgtatacctgcctcctagccaggaagagatgaccaag aaccaagtgtccctgacctgcctgggtgaagggtattctacctccgac atcgccgtggagtgggagagcaatggccagcccgagaacaactacaaa acaacccctcccgctgctgatagcgagcggcagcttcttctctacagc cggctgacagtggaacaagagcaggtggcaggagggaacgtgttctcc tgttccgtgatgcacgagggccctgcacaatcactacaccagaagagc ctctccctgtccctgggcaag
201	IgG4 heavy chain constant region-IgG4-PE	Heavy Chain Constant Region Nucleotide Sequence-Synthetic Version C	gccagcaccacaggcccttccgtgttccccctggcccccttgacagcagg agcacctccgaatccacagctgcctgggtgtctggtgaaggactac ttccccgagcccgtagcgtgagctggaacagcggcgctctgacatcc ggcggtccacaccttctcctgcctcctgcagtcctccggcctctactcc ctgtcctccgtggtgaccgtgcctagctcctccctcggcaccacgaagc tacacctgtaacgtggaccacaaacccctccaacaccaaggtggacaaa cgggtcgagagcaagtagcggccctccctgccctccttgtcctgcccc gagttcgaaggcggacccagcgtgttctgttccctcctaaagcccaag gacacctcatgatcagccggacaccccgaggtgacctgcgtgggtgggtg gatgtgagccaggaggacccctgaggtccagttcaactggtatgtggat ggctggaggtgcacaacgccaagacaaagcccggaagagcagttc aactccacctacagggtggtcagcgtgctgacctgctgcatcaggac tgggtgaacggcaaggagtacaagtgaaggtcagcaatgaaggactg

TABLE S1-continued

SEQ ID NOS: 1-342			
SEQ ID NO:	Name	Description	Sequence
			cccagcagcatcgagaagaccatctccaaggctaaaggccagccccggaacctcaggtgtacacctgcctcccagccaggaggagatgaccaagaaccaggtgagcctgacctgcctggtgaagggtattctaccttccgacatcgccgtggagtgagggtccaacggccagcccgagaacaattataagaccacccctcccgtcctcgacagcgacggatccttcttctgtactccaggctgaccgtggataagtcagggtggcagggaaggcaacgtgttcagctgctccgtgatgcacgagggccctgcacaatcactacaccagaagtccctgagcctgtccctgggaaag
202	IgG4 heavy chain constant region	Heavy Chain Constant Region Nucleotide Sequence-Synthetic Version D	gctctccaccaagggcccatccgtcttccccctggcgccctgctccaggagcacctccgagagcacggccgcccctgggtgcctggccaaggactacttccccgaaccagtgacgggtgtcgtggaactcaggcgccctgaccagcggcgtgcacaccttcccggtgtcctacagtcctcaggactctactccctcagcagcgtggtgacctgcctccagcagcttgggcacgaagaccacacctgcaacgtagatcacaagcccagcaacaccaagggtggacaagaagttgagtcctcaaatatgggtccccatgccaccatgccagcgctccagttgccccggggaccatcagttctcctgttccccccaaaacccaaggacactctcatgatctccggaccctgaggtcacgtgcgtgggtgggtgacgtgagccaggaagaccccaggtccagttcaactggtacgtggatggcgtggaggtgcataatgccaagacaaagccgaggaggagcagttcaacagcagctaccgtgtgggtcagcgtcctcaccgtcctgcaccaggagtggtggaacggcaaggagtacaagtgcaaggtctccaacaaggcctccgtcatcgatcgagaaaaccatctccaagccaaagggcagccccgagagccacaggtgtacacctgcccccatcccaggaggagatgaccaagaaccaggtcagcctgacctgcctgggtcaaggcttctaccccagcgacatcgccgtggagtgaggagcaatgggcagccggagagaacaactacaagaccagcctcccgtgctggactccgacggatccttcttctctacagcaggctaaccgtggacaagagcaggtggcaggagggaatgtcttctcagtctccgtgatgcatgaggtctgacacaaccactacacacagaagagcctctccctgtctctgggtaaa
203		Heavy Chain Constant Region Amino Acid Sequence-encoded by Synthetic Version D	ASTKGPSVFPLAPCSRSTSESTAALGCLVKDYFPEPVTVSWNSGALTS GVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDK RVESKYGPPCPPCPAPPVAGGPSVFLFPPKPKDTLMISRTPEVTCVVV DVSQEDPEVQFNWYVDGVEVHNAKTKPREEQFNSTYRVVSVLTVLHQD WLNGKEYKCKVSNKGLPSSIEKTIISKAKGQPREPQVYTLPPSQEEMTK NQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTPPVLDSDGSFFLYS RLTVDKSRWQEGNVFSCSVMEALHNHYTQKSLSLSPGK
204	Disabled Human IgG1 heavy chain constant region	Disabled IGHG1 Heavy Chain Constant Region Nucleotide Sequence	gctctccaccaagggcccatccgtcttccccctggcacccctcctccaaggacacctctggggggcacagcggccctgggtgcctggccaaggactacttccccgaaccgggtgacgggtgtcgtggaactcaggcgccctgaccagcggcgtgcacaccttcccggtgtcctacagtcctcaggactctactccctcagcagcgtggtgacctgcctccagcagcttgggcacccagaccacatctgcaacgtgaatcacaagcccagcaacaccaagggtggacaagaagtgaggcccaaatcttgtgacaaaactcacacatgccaccgtgcacagcactgaactcgccggggcacccgtcagttctcctcttcccccca aaacccaaggacacctcatgatctccgggacccctgaggtcacatgctgggtgggtggacgtgagccacgaagacctgaggtcaagttcaactgtacgtggacggcgtggaggtgcataatgccaagacaaagccgaggagagagcagtagcaacagcagctaccgtgtgggtcagcgtcctcaccgtcctgcaccaggactggctgaatggcaaggagtacaagtgcaaggtctccaacaaagccctcccagcccccatcgagaaaaccatctccaagccaaagggcagccccgagaaccacaggtgtacacctgcccccatcccggatgagctgaccaagaaccaggtcagcctgacctgcctgggtcaaggcttctatcccagcgacatcgccgtggagtgaggagcaatgggcagccggagagaacaaactacaagaccacgcctcccgtgctggactccgacggctccttcttctctacagcaagctcaccgtggacaagagcaggtggcagcaggggaagctcttctcatgctccgtgatgcatgaggtctgacacaaccactacacgcagaagagcctctccctgtctccgggtaaa
205		Heavy Chain Constant Region Amino Acid Sequence (Two residues that differ from the wild-type sequence are identified in bold)	ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTS GVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDK KVEPKSCDKTHTCPPCPAPELAGAPSVFLFPPKPKDTLMISRTPEVTC VVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQFNSTYRVVSVLTVL HQDWLNGKEYKCKVSNKALPAPIEKTIISKAKGQPREPQVYTLPPSRDE LTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTPPVLDSDGSFF LYSKLTVDKSRWQGNVFSVMEALHNHYTQKSLSLSPGK

TABLE S1-continued

SEQ ID NOS: 1-342			
SEQ ID NO:	Name	Description	Sequence
206	Human Ck constant region	IGKC*01 Ck Light Chain Constant Region Nucleotide Sequence	cgtacgggtggcgcgtccctccgtgttcattcttcccaccttccgacgag cagctgaagtccggcaccgcttctgtcgtgtgcctgctgaacaacttc tccccccgcgaggccaaggtgcagtggaaggtggacaacgcctgcag tccggcaactcccaggaatccgtgaccgagcaggactccaaggacagc acctactccctgtcctccacctgacctgtccaaggccgactacgag aagcacaaggtgtacgcctgcgaagtgaccaccagggcctgtctagc cccgtagaccaagtctttcaaccggggcgagtg
207		Ck Light Chain Constant Region Amino Acid Sequence	RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQ SGNSQESVTEQDSKSDSTYLSSTLTLSKADYEKHKVYACEVTHQGLSS PVTKSFNRGEC
208	Human Ck constant region	IGKC*02 Ck Light Chain Constant Region Nucleotide Sequence	cgaactgtggctgcaccatctgtcttcatcttcccgccatctgatgag cagttgaaatctggaactgcctctgttgtgtgcctgctgaataacttc tatcccagagaggccaaggtacagtggaaggtggataacgcctccaa tcgggtaactcccaggagagtggtcacagagcaggagagcaaggacagc acctacagcctcagcagcaccctgacgtgagcaagcagactacgag aaacacaaagtctacgcctgcgaagtaccccatcagggcctgagctcg cccgtcacaaagagcttcaacaggggagagtg
209		Ck Light Chain Constant Region Amino Acid Sequence	RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQ SGNSQESVTEQESKSDSTYLSSTLTLSKADYEKHKVYAGEVTHQGLSS PVTKSFNRGEC
210	Human Ck constant region	IGKC*03 Ck Light Chain Constant Region Nucleotide Sequence	cgaactgtggctgcaccatctgtcttcatcttcccgccatctgatgag cagttgaaatctggaactgcctctgttgtgtgcctgctgaataacttc tatcccagagaggccaaggtacagcggaaggtggataacgcctccaa tcgggtaactcccaggagagtggtcacagagcaggagagcaaggacagc acctacagcctcagcagcaccctgacgtgagcaagcagactacgag aaacacaaagtctacgcctgcgaagtaccccatcagggcctgagctcg cccgtcacaaagagcttcaacaggggagagtg
211		Ck Light Chain Constant Region Amino Acid Sequence	RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQ SGNSQESVTEQESKSDSTYLSSTLTLSKADYEKHKVYACEVTHQGLSS PVTKSFNRGEC
212	Human Ck constant region	IGKC*04 Ck Light Chain Constant Region Nucleotide Sequence	cgaactgtggctgcaccatctgtcttcatcttcccgccatctgatgag cagttgaaatctggaactgcctctgttgtgtgcctgctgaataacttc tatcccagagaggccaaggtacagtggaaggtggataacgcctccaa tcgggtaactcccaggagagtggtcacagagcaggagagcaaggacagc acctacagcctcagcagcaccctgacgtgagcaagcagactacgag aaacacaaactctacgcctgcgaagtaccccatcagggcctgagctcg cccgtcacaaagagcttcaacaggggagagtg
213		Ck Light Chain Constant Region Amino Acid Sequence	RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQ SGNSQESVTEQDSKSDSTYLSSTLTLSKADYEKHKLYACEVTHQGLSS PVTKSFNRGEC
214	Human Ck constant region	IGKC*05 Ck Light Chain Constant Region Nucleotide Sequence	cgaactgtggctgcaccatctgtcttcatcttcccgccatctgatgag cagttgaaatctggaactgcctctgttgtgtgcctgctgaataacttc tatcccagagaggccaaggtacagtggaaggtggataacgcctccaa tcgggtaactcccaggagagtggtcacagagcaggagagcaaggacagc acctacagcctcagcaacaccctgacgtgagcaagcagactacgag aaacacaaagtctacgcctgcgaagtaccccatcagggcctgagctcg cccgtcacaaagagcttcaacaggggagagtg
215		Ck Light Chain Constant Region Amino Acid Sequence	RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQ SGNSQESVTEQDSKSDSTYLSNTLTLSKADYEKHKVYACEVTHQGLSS PVTKSFNRGEC

TABLE S1-continued

SEQ ID NOS: 1-342				
SEQ ID NO:	Name	Description	Sequence	
216	Human Cλ constant region	IGCλ1*01	Cλ Light Chain Constant Region Nucleotide Sequence	
217			Cλ Light Chain Constant Region Amino Acid Sequence	
218	Human Cλ constant region	IGCλ1*02	Cλ Light Chain Constant Region Nucleotide Sequence	
219			Cλ Light Chain Constant Region Amino Acid Sequence	
220	Human Cλ constant region	IGCλ2*01	Cλ Light Chain Constant Region Nucleotide Sequence- Version A	
221			Cλ Light Chain Constant Region Nucleotide Sequence- Version B	
222			Cλ Light Chain Constant Region Nucleotide Sequence- Version C	
223			Cλ Light Chain Constant Region Amino Acid Sequence- Encoded by Version A, B & C	
224	Human Cλ constant region	IGCλ2*02 & IGLC2*03	Cλ Light Chain Constant Region Nucleotide Sequence	
225			Cλ Light Chain Constant Region Amino Acid Sequence	

TABLE S1-continued

SEQ ID NOS: 1-342				
SEQ ID NO:	Name	Description	Sequence	
226	Human Cλ constant region	IGCλ3*01 Cλ Light Chain Constant Region Nucleotide Sequence	cccaaggctgccccctcggtcactctgttcccaccctcctctgaggag cttcaagccaacaagggcacactgggtgtgtctcataagtgaacttctac ccgggagccgtgacagttgcctggaaggcagatagcagccccgtcaag gcgggggtggagaccaccacaccctccaaacaaagcaacaacaagtac gcggccagcagctacctgagcctgacgcctgagcagtggaagtccac aaaagctacagctgccaggtcacgcatgaaggggagcaccgtggagaag acagttgccccctacggaatgttca	
227		Cλ Light Chain Constant Region Amino Acid Sequence	PKAAPSVTLFPPSSEELQANKATLVCLISDFYPGAVTVAWKADSSPVK AGVETTTTPSKQSNKNKYAASSYLSLTPEQWKSHKSYSCQVTHEGSTVEK TVAPTECS	
228	Human Cλ constant region	IGCλ3*02 Cλ Light Chain Constant Region Nucleotide Sequence	ggtcagcccaaggctgccccctcggtcactctgttcccaccctcctct gaggagcttcaagccaacaagggcacactgggtgtgtctcataagtga ttctacccgggggagcagtgacagttgcctggaaggcagatagcagcccc gtcaaggcgggggtggagaccaccacaccctccaaacaaagcaacaac aagtacgcggccagcagctacctgagcctgacgcctgagcagtggaag tcccacaaaagctacagctgccaggtcacgcatgaaggggagcaccgtg gagaagacagtgggccccctacggaatgttca	
229		Cλ Light Chain Constant Region Amino Acid Sequence	GQPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGPVTVAWKADSSP VKAGVETTTTPSKQSNKNKYAASSYLSLTPEQWKSHKSYSCQVTHEGSTV EKTVPATECS	
230	Human Cλ constant region	IGCλ3*03 Cλ Light Chain Constant Region Nucleotide Sequence	ggtcagcccaaggctgccccctcggtcactctgttcccaccctcctct gaggagcttcaagccaacaagggcacactgggtgtgtctcataagtga ttctacccgggggagccgtgacagttggcctggaaggcagatagcagcccc gtcaaggcggggagtgagaccaccacaccctccaaacaaagcaacaac aagtacgcggccagcagctacctgagcctgacgcctgagcagtggaag tcccacaaaagctacagctgccaggtcacgcatgaaggggagcaccgtg gagaagacagtgggccccctacagaatgttca	
231		Cλ Light Chain Constant Region Amino Acid Sequence	GQPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGAVTVAWKADSSP VKAGVETTTTPSKQSNKNKYAASSYLSLTPEQWKSHKSYSCQVTHEGSTV EKTVPATECS	
232	Human Cλ constant region	IGCλ3*04 Cλ Light Chain Constant Region Nucleotide Sequence	ggtcagcccaaggctgccccctcggtcactctgttcccgcctcctct gaggagcttcaagccaacaagggcacactgggtgtgtctcataagtga ttctacccggggagccgtgacagttggcctggaaggcagatagcagcccc gtcaaggcggggagtgagaccaccacaccctccaaacaaagcaacaac aagtacgcggccagcagctacctgagcctgacgcctgagcagtggaag tcccacagaagctacagctgccaggtcacgcatgaaggggagcaccgtg gagaagacagtgggccccctacagaatgttca	
233		Cλ Light Chain Constant Region Amino Acid Sequence	GQPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGAVTVAWKADSSP VKAGVETTTTPSKQSNKNKYAASSYLSLTPEQWKSHRSYSCQVTHEGSTV EKTVPATECS	
234	Human Cλ constant region	IGCλ6*01 Cλ Light Chain Constant Region Nucleotide Sequence	ggtcagcccaaggctgccccctcggtcactctgttcccgcctcctct gaggagcttcaagccaacaagggcacactgggtgtgtcctgatcagtgac ttctacccggggagcgtgtgaaagtggcctggaaggcagatggcagcccc gtcaacacggggagtgagaccaccacaccctccaaacagagcaacaac aagtacgcggccagcagctacctgagcctgacgcctgagcagtggaag tcccacagaagctacagctgccaggtcacgcatgaaggggagcaccgtg gagaagacagtgggccccctgcagaatgttca	
235		Cλ Light Chain Constant Region Amino Acid Sequence	GQPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGAVKVAWKADGSP VNTGVETTTTPSKQSNKNKYAASSYLSLTPEQWKSHRSYSCQVTHEGSTV EKTVPABECS	
236	Human Cλ constant region	IGLC7*01 & IGCλ7*02 Cλ Light Chain Constant Region	ggtcagcccaaggctgccccctcggtcactctgttcccaccctcctct gaggagcttcaagccaacaagggcacactgggtgtgtctcgtaaagtga ttctacccggggagccgtgacagttggcctggaaggcagatggcagcccc gtcaagggtggagtgagaccaccaaacccctccaaacaaagcaacaac	

TABLE S1-continued

SEQ ID NOS: 1-342			
SEQ ID NO:	Name	Description	Sequence
237		Nucleotide Sequence Cλ Light Chain Constant Region Amino Acid Sequence	aagtatgcgccagcagctacctgagcctgacgcccagcagtggaag tcccacagaagctacagctgccgggtcacgcatgaaggagcaccgtg gagaagacagtgcccctgcagaatgctct GQPKAAPSVTLFPPSSEELQANKATLVCLVSDFYPGA VTVAWKADGSP VKVGVETTKPSKQSNKYAASSYLSLTPEQWKSHRSYSCRVTHEGSTV EKTVPAPACS
238	413G05-CDRH1 (IMGT)	Amino acid sequence of CDRH1 of 413G05 using IMGT	GFTFSDYY
239	413G05-CDRH2 (IMGT)	Amino acid sequence of CDRH2 of 413G05 using IMGT	ISTSGSTI
240	413G05-CDRH3 (IMGT)	Amino acid sequence of CDRH3 of 413G05 using IMGT	ARGITGTFNYHYGLGV
241	413G05-CDRH1 (Kabat)	Amino acid sequence of CDRH1 of 413G05 using Kabat	DYYMS
242	413G05-CDRH2 (Kabat)	Amino acid sequence of CDRH2 of 413G05 using Kabat	YISTSGSTIYYADSVKG
243	413G05-CDRH3 (Kabat)	Amino acid sequence of CDRH3 of 413G05 using Kabat	GITGTFNYHYGLGV
244	413G05-Heavy chain variable region	Amino acid sequence of V _H of 413G05	QVQLVESGGGLVKPGGSLRLSCAASGFTFSDYYMSWIRQVPGKGLEWV SYISTSGSTIYYADSVKGRFTISRDNAKNSLYLQMNSLRAEDAAVYHC ARGITGTFNYHYGLGVWGQTTVTSS
245	413G05-Heavy chain variable region	Nucleic acid sequence of V _H of 413G05	CAGGTGCAGCTGGTGGAGTCTGGGGGAGGCTTGGTCAAGCCTGGAGGG TCCCTGAGACTCTCCTGTGCAGCCTCTGGATTACCTTCAGTGACTAC TACATGAGCTGGATCCGCCAGGTTCCAGGGAAGGGCTGGAGTGGGTT TCATACATTAGTACTAGTGGTAGTACCATATACTACGCAGACTCTGTG AAGGGCCGATTACCATCTCCAGGGACAACGCCAAGAACTCACTGTAT CTACAAATGAACAGCCTGAGAGCCGAGGACGCGGCCGTGTATCACTGT GCGAGAGGTATAACTGGAACCTAATTCTACCACTACGGTTTGGGCGTC TGGGGCCAAGGGACACGGTCAACCGTCTCCTCAG
246	413G05-full heavy chain sequence	Amino acid sequence of 413G05 heavy chain	QVQLVESGGGLVKPGGSLRLSCAASGFTFSDYYMSWIRQVPGKGLEWV SYISTSGSTIYYADSVKGRFTISRDNAKNSLYLQMNSLRAEDAAVYHC ARGITGTFNYHYGLGVWGQTTVTSSASTKGPSVFPLAPSSKSTSGG TAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSV TVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKHTHTCPPCPAPEL LGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGV EVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPA PIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGFYPSDIA VEWESNGQPENNYKTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFCSS VMHEALHNYHTQKSLSLSPGK
247	413G05-full heavy chain sequence	Nucleic acid sequence of 413G05 heavy chain	CAGGTGCAGCTGGTGGAGTCTGGGGGAGGCTTGGTCAAGCCTGGAGGG TCCCTGAGACTCTCCTGTGCAGCCTCTGGATTACCTTCAGTGACTAC TACATGAGCTGGATCCGCCAGGTTCCAGGGAAGGGCTGGAGTGGGTT TCATACATTAGTACTAGTGGTAGTACCATATACTACGCAGACTCTGTG AAGGGCCGATTACCATCTCCAGGGACAACGCCAAGAACTCACTGTAT CTACAAATGAACAGCCTGAGAGCCGAGGACGCGGCCGTGTATCACTGT GCGAGAGGTATAACTGGAACCTAATTCTACCACTACGGTTTGGGCGTC TGGGGCCAAGGGACACGGTCAACCGTCTCCTCAGCCAGCACCAAGGGC CCCTCTGTGTTCCCTCTGGCCCCTTCCAGCAAGTCCACCTCTGGCGGA ACAGCCGCTCTGGGCTGCCTCGTGAAGGACTACTTCCCGAGCCTGTG ACCGTGTCTGGAACCTCTGGCGCTCTGACCAAGCGAGTGACACCTTC CCTGCTGTGCTGCAGTCTCCGGCCTGTACTCCCTGTCTCCGTCTGTG ACCGTGCCTTCCAGCTCTCTGGGCAACCCAGACCTACATCTGCAACGTG AACCACAAGCCCTCAACACCAAGGTGGACAAGAAGGTGGAACCAAG

TABLE S1-continued

SEQ ID NOS: 1-342			
SEQ ID NO:	Name	Description	Sequence
			TCCTGCGACAAGACCCACACCTGTCCCCCTTGTCCTGCCCCGAACTGCTGGGCGGACCTTCCGTGTTCCTGTTCCTCCCAAGCCCAAGGACACCCTGATGATCTCCCGGACCCCCGAAGTGACCTGCGTGGTGGTGGATGTGTCCACGAGGACCCCTGAAGTGAAGTTCATTGGTACGTGGACGGCGTGAAGTGCACAACGCCAAGACCAAGCCTAGAGAGGAACAGTACAACCTCCACCTACCGGGTGGTGTCCGTGCTGACCGTGCTGCACCAGGATTGGCTGAACGGCAAAGAGTACAAGTGAAGGTGTCCAACAAGGCCCTGCCTGCCCCATCGAAAGACCATCTCCAAGGCCAAGGGCCAGCCCCGGGAACCCAGGGTGACACACTGCCCCCTAGCAGGGACGAGCTGACCAAGAACCCAGGTGTCCTGACCTGTCTCGTGAAAGGCTTCTACCCCTCCGATATCGCCGTGGAATGGGAGTCCAACGGCCAGCCTGAGAACAACACAGACCACCCCCCTGTGCTGGACTCCGACGGCTCATTCTTCTGTACAGCAAGCTGACAGTGGACAAGTCCCGGTGGCAGCAGGGCAACGTGTTCTCTGTCTCCGTGATGCACGAGGCCCTGCACAACCACTACACCCAGAAGTCCCTGTCCCTGAGCCCCGGCAAG
248	413G05-CDRL1 (IMGT)	Amino acid sequence of CDRL1 of 413G05 using IMGT	QGINSW
249	413G05-CDRL2 (IMGT)	Amino acid sequence of CDRL2 of 413G05 using IMGT	AAS
250	413G05-CDRL3 (IMGT)	Amino acid sequence of CDRL3 of 413G05 using IMGT	QQVNSFPLT
251	413G05-CDRL1 (Kabat)	Amino acid sequence of CDRL1 of 413G05 using Kabat	RASQGINSWLA
252	413G05-CDRL2 (Kabat)	Amino acid sequence of CDRL2 of 413G05 using Kabat	AASTLQS
253	413G05-CDRL3 (Kabat)	Amino acid sequence of CDRL3 of 413G05 using Kabat	QQVNSFPLT
254	413G05-Light chain variable region	Amino acid sequence of V_L of 413G05	DIQMTQSPSSVSASVGDRTITCRASQGINSWLAWYQQKPGKAPKLLIYAASLTQSGVPSRFRSGSGSGADFTLTISLQPEDFATYYCQQVNSFPLTFGGGTKVEIK
255	413G05-Light chain variable region	Nucleic acid sequence of V_L of 413G05	GACATCCAGATGACCCAGTCTCCATCTTCCGTGTCTGCATCTGTAGGAGACAGAGTCACCATCACTTGTCTGGGCGAGTCAGGGTATTAAACAGCTGGTTAGCCTGGTATCAGCAGAAACAGGGAAAGCCCCCTAAGCTCCTGATCTATGCTGCATCCACTTTGCAAAGTGGGGTCCCATCAAGGTTCAAGCGGCAAGTGGGTCTGGGGCAGATTTCACTCTCACCATCAGCAGCCTGCAGCCTGAAGATTTTGCAACTTACTATTGTCAACAGGTTAACAGTTTCCCGCTCACTTTCGGCGGAGGACCAAGGTGGAGATCAAAC
256	413G05-full light chain sequence	Amino acid sequence of 413G05 light chain	DIQMTQSPSSVSASVGDRTITCRASQGINSWLAWYQQKPGKAPKLLIYAASLTQSGVPSRFRSGSGSGADFTLTISLQPEDFATYYCQQVNSFPLTFGGGTKVEIKRTVAAPSVFIFPPSDEQLKSGTASVVCCLNNFYPREAKVQWKVDNALQSGNSQESVTEGDSKDSSTYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC
257	413G05-full light chain sequence	Nucleic acid sequence of 413G05 light chain	GACATCCAGATGACCCAGTCTCCATCTTCCGTGTCTGCATCTGTAGGAGACAGAGTCACCATCACTTGTCTGGGCGAGTCAGGGTATTAAACAGCTGGTTAGCCTGGTATCAGCAGAAACAGGGAAAGCCCCCTAAGCTCCTGATCTATGCTGCATCCACTTTGCAAAGTGGGGTCCCATCAAGGTTCAAGCGGCAAGTGGGTCTGGGGCAGATTTCACTCTCACCATCAGCAGCCTGCAGCCTGAAGATTTTGCAACTTACTATTGTCAACAGGTTAACAGTTTCCCGCTCACTTTCGGCGGAGGACCAAGGTGGAGATCAAACGTACGGTGGCCGCTCCCTCCGTGTTTCTTCCACCTTCCGACGAGCAGCTGAAGTCCGGCACCGCTTCTGTCGTGTGCTGCTGAACAACCTTCTACCCCGCGAGGGCAAGGTGCAAGTGAAGGTGGACAACCCCTGCAGTCCGGCAACTCCAGGAATCCGTGACCGAGCAGGACTCCAAGGACAGCACCTACTCCCTGTCTCCACCCTGACCCTGTCCAAGGCCGACTACGAGAAGCACAAAGGTGTACGCCTGCGAAGTGACCCACAGGGCCTGTCTAGCCCCGTGACCAAGTCTTCAACCGGGGCGAGTGT

TABLE S1-continued

SEQ ID NOS: 1-342			
SEQ ID NO:	Name	Description	Sequence
258	413F09-CDRH1 (IMGT)	Amino acid sequence of CDRH1 of 413F09 using IMGT	GFTFSYYA
259	413F09-CDRH2 (IMGT)	Amino acid sequence of CDRH2 of 413F09 using IMGT	ISGGGGNT
260	413F09-CDRH3 (IMGT)	Amino acid sequence of CDRH3 of 413F09 using IMGT	AKDRMKQLVRAYYFDY
261	413F09-CDRH1 (Kabat)	Amino acid sequence of CDRH1 of 413F09 using Kabat	YYAMS
262	413F09-CDRH2 (Kabat)	Amino acid sequence of CDRH2 of 413F09 using Kabat	TISGGGGNTHYADSVKG
263	413F09-CDRH3 (Kabat)	Amino acid sequence of CDRH3 of 413F09 using Kabat	DRMKQLVRAYYFDY
264	413F09-Heavy chain variable region	Amino acid sequence of V_H of 413F09	EVPLVESGGGLVQPGGSLRLSCAASGFTFSYYAMSWVRQAPGKGLDWV STISGGGGNTHYADSVKGRFTISRDNKNTLYLHMNSLRAEDTAVYYC AKDRMKQLVRAYYFDYWGGTLVTVSS
265	413F09-Heavy chain variable region	Nucleic acid sequence of V_H of 413F09	GAGGTGCCGCTGGTGGAGTCTGGGGGAGGCTTGGTACAGCCTGGGGG TCCCTGAGACTCTCCTGTGCAGCCTCTGGATTACGTTTAGCTACTAT GCCATGAGCTGGGTCCGTCAGGCTCCAGGGAAGGGGCTGGACTGGGTC TCAACTATTAGTGGTGGTGGTAAACACACACTACGCAGACTCCGTG AAGGGCCGATTCACTATATCCAGAGACAATTCCAAGAACACGCTGTAT CTGCACATGAACAGCCTGAGAGCCGAAGACACGGCCGCTATTACTGT GCGAAGGATCGGATGAAACAGCTCGTCCGGGCTACTACTTTGACTAC TGGGGCCAGGGAACCTGGTCACCGTCTCCTCAG
266	413F09-full heavy chain sequence	Amino acid sequence of 413F09 heavy chain	EVPLVESGGGLVQPGGSLRLSCAASGFTFSYYAMSWVRQAPGKGLDWV STISGGGGNTHYADSVKGRFTISRDNKNTLYLHMNSLRAEDTAVYYC AKDRMKQLVRAYYFDYWGGTLVTVSSASTKGPSVFPLAPSSKSTSGG TAALGCLVKDYFPEPTVSWNSGALTSGVHTFPAVLQSSGLYSLSSV TVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKHTHTCPPCPAPEL LGGPSVFLFPPKPKDRLMISRTPEVTCVVVDVSHEDPEVKFNWYVDG EVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPA PIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGFYPSDIA VEWESNGQPENNYKTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFCSS VMHEALHNYHTQKSLSLSPGK
267	413F09-full heavy chain sequence	Nucleic acid sequence of 413F09 heavy chain	GAGGTGCCGCTGGTGGAGTCTGGGGGAGGCTTGGTACAGCCTGGGGG TCCCTGAGACTCTCCTGTGCAGCCTCTGGATTACGTTTAGCTACTAT GCCATGAGCTGGGTCCGTCAGGCTCCAGGGAAGGGGCTGGACTGGGTC TCAACTATTAGTGGTGGTGGTAAACACACACTACGCAGACTCCGTG AAGGGCCGATTCACTATATCCAGAGACAATTCCAAGAACACGCTGTAT CTGCACATGAACAGCCTGAGAGCCGAAGACACGGCCGCTATTACTGT GCGAAGGATCGGATGAAACAGCTCGTCCGGGCTACTACTTTGACTAC TGGGGCCAGGGAACCTGGTCACCGTCTCCTCAGCCAGCACCAAGGGC CCCTCTGTGTTCCCTCTGGCCCTTCCAGCAAGTCCACCTCTGGCGGA ACAGCCGCTCTGGGCTGCCTCGTGAAGGACTACTTCCCCGAGCCTGTG ACCGTGTCTTGAAGTCTGGCGCTCTGACCAGCGGAGTGACACACCTTC CCTGTGTGCTGCAGTCTCCTGGGCTGTACTCCCTGTCTCCGTCGTG ACCGTGCCTTCCAGCTCTCTGGGCACCCAGACCTACATCTGCAACGTG AACCACAAGCCCTCCAACACCAAGGTGGACAAGAAGGTGGAACCCAAG TCCTGCGACAAGACCCACACCTGTCCCCTTGTCTGCCCCGTAAGTCT CTGGGCGGACCTTCCGTGTTCTGTTCCTGTTCCCCCAAAGCCCAAGGAC CTGATGATCTCCCGGACCCCGAAGTGACCTGCGTGGTGGTGGATGTG TCCCACGAGGACCTGAAGTGAAGTTCAATTGGTACGTGGACGGCGTG GAAGTGACACAAGCCAAGACCAAGCCTAGAGAGGAACAGTACAACCTCC ACCTACCGGGTGGTGTCCGTGCTGACCGTGTGTCACCAAGGATTGGCTG AACGGCAAGAGTACAAGTGAAGGTGTCCAACAGGCCCTGCTGCCC CCCATCGAAAAGACCATCTCCAAGGCCAAGGCCAGCCCCGGGAACCC CAGGTGTACACACTGCCCTTAGCAGGACGAGCTGACCAAGAACCCAG

TABLE S1-continued

SEQ ID NOS: 1-342			
SEQ ID NO: Name	Description	Sequence	
		GTGTCCTGACCTGTCTCGTGAAAGGCTTCTACCCCTCCGATATCGCC GTGGAATGGGAGTCCAACGGCCAGCTGAGAACAACACAGACCACC CCCCCTGTGCTGGACTCCGACGGCTCATTCTTCCTGTACAGCAAGCTG ACAGTGGACAAGTCCCGGTGGCAGCAGGGCAACGTGTTCTCTGTCTCC GTGATGCACAGGCCCTGCACAACCACTACACCAGAGTCCCTGTCC CTGAGCCCCGGCAAG	
268 413F09-CDRL1 (IMGT)	Amino acid sequence of CDRL1 of 413F09 using IMGT	QDISTY	
269 413F09-CDRL2 (IMGT)	Amino acid sequence of CDRL2 of 413F09 using IMGT	GTS	
270 413F09-CDRL3 (IMGT)	Amino acid sequence of CDRL3 of 413F09 using IMGT	QQLHTDPIT	
271 413F09-CDRL1 (Kabat)	Amino acid sequence of CDRL1 of 413F09 using Kabat	WASQDISTYLG	
272 413F09-CDRL2 (Kabat)	Amino acid sequence of CDRL2 of 413F09 using Kabat	GTSSLQS	
273 413F09-CDRL3 (Kabat)	Amino acid sequence of CDRL3 of 413F09 using Kabat	QQLHTDPIT	
274 413F09-Light chain variable region	Amino acid sequence of V_L of 413F09	DIQLTQSPSFLSASVGDVRTITCWASQDISTYLGWYQQKPGKAPKLLI YGTSSLQSGVPSRFSGSGSGTEFTLTISLQPEDFATYYCQQLHTDPI TFGQGRLEIK	
275 413F09-Light chain variable region	Nucleic acid sequence of V_L of 413F09	GACATCCAGTTGACCCAGTCTCCATCCTTCCTGTCTGCATCTGTAGGA GACAGAGTCACCATCACTTGCTGGGCCAGTCAGGACATTAGCACTTAT TTAGGCTGGTATCAGCAAAAACAGGGAAAGCCCTAAGCTCCTGATC TATGGTACATCCAGTTTGCAAAGTGGGGTCCCATCAAGGTTTCAGCGGC AGTGGATCTGGGACAGAATTCACTCTCACAATCAGCAGCCTGCAGCCT GAAGATTTTGCAACTTATTACTGTCAACAGCTTCATACTGACCCGATC ACCTTCGGCCAAGGGACACGACTGGAGATCAAAC	
276 413F09-full light chain sequence	Amino acid sequence of 413F09 light chain	DIQLTQSPSFLSASVGDVRTITCWASQDISTYLGWYQQKPGKAPKLLI YGTSSLQSGVPSRFSGSGSGTEFTLTISLQPEDFATYYCQQLHTDPI TFGQGRLEIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREA KVQWKVDNALQSGNSQESVTEQDSKDSYLSSTLTLSKADYEKHKVY ACEVTHQGLSSPVTKSFNRGEC	
277 413F09-full light chain sequence	Nucleic acid sequence of 413F09 light chain	GACATCCAGTTGACCCAGTCTCCATCCTTCCTGTCTGCATCTGTAGGA GACAGAGTCACCATCACTTGCTGGGCCAGTCAGGACATTAGCACTTAT TTAGGCTGGTATCAGCAAAAACAGGGAAAGCCCTAAGCTCCTGATC TATGGTACATCCAGTTTGCAAAGTGGGGTCCCATCAAGGTTTCAGCGGC AGTGGATCTGGGACAGAATTCACTCTCACAATCAGCAGCCTGCAGCCT GAAGATTTTGCAACTTATTACTGTCAACAGCTTCATACTGACCCGATC ACCTTCGGCCAAGGGACACGACTGGAGATCAAACGTACGGTGGCCGCT CCCTCCGTGTTTATCTTCCACCTTCCGACGAGCAGCTGAAGTCCCGGC ACCGCTTCTGTCTGTGCTGCTGAACAACCTTCTACCCCGCGAGGGCC AAGGTGCAGTGAAGGTGGACAACGCCCTGCAGTCCGGCAACTCCAG GAATCCGTGACCGAGCAGGACTCCAAGGACAGCACCTACTCCCTGTCC TCCACCCTGACCCTGTCCAAGGCCGACTACGAGAAGCACAAGGTGTAC GCCTGCGAAGTGACCCACAGGGCCTGTCTAGCCCCGTGACCAAGTCT TTCAACCGGGGCGAGTGT	
278 414B06-CDRH1 (IMGT)	Amino acid sequence of CDRH1 of 414B06 using IMGT	GFTFSSYW	
279 414B06-CDRH2 (IMGT)	Amino acid sequence of CDRH2 of 414B06 using IMGT	IKQDGSEK	

TABLE S1-continued

SEQ ID NOS: 1-342			
SEQ ID NO:	Name	Description	Sequence
280	414B06-CDRH3 (IMGT)	Amino acid sequence of CDRH3 of 414B06 using IMGT	ARVRQWSDYSDY
281	414B06-CDRH1 (Kabat)	Amino acid sequence of CDRH1 of 414B06 using Kabat	SYWMN
282	414B06-CDRH2 (Kabat)	Amino acid sequence of CDRH2 of 414B06 using Kabat	NIKQDGSEKYYVDSVKG
283	414B06-CDRH3 (Kabat)	Amino acid sequence of CDRH3 of 414B06 using Kabat	VRQWSDYSDY
284	414B06-Heavy chain variable region	Amino acid sequence of V _H of 414B06	EVHLVESGGGLVQPGGSLRLSCAASGFTFSSYWMNWVRQAPGKGLEWV ANIKQDGSEKYYVDSVKGRFTVSRDNAKNSLYLQMNSLRAEDTAVYYC ARVRQWSDYSDYWGQGTPTVTVSS
285	414B06-Heavy chain variable region	Nucleic acid sequence of V _H of 414B06	GAGGTGCACCTGGTGGAGTCTGGGGGAGGCTTGGTCCAGCCTGGGGGG TCCCTGAGACTCTCCTGTGCAGCCTCTGGATTCACCTTAGTAGCTAT TGGATGAAC TGGGTCCGCCAGGCTCCAGGGAAGGGGCTGGAGTGGGTG GCCAACATAAAGCAAGATGGAAGTGAAGAACTACTATGTGGACTCTGTG AAGGGCCGCTTACCGTCTCCAGAGACAACGCCAAGAACTCACTGTAT CTGCAATGAACAGCCTGAGAGCCGAGGACACGGCTGTGTATTACTGT GCGAGAGTTCGACAATGGTCCGACTACTCTGACTACTGGGGCCAGGGA ACCCCGGTACCGTCTCCTCAG
286	414B06-full heavy chain sequence	Amino acid sequence of 414B06 heavy chain	EVHLVESGGGLVQPGGSLRLSCAASGFTFSSYWMNWVRQAPGKGLEWV ANIKQDGSEKYYVDSVKGRFTVSRDNAKNSLYLQMNSLRAEDTAVYYC ARVRQWSDYSDYWGQGTPTVTVSSASTKGPSVFPLAPSSKSTSGGTAAL GCLVKDYFPEPTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPS SSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKHTCTPPCPAPPELLGGP SVFLFPPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHN AKTKPREEQYNSTYRVVSVLTVLHQDNLNGKEYKCKVSNKALPAPIEK TISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWE SNGQPENNYKTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFSQCSVMHE ALHNHYTQKSLSLSPGK
287	414B06-full heavy chain sequence	Nucleic acid sequence of 414B06 heavy chain	GAGGTGCACCTGGTGGAGTCTGGGGGAGGCTTGGTCCAGCCTGGGGGG TCCCTGAGACTCTCCTGTGCAGCCTCTGGATTCACCTTAGTAGCTAT TGGATGAAC TGGGTCCGCCAGGCTCCAGGGAAGGGGCTGGAGTGGGTG GCCAACATAAAGCAAGATGGAAGTGAAGAACTACTATGTGGACTCTGTG AAGGGCCGCTTACCGTCTCCAGAGACAACGCCAAGAACTCACTGTAT CTGCAATGAACAGCCTGAGAGCCGAGGACACGGCTGTGTATTACTGT GCGAGAGTTCGACAATGGTCCGACTACTCTGACTACTGGGGCCAGGGA ACCCCGGTACCGTCTCCTCAGCCAGCACCAAGGGCCCCCTCTGTGTTT CCTCTGGCCCCCTCCAGCAAGTCCACCTCTGGCGGAACAGCCGCTCTG GGCTGCCTCGTGAAGGACTACTTCCCCGAGCCTGTGACCGTGTCTCTGG AACTCTGGCGCTCTGACCAGCGGAGTGACACCTTCCCTGCTGTCTGTG CAGTCTCTCCGGCTGTACTCCCTGTCTCCGTCGTGACCGTGCCTTCC AGCTCTCTGGGCACCCAGACCTACATCTGCAACGTGAACCAAGCCCC TCAACACCAAGGTGGACAAGAAGGTGAACCAAGTCTTGCACCAAG ACCCACACCTGTCCCCCTTGCTCTGCCCTGAAGTGTGGGCGGACCT TCCGTGTTTCTGTTCCTTCCCCCAAGCCCAAGGACACCTGATGATCTCC CGGACCCCCGAAGTGACCTGCGTGGTGGATGTGTCCACGAGGAC CCTGAAGTGAAGTTCAATTGGTACGTGGACGCGTGAAGTGACACAAC GCCAAGACCAAGCCTAGAGAGGAACAGTACAACCTCACTTACCGGGTG GTGTCCGTGCTGACCGTGTGACACAGGATTGGCTGAACGGCAAGAG TACAAGTGCAAGGTGTCAACAAGGCCCTGCCTGCCCATCGAAAG ACCATCTCCAAGGCCAAGGGCCAGCCCCGGGAACCCAGGTGTACACA CTGCCCTTAGCAGGGACGAGCTGACCAAGAACAGGTGTCTCTGACC TGCTCTCGTGAAAGGCTTCTACCCCTCCGATATCGCCGTGGAATGGGAG TCCAACGGCCAGCCTGAGAACAACACAAGACACCCCCCTGTGCTG GACTCCGACGGCTCATTTCTCTGTACAGCAAGCTGACAGTGGACAAG TCCCGGTGGCAGCAGGGCAACGTGTTCTCTGCTCCGTGATGCACGAG GCCCTGCACAACCACTACACCCAGAAGTCCCTGTCTCTGAGCCCCGGC AAG

TABLE S1-continued

SEQ ID NOS: 1-342			
SEQ ID NO:	Name	Description	Sequence
288	414B06-CDRL1 (IMGT)	Amino acid sequence of CDRL1 of 414B06 using IMGT	QGISSW
289	414B06-CDRL2 (IMGT)	Amino acid sequence of CDRL2 of 414B06 using IMGT	AAS
290	414B06-CDRL3 (IMGT)	Amino acid sequence of CDRL3 of 414B06 using IMGT	QQANSFPFT
291	414B06-CDRL1 (Kabat)	Amino acid sequence of CDRL1 of 414B06 using Kabat	RASQGISSWLA
292	414B06-CDRL2 (Kabat)	Amino acid sequence of CDRL2 of 414B06 using Kabat	AASSLQS
293	414B06-CDRL3 (Kabat)	Amino acid sequence of CDRL3 of 414B06 using Kabat	QQANSFPFT
294	414B06-Light chain variable region	Amino acid sequence of V_L of 414B06	DIQMTQSPSSVSASVGDRTITCRASQGISSWLAWYQQKPGKAPKLLIYAASSLQSGVPSRFSGSGSGTDFTLTISLQPEDFATYYCQQANSFPFTFGPGTKVDIK
295	414B06-Light chain variable region	Nucleic acid sequence of V_L of 414B06	GACATCCAGATGACCCAGTCTCCATCTTCGGTGTCTGCATCTGTAGGA GACAGAGTCACCATCACTTGTCTGGGCGAGTCAGGGTATTAGCAGCTGG TTAGCCTGGTATCAGCAGAAACAGGGAAAGCCCCTAAGCTCCTGATC TATGCTGCATCCAGTTTGCAAAGTGGGTCCCATCAAGGTTTCAGCGGC AGTGGATCTGGGACAGATTTCACTCTCACCATCAGCAGCCTGCAGCCT GAAGATTTTGCAACTTACTATTGTCAACAGGCTAACAGTTTCCCATTC ACTTTCGGCCCTGGGACCAAAGTGGATATCAAAC
296	414B06-full light chain sequence	Amino acid sequence of 414B06 light chain	DIQMTQSPSSVSASVGDRTITCRASQGISSWLAWYQQKPGKAPKLLIYAASSLQSGVPSRFSGSGSGTDFTLTISLQPEDFATYYCQQANSFPFTFGPGTKVDIKRTVAAPSVFIPPPSDEQLKSGTASVVCLLNNFYPREA KVQWKVDNALQSGNSQESVTEQDSKDSYLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC
297	414B06-full light chain sequence	Nucleic acid sequence of 414B06 light chain	GACATCCAGATGACCCAGTCTCCATCTTCGGTGTCTGCATCTGTAGGA GACAGAGTCACCATCACTTGTCTGGGCGAGTCAGGGTATTAGCAGCTGG TTAGCCTGGTATCAGCAGAAACAGGGAAAGCCCCTAAGCTCCTGATC TATGCTGCATCCAGTTTGCAAAGTGGGTCCCATCAAGGTTTCAGCGGC AGTGGATCTGGGACAGATTTCACTCTCACCATCAGCAGCCTGCAGCCT GAAGATTTTGCAACTTACTATTGTCAACAGGCTAACAGTTTCCCATTC ACTTTCGGCCCTGGGACCAAAGTGGATATCAAACGTACGGTGGCCGCT CCCTCCGTGTTTCATCTTCCACCTTCCGACGAGCAGCTGAAGTCCGGC ACCGCTTCTGTCGTGTGCTGCTGAACAACCTTCTACCCCGCAGAGGCC AAGGTGCAGTGAAGGTGGACAACGCTTGCAGTCCGGCAACTCCAG GAATCCGTGACCGAGCAGGACTCCAAGGACAGCACCTACTCCCTGTCC TCCACCCTGACCTGTCCAAGGCCGACTACGAGAAGCACAAGGTGTAC GCCTGCGAAGTGACCCACAGGGCCTGTCTAGCCCGTGACCAAGTCT TTCAACCGGGGCGAGTGT
298	Mutated 1D05-LC mutant 3	Amino acid sequence of 1D05 kappa light chain with V to Y mutation in CDRL2 highlighted	DIQMTQSPSSLSASVGDRTITCRASQSISSYLNWYQQKPGKAPKLLIYYASSLQSGVPSRFSGSGSGTDFTLTISLQPEDFATYYCQQQSYSTPI TFGQGTRELEIKRTVAAPSVFIPPPSDEQLKSGTASVVCLLNNFYPREA KVQWKVDNALQSGNSQESVTEQDSKDSYLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC
299	1D05-heavy chain disabled IgG1 Fc	Amino acid sequence of IgG1 disabled variant of 1D05	EVQLVESGGGLVQPGSRSLRLSCAASGFTFDDYAMHWVRQVPGKGLEWV SGISWIRTGIGYADSVKGRFTIFRDNAKNSLYLQMNSLRAEDTALYYC AKDMKSGSGTYGGWFDTWGQGLVTVSSASTKGPSVFPLAPSSKSTSGV TAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSV TVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHITCPCPAPEL AGAPSVFLFPPPKDITLMISRTPEVTVVVDVSHEDPEVKFNWYVDG EVHNAKTKPREQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAP PIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGFYPSDIA

TABLE S1-continued

SEQ ID NOS: 1-342			
SEQ ID NO:	Name	Description	Sequence
			VEWESNGQPENNYKTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCS VMHEALHNHYTQKSLSLSPGK
300	1D05-light chain IL-2 fusion	1D05 Light chain sequence fused to wild-type human IL-2 sequence (IL-2 amino acid sequence is underlined and region to be varied is shown in bold)	DIQMTQSPSSLSASVGRVTITCRASQSISSYLNWYQKPKGKAPKLLI YVASSLQSGVPSRFRSGSGSDFTLTISLQPEDFATYYCQQSYSTPI TFGQGRTRLEIKRTVAAPSVFIFPPPSDEQLKSGTASVVCLLNNFYPREA KVQWKVDNALQSGNSQESVTEQDSKDSYSTLSSTLTLSKADYEKHKVY ACEVTHQGLSSPVTKSFNRGEC APTSSSTKKTQLQLEHLLLDLQ MILN <u>GINNYKNPKLTRMLTFKFYMPKKATELKHLCLEELKPLEEVLNLAQ</u> <u>SKNFHLRPRDLISNINIVIVLELKGSETTFMCEYADETATIVEEFLNRWI</u> <u>TFCQSIISTLT</u>
301	Human IL-2	Uniprot number: P60568 Full length amino acid sequence of human IL-2 (minus signal sequence)	APTSSSTKKTQLQLEHLLLDLQ MILNGINNYKNPKLTRMLTFKFYMPK KATELKHLCLEELKPLEEVLNLAQSKNFHLRPRDLISNINIVIVLEL KGSETTFMCEYADETATIVEEFLNRWITFCQSIISTLT
302	Control 1D05 immunocy tokine HC C-terminal fusion	Heavy chain 1D05 IgG1 variant fused at the N-terminus to wild-type human IL2 sequence (control)	EVQLVESGGGLVQPGRSRLRSCAASGFTFDDYAMHWVRQVPKGLEWV SGISWIRTGIGYADSVKGRFTIFRDNAKNSLYLQMNSLRAEDTALYYC AKDMKSGSGTYGGWFDTWGQGLTVTVSSASTKGPSVFPLAPSSKSTSGG TAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVV TVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCPPCPAPEL <u>AGAPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGV</u> EVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPA PIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGFYPSDIA VEWESNGQPENNYKTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCS VMHEALHNHYTQKSLSLSPGKAPTSSSTKKTQLQLEHLLLDLQ MILNGINNYKNPKLTRMLTFKFYMPKKATELKHLCLEELKPLEEVLNLAQ SKNFHLRPRDLISNINIVIVLELKGSETTFMCEYADETATIVEEFLNRWIT FCQSIISTLT
303	IL-2 D5-9	IL-2 IC45 (Del 5-9) N terminal IL-2 sequence	APTSTQLQLELLLD
304	IL-2 D1-9	IL-2 IC46 (Del 1-9) N terminal IL-2 sequence	TQLQLEHLLLD
305	IL-2 D5-7	IL-2 IC64 (Del 5-7) N terminal IL-2 sequence	APTSKKTQLQLEHLLLD
306	IL-2 D1	IL-2 D1 N terminal IL-2 sequence	PTSSSTKKTQLQLEHLLLD
307	IL-2 D1-2	IL-2 D1-2 N terminal IL-2 sequence	TSSSTKKTQLQLEHLLLD
308	IL-2 D1-3	IL-2 D1-3 N terminal IL-2 sequence	SSSTKKTQLQLEHLLLD
309	IL-2 D1-4	IL-2 D1-4 N terminal IL-2 sequence	SSTKKTQLQLEHLLLD
310	IL-2 D1-5	IL-2 D1-5 N terminal IL-2 sequence	STKKTQLQLEHLLLD
311	IL-2 D1-6	IL-2 D1-6 N terminal IL-2 sequence	TKKTQLQLEHLLLD
312	IL-2 D1-7	IL-2 D1-7 N terminal IL-2 sequence	KKTQLQLEHLLLD

TABLE S1-continued

SEQ ID NOS: 1-342			
SEQ ID NO:	Name	Description	Sequence
313	IL-2 D1-8	IL-2 D1-8 N terminal IL-2 sequence	KTQLQLEHLLLD
314	IL-2 D9	IL-2 D9 N terminal IL-2 sequence	APTSSSTKTQLQLEHLLLD
315	IL-2 D9-8	IL-2 D9-8 N terminal IL-2 sequence	APTSSSTTQLQLEHLLLD
316	IL-2 D9-7	IL-2 D9-7 N terminal IL-2 sequence	APTSSSTQLQLEHLLLD
317	IL-2 D9-6	IL-2 D9-6 N terminal IL-2 sequence	APTSSSTQLQLEHLLLD
318	IL-2 D9-4	IL-2 D9-4 N terminal IL-2 sequence	APTTQLQLEHLLLD
319	IL-2 D9-3	IL-2 D9-3 N terminal IL-2 sequence	APTQLQLEHLLLD
320	IL-2 D9-2	IL-2 D9-2 N terminal IL-2 sequence	ATQLQLEHLLLD
321	IL-2 D2-6	IL-2 D2-6 N terminal IL-2 sequence	ATKKTQLQLEHLLLD
322	IL-2 D3-7	IL-2 D3-7 N terminal IL-2 sequence	APKKTQLQLEHLLLD
323	IL-2 D4-8	IL-2 D4-8 N terminal IL-2 sequence	APTKTQLQLEHLLLD
324	C-terminal amino acid sequence of hIL-2	Amino acids 21 to 133 of hIL-2	LQMILNGINNYKNPKLTRMLTFKFPMPKKATELKHLCLEBELKPLEE VLNLAQSKNFHLRPRDLISNINVIIVLELKGSETTFMCEYADETATIVE FLNRWITFCQSIISTLT
325	Mouse PD-L1	Uniprot number: Q9EP73 (ECD highlighted in BOLD, and cytoplasmic domain underlined)	MRIFAGIIIFTACCHLLRAFTITAPEDLYVVEYGSNVTMECREPVEREL DLLALVVYWEKEDEQVIQFVAGEEDLEPQHSNERGRASLPEDQLLEGN AALQITDVELQDAGVYCCIIISYGGADYKRITLEVNAFYRKINQRIQSV PATSEHELICQAEGYPEAEVIWNTSDHQPVSGERSVTTSRTGMLLNVT TSSLRVNATANDVFYCTFWRSQPGQNHTAELIIPELPATHPPQNRTHW VLLGSILLFLIVVSTVLLFLRKQVRMLDVEKCGVEDTSSKNRNDTQFE ET
326	Mouse PD-L1 ECD His	Mouse PD-L1 extracellular domain with his tag	FTITAPKDLVVEYGSNVTMECRFPVERELDLLALVVYWEKEDEQVIQ FVAGEEDLKPQHSNFRGRASLPKDLQLKGNAALQITDVKLQDAGVYCC IISYGGADYKRITLKVNAFYRKINQRIQSVDPATSEHELICQAEGYPEA EVIWNTSDHQPVSGERSVTTSRTGMLLNVTSSLRVNATANDVFYCTF WRSQPGQNHTAELIIPELPATHPPQNRTHHHHHH
327	Human IL-2R α chain	Human IL-2 receptor alpha chain	ELCDDDPPEIPHATFKAMAYKEGTMNLNCECKRGFRRIKSGSLYMLCTG NSSHSSWDNQCCQTSSATRNNTKQVTPQPEEQKERKTTEMQSPMQPVD QASLPGHCREPPPWENEATERIYHFVVGQMVYYQCVQGYRALHRGP SVCMTMTHGKTRWTQPLICTGEMETSQFPGEKQPASPEGRPESETSC LVTTDFQIQTEMAATMETSIFTTEYQVAVAGCVFLLISVLLLSGLTW QRRQRKSRRTI

TABLE S1-continued

SEQ ID NOS: 1-342			
SEQ ID NO:	Name	Description	Sequence
328	Human IL-2R β chain	Human IL-2 receptor beta chain	AVNGTSQFTCFYNSRANISCVWSQDQALQDTSQVHAWPDRRRWNQTC ELLPVSQASWACNLILGAPDSQKLTIVDITLRLVLCREGVRWRVMAIQ DFKPFENLRMAPISLQVVHVETHRCNISWEISQASHYFERHLEFEAR TLPSPGHTWEEAPLLTLKQKQEWICLETLPDTPQYEFQVRVKPLQGEFT TWSPWSQPLAFRTKPAALGKDTIPWLGHLVLGSGAGFIIILVYLLIN CRNTGPWLKKVLKCNTPDPSKFSQLSSEHGGDVQKWLSSPFPSSSFS PGGLAPEISPLEVLERDKVTQLLLQQDKVPEPASLSSNHSLSCTFTNP GYFFPHLPDALEIEACQVYFTYDPYSEEDPDEGVAGAPTGSSPQLQP LSGEDDAYCTFPSRDDLLFSPSLGGPSPSTAPGGSGAGEERMPPS LQERVPRDWPQPLGPPTPGVPLVDVFPPELVLEAGEEVPDAGPR EGVSFPWSRPPGQGEFRALNARLPNTDAYLSLQELQGQDPTHLV
329	Human IL-2R γ chain	Human IL-2 receptor common gamma chain	LNNTILTPNGNEDTTADFFLTMTPTDSLVSSTLPLPEVQCQCFVFNVEYM NCTWNSSSEPQPTNLTLYHYKNSDNDKVQKCSHYLFSEETSGCQLQ KKEIHLYQTFFVVLQDQPREPRQATQMLKLQNLVLPWAPENLTLLKLS ESQLELNWNRRFLNHCLEHLVQYRTDWDHSWTEQSVDIRHKSFLPSVD GQKRYTFRVRSRFLNLCGSAQHWSEWSHPIHWGNSNTSKENPFLFALEA VVISVGSMLIIISLLCVYFWLERTMPRIPTLKNLEDLVTEYHGNFSAW SGVSKGLAESLQPDYSERLCLVSEIPKGGALGEGPGASPCNQHSPIYW APPCYTLKPET
330	IL-7	Human IL-7 amino acid sequence	DCDIEGKDGKQYESVLMVSIQQLLDSMKEIGSNCLNNEFNFFKRHICD ANKEGMFLFRAARKLRQFLKMNSTGDFDLHLKLVSEGTIILNCTGQV KGRKPAALGEAQPTKSLEENKSLKEQKKLNDLCFLKRLQEIKTQWKN ILMGTKEH
331	IL-15	Human IL-15 amino acid sequence	GIHVFILGCFSGAGLPKTEANWVNVISDLKKIEDLIQSMHIDATLYTES DVHPSCVKVTAMKCFLELQVISLESQDASIHDTVENLII LANNSLSSN GNVTESGCKECELEEKNIKEFLQSFVHVQMFINTS
332	IL-21	Human IL-21 amino acid sequence	QGQDRHMIRMRLIDIVDQLKNYVNDLVPEFLPAPEDVETNCEWSAFS CFQKAQLKSANTGNNERI INVS IKKLKRKPPSTNAGRQKHRLTCPS DSYEKKPKPEFLERFKSLQKMIHQHLSRTHGSEDS
333	GM-CSF	Human GM-CSF amino acid sequence	APARSPSPSTQPEWHVNAIQEARRLLNLSRDTAAEMNETVEVISEMFD LQEPCTCLQTRLELYKQGLRGSLTKLKGPLTMASHYKQHCPTPETSC ATQIIITFESFKENLKDFFLVIPFDCWEPVQE
334	IFN α	Human IFN- α amino acid sequence	CDLPQNHGLLSRNTLVLLHQMRRI SPFLCLKDRDRFRFPQEMVKGSQL QKAHVMSVLHEMLQQIFSLFHTERSAANMTLLDQLHTELHQQLQHL ETCLLQVVGEGESAGAISSPALTLRRYFQGI RYVLYKEKKYSDCAWEVV RMEIMKSLFLSTNMQERLRSKDRDLGS
335	TNF α	Extracellular portion of human TNF- α amino acid sequence	GPQREEFPRDLSLISPLAQAVRSSRTPSDKPVAVVVPQAEQQLQW LNRANALLANGVELRDNLVVPSEGLYLIYSQVLFKGQCPSTHVL THTISRIAVSYQTKVNLLSAIKSPCQRETPEGAEAKPWYEPIYLGGVF QLEKGDRLSAEINRPDYLDFAESGQVYFGIIAL
336	IL-12 α	Alpha chain of human IL-12 amino acid sequence	RNLPVATPDGMFPCPLHHSQNLRAVSNMLQKARQTLEFYPTSEEID HEDITKDKTSTVEACLPLELTKNESCLNSRETSFITNGSCLASRKT MMALCLSSIYEDLKMYQVEFKTMNAKLLMDPKRQIFLDQNMLAVIDEL MQALNFNSETVPQKSSLEEDPFYKTKIKLCILLHAFRIRAVTIDRVMS YLNAS
337	IL-12 β	Beta chain of human IL-12 amino acid sequence	IWELKKDVYVVELDWYPDAPGEMVLTCDTPEEDGITWTLDQSSEVLG SGKTLTIQVKEFGDAGQYTKHGGEVLSHSLLLHKKEDGIWSTDILK DQKEPKNKTLFLRCEAKNYSGRFTCWLLTITISDITFSVKSSRGSSDPQ GVTCGAATLSAERVGRDNKEYEYSVEQCQEDSACPAAEESLPPIEVMDA VHKLYENYTSFFIRDI IKPDPPKNLQKLPKNSRQVEVSWEYEDTW STPHSYFSLTFCVQVQKSKREKKDRVFTDKTSATVICRKNASISVRA QDRYSSSWSEWASVPCS
338	CXCL9	Human CXCL-9 amino acid sequence	TPVVRKGRCSISTNQGTIHLQSLKDLKQFAPSPSCEKIEIIATLKNG VQTCNLNPDADVKELIKKWEKQVSQKKKQKNGKKHKKVVKVRKSQR SRQKKT
339	CXCL10	Human CXCL-10 amino acid sequence	VPLSRTVRCTCISISNQPVNPRSLEKLEIIPASQFCPRVEIIATMKKK GEKRCNLNPESKAIKNLLKAVSKERSKRSP

TABLE S1-continued

SEQ ID NOS: 1-342			
SEQ ID			
NO: Name	Description	Sequence	
340	Human WT IgG1 constant region	IGHG1*01 & IGHG1*02 & IGHG1*05 (IgG1) sequence	WT human IgG1 amino acid sequence
341		WT human IgG1 nucleic acid sequence	
342	Mutated 1D05-HC mutant 2	Amino acid sequence of 1D05 heavy chain with V to A and F to S back-mutation in framework region to germline highlighted with IgG1 disabled (LAGA) constant region	

TABLE S2

SEQ ID NOS: 343-538			
SEQ ID			
NO: Name	Description	Sequence	
343	416E01 - CDRH1 (IMGT)	Amino acid sequence of CDRH1 of 416E01 using IMGT	GFTFSNYA
344	416E01 - CDRH2 (IMGT)	Amino acid sequence of CDRH2 of 416E01 using IMGT	ISFSGGTT
345	416E01 - CDRH3 (IMGT)	Amino acid sequence of CDRH3 of 416E01 using IMGT	AKDEAPAGATFFDS
346	416E01 - CDRH1 (Kabat)	Amino acid sequence of CDRH1 of 416E01 using Kabat	NYAMS
347	416E01 - CDRH2 (Kabat)	Amino acid sequence of CDRH2 of 416E01 using Kabat	AISFSGGTTYADSVKG

TABLE S2-continued

SEQ ID NOS: 343-538		
SEQ ID NO: Name	Description	Sequence
348 416E01 - CDRH3 (Kabat)	Amino acid sequence of CDRH3 of 416E01 using Kabat	DEAPAGATFFDS
349 416E01 - Heavy chain variable region	Amino acid sequence of V_H of 416E01 (mutations from germline are shown in bold letters)	EVQLAESGGGLVQPGGSLRLSCAASGFTFS NY AMSWVRQTPGKGLEWVSAIS FS GGTTYADSVKGRFTISRDN SKNTLYLHMNSLR ADDTAVYYCAK DEAPAGATFFDS WGQGLTVTVSS
350 416E01 - Heavy chain variable region	Nucleic acid sequence of V_H of 416E01	GAAGTGCAACTGGCGGAGTCTGGGGGAGGCTTGGTACAGCCGGGGGGTCCCTGAGACTCTCCTGTGCAGCCTCTGGATTACCTTTAGCAACTATGCCATGAGTTGGGTCCGCCAGACTCCAGGAAAGGGGCTGGAGTGGGTCTCAGCTATTAGTTTTAGTGGTGGTACTACATACTACGCTGACTCCGTGAAGGGCCGGTTCAACATCTCCAGAGACAATTCCAAGAACACGCTGTATTGTCACATGAACAGCCTGAGAGCCGATGACACGGCCGTATATTACTGTGCGAAAGATGAGGCACACGCTGGCGCAACCTTCTTTGACTCCTGGGGCCAGGGAACGCTGGTCACCGTCTCCTCAG
351 416E01 - full heavy chain sequence	Amino acid sequence of 416E01 heavy chain	EVQLAESGGGLVQPGGSLRLSCAASGFTFS NY AMSWVRQTPGKGLEWVSAIS FS GGTTYADSVKGRFTISRDN SKNTLYLHMNSLR ADDTAVYYCAK DEAPAGATFFDS WGQGLTVTVSSASTKGPSVFPLAPCSRSTSESTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTTKTYTCNVDPKPKNTKVDKRVESKYGPPCPPCPAPEFEGGSPVFLFPPKPKDTLMISRTPEVTCVVVDVSDPEVQFNWYVDGVEVHNAKTKPREEQFNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKGLPSSIEKTI SKAKGQ PREPQVYTLPPSQEEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTPPVLDSDGSFFLYSRLTVDKSRWQEGNVFSCSVMEALHNHYTQKSLSLSLGK
352 416E01 - full heavy chain sequence	Nucleic acid sequence of 416E01 heavy chain	GAAGTGCAACTGGCGGAGTCTGGGGGAGGCTTGGTACAGCCGGGGGGTCCCTGAGACTCTCCTGTGCAGCCTCTGGATTACCTTTAGCAACTATGCCATGAGTTGGGTCCGCCAGACTCCAGGAAAGGGGCTGGAGTGGGTCTCAGCTATTAGTTTTAGTGGTGGTACTACATACTACGCTGACTCCGTGAAGGGCCGGTTCAACATCTCCAGAGACAATTCCAAGAACACGCTGTATTGTCACATGAACAGCCTGAGAGCCGATGACACGGCCGTATATTACTGTGCGAAAGATGAGGCACACGCTGGCGCAACCTTCTTTGACTCCTGGGGCCAGGGAACGCTGGTCACCGTCTCCTCAGCCAGCACCAAGGGCCCTTCCGTGTTCCCCCTGGCCCTTGACAGCAGGAGCACCTCCGAATCCACAGCTGCCCTGGGCTGTCTGGTGAAGGACTACTTCCCGAGCCCGTGACCGTGAGCTGGAACAGCGGCGCTCTGACATCGGCGTCCACACCTTTCTCGCGTCTCTGCAGTCTCCCGGCCTCTACTCCCTGTCTCCGTGGTGACCGTGCTAGCTCCTCCCTCGGCACCAAGACCTACACCTGTAACGTGGACCACAAACCTTCAACACCAAGGTGGACAAACGGGTGAGAGCAAGTACGGCCCTCCTGCGCTCCTTGTCTGCCCCGAGTTTCGAAGGCGGACCCAGCGTGTCTCTGTTCCCTCTAAGCCCAAGGACACCCCTCATGATCAGCCGGACACCCGAGGTGACCTGCGTGGTGGTGGATGTGAGCCAGGAGGACCTGAGGTCAGTTCAACTGGTATGTGGATGGCGTGGAGGTGCACAACGCCAAGACAAGCCCCGGGAAGAGCAGTTCAACTCCACCTACAGGGTGGTCAGCGTGTGACCGTGTCTGCATCAGGACTGGCTGAACGGCAAGGAGTACAAGTGC AAGGTACGCAATAAGGGACTGCCAGCAGCATCGAGAAGACCATCTCCAAGGCTAAAGGCCAGCCCCGGGAACCTCAGGTGTACACCTTGCC TCCAGCCAGGAGGAGATGACCAAGAACCAGGTGAGCCTGACCTGCTGCTGGTGAAGGATTTACCTTCCGACATCGCGTGGAGTGGGAGTCCAACGGCCAGCCCGAGAAACAATTATAAGACCACCCCTCCCGTCCTCGACAGCGACGGATCCTTCTTTCTGTACTCCAGGCTGACCGTGGAT AAGTCCAGGTGGCAGGAAGGCAACGTGTTCAAGTGTCTCCGTGATGCACGAGGCCCTGCACAATCACTACACCCAGAGTCCCTGAGCCTGTGCCTGGGAAAG
353 416E01 - CDRL1 (IMGT)	Amino acid sequence of CDRL1 of 416E01 using IMGT	QGIRRW
354 416E01 - CDRL2 (IMGT)	Amino acid sequence of CDRL2 of 416E01 using IMGT	GAS
355 416E01 - CDRL3 (IMGT)	Amino acid sequence of CDRL3 of 416E01 using IMGT	QQANSFPIT

TABLE S2-continued

SEQ ID NOS: 343-538			
SEQ ID NO:	Name	Description	Sequence
356	416E01 - CDR11 (Kabat)	Amino acid sequence of CDR11 of 416E01 using Kabat	RASQGIRRWLA
357	416E01 - CDR12 (Kabat)	Amino acid sequence of CDR12 of 416E01 using Kabat	GASSLQS
358	416E01 - CDR13 (Kabat)	Amino acid sequence of CDR13 of 416E01 using Kabat	QQANSFPIT
359	416E01 - Light chain variable region	Amino acid sequence of V_L of 416E01 (mutations from germline are shown in bold letters)	DIQMTQSPSSVSASVGDRTITCRASQGIRRWLAWYQQKPGKAPKLLIS G ASSLQSGVPSRFSGSGSGTDFTLLIITSLQPEDFATYYCQQANSPFITFGQGRLEIK
360	416E01 - Light chain variable region	Nucleic acid sequence of V_L of 416E01	GACATCCAGATGACCCAGTCTCCATCTTCCGTGTCTGCATCTGTAGGAGACAGAGTCACCATCACTTGTCTGGGCGAGTCAGGGTATTAGGAGGTGGTTAGCCTGGTATCAGCAGAAACCAGGAAAGCCCCTAAACTCTGATCTCTGGTGCATCCAGTTTGCAAAGTGGGGTCCCATCAAGGTTACAGCGGCAGTGGATCTGGGACAGATTTCACTCTCATCATTACCAGTCTGCAGCCTGAAGATTTTGCAACTTACTATTGTCAACAGGCTAACAGTTTCCCGATCACCTTCGGCCAAGGGACACGACTGGAGATCAAAC
361	416E01 - full light chain sequence	Amino acid sequence of 416E01 light chain	DIQMTQSPSSVSASVGDRTITCRASQGIRRWLAWYQQKPGKAPKLLISGASSLQSGVPSRFSGSGSGTDFTLLIITSLQPEDFATYYCQQANSFPITFGQGRLEIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQSGNSQESVTEQDSKSTYSLSSLTSLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC
362	416E01 - full light chain sequence	Nucleic acid sequence of 416E01 light chain	GACATCCAGATGACCCAGTCTCCATCTTCCGTGTCTGCATCTGTAGGAGACAGAGTCACCATCACTTGTCTGGGCGAGTCAGGGTATTAGGAGGTGGTTAGCCTGGTATCAGCAGAAACCAGGAAAGCCCCTAAACTCTGATCTCTGGTGCATCCAGTTTGCAAAGTGGGGTCCCATCAAGGTTACAGCGGCAGTGGATCTGGGACAGATTTCACTCTCATCATTACCAGTCTGCAGCCTGAAGATTTTGCAACTTACTATTGTCAACAGGCTAACAGTTTCCCGATCACCTTCGGCCAAGGGACACGACTGGAGATCAAACGTACGGTGGCCGCTCCCTCCGTGTTTCATCTTCCCACCTTCCGACGACGAGCTGAAGTCCGGCACCCGCTTCTGTCGTGTGCCTGCTGAACAACCTTCTACCCCGCGAGGCCAAGGTGCAGTGGAAAGGTGGACAACGCCCCTGCACTCCGGCAACTCCAGGAATCCGTGACCGAGCAGGACTCCAAAGGACAGCACCTACTCCCTGTCTCCACCCCTGACCCCTGTCCAAGGCCGACTACGAGAAGCACAAAGGTGTACGCCTGCGAAGTGACCCACCAGGCGCTGTCTAGCCCCGTGACCAAGTCTTTCAACCGGGCGAGTGT
363	STIM001 - CDRH1	Amino acid sequence of CDRH1 of STIM001 using IMGT	GYTFSTFG
364	STIM001 - CDRH2	Amino acid sequence of CDRH2 of STIM001 using IMGT	ISAYNGDT
365	STIM001 - CDRH3	Amino acid sequence of CDRH3 of STIM001 using IMGT	ARSSGHYYYYGMDV
366	STIM001 - Heavy chain variable region	Amino acid sequence of V_H of STIM001	QVQVVQSGAEVKKPGASVKVSKASGYTFSTFGITWVRQAPGQGLEWMGWISAYNGDTNYAQNLRGRVIMTTDTSTSTAYMELRSLRSDDTAVYYCARSSGHYYYYGMDVWGQGTITVTVSS
367	STIM001 - Heavy chain variable region	Nucleic acid sequence of V_H of STIM001	CAGGTTCAAGTGGTGCAGTCTGGAGCTGAGGTGAAGAAGCCTGGGGCCTCAGTGAAGGTCTCTGCAAGGCTTCTGGTTACACCTTTTCCACTTTGGTATCACCTGGGTGCGACAGGCCCTGGACAAGGGCTTGAAATGGATGGGATGGATCAGCGCTTACAATGGTGACACAACTATGCACAGAATCTCCAGGGCAGAGTCATCATGACCACAGACACATCCACGACACAGCCTACATGGAGCTGAGGAGCCTGAGATCTGACGACACGGCGTTTATTACTGTGCGAGGAGCAGTGGCCACTACTACTACTACGGTATGGACGTCTGGGGCCAAGGACACCGGTACCGTCTCCTCA

TABLE S2-continued

SEQ ID NOS: 343-538			
SEQ ID NO: Name	Description	Sequence	
368 STIM001 - full heavy chain sequence	Amino acid sequence of STIM001 heavy chain	QVQVVQSGAEVKKPGASVKVSCKASGYTFSFGITWVRQAPGQGLE WMGWISAYNGDTNYAQNLRVIMTTDTSTSTAYMELRSLRSDDTA VYYCARSSGHYYYGMDVWGQGTIVTVSSASTKGPSVFPLAPSSKS TSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLY SLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKHTHC PPCPAPELLGGPSVFLFPPPKDITLMISRTPEVTCVVVDVSHEDPE VKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKE YKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVS LTKLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFPLYSKL TVDKSRWQGNVFSQVMHEALHNHYTQKSLSLSPGK	
369 STIM001 - full heavy chain sequence	Nucleic acid sequence of STIM001 heavy chain	CAGGTTCCAGGTGGTGCAGTCTGGAGCTGAGGTGAAGAAGCCTGGGG CCTCAGTGAAGGTCTCCTGCAAGGCTTCTGGTTACACCTTTTCCAC CTTTGGTATCACCTGGGTGCGACAGGCCCTTGGACAAGGGCTTGAA TGGATGGGATGGATCAGCGCTTACAAATGGTGACACAACTATGCAC AGAATCTCCAGGGCAGAGTCATCATGACCACAGACACATCCACGAG CACAGCTACATGGAGCTGAGGAGCCTGAGATCTGACGACACGGCC GTTTATTACTGTGCGAGGAGCAGTGGCCACTACTACTACTACGGTA TGACAGTCTGGGGCCAAGGGACACCGTCAACCGTCTCCTCAGCCAG CACCAAGGGCCCCCTCTGTGTTCCCTCTGGCCCCCTTCCAGCAAGTCC ACCTCTGGCGGAACAGCCGCTCTGGGCTGCCTCGTGAAGGACTACT TCCCCGAGCCTGTGACCGTGTCTGGAACTCTGGCGCTCTGACCAG CGGAGTGCACACCTTCCCTGCTGTGCTGCAGTCTCCGGCCTGTAC TCCCTGTCTCTCCGTCTGACCGTGCCTTCCAGCTCTCTGGGCACCC AGACCTACATCTGCAACGTGAACCAAGCCCTCCAACACCAAGGT GGACAAGAAGGTGAACCCAGTCTTGCACAAAGCCACACCTGT CCCCCCTGTCTCTGCCCCCTGAAGTCTGGGCGGACCTTCCGTGTTC TGTCCCCCCCCAAGGCCAAGGACACCTGATGATCTCCCGGACCCC CGAAGTGACCTGCGTGGTGGTGGATGTGTCCACGAGGACCCGTAA GTGAAGTTCAATTGGTACGTGGACGGCGTGAAGTGACAAACGCCA AGACCAAGCCTAGAGAGGAACAGTACAACCTCCACCTACCGGGTGGT GTCCGTGCTGACCGTGTGACACAGGATTGGCTGAACGGCAAGAG TACAAGTGCAAGGTGTCCAACAAGGCCCTGCCTGCCCCCATCGAAA AGACCATCTCCAAGGCCAAGGGCCAGCCCCGGGAACCCAGGTGTA CACACTGCCCCCTAGCAGGGACGAGCTGACCAAGAACCAGGTGTCC CTGACCTGTCTCTGTAAAGGCTTCTACCCCTCCGATATCGCCGTGG AATGGGAGTCCAACGGCCAGCCTGAGAACAACACAGACACCCCC CCCTGTGCTGGACTCCGACGGCTCATTTCTTCTGTACAGCAAGCTG ACAGTGGACAAGTCCCGTGGCAGCAGGGCAACGTGTTCTCCTGCT CCGTGATGCACGAGGCCCTGCACAACCACTACACCCAGAAGTCCCT GTCCTGAGCCCCGGCAAGTGATGA	
370 STIM001 - CDRL1	Amino acid sequence of CDRL1 of STIM001 using IMGT	QSLLSNEYNY	
371 STIM001 - CDRL2	Amino acid sequence of CDRL2 of STIM001 using IMGT	LGS	
372 STIM001 - CDRL3	Amino acid sequence of CDRL3 of STIM001 using IMGT	MQSLQTPLT	
373 STIM001 - Light chain variable region	Amino acid sequence of V_L of STIM001	DIVMTQSPLSLPVTGPGEPAISCRSSQSLLSNEYNYLDWYLQKPG QSPQLLIFLGSNRASGVDPDRFSGSGSDFTLKIITRVEADVGIYY CMQSLQTPLTFGGGTKVEIK	
374 STIM001 - Light chain variable region	Nucleic acid sequence of V_L of STIM001	GATATTGTGATGACTCAGTCTCCACTCTCCCTGCCCGTCACCCCTG GAGAGCCGGCCTCCATCTCCTGCAGGTCTAGTCAGAGCCTCCTGCA TAGTAATGAATACAACCTATTGGATTGGTACCTGCAGAAAGCCAGGG CAGTCTCCACAGCTCCTGATCTTTTGGGTCTAATCGGGCCTCCG GGTCCCTGCAGGTTCACTGGCAGTGATCAGGCACAGATTTTAC ACTGAAAATCACCAGAGTGGAGCTGAGGATGTTGGAATTTATTAC TGCATGCAATCTTACAAACTCCGCTCACTTTCGGCGAGGGACCA AGGTGGAGATCAAA	

TABLE S2-continued

SEQ ID NOS: 343-538			
SEQ ID NO:	Name	Description	Sequence
375	STIM001 - full light chain sequence	Amino acid sequence of STIM001 light chain	DIVMTQSPFLSLPVTGPGEPAISCRSSQSLLSNEYNYLDWYLQKPG QSPQLLIFLGSNRASGVPDRFSGSGSDFTLKI TRVEAEDVGIYY CMQSLQTPPLTFGGGTKVEIK RTVAAPSVFIFPPSDEQLKSGTASVVCCLNNFYPREAKVQWKVDNA LQSGNSQESVTEQDSKSTYSLSSSTLTLSKADYEKHKVYACEVTHQ GLSSPVTKSFNRGEC
376	STIM001 - full light chain sequence	Nucleic acid sequence of STIM001 light chain	GATATTGTGATGACTCAGTCTCCACTCTCCCTGCCCGTCACCCCTG GAGAGCCGGCCTCCATCTCCTGCAGGTCTAGTCAGAGCCTCCTGCA TAGTAATGAATACAACTATTGGATTGGTACCTGCAGAAAGCAGGG CAGTCTCCACAGCTCCTGATCTTTTGGGTTCTAATCGGGCCTCCG GGTCCCTGACAGGTTCAGTGGCAGTGGATCAGGCACAGATTTTAC ACTGAAATCACCAGAGTGGAGGCTGAGGATGTTGGATTATTTAC TGCATGCAATCTCTACAACTCCGCTCACTTTCGGCGGAGGGACCA AGGTGGAGATCAAACgtacggtggccgctccctccgtgttcattctt cccaccttccgacgagcagctgaagtccggcaccgcttctgtcgtg tgctgtgtaacaacttctacccccgcgagggccaaggtgcagtgga aggtggacaacgccttgcagtcgggcaactcccaggaatccgtgac cgagcaggactccaaggacagcacctactccctgtcctccacccctg acctgtccaaggccgactacgagaagcacaaggtgtacgcctgcg aagtgacccaccaggccctgtctagccccgtgaccaagtctttcaa ccggggcgagtg
377	STIM002 - CDRH1	Amino acid sequence of CDRH1 of STIM002 using IMGT	GYTFTSYG
378	STIM002 - CDRH2	Amino acid sequence of CDRH2 of STIM002 using IMGT	ISAYNGNT
379	STIM002 - CDRH3	Amino acid sequence of CDRH3 of STIM002 using IMGT	ARSTYFYGSGTLYGMDV
380	STIM002 - Heavy chain variable region	Amino acid sequence of V_H of STIM002	QVQLVQSGGEVKKPGASVKVSKASGYTFTSYGFSWVRQAPGQGLE WMGWISAYNGNTNYAQLQGRVTMTDTSTSTAYMELRSLRSDDTA VYYCARSTYFYGSGTLYGMDVWGQGTVTTVSSASTKGPSVFPPLAPS SKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSS GLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKT HTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVDVDSHE DPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLN GKEYKCKVSNKALPAPIEKTI SAKAGQPREPQVYTLPPSRDELTKN QVSLTCLVKGFPYSDIAVEWESNGQPENNYKTPPVLDSDGSFFLY SKLTVDKSRWQQGNVFCSCVMHEALHNHYTQKSLSLSPGK
381	STIM002 - Heavy chain variable region	Nucleic acid sequence of V_H of STIM002	CAGGTTCAACTGGTGAGTCTGGAGGTGAGGTGAAGAAGCCTGGGG CCTCAGTGAAGGTCTCCTGCAAGGCTTCTGGTTACACCTTTACCAG CTATGGTTTCAGCTGGGTGCGACAGGCCCTTGACAAAGGACTAGAG TGATGGGATGGATCAGCGCTTACAATGGTAACACAACTATGCAC AGAAGCTCCAGGGCAGAGTACCATGACCACAGACACATCCACGAG CACAGCCTACATGGAGCTGAGGAGCTTGAGATCTGACGACACGGCC GTGTATTACTGTGCGAGATCTACGTATTTCTATGGTTCGGGGACCC TCTACGGTATGGACGTCTGGGGCCAAAGGACCACGGTCACCGTCTC CTCA
382	STIM002 - full heavy chain sequence	Amino acid sequence of STIM002 heavy chain	QVQLVQSGGEVKKPGASVKVSKASGYTFTSYGFSWVRQAPGQGLE WMGWISAYNGNTNYAQLQGRVTMTDTSTSTAYMELRSLRSDDTA VYYCARSTYFYGSGTLYGMDVWGQGTVTTVSSASTKGPSVFPPLAPS SKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSS GLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKT HTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVDVDSHE DPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLN GKEYKCKVSNKALPAPIEKTI SAKAGQPREPQVYTLPPSRDELTKN QVSLTCLVKGFPYSDIAVEWESNGQPENNYKTPPVLDSDGSFFLY SKLTVDKSRWQQGNVFCSCVMHEALHNHYTQKSLSLSPGK
383	STIM002 - full heavy chain sequence	Nucleic acid sequence of STIM002 heavy chain	CAGGTTCAACTGGTGAGTCTGGAGGTGAGGTGAAGAAGCCTGGGG CCTCAGTGAAGGTCTCCTGCAAGGCTTCTGGTTACACCTTTACCAG CTATGGTTTCAGCTGGGTGCGACAGGCCCTTGACAAAGGACTAGAG TGATGGGATGGATCAGCGCTTACAATGGTAACACAACTATGCAC AGAAGCTCCAGGGCAGAGTACCATGACCACAGACACATCCACGAG CACAGCCTACATGGAGCTGAGGAGCTTGAGATCTGACGACACGGCC GTGTATTACTGTGCGAGATCTACGTATTTCTATGGTTCGGGGACCC TCTACGGTATGGACGTCTGGGGCCAAAGGACCACGGTCACCGTCTC CTCA GCCAGCACCAAGGGCCCTCTGTGTTCCCTCTGGCCCTTCCAGCA AGTCCACCTCTGGCGGAACAGCCGCTCTGGGCTGCCTCGTGAAGGA

TABLE S2-continued

SEQ ID NOS: 343-538		
SEQ ID NO: Name	Description	Sequence
391 STIM002-B - CDRH1	Amino acid sequence of CDRH1 of STIM002-B using IMGT	GYTFTSYG
392 STIM002-B - CDRH2	Amino acid sequence of CDRH2 of STIM002-B using IMGT	ISAYNGNT
393 STIM002-B - CDRH3	Amino acid sequence of CDRH3 of STIM002-B using IMGT	ARSTYFYGSGTLYGMDV
394 STIM002-B - Heavy chain variable region	Amino acid sequence of V_H of STIM002-B	QVQLVQSGGEVKKPGASVKVSKASGYTFTSYGFSWVRQAPGQGLE WMGWISAYNGNTNYAQLQGRVTMTDTSTSTAYMELRSLRSDDTA VYYCARSTYFYGSGTLYGMDVWGQGTIVTVSS
395 STIM002-B - Heavy chain variable region	Nucleic acid sequence of V_H of STIM002-B	CAGGTTCAACTGGTGCAGTCTGGAGGTGAGGTGAAGAAGCCTGGGG CCTCAGTGAAGGTCTCCTGCAAGGCTTCTGGTTACACCTTTACCAG CTATGGTTTCAGCTGGGTGCGACAGGCCCTGGACAAAGGACTAGAG TGGATGGGATGGATCAGCGCTTACAATGGTAACACAAACTATGCAC AGAAGCTCCAGGGCAGAGTCACCATGACCACAGACACATCCACGAG CACAGCCTACATGGAGCTGAGGAGCTTGAGATCTGACGACACGCCC GTGTATTACTGTGCGAGATCTACGTATTCTATGGTTCGGGGACCC TCTACGGTATGGACGTCTGGGGCCAAGGGACCACGGTCACCGTCTC CTCA
396 STIM002-B - full heavy chain sequence	Amino acid sequence of STIM002-B heavy chain	QVQLVQSGGEVKKPGASVKVSKASGYTFTSYGFSWVRQAPGQGLE WMGWISAYNGNTNYAQLQGRVTMTDTSTSTAYMELRSLRSDDTA VYYCARSTYFYGSGTLYGMDVWGQGTIVTVSSASTKGPSVFPLAPS SKTSSGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSS GLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKT HTCPPCPAPELLGGPSVFLFPPKPKDILMISRTPETCVVVDVSH DPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLN GKEYCKKVSINKALPAPIEKTI SKAKGQPREPQVYTLPPSRDELTKN QVSLTCLVKGFPYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLY SKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK
397 STIM002-B - full heavy chain sequence	Nucleic acid sequence of STIM002-B heavy chain	CAGGTTCAACTGGTGCAGTCTGGAGGTGAGGTGAAGAAGCCTGGGG CCTCAGTGAAGGTCTCCTGCAAGGCTTCTGGTTACACCTTTACCAG CTATGGTTTCAGCTGGGTGCGACAGGCCCTGGACAAAGGACTAGAG TGGATGGGATGGATCAGCGCTTACAATGGTAACACAAACTATGCAC AGAAGCTCCAGGGCAGAGTCACCATGACCACAGACACATCCACGAG CACAGCCTACATGGAGCTGAGGAGCTTGAGATCTGACGACACGCCC GTGTATTACTGTGCGAGATCTACGTATTCTATGGTTCGGGGACCC TCTACGGTATGGACGTCTGGGGCCAAGGGACCACGGTCACCGTCTC CTCAGCCAGACCAAGGGCCCTCTGTGTTCCCTCTGGCCCTTCC AGCAAGTCCACCTCTGGCGGAACAGCCGCTCTGGGCTGCCTCGTGA AGGACTACTTCCCGAGCCTGTGACCGTGTCTGGAACTCTGGCGC TCTGACCAGCGAGTGACACCTTCCCTGCTGTGCTGCAGTCTCTCC GGCCTGTACTCCCTGTCTCCGTCGTGACCGTGCCCTCCAGCTCTC TGGGCACCCAGACCTACATCTGCAACGTGAACCAAGCCCTCCAA CACCAGGTGGACAAGAAGGTGAACCAAGTCTGCGACAAGACC CACACCTGTCCCCCTTGTCCTGCCCTGAACTGCTGGGCGGACCTT CCGTGTTCCCTGTTCCCCCAAGCCCAAGGACACCTGATGATCTC CCGACCCCCGAAGTGACCTGCGTGGTGGATGTGTCCACGAG GACCCCTGAAGTGAAGTCAATTGGTACGTGGACGGCGTGAAGTGC ACAACGCCAAGACCAAGCCTAGAGAGGAACAGTACAACCTCACTA CCGGGTGGTGTCCGTGCTGACCGTGTGCAACAGGATTGGCTGAAC GGCAAAGAGTACAAGTGCAAGGTGTCCAACAAGGCCCTGCCTGCC CCATCGAAAAGACCATCTCCAAGGCCAAGGGCCAGCCCCGGGAACC CCAGGTGTACACACTGCCCTTAGCAGGGACGAGCTGACCAAGAAC CAGGTGTCCCTGACCTGTCTCGTGAAAGGCTTCTACCCCTCCGATA TCGCCGTGGAATGGGAGTCCAACGGCCAGCCTGAGAACAACCTACA GACCACCCCCCTGTGCTGGACTCCGACGGCTCATTCTTCTGTAC AGCAAGCTGACAGTGGACAAGTCCCGTGGCAGCAGGGCAACGTGT TCTCTGCTCCGTGATGCACGAGGCCCTGCACAACCACTACACCCA GAAGTCCCTGTCCCTGAGCCCCGGCAAGTGTATGA

TABLE S2-continued

SEQ ID NOS: 343-538			
SEQ ID NO: Name	Description	Sequence	
398 STIM002-B - CDRL1	Amino acid sequence of CDRL1 of STIM002-B using IMGT	QSLHSDGYNC	
399 STIM002-B - CDRL2	Amino acid sequence of CDRL2 of STIM002-B using IMGT	LGS	
400 STIM002-B - CDRL3	Amino acid sequence of CDRL3 of STIM002-B using IMGT	MQALQTPCS	
401 STIM002-B - Light chain variable region	Amino acid sequence of V_L of STIM002-B	DIVMTQSPSLPVPVTPGEPASISCRSSQSLHSDGYNC LDWYLQKPG QSPQLLIYLGSTRASGFPDRFSGSGSGTDFTLKI SRVEAEDVG VYY CMQALQTPCSFGQGTKLEIK	
402 STIM002-B - Light chain variable region	Nucleic acid sequence of V_L of STIM002-B	GATATTGTGATGACTCAGTCTCCACTCTCCCTGCCCGTCACCCCTG GAGAGCCGGCCTCCATCTCCTGCAGGTCTAGTCAGAGCCTCCTGCA TAGTGATGGATACAACGTGTTGGATTGGTACCTGCAGAGCCAGGG CAGTCTCCACAGCTCCTGATCTATTTGGGTCTACTCGGGCCTCCG GGTTCCTGACAGGTTTCAGTGGCAGTGGATCAGGCACAGATTTTAC ACTGAAAATCAGCAGAGTGGAGGCTGAGGATGTTGGGGTTTATTAC TGCATGCAAGCTCTACAACTCCGTGCAGTTTGGCCAGGGGACCA AGCTGGAGATCAAA	
403 STIM002-B - full light chain sequence	Amino acid sequence of STIM002-B light chain	DIVMTQSPSLPVPVTPGEPASISCRSSQSLHSDGYNC LDWYLQKPG QSPQLLIYLGSTRASGFPDRFSGSGSGTDFTLKI SRVEAEDVG VYY CMQALQTPCSFGQGTKLEIK RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA LQSGNSQESVTEQDSKDSYLSSTLTLSKADYEKHKVYACEVTHQ GLSSPVTKSFNRGEC	
404 STIM002-B - full light chain sequence	Nucleic acid sequence of STIM002-B light chain	GATATTGTGATGACTCAGTCTCCACTCTCCCTGCCCGTCACCCCTG GAGAGCCGGCCTCCATCTCCTGCAGGTCTAGTCAGAGCCTCCTGCA TAGTGATGGATACAACGTGTTGGATTGGTACCTGCAGAGCCAGGG CAGTCTCCACAGCTCCTGATCTATTTGGGTCTACTCGGGCCTCCG GGTTCCTGACAGGTTTCAGTGGCAGTGGATCAGGCACAGATTTTAC ACTGAAAATCAGCAGAGTGGAGGCTGAGGATGTTGGGGTTTATTAC TGCATGCAAGCTCTACAACTCCGTGCAGTTTGGCCAGGGGACCA AGCTGGAGATCAAAcgtacggtggccgctccctccgtgttcattt cccaccttccgacgagcagctgaagtcggcaccgcttctgtcgtg tgctgtgtaacaacttctacccccgcaggccaaggtgcagtgga aggtggacaacgccctgcagtcgggcaactcccaggaatccgtgac cgagcaggactccaaggacagcacctactccctgtcctccaccctg accctgtccaaggccgactacgagaagcacaaggtgtacgcctgcg aagtgacccaccaggccctgtctagccccgtgaccaagtccttcaa ccggggcgagtggt	
405 STIM003 - CDRH1	Amino acid sequence of CDRH1 of STIM003 using IMGT	GVTFDDYG	
406 STIM003 - CDRH2	Amino acid sequence of CDRH2 of STIM003 using IMGT	INWNGGDT	
407 STIM003 - CDRH3	Amino acid sequence of CDRH3 of STIM003 using IMGT	ARDFYSGSYHYHVPFDY	
408 STIM003 - Heavy chain variable region	Amino acid sequence of V_H of STIM003	EVQLVESGGGVVVRPGGSLRLSCVASGVTFDYGMSSWRQAPGKGLE WSGINWNGGDTDYSDSVKGRFTISRDNKNSLYLQMNSLRAEDTA LYYCARDFYSGSYHYHVPFDYWGQGILVTVSS	

TABLE S2-continued

SEQ ID NOS: 343-538			
SEQ ID NO: Name	Description	Sequence	
409 STIM003 - Heavy chain variable region	Nucleic acid sequence of V_H of STIM003	GAGGTGCAGCTGGTGGAGTCTGGGGGAGGTGTGGTACGGCCTGGGG GGTCCCTGAGACTCTCCTGTGTAGCCTCTGGAGTCACCTTTGATGA TTATGGCATGAGCTGGGTCCGCCAAGCTCCAGGGAAGGGCTGGAR TGGGTCTCTGGTATTAATTGGAATGGTGGCGACACAGATTATTTCAG ACTCTGTGAAGGGCCGATTACCATCTCCAGAGACACGCCAAGAA CTCCCTGTATCTACAAATGAATAGTCTGAGAGCCGAGGACACGGCC TTGTATTACTGTGCGAGGGATTCTATGGTTCGGGGAGTTATTATC ACGTTCTCTTTGACTACTGGGGCCAGGGAATCCTGGTCACCGCTCTC CTCA	
410 STIM003 - full heavy chain sequence	Amino acid sequence of STIM003 heavy chain	EVQLVESGGGVVVRPGGSLRLSCVASGVTFDYGMSSWRQAPGKGLE WVSGINWNGDSTDYSDSVKGRFTISRDNAKNSLYLQMNSLRAEDTA LYYCARDFYGSGSYHVPFDYWGQILVTVSSASTKGPSVFPPLAPS SKSTSGGTAAALGCLVKDYFPEPVTWNSGALTSGVHTFPAVLQSS GLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKEPKSCDKT HTCPPCPAPELLGGPSVFLFPPKPKDLMISRTPEVTCVVDVSHE DPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLN GKEYKCKVSNKALPAPIEKTI SKAKGQPREPQVYITLPPSRDELTKN QVSLTCLVKGFPYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLY SKLTVDKSRWQQGNVFCSCVMHEALHNHYTQKSLSLSPGK	
411 STIM003 - full heavy chain sequence	Nucleic acid sequence of STIM003 heavy chain	GAGGTGCAGCTGGTGGAGTCTGGGGGAGGTGTGGTACGGCCTGGGG GGTCCCTGAGACTCTCCTGTGTAGCCTCTGGAGTCACCTTTGATGA TTATGGCATGAGCTGGGTCCGCCAAGCTCCAGGGAAGGGCTGGAR TGGGTCTCTGGTATTAATTGGAATGGTGGCGACACAGATTATTTCAG ACTCTGTGAAGGGCCGATTACCATCTCCAGAGACACGCCAAGAA CTCCCTGTATCTACAAATGAATAGTCTGAGAGCCGAGGACACGGCC TTGTATTACTGTGCGAGGGATTCTATGGTTCGGGGAGTTATTATC ACGTTCTCTTTGACTACTGGGGCCAGGGAATCCTGGTCACCGCTCTC CTCAGCCAGCACCAAGGGCCCTCTGTGTTCCCTCTGGCCCTTCC AGCAAGTCCACCTCTGGCGGAACAGCCGCTCTGGGCTGCCTCGTGA AGGACTACTTCCCGAGCCTGTGACCGTGTCTTGGAACTCTGGCGC TCTGACCAGCGGAGTGACACCTTCCCTGCTGTGCTGACGTCTCTCC GGCTGTACTCCCTGTCTCCGTCTGACCGTGCTTCCAGCTCTC TGGGCACCCAGACCTACATCTGAACGTGAACCACAAGCCCTCCAA CACCAAGGTGGACAAGAAGGTGGAACCAAGTCTCGCACAAGACC CACACCTGTCCCTTGTCTGCCCCGAACTGCTGGGCGGACCTT CCGTGTCTCTGTTCCCCCAAGCCCAAGGACACCTGTATGATCTC CCGGACCCCGAAGTGACCTGCGTGGTGGTGGATGTGTCCACAGAG GACCTGAAGTGAAGTTCAATTGGTACGTGGACGGCGTGGAAAGTGC ACAACGCCAAGACCAAGCCTAGAGAGGAACAGTACAACCTCACCTA CCGGGTGGTGTCCGTGTGACCGTGTGACACAGGATTGGCTGAAC GGCAAAGAGTACAAGTGCAAGGTGTCCACAAGGCCCTGCCTGCC CCATCGAAAAGACCATCTCCAAGGCCAAGGGCCAGCCCCGGGAACC CCAGGTGTACACACTGCCCTTAGCAGGGACGAGCTGACCAAGAAC CAGGTGTCCCTGACCTGTCTCGTGAAAGGCTTCTACCCCTCCGATA TCGCCGTGGAATGGGAGTCCAACGGCCAGCCTGAGAACAACACAA GACCACCCCCCTGTGCTGGACTCCGACGGCTCATTCTTCTGTGAC AGCAAGCTGACAGTGGACAAGTCCCGGTGGCAGCAGGGCAACGTGT TCTCTGCTCCGTGATGCACGAGGCCCTGCACAACCACTACACCCA GAAGTCCCTGTCTGAGCCCCGGCAAGTATGA	
412 STIM003 - CDRL1	Amino acid sequence of CDRL1 of STIM003 using IMGT	QSVRSY	
413 STIM003 - CDRL2	Amino acid sequence of CDRL2 of STIM003 using IMGT	GAS	
414 STIM003 - CDRL3	Amino acid sequence of CDRL3 of STIM003 using IMGT	HQYDMSFFT	
415 STIM003 - Light chain variable region	Amino acid sequence of V_L of STIM003	EIVLTQSPGTLISLSPGERATLSCRASQSVRSYLAHWYQQKRGQAPR LLIYGASSRATGIPDRFSGDGSFTDLTSLISRLPEDFAVVYCHQY DMSFFTFGPGTKVDIK	

TABLE S2-continued

SEQ ID NOS: 343-538		
SEQ ID NO: Name	Description	Sequence
416 STIM003 - Light chain variable region	Nucleic acid sequence of V_L of STIM003	GAAATTGTGTGACGCAGTCTCCAGGGACCTGTCTTTGTCTCCAG GGGAAAGAGCCACCCCTCTCCTGCAGGGCCAGTCAGAGTGTTAGCAG AAGCTACTTAGCCTGGTACCAGCAGAAACGTGGCCAGGCTCCAGG CTCCTCATCTATGGTGCATCCAGCAGGGCCACTGGCATCCCAGACA GGTTCAGTGGCGATGGGTCTGGGACAGACTTCACTCTCTCCATCAG CAGACTGGAGCCTGAAGATTTGCAGTGTATTACTGTCACCAAGTAT GATATGTCACCATTCACTTTCCGGCCCTGGGACCAAAGTGGATATCA AA
417 STIM003 - full light chain sequence	Amino acid sequence of STIM003 light chain	EIVLTQSPGTLSSLSPGERATLSCRASQSVRSYLAWYQQKRGQAPR LLIYGASSRATGIPDRFSGDGSFTDFTLSISRLEPEDFAVYCHQY DMSPFTFGPGTKVDIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLN NFYPREAKVQWKVDNALQSGNSQESVTEQDSKDSSTYLSSTLTLSK ADYEKHKVYACEVTHQGLSSPVTKSFNRGEC
418 STIM003 - full light chain sequence	Nucleic acid sequence of STIM003 light chain	GAAATTGTGTGACGCAGTCTCCAGGGACCTGTCTTTGTCTCCAG GGGAAAGAGCCACCCCTCTCCTGCAGGGCCAGTCAGAGTGTTAGCAG AAGCTACTTAGCCTGGTACCAGCAGAAACGTGGCCAGGCTCCAGG CTCCTCATCTATGGTGCATCCAGCAGGGCCACTGGCATCCCAGACA GGTTCAGTGGCGATGGGTCTGGGACAGACTTCACTCTCTCCATCAG CAGACTGGAGCCTGAAGATTTGCAGTGTATTACTGTCACCAAGTAT GATATGTCACCATTCACTTTCCGGCCCTGGGACCAAAGTGGATATCA AAcgtacggtggcgcgtccctccgtgttcatctcccaccttcga cgagcagctgaagtcgggacccgcttctgctggtgctgctgaac aacttctacccccgcgagggccaaggtgcagtggaaggtggacaacg ccctgcagtcgggcaactcccaggaatccgtgaccgagcaggactc caaggacagcacctactccctgtcctccaccctgaccctgtccaag gccgactacgagaagcacaaggtgtacgcctgcgaagtgacccacc agggcctgtctagccccgtgaccaagtctttcaaccggggcgagtg t
419 STIM004 - CDRH1	Amino acid sequence of CDRH1 of STIM004 using IMGT	GLTFDDYG
420 STIM004 - CDRH2	Amino acid sequence of CDRH2 of STIM004 using IMGT	INWNGDNT
421 STIM004 - CDRH3	Amino acid sequence of CDRH3 of STIM004 using IMGT	ARDYYGSGSYYNVPFDY
422 STIM004 - Heavy chain variable region	Amino acid sequence of V_H of STIM004	EVQLVESGGGVVRPGGSLRLSCAASGLTFDDYGMSWVRQVPGKGLE WVSGINWNGDNTDYADSVKGRFTISRDNAKNSLYLQMNSLRAEDTA LYYCARYDYGSGSYYNVPFDYWQGTLVTVSS
423 STIM004 - Heavy chain variable region	Nucleic acid sequence of V_H of STIM004	GAGGTGCAGCTGGTGGAGTCTGGGGGAGGTGTGGTACGGCCTGGGG GGTCCCTGAGACTCTCCTGTGCAGCCTCTGGACTCACCTTTGATGA TTATGGCATGAGCTGGGTCCGCCAAGTTCAGGGAAGGGCTGGAG TGGGTCTCTGGTATTAATTGGAATGGTGATAACACAGATTATGCAG ACTCTGTGAAGGGCCGATTCAACATCTCCAGAGACACGCCAAGAA CTCCCTGTATCTGCAATGAACAGTCTGAGAGCCGAGGACACGGCC TTGTATTACTGTGCGAGGGATTACTATGGTTCGGGGAGTTATTATA ACGTTCTCTTTGACTACTGGGGCCAGGGAACCTGGTCACCGTCTC CTCA
424 STIM004 - full heavy chain sequence	Amino acid sequence of STIM004 heavy chain	EVQLVESGGGVVRPGGSLRLSCAASGLTFDDYGMSWVRQVPGKGLE WVSGINWNGDNTDYADSVKGRFTISRDNAKNSLYLQMNSLRAEDTA LYYCARYDYGSGSYYNVPFDYWQGTLVTVSSASTKGPSVFPLAPS SKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSS GLYSLSSVTVTPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKT HTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSH DPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLN GKEYKCKVSNKALPAPIEKTIKAKGQPREPQVYTLPPSRDELTKN QVSLTCLVKGFPYPSDIAVEWESNGQPENNYKTPPVLDSDGSFPLY SKLTVDKSRWQQGNVFCSCVMHEALHNHYTQKSLSLSPGK

TABLE S2-continued

SEQ ID NOS: 343-538		
SEQ ID		
NO: Name	Description	Sequence
425 STIM004 - full heavy chain sequence	Nucleic acid sequence of STIM004 heavy chain	GAGGTGCAGCTGGTGGAGTCTGGGGGAGGTGTGGTACGGCCTGGGG GGTCCCTGAGACTCTCCTGTGCAGCCTCTGGACTCACCTTTGATGA TTATGGCATGAGCTGGGTCCGCCAAGTTCAGGGAAGGGGCTGGAG TGGGTCTCTGGTATTAAATTGGAATGGTGATAACACAGATTATGCAG ACTCTGTGAAGGGCCGATTACCATCTCCAGAGACAACGCCAAGAA CTCCCTGTATCTGCAATGAACAGTCTGAGAGCCGAGGACACGGCC TTGTATTACTGTGCGAGGGATTACTATGGTTCGGGGAGTTATTATA ACGTTCTCTTTGACTACTGGGGCCAGGGAACCCCTGGTCACCGTCTC CTCAGCCAGCACCAAGGGCCCTCTGTGTTCCCTCTGGCCCCCTCC AGCAAGTCCACCTCTGGCGGAACAGCCGCTCTGGGCTGCCTCGTGA AGGACTACTTCCCCGAGCCTGTGACCGTGTCTGGAACCTCTGGCGC TCTGACCAGCGGAGTGCACACCTTCCCTGCTGTGCTGCAGTCTCTCC GGCCTGTACTCCCTGTCTCCGTCGTGACCGTGCCTTCCAGCTCTC TGGGCACCCAGACCTACATCTGCAACGTGAACCACAAGCCCTCCAA CACCAAGGTGGACAAGAAGGTGGAACCCAAGTCTTGCACAAGACC CACACCTGTCCCCCTTGTCTGCCCTGAACTGCTGGGCGGACCTT CCGTGTTCTCTGTCCCCCAAGCCCAAGGACACCCCTGATGATCTC CCGGACCCCCGAAGTGACCTGCGTGGTGGTGGATGTGTCCACGAG GACCTGAAGTGAAGTTCAATTGGTACGTGGACGGCGTGAAGTGC ACAACGCCAAGACCAAGCCTAGAGAGGAACAGTACAACCTCCACCTA CCGGTGGTGTCCGTGCTGACCGTGTGCACCAAGGATTGGCTGAAC GGCAAAGAGTACAAGTGCAAGGTGTCCAACAAGGCCCTGCCTGCC CCATCGAAAAGACCATCTCCAAGGCCAAGGGCCAGCCCCGGGAACC CCAGGTGTACACACTGCCCTTAGCAGGGACGAGCTGACCAAGAAC CAGGTGTCTCTGACCTGTCTCGTGAAAGGCTTCTACCCCTCCGATA TCGCCGTGGAATGGGAGTCCAACGGCCAGCCTGAGAACAACCTACAA GACCACCCCTGTGCTGGACTCCGACGGCTCATTTCTTCTGTAC AGCAAGCTGACAGTGGACAAGTCCCGTGGCAGCAGGGCAACGTGT TCTCTGTCTCCGTGATGCACGAGGCCCTGCACAACCACTACACCCA GAAGTCCCTGTCCCTGAGCCCCGGCAAGTGATGA
426 STIM004 - CDRL1	Amino acid sequence of CDRL1 of STIM004 using IMGT	QSVSSSY
427 STIM004 - CDRL2	Amino acid sequence of CDRL2 of STIM004 using IMGT	GAS
428 STIM004 - CDRL3	Amino acid sequence of CDRL3 of STIM004 using IMGT	QQYGSSPF
429 STIM004 of Corrected light chain variable region	Amino acid sequence corrected V _L of STIM004	EIVLTQSPGTLSSLSPGERATLSCRASQSVSSSYLAWYQQKPGQAPR LLIYGASSRATGIPDRFSGSGSDFTLTIRLEPEDFAVYYCQQY GSSPFFGPGTKVDIK
430 STIM004 - Corrected light chain variable region	Nucleic acid sequence of corrected V _L of STIM004	GAAATTGTGTTGACGCAGTCTCCAGGCACCCCTGTCTTTGTCTCCAG GGGAAAGAGCCACCCTCTCCTGCAGGGCCAGTCAGAGTGTAGCAG CAGCTACTTAGCCTGGTACCAGCAGAAACCTGGCCAGGCTCCAGG CTCCTCATATATGGTGCATCCAGCAGGGCCACTGGCATCCCAGACA GGTTCAGTGGCAGTGGGTCTGGGACAGACTTCACTCTCACCATCAG AAGACTGGAGCCTGAAGATTTTGCAAGTGTATTACTGTACAGCATAT GGTAGTTCACCATTCCTTCGGCCCTGGGACCAAGTGGATATCAAA
431 STIM004 - Light chain variable region	Nucleic acid sequence of V _L of STIM004	GAAATTGTGTTGACGCAGTCTCCAGGCACCCCTGTCTTTGTCTCCAG GGGAAAGAGCCACCCTCTCCTGCAGGGCCAGTCAGAGTGTAGCAG CAGCTACTTAGCCTGGTACCAGCAGAAACCTGGCCAGGCTCCAGG CTCCTCATATATGGTGCATCCAGCAGGGCCACTGGCATCCCAGACA GGTTCAGTGGCAGTGGGTCTGGGACAGACTTCACTCTCACCATCAG AAGACTGGAGCCTGAAGATTTTGCAAGTGTATTACTGTACAGCATAT GGTAGTTCACCATTCCTTCGGCCCTGGGACCAAGTGGATATCAAA A

TABLE S2-continued

SEQ ID NOS: 343-538		
SEQ ID NO: Name	Description	Sequence
432 STIM004 - full corrected light chain sequence	Amino acid sequence of STIM004 light chain	EIVLTQSPGTL _{SL} SPGERATL _{SCR} ASQSVSSSYLAWYQQKPGQAPR LLIYGASSRATGIPDRFSGSGSGTD _{FTLT} IRRLEPEDFAVYYCQQY GSSPFFPGPKVDIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNN FYPREAKVQWKVDNALQSGNSQESVTEQDSKDSSTYSLSSTLTLSKA DYEKHKVYACEVTHQGLSSPVTKSFNRGEC
433 STIM004 - full corrected light chain sequence	Nucleic acid sequence of corrected STIM004 light chain	GAAATTGTGTTGACGCAGTCTCCAGGCACCCCTGTCTTTGTCTCCAG GGGAAAGAGCCACCCCTCTCCTGCAGGGCCAGTCAGAGTGTTAGCAG CAGCTACTTAGCCTGGTACCAGCAGAAACCTGGCCAGGCTCCAGG CTCCTCATATATGGTGCATCCAGCAGGGCCACTGGCATCCCAGACA GGTTCAGTGGCAGTGGGTCTGGGACAGACTTCACTCTCACCATCAG AAGACTGGAGCCTGAAGATTTGCAGTGTATTACTGTCAGCAGTAT GGTAGTTCACCATTCCTCGGCCCTGGGACCAAAGTGGATATCAAAC gtacgggtggcgcgtccctccgtgttcacatctcccaccttccgacga gcagctgaagtccggcaccgcttctgtcgtgtgctgctgaacaac ttctacccccgcgaggccaaggtgcagtggaaggtggacaacgccc tgagtcgagcgaactcccaggaatccgtgaccgagcaggactccaa ggacagcacctactccctgtcctccacccctgacctgtccaaggcc gactacgagaagcacaaggtgtacgcctgcgaagtgaccaccagg gcctgtctagccccgtgaccaagtctttcaaccggggcgagtgt
434 STIM004 - full light chain sequence	Nucleic acid sequence of STIM004 light chain	GAAATTGTGTTGACGCAGTCTCCAGGCACCCCTGTCTTTGTCTCCAG GGGAAAGAGCCACCCCTCTCCTGCAGGGCCAGTCAGAGTGTTAGCAG CAGCTACTTAGCCTGGTACCAGCAGAAACCTGGCCAGGCTCCAGG CTCCTCATATATGGTGCATCCAGCAGGGCCACTGGCATCCCAGACA GGTTCAGTGGCAGTGGGTCTGGGACAGACTTCACTCTCACCATCAG AAGACTGGAGCCTGAAGATTTGCAGTGTATTACTGTCAGCAGTAT GGTAGTTCACCATTCACCTCGGCCCTGGGACCAAAGTGGATATCAA Aagtcacgggtggcgcgtccctccgtgttcacatctcccaccttccgac gagcagctgaagtccggcaccgcttctgtcgtgtgctgctgaaca acttctacccccgcgaggccaaggtgcagtggaaggtggacaacgc cctgcagtcgagcgaactcccaggaatccgtgaccgagcaggactcc aaggacagcacctactccctgtcctccacccctgacctgtccaagg ccgactacgagaagcacaaggtgtacgcctgcgaagtgaccaccaca ggcgctgtctagccccgtgaccaagtctttcaaccggggcgagtgt
435 STIM005 - CDRH1	Amino acid sequence of CDRH1 of STIM005 using IMGT	GYTFNSYG
436 STIM005 - CDRH2	Amino acid sequence of CDRH2 of STIM005 using IMGT	ISVHNGNT
437 STIM005 - CDRH3	Amino acid sequence of CDRH3 of STIM005 using IMGT	ARAGYDILTD _{FS} DAFDI
438 STIM005 - Heavy chain variable region	Amino acid sequence of V_H of STIM005	QVQLVQSGAEVKKPGASVKVSKASGYTFNSYGI IWRQAPGQGLE WMGWI _{SVHNGNT} NC _Q KLQGRVTMTTDTSTSTAYMELRS _{LRT} DDTA VYYCARAGYDILTD _{FS} DAFDI _I WGHGTMVTVSS
439 STIM005 - Heavy chain variable region	Nucleic acid sequence of V_H of STIM005	CAGGTTCA _{GT} TGGTGCA _{GT} CTGGAGCTGAGGTGAAGAAGCCTGGGG CCTCAGTGAAGGTCTCCTGCAAGGCTTCTGGTTACACCTTAAATAG TTATGGTATCATCTGGGTGCGACAGGCCCTGGACAGGGCTTGAG TGGATGGGATGGATCAGCGTTCAATGGTAACACAAACGTGCAC AGAGCTCCAGGGTAGAGTCACCATGACCACAGACATCCACGAG CACAGCCTACATGGAGCTGAGGAGCCTGAGAACTGACGACACGCCC GTGTATTACTGTGCGAGAGCGGTTACGATATTTGACTGATTTT CCGATGCTTTTGATATCTGGGGCCACGGGACAATGGTCACCGTCTC TTCA
440 STIM005 - full heavy chain sequence	Amino acid sequence of STIM005 heavy chain	QVQLVQSGAEVKKPGASVKVSKASGYTFNSYGI IWRQAPGQGLE WMGWI _{SVHNGNT} NC _Q KLQGRVTMTTDTSTSTAYMELRS _{LRT} DDTA VYYCARAGYDILTD _{FS} DAFDI _I WGHGTMVTVSSASTKGPSVFPLAPS SKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSS GLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKT HTCPPCPAPPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVDVSHE DPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWL NGKEYCKVSNKALPAPIEKTI _{SK} AKGQPREPQVYTLPPSRDELTKN

TABLE S2-continued

SEQ ID NOS: 343-538		
SEQ ID NO: Name	Description	Sequence
		QVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLY SKLTVDKSRWQQGNVFCSCVMHEALHNHYTQKSLSLSPGK
441 STIM005 - full heavy chain sequence	Nucleic acid sequence of STIM005 heavy chain	CAGGTTTCAGTTGGTGCAGTCTGGAGCTGAGGTGAAGAAGCCTGGGG CCTCAGTGAAGGTCTCCTGCAAGGCTTCTGGTTACACCTTTAATAG TTATGGTATCATCTGGGTGCGACAGGCCCTGGACAAGGGCTTGAG TGGATGGGATGGATCAGCGTTTCAATGGTAACACAACTGTGCAC AGAAGCTCCAGGGTAGAGTACCATGACCACAGACACATCCACGAG CACAGCCTACATGGAGCTGAGGAGCCTGAGAACTGACGACACGGCC GTGTATTACTGTGCGAGAGCGGTTACGATATTTTGACTGATTTT CCGATGCTTTTGATATCTGGGGCCACGGGACAATGGTACCCGTCTC TTCA GCCAGCACCAAGGGCCCTCTGTGTTCCCTCTGGCCCTTCCAGCA AGTCCACCTCTGGCGGAACAGCCGCTCTGGGCTGCCTCGTGAAGGA CTACTTCCCCGAGCCTGTGACCGTGTCTTGGAACTCTGGCGCTCTG ACCAGCGGAGTGACACCTTCCCTGCTGTGCTGCAGTCTCCCGCC TGTAATCCCTGTCTCCGTGCTGACCGTGCCTTCCAGCTCTCTGGG CACCCAGACCTACATCTGCAACGTGAACCACAAGCCCTCCAACACC AAGGTGGACAAGAAGGTGGAACCAAGTCTTGGCACAAGACCCACA CCTGTCCCCCTTGTCTGCCCCTGAAGTGTGGGCGGACCTTCCGT GTTCTGTTCCTCCCAAGGCCAAGGACACCCCTGATGATCTCCCGG ACCCCGGAAGTGACCTGCGTGGTGGTGGATGTGTCCACGAGGACC CTGAAGTGAAGTTCAATTGGTACGTGGACGGCGTGAAGTGACAAA CGCCAAGACCAAGCCTAGAGAGGAACAGTACAACCTCCACCTACCCG GTGGTGTCTGCTGACCGTGTGCACAGGATTGGCTGAACGGCA AAGAGTACAAGTGCAAGGTGTCCAACAAGGCCCTGCTGCCCCAT CGAAAAGACCATCTCCAAGGCCAAGGGCCAGCCCCGGGAACCCAG GTGTACACACTGCCCCCTAGCAGGGACGAGCTGACCAAGAACCAGG TGTCCTTGACCTGTCTCGTGAAGGCTTCTACCCCTCCGATATCGC CGTGAATGGGAGTCCAACGGCCAGCCTGAGAACAACTACAAGACC ACCCCTCTGTGCTGGACTCCGACGGCTCATTCTTCTGTACAGCA AGCTGACAGTGGACAAGTCCCGGTGGCAGCAGGGCAACGTGTTCTC CTGCTCCGTGATGCACGAGGCCCTGCACAACCACTACACCCAGAAG TCCCTGTCCCTGAGCCCCGGCAAGTGATGA
442 STIM005 - CDRL1	Amino acid sequence of CDRL1 of STIM005 using IMGT	QNINNF
443 STIM005 - CDRL2	Amino acid sequence of CDRL2 of STIM005 using IMGT	AAS
444 STIM005 - CDRL3	Amino acid sequence of CDRL3 of STIM005 using IMGT	QQSYGIPW
445 STIM005 - Light chain variable region	Amino acid sequence of V_L of STIM005	DIQMTQSPSSLSASVGDRTITCRASQNINNFNLNWFYQQKEGKPKL LIYAASLQRGIPSTFSGSGSDFTLTISLQPEDFATYICQQSY GIPWVGQGTKVEIK
446 STIM005 - Light chain variable region	Nucleic acid sequence of V_L of STIM005	GACATCCAGATGACCCAGTCTCCATCCTCCCTGTCTGCATCTGTAG GAGACAGAGTCACCATCACTTGCCGGGCAAGTCAGAACATTAATAA CTTTTAAATGGTATCAGCAGAAAGAGGAAAGGCCCTAAGCTC CTGATCTATGCAGCATCCAGTTTGCAAGAGGGATACCATCAACGT TCAGTGGCAGTGGATCTGGGACAGACTTCACTCTCACCATCAGCAG TCTGCAACCTGAAGATTTTGCAACTTACATCTGTCAACAGAGCTAC GGTATCCCGTGGGTCGGCCAAAGGGACCAAGGTGGAATCAAA
447 STIM005 - full light chain sequence	Amino acid sequence of STIM005 light chain	DIQMTQSPSSLSASVGDRTITCRASQNINNFNLNWFYQQKEGKPKL LIYAASLQRGIPSTFSGSGSDFTLTISLQPEDFATYICQQSY GIPWVGQGTKVEIK RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA LQSGNSQESVTEQDSKDSYSLSSLTLSKADYEKHKVYACEVTHQ GLSPVTKSFNRGEC
448 STIM005 - full light chain sequence	Nucleic acid sequence of STIM005 light chain	GACATCCAGATGACCCAGTCTCCATCCTCCCTGTCTGCATCTGTAG GAGACAGAGTCACCATCACTTGCCGGGCAAGTCAGAACATTAATAA CTTTTAAATGGTATCAGCAGAAAGAGGAAAGGCCCTAAGCTC CTGATCTATGCAGCATCCAGTTTGCAAGAGGGATACCATCAACGT TCAGTGGCAGTGGATCTGGGACAGACTTCACTCTCACCATCAGCAG TCTGCAACCTGAAGATTTTGCAACTTACATCTGTCAACAGAGCTAC

TABLE S2-continued

SEQ ID NOS: 343-538		
SEQ ID NO: Name	Description	Sequence
		GGTATCCCGTGGGTCGGCCCAAGGGACCAAGGTGGAAATCAAACgta cggtggcgcgtccctccgtgttcattctcccacctccgacgagca gctgaagtccggcaccgcttctgtcgtgtgcctgctgaacaacttc taccctcgagagccaaagtgagtggaagtggaacacgcctgc agtccggcaactcccaggaatccgtgaccgagcaggactccaagga cagcaactactccctgtcctccacccctgacctgtccaaggccgac tacgagaagcacaaggtgtacgcctgcgaagtgaccaccagggcc tgtctagcccggtgaccaagtctttcaaccggggcgagtgt
449 STIM006 - CDRH1	Amino acid sequence of CDRH1 of STIM006 using IMGT	GFTFSDYF
450 STIM006 - CDRH2	Amino acid sequence of CDRH2 of STIM006 using IMGT	ISSSGSTI
451 STIM006 - CDRH3	Amino acid sequence of CDRH3 of STIM006 using IMGT	ARDHYDGSgiypllyyygldv
452 STIM006 - Heavy chain variable region	Amino acid sequence of V_H of STIM006	QVQLVESGGGLVKPGGSLRLSCAASGFTFSDYFMSWIRQAPGKGLE WISYISSSGSTIYYADSVRGRFTISRDNAKYSLYLQMSLRSEDTA VYYCARDHYDGSgiypllyyygldvWGQGTTVTVSS
453 STIM006 - Heavy chain variable region	Nucleic acid sequence of V_H of STIM006	CAGGTGCAGCTGGTGGAGTCTGGGGGAGGCTTGGTCAAGCCTGGAG GGTCCCTGAGACTCTCCTGTGCAGCCTCTGGATTACCTTCAGTGA CTACTTCATGAGCTGGATCCGCCAGGCGCCAGGGAAGGGCTGGAG TGGATTTTCATACATTAGTTCTAGTGGTAGTACCATATACACGAG ACTCTGTGAGGGGCCGATTACCATCTCCAGGGACACGCCAAGTA CTCACTGTATCTGCAATGAACAGCCTGAGATCCGAGGACACGGCC GTGTATTACTGTGCGAGAGATCACTACGATGGTTCGGGGATTTATC CCCTCTACTACTATTACGGTTTGGACGTCTGGGGCCAGGGGACCAC GGTCACCGTCTCCTCA
454 STIM006 - full heavy chain sequence	Amino acid sequence of STIM006 heavy chain	QVQLVESGGGLVKPGGSLRLSCAASGFTFSDYFMSWIRQAPGKGLE WISYISSSGSTIYYADSVRGRFTISRDNAKYSLYLQMSLRSEDTA VYYCARDHYDGSgiypllyyygldvWGQGTTVTVSSASTKGPSVFP LAPSSKSTSGGTAAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAV LQSSGLYSLSSVTVPSLSLGTQTYICNVNHKPSNTKVDKKVEPKS CDKTHTCPAPPELLGGPSVFLFPPKPKDTLMI SRTEVTCVVVD VSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQ DWLNGKEYKCKVSNKALPAPI EKTISKAKGQPREPQVYTLPPSRDE LTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTPPVLDSDGS FFLYSKLTVDKSRWQQGNVFSQSVMHREALHNYTQKSLSLSPGK
455 STIM006 - full heavy chain sequence	Nucleic acid sequence of STIM006 heavy chain	CAGGTGCAGCTGGTGGAGTCTGGGGGAGGCTTGGTCAAGCCTGGAG GGTCCCTGAGACTCTCCTGTGCAGCCTCTGGATTACCTTCAGTGA CTACTTCATGAGCTGGATCCGCCAGGCGCCAGGGAAGGGCTGGAG TGGATTTTCATACATTAGTTCTAGTGGTAGTACCATATACACGAG ACTCTGTGAGGGGCCGATTACCATCTCCAGGGACACGCCAAGTA CTCACTGTATCTGCAATGAACAGCCTGAGATCCGAGGACACGGCC GTGTATTACTGTGCGAGAGATCACTACGATGGTTCGGGGATTTATC CCCTCTACTACTATTACGGTTTGGACGTCTGGGGCCAGGGGACCAC GGTCACCGTCTCCTCAGCCAGCACCAAGGGCCCTCTGTGTTCCCT CTGGCCCCCTCCAGCAAGTCCACCTCTGGCGGAACAGCCGCTCTGG GCTGCCTCGTGAAGGACTACTTCCCCGAGCCTGTGACCGTGTCTGTG GAACCTCTGGCGCTCTGACCAGCGGAGTGACACCTTCCCTGTCTGTG CTCAGTCCCTCCGGCCTGTACTCCCTGTCTCCGTCTGTGACCGTGC CTTCCAGCTCTCTGGGCACCCAGACCTACATCTGCAACGTGAACCA CAAGCCCTCCAACACCAAGGTGGACAAGAAGGTGGAACCCAAAGTCC TGGCACAAGACCCACACCTGTCCCCCTTGTCTGCCCCCTGAAGTGC TGGGCGGACCTTCCGTGTTCTGTTCCTCCCAAGCCCAAGGACAC CCTGATGATCTCCCGGACCCCGAAGTGACCTGCGTGGTGGTGGAT GTGTCACGAGGACCTGAAGTGAAGTTCAATTGGTACGTGGACG GCGTGAAGTGCAACGCAAGACCAAGCCTAGAGAGGAACAGTA CAACTCCACCTACCGGTGGTGTCCGTGTGACCGTGTGTCACACG GATGGCTGAACGGCAAGAGTACAAGTGCAAGGTGTCCAACAAG CCCTGCCTGCCCCATCGAAAAGACCATCTCAAGGCCAAGGGCCA GCCCCGGGAACCCAGGTGTACACACTGCCCCCTAGCAGGACGAG CTGACCAAGAACCAGGTGTCCCTGACCTGTCTCGTGAAGGCTTCT ACCCCTCCGATATCGCCGTGGAATGGGAGTCCAACGCCAGCCTGA

TABLE S2-continued

SEQ ID NOS: 343-538		
SEQ ID NO: Name	Description	Sequence
		GAACAACATAAGACACCCCCCTGTGCTGGACTCCGACGGCTCA TTCTTCCTGTACAGCAAGCTGACAGTGGACAGTCCCGGTGGCAGC AGGGCAACGTGTTCTCCTGTCTCCGTGATGCACGAGGCCCTGCACAA CCACTACACCCAGAAGTCCCTGTCCCTGAGCCCCGGCAAGTGATGA
456 STIM006 - CDRL1	Amino acid sequence of CDRL1 of STIM006 using IMGT	QSLHSNGYNY
457 STIM006 - CDRL2	Amino acid sequence of CDRL2 of STIM006 using IMGT	LGS
458 STIM006 - CDRL3	Amino acid sequence of CDRL3 of STIM006 using IMGT	MQALQTPRS
459 STIM006 - Light chain variable region	Amino acid sequence of V_L of STIM006	IVMTQSPSLPVPVTPGEPASISCRSSQSLHSNGYNYLDYYLQKPGQ SPQLLIYLGSYRASGVDPDRFSGSGSDTFTLKISRVEAEDVGVYYC MQALQTPRSFGQGTTLLEIK
460 STIM006 - Light chain variable region	Nucleic acid sequence of V_L of STIM006	ATTGTGATGACTCAGTCTCCACTCTCCCTACCCGTCACCCCTGGAG AGCCGGCCTCCATCTCCTGCAGGTCTAGTCAGAGCCTCCTGCATAG TAATGGATACAACATATTGGATTATTACCTGCAGAACCCAGGGCAG TCTCCACAGCTCCTGATCTATTTGGGTTCTTATCGGGCTCCGGGG TCCCTGACAGGTTTCAGTGGCAGTGGATCAGGCACAGATTTTACACT GAAATCAGCAGAGTGGAGGCTGAGGATGTTGGGTTTATTACTGC ATGCAAGCTCTACAACTCCTCGCAGTTTGGCCAGGGACCACGC TGGAGATCAAA
461 STIM006 - full light chain sequence	Amino acid sequence of STIM006 light chain	IVMTQSPSLPVPVTPGEPASISCRSSQSLHSNGYNYLDYYLQKPGQ SPQLLIYLGSYRASGVDPDRFSGSGSDTFTLKISRVEAEDVGVYYC MQALQTPRSFGQGTTLLEIKRTVAAPSVFIFPPSDEQLKSGTASVVC LLNNFYPREAKVQWKVDNALQSGNSQESVTEQDSKDSSTYLSLSTLT LSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC
462 STIM006 - full light chain sequence	Nucleic acid sequence of STIM006 light chain	ATTGTGATGACTCAGTCTCCACTCTCCCTACCCGTCACCCCTGGAG AGCCGGCCTCCATCTCCTGCAGGTCTAGTCAGAGCCTCCTGCATAG TAATGGATACAACATATTGGATTATTACCTGCAGAACCCAGGGCAG TCTCCACAGCTCCTGATCTATTTGGGTTCTTATCGGGCTCCGGGG TCCCTGACAGGTTTCAGTGGCAGTGGATCAGGCACAGATTTTACACT GAAATCAGCAGAGTGGAGGCTGAGGATGTTGGGTTTATTACTGC ATGCAAGCTCTACAACTCCTCGCAGTTTGGCCAGGGACCACGC TGGAGATCAAAcgtacggtgcccgtccctccgtgttcattcttccc accttccgacgagcagctgaagtcggcaccgcttctgtcgtgtgc ctgctgaacaacttctacccccgcgagggcaagggtgcagtggaagg tggaacaacgcccctgcagtcgggcaactcccagggaatccgtgaccga gcaggactccaaggacagcacctactccctgtcctccaccctgacc ctgtccaaggccgactacgagaagcacaagggtgtacgcctgcgaag tgaccaccagggcctgtctagccccgtgaccaagtccttcaaccg ggcgagtggt
463 STIM007 - CDRH1	Amino acid sequence of CDRH1 of STIM007 using IMGT	GFSLSTTGVG
464 STIM007 - CDRH2	Amino acid sequence of CDRH2 of STIM007 using IMGT	IYWDDDK
465 STIM007 - CDRH3	Amino acid sequence of CDRH3 of STIM007 using IMGT	THGYGSASYHYGMDV
466 STIM007 - Heavy chain variable region	Amino acid sequence of V_H of STIM007	QITLKESGPTLVKPTQTLTLTCTFSGFSLSTTGVGVGWIQPPGKA LEWLAVIYWDDDKRYSPLKSRITITKDTSKNQVLTMTNMDPVDT ATYFCTHGYGSASYHYGMDVWQGQTTVTVSS

TABLE S2-continued

SEQ ID NOS: 343-538		
SEQ ID NO: Name	Description	Sequence
467 STIM007 - Heavy chain variable region	Nucleic acid sequence of V_H of STIM007	CAGATCACCTTGAAGGAGTCTGGTCTACGCTGGTGAACCCACAC AGACCCCTCACGCTGACCTGCACCTTCTCTGGGTCTCTCACTCAGCAC TACTGGAGTGGGTGTGGGCTGGATCCGTAGCCCCCAGGAAAGGCC CTGGAGTGGCTTGCACTCATTATTTGGGATGATGATAAGCGCTACA GCCCCATCTCTGAAGAGCAGACTCACCATCACCAGGACACCTCCAA AAACCAGGTGGTCTTACAATGACCAACATGGACCTGTGGACACA GCCACATATTTCTGTACACACGGATATGGTTCGGCGAGTTATTACC ACTACGGTATGGACGTCTGGGGCCAAGGGACCACGGTCACCGTCTC CTCA
468 STIM007 - full heavy chain sequence	Amino acid sequence of STIM007 heavy chain	QITLKESGPTLVKPTQLTLTLCTFSGFSLSTTGVGVWIRQPPGKA LEWLAVIYWDDDKRYSPLKSLRTITKDTSKNQVVLTMNMDPVD ATYFCTHGYGSASYHYGMDVWGQGTVTVTSSASTKGPSVPLAPS SKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSQVHTFPVAVLQSS GLYSLSSVTVPSSSLGTQTYICNVNHPKPNKVDKKEPKSCDKT HTCPPCPAPELLGGPSVFLFPPPKPDTLMISRTPEVTCVVVDVSHE DPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLN GKEYKCKVSNKALPAPIEKTI SKAKGQPREPQVYTLPPSRDELTKN QVSLTCLVKGFPYSDIAVEWESNGQPENNYKTTTPVLDSDGSFPLY SKLTVDKSRWQQGNVPSCSVMHEALHNHYTQKSLSLSPGK
469 STIM007 - full heavy chain sequence	Nucleic acid sequence of STIM007 heavy chain	CAGATCACCTTGAAGGAGTCTGGTCTACGCTGGTGAACCCACAC AGACCCCTCACGCTGACCTGCACCTTCTCTGGGTCTCTCACTCAGCAC TACTGGAGTGGGTGTGGGCTGGATCCGTAGCCCCCAGGAAAGGCC CTGGAGTGGCTTGCACTCATTATTTGGGATGATGATAAGCGCTACA GCCCCATCTCTGAAGAGCAGACTCACCATCACCAGGACACCTCCAA AAACCAGGTGGTCTTACAATGACCAACATGGACCTGTGGACACA GCCACATATTTCTGTACACACGGATATGGTTCGGCGAGTTATTACC ACTACGGTATGGACGTCTGGGGCCAAGGGACCACGGTCACCGTCTC CTCA GCCAGCACCAAGGGCCCTCTGTGTTCCCTCTGGCCCTTCCAGCA AGTCCACCTCTGGCGGAACAGCCGCTCTGGGCTGCCTCGTGAAGGA CTACTTCCCCGAGCCTGTGACCGTGTCTTGAACCTCTGGCGCTCTG ACCAGCGGAGTGACACCTTCCCTGCTGTGCTGCAGTCTCTCCGGCC TGTACTCCCTGCTCCTCCGTGCTGACCGTGCCTTCCAGCTCTCTGGG CAGCCAGACCTACATCTGCAACGTGAACCAAGCCCTCCAACACC AAGTGGACAAGAAGGTGGAACCAAGTCTCTGCGACAAGACCCACA CTGTCCCCCTTGTCTGCCCCCTGAACCTGCTGGGCGACCTTCCGT GTTCTGTCTCCCCCAAGCCCAAGGACACCTGATGATCTCCCGG ACCCCGAAGTGACCTGCGTGGTGGTGGATGTGTCCACGAGGACC CTGAAGTGAAGTTCAATTGGTACGTGGACGGCGTGAAGTGACAA CGCCAAGACCAAGCCTAGAGAGGAACAGTACAACCTCCACCTACCGG GTGGTGTCCGTGCTGACCGTGTGACACAGGATTGGCTGAACGGCA AAGAGTACAAGTGCAAGGTGTCCAACAAGGCCCTGCCTGCCCCCAT CGAAAAGACCATCTCCAAGGCCAAGGGCCAGCCCCGGGAACCCAG GTGTACACACTGCCCCCTAGCAGGGACGAGCTGACCAAGAACAGG TGTCCCTGACCTGTCTCTGTAAAGGCTTCTACCCCTCCGATATCGC CGTGGAAATGGGAGTCCAACGGCCAGCCTGAGAACAACACAAGAAC ACCCCCCTGTGCTGGACTCCGACGGCTCATTCTTCTGTACAGCA AGCTGACAGTGGACAAGTCCCGGTGGCAGCAGGGCAACGTGTTCTC CTGCTCCGTGATGCACGAGGCCCTGCACAACCACTACACCCAGAAG TCCCTGTCCCTGAGCCCCGCAAGTGATGA
470 STIM007-CDRL1	Amino acid sequence of CDRL1 of STIM007 using IMGT	QSVTNY
471 STIM007-CDRL2	Amino acid sequence of CDRL2 of STIM007 using IMGT	DAS
472 STIM007-CDRL3	Amino acid sequence of CDRL3 of STIM007 using IMGT	QHRSNWPLT
473 STIM007 - Light chain variable region	Amino acid sequence of V_L of STIM007	EIVLTQSPATLSLSPGERATLSCRASQSVNTYLAWHQQKPGQAPRL LIYDASNRATGIPARFSGSGSDFTLTISSELPEDFAVYYCQHR NWPLTFGGGTKVEIK

TABLE S2-continued

SEQ ID NOS: 343-538			
SEQ ID NO:	Name	Description	Sequence
474	STIM007 - Light chain variable region	Nucleic acid sequence of V_L of STIM007	GAAATTGTATTGACACAGTCTCCAGCCACCTGTCTTTGTCTCCAG GGGAAAGAGCCACCTCTCCTGCAGGGCCAGTCAGAGTGTTACCAA CTACTTAGCCTGGCACCACAGAAACCTGGCCAGGCTCCAGGCTC CTCATCTATGATGCATCCAACAGGGCCACTGGCATCCCAGCCAGGT TCAGTGGCAGTGGGTCTGGGACAGACTTCACTCTCACCATCAGCAG CCTAGAGCCTGAAGATTTGCAGTTTATTACTGTGAGCACCCTAGC AACTGGCCTCTCACTTTCCGCGGAGGGACCAAGGTGGAGATCAAA
475	STIM007 - full light chain sequence	Amino acid sequence of STIM007 light chain	EIVLTQSPATLSLSPGERATLSCRASQSVTNYLAWHQKPKGPQAPRL LIYDASNRTGIPARFSGSGSGTDFTLTISLLEPEDFAVYYCQHR NWPLTFGGGKVEIKRTVAAPSVFIFPPSDEQLKSGTASVVCCLNN FYPREAKVQWKVDNALQSGNSQESVTEQDSKDSSTLSSTLTLSKA DYEKHKVYACEVTHQGLSSPVTKSFNRGEC
476	STIM007 - full light chain sequence	Nucleic acid sequence of STIM007 light chain	GAAATTGTATTGACACAGTCTCCAGCCACCTGTCTTTGTCTCCAG GGGAAAGAGCCACCTCTCCTGCAGGGCCAGTCAGAGTGTTACCAA CTACTTAGCCTGGCACCACAGAAACCTGGCCAGGCTCCAGGCTC CTCATCTATGATGCATCCAACAGGGCCACTGGCATCCCAGCCAGGT TCAGTGGCAGTGGGTCTGGGACAGACTTCACTCTCACCATCAGCAG CCTAGAGCCTGAAGATTTGCAGTTTATTACTGTGAGCACCCTAGC AACTGGCCTCTCACTTTCCGCGGAGGGACCAAGGTGGAGATCAAA cgtaacgggtggcgcgtccctccgtgttcattcttccaccttcgcagc agcagctgaagtcggccaccgcttctgtcgtgtgcctgctgaacaa cttctacccccgcgagggcaaggtgcagtggaaggtggacaacgcc ctgcagtcgggcaactcccaggaatccgtgaccgagcaggactcca aggacagcacctactccctgtcctccacctgacctgtccaagggc cgactacgagaagcacaaggtgtacgcctgcgaagtgacccaccag ggcctgtctagccccgtgaccaagtcctttcaaccggggcgagtg
477	STIM008- CDRH1	Amino acid sequence of CDRH1 of STIM008 using IMGT	GFSLSTSGVG
478	STIM008- CDRH2	Amino acid sequence of CDRH2 of STIM008 using IMGT	IYWDDDK
479	STIM008- CDRH3	Amino acid sequence of CDRH3 of STIM008 using IMGT	THGYGSASYHYGMDV
480	STIM008 - Heavy chain variable region	Amino acid sequence of V_H of STIM008	QITLKESGPTLVKPTQTLTLTCTFSGFSLSTSGVGVGWIRQPPGKA LEWLAVIYWDDDKRYSPLKSLRLTITKDTSKNQVVLMTNMDPVDT ATYFCTHGYGSASYHYGMDVWGQGTIVTVSS
481	STIM008 - Heavy chain variable region	Nucleic acid sequence of V_H of STIM008	CAGATCACCTTGAAGGAGTCTGGTCCTACGCTGGTGAAACCCACAC AGACCTTCACGCTGACCTGCACCTTCTCTGGGTTCTCACTCAGCAC TAGTGGAGTGGGTGTGGCTGGATCCGTGAGCCCCAGGAAAGGCC CTGGAGTGGCTTGCACTATTTATTTGGGATGATGATAAGCGCTACA GCCCCATCTCTGAAGAGCAGGCTCACCATCACCAGGACACCTCCAA AAACCAGGTGGTCCTTACAATGACCAACATGGACCCCTGTGGACACA GCCACATATTTCTGTACACACGGATATGGTTCGGCGAGTTATTACC ACTACGGTATGGACGTCTGGGGCCAAAGGACACCGGTACCGTCTC CTCA
482	STIM008 - full heavy chain sequence	Amino acid sequence of STIM008 heavy chain	QITLKESGPTLVKPTQTLTLTCTFSGFSLSTSGVGVGWIRQPPGKA LEWLAVIYWDDDKRYSPLKSLRLTITKDTSKNQVVLMTNMDPVDT ATYFCTHGYGSASYHYGMDVWGQGTIVTVSSASTKGPSVFPLAPS SKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSS GLYSLSVSVTVPSSSLGTQTYICNVNHKPSNTKVDKKEPKSCDKT HTCPPCPAPELLGGPSVFLPPPKPKDTLMI SRTPEVTCVVVDVSH DPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLN GKEYKCKVSNKALPAPIEKTI SKAKGQPREPQVYTLPPSRDELTKN QVSLTCLVKGFPYSDIAVEWESNGQPENNYKTPPVLDSDGSFFLY SKLTVDKSRWQQGNVFC SSVMEALHNHYTQKSLSLSPGK

TABLE S2-continued

SEQ ID NOS: 343-538		
SEQ ID NO: Name	Description	Sequence
483 STIM008 - full heavy chain sequence	Nucleic acid sequence of STIM008 heavy chain	CAGATCACCTTGAAGGAGTCTGGTCTACGCTGGTGAACCCACAC AGACCCCTCACGCTGACCTGCACCTTCTCTGGGTTCTCACTCAGCAC TAGTGGAGTGGGTGTGGGCTGGATCCGTCAGCCCCAGGAAAGGCC CTGGAGTGGCTTGCACTATTATTTGGGATGATGATAAGCGCTACA GCCCCATCTCTGAAGAGCAGGCTCACCATCACCAAGGACACCTCCAA AAACCAGGTGGTCTTACAATGACCAACATGGACCTGTGGACACA GCCACATATTCTGTACACACGGATATGGTTCGGCGAGTTATTACC ACTACGGTATGGACGTCTGGGGCCAAAGGACCACGGTACCGGTCTC CTCAGCCAGCACCAAGGGCCCCCTGTGTTCCCTCTGGCCCCCTCC AGCAAGTCCACCTCTGGCGGAACAGCCGCTCTGGGCTGCCTCGTGA AGGACTACTTCCCCGAGCCTGTGACCGTGTCTTGGAACTCTGGCGC TCTGACCAGCGAGTGCACACCTTCCCTGCTGTGCTGCAGTCTCTCC GGCCTGTACTCCCTGTCTCCGTCTGACCGTGCCTTCCAGCTCTC TGGGCAACCAGACCTACATCTGCAACGTGAACCACAAGCCCTCCAA CACCAAGGTGGACAAGAAGGTGGAACCAAGTCTTGCACAAAGACC CACACCTGTCCCCCTGTCTGCCCCTGAAGTCTGTGGCGGACCTT CCGTGTTCTGTTCCTCCCCAAAGCCCAAGGACACCTGTATGATCTC CCGGACCCCGAAGTGACCTGCGTGGTGGTGGATGTGTCCACGAG GACCTGAAGTGAAGTTCAATTGGTACGTGGACGGCGTGAAGTGC ACAACGCCAAGACCAAGCCTAGAGAGGAACAGTACAACCTCCACCTA CCGGGTGGTGTCTGCTGACCGTGTGACCCAGGATGGCTGAAC GGCAAGAGTACAAGTGCAAGGTGTCCAACAAGGCCCTGCCTGCCC CCATCGAAAAGACCATCTCCAAGGCCAAGGGCCAGCCCCGGGAACC CCAGGTGTACACACTGCCCCCTAGCAGGGACGAGCTGACCAAGAAC CAGGTGTCTCTGACCTGTCTCGTGAAGGCTTCTACCCCTCCGATA TCGCCGTGGAATGGGAGTCCAACGGCCAGCCTGAGAACACTACAA GACCACCCCTGTGCTGGACTCCGACGGCTATTCTTCTGTAC AGCAAGCTGACAGTGGACAAGTCCCGGTGGCAGCAGGGCAACGTGT TCTCTGCTCCGTGATGCACGAGGCCCTGCACAACCACTACACCCA GAAGTCCCTGTCTCTGAGCCCCGGCAAGTGATGA
484 STIM008- CDRL1	Amino acid sequence of CDRL1 of STIM008 using IMGT	QSVTNY
485 STIM008- CDRL2	Amino acid sequence of CDRL2 of STIM008 using IMGT	DAS
486 STIM008- CDRL3	Amino acid sequence of CDRL3 of STIM008 using IMGT	QQRSNWPLT
487 STIM008 - Light chain variable region	Amino acid sequence of V_L of STIM008	EIVLTQSPATLSLSPGERATLSCRASQSVTNYLAWHQKPGQAPRL LIYDASNRATGIPARFSGSGSDFTLTISLEPEDFAVYYCQQRS NWPLTFGGGTKVEIK
488 STIM008 - Light chain variable region	Nucleic acid sequence of V_L of STIM008	GAAATTGTGTGACACAGTCTCCAGCCACCCTGTCTTTGTCTCCAG GGGAAAGAGCCACCCTCTCTGCAAGGCCAGTCAGAGTGTACCAA CTACTTAGCCTGGCACCAACAGAAACCTGGCCAGGCTCCAGGCTC CTCATCTATGATGCATCCAACAGGGCCACTGGCATCCAGCCAGGT TCAGTGGCAGTGGGTCTGGGACAGACTTCACTCTCACCATCAGCAG CCTAGAGCCTGAAGATTTGCAGTTTATTACTGTGACAGCGTAGC AACTGGCCTCTCACTTTCCGGCGGAGGACCAAGGTGGAGATCAAA
489 STIM008 - full light chain sequence	Amino acid sequence of STIM008 light chain	EIVLTQSPATLSLSPGERATLSCRASQSVTNYLAWHQKPGQAPRL LIYDASNRATGIPARFSGSGSDFTLTISLEPEDFAVYYCQQRS NWPLTFGGGTKVEIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNN FYPREAKVQWKVDNALQSGNSQESVTEQDSKDSSTYSLSSTLTLSKA DYEKHVKYACEVTHQGLSSPVTKSFNRGEC
490 STIM008 - full light chain sequence	Nucleic acid sequence of STIM008 light chain	GAAATTGTGTGACACAGTCTCCAGCCACCCTGTCTTTGTCTCCAG GGGAAAGAGCCACCCTCTCTGCAAGGCCAGTCAGAGTGTACCAA CTACTTAGCCTGGCACCAACAGAAACCTGGCCAGGCTCCAGGCTC CTCATCTATGATGCATCCAACAGGGCCACTGGCATCCAGCCAGGT TCAGTGGCAGTGGGTCTGGGACAGACTTCACTCTCACCATCAGCAG CCTAGAGCCTGAAGATTTGCAGTTTATTACTGTGACAGCGTAGC AACTGGCCTCTCACTTTCCGGCGGAGGACCAAGGTGGAGATCAAAc gtacggtggccgctccctccgtgttcatcttccaccttccgacga gcagctgaagtcgggcaaccgcttctgtcgtgtgcctgctgaacaac ttctacccccgcgagggccaaggtgcagtggaaggtggacaacgcc

TABLE S2-continued

SEQ ID NOS: 343-538		
SEQ ID NO: Name	Description	Sequence
		tgcagtcggaactcccaggaatccgtgaccgagcaggactccaa ggacagcacctactccctgtcctccaccctgacctgtccaaggcc gactacgagaagcacaaggtgtacgcctgcgaagtgaccaccagg gcctgtctagccccgtgaccaagtctttcaaccggggcagtg
491 STIM009- CDRH1	Amino acid sequence of CDRH1 of STIM009 using IMGT	GFTFSDYY
492 STIM009- CDRH2	Amino acid sequence of CDRH2 of STIM009 using IMGT	ISSSGSTI
493 STIM009- CDRH3	Amino acid sequence of CDRH3 of STIM009 using IMGT	ARDFYDILTDSPIFYGV
494 STIM009 - Heavy chain variable region	Amino acid sequence of V_H of STIM009	QVQLVESGGGLVKGPGSLRLSCAASGFTFSDYYMSWIRQAPGKGLE WVSYISSSGSTIYADSVKGRFTISRDNKNSLYLQINSLRAEDTA VYYCARDFYDILTDSPIFYGVVDVWGQGTITVTVSS
495 STIM009 - Heavy chain variable region	Nucleic acid sequence of V_H of STIM009	CAGGTGCAGCTGGTGGAGTCTGGGGGAGGCTTGGTCAAGCCTGGAG GGTCCCTGAGACTCTCCTGTGCAGCCTCTGGATTACCTTCAGTGA CTACTACATGAGCTGGATCCGCCAGGCTCCAGGGAAGGGCTGGAG TGGGTTTCATACATTAGTAGTAGTGGTAGTACCATATACACGAG ACTCTGTGAAGGGCCGATTACCATCTCCAGGGACACGCCAAGAA CTCACTGTATCTGCAATTAAACAGCCTGAGAGCCGAGGACACGCC GTGTATTACTGTGCGAGAGATTTTACGATATTTGACTGATAGTC CGTACTTCTACTACGGTGTGGACGCTCGGGGCCAAGGACACGGT CACCGTCTCCTCA
496 STIM009 - full heavy chain sequence	Amino acid sequence of STIM009 heavy chain	QVQLVESGGGLVKGPGSLRLSCAASGFTFSDYYMSWIRQAPGKGLE WVSYISSSGSTIYADSVKGRFTISRDNKNSLYLQINSLRAEDTA VYYCARDFYDILTDSPIFYGVVDVWGQGTITVTVSSASTKGPSVFPL APSSKSTSGGTAALGLVKDYFPEPVTVSWNSGALTSGVHTFPAVL QSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKEPKSC DKHTCPPCPAPELLGGPSVFLFPPPKKDTLMISRTPEVTCVVVDV SHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQD WLNKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSRDEL TKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSF FLYSKLTVDKSRWQQGNVFSQVMHEALHNHYTQKSLSLSPGK
497 STIM009 - full heavy chain sequence	Nucleic acid sequence of STIM009 heavy chain	CAGGTGCAGCTGGTGGAGTCTGGGGGAGGCTTGGTCAAGCCTGGAG GGTCCCTGAGACTCTCCTGTGCAGCCTCTGGATTACCTTCAGTGA CTACTACATGAGCTGGATCCGCCAGGCTCCAGGGAAGGGCTGGAG TGGGTTTCATACATTAGTAGTAGTGGTAGTACCATATACACGAG ACTCTGTGAAGGGCCGATTACCATCTCCAGGGACACGCCAAGAA CTCACTGTATCTGCAATTAAACAGCCTGAGAGCCGAGGACACGCC GTGTATTACTGTGCGAGAGATTTTACGATATTTGACTGATAGTC CGTACTTCTACTACGGTGTGGACGCTCGGGGCCAAGGACACGGT CACCGTCTCCTCAGCAGCACCAGGGCCCTCTGTGTTCCCTCTG GCCCCTTCCAGCAAGTCCACCTCTGGCGGAACAGCCGCTCTGGGCT GCCTCGTGAAGGACTACTTCCCGAGCCTGTGACCGTGTCTGGAA CTCTGGCGCTCTGACCAGCGGAGTGACACCTTCCCTGCTGTGCTG CAGTCCCTCCGGCCTGTACTCCCTGTCTCCGTGCTGACCGTGCTCT CCAGCTCTCTGGGCACCCAGACCTACATCTGCAACGTGAACCAAA GCCCTCCAACACCAAGGTGACCAAGAGGTGGAACCCAAAGTCTCTGC GACAAGACCCACACCTGTCCCCCTGTCTGCCCCCTGAAGTGTCTGG GCGGACCTTCCGTGTTCTGTTCCTCCCAAGGCCAAGGACACCT GATGATCTCCCGGACCCCGAAGTGACCTGCGTGGTGGTGGATGTG TCCCACGAGGACCTGAAGTGAAGTTCAATTGGTACGTGGACGGCG TGGAAAGTGCACAACGCCAAGACCAAGCCTAGAGAGGAACAGTACAA CTCCACCTACCGGGTGGTGTCCGTGTGACCGTGTGACACAGGAT TGGCTGAACGGCAAGAGTACAAGTGAAGGTGCCAACAGGCC TGCTTGCCCCCATCGAAAAGACCATCTCAAGGCCAAGGGCCAGCC CCGGGAACCCAGGTGTACACACTGCCCCCTAGCAGGGACGAGCTG ACCAAGAACCAGGTGTCTCTGACCTGTCTCGTGAAAGGCTTCTAC CTCCGATATCGCCGTGGAATGGGAGTCCACGGCCAGCCTGAGAA CAACTACAAGACCACCCCTGTGTGGACTCCGACGGCTCATTC TTCTGTACAGCAAGCTGACAGTGGACAAGTCCCGGTGGCAGCAGG GCAACGTGTTCTCTGCTCCGTGATGACAGGCGCTGCACAACCA CTACACCCAGAAGTCCCTGTCTGAGCCCCGGCAAGTGATGA

TABLE S2-continued

SEQ ID NOS: 343-538		
SEQ ID NO: Name	Description	Sequence
498 STIM009-CDRL1	Amino acid sequence of CDRL1 of STIM009 using IMGT	QSLHNSNGYNY
499 STIM009-CDRL2	Amino acid sequence of CDRL2 of STIM009 using IMGT	LGS
500 STIM009-CDRL3	Amino acid sequence of CDRL3 of STIM009 using IMGT	MQALQTPRT
501 STIM009 - Light chain variable region	Amino acid sequence of V_L of STIM009	DIVMTQSPLSLPVTGPGEPAISCRSSQSLHNSNGYNYLDWYLQKPG QSPQLLIYLGNSNRASGVDPDRFSGSGSGTDFTLKI SRVEAEDVGVYY CMQALQTPRTFQGQTKVEIK
502 STIM009 - Light chain variable region	Nucleic acid sequence of V_L of STIM009	GATATTGTGATGACTCAGTCTCCACTCTCCCTGCCCGTCACCCCTG GAGAGCCGGCCTCCATCTCCTGCAGGTCTAGTCAGAGCCTCCTGCA TAGTAATGGATACAACTATTGGATTGGTACCTGCAGAAGCCAGGG CAGTCTCCACAGCTCCTGATCTATTTGGGTCTTAATCGGGCCTCCG GGGTCCCTGCAGGTTTCAGTGGCAGTGGATCAGGCACAGATTTTAC ACTGAAAATCAGCAGAGTGGAGGCTGAGGATGTTGGGTTTATTAC TGCATGCAAGCTCTACAACTCCTCGGACGTTCCGCCAAGGGACCA AGGTGGAATCAAA
503 STIM009 - full light chain sequence	Amino acid sequence of STIM009 light chain	DIVMTQSPLSLPVTGPGEPAISCRSSQSLHNSNGYNYLDWYLQKPG QSPQLLIYLGNSNRASGVDPDRFSGSGSGTDFTLKI SRVEAEDVGVYY CMQALQTPRTFQGQTKVEIKRTVAAPSVFIFPPSDEQLKSGTASV CLLNRFYPREAKVQWKVDNALQSGNSQESVTEQDSKDSSTYSLSSTL TLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC
504 STIM009 - full light chain sequence	Nucleic acid sequence of STIM009 light chain	GATATTGTGATGACTCAGTCTCCACTCTCCCTGCCCGTCACCCCTG GAGAGCCGGCCTCCATCTCCTGCAGGTCTAGTCAGAGCCTCCTGCA TAGTAATGGATACAACTATTGGATTGGTACCTGCAGAAGCCAGGG CAGTCTCCACAGCTCCTGATCTATTTGGGTCTTAATCGGGCCTCCG GGGTCCCTGCAGGTTTCAGTGGCAGTGGATCAGGCACAGATTTTAC ACTGAAAATCAGCAGAGTGGAGGCTGAGGATGTTGGGTTTATTAC TGCATGCAAGCTCTACAACTCCTCGGACGTTCCGCCAAGGGACCA AGGTGGAATCAAAcgtacggtggcgcgtccctccgtgttcatctt cccaccttccgacgagcagctgaagtcgagcaccgcttctgtcgtg tgccctgctgaacaacttctaccccccgaggccaaggtgcagtgga aggtggacaacgcctgcagtcgggcaactcccagggaatccgtgac cgagcaggactccaaggacagcacctactccctgtcctccaccctg accctgtccaaggccgactacgagaagcacaaggtgtacgcctgag aagtgacccaccaggccctgtctagccccgtgaccaagctcttcaa ccggggcgagtg
505 Human PD-L1 Flag His (KYROT2 86)	Amino acid sequence of KYROT286 with FLAG tag in bold and underlined and histidine tag in bold	FTVTVPKDLVVEYGSNMTIECKFPVEKQLDLAALIVWEMEDKNI IQFVHGEEDLVQHSYRQRARLLKDQLSLGNAALQITDVKLQDAG VYRCMISYGGADYKRITVKVNAPYNKINQRIILVDPVTSEHETCQ AEGYPKAEVIWTSDDHQLVSGKTTTNSKREEKLFNVTSLRINTT TNEIFYCTFRRLDPEENHTAELVPELPLAHPNERTIEGR <u>DDKDD</u> <u>DDKHHHHH</u>
506 Mature human ICOS	Mature amino acid sequence of human ICOS	EINGSANYEMFIFHNGGVQILCKYPDIVQQFKMQLLKGGQILCDLT KTKGSGNTVSIKSLKFCHSLSNNSVSFFLYNLDSHANYFCNLS IFDPPPPKVTLTGGYLHIYESQLCCQLKFWLPICAAAFVVVCLIGC ILICWLTKKKYSSSVHDPNGEYMFMRVNTAKKSRLTDVTL
507 Human ICOS extracellular domain	Amino acid sequence of human ICOS extracellular domain	EINGSANYEMFIFHNGGVQILCKYPDIVQQFKMQLLKGGQILCDLT KTKGSGNTVSIKSLKFCHSLSNNSVSFFLYNLDSHANYFCNLS IFDPPPPKVTLTGGYLHIYESQLCCQLKF
508 Human ICOS with signal peptide	Amino acid sequence of human ICOS (signal peptide is underlined)	MKSGLWYFFLFCLRIKVLTEINGSANYEMFIFHNGGVQILCKYPD IVQQFKMQLLKGGQILCDLT KTKGSGNTVSIKSLKFCHSLSNNSV SFFLYNLDSHANYFCNLSIFDPPPPKVTLTGGYLHIYESQLCCQ LKFWLPICAAAFVVVCLIGCILICWLTKKKYSSSVHDPNGEYMFMR AVNTAKKSRLTDVTL

TABLE S2-continued

SEQ ID NOS: 343-538			
SEQ ID NO:	Name	Description	Sequence
509	Isoform of human ICOS (Q9Y6W8-2)	Amino acid sequence of a human ICOS isoform	The sequence of this isoform differs from the canonical sequence in its cytoplasmic domain as follows: 168-199: KYSSSVHDPNGEYMFMRVNTAKKSRLTDVTLM
510	Mature mouse ICOS	Mature amino acid sequence of mouse ICOS	EINGSADHRMFSFHNGGVQISCKYPETVQQLKMRLFREREVLCELT KTKGSGNAVSIKNPMLCLYHLSNNSVSFFLNNPDSSQGSYYFCSL IFDPPPFQERNLSGGYLHIYESQLCCQLKIVVQVTE
511	Mouse ICOS extracellular domain	Amino acid sequence of the extracellular domain of mouse ICOS	EINGSADHRMFSFHNGGVQISCKYPETVQQLKMRLFREREVLCELT KTKGSGNAVSIKNPMLCLYHLSNNSVSFFLNNPDSSQGSYYFCSL IFDPPPFQERNLSGGYLHIYESQLCCQLK
512	Mouse ICOS with signal peptide	Amino acid sequence of mouse ICOS (signal peptide is underlined)	MGWSCIIILFLVATATGVHSEINGSADHRMFSFHNGGVQISCKYPET VQQLKMRLFREREVLCELT KTKGSGNAVSIKNPMLCLYHLSNNSVS FFLNNPDSSQGSYYFCSL IFDPPPFQERNLSGGYLHIYESQLCCQLKIVVQVTE
513	Cynomolgus ICOS with signal peptide	Amino acid sequence of cynomolgus ICOS (signal peptide is underlined)	MKSGLWYFPL FCLHMKVLTG EINGSANYEM FIFHNGGVQI LCKYPDIVQQ FKMQLLKGGQILCDLT KTKGSGNKVSIKSLKFCHS QLSNNSVSFFL YNLD RSHANYFYCNLSIFDPPPFKVTLTGGYLHIYESQLCCQLKFWLP IGCATF VVVICIFGCILICWLTKKKYSSTVHDPNGEYMFMRVNTAKKSRLT GTTP
514	Cynomolgus ICOS extracellular domain	Amino acid sequence of cynomolgus ICOS extracellular domain	EINGSANYEMFIFHNGGVQILCKYPDIVQQFKMQLLKGGQILCDLT KTKG SGNKVSIKSLKFCHS QLSNNSVSFFL YNLDRSHANYFYCNLSIFDPPPFKVTLTGGYLHIYESQLCCQLK
515	Human ICOS ligand	Amino acid sequence of human ICOS ligand comprising extracellular domain	DTQEKEVRAMVGSDELSCACPEGSRFDLNDVYVYWQTSSEKTVVT YHIPPQNSSLENVDSRYRNRALMSPAGMLRGDFSLRLFNVT PQDEQK FHCLVLSQSLGQFQEVLSVEVTLHVAANFSVPVVSAPHSPS QDELTF TCTSIINGYPRPNVYVWINKTDNSLDDQALQNDTVFLNMRGLYDVVSV LRIARTPSVNI GCCIEENVLLQQLNLTGVSQTGNDIGERDKITENPVSTGEKNAATWS
516	Human ICOS ligand	Amino acid sequence of human ICOS ligand including signal peptide	MRLGSPGLLFLFSSSLRADTQEKEVRAMVGSDELSCACPEGSRFDLNDVYVYWQTSSEKTVVTYHI PQNSSLENVDSRYRNRALMSPAGMLRGDFSLRLFNVT PQDEQK FHCLVLSQSLGQFQEVLSVEVTLHVAANFSVPVVSAPHSPS QDELTF TCTSIINGYPRPNVYVWINKTDNSLDDQALQNDTVFLNMRGLYDVVSVLRIARTPSVNI GCCIEENVLLQQLNLTGVSQTGNDIGERDKITENPVSTGEKNAATWS ILAVLCLLVVVAIGV VCRDRCLQHSYAGAWAVSPETELTGHV
SEQ ID NO: 610 ICOSL-Fc DTQEKEVRAMVGSDELSCACPEGSRFDLNDVYVYWQTSSEKTVVTYHI PQNSSLENVDSRYRNRALMSPAGMLRGDFSLRLFNVT PQDEQK FHCLVLSQSLGQFQEVLSVEVTLHVAANFSVPVVSAPHSPS QDELTF TCTSIINGYPRPNVYVWINKTDNSLDDQALQNDTVFLNMRGLYDVVSVLRIARTPSVNI GCCIEENVLLQQLNLTGVSQTGNDIGERDKITENPVSTGEKNAATWS <u>DI</u> <u>EGRM</u> DPKSCDKTHTCP PCPAPPELLGGGSPVFLFPPPKPKDTLMISRTPEVTCVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDW LNKKEYKCKVSNKALPAPIEKTI SKAKGQPREPQVYTLPPSRDELTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPPVLD SDGSFFLYSKLTVDKSRWQQGNVFSQSVMEALHNHYTQKSLSLSPGK Linker is underlined and in bold. Sequence preceding linker is human ICOSL (B7-H2). Sequence following linker is human IgG1 Fc.			
517	C-terminal amino acid sequence of 133 of hIL-2 with hIL-2	Amino acids 21 to 133 of hIL-2 with R38W mutation (<u>bold & underlined</u>)	LQMILNGINNYKNPKLT <u>AM</u> LTFFKPYMPKKATELKHLCQL EEELKPLEEVLNLAQSKNFHLRPRDLISNINIVLELKG SETTFMCEYADETATIVESFLNRWITFCQSIISTLT
518	C-terminal amino acid sequence of 133 of hIL-2 with hIL-2	Amino acids 21 to 133 of hIL-2 with R38Q mutation (<u>bold & underlined</u>)	LQMILNGINNYKNPKLT <u>QM</u> LTFFKPYMPKKATELKHLCQL EEELKPLEEVLNLAQSKNFHLRPRDLISNINIVLELKG SETTFMCEYADETATIVESFLNRWITFCQSIISTLT

TABLE S2-continued

SEQ ID NOS: 343-538		
SEQ ID NO: Name	Description	Sequence
519 STIM002 Corrected Light chain variable region	Nucleic acid sequence of corrected V_L of STIM002	GATATTGTGATGACTCAGTCTCCACTCTCCCTGCCCGTC ACCCCTGGAGAGCCGGCCTCCATCTCCTGCAGGTCTAGT CAGAGCCTCCTGCATAGTGTGGATACAACATTTGGAT TGGTACCTGCAGAAGCCAGGGCAGTCTCCACAGCTCCTG ATCTATTTGGGTTCTACTCGGGCTCCGGGTTCCCTGAC AGGTTCACTGGCAGTGGATCAGGCACAGATTTTACACTG AAAATCAGCAGAGTGGAGGCTGAGGATGTTGGGGTTTAT TACTGCATGCAAGCTCTACAACTCCGCTCAGTTTGGC CAGGGGACCAAGCTGGAGATCAAA
520 STIM002 - Corrected full light chain sequence	Nucleic acid sequence of corrected STIM002 light chain	GATATTGTGATGACTCAGTCTCCACTCTCCCTGCCCGTC ACCCCTGGAGAGCCGGCCTCCATCTCCTGCAGGTCTAGT CAGAGCCTCCTGCATAGTGTGGATACAACATTTGGAT TGGTACCTGCAGAAGCCAGGGCAGTCTCCACAGCTCCTG ATCTATTTGGGTTCTACTCGGGCTCCGGGTTCCCTGAC AGGTTCACTGGCAGTGGATCAGGCACAGATTTTACACTG AAAATCAGCAGAGTGGAGGCTGAGGATGTTGGGGTTTAT TACTGCATGCAAGCTCTACAACTCCGCTCAGTTTGGC CAGGGGACCAAGCTGGAGATCAAAcgtacggtggcgcgt ccctccgtgttcattcttccaccttccgacgagcagctg aagtcgggacccgcttctgtcgtgtgcctgctgaacaac ttctacccccgcgagggccaaggtgcagtggaaagtgagac aacgcccgtgcagtcggcaactcccaggaatccgtgacc gagcaggactccaaggacagcactactccctgtcctcc accctgaccctgtccaaggccgactacgagaagcacaag gtgtacgctgcgaagtgaaccacagggtcgtctagc cccgtgaccaagtcttcaaccggggcgagtgt
521 STIM003 - Corrected heavy chain variable region	Nucleic acid sequence of corrected V_H of STIM003	GAGGTGCAGCTGGTGGAGTCTGGGGGAGGTGTGGTACGG CCTGGGGGGTCCCAGAGACTCTCCTGTGTAGCCTCTGGA GTACACCTTTGATGATTATGGCATGAGCTGGGTCCGCCAA GCTCCAGGGAAGGGGCTGGAGTGGGTCTCTGGTATTAAAT TGGAATGGTGGCGACACAGATTATTCAGACTCTGTGAAG GGCCGATTACCATCTCCAGAGACAACGCCAAGAACTCC CTGTATCTACAAATGAATAGTCTGAGAGCCGAGGACACG GCCTTGATTACTGTGCGAGGGATTTCTATGGTTCGGGG AGTTATTATCACGTTCTTTTACTACTGGGGCCAGGGA ATCCTGGTCAACCTCTCTCTAGCCAGCACCAAGGCCCC TCTGTGTTCCCTCTGGCCCCCTCCAGCAAGTCCACCTCT GGCGGAACAGCCGCTCTGGGCTGCCTCGTGAAGGACTAC TTCCCGAGCCTGTGACCGTGTCTGGAACCTCTGGCGCT CTGACCAGCGAGTGCACACCTTCCCTGCTGTCTGCAG TCCTCCGGCCTGTACTCCTGTCTCCTCGTCTGACCGTG CCTTCCAGCTCTCTGGGCAACCCAGACTACATCTGCAAC GTGAACCACAAGCCTTCAACACCAAGGTGGACAAGAAG GTGAACCCCAAGTCTGCGACAAGACCCACACCTGTCCC CCTTGTCTTCCCCCTGAAGTGTGGGCGGACCTTCCGTG TTCTGTGTTCCCCCAAAGCCCAAGGACACCTGATGATC TCCCGGACCCCCAAGTGACCTGCGTGGTGGTGGATGTG TCCACAGAGGACCTGAAGTGAAGTTCAATTGGTACGTG GACGGCTGGAAGTGACAACGCCAAGACCAAGCCTAGA GAGGAACAGTACAACCTCACTACCGGGTGGTGTCCGTG CTGACCGTGTGACACAGGATTGGCTGAACGGCAAGAG TACAAGTGCAAGGTGTCCAACAAGGCCCTGCCTGCCCCC ATCGAAAAGACCATCTCAAGGCCAAGGGCCAGCCCCGG GAACCCAGGTGTACACACTGCCCCCTAGCAGGGACGAG CTGACCAAGAACCAGGTGTCCCTGACCTGTCTCGTGAAA GGCTTCTACCCCTCCGATATCGCCGTGGAATGGGAGTCC AACGGCCAGCCTGAGAACAACTACAAGACCAACCCCCCT GTGTGGAAGTCCGACGGCTCATCTTCTGTACAGCAAG
522 STIM003 - Corrected full heavy chain sequence	Nucleic acid sequence of corrected STIM003 heavy chain	GAGGTGCAGCTGGTGGAGTCTGGGGGAGGTGTGGTACGG CCTGGGGGGTCCCAGAGACTCTCCTGTGTAGCCTCTGGA GTACACCTTTGATGATTATGGCATGAGCTGGGTCCGCCAA GCTCCAGGGAAGGGGCTGGAGTGGGTCTCTGGTATTAAAT TGGAATGGTGGCGACACAGATTATTCAGACTCTGTGAAG GGCCGATTACCATCTCCAGAGACAACGCCAAGAACTCC CTGTATCTACAAATGAATAGTCTGAGAGCCGAGGACACG GCCTTGATTACTGTGCGAGGGATTTCTATGGTTCGGGG AGTTATTATCACGTTCTTTTACTACTGGGGCCAGGGA ATCCTGGTCAACCTCTCTCTAGCCAGCACCAAGGCCCC TCTGTGTTCCCTCTGGCCCCCTCCAGCAAGTCCACCTCT GGCGGAACAGCCGCTCTGGGCTGCCTCGTGAAGGACTAC TTCCCGAGCCTGTGACCGTGTCTGGAACCTCTGGCGCT CTGACCAGCGAGTGCACACCTTCCCTGCTGTCTGCAG TCCTCCGGCCTGTACTCCTGTCTCCTCGTCTGACCGTG CCTTCCAGCTCTCTGGGCAACCCAGACTACATCTGCAAC GTGAACCACAAGCCTTCAACACCAAGGTGGACAAGAAG GTGAACCCCAAGTCTGCGACAAGACCCACACCTGTCCC CCTTGTCTTCCCCCTGAAGTGTGGGCGGACCTTCCGTG TTCTGTGTTCCCCCAAAGCCCAAGGACACCTGATGATC TCCCGGACCCCCAAGTGACCTGCGTGGTGGTGGATGTG TCCACAGAGGACCTGAAGTGAAGTTCAATTGGTACGTG GACGGCTGGAAGTGACAACGCCAAGACCAAGCCTAGA GAGGAACAGTACAACCTCACTACCGGGTGGTGTCCGTG CTGACCGTGTGACACAGGATTGGCTGAACGGCAAGAG TACAAGTGCAAGGTGTCCAACAAGGCCCTGCCTGCCCCC ATCGAAAAGACCATCTCAAGGCCAAGGGCCAGCCCCGG GAACCCAGGTGTACACACTGCCCCCTAGCAGGGACGAG CTGACCAAGAACCAGGTGTCCCTGACCTGTCTCGTGAAA GGCTTCTACCCCTCCGATATCGCCGTGGAATGGGAGTCC AACGGCCAGCCTGAGAACAACTACAAGACCAACCCCCCT GTGTGGAAGTCCGACGGCTCATCTTCTGTACAGCAAG

TABLE S2-continued

SEQ ID NOS: 343-538		
SEQ ID NO: Name	Description	Sequence
		CTGACAGTGGACAAGTCCCGGTGGCAGCAGGGCAACGTG TTCTCCTGCTCCGTGATGCACGAGGCCCTGCACAACCAC TACACCCAGAAGTCCCTGTCCCTGAGCCCGGCAAGTGA TGA
523 Human IgG1 constant region	IGHG Human Heavy Chain 1*03 Constant Region (IGHG1*03) Nucleotide Sequence	gcctccaccaagggcccatcggtcttccccctggcacc tcctccaagagcacctctgggggcacagcgccctgggc tgcttggtcaaggactacttccccgaaccggtgacggtg tcgtggaactcaggcgccctgaccagcggtgcacacc ttcccggtgttctacagtcctcaggactctactccctc agcagcgtggtgaccgtgccctccagcagcttgggcacc cagacctacatctgcaacgtgaatcacaagcccagcaac accaaggtggacaagagagttgagcccaaatcttgtag aaaactcacacatgccaccgtgccagcacctgaactc ctggggggacgcgtcagttctctcttcccccaaaaccc aaggacacctcatgatctcccgaccctgaggtcaca tgcttggtggtggacgtgagccacgaagacctgaggtc aagttcaactggtacgtggacggcgtgaggtgcataat gccaagacaaagccgcgaggagcagtagacaagcagcag taccgtgtggtcagcgtcctcaccgtcctgcaccaggac tggtgaatggcaaggagtagaagtgcaaggtctccaac aaagccctcccagccccatcgagaaaacctctccaaa gccaaagggcagccccgagaaccacaggtgtacacctg cccccatccccgggagtagtagacaagaaccaggtcagc ctgacctgcttggtcaaaggcttctatcccagcgacatc gccgtggagtgggagagcaatgggcagccggagaaacac tacaagaccacgcctcccgtgctggactccgacggctcc ttcttctctatagcaagctcaccgtggacaagagcagg tggcagcaggggaacgtcttctcatgctccgtgatgcat gaggtctctgcacaaccactacacgcagaagagcctctcc ctgtccccgggtaaa
524	Human Heavy Chain Constant Region (IGHG1*03) Protein Sequence	A S T K G P S V F P L A P S S K S T S G G T A A L G C L V K D Y F P E P V T V S W N S G A L T S G V H T F P A V L Q S S G L Y S L S S V V T V P S S S L G T Q T Y I C N V N H K P S N T K V D K R V E P K S C D K T H T C P P C P A P E L L G G P S V F L F P P K P K D T L M I S R T P E V T C V V V D V S H E D P E V K F N W Y V D G V E V H N A K T K P R E E Q Y N S T Y R V V S V L T V L H Q D W L N G K E Y K C K V S N K A L P A P I E K T I S K A K G Q P R E P Q V Y T L P P S R E E M T K N Q V S L T C L V K G F Y P S D I A V E W E S N G Q P E N N Y K T T P P V L D S D G S F F L Y S K L T V D K S R W Q Q G N V F S C S V M H E A L H N H Y T Q K S L S L S P G K
525 Human IgG1 constant region	IGHG Human Heavy Chain 1*04 Constant Region (IGHG1*04) Nucleotide Sequence	gcctccaccaagggcccatcggtcttccccctggcacc tcctccaagagcacctctgggggcacagcgccctgggc tgcttggtcaaggactacttccccgaaccggtgacggtg tcgtggaactcaggcgccctgaccagcggtgcacacc ttcccggtgttctacagtcctcaggactctactccctc agcagcgtggtgaccgtgccctccagcagcttgggcacc cagacctacatctgcaacgtgaatcacaagcccagcaac accaaggtggacaagaaagttgagcccaaatcttgtag aaaactcacacatgccaccgtgccagcacctgaactc ctggggggacgcgtcagttctctcttcccccaaaaccc aaggacacctcatgatctcccgaccctgaggtcaca tgcttggtggtggacgtgagccacgaagacctgaggtc aagttcaactggtacgtggacggcgtgaggtgcataat gccaagacaaagccgcgaggagcagtagacaagcagcag taccgtgtggtcagcgtcctcaccgtcctgcaccaggac tggtgaatggcaaggagtagaagtgcaaggtctccaac aaagccctcccagccccatcgagaaaacctctccaaa gccaaagggcagccccgagaaccacaggtgtacacctg cccccatccccgggatgagctgaccaagaaccaggtcagc ctgacctgcttggtcaaaggcttctatcccagcgacatc gccgtggagtgggagagcaatgggcagccggagaaacac tacaagaccacgcctcccgtgctggactccgacggctcc ttcttctctacagcaagctcaccgtggacaagagcagg tggcagcaggggaacatcttctcatgctccgtgatgcat

TABLE S2-continued

SEQ ID NOS: 343-538		
SEQ ID NO: Name	Description	Sequence
		gaggctctgcacaaccactacacgcagaagagcctctcc ctgtctccgggtaaa
526	Human Heavy Chain Constant Region (IGHG1*04) Protein Sequence	ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTW SWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGT QTYICNVNHKPSNTKVDKKVEPKSCDKHTHTCPPCPAPEL LGGPSVFLFPPPKDLMISRTPEVTCVVVDVSHEDPEV KFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQD WLNQKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTL PPSRDELTKNQVSLTCLVKGFYPSDIAVEWESNGQPENN YKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNIFSCSVMH EALNHHTYQKSLSLSPGK
527 Human IgG2 constant region	IGHG Human Heavy Chain 2*01 Constant Region & (IGHG2*01) IGHG Nucleotide 2*03 Sequence & IGHG 2*05	gcctccaccaagggcccatcggtcttccccctggcgccc tgctccaggagcaccctccgagagcacagccgcccctgggc tgccctgggtcaaggactacttccccgaaccggtgacggtg tcgtggaaactcaggcgctctgaccagcgcggtgcacacc ttcccagctgtctctacagtcctcaggactctactccctc agcagcgtggtgaccgtgccctccagcaacttcggcacc cagacctacacctgcaacgtagatcacaaagcccagcaac accaaggtggacaagacagttgagcgcaaatgttggtgc gagtgcccaccggtgccagcaccacctgtggcaggaccg tcagtcttccctcttcccccaaaacccaaggacaccctc atgatctccccgacccctgaggtcagctgcgtgggtggg gacgtgagccacgaagaccccgaggtccagttcaactgg tacgtggacggcggtggaggtgcataatgccaaagacaaag ccacgggaggagcagttcaacagcagcttccgtgtgggtc agcgtcctcaccgttgtgcaccaggactggctgaacggc aaggagtacaagtgcgaaggtctccaacaaaaggcctccca gcccccatcgagaaaaccatctccaaaaccaaaggcgag ccccgagaaccacaggtgtacacccctgcccccatcccg gaggagatgaccaagaaccaggtcagcctgacctgcctg gtcaaaggcttctacccagcgacatcgccgtggagtg gagagcaatgggcagccggagaaacactacaagaccaca ctcccatgctggactccgacggctccttcttctctac agcaagctcacctgggacaagagcaggtggcagcagggg aacgtcttctcatgctcgtgatgcatagaggtctgcac aaccactacacgcagaagagcctctccctgtctccgggt aaa
528	Human Heavy Chain Constant Region (IGHG2*01) Protein Sequence	ASTKGPSVFPLAPCSRSTSESTAALGCLVKDYFPEPVTW SWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSNFGT QTYTCNVNHHKPSNTKVDKTVKRCCEPCCPAPPVAGP SVFLFPPPKDLMISRTPEVTCVVVDVSHEDPEVQFNW YVDGVEVHNAKTKPREEQFNSTFRVSVLTVVHQDWLNG KEYKCKVSNKGLPAPIEKTISKAKGQPREPQVYTLPPSR EEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTT PPMLDSDGSFFLYSKLTVDKSRWQQGNVFCSCVMHEALH NHYTQKSLSLSPGK
529 Human IgG2 constant region	IGHG Human Heavy Chain 2*02 Constant Region (IGHG2*02) Nucleotide Sequence	GCCTCCACCAAGGGCCCATCGGTCTTCCCCCTGGCGCCC TGCTCCAGGAGCACCTCCGAGAGCACAGCGGCCCTGGGC TGCTTGGTCAAGGACTACTTCCCCGAACCGGTGACGGTG TCGTGGAACCTCAGGCGCTCTGACCAGCGCGGTGCACACC TTCCCGGTGTCTACAGTCTCAGGACTCTACTCCCTC AGCAGCGTGGTGACCGTGACCTCCAGCAACTTCGGCACC CAGACCTACACCTGCAACGTAGATCACAAGCCAGCAAC ACCAAGGTGGACAAGACAGTTGAGCGCAATGTTGTGTC GAGTGCCACCGTGCCAGCACCACCTGTGGCAGGACCG TCAGTCTTCTCTTCCCCCAAAACCAAGGACACCTC ATGATCTCCCGGACCCCTGAGGTACGTGCGTGGTGGTG GACGTGAGCCACGAAGACCCGAGGTCCAGTTCAACTGG TACGTGGACGGCATGGAGGTGCATAATGCCAAGACAAAG CCACGGGAGGAGCAGTTCAACAGCAGTTCCGTGTGGTC AGCGTCCTCACCGTCTGTGCACAGGACTGGCTGAACGGC AAGGAGTACAAGTGCAAGGTCTCCAACAAGGCCTCCCA GCCCCATCGAGAAAACCATCTCCAAAACCAAGGGCAG CCCCGAGAACCACAGGTGTACACCTGCCCCCATCCCG GAGGAGATGACCAAGAACCAGGTGACCTGACCTGCCTG GTCAAAGGCTTCTACCCAGCGACATCGCCGTGGAGTGG GAGAGCAATGGGACGCCGAGAACTACAAGACCACA CCTCCCATGCTGGACTCCGACGGCTCTTCTTCTCTAC AGCAAGCTCACCGTGGACAAGAGCAGGTGGCAGCAGGG AACGTCTTCTCATGCTCCGTGATGCATGAGGCTCTGCAC

TABLE S2-continued

SEQ ID NOS: 343-538		
SEQ ID NO: Name	Description	Sequence
		AACCACTACACACAGAAGAGCCTCTCCCTGTCTCCGGGT AAA
530	Human Heavy Chain Constant Region (IGHG2*02) Protein Sequence	ASTKGPSVFPLAPCSRSTSESTAALGCLVKDYFPEPVTW SWNSGALTSGVHTFPAVLQSSGLYSLSSVVTSTSSNFGT QTYTCNVDHKPSNTKVDKTKVERKCCVECPPCPAPPVAGP SVFLFPPKPKDITLMI SRTPEVTCVVVDVSHEDPEVQFNW YVDGMEVHNAKTKPREEQFNSTFRVVSVLTVVHQDWLNG KEYKCKVSNKGLPAPIEKTISKTKGQPREPQVYITLPPSR EEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTT PPMLDSGSPFLYSLKLTVDKSRWQQGNVFCSSVMHEALH NHYTQKSLSLSPGK
531 Human IgG2 constant region	IGHG Human Heavy Chain 2*04 Constant Region (IGHG2*04) Nucleotide Sequence	gcctccaccaagggcccatcggtcttccccctggcgccc tgctccaggagcactccgagagcacagcgccctgggc tgcttggtcaaggactacttccccgaaccggtgacggtg tcgtggaactcaggcgctctgaccagcggtgacaccc ttccagctgtctacagtcctcaggactctactccctc agcagcgtggtgacctgacctccagcagcttgggcacc cagacctacacctgcaacgtagatcacaaagccagcaac accaaggtggacaagacagttgagcgcaaatgttgtgtc gagtgcccaaccgtgcccagcaccacctgtggcaggaccg tcagtcttctcttcccccaaaacccaaggacacctc atgatctcccgaccctcgaggtcagtgcggtggtggtg gacgtgagccacgaagaccccgaggtccagttcaactgg tacgtggacggcggtggaggtgcataatgccaaagacaaag ccacgggagggagcagttcaacagcagcttccgtgtggtc agcgtcctcaccgttgtgcaccaggactggctgaacggc aaggagtacaagtgcagggtctccaaacaaaggcctccca gcccccatcgagaaaacctctccaaaacaaaggcgag ccccgagaaccacaggtgtacacctgcccccatcccg gaggagatgaccaagaaccaggtcagcctgacctgcctg gtcaaaaggtctctacccagcgacatcgccgtggagtg gagagcaatgggcagccggagaaacactacaagaccaca ctcccatgctggactccgacggctccttcttctctac agcaagctcaccgtggacaagagcaggtggcagcagggg aacgtcttctcatgctcgtgagtgatgaggtctgtcac aaccactacacgcagaagagcctctcctctgtctccgggt aaa
532	Human Heavy Chain Constant Region (IGHG2*04) Protein Sequence	ASTKGPSVFPLAPCSRSTSESTAALGCLVKDYFPEPVTW SWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGT QTYTCNVDHKPSNTKVDKTKVERKCCVECPPCPAPPVAGP SVFLFPPKPKDITLMI SRTPEVTCVVVDVSHEDPEVQFNW YVDGVEVHNAKTKPREEQFNSTFRVVSVLTVVHQDWLNG KEYKCKVSNKGLPAPIEKTISKTKGQPREPQVYITLPPSR EEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTT PPMLDSGSPFLYSLKLTVDKSRWQQGNVFCSSVMHEALH NHYTQKSLSLSPGK
533 Human IgG2 constant region	IGHG Human Heavy Chain 2*06 Constant Region (IGHG2*06) Nucleotide Sequence	GCCTCCACCAAGGGCCCATCGGTCTTCCCCCTGGCGCCC TGCTCCAGGAGCACCTCCGAGAGCACAGCGGCCCTGGGC TGCTTGGTCAAGGACTACTTCCCCGAACCGGTGACGGTG TCGTGGAACCTCAGGCGCTCTGACCAGCGCGTGCACACC TTCCCGCTGTCTACAGTCTCAGGACTCTACTCCCTC AGCAGCGTGGTGACCGTGCCCTCCAGCAACTTCGGCACC CAGACCTACACCTGCAACGTTAGATCACAAGCCAGCAAC ACCAAGGTGGACAAGACAGTTGAGCGCAATGTTGTGTC GAGTGCCACCGTGCCAGCACCCTGTGGCAGGACCG TCAGTCTTCTCTTCCCCCAAAACCAAGGACACCTC ATGATCTCCCGGACCCCTGAGGTACGTGCGTGGTGGTG GACGTGAGCCACGAAGACCCGAGGTCCAGTTCAACTGG TACGTGGACGGCGTGGAGGTGCATAATGCCAAGACAAAG CCACGGGAGGAGCAGTTCAACAGCAGTTCCGTGTGGTC AGCGTCTCACCCTCGTGCACAGGACTGGCTGAACGGC AAGGAGTACAAGTGCAAGGTCTCCAACAAGGCCTCCCA GCCCCATCGAGAAAACCATCTCCAAAACCAAGGGCAG CCCCGAGAACCACAGGTGTACACCTGCCCCCATCCCG GAGGAGATGACCAAGAACCAGGTGACCTGACCTGCCTG GTCAAAGGCTTCTACCCAGCGACATCTCCGTGGAGTGG GAGAGCAATGGGACGCCGAGAACTACAAGACCACA CCTCCCATGCTGGACTCCGACGGCTCTTCTTCTCTAC AGCAAGCTCACCCTGGACAAGAGCAGGTGGCAGCAGGG AACGTCTTCTCATGCTCCGTGATGCATGAGGCTCTGCAC

TABLE S2-continued

SEQ ID NOS: 343-538		
SEQ ID NO: Name	Description	Sequence
		AACCACTACACACAGAAGAGCCTCTCCCTGTCTCCGGGT AAA
534	Human Heavy Chain Constant Region (IGHG2*06) Protein Sequence	ASTKGPSVFPLAPCSRSTSESTAALGCLVKDYFPEPVTV SWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSNFGT QTYTCNVDPKPSNTKVDKTKVERKCCVECPAPPVAGP SVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVQFNW YVDGVEVHNAKTKPREEQFNSTFRVSVLTVVHQDNLNG KEYKCKVSNKGLPAPIEKTISKTKGQPREPQVYTLPPSR EEMTKNQVSLTCLVKGFYPSDISEWESNGQPENNYKTT PPMLDSDGSPFLYSKLTVDKSRWQQGNVFSQSVMHEALH NHYTQKSLSLSPGK
535 Human Cλ constant region	IGLC Cλ Light Chain 7*03 Constant Region (IGLC7*03) Nucleotide Sequence	GGTCAGCCCAAGGCTGCCCCCTCGGTCACTCTGTTCCTCA CCCTCCTCTGAGGAGCTTCAAGCCAAACAGGCCACACTG GTGTGTCTCGTAAGTGACTTCAACCCGGGAGCCGTGACA GTGGCCTGGAAGGCAGATGGCAGCCCGCTCAAGGTGGGA GTGGAGACCACCAACCTCCAAACAAGCAACAACAAG TATGCGGCAGCAGCTACCTGAGCCTGACGCCCGAGCAG TGGAAGTCCACAGAAGCTACAGCTGCCGGGTACGCAT GAAGGGAGCACCGTGGAGAAGACAGTGGCCCTGCAGAA TGCTCT
536	Cλ Light Chain Constant Region (IGLC7*03) Amino Acid Sequence	GQPKAAPSVTLFPPSSEELQANKATLVCLVSDFNPGAVT VAWKADGSPVKGVETTKPSKQSNNKYAASSYLSLTPEQ WKSRSYSQVTHEGSTVEKTVAPAECS
537 Human WT IgG1 constant region	IGHG WT human IgG1 1*01 nucleotide & sequence #2 IGHG 1*05 (IgG 1)	gcctccaccaagggcccatcggtcttccccctggcaccc tcctccaagagcactctgggggacacagcgccctgggc tgcttggtcaaggactacttccccgaaccggtgacgggtg tcgtgggaactcaggcgccctgaccagcggtgcacacc ttcccggtgtcctacagtcctcaggactctactccctc agcagcgtggtgacctgacctccagcagcttgggcacc cagacctacatctgcaacgtgaatcacaagccagcaac accaaggtggacaagaagttgagcccaaatcttgtagc aaaactcacacatgcccaccgtgcccagcacctgaactc ctgggggggacgtcagctcttctcttccccccaaaacc aaggacacctcatgatctcccgaccctgaggtcaca tgcttggtggtggacgtgagccacgaagacctgaggtc aagtccaactggtacgtggacggcgtggaggtgcataat gccaagacaagcgcgggaggagcagtaaacagcacg taccgggtggtcagcgtcctcaccgtcctgcaccaggac tggctgaatggcaaggagtacaagtgcaaggtctccaac aaagccctcccagcccccatcgagaaaaccatctccaaa gccaaggggcagccccgagaaccacaggtgtacacctg cccccatcccgggatgagctgaccaagaaccaggtcagc ctgacctgctggtcaaaggcttctatcccagcgacatc gccgtggagtgggagagcaatgggcagccggagaacaac tacaagaccacgcctcccgtgctggactccgacgggtcc ttcttctctacagcaagctcaccgtggacaagagcagg tggcagcaggggaacgtcttctcatgctccgtgatgcat gaggctctgcacaaccactacacgcagaagagcctctcc ctgtctccgggtaaa
538 Human Cλ constant region	IGLC Cλ Light Chain 2*01 Constant Region Amino Acid Sequence #2 - Encoded by nucleotide sequence version A & B	GQPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGAVT VAWKADSSPVKAGVETTPSKQSNNKYAASSYLSLTPEQ WKSRSYSQVTHEGSTVEKTVAPTECS

TABLE S3

SEQ ID NOS: 539-562		
Sequence		
hIgG1 FIT-Ig bispecific 1a		
Antibody A	anti-ICOS STIM003	
Antibody B	anti-PD-L1 84G09	
FIT-Ig Construct #1	SEQ ID NO: 539	DIQMTQSPASLSASLGETVTIQCRASEDIYSGLAWFQQKPGKSPQLLIYGASS LQDGVPSRFSGSGSGTQYSLKISSMQTEDEGVYFCQQGLKYPPTFGSGTKLEI KRTVAAPSVFIFPPSDEQLKSGTASVVCCLNNFYPREAKVQWKVDNALQSGNS QESVTEQDSKDSSTYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRG ECEVQLVESGGGLTQPGKSLKLSCEASGFTFSSFTMHWVRQSPGKGLEWVAFI RSGSGIVFYADAVRGRFTISRDNALNLLFLQMNLDKSEDAMYYCARRPLGHN TFDSWGQGTLLVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPV VSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSN TKVDKKVEPKSCDKTHTCPPCPAPPELLGGPSVFLFPPKPKDTLMISRTPEVTC VVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWL NGKEYCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQVSLTCL LVKGFYPSDIAVEWESNGQPENNYKTPPVLDSDGSFFLYSKLTVDKSRWQQG NVFSCSVMHEALHNHYTQKSLSLSPGK
FIT-Ig Construct #2	SEQ ID NO: 540	EVQLVESGGGLVQPGRSLKLSCAASGFTFSDFYMAWVRQAPKKGLEWVASISY EGSSTYYGDSVMGRFTISRDNASTLYLQMNLSRSEDATYYCARQREANWED WGQGMVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWN SGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVD KKV
FIT-Ig Construct #3	SEQ ID NO: 541	DIVMTQSPSSSLAVSPGEKVTMTCKSSQSLYSGVKENLLAWYQQKPGQSPKLL IYYASIRFTGVPRFTGSGSGTDYTLTIISVQAEDMGQYFCQQGINNPLTFGD GTKLEIKRTVAAPSVFIFPPSDEQLKSGTASVVCCLNNFYPREAKVQWKVDNA LQSGNSQESVTEQDSKDSSTYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPV TSFNRRGEC
hIgG1 FIT-Ig bispecific 1b		
Antibody A	anti-PD-L1 84G09	
Antibody B	anti-ICOS STIM003	
FIT-Ig Construct #1	SEQ ID NO: 542	DIVMTQSPSSSLAVSPGEKVTMTCKSSQSLYSGVKENLLAWYQQKPGQSPKLL IYYASIRFTGVPRFTGSGSGTDYTLTIISVQAEDMGQYFCQQGINNPLTFGD GTKLEIKRTVAAPSVFIFPPSDEQLKSGTASVVCCLNNFYPREAKVQWKVDNA LQSGNSQESVTEQDSKDSSTYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPV TSFNRRGEC ECEVQLVESGGGLVQPGRSLKLSCAASGFTFSDFYMAWVRQAPKKGL EWVASISYEGSSTYYGDSVMGRFTISRDNASTLYLQMNLSRSEDATYYCAR QREANWEDWGQGMVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFP EPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNH KPSNTKVDKKVEPKSCDKTHTCPPCPAPPELLGGPSVFLFPPKPKDTLMISRT PEVTCVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLH QDWLNGKEYCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSRDELTKNQV SLTCLVKGFYPSDIAVEWESNGQPENNYKTPPVLDSDGSFFLYSKLTVDKSR WQQGNVPSCSVMHEALHNHYTQKSLSLSPGK
FIT-Ig Construct #2	SEQ ID NO: 543	EVQLVESGGGLTQPGKSLKLSCEASGFTFSSFTMHWVRQSPGKGLEWVAFIRS GSGIVFYADAVRGRFTISRDNALNLLFLQMNLDKSEDAMYYCARRPLGHNTF DSWGQGTLLVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVS WNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTK VDKKV
FIT-Ig Construct #3	SEQ ID NO: 544	DIQMTQSPASLSASLGETVTIQCRASEDIYSGLAWFQQKPGKSPQLLIYGASS LQDGVPSRFSGSGSGTQYSLKISSMQTEDEGVYFCQQGLKYPPTFGSGTKLEI KRTVAAPSVFIFPPSDEQLKSGTASVVCCLNNFYPREAKVQWKVDNALQSGNS QESVTEQDSKDSSTYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRG EC
hIgG1 FIT-Ig bispecific 2a		
Antibody A	anti-ICOS STIM001	
Antibody B	anti-PD-L1 1D05	
FIT-Ig Construct #1	SEQ ID NO: 545	DIQMTQSPASLSASLGETVTIQCRASEDIYSGLAWFQQKPGKSPQLLIYGASS LQDGVPSRFSGSGSGTQYSLKISSMQTEDEGVYFCQQGLKYPPTFGSGTKLEI KRTDAAPTQSVIFPPSDEQLTSGGASVVCFLNNFYPKDINVWKIDGSRQNGV

TABLE S3-continued

SEQ ID NOS: 539-562		
Sequence		
		LNSWTDQDSKSDSTYSMSSTLTLTKEYERHNSYTCEATHKTSTSPIVKSFNRN ECEVQLVESGGGLTQPGKSLKLSCEASGFTFSSFTMHWVRQSPGKGLEWVAFI RSGSGIVFYADAVRGRFTISRDNANKLLFLQMNDLKSEDTAMYYCARRPLGHN TFDSWGQGTLLVTSSAKTTAPSVYPLAPVCGDTGSSVTLGCLVKGYFPEPV LTWNSGSLSSGVHTFPAVLQSDLYTLSSSVTVTSSTWPSQSITCNVAHPASST KVDKKIEPRGPTIKCPPCKCPAPNLLGGPSVFI FPPKIKDVLMLISLSPIVTC VVVDVSEDDPDVQISWVFNNEVHTAQTQTHREDYNSTLRVVSALPIQHODWM SGKEFKCKVNNKDLPAPIERTISKPKGSVRAPQVYVLPPEEEMTKKQVTLTC MVTDFMPEDIYVEWTNNGKTELNYKNTEPVLDSGGSYFMYSKLRVEKKNWVER NSYSCSVVHEGLHNHHTTKSFRTPGK
FIT-Ig Construct #2	SEQ ID NO: 546	EVQLVESGGGLVQPGRSLKLSAASGFTFSDFYMAWVRQAPKKGLEWVASISY EGSSTYYGDSVMGRFTISRDNASTLYLQMNLSRSEDATYYCARQREANWED WGQGVMTVSSAKTTAPSVYPLAPVCGDTGSSVTLGCLVKGYFPEPVTLTN SGLSSGVHTFPAVLQSDLYTLSSSVTVTSSTWPSQSITCNVAHPASSTKVDK KI
FIT-Ig Construct #3	SEQ ID NO: 547	DIVMTQSPSSLAVSPGEKVTMTCKSSQSLYSGVKENLLAWYQQKPGQSPKLL IYYASIRFTGVPDRFTGSGSGTDYTLTITSVQAEDMGQYFCQQGINNPLTFGD GTKLEIKRTDAAPTIVSIFPPSSSEQLTSGGASVVCFLNNFYPKIDINVWKIDGS ERQNGVLNSWTDQDSKSDSTYSMSSTLTLTKEYERHNSYTCEATHKTSTSPIV KSFNRNEC
hIgG1 FIT-Ig bispecific 2b		
Antibody A	anti-PD-L1 1D05	
Antibody B	anti-ICOS STIM001	
FIT-Ig Construct #1	SEQ ID NO: 548	DIVMTQSPSSLAVSPGEKVTMTCKSSQSLYSGVKENLLAWYQQKPGQSPKLL IYYASIRFTGVPDRFTGSGSGTDYTLTITSVQAEDMGQYFCQQGINNPLTFGD GTKLEIKRTDAAPTIVSIFPPSSSEQLTSGGASVVCFLNNFYPKIDINVWKIDGS ERQNGVLNSWTDQDSKSDSTYSMSSTLTLTKEYERHNSYTCEATHKTSTSPIV KSFNRNECEVQLVESGGGLVQPGRSLKLSAASGFTFSDFYMAWVRQAPKKGL EWVASISYEGSSTYYGDSVMGRFTISRDNASTLYLQMNLSRSEDATYYCAR QREANWEDWGQGVMTVSSAKTTAPSVYPLAPVCGDTGSSVTLGCLVKGYF EPVTLTNWNSGSLSSGVHTFPAVLQSDLYTLSSSVTVTSSTWPSQSITCNVAHP ASSTKVDKKIEPRGPTIKCPPCKCPAPNLLGGPSVFI FPPKIKDVLMLISLS IVTCVVVDVSEDDPDVQISWVFNNEVHTAQTQTHREDYNSTLRVVSALPIQH QDWMSGKEFKCKVNNKDLPAPIERTISKPKGSVRAPQVYVLPPEEEMTKKQV TLTCMVTDFMPEDIYVEWTNNGKTELNYKNTEPVLDSGGSYFMYSKLRVEKKN WVERNSYSCSVVHEGLHNHHTTKSFRTPGK
FIT-Ig Construct #2	SEQ ID NO: 549	EVQLVESGGGLTQPGKSLKLSCEASGFTFSSFTMHWVRQSPGKGLEWVAFIRS GSGIVFYADAVRGRFTISRDNANKLLFLQMNDLKSEDTAMYYCARRPLGHNTF DSWGQGTLLVTSSAKTTAPSVYPLAPVCGDTGSSVTLGCLVKGYFPEPVTLT WNSGSLSSGVHTFPAVLQSDLYTLSSSVTVTSSTWPSQSITCNVAHPASSTKV DKKI
FIT-Ig Construct #3	SEQ ID NO: 550	DIQMTQSPASLSASLGSETVTIQCRASEDIYSGLAWFQQKPGKSPQLLIYGASS LQDGVPSRFGSGSGTQYSLKISSMQTEDEGVYFCQQGLKYPPTFGSGTKLEI KRTDAAPTIVSIFPPSSSEQLTSGGASVVCFLNNFYPKIDINVWKIDGSE RQNGV LNSWTDQDSKSDSTYSMSSTLTLTKEYERHNSYTCEATHKTSTSPIVKSFNRN EC
hIgG1 FIT-Ig bispecific 3a		
Antibody A	anti-ICOS STIM003	
Antibody B	anti-PD-L1 1D05	
FIT-Ig Construct #1	SEQ ID NO: 551	DIQMTQSPASLSASLGSETVTIQCRASEDIYSGLAWFQQKPGKSPQLLIYGASS LQDGVPSRFGSGSGTQYSLKISSMQTEDEGVYFCQQGLKYPPTFGSGTKLEI KRTVAAPSVFI FPPSDEQLKSGTASVVCFLNNFYPREAKVQWKVDNALQSGNS QESVTEQDSKSDSTYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRG ECEVQLVESGGGLTQPGKSLKLSCEASGFTFSSFTMHWVRQSPGKGLEWVAFI RSGSGIVFYADAVRGRFTISRDNANKLLFLQMNDLKSEDTAMYYCARRPLGHN TFDSWGQGTLLVTSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPV VSWNSGALTSGVHTFPAVLQSSGLYSLSSSVTVPSSSLGTQTYICNVNHKPSN TKVDKKVEPKSCDKHTCTPPNLLGGPSVFI FPPKIKDVLMLISLSPIVTCVVVD VSEDDPDVQISWVFNNEVHTAQTQTHREDYNSTLRVVSALPIQHODWMSGKE FKCKVNNKDLPAPIERTISKPKGSVRAPQVYVLPPEEEMTKKQVTLTCMVTDF MPEDIYVEWTNNGKTELNYKNTEPVLDSGGSYFMYSKLRVEKKNWVERNSY SCVVHEGLHNHHTTKSFRTPGK

TABLE S3-continued

SEQ ID NOS: 539-562		
Sequence		
FIT-Ig Construct #2	SEQ ID NO: 552	EVQLVESGGGLVQPGRSLKLSCAASGFTFSDFYMAWVRQAPKKGLEWVASISY EGSSTYYGDSVMGRFTISRDNASTLYLQMNLSRSEDATYYCARQREANWED WGQGVMTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWN SGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVD KKV
FIT-Ig Construct #3	SEQ ID NO: 553	DIVMTQSPSSSLAVSPGKEVTMTCKSSQSLYYSGVKENLLAWYQQKPGQSPKLL IYYASIRFTGVDPDRFTGSGSGTDYTLTITSVQAEDMGQYFCQQGINNPLTFGD GTKLEIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA LQSGNSQESVTEQDSKDYSLSSSTLTLSKADYEKHKVYACEVTHQGLSSPVT KSFNRGEC
hIgG1 FIT-Ig bispecific 3b Antibody A	anti-PD-L1 1D05	
Antibody B	anti-ICOS STIM003	
FIT-Ig Construct #1	SEQ ID NO: 554	DIVMTQSPSSSLAVSPGKEVTMTCKSSQSLYYSGVKENLLAWYQQKPGQSPKLL IYYASIRFTGVDPDRFTGSGSGTDYTLTITSVQAEDMGQYFCQQGINNPLTFGD GTKLEIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA LQSGNSQESVTEQDSKDYSLSSSTLTLSKADYEKHKVYACEVTHQGLSSPVT KSFNRGECEVQLVESGGGLVQPGRSLKLSCAASGFTFSDFYMAWVRQAPKKGL EWVASISYEGSSTYYGDSVMGRFTISRDNASTLYLQMNLSRSEDATYYCAR QREANWEDWGQGVMTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPE PVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNH KPSNTKVDKKVEPKSCDKTHTCPPNLLGGPSVFIFPPKIKDVLMISSLPIVTC VVVDVSEDDPDVQISWVFNNEVHTAQTQTHREDYNSTLRVVSALPIQHODWM SGKEFKCKVNNKDLPAPIERTISKPKGSVRAPQVYVLPPEEEMTKQVTLTC MVTDFMPEDIYVEWTNNGKTELNYKNTEPVLDSGGSYFMYSKLRVEKKNWVER NSYSCSVVHEGLHNHHTTKSFSRTPGK
FIT-Ig Construct #2	SEQ ID NO: 555	EVQLVESGGGLTQPGKSLKLSCEASGFTFSSTMHWRQSPGKLEWVAFIRS GSGIVFYADAVRGRFTISRDNAKNLLFLQMNDLKS EDTAMYYCARRPLGHNTF DSWGQGTLVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVS WNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTK VDKKV
FIT-Ig Construct #3	SEQ ID NO: 556	DIQMTQSPASLSASLGETVTIQCRASEDIYSGLAWFQQKPGKSPQLLIYGASS LQDGVPSRFGSGSGTQYSLKISSMQTEDEGVYFCQQGLKYPPTFGSGTKLEI KRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQSGNS QESVTEQDSKDYSLSSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRG EC
hIgG1 FIT-Ig bispecific 4a Antibody A	anti-ICOS STIM001	
Antibody B	anti-PD-L1 84G09	
FIT-Ig Construct #1	SEQ ID NO: 557	DIQMTQSPASLSASLGETVTIQCRASEDIYSGLAWFQQKPGKSPQLLIYGASS LQDGVPSRFGSGSGTQYSLKISSMQTEDEGVYFCQQGLKYPPTFGSGTKLEI KRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQSGNS QESVTEQDSKDYSLSSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRG ECEVQLVESGGGLTQPGKSLKLSCEASGFTFSSTMHWRQSPGKLEWVAFI RSGSGIVFYADAVRGRFTISRDNAKNLLFLQMNDLKS EDTAMYYCARRPLGHN TFDSWGQGTLVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT VSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSN TKVDKKVEPKSCDKTHTCPPNLLGGPSVFIFPPKIKDVLMISSLPIVTCVVVD VSEDDPDVQISWVFNNEVHTAQTQTHREDYNSTLRVVSALPIQHODWMSGKE FKCKVNNKDLPAPIERTISKPKGSVRAPQVYVLPPEEEMTKQVTLTCMVT DFMPEDIYVEWTNNGKTELNYKNTEPVLDSGGSYFMYSKLRVEKKNWVERNSY SCSVVHEGLHNHHTTKSFSRTPGK
FIT-Ig Construct #2	SEQ ID NO: 558	EVQLVESGGGLVQPGRSLKLSCAASGFTFSDFYMAWVRQAPKKGLEWVASISY EGSSTYYGDSVMGRFTISRDNASTLYLQMNLSRSEDATYYCARQREANWED WGQGVMTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWN SGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVD KKV
FIT-Ig Construct #3	SEQ ID NO: 559	DIVMTQSPSSSLAVSPGKEVTMTCKSSQSLYYSGVKENLLAWYQQKPGQSPKLL IYYASIRFTGVDPDRFTGSGSGTDYTLTITSVQAEDMGQYFCQQGINNPLTFGD GTKLEIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA

TABLE S3-continued

SEQ ID NOS: 539-562	
Sequence	
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVT KSFNRGEC	
hIgG1 FIT-Ig bispecific 4b	
Antibody A	anti-PD-L1 84G09
Antibody B	anti-ICOS STIM001
FIT-Ig Construct #1	SEQ ID NO: 560 DIVMTQSPSSSLAVSPGKEVTMTCKSSQSLYYSGVKENLLAWYQQKPGQSPKLL IYYASIRFTGVDPDRFTGSGSDTYTLTIISVQAEDMGQYFCQQGINNPLTFGD GTKLEIKRTVAAPSVFIFPPSDEQLKSGTASVVCCLNNFYPREAKVQWKVDNA LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVT KSFNRGECVQLVESGGGLVQPGRSLKLSAASGFTFSDFYMAWVRQAPKKG L EWVASISYEGSSTYYGDSVMGRFTISRDNAKSTLYLQMNLSRSEDATYYCAR QREANWEDWGQGMVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF P EPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNH KPSNTKVDKKVEPKSCDKHTCTPPNLLGGPSVFIFPPKIKDVLIMISLPIVTC VVVDVSEDDPDVQISWVFNNEVHTAQTQTHREDYNSLTFRVVSALPIQHQQDWM SGKEFKCKVNNKDLPAPIERTISKPKGSVRAPQVYVLPPEEEMTKQVTLTC MVTDFMPEDIYVEWTNNGKTELNYKNTEPVLDSGSYFMYSKLRVEKKNWVER NSYSCSVVHEGLHNHHTTKFSRTPGK
FIT-Ig Construct #2	SEQ ID NO: 561 EVQLVESGGGLTQPGKSLKLSCEASGFTFSSFTMHWRQSPGKLEWVAFIRS GSGIVFYADAVRGRFTISRDNAKNLLFLQMNLDKSEDTAMYYCARRPLGHNTF DSWQQGTLVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVS WNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTK VDKKV
FIT-Ig Construct #3	SEQ ID NO: 562 DIQMTQSPASLSASLGETVTIQCRASEDIYSGLAWFQKPKSPQLLIYGASS LQDGVPSRFSGSGSGTQYSLKISSMQTEDEGVYFCQQGLKYPPTFGSGTKLEI KRTVAAPSVFIFPPSDEQLKSGTASVVCCLNNFYPREAKVQWKVDNALQSGNS QESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRG EC

TABLE S4

Sequences of antibody heavy chain variable regions obtained from additional clones CDRs are defined according to IMGT.					
CLONE_ID	VH_NUCLEOTIDE_SEQUENCE	VH_AMINO_ACID_SEQ	HCDR1	HCDR2	HCDR3
CL-61091	CAGGTTCAACTGATGCAGTCTGGAAGCTGAGGTGAAGAAGCCTGGG	QVQLMQSGTEVKKPGASV	GYTFTTYG	ISAYS GDT	ARSSGWP
STIM017	GCCTCAGTGAAGGTCTCCTGCAAGACTTCTGGTTACACCTTTACC ACCTATGTTATCACTTGGGTGCGACAGGCCCTGGACAAGGGCTT GAGTGGATGGGATGGATCAGCGCTTACAGTGGTGACACAGACTAT GCACAGAAGTCCAGGGCAGAGTCACCGTGACAACAGACACATCC ACGAACACAGCCTACATGGAGTTGAGGAGCCTGAAATCTGACGAC ACGGCCGTGTATTATGTGCGAGAAGTAGTGGCTGGCCCCACCAC TACGGTATGGACGCTCTGGGGCCAAGGGACCACGGTCAACCGTCTCC TCAG SEQ ID NO: 563	KVSCKTSGYFTTYGIGTW VRQAPGQGLEWMGWISAY SGD TDYAKFQGRVTVTT DTSNTAYMELRSLKSDD TAVYYCARSSGWP HHYGM DVWGQGT TVTVSS SEQ ID NO: 564	SEQ ID NO: 565	SEQ ID NO: 566	HYGMDV SEQ ID NO: 567
CL-64536	CAGGTTCAACTGGTGCAGTCTGGAGGTGAGGTGAAAAAGCCTGGG GCCTCAGTGAAGGTCTCCTGCAAGGCTTCTGGTTACACCTTTACC AGCTATGTTTTCAGTGGGTGCGACAGGCCCTGGACAAGGACTA GAGTGGATGGGATGGATCAGCGCTTACAATGGTAACACAACTAT GCACAGAAGTCCAGGGCAGAGTCTCCATGACCACAGACACATCC ACGAGCACAGCCTACATGGAGCTGAGGAGCTTGAGATCTGACGAC ACGGCCGTGTATTCTGTGCGCGATCTACGCTTACTATGGTTCG GGGACCTTATACGGTATGGACGCTCTGGGGCCAAGGGACCACGGTC ACCGTCTCCTCAG SEQ ID NO: 568	QVQLVQSGGEVKKPGASV KVSCKASGYFTTSYGFSW VRQAPGQGLEWMGWISAY NGNTNYAQKLQGRVSMIT DTSNTAYMELRSLRSD TAVYFCARSTSYGSGTL YGM DVWGQGT TVTVSS SEQ ID NO: 569	GYTFTSYG	ISAYNGNT	ARSTSYYS GTLYGMDV SEQ ID NO: 570
CL-64837	CAGGTTCAACTGGTGCAGTCTGGAGGTGAGGTGAAGAAGCCTGGG GCCTCAGTGAAGGTCTCCTGCAAGGCTTCTGGTTACACCTTTACC AGCTATGTTTTCAGTGGGTGCGACAGGCCCTGGACAAGGACTA GAGTGGATGGGATGGATCAGCGCTTACAATGGTAACACAACTAT GCACAGAAGTCCAGGGCAGAGTCTCCATGACCACAGACACATCC ACGAGCACAGCCTACATGGAGCTGAGGAGCTTGAGATCTGACGAC ACGGCCGTGTATTCTGTGCGCGATCTACGCTTACTATGGTTCG GGGACCTTATACGGTATGGACGCTCTGGGGCCAAGGGACCACGGTC ACCGTCTCCTCAG SEQ ID NO: 571	QVQLVQSGGEVKKPGASV KVSCKASGYFTTSYGFSW VRQAPGQGLEWMGWISAY NGNTNYAQKLQGRVSMIT DTSNTAYMELRSLRSD TAVYFCARSTSYGSGTL YGM DVWGQGT TVTVSS SEQ ID NO: 572	GYTFTSYG	ISAYNGNT	ARSTSYYS GTLYGMDV SEQ ID NO: 570

TABLE S4-continued

Sequences of antibody heavy chain variable regions obtained from additional clones CDRs are defined according to IMGT.					
CLONE_ID	VH_NUCLEOTIDE_SEQUENCE	VH_AMINO_ACID_SEQ	HCDR1	HCDR2	HCDR3
CL-64841	CAGGTTCAACTGGTGCAGTCTGGAGGTGAGGTGAAAAAGCCTGGG GCCTCAGTGAAGGTCTCCTGCAAGGCTTCTGGTTACACCTTTACC AGCTATGGTTTCAGCTGGGTGCGACAGGCCCTGGACAAGGACTA GAGTGGATGGGATGGATCAGCGCTTACAATGGTAACACAAACTAT GCACAGAAGCTCCAGGGCAGAGTCTCCATGACCACAGACACATCC ACGAGCACAGCCTACATGGAGCTGAGGAGCTTGAGATCTGACGAC ACGGCCGTGTATTCTGTGCGCGATCTACGTCTTACTATGGTTCG GGGACCCCTATACGGTATGGACGTCTGGGGCCAAGGGACCACGGTC ACCGTCTCCTCAG SEQ ID NO: 573	QVQLVQSGGEVKKPGASV KVSKKASGYFTTSYGFWS VRQAPGQGLEWMGWISAY NGNTNYAQLQGRVSMIT DTSTSTAYMELRSLRSD TAVYFCARSTSYSGT YGMVWVGQGTITVSS SEQ ID NO: 574	GYTFTSYG SEQ ID NO: 377	ISAYNGNT SEQ ID NO: 378	ARSTSYYS GTLYGMDV SEQ ID NO: 570
CL-64912	CAGGTTCAACTGGTGCAGTCTGGAGGTGAGGTGAAAAAGCCTCGG GCCTCAGTGAAGGTCTCCTGCAAGGCTTCTGGTTACACCTTTACC AGCTATGTGTTTCAGCTGGGTGCGACATGCCGCTGGACAAGGACTA GAGTGGATGGGATGGATCAGCGGTTACAATGGTAACACAACTAT GCACAGAAGCTCCAGTGGCGAGTCTCGATGACCCGAGACACATCC ACGAGCACAGCCTACATGGAGCTGAGGAGCTTGAGATCTGACGAC ACGGCCGTGTATTCTGTGCGCGATCTACGTCTTACTATGGTTCG GGGACCCCTATACGGTATGGACGTCTGGGGCCAAGGGACCACGGTC ACCGTCTCCTCAG SEQ ID NO: 575	QVQLVQSGGEVKKPRASV KVSKKASGYFTTSYVFSW VRHAAGQGLEWMGWISGY NGNTNYAQLQCGVSMIT DTSTSTAYMELRSLRSD TAVYFCARSTSYGAGT YGMVWVGQGTITVSS SEQ ID NO: 576	GYTFTSYV SEQ ID NO: 577	ISGYNGNT SEQ ID NO: 578	ARSTSYYG GTLYGMDV SEQ ID NO: 579
CL-71642	GAGGTGCAGCTGGTGGAGTCTGGGGGAGGTGTGGTACGGCCTGGG GGGTCCCTGAGACTCTCCTGTGCGAGCCTCTGGATTACCTTTGAT GATTATGGCATGAGCTGGGTCCGCCAAGCTCCAGGGAAGGGGCTG GAGTGGGTCTCTGTTATTAATTGGAATGGTGGTAGCACAGGTTAT GCAGACTCTGTGAAGGGCCGATTACCATCTCCAGAGACAACGCC AAGAACTCCCTGTATCTGCAAAATGAACAGTCTGAGAGCCGAGGAC ACGGCCTTGTATTACTGTGCGGCCGATTACTATGGTTCGGGGAGT TATTATAACGTCCTCTTTGACTACTGGGGCCAGGGAACCTGGTC ACCGTCTCCTCAG SEQ ID NO: 580	EVQLVESGGGVVRPGGSL RLSCAASGFTFDDYGMWS VRQAPGKLEWVSGINWN GGSTGYADVSKGRFTISR DNAKNSLYLQMNSLRAED TALYYCAADYYSYSGSYN VPPDYWGQGLTVTVSS SEQ ID NO: 581	GFTFDDYG SEQ ID NO: 582	INWNGGST SEQ ID NO: 583	AADYYSGS YYNVPFDY SEQ ID NO: 584
CL-74570	GAGGTGCAGCTGGTGGAGTCTGGGGGAGGTGTGATACGGCCTGGG GGGTCCCTGAGACTCTCCTGTGCGAGCCTCTGGATTACCTTTGAT GATTATGGCATGAGCTGGGTCCGCCAAGCTCCAGGGAAGGGGCTG GAGTGGGTCTCTGTTATTAATTGGAATGGTGGTAGAACACAGATTAT GCAGACTCTGTGAAGGGCCGATTACCATCTCCAGAGACAACGCC AAGAACTCCCTATATCTGCAAAATGAACAGTCTGAGAGCCGAGGAC ACGGCCTTGTATTACTGTGCGAGAGATTACTTTGGTTCGGGGAGT TATTATAACGTTCCCTTTGACTACTGGGGCCAGGGAACCTGGTC ACCGTCTCCTCAG SEQ ID NO: 585	EVQLVESGGGVIRPGGSL RLSCAASGFTFDDYGMWS VRQAPGKLEWVSGINWI GDNTDYADVSKGRFTISR DNAKNSLYLQMNSLRAED TALYYCARDYFSGSYN VPPDYWGQGLTVTVSS SEQ ID NO: 586	GFTFDDYG SEQ ID NO: 582	INWIGDNT SEQ ID NO: 587	ARDYFSGS YYNVPFDY SEQ ID NO: 588

TABLE S5

Sequences of antibody light chain variable regions obtained from additional clones N terminal E and 5' nucleotide additions in CL-71642 are shown in bold. These were not recovered in sequencing but were determined to be present in the sequence by comparison against the related clones as shown in FIG. 36. CDRs are defined according to IMGT.					
CLONE_ID	VL_NUCLEOTIDE_SEQUENCE	VL_AMINO_ACID_SEQ	LCDR1	LCDR2	LCDR3
CL-61091 STIM017	GATATTGTGATGACTCAGTCTCCACTCTCCCTGCCCGTCACCCCT GGAGAGCCGGCCTCCATCTCCTGCGAGGTCTAGTCAGAGCCTCCTG CATAGTAATGGATTCAACTATTTCGATTGGTACCTGCAGAAGCCA GGACAGTCTCCACAGCTCCTGATCTTTTGGTTTCTAATCGGGCC TCCGGGGTCCCTGACAGGTTCACTGGCAGTGGATCAGGCACAGAT TTTACTACTGAAATCAGCAGAGTGGAGGCTGAGGATGTTGGGATT TATTACTGCATGCAAGCTCTACAACTCCGCTCACTTTTCGGCGGA GGGACCAAGGTGGAGATCAAC SEQ ID NO: 589	DIVMTQSPSLPVPTEPEA SISCRSSQSLHNSGNFYF DWYLQKPGQSPQLLIFLVS NRASGVPDRFSGSGSTDF TLKISRVEAEDVGIYYCMQ ALQTPLTFGGGTKVEIK SEQ ID NO: 590	QSLHNSGNFY SEQ ID NO: 591	LVS SEQ ID NO: 592	MQALQTPLT SEQ ID NO: 593

TABLE S5-continued

Sequences of antibody light chain variable regions obtained from additional clones N terminal E and 5' nucleotide additions in CL-71642 are shown in bold. These were not recovered in sequencing but were determined to be present in the sequence by comparison against the related clones as shown in FIG. 36. CDRs are defined according to IMGT.					
CLONE_ID	VL_NUCLEOTIDE_SEQUENCE	VL_AMINO_ACID_SEQ	LCDR1	LCDR2	LCDR3
CL-64536	GATATTGTGATGACTCAGTCTCCACTCTCCCTGCCCGTCACCCCT GGAGAGCCGGCCTCCATCTCCTGCAGGTCTAGTCAGAGCCTCCTG CATAGTAATGGATACAACCTGTTGGATTGGTACCTGCAGAAGCCA GGGCAGTCTCCACAGCTCCTGATCTATTGGGTTCTACTCGGGCC TCCGGGTTCCTTGACAGGTTCACTGGCAGTGGATCAGGCACAGAT TTTACACTGAAAATCAGCAGAGTGGAGGCTGAGGATGTTGGGGTT TATTACTGCATGCAAGCTCTACAACTCCGTGCAGTTTGGCCAG GGGACCAAGCTGGAGATCAAAAC SEQ ID NO: 594	DIVMTQSPSLSLPVTGPGEPA QSLHNSGYNC LGS MQALQTPCS SISCRSSQSLHNSGYNCL SEQ ID NO: SEQ SEQ ID DWYLQKPGQSPQLLIYLG 596 ID NO: NO: 400 TRASGFPDRFSGSGSGTDF 371 TLKISRVEAEDVGYYCMQ ALQTPCSFGQGTKLEIK SEQ ID NO: 595			
CL-64837	GATATTGTGATGACTCAGTCTCCACTCTCCCTGCCCGTCACCCCT GGAGAGCCGGCCTCCATCTCCTGCAGGTCTAGTCAGAGCCTCCTG CATAGTAATGGATACAACCTGTTGGATTGGTACCTGCAGAAGCCA GGGCAGTCTCCACAGCTCCTGATCTATTGGGTTCTACTCGGGCC TCCGGGTTCCTTGACAGGTTCACTGGCAGTGGATCAGGCACAGAT TTTACACTGAAAATCAGCAGAGTGGAGGCTGAGGATGTTGGGGTT TATTACTGCATGCAAGCTCTACAACTCCGTGCAGTTTGGCCAG GGGACCAAGCTGGAGATCAAAAC SEQ ID NO: 597	DIVMTQSPSLSLPVTGPGEPA QSLHNSGYNC LGS MQALQTPCS SISCRSSQSLHNSGYNCL SEQ ID NO: SEQ SEQ ID DWYLQKPGQSPQLLIYLG 596 ID NO: NO: 400 TRASGFPDRFSGSGSGTDF 371 TLKISRVEAEDVGYYCMQ ALQTPCSFGQGTKLEIK SEQ ID NO: 598			
CL-64841	GATATTGTGATGACTCAGTCTCCACTCTCCCTGCCCGTCACCCCT GGAGAGCCGGCCTCCATCTCCTGCAGGTCTAGTCAGAGCCTCCTG CATAGTAATGGATACAACCTGTTGGATTGGTACCTGCAGAAGCCA GGGCAGTCTCCACAGCTCCTGATCTATTGGGTTCTACTCGGGCC TCCGGGTTCCTTGACAGGTTCACTGGCAGTGGATCAGGCACAGAT TCTACACTGAAAATCAGCAGAGTGGAGGCTGAGGATGTTGGGGTT TATTACTGCATGCAAGCTCTACAACTCCGTGCAGTTTGGCCAG GGGACCAAGCTGGAGATCAAAAC SEQ ID NO: 599	DIVMTQSPSLSLPVTGPGEPA QSLHNSGYNC LGS MQALQTPCS SISCRSSQSLHNSGYNCL SEQ ID NO: SEQ SEQ ID DWYLQKPGQSPQLLIYLG 596 ID NO: NO: 400 TRASGFPDRFSGSGSGTDS 371 TLKISRVEAEDVGYYCMQ ALQTPCSFGQGTKLEIK SEQ ID NO: 600			
CL-64912	GATATTGTGATGACTCAGTCTCCACTCTCCCTGCCCGTCACCCCT GGAGAGCCGGCCTCCATCTCCTGCAGGTCTAGTCAGAGCCTCCTG CATAGTAATGGATACAACCTGTTGGATTGGTACCTGCAGAAGCCA GGGCAGTCTCCACAGCTCCTGATCTATTGGGTTCTACTCGGGCC TCCGGGTTCCTTGACAGGTTCACTGGCAGTGGATCAGGCACAGAT TTTACACTGAAAATCAGCAGAGTGGAGGCTGAGGATGTTGGGGTT TATTACTGCATGCAAGCTCTACAACTCCGTGCAGTTTGGCCAG GGGACCAAGCTGGAGATCAAAAC SEQ ID NO: 601	DIVMTQSPSLSLPVTGPGEPA QSLHNSGYNC LGS MQALQTPCS SISCRSSQSLHNSGYNCL SEQ ID NO: SEQ SEQ ID DWYLQKPGQSPQLLIYLG 596 ID NO: NO: 400 TRASGFPDRFSGSGSGTDF 371 TLKISRVEAEDVGYYCMQ ALQTPCSFGQGTKLEIK SEQ ID NO: 602			
CL-71642	GAA ATTGTGTTGACGCGAGTCTCCAGGCACCCCTGTCTTTGTCTCCA GGGGAAGAGCCACCCCTCTCCTGCAGGGCCAGTCAGAGTGTAGC AGCAGCTACTTAGCCTGGTACCAGCAGAAACCTGGCCAGGCTCCC AGGCTCCTCATCTATGGTGCATCCAGCAGGGCCACTGGCATCCCA GACAGGTTCACTGGCAGTGGGTCTGGGACAGACTTCACTCTCACC ATCAGCAGACTGGAGCCTGAAGATTTTGCAGTGATTACTGTCAG CAGTATGGTAGCTCACCTTTCACTTTCGGCCCTGGGACCAAGTG GATATCAAAAC SEQ ID NO: 603	EIV LTSQSPGTLSPGERA QSVSSSY GAS QYQSSPFT TLSCRASQSVSSSYLAWYQ SEQ ID NO: SEQ SEQ ID QKPGQAPRLLIYGASSRAT 426 ID NO: NO: 605 GIPDRFSGSGSGTDFTLTI 413 SRLEPEDFAVYYCQYQSS PFTFGPGTKVDIK SEQ ID NO: 604			
CL-74570 STIM022	GAA ATTGTGTTGACGCGAGTCTCCAGGCACCCCTGTCTTTGTCTCCA GGGGAAGAGCCACCCCTCTCCTGCAGGGCCAGTCAGAGTGTAGC AGCAGCTACTTAGCCTGGTACCAGCAGAAACCTGGCCAGGCTCCC AGGCTCCTCATCTATGGTGCATCCAGCAGGGCCACTGGCATCCCA GACAGGTTCACTGGCAGTGGGTCTGGGACAGACTTCACTCTCACC ATCAGCAGACTGGAACCTGAAGATTTTGCAGTATATTACTGTCAC CAGTATGGTAATTACCATTTCACTTTCGGCCCTGGGACCAAGTG GATATCAAAAC SEQ ID NO: 606	EIV LTSQSPGTLSPGERA QSVSSSY GAS HQYGNPFT TLSCRASQSVSSSYLAWYQ SEQ ID NO: SEQ SEQ ID QKPGQAPRLLIYGASSRAT 426 ID NO: NO: 608 GIPDRFSGSGSGTDFTLTI 413 SRLEPEDFAVYYCHQYGN PFTFGPGTKVDIK SEQ ID NO: 607			

SEQUENCE LISTING

The patent contains a lengthy sequence listing. A copy of the sequence listing is available in electronic form from the USPTO web site (<https://seqdata.uspto.gov/?pageRequest=docDetail&DocID=US12404330B2>). An electronic copy of the sequence listing will also be available from the USPTO upon request and payment of the fee set forth in 37 CFR 1.19(b)(3).

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What is claimed is:

1. An isolated antibody that binds the extracellular domain of human and mouse ICOS, wherein the antibody comprises a VH domain comprising a set of heavy chain complementarity determining regions (HCDRs) HCDR1, HCDR2 and HCDR3, and a VL domain comprising a set of light chain complementarity determining regions (LCDRs) LCDR1, LCDR2 and LCDR3, wherein the HCDRs and LCDRs have amino acid sequences as follows:
 - HCDR1 comprises an amino acid sequence identical to SEQ ID NO: 992 or SEQ ID NO: 995;
 - HCDR2 comprises an amino acid sequence identical to SEQ ID NO: 993 or SEQ ID NO: 996;
 - HCDR3 comprises an amino acid sequence identical to SEQ ID NO: 994 or SEQ ID NO: 997;
 - LCDR1 comprises an amino acid sequence identical to SEQ ID NO: 1000 or SEQ ID NO: 1003;
 - LCDR2 comprises an amino acid sequence identical to SEQ ID NO: 1001 or SEQ ID NO: 1004; and
 - LCDR3 comprises an amino acid sequence identical to SEQ ID NO: 1002.
2. The antibody according to claim 1, wherein the antibody comprises a VH domain which comprises the amino acid sequence of the STIM021 VH domain or which comprises that amino acid sequence with 1, 2, 3, 4 or 5 conservative amino acid substitutions, and/or wherein the antibody comprises a VL domain which comprises the amino acid sequence of the STIM021 VL domain or which comprises that amino acid sequence with 1, 2, 3, 4 or 5 conservative amino acid substitutions.
3. The antibody according to claim 1, wherein the antibody comprises a VH domain having at least 95% sequence identity to the amino acid sequence of the STIM021 VH domain, and wherein the antibody comprises a VL domain having at least 95% sequence identity to the amino acid sequence of the STIM021 VL domain.
4. The antibody according to claim 1, comprising the STIM021 VH domain and the STIM021 VL domain.
5. The antibody according to claim 1, wherein the VH domain comprises an amino acid sequence of SEQ ID NO: 998 and the VL domain comprises an amino acid sequence of SEQ ID NO: 1005.
6. A method of modulating the balance of regulatory T cells (Tregs) to effector T cells (Teffs) to increase Teff response in a patient, comprising administering an antibody according to claim 1 to the patient.
7. A method of treating a disease or condition amenable to therapy by depleting regulatory T cells (Tregs) and/or increasing effector T cell (Teff) response in a patient, the method comprising administering an antibody according to claim 1 to the patient.
8. A method of treating cancer in a human patient, comprising administering an antibody according to claim 1 to the patient.

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9. The method according to claim 6, wherein the method comprises administering the antibody and another therapeutic agent and/or radiation therapy to the patient.
10. The method according to claim 6, wherein the anti-ICOS antibody is conjugated to a pro-drug, and wherein the method comprises administering the anti-ICOS antibody to a patient; and selectively activating the pro-drug at a target tissue site.
11. A method of treating a patient, the method comprising administering an anti-ICOS antibody according to claim 1 to a patient who has an increased level of ICOS-positive regulatory T cells following treatment with another therapeutic agent.
12. A method of treating cancer in a patient by in vivo vaccination of the patient against their cancer cells, the method comprising treating the patient with a therapy that causes immunological cell death of the cancer cells, resulting in presentation of antigen to antigen-specific effector T cells, and administering an anti-ICOS antibody according to claim 1 to the patient, wherein the anti-ICOS antibody enhances the antigen-specific effector T cell response.
13. A method of treating cancer in a patient by in vivo vaccination of the patient against their cancer cells, the method comprising administering an anti-ICOS antibody according to claim 1, wherein the patient is one who has been previously treated with a therapy that causes immunological cell death of the cancer cells, resulting in presentation of antigen to antigen-specific effector T cells, and wherein the anti-ICOS antibody enhances the antigen-specific effector T cell response.
14. A method of treating a cancer in a patient, wherein the cancer is or has been characterised as being positive for expression of ICOS ligand and/or FOXP3, the method comprising administering an anti-ICOS antibody according to claim 1 to the patient.
15. A method of treating a cancer in a patient, wherein the cancer is or has been characterised as being refractory to treatment with an immunooncology drug, the method comprising administering an anti-ICOS antibody according to claim 1 to the patient.
16. The method according to claim 15, wherein the method comprises treating the patient with an anti-CD20 antibody;
 - determining that the cancer is not responsive to the anti-CD20 antibody;
 - testing a sample from a patient to determine that the cancer expresses ICOS ligand;
 - selecting the patient for treatment with the anti-ICOS antibody; and
 - administering the anti-ICOS antibody to the patient.

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